

Master MapLibre Components-Functions-Features.pdf

Cumulative KFM MapLibre, Tile, Renderer, Style, UI, Evidence Drawer, and Release Architecture
Atlas

Field	Value
Version	v1.9 cumulative update
Run date	2026-05-10
Attached source count	41 PDFs visible in /mnt/data, including prior master and new user-uploaded New Ideas.pdf. Current prompt is SRC-RUN-PROMPT.
Prior master PDF detected	Yes - v1.8 copied as SRC-MASTER-PREV before overwrite; prior pages 504; sha256 prefix 7e0da1ddcf77c592.
Output mode	PDF only. No ZIP, tarball, scaffold folder, repository patch, loose source tree, or extra user-facing deliverable.
Truth posture	CONFIRMED source scans and retained doctrine; PROPOSED implementation guidance; UNKNOWN current repo/runtime depth; NEEDS VERIFICATION for versions, hosting, rights, source endpoints, policy enforcement, performance budgets, mobile/native parity, and path placement.
Evidence limits	No mounted KFM repo, tests, workflows, dashboards, logs, release manifests, or runtime traces were available in this run. Paths in sources are source-reported and PROPOSED unless future repo evidence verifies them.
Source ledger count	42 current-run crosswalk rows: SRC-RUN-PROMPT, SRC-MASTER-PREV, 39 retained non-master source files from v1.8, and NEW SRC-064. The full v1.8 source ledger remains appended in Appendix A.
Idea count	2778 cumulative idea records: 2658 retained from v1.8 baseline + 120 v1.9 delta records.
New discoveries count	120 delta records from SRC-064.
Expanded ideas count	25 category clusters expanded or sharpened; F and D are retained with no new SRC-064 material.
Superseded ideas	0 - no attached evidence invalidated a prior idea.
Open verification items	505 cumulative estimate: prior 445 + 60 new or sharpened verification items.

Core determination: MapLibre remains a downstream renderer and interaction runtime. Tiles, PMTiles, MVT, MLT, COGs, style JSON, popups, screenshots, Story Nodes, 3D scenes, graph projections, catalog records, and AI answers remain downstream carriers, not sovereign truth.

1. Table of Contents

- 1. Cover Page and run metadata
- 2. Executive Determination
- 3. Continuity and Delta Since Last Run
- 4. Source Ledger
- 5. Master Idea Index - v1.9 Delta
- 6. Category Atlas
- 7. Detailed Category Chapters
- 8. Components-Functions-Features Matrix
- 9. Tile Strategy Chapter
- 10. Governance and Trust-Membrane Chapter
- 11. Implementation-Ready Object Map
- 12. Validation and Test Plan
- 13. Anti-Patterns Register
- 14. Open Questions and Verification Backlog
- 15. How to Continue This Master PDF Next Run
- Final PDF Self-Check
- Appendix A. Full retained v1.8 Master PDF baseline (SRC-MASTER-PREV, appended page-for-page)

Navigation note: this update packet is current-run specific. The full v1.8 baseline follows as Appendix A so prior source ledger entries, category chapters, matrices, idea IDs, and open questions are not silently deleted.

2. Executive Determination

CONFIRMED from SRC-MASTER-PREV and retained doctrine: MapLibre is a disciplined 2D renderer and interaction runtime inside a governed KFM shell. It is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority, or AI authority. Public surfaces use governed APIs, released artifacts, catalog records, tile services, EvidenceBundle resolution, and release manifests.

CONFIRMED from this run: the prior v1.8 master baseline was detected, copied before overwrite, and appended page-for-page. One new source was analyzed: SRC-064 New Ideas.pdf (634 pages, sha256 prefix 2bdc40e3df70cea6).

The strongest SRC-064 additions are OCI/ORAS publication of PMTiles/GeoParquet/COG with Cosign, STAC/DCAT/PROV mapping for tile/raster/vector artifacts, WZDx and KGS M-118 PMTiles pipelines, deterministic PMTiles build rules, IIIF/Allmaps historic-map overlays, canonical digest and attestation patterns for PMTiles/COGs, MapLibre ecosystem watchlist items including MLT/Martin/Native Vulkan, Pydantic/JSON-Schema contracts for map registries, offline-first capture to STAC/PROV/PMTiles/COG, LiDAR/GLO terrain and historic overlay workflows, and Focus Mode/Story Node governance constraints.

PROPOSED synthesis: v1.9 should be treated as an artifact-provenance, PMTiles/COG verification, source-watch, field-capture, historic-overlay, terrain/Cesium, Focus-Mode, and contract-governance update. It does not reorganize the prior master. It adds one source, preserves all prior records, and sharpens object maps, test gates, tile strategy, and release backlogs.

UNKNOWN: actual repo topology, package manager, schema home, UI component homes, MapLibre GL JS target version, MLT implementation status, PMTiles tooling, PMTiles/COG hosting headers, browser/device performance, mobile/native parity, plugin allowlist, source rights, policy enforcement, release workflow, and branch protection remain unverified.

NEEDS VERIFICATION: every package version, tool version, source URL, feed schema, hosted header, artifact digest, CARE consent assertion, OCI referrer, MLT support claim, local Focus Mode setting, and proposed file path before operational use.

Highest-risk anti-pattern: loading a public map source, PMTiles archive, COG, MLT tile, STAC search result, field note layer, historic overlay, 3D scene, Story Node, popup, or Focus Mode answer before EvidenceBundle, manifest, PolicyDecision, PromotionDecision, citation validation, proof object, and rollback target exist.

Metric	Value
Previous version	v1.8 cumulative update
Current version	v1.9 cumulative update
Prior ideas carried forward	2658
New SRC-064 delta ideas	120
Total cumulative idea count	2778
Superseded ideas	0
Open verification items	505 estimate

3. Continuity and Delta Since Last Run

SRC-MASTER-PREV is the prior v1.8 Master MapLibre Components-Functions-Features.pdf. It remains the cumulative memory baseline and is appended page-for-page. It does not outrank original evidence. Current-run source evidence can refine, expand, conflict with, or supersede its entries. No supersession occurred in this run.

Delta item	Result
Prior version detected	v1.8, 504 pages, sha256 prefix 7e0da1ddcf77c592
Version rule applied	+0.1 ordinary cumulative update. No fundamental structural reorganization.
New sources added	SRC-064 New Ideas.pdf.
Ideas retained	All 2658 prior idea records preserved by appending SRC-MASTER-PREV unchanged.
New records	120 ML-064-* delta records.
Merged / superseded	0 merged, 0 superseded.
Conflicts discovered	No direct source conflict. Main tension remains implementation maturity: SRC-064 includes implementation-style paths and code snippets, but no mounted repo/runtime proof was available.
Open questions carried forward	Package versions, schema home, source rights, MLT status, mobile parity, hosting headers, sensitive geometry, release manifest home, CI workflow, accessibility and performance budgets.
New open questions	ORAS/OCI refererrs; PMTiles tilehash manifests; COG internal chunk manifests; Allmaps plugin governance; field-capture redaction receipts; GLO/LiDAR sensitivity; MapLibre Native Vulkan parity; MLT production readiness; WMS/WMTS source controls.

Prior idea range / cluster	Current status	Source that changed status	Note
All prior idea IDs through v1.8	RETAINED	SRC-MASTER-PREV	No prior idea was deleted or superseded.
PMTiles / static delivery clusters	EXPANDED	SRC-064 pp.1-5, 88-92, 214-217, 287-290	Adds OCI/ORAS, deterministic build, offline field capture, Range/CDN validation.
COG / raster / terrain clusters	EXPANDED	SRC-064 pp.60-64, 225-229, 534-542, 618-627	Adds COG sidecar hashes, LiDAR derivatives, Sentinel-1/HydroGNSS governance.
MapLibre UI / view registry clusters	EXPANDED	SRC-064 pp.101-114, 216, 238-242	Adds view registry constraints, render hints, preview renderer, source-layer checks.
Focus Mode / Story Node clusters	EXPANDED	SRC-064 pp.363-369, 521-522, 537-538, 552-567	Adds graph health, docs lint, LiDAR/GLO and explainability constraints.
Sensitive geometry / rights clusters	EXPANDED	SRC-064 pp.92-96, 287-290, 537-538, 620-627	Adds sovereignty label sync, redacted field capture, SAR/HydroGNSS H3 generalization.

4. Source Ledger

This ledger is a control surface, not a bibliography. It states what each source can support and what it cannot prove. Current external facts embedded in source PDFs remain source-supported but NEEDS VERIFICATION before operational package pins, access decisions, source activation, or release claims.

source_id	file / document title	source type	status	role in this PDF	MapLibre / tile relevance	limitations
SRC-RUN-PR OMPT	Current user prompt	Prompt	CONFIRMED	Controls output shape, truth labels, source handling, PDF-only rule, and KFM invariants.	High requirements source.	Instruction source; not implementation proof.
SRC-MASTER -PREV	Master MapLibre Components-Fun ctions-Features.pdf v1.8	Prior master PDF	CONFIRMED / RETAINED	Cumulative baseline appended page-for-page as Appendix A.	Highest continuity source; 2658 prior ideas retained.	Does not outrank original evidence.
SRC-064	New Ideas.pdf	New Ideas packet	NEW	Adds v1.9 delta records and new MapLibre/tile/Focus/COG/PMTiles governance ideas.	Very high: PMTiles, COG, MapLibre, MLT, Martin, Focus Mode, Story Nodes, LiDAR/GLO, field capture.	No mounted repo proof; external facts and version-sensitive statements were not revalidated.
SRC-003	KFM_MapLibre_Operating_Archite cture_Governed_UI_AI_Interaction _Manual_REVISED.pdf	Map/UI doctrine	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Renderer boundary, governed shell, Evidence Drawer, Focus Mode.	Current implementatio n remains UNKNOWN.
SRC-004	Directory Rules.pdf	Directory doctrine	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Responsibility-root, schema-home, lifecycle, ADR/drift rules.	Current implementatio n remains UNKNOWN.
SRC-005	Kansas_Frontier_Matrix_Pipeline_L iving_Implementation_Manual_v0.3. pdf	Pipeline manual	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Lifecycle law, receipts, promotion, query/save/recompile loop.	Current implementatio n remains UNKNOWN.
SRC-006	Kansas_Frontier_Matrix_Definitive Greenfield_Building_Plan_v1_1.pdf	Build plan	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Trust spine, pre-RAW events, public-safe defaults, map architecture.	Current implementatio n remains UNKNOWN.
SRC-007	KFM_Whole_UI_Governed_AI_Exp ansion_Report.pdf	UI + AI plan	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Whole UI, governed API, MapLibre boundary.	Current implementatio n remains UNKNOWN.
SRC-008	KFM_Governed_AI_Extended_Pro _Source_Ledger_PDF_Only_Archit ecture_Report_2026-04-20.pdf	Governed AI plan	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Provider-neutral AI, finite outcomes, citation validation.	Current implementatio n remains UNKNOWN.
SRC-009	Ollama & Ubuntu Information.pdf	Runtime guide	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Local model behind governed API; no direct public model path.	Current implementatio n remains UNKNOWN.
SRC-010	kfm_encyclopedia.pdf	Capability encyclopedia	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	System-wide KFM domain and capability atlas.	Current implementatio n remains UNKNOWN.
SRC-011	# Kansas Frontier Matrix Implementation Reference.pdf	Implementation reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Prior public-repo summary and MapLibre/PMTiles stack signals.	Current implementatio n remains UNKNOWN.
SRC-012	a-primer-of-gis-fundamental-geogra phic-and-cartographic-concepts.pdf	GIS/cartography reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Representation/projection/map communication concepts.	Current implementatio n remains UNKNOWN.
SRC-013	Advanced-SQL-Concepts.pdf	SQL reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Analytic SQL concepts; low direct map relevance.	Current implementatio n remains UNKNOWN.
SRC-014	AI_Concepts_Using_Python.pdf	AI reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	AI/data workflow background; low direct map relevance.	Current implementatio n remains UNKNOWN.
SRC-015	Archaeological 3D GIS.pdf	3D GIS reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	3D GIS and archaeological visualization context.	Current implementatio n remains UNKNOWN.
SRC-017	Designing Great Web APIs.pdf	API reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	API contract and resource design.	Current implementatio n remains UNKNOWN.
SRC-018	developing-time-oriented-database- applications-in-sql.pdf	Temporal DB reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Valid/transaction time concepts.	Current implementatio n remains UNKNOWN.
SRC-019	Domain-Driven Design Reference.pdf	DDD reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Bounded context and published language.	Current implementatio n remains UNKNOWN.

source_id	file / document title	source type	status	role in this PDF	MapLibre / tile relevance	limitations
SRC-020	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf	GIS lab/textbook	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Environmental GIS/hazards/climate layer context.	Current implementation remains UNKNOWN.
SRC-022	GIS Succinctly.pdf	GIS reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Spatial database and geometry basics.	Current implementation remains UNKNOWN.
SRC-023	GIS in Sustainable Urban Planning and Management.pdf	Urban GIS reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Planning support, indicators, resilience.	Current implementation remains UNKNOWN.
SRC-024	instant-web-scraping-with-java.pdf	Scraping reference	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Web acquisition and error handling background.	Current implementation remains UNKNOWN.
SRC-025	KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Receipts, deterministic identity, promotion gates.	Current implementation remains UNKNOWN.
SRC-026	KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	PMTiles, MLT, map stack guidance.	Current implementation remains UNKNOWN.
SRC-027	KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Pre-RAW, watchers, catalog integrity.	Current implementation remains UNKNOWN.
SRC-056	KFM_Pass_15_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Runtime proof lanes and map-serving proof parity.	Current implementation remains UNKNOWN.
SRC-057	KFM_Pass_16_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Geospatial delivery, PMTiles, MapLibre default, promotion gates.	Current implementation remains UNKNOWN.
SRC-028	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Agriculture layers, catalog closure, MapLibre/Evidence Drawer.	Current implementation remains UNKNOWN.
SRC-029	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Archaeology exact-location denial and generalized map outputs.	Current implementation remains UNKNOWN.
SRC-030	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Air/smoke/EO layers and knowledge-character labels.	Current implementation remains UNKNOWN.
SRC-031	KFM_Fauna_Architecture_PDF_Only_Report.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Fauna occurrence geoprivacy and public-safe products.	Current implementation remains UNKNOWN.
SRC-032	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Rare plant controls and occurrence redaction.	Current implementation remains UNKNOWN.
SRC-033	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Geology/resource layers and public-safe geometry.	Current implementation remains UNKNOWN.
SRC-034	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Habitat assignment proof slice and drawer payloads.	Current implementation remains UNKNOWN.
SRC-035	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Habitat patches, suitability, connectivity.	Current implementation remains UNKNOWN.
SRC-036	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Hazards boundary; no emergency alerting.	Current implementation remains UNKNOWN.

source_id	file / document title	source type	status	role in this PDF	MapLibre / tile relevance	limitations
SRC-037	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Hydrology proof lane and NFHL/USGS/HUC layers.	Current implementation remains UNKNOWN.
SRC-038	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Living-person/DNA restrictions and temporal land assertions.	Current implementation remains UNKNOWN.
SRC-039	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Transport/corridor layers, graph projections, drawer payloads.	Current implementation remains UNKNOWN.
SRC-040	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Settlement/infrastructure layer controls and sensitive assets.	Current implementation remains UNKNOWN.
SRC-041	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	Domain blueprint	RETAINED	Retained via SRC-MASTER-PREV; not re-extracted as new delta.	Soil maps, SSURGO/gSSURGO, PMTiles/COG lineage.	Current implementation remains UNKNOWN.

5. Master Idea Index - v1.9 Delta

The full cumulative v1.8 idea index of 2658 prior ideas is retained in Appendix A via SRC-MASTER-PREV. The records below add the v1.9 delta from SRC-064. Implementation and validation consequences are PROPOSED unless a future mounted-repo inspection verifies them.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-001	WaterWatch percentiles as hydrology layer context	P. Time-Aware Map Interaction and Timeline	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 1	Kansas streamflow percentile context can explain whether low-flow sensor flags reflect real hydrologic conditions or sensor/data drift.	Expose hydrology anomaly layers as released context, not as raw gauge pulls.	Cross-check HUC percentiles with drought indicators and station metadata.
ML-064-002	OCI registry as storage for PMTiles, GeoParquet, and COG	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 1-5	The source proposes ORAS + Cosign publication of PMTiles, GeoParquet, and COG artifacts with tags, digests, referrers, SBOM, and PROV.	Allow tile/raster/vector artifacts to be fetched reproducibly from OCI only when digest and provenance are in the catalog.	Validate mediaType, digest, Cosign verification, and PROV referrers.
ML-064-003	Digest-pinned artifact pulls support rollback	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 3-5	OCI pulls by sha256 digest are treated as immutable; tags are friendly but mutable.	MapReleaseManifest should prefer digest-pinned PMTiles/COG/GeoParquet references.	Test that production manifests include digest and rollback target.
ML-064-004	STAC/DCAT catalog entries carry OCI distribution details	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 3-4	The mapping records registry, repository, tag, digest, artifact media types, sizes, and provenance_ref in catalog records.	TileArtifactManifest should align its artifact list with STAC/DCAT OCI distribution metadata.	Catalog validator checks digest, size, mediaType, and provenance_ref.
ML-064-005	Cosign referrers link SBOM and PROV to geospatial artifacts	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 4-5	The source uses OCI referrers for SBOM and PROV JSON-LD attachments.	Release gates should verify SBOM and PROV referrers for public tiles and rasters.	Cosign verify and referrer-presence CI checks.
ML-064-006	Energy and CO2 promotion policy affects map artifact release	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 5-8	Promotion can be denied unless catalog items include energy/CO2 telemetry and Cosign-verified provenance.	Map tile/COG release manifests can include sustainability telemetry without making it a map truth source.	OPA/Rego gate for energy_kwh, co2_kg, signature, commit_sha, and SBOM ref.
ML-064-007	CI provenance JSON-LD becomes run evidence for map builds	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 9-13	Each ETL run can emit signed JSON-LD with workflow, job id, commit, artifacts, hashes, SBOM, and energy telemetry.	Tile builds should produce RunReceipt/PROV records that Evidence Drawer can reference.	Schema check for provenance JSON-LD; signature artifact verification.
ML-064-008	Dagster asset model treats tiles as governed artifacts	Y. Implementation Backlog and PR Plan	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 34-36	The source frames tables, Parquet, tiles, and JSON as assets with post-materialization checks.	Treat PMTiles/COG/GeoParquet outputs as assets with explicit checks, not incidental files.	Asset check for row counts, tile coverage, schema, and index health.
ML-064-009	STAC/DCAT/PROV mapping for GeoParquet, COG, and PMTiles	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 60-64	Domain fields map into STAC geometry/properties/assets, DCAT distributions, and PROV activities for clean publication of GeoParquet/COG/PMTiles.	LayerManifest and TileArtifactManifest should inherit asset roles and provenance from catalog records.	STAC/DCAT/PROV validation with mediaType and roles.
ML-064-010	COG and PMTiles have explicit STAC asset roles	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 61-64	COGs are image/tiff GeoTIFF visual/data assets; PMTiles are application/vnd.pmtiles map/tiles assets; TileJSON is optional metadata.	Use roles to tell the MapLibre shell which assets are renderable, quicklook, data, or metadata.	Asset-role validation and disallow unknown media types.
ML-064-011	GeoJSON geometry remains catalog geometry, not proof	G. GeoJSON and Runtime Data	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 61-63	GeoJSON geometry is mapped to STAC geometry and optional bbox, but remains a carrier.	Click resolution should use feature IDs to fetch EvidenceBundle, not trust GeoJSON alone.	GeoJSON schema and EvidenceBundle resolution tests.
ML-064-012	Frictionless GeoJSON/TopoJSON gating catches manifest errors early	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 78	The source points to field-level validation for GeoJSON/TopoJSON in manifests and DCAT gates.	Add early schema gates before tile generation or catalog release.	Validate geometry type, CRS, bbox, and field type before materialization.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-013	WZDx pipeline produces PMTiles for fast tiled visualization	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 88-92	WZDx event snapshots are canonicalized, packaged to PMTiles, and represented in STAC/PROV/checksum artifacts.	Road-work or incident layers can be previewed via PMTiles only after snapshot validation.	Validate event namespace/version, PMTiles asset, prov.jsonld, manifest_ref, and SBOM refs.
ML-064-014	PMTiles generation from COG or GeoParquet uses tippecanoe/tegola	H. Vector Tiles: MVT and MLT	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 90	The source describes PMTiles generation from COG/GeoParquet via a tiling pipeline.	Choose tiling backend per data type, but output must be manifest-bound.	Tiler version, flags, zoom range, and input digest checks.
ML-064-015	Static CDN PMTiles avoids tile server for previews	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 90-92	The source says MapLibre can consume PMTiles directly from static CDN without a tile server.	Prefer serverless PMTiles for stable, public-safe layers with Range-capable hosting.	Range/CORS probe, cache headers, and checksum verification.
ML-064-016	MapLibre addSource/addLayer PMTiles wiring is downstream only	B. MapLibre GL JS Web Shell	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 92	The source provides map.addSource and map.addLayer snippets for PMTiles.	UI wiring should be generated from released manifests, not hand-coded raw URLs.	No unreleased tile URL in style/source registry; click opens STAC metadata/Evidence Drawer.
ML-064-017	Cached PMTiles can be degradation fallback	S. Accessibility, UX, and Trust-Visible States	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 91	The source proposes degrading to cached PMTiles if publish path is impaired and budgets burn.	UI may show cached/stale layer state when publication is impaired.	Stale-source badge, cache age, and ABSTAIN behavior tests.
ML-064-018	OCI plus PMTiles supports portable registry/CDN delivery	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 92	The source states OCI and PMTiles work across registries/CDNs and give serverless map previews.	Use portable artifact references in release manifests with checksums.	Resolve OCI snapshot and PMTiles URL from catalog; verify both.
ML-064-019	Sovereignty labels must stay synchronized across STAC/DCAT/PROV	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 92-96	A gate enforces that indigenous_constraint values do not drift across STAC assets, DCAT distributions, and PROV activities.	Layer manifests must carry policy labels from the catalog; MapLibre must receive already-redacted artifacts.	OPA gate for label enum, exceptions, approver, and signed_at.
ML-064-020	View registry forbids direct Neo4j handles in map UI	A. Renderer Boundary and KFM Trust Law	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 101-114	The source defines view metadata that points to STAC/DCAT/PMTiles/COG and explicitly avoids raw DB handles.	MapLibre views should receive layer pointers and render hints, not database query handles.	Ajv/JSON Schema plus OPA gate for no direct DB fields.
ML-064-021	Route chunks and lazy hydration support governed map shells	B. MapLibre GL JS Web Shell	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 101-114	Vite routeChunk, lazyLoad, and view registry validation are used for map routes.	MapLibre shells should hydrate route-specific chunks after governance state is available.	Validate view registry and chunk names in CI.
ML-064-022	Layer pointers use datasources not queries	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 101-114	datasources holds STAC/DCAT/PMTiles endpoints; renderHints describe tileProtocol and worker options.	LayerManifest should separate catalog sources from render hints.	Schema requires stac/dcat/pmtiles/cog fields when applicable.
ML-064-023	tileProtocol enum covers PMTiles, XYZ, WMTS, and WMS	L. WMS / WMTS / External Map Services	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 102-110	View specs allow tileProtocol values pmtiles, xyz, wmts, wms.	External service layers need explicit protocol and access/governance posture.	Protocol enum validation; WMS/WMTS attribution and terms checks.
ML-064-024	Performance and accessibility budgets are registry data	S. Accessibility, UX, and Trust-Visible States	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 106-114	View registry passes a11y and performance budgets to Map components and can fail Ajv gates.	Budget thresholds should be release criteria for heavy map layers.	Preview build measures tile load, route load, and accessibility score.
ML-064-025	KGS M-118 can become topology-safe serverless PMTiles	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 114-116	The source proposes fetching KGS M-118 as GeoJSON, repairing topology, storing GeoParquet, then tiling to PMTiles/MVT.	Surficial geology layers should become processed, cataloged artifacts before MapLibre use.	Validate topology, GeoParquet schema, PMTiles digest, and HTTP Range hosting.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-026	ArcGIS FeatureLayers can be ETL inputs, not public client dependencies	G. GeoJSON and Runtime Data	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 114-115	FeatureLayers can return GeoJSON for straight-through ETL.	Use ArcGIS/GeoJSON as RAW/WORK intake, then publish governed outputs.	SourceDescriptor and rights checks before ingest; no browser direct source.
ML-064-027	Topology-safe simplification avoids shared-boundary cracks	H. Vector Tiles: MVT and MLT	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 116-119	Mapshaper/TopoJSON shared-arc simplification is recommended to preserve boundaries before web tiling.	Use topology-aware simplification prior to PMTiles/MVT build.	Area/perimeter/Hausdorff and shared-border regression tests.
ML-064-028	Normalized GeoJSON geometry checksums support stable feature identity	G. GeoJSON and Runtime Data	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 125-140	Geometry checksum is computed from normalized GeoJSON with CRS/precision controls.	Feature IDs and promoted should incorporate stable geometry or record hashes where appropriate.	Checksum recomputation and precision/CRS tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-029	PostGIS sinks are internal, not UI truth	J. Martin and Server-Mediated Tile Serving	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 127	The source mentions upserts to Kafka, PostGIS, REST, or GraphQL sinks.	If PostGIS backs tiles or APIs, it remains behind governed services.	No direct browser connection; role and query scope tests.
ML-064-030	Pre-commit canonicalizes changed JSON/GeoJSON/Parquet	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 138	Changed structured assets are re-canonicalized and re-hashed before commits.	Use pre-commit or CI to prevent non-deterministic layer manifests and catalog files.	Canonicalization diff test.
ML-064-031	PMTiles deterministic pipeline begins with canonical GeoParquet	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 214-217	A page diagram and recipe show GeoParquet -> Tippecanoe/Planetiler -> PMTiles -> MapLibre.	Treat GeoParquet as canonical processed input for vector tile builds.	Input schema lock, sorted rows, fixed precision, fixed field order.
ML-064-032	Tiler versions and flags must be pinned	R. Plugin, Wrapper, and Dependency Governance	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 215-217	Tippecanoe/Planetiler versions, flags, and zoom strategies are part of provenance.	TileArtifactManifest should record tiler, version, flags, zooms, and input STAC links.	Rebuild comparison with same flags and inputs.
ML-064-033	Attribute whitelisting keeps vector tiles small and safer	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 215-217	The source warns that unneeded attributes bloat tiles at all zooms.	Layer manifests should declare allowed tile attributes.	Tile size and attribute leak tests.
ML-064-034	PMTiles sha256 belongs in STAC records	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 216-217	The source says to keep PMTiles sha256 in the STAC record referenced by the UI.	MapLibre fetch verification should compare against catalog checksum.	Client or prefetch checksum probe; manifest digest check.
ML-064-035	Native and embedded apps can ship single PMTiles files	C. MapLibre Native / Mobile / Platform Parity	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 216-217	The source notes native/embedded demos can ship the single .pmtiles file or use signed blob storage.	Mobile/native parity can use offline PMTiles only with the same EvidenceBundle and release metadata.	Offline package digest, expiry, and release-state checks.
ML-064-036	IIIF plus Allmaps overlays historic maps into MapLibre/Leaflet	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 217-219	IIIF manifests and Allmaps georeference annotations can warp historic maps into web maps without rehosting.	Historic map Story Nodes can reference original rights/metadata and render overlays.	Validate IIIF manifest rights, GCP provenance, and annotation digest.
ML-064-037	Allmaps WarpedMapLayer belongs in Story Node map panels	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 218-219	The source mentions Allmaps MapLibre docs and WarpedMapLayer rendering in a Story Node map panel.	Use as a conditional overlay plugin with dependency governance.	Plugin allowlist and attribution/rights test.
ML-064-038	Artifact canonicalization covers GeoJSON, PMTiles, and COG	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 225-229	The source defines deterministic geometry, artifact checksums, PMTiles tilehash, and COG tile/chunk hashes.	All public map artifacts need canonical digests and sidecar manifests.	Recompute geom_sha256/record_sha256 and sha256sum PMTiles/COG.
ML-064-039	PMTiles tilehash sidecar supports offline verification	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 226-229	PMTiles byte ranges or tile indexes can be hashed into sidecar manifests.	TileArtifactManifest can include per-tile or directory hashes for high-integrity releases.	Verify tilehash sidecar against PMTiles file.
ML-064-040	COG internal tiles and overviews get sidecar hashes	K. Raster, COG, DEM, Terrain, Hillshade	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 226-229	COG byte ranges from TIFF IFDs are hashed to sidecar manifests.	Raster release should support file-level and internal-chunk verification.	COG tilehash and overview verification.
ML-064-041	In-toto and Cosign attest map artifacts	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 227-229	The source signs an attestation predicate over tiles.pmtiles and image.tif with tool versions and canonicalization steps.	Release manifests should reference attestations for tiles, COGs, and build snapshots.	cosign verify-attestation and predicate schema tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-042	STAC integration stores geom and record digests	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 228-229	geom_sha256 and record_sha256 are placed in Feature properties; asset checksums and attestation bundles are linked.	Evidence Drawer can show feature digest and artifact digest.	Digest presence and recomputation tests.
ML-064-043	Delta-aware CI skips noiseless PRs but fails closed on material changes	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 229-234	The source proposes material path detection, policy logs, and fail-closed Contest gates.	Map style/tile/catalog CI can short-circuit only when no material release inputs changed.	Material path fixture; no-change and changed-path tests.
ML-064-044	WZDx GeoJSON is a transport layer source family	G. GeoJSON and Runtime Data	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 235, 611-621	The source treats WZDx as a harmonized GeoJSON work-zone/road-events model.	Transport event layers should carry source family, schema version, and temporal fields.	GeoJSON geometry, WZDx schema, and event freshness tests.
ML-064-045	USGS GeoJSON feeds are STAC-ready event sources	G. GeoJSON and Runtime Data	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 236-237	The source notes real-time USGS GeoJSON feeds can be filtered and mapped.	Event point layers can be cataloged and rendered after freshness and source checks.	Feed schema and stale/expired feed tests.
ML-064-046	Preview renderer is STAC-driven and MapLibre-headless	V. Exports, Screenshots, Reports, and Citation Preservation	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 238-242	A KFM Preview Renderer uses MapLibre GL/Puppeteer or rasterizer, with style JSON, source layers, and camera.	Generate review screenshots from pinned styles and released or candidate artifacts.	Renderer version, style digest, camera, and output hash checks.
ML-064-047	Preview renderer triggers on STAC, style, or renderer changes	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 241	Renderer CI is triggered by changes to STAC Items, styles, or renderer code.	Visual regression should run when inputs or rendering logic change.	Pixel diff/Screenshot diff threshold and provenance.
ML-064-048	Story Nodes and Focus Mode consume preview evidence	N. Evidence Drawer Payloads and Click Resolution	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 242	Preview renderer wiring supports Story Nodes and Focus Mode, with deterministic renderer/style pins.	Evidence Drawer can link preview screenshot, style digest, and artifact digest.	Click-to-drawer and preview hash tests.
ML-064-049	MapLibre ecosystem performance changes are watchlist items	Z. Open Questions and Verification Backlog	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 248	The source notes evolving Native Vulkan, MLT, and Martin improvements.	Treat version-specific ecosystem claims as NEEDS VERIFICATION before adoption.	Quarterly ecosystem watchlist and official-source verification.
ML-064-050	MapLibre Native Android Vulkan is a parity opportunity	C. MapLibre Native / Mobile / Platform Parity	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 248	The source says the Android SDK has a Vulkan backend for performance and future-proofing.	Native/mobile parity should be tested against web style/profile and evidence-drawer behavior.	Android render, style subset, offline PMTiles, and accessibility tests.
ML-064-051	MLT is a pilot candidate, not default production format	H. Vector Tiles: MVT and MLT	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 248	The source describes MapLibre Tile as stable and smaller/faster than MVT, but this run did not verify official status.	Keep MLT in pilot path until tooling, renderer support, and validation mature.	MVT-vs-MLT compatibility, decode, visual, and fallback tests.
ML-064-052	Martin improvements justify server-mediated tile serving review	J. Martin and Server-Mediated Tile Serving	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 248	The source mentions Martin tile server improvements and smaller MapLibre vector tile roadmap.	Evaluate Martin where dynamic PostGIS/PMTiles/MBTiles mediation is needed.	Martin config, access, source, and performance tests.
ML-064-053	JSON Schema and Pydantic align layer manifests and telemetry	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 249-252	Typed machine-parseable I/O keeps ETL jobs, validators, UI registries, and telemetry aligned.	Author schema-backed LayerManifest, StyleManifest, TileArtifactManifest, telemetry, and ViewRegistry objects.	Schema export, schema lint, fixture validation, and Contest gate.
ML-064-054	LLM structured outputs must validate against map contracts	O. Focus Mode and Governed AI Map Context	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 250-252	LLM tool outputs should be parsed into Pydantic/JSON Schema and fail closed if invalid.	Focus Mode map context responses should use schema validation and citation validation.	ANSWER/ABSTAIN/DENY/ERROR fixture tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-055	UI registries and analytics events use same contracts	R. Plugin, Wrapper, and Dependency Governance	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 252	The source says UI registries and telemetry should be Pydantic-backed and contract-aligned.	MapLibre configs and telemetry events should share schema semantics with backend.	Front-end boundary validation and telemetry schema tests.
ML-064-056	Offline-first field capture can deliver PMTiles and COG	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 287-290	Field teams capture offline, redact on-device, emit STAC/PROV, pack vector notes into PMTiles, and quicklooks into COG.	Field maps should publish only after redaction, classification, and STAC/PROV gates.	EXIF/PII redaction, STAC schema, PROV, PMTiles, COG, alt-text tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-057	PMTiles basemaps support offline mobile data collection	C. MapLibre Native / Mobile / Platform Parity	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 287-290	The source suggests MBTiles/PMTiles packages on devices for offline basemaps.	Mobile/native packs need release metadata, checksum, expiry, and policy labels.	Offline package validation and stale badge tests.
ML-064-058	On-device redaction is a pre-publication map gate	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 287-290	Face blur, logo/insignia blur, EXIF stripping, deterministic filenames, and consent flags are part of field publication.	Map notes/photos enter KFM only after redaction receipt.	No EXIF, redaction confirmation, consent flag, and deterministic filename tests.
ML-064-059	Accessibility metadata belongs in field STAC assets	S. Accessibility, UX, and Trust-Visible States	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 288-290	STAC assets include kfm:alt_text, captions, and sidecar YAML.	Evidence Drawer and map popups should expose accessible descriptions.	Alt-text presence and screen-reader checks.
ML-064-060	Conditional HTTP watchers reduce false map update churn	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 290-300	The source describes ETag/If-None-Match and Last-Modified polling that opens PRs only when changed.	Source watchers should avoid tile rebuilds and cache invalidations when source content is unchanged.	304/no-change fixture and digest comparison.
ML-064-061	Graph fleet health controls Story Node and Focus resilience	O. Focus Mode and Governed AI Map Context	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 363-369	Graph governance/telemetry feeds Story Node and Focus Mode readiness, with degraded fallbacks.	Map/Focus UI should know whether graph-backed context is healthy or degraded.	Graph-health baseline, latency SLO, and degraded-state tests.
ML-064-062	STAC/DCAT to graph mapping remains separate from cluster management	A. Renderer Boundary and KFM Trust Law	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 367	Fleet Manager does not alter STAC/DCAT mapping logic; it ensures target graph schemas and clusters are ready.	MapLibre should consume API/catalog outputs, not cluster lifecycle signals directly.	Graph target readiness and mapping contract tests.
ML-064-063	Sovereign graph tiers require governed replication	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 368	Sovereignty-sensitive clusters require documented replication targets or no replication.	Sensitive map-context queries must know access tier and deny exact geometry where required.	Replication policy and role tests.
ML-064-064	Explainability reports can become Story Nodes but not decisions	O. Focus Mode and Governed AI Map Context	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 512-522	AI explainability workflow outputs can feed Story Nodes and Focus Mode, but Focus must not alter metrics or infer governance decisions.	Focus Mode can summarize model audit evidence only with citations and finite outcomes.	Schema validation, telemetry, PROV, and guardrail threshold tests.
ML-064-065	LiDAR and GLO workflow produces terrain overlays and historic features	K. Raster, COG, DEM, Terrain, Hillshade	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 534-542	LiDAR DEMs and GLO plats derive hillshade, MDI, SVF, slope, contours, georeferenced plats, digitized vectors, and tiles.	MapLibre/Cesium can show released terrain/historic overlays with provenance and CARE tags.	CRS, RMS, terrain derivative, STAC/DCAT, Neo4j ingest, Focus summary tests.
ML-064-066	MapLibre/Cesium consume PMTiles and GeoJSON from outputs tiles web	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 535-536	The workflow states frontends consume PMTiles and GeoJSON from outputs/tiles/web.	Expose only published, public-safe outputs; do not use WORK/staging outputs.	No public raw/work path test; release manifest source path check.
ML-064-067	Focus Mode may summarize LiDAR/GL O evidence but must not reveal sensitive coordinates	O. Focus Mode and Governed AI Map Context	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 537-538	Focus Mode can summarize evidence, show temporal evolution, and surface CARE notes, but must not reveal sensitive coordinates or infer unsupported historical claims.	Map clicks on sensitive historical features should return DENY/generalized answer.	Sensitive coordinate deny fixture and citation validation.
ML-064-068	LiDAR/GLO Story Nodes link STAC/DCA T, Neo4j, and document IDs	N. Evidence Drawer Payloads and Click Resolution	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 537-538	Feature, area synthesis, and methodology Story Nodes link to catalog records, graph nodes, and semantic_document_id.	Evidence Drawer payloads should include catalog, graph, and documentation references.	Story Node link-resolves-to-source tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-069	LiDAR/GLO telemetry records tiles, features, sensitivity counts, energy and carbon	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 538-539	Per-run telemetry includes tiles_processed, features_detected/accepted, sensitive_features_count, duration, energy, and carbon.	Map release telemetry should support governance and performance review.	Telemetry schema and missing/invalid metrics tests.
ML-064-070	Layer registry includes LiDAR derivatives and GLO vectors	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 539-542	Layer registry entries include kfm-lidar and specific layers such as dtm, hillshade_mdi, svf, slope, contours, glo-plat, and glo-vectors.	LayerCatalog should distinguish raster derivatives, vectors, and interpretive overlays.	Layer ID uniqueness, roles, source, and evidence checks.
ML-064-071	COG, GPKG, GeoJSON, and PMTiles are interoperable outputs	K. Raster, COG, DEM, Terrain, Hillshade	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 541-542	The workflow lists COGs, GeoPackages, GeoJSON, PMTiles, and QA reports as interoperable outputs.	TileArtifactManifest should document which output is source-of-render, data, or QA.	Media type, CRS, and QA report linkage tests.
ML-064-072	Documentation lint protects Focus Mode source text	O. Focus Mode and Governed AI Map Context	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 549-567	Docs-lint ensures Markdown input to catalogs, graph, and Focus Mode is structurally valid.	Focus Mode source documents should be machine-extractable and linted before use.	Markdown lint, metadata-check, accessibility, provenance, and secret-scan tests.
ML-064-073	Focus Mode has explicit MAY/MUST NOT rules for docs health	O. Focus Mode and Governed AI Map Context	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 552, 567	Focus Mode can summarize documentation health and coverage but must not change content or override governance.	UI should show Focus outcomes with clear source status and no hidden modification path.	Focus Mode negative-state tests for prohibited actions.
ML-064-074	Story Node v3 and Focus Mode v3 require derived-layer evidence metadata	N. Evidence Drawer Payloads and Click Resolution	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 578-612	Derived layers used in narrative layers must carry evidence, QA, provenance, and governance labels.	EvidenceDrawerPayload should include derived_layer_id, QA status, PROV, and policy notes.	Derived layer metadata completeness tests.
ML-064-075	Sentinel-1 SAR layers need sovereignty masking and STAC generation	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 618-621	Sentinel-1 workflows propagate h3_sensitive, mask_required, care_label, sovereignty uncertainty, and PROV lineage before release.	SAR flood/deformation/wetland layers must be generalized or denied where sensitive.	FAIR/CARE, STAC, JSON-LD, and mask tests.
ML-064-076	Sentinel-1 outputs support MapLibre/Cesium environmental evidence	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 621	Applications include Focus Mode environmental evidence, Story Node context, and MapLibre/Cesium multi-layer rendering.	2D/3D environmental overlays need same trust path.	Layer release, sovereignty review, and evidence-citation tests.
ML-064-077	HydroGNSS products become FAIR/CARE STAC Items and Focus layers	K. Raster, COG, DEM, Terrain, Hillshade	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 624-627	HydroGNSS variables such as soil moisture, inundation, freeze-thaw, and biomass are harmonized into STAC/DCAT/PROV with CARE flags.	HydroGNSS layers should be time-aware, uncertainty-aware, and public-safe before map rendering.	QA masks, uncertainty thresholds, cross-compare, data-use notice tests.
ML-064-078	Dynamic H3 generalization applies to sensitive remote-sensing layers	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 627	HydroGNSS governance uses dynamic H3 generalization and no raw high-precision anomalies in sensitive areas.	MapLibre layers should receive generalized tiles or denied results, not raw sensitive anomalies.	H3 generalization and no-raw-anomaly tests.
ML-064-079	Source rights text must be embedded literally for tribal/cadastral overlays	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 212-214, 92-96	The source emphasizes provider rights/disclaimers and permission attestations for intersecting tribal/cadastral data.	Layer manifests should include rights_statement and obligations before map release.	Rights text and permission attestation tests.
ML-064-080	Map popups must open catalog and OCI snapshots read-only	N. Evidence Drawer Payloads and Click Resolution	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 92	WZDx viewer click shows STAC metadata and links to read-only OCI snapshot.	Popup should be a launch point to Evidence Drawer, not the evidence substitute.	Click -> STAC/EvidenceBundle resolution test.
ML-064-081	Policy gates validate event namespace, asset type, provenance and SBOM	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 91-92	OPA checks event namespace/version, oci-snapshot/pmtiles asset types, prov.jsonld, manifest_ref, and SBOM refs.	Tile release policy should validate both content and governance attachments.	Negative fixtures for each missing requirement.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-082	SLOs and error budgets govern publication health	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 90-92	Availability and latency SLOs drive backoff, degraded cached PMTiles, and promotion gating.	Map operations should expose load/interaction latency budgets.	OTEL metrics, p95/p99 latency, and budget burn tests.
ML-064-083	HTTP Range behavior is a release requirement for PMTiles and COG	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 90, 102-115, 289	PMTiles and COG delivery depends on Range-capable HTTP/S3/CDN hosting.	Hosting manifest must record Range/CORS support and cache behavior.	HEAD/GET Range and CORS probes in CI.
ML-064-084	Tile protocol renderHints are governance-significant configuration	E. Style Specification, Sources, Layers, Expressions	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 102-114	renderHints.worker and tileProtocol choose PMTiles HTTP Range vs XYZ/WMTS/WMS.	StyleManifest should treat renderHints as versioned config, not ad hoc UI props.	RenderHint schema, allowed values, and performance tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-085	Style JSON version and digest pin visual regression	E. Style Specification, Sources, Layers, Expressions	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 238-242	Preview renderer inputs include pinned style JSON, source layers, camera, and renderer version.	StyleManifest should store version, digest, and camera presets for review images.	Style digest and screenshot regression tests.
ML-064-086	Source-layer is required for PMTiles vector layers	E. Style Specification, Sources, Layers, Expressions	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 216	The PMTiles MapLibre snippet uses source-layer roads when adding a vector layer.	Layer definitions must declare source-layer for vector tiles.	Style/layer validator checks source-layer exists in tile metadata.
ML-064-087	MapLibre wrapper imports remain proposed paths	B. MapLibre GL JS Web Shell	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 106-114	The source uses example imports like web/ui/Map and route files, but no repo was mounted.	Treat UI component paths as PROPOSED until Directory Rules and repo evidence verify homes.	Repo topology inspection before PR.
ML-064-088	Directory Rules warn against path claims without repo verification	Y. Implementation Backlog and PR Plan	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. SRC-004; SRC-064 path snippets	Current source snippets propose many paths, but Directory Rules require responsibility-root verification.	All paths in this update remain PROPOSED unless verified in a mounted repo.	Path-home ADR/drift check.
ML-064-089	MapLibre can render quicklook COGs through raster source or tile endpoint	K. Raster, COG, DEM, Terrain, Hillshade	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 289	Quicklook imagery can be converted to COG and rendered via MapLibre raster source or tile endpoint.	Raster quicklooks should have COG manifest, attribution, and preview use limits.	COG validity, bounds, and quicklook role tests.
ML-064-090	MBTiles remains an offline basemap packaging option	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 287-290, 215-217	The source mentions MBTiles/PMTiles on devices and converting MBTiles to PMTiles.	Use MBTiles for local/mobile cases only when manifest and release controls remain intact.	MBTiles schema/digest and conversion tests.
ML-064-091	Privacy-safe field media must populate alt text for map access	S. Accessibility, UX, and Trust-Visible States	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 288-290	Redacted field photos include kfm:alt_text and captions.	Evidence Drawer image cards should use alt text and captions from source metadata.	Accessibility regression and missing-alt deny/warn policy.
ML-064-092	Telemetry schemas connect artifact cost to governance	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 5-13, 538-539	Energy/carbon telemetry appears both in promotion gates and LiDAR/GLO run telemetry.	MapReleaseManifest should record relevant telemetry refs without overloading map UI.	Telemetry schema and bounds tests.
ML-064-093	Promotion is still a state transition for map artifacts	A. Renderer Boundary and KFM Trust Law	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 1-5, 90-92, 225-229	Across ORAS, WZDx, and attestation sections, release happens after validation, signing, cataloging, and promotion.	No tile/style/raster artifact should be considered published by file presence.	No-public-unreleased-tile-load test.
ML-064-094	Catalog records are not sovereign truth either	A. Renderer Boundary and KFM Trust Law	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 60-64, 92, 536-538	Catalog records connect assets to provenance and review, but public claims still resolve to EvidenceBundle.	Map UI should use catalog to find artifacts and EvidenceBundle to support claims.	Catalog-to-EvidenceBundle resolution test.
ML-064-095	Graph projections support Focus Mode but do not replace evidence	A. Renderer Boundary and KFM Trust Law	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 363-369, 7939-8048	Graph artifact nodes and Focus Mode resolution support citations and Story Nodes.	Graph-derived overlays remain downstream projections.	Graph node crosswalk to artifact and evidence tests.
ML-064-096	Time-aware map interaction needs valid source and release times	P. Time-Aware Map Interaction and Timeline	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 1, 44, 537, 521	Hydrology percentiles, STAC datetimes, run timelines, and explainability timelines show the need for time controls.	Timeline/time slider should separate source time, retrieval time, processing time, and release time.	Temporal field validation and stale/expired display tests.
ML-064-097	MapReleaseManifest should link cache invalidation to artifact digest	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRME D source evidence; implementation PROPOSED	SRC-064 pp. 90-92, 229-234	PMTiles/COG build and CI gates imply cache changes should be tied to material artifact changes.	Cache invalidation records should cite old/new digests and release ID.	Noisy/no-change and changed-digest invalidation tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-098	Dynamic source watchers open PRs rather than publish directly	Y. Implementation Backlog and PR Plan	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 290-300	Watchers use conditional GET and GitHub automation to open PRs only when material change exists.	Source updates should enter review and validation lanes before tile rebuild.	PR artifact, source diff, and gate checks.
ML-064-099	LayerCatalog search should point to catalog records, not raw DB	B. MapLibre GL JS Web Shell	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 101-114	View registry/caches hold route and layer pointers to cataloged artifacts.	Layer catalog UI should be a governed index, not a database explorer.	Layer search result resolves to LayerManifest and EvidenceBundle.
ML-064-100	Public-safe tiles must be derived after redaction/generalization	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 287-290, 537-538, 620-627	Field capture, LiDAR/GLO, Sentinel-1, and HydroGNSS sections all require masking/redaction before public release.	Create public-safe PMTiles/COGs from transformed artifacts only.	Redaction receipt, generalization, and no exact geometry tests.
ML-064-101	3D/Cesium remains conditional around evidence burden	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 534-536, 621	The source pairs MapLibre/Cesium for LiDAR/GLO and Sentinel-1 contexts.	Use Cesium/3D only when vertical/terrain context materially improves evidence review.	2D/3D synchronization and Evidence Drawer parity tests.
ML-064-102	Hillshade/MDI/SVF/slope are evidence context layers	K. Raster, COG, DEM, Terrain, Hillshade	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 534-542	Terrain derivatives support feature discovery and landscape interpretation.	Render derivative layers with method, source DEM, CRS, and QA metadata.	Derivative provenance, visual inspection, and bounds tests.
ML-064-103	GLO plat georeferencing requires RMS and control-point receipts	V. Exports, Screenshots, Reports, and Citation Preservation	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 536-538	The workflow requires PLSS georeferencing with RMS tolerance and control point records.	Export reports/screenshots with georeference QA and citation preservation.	RMS threshold and control point completeness tests.
ML-064-104	Sensitive feature counts belong in governance telemetry	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 538-539	Telemetry includes sensitive_features_count for LiDAR/GLO.	Expose counts only in steward/governance views if sensitive.	Access control and aggregation threshold tests.
ML-064-105	Focus Mode cannot override CARE or sovereignty tags	O. Focus Mode and Governed AI Map Context	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 537-538, 567, 627	Multiple sections state Focus Mode must not override governance tags or remove warnings.	Focus responses must carry PolicyDecision and deny/abstain when tags require it.	PolicyDecision and FocusModeResponse tests.
ML-064-106	AI-generated schemas or manifests need structured validation	O. Focus Mode and Governed AI Map Context	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 249-252	The source applies structured outputs and validation/re-ask to LLM-produced objects.	Generated LayerManifest or STAC objects require schema validation and human review before promotion.	Invalid-output, repair, fail-closed, and review-state tests.
ML-064-107	Public release requires source rights, attribution, and license metadata	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 59-64, 90-96, 212-214	KGS, SSURGO, USGS, sovereignty labels, and rights statements are all treated as release inputs.	LayerManifest should include attribution, license_spdx, rights_statement, and review status.	Rights unknown blocks release; attribution display tests.
ML-064-108	Map artifacts require SBOM references when toolchains matter	R. Plugin, Wrapper, and Dependency Governance	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 1-5, 9-13, 225-229	ORAS/Cosign and CI provenance sections include SBOM references for build/tool chains.	TileArtifactManifest or RunReceipt should include SBOM refs for tilers, renderers, and converters.	SBOM presence and tool-version pin tests.
ML-064-109	Visual regression belongs beside schema and policy gates	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 238-242	Preview renderer supports image outputs when STAC, style, or renderer code changes.	Style releases should include screenshots and diffs for review.	Pixel diff, threshold, and review sign-off tests.
ML-064-110	Tile load and interaction latency budgets are first-class test targets	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 90-92, 106-114	WZDx SLOs and ViewRegistry budgets establish performance targets.	MapReleaseManifest should state tile-load and interaction budgets for high-volume layers.	Resource timing, p95 tile load, and drawer latency tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-111	MapLibre protocol adapter for PMTiles must be dependency-governed	R. Plugin, Wrapper, and Dependency Governance	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 216	The PMTiles protocol adapter is used via maplibregl.addProtocol.	Plugin/protocol adapters need allowlist, version pin, license/security review.	Dependency health and license scan tests.
ML-064-112	WMTS/WMS are allowed protocols but need extra source governance	L. WMS / WMTS / External Map Services	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 102-114	tileProtocol supports wmts and wms besides pmtiles and xyz.	External map services require SourceDescriptor, rights, attribution, and cache policy.	WMS/WMTS service metadata and attribution tests.

idea_id	idea title	category	status	source_ids	short normalized statement	implementation consequence	validation consequence
ML-064-113	MVT remains the practical default vector tile encoding	H. Vector Tiles: MVT and MLT	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 115, 248	The source tiles to PMTiles (MVT) and contrasts MLT with traditional MVT.	Use MVT/PMTiles as default until MLT pilot passes validation.	MVT tile metadata and MLT fallback tests.
ML-064-114	MLT validation must include renderer support and ecosystem tooling	H. Vector Tiles: MVT and MLT	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 248	MLT performance claims are source-reported and version-sensitive.	Create a separate MLT pilot manifest with rollback to MVT.	MLT decoder/render parity, source encoding, and style compatibility tests.
ML-064-115	Martin should be evaluated for dynamic PostGIS tiles only behind policy	J. Martin and Server-Mediated Tile Serving	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 248, 127	Martin/server tile serving is relevant where dynamic server mediation is needed.	Keep dynamic tile serving behind governed API/access model and release policy.	PostGIS role, tile SQL, cache, and deny tests.
ML-064-116	Exports and reports must retain citations and artifact digests	V. Exports, Screenshots, Reports, and Citation Preservation	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 225-229, 536-539	Build snapshots, release notes, georeference QA, and telemetry should preserve digests and links.	Screenshot/report export must include citations, catalog links, and digest references.	Export citation validation and digest inclusion tests.
ML-064-117	Layer toggle is not publication	X. Anti-Patterns and Failure Modes	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 90-92, 101-114, 225-229	Multiple sections require validation, signatures, catalog, and promotion before map availability.	UI layer lists should show release state and block unreleased layers.	Unreleased tile load and layer toggle tests.
ML-064-118	Dense GeoJSON remains intake or small-runtime data, not production default	X. Anti-Patterns and Failure Modes	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 114-119, 215-217, 287-290	Large vector data is normalized then tiled to PMTiles/MVT; GeoJSON is useful for ETL and small notes.	Use vector tiles for large public layers and GeoJSON for small candidate/runtime overlays.	Tile size and feature-count threshold tests.
ML-064-119	Popups must not substitute for Evidence Drawer	X. Anti-Patterns and Failure Modes	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. 92, 537-538	Click metadata and Story Nodes need links to catalog/evidence; Focus must cite or abstain.	Popup content should be short and link to Evidence Drawer with policy/citation state.	Popup citation and drawer-resolution tests.
ML-064-120	Current implementation depth remains unknown without mounted repo	Z. Open Questions and Verification Backlog	NEW / CONFIRMED source evidence; implementation PROPOSED	SRC-064 pp. SRC-064 overall; SRC-MASTER-PREV	The new source contains implementation-style snippets and paths, but no current repo inspection was available.	All file paths, route names, package versions, and runtime behavior remain PROPOSED/NEEDS VERIFICATION.	Mount repo and run path/test/workflow inspection next run.

6. Category Atlas

Category	Title	v1.9 delta count	Current status	What changed since v1.8
A	Renderer Boundary and KFM Trust Law	5	EXPANDED / NEW delta records	5 SRC-064 idea records added or sharpened.
B	MapLibre GL JS Web Shell	4	EXPANDED / NEW delta records	4 SRC-064 idea records added or sharpened.
C	MapLibre Native / Mobile / Platform Parity	3	EXPANDED / NEW delta records	3 SRC-064 idea records added or sharpened.
D	MapLibre RS / Experimental Renderers	0	RETAINED	No new SRC-064 idea record; prior v1.8 baseline retained in Appendix A.
E	Style Specification, Sources, Layers, Expressions	3	EXPANDED / NEW delta records	3 SRC-064 idea records added or sharpened.
F	Sprites, Glyphs, Fonts, Design Tokens	0	RETAINED	No new SRC-064 idea record; prior v1.8 baseline retained in Appendix A.
G	GeoJSON and Runtime Data	5	EXPANDED / NEW delta records	5 SRC-064 idea records added or sharpened.
H	Vector Tiles: MVT and MLT	5	EXPANDED / NEW delta records	5 SRC-064 idea records added or sharpened.
I	PMTiles, MBTiles, Static Tiles, Serverless Delivery	9	EXPANDED / NEW delta records	9 SRC-064 idea records added or sharpened.
J	Martin and Server-Mediated Tile Serving	3	EXPANDED / NEW delta records	3 SRC-064 idea records added or sharpened.
K	Raster, COG, DEM, Terrain, Hillshade	6	EXPANDED / NEW delta records	6 SRC-064 idea records added or sharpened.
L	WMS / WMTS / External Map Services	2	EXPANDED / NEW delta records	2 SRC-064 idea records added or sharpened.
M	LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	8	EXPANDED / NEW delta records	8 SRC-064 idea records added or sharpened.
N	Evidence Drawer Payloads and Click Resolution	4	EXPANDED / NEW delta records	4 SRC-064 idea records added or sharpened.
O	Focus Mode and Governed AI Map Context	8	EXPANDED / NEW delta records	8 SRC-064 idea records added or sharpened.
P	Time-Aware Map Interaction and Timeline	2	EXPANDED / NEW delta records	2 SRC-064 idea records added or sharpened.
Q	Sensitive Geometry, Geoprivacy, Rights, and Policy	9	EXPANDED / NEW delta records	9 SRC-064 idea records added or sharpened.
R	Plugin, Wrapper, and Dependency Governance	4	EXPANDED / NEW delta records	4 SRC-064 idea records added or sharpened.
S	Accessibility, UX, and Trust-Visible States	4	EXPANDED / NEW delta records	4 SRC-064 idea records added or sharpened.
T	Performance, Caching, CDN, Range Requests, and Resource Timing	8	EXPANDED / NEW delta records	8 SRC-064 idea records added or sharpened.
U	Testing, CI, Validation, Rollback, and Proof Objects	12	EXPANDED / NEW delta records	12 SRC-064 idea records added or sharpened.
V	Exports, Screenshots, Reports, and Citation Preservation	3	EXPANDED / NEW delta records	3 SRC-064 idea records added or sharpened.
W	3D / Cesium / Deck.gl / Overlay Interoperability	5	EXPANDED / NEW delta records	5 SRC-064 idea records added or sharpened.
X	Anti-Patterns and Failure Modes	3	EXPANDED / NEW delta records	3 SRC-064 idea records added or sharpened.
Y	Implementation Backlog and PR Plan	3	EXPANDED / NEW delta records	3 SRC-064 idea records added or sharpened.
Z	Open Questions and Verification Backlog	2	EXPANDED / NEW delta records	2 SRC-064 idea records added or sharpened.

7. Detailed Category Chapters

Each category below distinguishes CONFIRMED source evidence from PROPOSED implementation consequences. For categories with no new SRC-064 idea records, the v1.8 baseline remains the source of retained content in Appendix A.

A. Renderer Boundary and KFM Trust Law

Defines the renderer/trust membrane: MapLibre, graph, catalog, and runtime outputs are downstream carriers.

idea_id	title	source	implementation / validation consequence
ML-064-020	View registry forbids direct Neo4j handles in map UI	SRC-064 pp. 101-114	MapLibre views should receive layer pointers and render hints, not database query handles. Validation: Ajv/JSON Schema plus OPA gate for no direct DB fields.
ML-064-062	STAC/DCAT to graph mapping remains separate from cluster management	SRC-064 pp. 367	MapLibre should consume API/catalog outputs, not cluster lifecycle signals directly. Validation: Graph target readiness and mapping contract tests.
ML-064-093	Promotion is still a state transition for map artifacts	SRC-064 pp. 1-5, 90-92, 225-229	No tile/style/raster artifact should be considered published by file presence. Validation: No-public-unreleased-tile-load test.
ML-064-094	Catalog records are not sovereign truth either	SRC-064 pp. 60-64, 92, 536-538	Map UI should use catalog to find artifacts and EvidenceBundle to support claims. Validation: Catalog-to-EvidenceBundle resolution test.
ML-064-095	Graph projections support Focus Mode but do not replace evidence	SRC-064 pp. 363-369, 7939-8048	Graph-derived overlays remain downstream projections. Validation: Graph node crosswalk to artifact and evidence tests.

Evidence basis: ML-064-020 SRC-064 pp. 101-114; ML-064-062 SRC-064 pp. 367; ML-064-093 SRC-064 pp. 1-5, 90-92, 225-229; ML-064-094 SRC-064 pp. 60-64, 92, 536-538; ML-064-095 SRC-064 pp. 363-369, 7939-8048.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

B. MapLibre GL JS Web Shell

Covers the web map shell, route chunks, view registry, and browser behavior around MapLibre GL JS.

idea_id	title	source	implementation / validation consequence
ML-064-016	MapLibre addSource/addLayer PMTiles wiring is downstream only	SRC-064 pp. 92	UI wiring should be generated from released manifests, not hand-coded raw URLs. Validation: No unreleased tile URL in style/source registry; click opens STAC metadata/Evidence Drawer.
ML-064-021	Route chunks and lazy hydration support governed map shells	SRC-064 pp. 101-114	MapLibre shells should hydrate route-specific chunks after governance state is available. Validation: Validate view registry and chunk names in CI.
ML-064-087	MapLibre wrapper imports remain proposed paths	SRC-064 pp. 106-114	Treat UI component paths as PROPOSED until Directory Rules and repo evidence verify homes. Validation: Repo topology inspection before PR.
ML-064-099	LayerCatalog search should point to catalog records, not raw DB	SRC-064 pp. 101-114	Layer catalog UI should be a governed index, not a database explorer. Validation: Layer search result resolves to LayerManifest and EvidenceBundle.

Evidence basis: ML-064-016 SRC-064 pp. 92; ML-064-021 SRC-064 pp. 101-114; ML-064-087 SRC-064 pp. 106-114; ML-064-099 SRC-064 pp. 101-114.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

C. MapLibre Native / Mobile / Platform Parity

Covers mobile/native parity and offline packages.

idea_id	title	source	implementation / validation consequence
ML-064-035	Native and embedded apps can ship single PMTiles files	SRC-064 pp. 216-217	Mobile/native parity can use offline PMTiles only with the same EvidenceBundle and release metadata. Validation: Offline package digest, expiry, and release-state checks.
ML-064-050	MapLibre Native Android Vulkan is a parity opportunity	SRC-064 pp. 248	Native/mobile parity should be tested against web style/profile and evidence-drawer behavior. Validation: Android render, style subset, offline PMTiles, and accessibility tests.
ML-064-057	PMTiles basemaps support offline mobile data collection	SRC-064 pp. 287-290	Mobile/native packs need release metadata, checksum, expiry, and policy labels. Validation: Offline package validation and stale badge tests.

Evidence basis: ML-064-035 SRC-064 pp. 216-217; ML-064-050 SRC-064 pp. 248; ML-064-057 SRC-064 pp. 287-290.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

D. MapLibre RS / Experimental Renderers

Covers MapLibre RS and experimental renderer boundaries.

Status: RETAINED. No new SRC-064 record was added in this category. See Appendix A for the v1.8 baseline category chapter, idea IDs, and open questions.

E. Style Specification, Sources, Layers, Expressions

Covers style JSON, source definitions, source-layer, render hints, filters, expressions, and visual state.

idea_id	title	source	implementation / validation consequence
ML-064-084	Tile protocol renderHints are governance-significant configuration	SRC-064 pp. 102-114	StyleManifest should treat renderHints as versioned config, not ad hoc UI props. Validation: RenderHint schema, allowed values, and performance tests.

idea_id	title	source	implementation / validation consequence
ML-064-085	Style JSON version and digest pin visual regression	SRC-064 pp. 238-242	StyleManifest should store version, digest, and camera presets for review images. Validation: Style digest and screenshot regression tests.
ML-064-086	Source-layer is required for PMTiles vector layers	SRC-064 pp. 216	Layer definitions must declare source-layer for vector tiles. Validation: Style/layer validator checks source-layer exists in tile metadata.

Evidence basis: ML-064-084 SRC-064 pp. 102-114; ML-064-085 SRC-064 pp. 238-242; ML-064-086 SRC-064 pp. 216.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

F. Sprites, Glyphs, Fonts, Design Tokens

Covers sprites, glyphs, fonts, and design tokens.

Status: RETAINED. No new SRC-064 record was added in this category. See Appendix A for the v1.8 baseline category chapter, idea IDs, and open questions.

G. GeoJSON and Runtime Data

Covers GeoJSON, source data, runtime features, checksums, and feature identity.

idea_id	title	source	implementation / validation consequence
ML-064-011	GeoJSON geometry remains catalog geometry, not proof	SRC-064 pp. 61-63	Click resolution should use feature IDs to fetch EvidenceBundle, not trust GeoJSON alone. Validation: GeoJSON schema and EvidenceBundle resolution tests.
ML-064-026	ArcGIS FeatureLayers can be ETL inputs, not public client dependencies	SRC-064 pp. 114-115	Use ArcGIS/GeoJSON as RAW/WORK intake, then publish governed outputs. Validation: SourceDescriptor and rights checks before ingest; no browser direct source.
ML-064-028	Normalized GeoJSON geometry checksums support stable feature identity	SRC-064 pp. 125-140	Feature IDs and promoteId should incorporate stable geometry or record hashes where appropriate. Validation: Checksum recomputation and precision/CRS tests.
ML-064-044	WZDx GeoJSON is a transport layer source family	SRC-064 pp. 235, 611-621	Transport event layers should carry source family, schema version, and temporal fields. Validation: GeoJSON geometry, WZDx schema, and event freshness tests.
ML-064-045	USGS GeoJSON feeds are STAC-ready event sources	SRC-064 pp. 236-237	Event point layers can be cataloged and rendered after freshness and source checks. Validation: Feed schema and stale/expired feed tests.

Evidence basis: ML-064-011 SRC-064 pp. 61-63; ML-064-026 SRC-064 pp. 114-115; ML-064-028 SRC-064 pp. 125-140; ML-064-044 SRC-064 pp. 235, 611-621; ML-064-045 SRC-064 pp. 236-237.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache

invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

H. Vector Tiles: MVT and MLT

Covers vector tile formats, especially MVT default and MLT pilot posture.

idea_id	title	source	implementation / validation consequence
ML-064-014	PMTiles generation from COG or GeoParquet uses tippecanoe/tegola	SRC-064 pp. 90	Choose tiling backend per data type, but output must be manifest-bound. Validation: Tiler version, flags, zoom range, and input digest checks.
ML-064-027	Topology-safe simplification avoids shared-boundary cracks	SRC-064 pp. 116-119	Use topology-aware simplification prior to PMTiles/MVT build. Validation: Area/perimeter/Hausdorff and shared-border regression tests.
ML-064-051	MLT is a pilot candidate, not default production format	SRC-064 pp. 248	Keep MLT in pilot path until tooling, renderer support, and validation mature. Validation: MVT-vs-MLT compatibility, decode, visual, and fallback tests.
ML-064-113	MVT remains the practical default vector tile encoding	SRC-064 pp. 115, 248	Use MVT/PMTiles as default until MLT pilot passes validation. Validation: MVT tile metadata and MLT fallback tests.
ML-064-114	MLT validation must include renderer support and ecosystem tooling	SRC-064 pp. 248	Create a separate MLT pilot manifest with rollback to MVT. Validation: MLT decoder/render parity, source encoding, and style compatibility tests.

Evidence basis: ML-064-014 SRC-064 pp. 90; ML-064-027 SRC-064 pp. 116-119; ML-064-051 SRC-064 pp. 248; ML-064-113 SRC-064 pp. 115, 248; ML-064-114 SRC-064 pp. 248.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

I. PMTiles, MBTiles, Static Tiles, Serverless Delivery

Covers PMTiles, MBTiles, static tiles, serverless delivery, offline delivery, and CDN use.

idea_id	title	source	implementation / validation consequence
ML-064-002	OCI registry as storage for PMTiles, GeoParquet, and COG	SRC-064 pp. 1-5	Allow tile/raster/vector artifacts to be fetched reproducibly from OCI only when digest and provenance are in the catalog. Validation: Validate mediaType, digest, Cosign verification, and PROV referrers.
ML-064-013	WZDx pipeline produces PMTiles for fast tiled visualization	SRC-064 pp. 88-92	Road-work or incident layers can be previewed via PMTiles only after snapshot validation. Validation: Validate event namespace/version, PMTiles asset, prov.jsonld, manifest_ref, and SBOM refs.
ML-064-015	Static CDN PMTiles avoids tile server for previews	SRC-064 pp. 90-92	Prefer serverless PMTiles for stable, public-safe layers with Range-capable hosting. Validation: Range/CORS probe, cache headers, and checksum verification.
ML-064-018	OCI plus PMTiles supports portable registry/CDN delivery	SRC-064 pp. 92	Use portable artifact references in release manifests with checksums. Validation: Resolve OCI snapshot and PMTiles URL from catalog; verify both.
ML-064-025	KGS M-118 can become topology-safe serverless PMTiles	SRC-064 pp. 114-116	Surficial geology layers should become processed, cataloged artifacts before MapLibre use. Validation: Validate topology, GeoParquet schema, PMTiles digest, and HTTP Range hosting.

idea_id	title	source	implementation / validation consequence
ML-064-031	PMTiles deterministic pipeline begins with canonical GeoParquet	SRC-064 pp. 214-217	Treat GeoParquet as canonical processed input for vector tile builds. Validation: Input schema lock, sorted rows, fixed precision, fixed field order.
ML-064-039	PMTiles tilehash sidecar supports offline verification	SRC-064 pp. 226-229	TileArtifactManifest can include per-tile or directory hashes for high-integrity releases. Validation: Verify tilehash sidecar against PMTiles file.
ML-064-056	Offline-first field capture can deliver PMTiles and COG	SRC-064 pp. 287-290	Field maps should publish only after redaction, classification, and STAC/PROV gates. Validation: EXIF/PII redaction, STAC schema, PROV, PMTiles, COG, alt-text tests.
ML-064-090	MBTiles remains an offline basemap packaging option	SRC-064 pp. 287-290, 215-217	Use MBTiles for local/mobile cases only when manifest and release controls remain intact. Validation: MBTiles schema/digest and conversion tests.

Evidence basis: ML-064-002 SRC-064 pp. 1-5; ML-064-013 SRC-064 pp. 88-92; ML-064-015 SRC-064 pp. 90-92; ML-064-018 SRC-064 pp. 92; ML-064-025 SRC-064 pp. 114-116.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

J. Martin and Server-Mediated Tile Serving

Covers Martin, PostGIS, and server-mediated tile serving.

idea_id	title	source	implementation / validation consequence
ML-064-029	PostGIS sinks are internal, not UI truth	SRC-064 pp. 127	If PostGIS backs tiles or APIs, it remains behind governed services. Validation: No direct browser connection; role and query scope tests.
ML-064-052	Martin improvements justify server-mediated tile serving review	SRC-064 pp. 248	Evaluate Martin where dynamic PostGIS/PMTiles/MBTiles mediation is needed. Validation: Martin config, access, source, and performance tests.
ML-064-115	Martin should be evaluated for dynamic PostGIS tiles only behind policy	SRC-064 pp. 248, 127	Keep dynamic tile serving behind governed API/access model and release policy. Validation: PostGIS role, tile SQL, cache, and deny tests.

Evidence basis: ML-064-029 SRC-064 pp. 127; ML-064-052 SRC-064 pp. 248; ML-064-115 SRC-064 pp. 248, 127.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

K. Raster, COG, DEM, Terrain, Hillshade

Covers raster/COG/DEM/terrain/hillshade/SAR/HydroGNSS derived layers.

idea_id	title	source	implementation / validation consequence
ML-064-040	COG internal tiles and overviews get sidecar hashes	SRC-064 pp. 226-229	Raster release should support file-level and internal-chunk verification. Validation: COG tilehash and overview verification.
ML-064-065	LiDAR and GLO workflow produces terrain overlays and historic features	SRC-064 pp. 534-542	MapLibre/Cesium can show released terrain/historic overlays with provenance and CARE tags. Validation: CRS, RMS, terrain derivative, STAC/DCAT, Neo4j ingest, Focus summary tests.
ML-064-071	COG, GPKG, GeoJSON, and PMTiles are interoperable outputs	SRC-064 pp. 541-542	TileArtifactManifest should document which output is source-of-render, data, or QA. Validation: Media type, CRS, and QA report linkage tests.
ML-064-077	HydroGNSS products become FAIR/CARE STAC Items and Focus layers	SRC-064 pp. 624-627	HydroGNSS layers should be time-aware, uncertainty-aware, and public-safe before map rendering. Validation: QA masks, uncertainty thresholds, cross-compare, data-use notice tests.
ML-064-089	MapLibre can render quicklook COGs through raster source or tile endpoint	SRC-064 pp. 289	Raster quicklooks should have COG manifest, attribution, and preview use limits. Validation: COG validity, bounds, and quicklook role tests.
ML-064-102	Hillshade/MDI/SVF/slope are evidence context layers	SRC-064 pp. 534-542	Render derivative layers with method, source DEM, CRS, and QA metadata. Validation: Derivative provenance, visual inspection, and bounds tests.

Evidence basis: ML-064-040 SRC-064 pp. 226-229; ML-064-065 SRC-064 pp. 534-542; ML-064-071 SRC-064 pp. 541-542; ML-064-077 SRC-064 pp. 624-627; ML-064-089 SRC-064 pp. 289.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

L. WMS / WMTS / External Map Services

Covers WMS, WMTS, XYZ, and external service protocols.

idea_id	title	source	implementation / validation consequence
ML-064-023	tileProtocol enum covers PMTiles, XYZ, WMTS, and WMS	SRC-064 pp. 102-110	External service layers need explicit protocol and access/governance posture. Validation: Protocol enum validation; WMS/WMTS attribution and terms checks.
ML-064-112	WMTS/WMS are allowed protocols but need extra source governance	SRC-064 pp. 102-114	External map services require SourceDescriptor, rights, attribution, and cache policy. Validation: WMS/WMTS service metadata and attribution tests.

Evidence basis: ML-064-023 SRC-064 pp. 102-110; ML-064-112 SRC-064 pp. 102-114.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest

Covers manifests and catalog alignment for layers, styles, tiles, and releases.

idea_id	title	source	implementation / validation consequence
ML-064-004	STAC/DCAT catalog entries carry OCI distribution details	SRC-064 pp. 3-4	TileArtifactManifest should align its artifact list with STAC/DCAT OCI distribution metadata. Validation: Catalog validator checks digest, size, mediaType, and provenance_ref.
ML-064-009	STAC/DCAT/PROV mapping for GeoParquet, COG, and PMTiles	SRC-064 pp. 60-64	LayerManifest and TileArtifactManifest should inherit asset roles and provenance from catalog records. Validation: STAC/DCAT/PROV validation with mediaType and roles.
ML-064-010	COG and PMTiles have explicit STAC asset roles	SRC-064 pp. 61-64	Use roles to tell the MapLibre shell which assets are renderable, quicklook, data, or metadata. Validation: Asset-role validation and disallow unknown media types.
ML-064-022	Layer pointers use datasources not queries	SRC-064 pp. 101-114	LayerManifest should separate catalog sources from render hints. Validation: Schema requires stac/dcat/pmtiles/cog fields when applicable.
ML-064-034	PMTiles sha256 belongs in STAC records	SRC-064 pp. 216-217	MapLibre fetch verification should compare against catalog checksum. Validation: Client or prefetch checksum probe; manifest digest check.
ML-064-042	STAC integration stores geom and record digests	SRC-064 pp. 228-229	Evidence Drawer can show feature digest and artifact digest. Validation: Digest presence and recomputation tests.
ML-064-053	JSON Schema and Pydantic align layer manifests and telemetry	SRC-064 pp. 249-252	Author schema-backed LayerManifest, StyleManifest, TileArtifactManifest, telemetry, and ViewRegistry objects. Validation: Schema export, schema lint, fixture validation, and Confest gate.
ML-064-070	Layer registry includes LiDAR derivatives and GLO vectors	SRC-064 pp. 539-542	LayerCatalog should distinguish raster derivatives, vectors, and interpretive overlays. Validation: Layer ID uniqueness, roles, source, and evidence checks.

Evidence basis: ML-064-004 SRC-064 pp. 3-4; ML-064-009 SRC-064 pp. 60-64; ML-064-010 SRC-064 pp. 61-64; ML-064-022 SRC-064 pp. 101-114; ML-064-034 SRC-064 pp. 216-217.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

N. Evidence Drawer Payloads and Click Resolution

Covers Evidence Drawer, Story Nodes, click resolution, and payload link integrity.

idea_id	title	source	implementation / validation consequence
ML-064-048	Story Nodes and Focus Mode consume preview evidence	SRC-064 pp. 242	Evidence Drawer can link preview screenshot, style digest, and artifact digest. Validation: Click-to-drawer and preview hash tests.

idea_id	title	source	implementation / validation consequence
ML-064-068	LiDAR/GLO Story Nodes link STAC/DCAT, Neo4j, and document IDs	SRC-064 pp. 537-538	Evidence Drawer payloads should include catalog, graph, and documentation references. Validation: Story Node link-resolves-to-source tests.
ML-064-074	Story Node v3 and Focus Mode v3 require derived-layer evidence metadata	SRC-064 pp. 578-612	EvidenceDrawerPayload should include derived_layer_id, QA status, PROV, and policy notes. Validation: Derived layer metadata completeness tests.
ML-064-080	Map popups must open catalog and OCI snapshots read-only	SRC-064 pp. 92	Popup should be a launch point to Evidence Drawer, not the evidence substitute. Validation: Click -> STAC/EvidenceBundle resolution test.

Evidence basis: ML-064-048 SRC-064 pp. 242; ML-064-068 SRC-064 pp. 537-538; ML-064-074 SRC-064 pp. 578-612; ML-064-080 SRC-064 pp. 92.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

O. Focus Mode and Governed AI Map Context

Covers Focus Mode, governed AI map context, structured outputs, and finite outcomes.

idea_id	title	source	implementation / validation consequence
ML-064-054	LLM structured outputs must validate against map contracts	SRC-064 pp. 250-252	Focus Mode map context responses should use schema validation and citation validation. Validation: ANSWER/ABSTAIN/DENY/ERROR fixture tests.
ML-064-061	Graph fleet health controls Story Node and Focus resilience	SRC-064 pp. 363-369	Map/Focus UI should know whether graph-backed context is healthy or degraded. Validation: Graph-health baseline, latency SLO, and degraded-state tests.
ML-064-064	Explainability reports can become Story Nodes but not decisions	SRC-064 pp. 512-522	Focus Mode can summarize model audit evidence only with citations and finite outcomes. Validation: Schema validation, telemetry, PROV, and guardrail threshold tests.
ML-064-067	Focus Mode may summarize LiDAR/GLO evidence but must not reveal sensitive coordinates	SRC-064 pp. 537-538	Map clicks on sensitive historical features should return DENY/generalized answer. Validation: Sensitive coordinate deny fixture and citation validation.
ML-064-072	Documentation lint protects Focus Mode source text	SRC-064 pp. 549-567	Focus Mode source documents should be machine-extractable and linted before use. Validation: Markdown lint, metadata-check, accessibility, provenance, and secret-scan tests.
ML-064-073	Focus Mode has explicit MAY/MUST NOT rules for docs health	SRC-064 pp. 552, 567	UI should show Focus outcomes with clear source status and no hidden modification path. Validation: Focus Mode negative-state tests for prohibited actions.
ML-064-105	Focus Mode cannot override CARE or sovereignty tags	SRC-064 pp. 537-538, 567, 627	Focus responses must carry PolicyDecision and deny/abstain when tags require it. Validation: PolicyDecision and FocusModeResponse tests.
ML-064-106	AI-generated schemas or manifests need structured validation	SRC-064 pp. 249-252	Generated LayerManifest or STAC objects require schema validation and human review before promotion. Validation: Invalid-output, repair, fail-closed, and review-state tests.

Evidence basis: ML-064-054 SRC-064 pp. 250-252; ML-064-061 SRC-064 pp. 363-369; ML-064-064 SRC-064 pp. 512-522; ML-064-067 SRC-064 pp. 537-538; ML-064-072 SRC-064 pp. 549-567.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

P. Time-Aware Map Interaction and Timeline

Covers time sliders, event/release/source times, and temporal interpretation.

idea_id	title	source	implementation / validation consequence
ML-064-001	WaterWatch percentiles as hydrology layer context	SRC-064 pp. 1	Expose hydrology anomaly layers as released context, not as raw gauge pulls. Validation: Cross-check HUC percentiles with drought indicators and station metadata.
ML-064-096	Time-aware map interaction needs valid source and release times	SRC-064 pp. 1, 44, 537, 521	Timeline/time slider should separate source time, retrieval time, processing time, and release time. Validation: Temporal field validation and stale/expired display tests.

Evidence basis: ML-064-001 SRC-064 pp. 1; ML-064-096 SRC-064 pp. 1, 44, 537, 521.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

Q. Sensitive Geometry, Geoprivacy, Rights, and Policy

Covers sensitive geometry, rights, sovereignty, redaction, consent, and policy labels.

idea_id	title	source	implementation / validation consequence
ML-064-019	Sovereignty labels must stay synchronized across STAC/DCAT/PROV	SRC-064 pp. 92-96	Layer manifests must carry policy labels from the catalog; MapLibre must receive already-redacted artifacts. Validation: OPA gate for label enum, exceptions, approver, and signed_at.
ML-064-058	On-device redaction is a pre-publication map gate	SRC-064 pp. 287-290	Map notes/photos enter KFM only after redaction receipt. Validation: No EXIF, redaction confirmation, consent flag, and deterministic filename tests.
ML-064-063	Sovereign graph tiers require governed replication	SRC-064 pp. 368	Sensitive map-context queries must know access tier and deny exact geometry where required. Validation: Replication policy and role tests.
ML-064-075	Sentinel-1 SAR layers need sovereignty masking and STAC generation	SRC-064 pp. 618-621	SAR flood/deformation/wetland layers must be generalized or denied where sensitive. Validation: FAIR/CARE, STAC, JSON-LD, and mask tests.
ML-064-078	Dynamic H3 generalization applies to sensitive remote-sensing layers	SRC-064 pp. 627	MapLibre layers should receive generalized tiles or denied results, not raw sensitive anomalies. Validation: H3 generalization and no-raw-anomaly tests.

idea_id	title	source	implementation / validation consequence
ML-064-079	Source rights text must be embedded literally for tribal/cadastral overlays	SRC-064 pp. 212-214, 92-96	Layer manifests should include rights_statement and obligations before map release. Validation: Rights text and permission attestation tests.
ML-064-100	Public-safe tiles must be derived after redaction/generalization	SRC-064 pp. 287-290, 537-538, 620-627	Create public-safe PMTiles/COGs from transformed artifacts only. Validation: Redaction receipt, generalization, and no exact geometry tests.
ML-064-104	Sensitive feature counts belong in governance telemetry	SRC-064 pp. 538-539	Expose counts only in steward/governance views if sensitive. Validation: Access control and aggregation threshold tests.
ML-064-107	Public release requires source rights, attribution, and license metadata	SRC-064 pp. 59-64, 90-96, 212-214	LayerManifest should include attribution, license_spdx, rights_statement, and review status. Validation: Rights unknown blocks release; attribution display tests.

Evidence basis: ML-064-019 SRC-064 pp. 92-96; ML-064-058 SRC-064 pp. 287-290; ML-064-063 SRC-064 pp. 368; ML-064-075 SRC-064 pp. 618-621; ML-064-078 SRC-064 pp. 627.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

R. Plugin, Wrapper, and Dependency Governance

Covers dependencies, protocol adapters, SBOMs, wrappers, and plugin governance.

idea_id	title	source	implementation / validation consequence
ML-064-032	Tiler versions and flags must be pinned	SRC-064 pp. 215-217	TileArtifactManifest should record tiler, version, flags, zooms, and input STAC links. Validation: Rebuild comparison with same flags and inputs.
ML-064-055	UI registries and analytics events use same contracts	SRC-064 pp. 252	MapLibre configs and telemetry events should share schema semantics with backend. Validation: Front-end boundary validation and telemetry schema tests.
ML-064-108	Map artifacts require SBOM references when toolchains matter	SRC-064 pp. 1-5, 9-13, 225-229	TileArtifactManifest or RunReceipt should include SBOM refs for tilers, renderers, and converters. Validation: SBOM presence and tool-version pin tests.
ML-064-111	MapLibre protocol adapter for PMTiles must be dependency-governed	SRC-064 pp. 216	Plugin/protocol adapters need allowlist, version pin, license/security review. Validation: Dependency health and license scan tests.

Evidence basis: ML-064-032 SRC-064 pp. 215-217; ML-064-055 SRC-064 pp. 252; ML-064-108 SRC-064 pp. 1-5, 9-13, 225-229; ML-064-111 SRC-064 pp. 216.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

S. Accessibility, UX, and Trust-Visible States

Covers UX, accessible states, alt text, stale/degraded states, and trust-visible UI.

idea_id	title	source	implementation / validation consequence
ML-064-017	Cached PMTiles can be degradation fallback	SRC-064 pp. 91	UI may show cached/stale layer state when publication is impaired. Validation: Stale-source badge, cache age, and ABSTAIN behavior tests.
ML-064-024	Performance and accessibility budgets are registry data	SRC-064 pp. 106-114	Budget thresholds should be release criteria for heavy map layers. Validation: Preview build measures tile load, route load, and accessibility score.
ML-064-059	Accessibility metadata belongs in field STAC assets	SRC-064 pp. 288-290	Evidence Drawer and map popups should expose accessible descriptions. Validation: Alt-text presence and screen-reader checks.
ML-064-091	Privacy-safe field media must populate alt text for map access	SRC-064 pp. 288-290	Evidence Drawer image cards should use alt text and captions from source metadata. Validation: Accessibility regression and missing-alt deny/warn policy.

Evidence basis: ML-064-017 SRC-064 pp. 91; ML-064-024 SRC-064 pp. 106-114; ML-064-059 SRC-064 pp. 288-290; ML-064-091 SRC-064 pp. 288-290.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

T. Performance, Caching, CDN, Range Requests, and Resource Timing

Covers performance, cache, Range requests, telemetry, SLOs, and resource timing.

idea_id	title	source	implementation / validation consequence
ML-064-033	Attribute whitelisting keeps vector tiles small and safer	SRC-064 pp. 215-217	Layer manifests should declare allowed tile attributes. Validation: Tile size and attribute leak tests.
ML-064-060	Conditional HTTP watchers reduce false map update churn	SRC-064 pp. 290-300	Source watchers should avoid tile rebuilds and cache invalidations when source content is unchanged. Validation: 304/no-change fixture and digest comparison.
ML-064-069	LiDAR/GLO telemetry records tiles, features, sensitivity counts, energy and carbon	SRC-064 pp. 538-539	Map release telemetry should support governance and performance review. Validation: Telemetry schema and missing/invalid metrics tests.
ML-064-082	SLOs and error budgets govern publication health	SRC-064 pp. 90-92	Map operations should expose load/interaction latency budgets. Validation: OTEL metrics, p95/p99 latency, and budget burn tests.
ML-064-083	HTTP Range behavior is a release requirement for PMTiles and COG	SRC-064 pp. 90, 102-115, 289	Hosting manifest must record Range/CORS support and cache behavior. Validation: HEAD/GET Range and CORS probes in CI.
ML-064-092	Telemetry schemas connect artifact cost to governance	SRC-064 pp. 5-13, 538-539	MapReleaseManifest should record relevant telemetry refs without overloading map UI. Validation: Telemetry schema and bounds tests.
ML-064-097	MapReleaseManifest should link cache invalidation to artifact digest	SRC-064 pp. 90-92, 229-234	Cache invalidation records should cite old/new digests and release ID. Validation: Noisy/no-change and changed-digest invalidation tests.

idea_id	title	source	implementation / validation consequence
ML-064-110	Tile load and interaction latency budgets are first-class test targets	SRC-064 pp. 90-92, 106-114	MapReleaseManifest should state tile-load and interaction budgets for high-volume layers. Validation: Resource timing, p95 tile load, and drawer latency tests.

Evidence basis: ML-064-033 SRC-064 pp. 215-217; ML-064-060 SRC-064 pp. 290-300; ML-064-069 SRC-064 pp. 538-539; ML-064-082 SRC-064 pp. 90-92; ML-064-083 SRC-064 pp. 90, 102-115, 289.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

U. Testing, CI, Validation, Rollback, and Proof Objects

Covers testing, CI, validation, rollback, receipts, signatures, and proof objects.

idea_id	title	source	implementation / validation consequence
ML-064-003	Digest-pinned artifact pulls support rollback	SRC-064 pp. 3-5	MapReleaseManifest should prefer digest-pinned PMTiles/COG/GeoParquet references. Validation: Test that production manifests include digest and rollback target.
ML-064-005	Cosign referrers link SBOM and PROV to geospatial artifacts	SRC-064 pp. 4-5	Release gates should verify SBOM and PROV referrers for public tiles and rasters. Validation: Cosign verify and referrer-presence CI checks.
ML-064-006	Energy and CO2 promotion policy affects map artifact release	SRC-064 pp. 5-8	Map tile/COG release manifests can include sustainability telemetry without making it a map truth source. Validation: OPA/Rego gate for energy_kwh, co2_kg, signature, commit_sha, and SBOM ref.
ML-064-007	CI provenance JSON-LD becomes run evidence for map builds	SRC-064 pp. 9-13	Tile builds should produce RunReceipt/PROV records that Evidence Drawer can reference. Validation: Schema check for provenance JSON-LD; signature artifact verification.
ML-064-012	Frictionless GeoJSON/TopoJSON gating catches manifest errors early	SRC-064 pp. 78	Add early schema gates before tile generation or catalog release. Validation: Validate geometry type, CRS, bbox, and field type before materialization.
ML-064-030	Pre-commit canonicalizes changed JSON/GeoJSON/Parquet	SRC-064 pp. 138	Use pre-commit or CI to prevent non-deterministic layer manifests and catalog files. Validation: Canonicalization diff test.
ML-064-038	Artifact canonicalization covers GeoJSON, PMTiles, and COG	SRC-064 pp. 225-229	All public map artifacts need canonical digests and sidecar manifests. Validation: Recompute geom_sha256/record_sha256 and sha256sum PMTiles/COG.
ML-064-041	In-toto and Cosign attest map artifacts	SRC-064 pp. 227-229	Release manifests should reference attestations for tiles, COGs, and build snapshots. Validation: cosign verify-attestation and predicate schema tests.
ML-064-043	Delta-aware CI skips noiseless PRs but fails closed on material changes	SRC-064 pp. 229-234	Map style/tile/catalog CI can short-circuit only when no material release inputs changed. Validation: Material path fixture; no-change and changed-path tests.
ML-064-047	Preview renderer triggers on STAC, style, or renderer changes	SRC-064 pp. 241	Visual regression should run when inputs or rendering logic change. Validation: Pixel diff/Screenshot diff threshold and provenance.

Additional v1.9 records in this category: ML-064-081, ML-064-109

Evidence basis: ML-064-003 SRC-064 pp. 3-5; ML-064-005 SRC-064 pp. 4-5; ML-064-006 SRC-064 pp. 5-8; ML-064-007 SRC-064 pp. 9-13; ML-064-012 SRC-064 pp. 78.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport,

AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

V. Exports, Screenshots, Reports, and Citation Preservation

Covers exports, screenshots, preview renderer, reports, and citation preservation.

idea_id	title	source	implementation / validation consequence
ML-064-046	Preview renderer is STAC-driven and MapLibre-headless	SRC-064 pp. 238-242	Generate review screenshots from pinned styles and released or candidate artifacts. Validation: Renderer version, style digest, camera, and output hash checks.
ML-064-103	GLO plat georeferencing requires RMS and control-point receipts	SRC-064 pp. 536-538	Export reports/screenshots with georeference QA and citation preservation. Validation: RMS threshold and control point completeness tests.
ML-064-116	Exports and reports must retain citations and artifact digests	SRC-064 pp. 225-229, 536-539	Screenshot/report export must include citations, catalog links, and digest references. Validation: Export citation validation and digest inclusion tests.

Evidence basis: ML-064-046 SRC-064 pp. 238-242; ML-064-103 SRC-064 pp. 536-538; ML-064-116 SRC-064 pp. 225-229, 536-539.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

W. 3D / Cesium / Deck.gl / Overlay Interoperability

Covers 3D/Cesium/deck.gl/overlay interoperability and historic map warping.

idea_id	title	source	implementation / validation consequence
ML-064-036	IIIF plus Allmaps overlays historic maps into MapLibre/Leaflet	SRC-064 pp. 217-219	Historic map Story Nodes can reference original rights/metadata and render overlays. Validation: Validate IIIF manifest rights, GCP provenance, and annotation digest.
ML-064-037	Allmaps WarpedMapLayer belongs in Story Node map panels	SRC-064 pp. 218-219	Use as a conditional overlay plugin with dependency governance. Validation: Plugin allowlist and attribution/rights test.
ML-064-066	MapLibre/Cesium consume PMTiles and GeoJSON from outputs tiles web	SRC-064 pp. 535-536	Expose only published, public-safe outputs; do not use WORK/staging outputs. Validation: No public raw/work path test; release manifest source path check.

idea_id	title	source	implementation / validation consequence
ML-064-076	Sentinel-1 outputs support MapLibre/Cesium environmental evidence	SRC-064 pp. 621	2D/3D environmental overlays need same trust path. Validation: Layer release, sovereignty review, and evidence-citation tests.
ML-064-101	3D/Cesium remains conditional around evidence burden	SRC-064 pp. 534-536, 621	Use Cesium/3D only when vertical/terrain context materially improves evidence review. Validation: 2D/3D synchronization and Evidence Drawer parity tests.

Evidence basis: ML-064-036 SRC-064 pp. 217-219; ML-064-037 SRC-064 pp. 218-219; ML-064-066 SRC-064 pp. 535-536; ML-064-076 SRC-064 pp. 621; ML-064-101 SRC-064 pp. 534-536, 621.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

X. Anti-Patterns and Failure Modes

Covers anti-patterns and failure modes.

idea_id	title	source	implementation / validation consequence
ML-064-117	Layer toggle is not publication	SRC-064 pp. 90-92, 101-114, 225-229	UI layer lists should show release state and block unreleased layers. Validation: Unreleased tile load and layer toggle tests.
ML-064-118	Dense GeoJSON remains intake or small-runtime data, not production default	SRC-064 pp. 114-119, 215-217, 287-290	Use vector tiles for large public layers and GeoJSON for small candidate/runtime overlays. Validation: Tile size and feature-count threshold tests.
ML-064-119	Popups must not substitute for Evidence Drawer	SRC-064 pp. 92, 537-538	Popup content should be short and link to Evidence Drawer with policy/citation state. Validation: Popup citation and drawer-resolution tests.

Evidence basis: ML-064-117 SRC-064 pp. 90-92, 101-114, 225-229; ML-064-118 SRC-064 pp. 114-119, 215-217, 287-290; ML-064-119 SRC-064 pp. 92, 537-538.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

Y. Implementation Backlog and PR Plan

Covers implementation backlog, PR planning, watchers, and path discipline.

idea_id	title	source	implementation / validation consequence
ML-064-008	Dagster asset model treats tiles as governed artifacts	SRC-064 pp. 34-36	Treat PMTiles/COG/GeoParquet outputs as assets with explicit checks, not incidental files. Validation: Asset check for row counts, tile coverage, schema, and index health.
ML-064-088	Directory Rules warn against path claims without repo verification	SRC-064 pp. SRC-004; SRC-064 path snippets	All paths in this update remain PROPOSED unless verified in a mounted repo. Validation: Path-home ADR/drift check.
ML-064-098	Dynamic source watchers open PRs rather than publish directly	SRC-064 pp. 290-300	Source updates should enter review and validation lanes before tile rebuild. Validation: PR artifact, source diff, and gate checks.

Evidence basis: ML-064-008 SRC-064 pp. 34-36; ML-064-088 SRC-064 pp. SRC-004; SRC-064 path snippets; ML-064-098 SRC-064 pp. 290-300.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

Z. Open Questions and Verification Backlog

Covers unresolved verification backlog and unknowns.

idea_id	title	source	implementation / validation consequence
ML-064-049	MapLibre ecosystem performance changes are watchlist items	SRC-064 pp. 248	Treat version-specific ecosystem claims as NEEDS VERIFICATION before adoption. Validation: Quarterly ecosystem watchlist and official-source verification.
ML-064-120	Current implementation depth remains unknown without mounted repo	SRC-064 pp. SRC-064 overall; SRC-MASTER-PREV	All file paths, route names, package versions, and runtime behavior remain PROPOSED/NEEDS VERIFICATION. Validation: Mount repo and run path/test/workflow inspection next run.

Evidence basis: ML-064-049 SRC-064 pp. 248; ML-064-120 SRC-064 pp. SRC-064 overall; SRC-MASTER-PREV.

Required objects / contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, and rollback target remain the controlling families where applicable.

Public UI implications: map UI must show released/stale/degraded/denied state, must not fetch RAW/WORK/QUARANTINE or direct model outputs, and must route consequential claims through Evidence Drawer or Focus Mode finite outcomes.

Policy implications: rights, sensitivity, sovereignty/CARE labels, source roles, review state, and release state must be visible to gates before public exposure.

Validation tests: schema validation, catalog closure, digest/signature checks, no public raw path, no unreleased tile load, click-to-EvidenceBundle, citation validation, sensitive geometry denial, visual regression, accessibility, tile load budget, cache invalidation, and rollback replay.

Anti-patterns: treating MapLibre, tiles, screenshots, graph projections, popups, STAC records, or AI answers as sovereign truth; relying on style filters for sensitive geometry; or using source snippets as proof of current repo implementation.

Changed since prior master: v1.9 adds SRC-064-specific source evidence while retaining all v1.8 entries in Appendix A.

8. Components-Functions-Features Matrix

The matrix is split across two landscape tables to avoid cut-off text. Status values are CONFIRMED evidence for source ideas, PROPOSED implementation unless future repo evidence verifies existence.

component	function	feature	input artifacts	output artifacts	status
MapLibre GL JS	Render governed 2D web maps	Style/source/layer runtime, click context	StyleManifest, LayerManifest	Rendered map, click candidates	EXPANDED
MapLibre Native	Mobile/native rendering	Offline PMTiles packs, Vulkan parity candidate	Mobile release pack	Native map view	EXPANDED
MapLibre RS	Experimental renderer	Future Rust/WebGPU rendering	UNKNOWN	UNKNOWN	RETAINED
Style JSON	Define visual rendering	Pinned style, camera, source layers	StyleManifest	Style bundle	EXPANDED
source definition	Bind data source to style	Vector/raster/geojson/raster-dem /pmtiles	LayerManifest	Map source config	EXPANDED
layer definition	Render source-layer	Line/fill/symbol/raster etc.	StyleManifest	Layer render	EXPANDED
expressions	Compute visual style	Style filters/paint/layout	Style JSON	Rendered properties	RETAINED
filters	Limit rendered features	Visibility controls only	Style JSON	Filtered render	RETAINED
sprites	Icon sheets	Symbol rendering	sprite PNG/JSON	Icons	RETAINED
glyphs	Font glyph PBFs	Label rendering	glyphs URL	labels	RETAINED
MVT	Default vector tile encoding	Public vector tiles	GeoParquet, tile spec	MVT/PMTiles	EXPANDED
MLT	Pilot next-gen tile format	Potential smaller/faster tiles	pilot manifest	MLT tiles	EXPANDED
PMTiles	Single-file tile archive	Serverless/offline vector/raster delivery	TileArtifactManifest	PMTiles archive	EXPANDED
MBTiles	SQLite tile archive	Local/offline interim package	MBTiles file	Converted PMTiles	EXPANDED
COG	Cloud Optimized GeoTIFF	Raster/quicklook/terrain source	COG, metadata	COG asset	EXPANDED
Martin	Server-mediated tiles	Dynamic PostGIS/PMTiles serving	PostGIS/PMTiles	Tile endpoint	EXPANDED
Maputnik	Style authoring	Style editing and review	style JSON	candidate style	RETAINED
layer catalog	Expose released layers	Search/filter layer records	LayerManifest/STAC	catalog results	EXPANDED
Evidence Drawer	Resolve clicked evidence	EvidenceBundle display	EvidenceDrawerPayload	drawer state	EXPANDED
Focus Mode	Bounded AI map context	ANSWER/ABSTAIN/DENY/ERROR	MapContextEnvelope	Focus response	EXPANDED
timeline / time slider	Time-aware interaction	Filter by source/release/valid time	time metadata	time state	EXPANDED
MapContextEnvelope	Constrain AI context	Map view + selected features + evidence refs	map state	bounded context	EXPANDED
LayerManifest	Layer contract	Layer source, policy, catalog links	SourceDescriptor	governed layer	EXPANDED
StyleManifest	Style contract	Style JSON, sprites, glyphs, digests	style bundle	style release	EXPANDED
TileArtifactManifest	Tile/raster artifact contract	PMTiles/COG/MBTiles/MLT digest	tile artifacts	release assets	EXPANDED
MapReleaseManifest	Publishable map release	Version, artifacts, rollback	all manifests	release bundle	EXPANDED
PolicyDecision	Policy output	allow/deny/abstain reasons	policy input	decision object	EXPANDED
CitationValidationReport	Citation gate	Cite-or-abstain verification	citations	pass/fail report	RETAINED

component	function	feature	input artifacts	output artifacts	status
AIReceipt	AI run audit	Model/provider/context/output refs	Focus request	receipt	EXPANDED
rollback target	Reversal anchor	Prior release/digest/commands	release manifest	rollback record	EXPANDED
cache invalidation record	Cache governance	Old/new digests and invalidation scope	release manifest	cache record	EXPANDED

component	governed API dependency	evidence dependency	policy dependency	tile/style dependency	runtime/UI behavior	tests required
MapLibre GL JS	governed map API	EvidenceBundle	PolicyDecision	PMTiles/MVT/GeoJSON/COG	Interactive shell	style/render/click tests
MapLibre Native	governed mobile API	EvidenceBundle	offline policy	PMTiles/MBTiles	Offline/signed packages	mobile parity tests
MapLibre RS	governed API only	EvidenceBundle	policy gate	style/tile TBD	Not default	watchlist tests
Style JSON	style service/API	citation via layers	review state	sources/layers	visual state	visual regression
source definition	catalog API	EvidenceBundle	source policy	TileJSON/PMTiles/COG	source load	source validity
layer definition	catalog/layer API	EvidenceDrawer	policy labels	source-layer	layer toggle	layer validation
expressions	style API	no proof alone	policy safe	style spec	visual state	style expression tests
filters	style API	no proof alone	policy safe	style spec	visual state	sensitive geometry deny
sprites	static service	no proof alone	license	style dependency	symbol display	asset checks
glyphs	static service	no proof alone	font rights	style dependency	label display	glyph availability
MVT	tile service/CDN	EvidenceBundle	policy gate	PMTiles/MVT	vector layers	tile metadata
MLT	governed pilot API	EvidenceBundle	policy gate	MLT	pilot render	MVT fallback tests
PMTiles	CDN/OCI/gov API	EvidenceBundle	release policy	PMTiles	range fetch	Range/digest tests
MBTiles	local API	EvidenceBundle	offline policy	MBTiles/PMTiles	offline map	conversion tests
COG	CDN/API	EvidenceBundle	policy gate	COG/raster	raster layer	COG validate
Martin	governed tile API	EvidenceBundle	access policy	MVT/PMTiles/COG	tile server	access/cache tests
Maputnik	review API	no proof alone	review policy	style spec	style preview	style diff
layer catalog	catalog API	EvidenceBundle	policy decision	tile/style deps	layer list	catalog tests
Evidence Drawer	evidence API	EvidenceBundle	PolicyDecision	feature IDs	trust details	click resolution
Focus Mode	Focus API	EvidenceBundle	PolicyDecision	released layers	answer/abstain	citation tests
timeline / time slider	time API	temporal evidence	stale policy	time layers	slider	temporal tests
MapContextEnvelope	Focus API	EvidenceBundle refs	policy context	layer state	AI request	schema tests
LayerManifest	catalog API	EvidenceBundle refs	PolicyDecision	tile/style deps	layer load	manifest tests
StyleManifest	style API	layer citations	review policy	style deps	render	style tests
TileArtifactManifest	tile API/CDN	EvidenceBundle	release policy	tiles/raster	fetch/render	digest/range tests
MapReleaseManifest	release API	EvidenceBundle	PromotionDecision	tile/style deps	public map	rollback tests
PolicyDecision	policy API	evidence refs	policy engine	layer labels	UI state	policy tests
CitationValidationReport	citation API	EvidenceBundle	policy postcheck	n/a	Focus/export	citation tests
AIReceipt	AI API	EvidenceBundle	PolicyDecision	n/a	audit	receipt tests
rollback target	release API	evidence chain	promotion policy	tile/style deps	restore old state	rollback replay
cache invalidation record	CDN/API	evidence chain	release policy	PMTiles/COG	cache refresh	cache tests

9. Tile Strategy Chapter

Comparison	When it wins	When it fails	KFM trust consequence	Release / rollback consequence	Validation needed
GeoJSON vs vector tiles	GeoJSON wins for small candidates, field notes, and API feature results. Vector tiles win for large public layers.	GeoJSON fails at scale and can leak attributes; vector tiles fail if source-layer/metadata/digest missing.	Both are carriers; claims still resolve EvidenceBundle.	Vector tiles need TileArtifactManifest and rollback target.	GeoJSON schema, tile metadata, source-layer, digest, size budget.
MVT vs MLT	MVT wins as mature default. MLT may win when pilot proves smaller/faster decode.	MLT fails if renderer/tooling support is unverified.	Do not treat performance claim as truth or default.	MLT release needs fallback to MVT.	MVT/MLT parity, renderer support, visual diff, fallback.
PMTiles vs MBTiles	PMTiles wins for static/CDN/Range and offline web. MBTiles wins for local/mobile conversion or legacy tooling.	PMTiles fails without Range/CORS; MBTiles fails as unmanaged public archive.	Both require release manifest and checksum.	Conversion chain must be recorded.	Range/CORS, digest, conversion reproducibility.
Static tiles vs dynamic tile server	Static wins for stable public-safe releases. Dynamic server wins for steward-controlled, PostGIS-backed, role-aware layers.	Static fails for rapidly changing/restricted data; dynamic fails if access differs from policy.	Dynamic serving must remain behind governed policy.	Static releases rollback by manifest; dynamic needs config/query rollback.	Access tests, cache tests, SQL/source tests.
PMTiles serverless vs Martin server-mediated	PMTiles wins for simple public artifacts; Martin wins for dynamic PostGIS. MBTiles/PMTiles mediation, or operational server needs.	PMTiles fails when per-user filtering is needed; Martin fails if exposed as raw tile backend.	Serverless is not governance-free; Martin is not truth store.	PMTiles rollback by digest; Martin rollback by config and data snapshot.	Range/CORS, Martin config, policy checks.
Vector vs raster	Vector wins for interaction, labels, filters. Raster wins for imagery, quicklook, DEM derivatives.	Vector fails for dense continuous surfaces; raster fails for feature click semantics unless tied to metadata.	Both need catalog and evidence refs.	Raster/COG and vector tiles need separate artifact manifests.	COG validate, vector tile metadata, click fallback.
COG vs raster tiles	COG wins for source-like raster with Range and overviews. Raster tiles win for simple rendered basemap previews.	COG fails without proper tiling/overviews; raster tiles fail if they hide data/provenance.	COG is still a carrier; quicklook is not full data.	COG rollback by digest; rendered raster tiles by style/source digest.	COG profile, overviews, byte-range, metadata.
Terrain-enabled vs non-terrain	Terrain wins when elevation/vertical context carries evidence burden.	Terrain fails when it adds spectacle or performance cost without evidence value.	3D/terrain must preserve Evidence Drawer parity.	Terrain release needs DEM/derivative provenance and rollback.	DEM lineage, terrain visual regression, 2D fallback.
2D default vs 3D conditional	2D wins for clarity, accessibility, and broad support. 3D/Cesium wins for LiDAR/GLO, terrain, volume, or vertical inspection.	3D fails when it hides uncertainty or lacks click/evidence parity.	2D is default; 3D is conditional.	3D scenes need separate manifest and rollback.	3D picked-feature, screenshot, metadata, and accessibility tests.
Public-safe tiles vs restricted/internal tiles	Public-safe wins for ordinary UI. Restricted tiles support steward review or internal analysis.	Public tiles fail if exact sensitive geometry remains; internal tiles fail if accidentally exposed.	No sensitive geometry hidden only by style filters.	Public release requires redaction/generalization receipt.	Sensitive geometry deny, H3 generalization, no public raw path.

10. Governance and Trust-Membrane Chapter

The governed flow remains: released layer -> user click -> governed API -> EvidenceBundle -> Evidence Drawer -> Focus Mode answer / abstain / deny / error. SRC-064 reinforces this with PMTiles click-to-STAC, Story Node links, Focus Mode MAY/MUST NOT constraints, and graph-health/degraded-state patterns.

- No public RAW path. Public clients do not fetch RAW, WORK, QUARANTINE, unpublished candidates, direct source services, or canonical/internal stores.
- No direct model client. Focus Mode uses governed API, EvidenceBundle, PolicyDecision, CitationValidationReport, and AIReceipt.
- No canonical/internal client fetch. Neo4j/PostGIS/ArcGIS/OGC source endpoints stay behind governed services unless a public-safe released artifact is explicitly manifest-bound.
- No unreleased tile load. Layer toggles and addSource calls are not publication.
- No sensitive geometry hidden only by style. Sensitive geometry is generalized, redacted, delayed, restricted, or denied before tile generation.
- No popup as Evidence Drawer substitute. Popups may summarize and link, but claims resolve through Evidence Drawer.
- No uncited export. Screenshots, previews, reports, Story Nodes, and Focus Mode outputs preserve citations, digests, and artifact provenance.

SRC-064 adds a concrete click pattern: a viewer can add a PMTiles source and layer, but click behavior should show STAC metadata and a read-only OCI snapshot link only after validation. This is consistent with the KFM posture that rendered features identify candidates, not proof.

11. Implementation-Ready Object Map

Object family	Field intent / minimum content	Status	Path placement note
SourceDescriptor	source_id, source_family, endpoint, rights, cadence, role, license_spdx, sensitivity, authoritative_scope.	PROPOSED; required by SRC-064 watchers and source ETL.	Path PROPOSED until Directory Rules and repo evidence verify home.
LayerManifest	layer_id, source_id, catalog refs, policy labels, release_state, tile/style dependency, attribution.	EXPANDED by SRC-064 view registry and layer pointer ideas.	Use schema home per accepted ADR; do not create parallel homes.
StyleManifest	style_id, style_json_digest, renderer version, camera presets, sprite/glyph refs, visual QA.	EXPANDED by preview renderer ideas.	Style paths PROPOSED.
TileArtifactManifest	artifact_id, type, uri, sha256, sidecar hashes, tiler, flags, zooms, input digests, Range/CORS.	EXPANDED by PMTiles/CO G digest ideas.	Artifact location PROPOSED; release manifest governs exposure.
MapReleaseManifest	release_id, LayerManifest refs, StyleManifest refs, TileArtifactManifest refs, PromotionDecision, rollback target, cache invalidation.	EXPANDED.	Release home PROPOSED.
EvidenceBundle	claim/evidence refs, source role, quote/paraphrase, temporal/spatial scope, review state.	RETAINED / central.	Outranks generated language and tiles.
EvidenceDrawerPayload	clicked feature id, evidence refs, catalog refs, policy state, citations, digests, related Story Nodes.	EXPANDED by Story Node and click patterns.	UI DTO path PROPOSED.
MapContextEnvelope	viewport, selected feature IDs, released layers, time context, policy state, evidence refs.	EXPANDED by Focus Mode constraints.	Focus API path PROPOSED.
FocusModeRequest	question/scope, MapContextEnvelope, evidence refs, user role, requested transform.	EXPANDED.	No direct model client.
FocusModeResponse	ANSWER/ABSTAIN/DENY/ERROR, citations, policy decision, confidence, limitations, AIRceipt ref.	EXPANDED.	Must not reveal sensitive coordinates or override CARE tags.
AIRceipt	model/provider, prompt/context hash, evidence refs, policy decisions, output digest, citation report.	EXPANDED by structured output and explainability notes.	No private chain-of-thought persisted as truth.
CitationValidationReport	citations checked, missing evidence, unsupported claims, pass/fail.	RETAINED.	Required for Focus/exports.
PolicyDecision	allow/deny/abstain/error, reason codes, obligations, redaction/generalization profile, approver.	EXPANDED by sovereignty label sync.	Policy paths PROPOSED.
PromotionDecision	gate results, release allow/deny, proof refs, rollback target, reviewer.	EXPANDED.	Promotion is state transition.
RunReceipt	run_id, tool versions, inputs, outputs, hashes, commit, SBOM, telemetry, signature.	EXPANDED by ORAS/Cosign/provenance.	Receipts stay process memory.
rollback target	prior release id/digests, restore commands, cache undo, validation after restore.	EXPANDED.	Must be present before release.
stale-source fixture	old timestamp, ETag/Last-Modified behavior, stale badge expected.	EXPANDED by conditional watchers.	Use no-network fixture.
sensitive-geometry deny fixture	exact coordinate in sensitive class, public request denied or generalized.	EXPANDED by field capture, LiDAR/GLO, Sentinel/HydroGNSS.	Must fail closed.

12. Validation and Test Plan

- schema validation for SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceDrawerPayload, MapContextEnvelope, FocusModeResponse, RunReceipt, and PolicyDecision
- source ledger completeness and source_id stability
- style source/layer validity, source-layer presence, renderHints enum, style JSON digest
- PMTiles / MVT / MLT manifest checks, including MLT pilot fallback to MVT
- COG / range request / overview / byte-range sidecar requirements
- no public raw path and no direct canonical/internal store fetch
- no direct model client and no raw LLM output in UI
- no unreleased tile load or addSource from candidate artifacts
- sensitive geometry deny, redaction receipt, H3 generalization, and no style-only hiding
- source rights unknown blocks release; sovereignty labels stay in STAC/DCAT/PROV sync
- stale source badge, degraded graph state, and abstain behavior
- clicked feature resolves EvidenceBundle and EvidenceDrawerPayload
- citation validation pass/fail for Focus Mode, Story Nodes, exports, and preview captions
- Focus Mode answer with citations and Focus Mode abstain on missing evidence
- rollback restores prior manifest, old artifact digest, and old cache state
- cache invalidation recorded with old/new release ids and artifact digests
- visual regression for styles, previews, screenshots, and 3D/terrain views
- keyboard navigation / accessibility / alt-text and caption validation
- tile load budget, PMTiles Range latency, route load, and interaction-to-drawer latency budget
- OCI/ORAS digest, Cosign attestation, SBOM, PROV referrer, and RunReceipt verification
- field capture redaction, EXIF stripping, deterministic filename, and consent flag checks
- Allmaps/IIIF rights, GCP provenance, and georeferencing precision checks
- LiDAR/GLO CRS/RMS/control-point/terrain-derivative QA checks
- Sentinel/HydroGNSS QA masks, uncertainty thresholds, sovereignty review, and data-use notices

13. Anti-Patterns Register

Anti-pattern	Why it is dangerous	Mitigation / test
Treating MapLibre as truth authority	Renderer output is visual context, not proof.	EvidenceBundle resolution and cite-or-abstain tests.
Treating tiles as proof	Tiles are downstream carriers and may be simplified/redacted.	TileArtifactManifest + EvidenceBundle linkage.
Treating layer toggle as publication	Visibility is not release approval.	No unreleased tile load test.
Hiding exact sensitive geometry with style filters	Geometry still ships to client.	Pre-tile redaction/generalization and deny fixture.
Dense GeoJSON as production default for large datasets	Performance and attribute leakage risks.	Feature-count threshold and tile budget.
Uncontrolled style drift	Visual meaning changes without review.	Style digest and visual regression.
Plugin adoption without review	Maintenance, license, and security risks.	Dependency allowlist and health checks.
Direct browser access to model runtime	Bypasses policy, evidence, and audit.	No direct model client test.
AI answers based only on rendered map features	Rendered features are candidates, not evidence.	Focus Mode requires EvidenceBundle and citations.
Uncited popups or exports	Detached claims become uninspectable.	Citation validation on popups/exports.
MLT as default before KFM validation	Version/tooling support remains unsettled.	MLT pilot with MVT fallback.
3D/globe as default when 2D is clearer	Can obscure uncertainty and reduce accessibility.	2D default and 3D evidence-burden test.
Mapbox-era backports without legal/security review	License/security ambiguity.	Dependency review and ADR.
Mutable OCI tag used as production reference	Rollback and verification weaken.	Digest-required promotion gate.
Source snippets treated as repo facts	No mounted repo evidence.	Path-home verification and Directory Rules check.
Quicklook COG treated as full data	Preview assets can understate limitations.	Asset role and disclaimer validation.
Offline field pack without revocation/stale state	Disconnected maps can outlive permissions.	Offline pack expiry and release check.
Allmaps historic overlay treated as precise survey	Georeferencing uncertainty may be high.	GCP/RMS and uncertainty display.

14. Open Questions and Verification Backlog

- MapLibre GL JS target version and package pin.
- MapLibre Native Android Vulkan support, style subset, and Evidence Drawer parity.
- MapLibre RS/WebGPU maturity and whether it belongs in any KFM lane.
- MLT support status across GL JS, Native, tooling, Martin, and style/source encoding.
- PMTiles hosting Range/CORS behavior on chosen object store/CDN.
- OCI/ORAS registry selection, referrer support, retention, and access model.
- Martin deployment/access model for PostGIS-backed dynamic tiles.
- Plugin allowlist: pmtiles protocol, Allmaps WarpedMapLayer, any COG protocol adapters.
- Style/sprite/glyph hosting and cache invalidation model.
- Attribution and rights for KGS M-118, IIF maps, field media, Sentinel/HydroGNSS sources, WZDx feeds.
- Sensitive geometry transform rules: H3 resolution, redaction profiles, generalization thresholds, review roles.
- Release manifest location under actual repo Directory Rules and ADR state.
- CI workflow names and whether OPA/Conftest/Cosign are available and pinned.
- Accessibility acceptance thresholds, alt-text coverage, keyboard navigation, and Focus Mode a11y.
- Performance budgets for PMTiles load, COG quicklook load, route chunk hydration, and interaction-to-drawer latency.
- Whether tilehash sidecars are required for all PMTiles or only high-integrity releases.
- Whether COG internal tile/overview manifests are feasible for routine releases.
- How cache invalidation records are stored and surfaced to rollback.
- How field capture offline packs expire and revoke.
- How Focus Mode handles degraded graph health or stale catalog state.
- How Story Node v3 links back to STAC/DCAT, Neo4j, and semantic document IDs.
- How directory path snippets in SRC-064 are reconciled with Directory Rules and actual repo topology.

15. How to Continue This Master PDF Next Run

- Attach this PDF again with any new documents.
- The next run should read this PDF first as SRC-MASTER-PREV.
- Preserve all idea IDs unless merged or superseded.
- Update the delta table.
- Add new sources to the source ledger.
- Add new discoveries to the idea index.
- Recompute counts.
- Keep uncertainty labels visible.

Recommended next-run verification focus: mount the actual KFM repo, inspect Directory Rules conformance, verify schema home, inspect app/UI/tile/style paths, test any existing MapLibre package version, run source-ledger checks, and verify whether any SRC-064 path or workflow already exists before promoting it beyond PROPOSED.

Final PDF Self-Check

Check	Result
No obvious cut-off text	PASS after wrapped tables, landscape pages for wide matrices, and selected render verification.
No missing source ledger	PASS - Section 4 includes current prompt, prior master, new source, and retained source crosswalk.
No missing delta section	PASS - Section 3 includes continuity and delta since v1.8.
No missing open questions	PASS - Section 14 lists new and carried-forward verification items.
No unsupported implementation claims	PASS - all repo paths and code snippets from SRC-064 remain PROPOSED/NEEDS VERIFICATION.
No forgotten prior master-PDF ideas	PASS - v1.8 baseline appended page-for-page as Appendix A.
No external research silently used	PASS - no EXT sources added; version-sensitive facts from SRC-064 are marked NEEDS VERIFICATION.
PDF-only user-facing output	PASS - final deliverable is one PDF. Temporary files were used only for rendering/verification.

Appendix A follows: full retained v1.8 Master MapLibre Components-Functions-Features.pdf baseline, appended page-for-page as SRC-MASTER-PREV.

Master MapLibre Components-Functions-Features.pdf

Cumulative KFM MapLibre, Tile, Renderer, Style, UI, Evidence Drawer, and Release
Architecture Atlas

Field	Value
Version	v1.8 cumulative update
Run date	2026-05-10
Attached source count	42 PDFs visible in /mnt/data including prior master; 41 non-master source documents.
Prior master PDF detected	Yes - v1.7 copied as SRC-MASTER-PREV before overwrite; prior pages 468; original attachment sha256 prefix 9e91ea6237d710ea.
Output mode	PDF only. No ZIP, tarball, scaffold folder, repository patch, loose source tree, or extra user-facing artifact.
Truth posture	CONFIRMED doctrine/source scans; PROPOSED implementation guidance; UNKNOWN current repo/runtime depth; NEEDS VERIFICATION for versions, hosting, rights, source endpoints, policy enforcement, performance budgets, mobile/native parity, and path placement.
Evidence limits	No mounted KFM repo, tests, workflows, dashboards, logs, release manifests, or runtime traces were available. Paths in sources are source-reported and PROPOSED unless repo evidence later verifies them.
Source ledger count	43 current-run records: SRC-RUN-PROMPT, SRC-MASTER-PREV, 39 retained non-master source records, and NEW SRC-062 and SRC-063.
Idea count	2658 cumulative idea records: 2550 prior + 108 v1.8 delta records.
New discoveries count	108 delta records from SRC-062 and SRC-063.
Expanded category clusters	25 category clusters expanded by current-run source evidence; F is retained only.
Superseded ideas	0 - no attached evidence invalidated a prior idea.
Open verification items	445 cumulative estimate: prior 393 + 52 new or sharpened items.

Core determination: MapLibre remains a downstream renderer and interaction runtime. Tiles, PMTiles, MVT, MLT, COGs, style JSON, popups, screenshots, Story Nodes, 3D scenes, graph projections, catalog records, and AI answers remain downstream carriers, not sovereign truth.

1. Table of Contents

Navigation note. This update packet is current-run specific. The full v1.7 baseline follows as Appendix A so prior source ledger entries, category chapters, matrices, idea IDs, and open questions are not silently deleted.

- 1. Cover Page and run metadata
- 2. Executive Determination
- 3. Continuity and Delta Since Last Run
- 4. Source Ledger
- 5. Master Idea Index - v1.8 Delta
- 6. Category Atlas
- 7. Detailed Category Chapters
- 8. Components-Functions-Features Matrix
- 9. Tile Strategy Chapter
- 10. Governance and Trust-Membrane Chapter
- 11. Implementation-Ready Object Map
- 12. Validation and Test Plan
- 13. Anti-Patterns Register
- 14. Open Questions and Verification Backlog
- 15. How to Continue This Master PDF Next Run
- Final PDF Self-Check
- Appendix A. Full retained v1.7 Master PDF baseline (SRC-MASTER-PREV, appended page-for-page)

2. Executive Determination

CONFIRMED from SRC-MASTER-PREV and retained doctrine: MapLibre is a disciplined 2D renderer and interaction runtime inside a governed KFM shell. It is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority, or AI authority. Public surfaces use governed APIs, released artifacts, catalog records, tile services, EvidenceBundle resolution, and release manifests.

CONFIRMED from this run: the prior v1.7 master baseline was detected, copied before overwrite, and appended page-for-page. Two new source packets were analyzed: SRC-062 (101 pages, sha256 prefix b658b53dcdab4fc2) and SRC-063 (189 pages, sha256 prefix 21fd73dc14df5cab).

The strongest SRC-062 additions are STAC 1.1 alignment for layer catalogs, 3D Tiles Next/Cesium conditional support, typed CloudEvents deltas, governed hydrology/logistics intake patterns, soils/surficial geology COG/GeoParquet outputs, CARE-bound catalog metadata, zero-trust ingest, and PR-first change detectors.

The strongest SRC-063 additions are GitHub App orchestration, fail-closed acceptance harnesses, soil watcher COG/GeoParquet/PMTiles readiness, COG/Zarr deterministic packaging, geospatial quick QC, OCI/ORAS artifact provenance, pgSTAC to GeoParquet to PMTiles to OCI publication, Ollama/Focus runtime configuration risk, MLT as a pilot candidate, and watcher/run-receipt schemas.

PROPOSED synthesis: v1.8 should be treated as a catalog/provenance, reproducible-build, watcher/CI, COG/Zarr, PMTiles/OCI, MLT-pilot, 3D/Cesium, sensitive-policy, and Focus-runtime governance update. It does not reorganize the prior master. It adds two sources, preserves all prior records, and sharpens object maps, test gates, and release backlogs.

UNKNOWN: actual repo topology, package manager, schema home, UI component homes, MapLibre GL JS target version, MLT implementation status, PMTiles tooling, PMTiles/COG/Zarr/3D-Tiles hosting headers, browser/device performance, mobile/native parity, plugin allowlist, source rights, policy enforcement, release workflow, and branch protection remain unverified.

NEEDS VERIFICATION: every package version, tool version, source URL, feed schema, hosted header, release manifest, SBOM, telemetry file, artifact digest, CARE consent assertion, OCI referrer, MLT support claim, and local Focus Mode runtime setting before operational use.

Highest-risk anti-pattern: loading a public map source, PMTiles archive, COG, MLT tile, STAC search result, logistics incident layer, 3D scene, Story Node, popup, or Focus Mode answer before EvidenceBundle, manifest, PolicyDecision, PromotionDecision, citation validation, proof object, and rollback target exist.

Metric	Value
Previous version	v1.7 cumulative update
Current version	v1.8 cumulative update
Prior ideas carried forward	2550
New SRC-062 delta ideas	44
New SRC-063 delta ideas	64
Total cumulative idea count	2658
Superseded ideas	0
Open verification items	445 estimate

3. Continuity and Delta Since Last Run

SRC-MASTER-PREV is the prior v1.7 Master MapLibre Components-Functions-Features.pdf. It remains the cumulative memory baseline and is appended page-for-page. It does not outrank original evidence. Current-run source evidence can refine, expand, conflict with, or supersede its entries. No supersession occurred in this run.

Delta item	Result
Prior version detected	v1.7, 468 pages, original attachment sha256 prefix 9e91ea6237d710ea
Version rule applied	+0.1 ordinary cumulative update. No fundamental structural reorganization.
New sources added	SRC-062 New Ideas 12-29-2025.docx.pdf and SRC-063 New Ideas Feb-2026.pdf.
Ideas retained	All 2550 prior idea records preserved by appending SRC-MASTER-PREV unchanged.
New records	108 ML-062-* and ML-063-* delta records.
Merged / superseded	0 merged, 0 superseded.
Conflicts discovered	No direct source conflict. Main tension remains implementation maturity: source packets include implementation-style paths and code snippets, but no mounted repo/runtime proof was available.
Open questions carried forward	Package versions, schema home, source rights, MLT status, mobile parity, hosting headers, sensitive geometry, release manifest home, CI workflow, accessibility and performance budgets.
New open questions	MLT production readiness; OCI referrers; COG/Zarr tool pins; pgSTAC role model; watcher schema home; Ollama context-length controls; CARE transform rules; GitHub App/check-run enforcement.

3A. Prior idea comparison summary

prior idea range / cluster	prior title scope	current status	source that changed status	note
All prior idea IDs through v1.7	Full retained v1.7 idea index and category chapters	RETAINED	SRC-MASTER-PREV	No prior idea was deleted or superseded.
MVT / PMTiles clusters	Vector tile, PMTiles, static/dynamic delivery, runtime verification	EXPANDED	SRC-063 pp.150-153; pp.120-128	Adds pgSTAC->GeoParquet->PMTiles->OCI and ORAS/Cosign provenance.
COG / raster clusters	COG, Zarr, DEM, raster, remote-sensing time series	EXPANDED	SRC-063 pp.26-33; SRC-062 pp.64-76	Adds deterministic COG/Zarr packaging, quick QC, soil/geology raster gates.
3D / Cesium clusters	Conditional 3D/globe/terrain support	EXPANDED	SRC-062 pp.20-26	Adds 3D Tiles Next implicit tiling, metadata, digest tree and picked-feature QA.
Focus Mode clusters	Governed AI map context and finite outcomes	EXPANDED	SRC-063 pp.153-154; SRC-062 pp.98-101	Adds local context-length risk and Focus diff review.
Sensitive geometry / rights clusters	CARE, geoprivacy, restricted geometry	EXPANDED	SRC-062 pp.77-83; SRC-063 pp.112-116	Adds CARE-bound STAC/DCAT and rights text gates.
Open verification backlog	Versions, hosting, policy, source rights	EXPANDED	SRC-062/SRC-063	Adds OCI referrers, MLT pilot, watcher schemas, context length, pgSTAC roles.

4. Source Ledger

This ledger is a control surface, not a bibliography. It states what each source can support and what it cannot prove. Current external facts embedded in source PDFs remain source-supported but NEEDS VERIFICATION before operational package pins, access decisions, source activation, or release claims.

source_id	file / document title	source type	status	role in this PDF	MapLibre / tile relevance	limitations
SRC-RUN-PRO MPT	Current user prompt	Prompt	CONFIRMED	Controls output shape, truth labels, source handling, PDF-only rule, and KFM invariants.	High requirements source.	Instruction source; not implementation proof.
SRC-MASTER-P REV	Master MapLibre Components-Functions-Features.pdf v1.7	Prior master PDF	CONFIRMED / RETAINED	Cumulative baseline appended page-for-page as Appendix A.	Highest continuity source; 2550 prior ideas retained.	Does not outrank original evidence.
SRC-003	KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf	Map/UI doctrine	RETAINED	Renderer boundary, governed shell, Evidence Drawer, Focus Mode.	High.	No mounted repo proof.
SRC-004	Directory Rules.pdf	Directory doctrine	RETAINED	Responsibility-root, schema-home, lifecycle, ADR/drift rules.	High for proposed file homes.	Specific paths remain PROPOSED.
SRC-005	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf	Pipeline manual	RETAINED	Lifecycle law, receipts, promotion, query/save/recompile loop.	High governance relevance.	No runtime proof.
SRC-006	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan_v1_1.pdf	Build plan	RETAINED	Trust spine, pre-RAW events, public-safe defaults, map architecture.	High doctrine.	Greenfield/proposed-to-create posture.
SRC-007	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf	UI + AI plan	RETAINED	Whole UI, governed API, MapLibre boundary.	High UI lineage.	No repo proof.
SRC-008	KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf	Governed AI plan	RETAINED	Provider-neutral AI, finite outcomes, citation validation.	High Focus Mode relevance.	No runtime proof.
SRC-009	Ollama & Ubuntu Information.pdf	Runtime guide	RETAINED	Local model behind governed API; no direct public model path.	Medium Focus relevance.	Version/runtime facts need verification.
SRC-010	kfm_encyclopedia.pdf	Capability encyclopedia	RETAINED	System-wide KFM domain and capability atlas.	High doctrine.	Implementation depth unknown.
SRC-011	# Kansas Frontier Matrix Implementation Reference.pdf	Implementation reference	RETAINED	Prior public-repo summary and MapLibre/PMTiles stack signals.	Lineage signal.	Prior connector claims not reverified.
SRC-012	a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf	GIS/cartography reference	RETAINED	Representation/projection/map communication concepts.	Conceptual support.	Not KFM proof.
SRC-013	Advanced-SQL-Concepts.pdf	SQL reference	RETAINED	Analytic SQL concepts.	Low direct map relevance.	Not MapLibre-specific.
SRC-014	AI_Concepts_Using_Python.pdf	AI reference	RETAINED	AI/data workflow background.	Low direct map relevance.	Not KFM proof.
SRC-015	Archaeological 3D GIS.pdf	3D GIS reference	RETAINED	3D GIS and archaeological visualization context.	Supports conditional 3D.	External examples only.
SRC-017	Designing Great Web APIs.pdf	API reference	RETAINED	API contract and resource design.	Supports governed API dependency.	Not KFM proof.
SRC-018	developing-time-oriented-database-applications-in-sql.pdf	Temporal DB reference	RETAINED	Valid/transaction time concepts.	Supports timeline maps.	Not KFM proof.
SRC-019	Domain-Driven Design Reference.pdf	DDD reference	RETAINED	Bounded context and published language.	Supports responsibility boundaries.	Not KFM proof.
SRC-020	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf	GIS lab/textbook	RETAINED	Environmental GIS/hazards/climate layer context.	Conceptual support.	ArcGIS-centric.
SRC-022	GIS Succinctly.pdf	GIS reference	RETAINED	Spatial database and geometry basics.	Conceptual support.	Not KFM proof.
SRC-023	GIS in Sustainable Urban Planning and Management.pdf	Urban GIS reference	RETAINED	Planning support, indicators, resilience.	Context for planning layers.	Not KFM proof.
SRC-024	instant-web-scraping-with-java.pdf	Scraping reference	RETAINED	Web acquisition and error handling background.	Low direct map relevance.	Not KFM proof.

source_id	file / document title	source type	status	role in this PDF	MapLibre / tile relevance	limitations
SRC-025	KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Receipts, deterministic identity, promotion gates.	High governance lineage.	Prior synthesis.
SRC-026	KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	PMTiles, MLT, map stack guidance.	High map lineage.	Prior synthesis.
SRC-027	KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Pre-RAW, watchers, catalog integrity.	High map/data lineage.	Prior synthesis.
SRC-028	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf	Domain blueprint	RETAINED	Agriculture layers, catalog closure, MapLibre/Evidence Drawer.	Domain layer examples.	Paths PROPOSED.
SRC-029	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	Domain blueprint	RETAINED	Archaeology exact-location denial and generalized map outputs.	Strong sensitivity relevance.	Paths PROPOSED.
SRC-030	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Air/smoke/EO layers and knowledge-character labels.	Air layer relevance.	Paths PROPOSED.
SRC-031	KFM_Fauna_Architecture_PDF_Only_Report.pdf	Domain blueprint	RETAINED	Fauna occurrence geoprivacy and public-safe products.	Sensitive occurrence relevance.	Paths PROPOSED.
SRC-032	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	Domain blueprint	RETAINED	Rare plant controls and occurrence redaction.	Sensitive flora relevance.	Paths PROPOSED.
SRC-033	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Geology/resource layers and public-safe geometry.	Geology tile relevance.	Paths PROPOSED.
SRC-034	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	Domain blueprint	RETAINED	Habitat assignment proof slice and drawer payloads.	Habitat/fauna relevance.	Paths PROPOSED.
SRC-035	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	Domain blueprint	RETAINED	Habitat patches, suitability, connectivity.	Domain layer examples.	Paths PROPOSED.
SRC-036	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	Domain blueprint	RETAINED	Hazards boundary; no emergency alerting.	Hazard map relevance.	Paths PROPOSED.
SRC-037	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Hydrology proof lane and NFHL/USGS/HUC layers.	Hydrology map foundation.	Paths PROPOSED.
SRC-038	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	Domain blueprint	RETAINED	Living-person/DNA restrictions and temporal land assertions.	Sensitivity/parcel relevance.	Paths PROPOSED.
SRC-039	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	Domain blueprint	RETAINED	Transport/corridor layers, graph projections, drawer payloads.	Transport map relevance.	Paths PROPOSED.
SRC-040	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	Domain blueprint	RETAINED	Settlement/infrastructure layer controls and sensitive assets.	Infrastructure relevance.	Paths PROPOSED.
SRC-041	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	Domain blueprint	RETAINED	Soil maps, SSURGO/gSSURGO, PMTiles/COG lineage.	Soil tile/raster relevance.	Paths PROPOSED.
SRC-056	KFM_Pass_15_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Runtime proof lanes and map-serving proof parity.	High prior master relevance.	Already incorporated in baseline.
SRC-057	KFM_Pass_16_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Geospatial delivery, PMTiles, MapLibre default, promotion gates.	High prior master relevance.	Retained; no new scan conflict.
SRC-062	New Ideas 12-29-2025.docx.pdf	New Ideas packet	NEW	Adds STAC 1.1 alignment, 3D Tiles Next, CloudEvents deltas, hydrology/logistics intake, soil/geology catalog rules, CARE metadata, zero-trust ingest, PR-first change detectors.	High indirect MapLibre/tile relevance; pages 101; sha b658b53dcdb4fc2.	No mounted repo proof; external facts in packet not revalidated.

source_id	file / document title	source type	status	role in this PDF	MapLibre / tile relevance	limitations
SRC-063	New Ideas Feb-2026.pdf	New Ideas packet	NEW	Adds GitHub App orchestration, acceptance harnesses, soils watcher, COG/Zarr QC, OCI/ORAS, pgSTAC->GeoParquet->PMTiles->OCI, MLT pilot, watcher schemas.	Very high MapLibre/tile relevance; pages 189; sha 21fd73dc14df5cab.	No mounted repo proof; version/package claims require verification.

5. Master Idea Index - v1.8 Delta

The full cumulative v1.7 idea index of 2550 prior ideas is retained in Appendix A via SRC-MASTER-PREV. The records below add the v1.8 delta from SRC-062 and SRC-063. Implementation and validation consequences are PROPOSED unless a future mounted-repo inspection verifies them.

idea_id	idea title	category	status / source	normalized statement / evidence	implementation consequence	validation / risk
ML-062-001	Neo4j/GDS version coupling protects map-graph projections	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-062 pp.1-8	Neo4j/GDS migration content treats graph analytics as a coupled, pinned operational component; MapLibre graph overlays should consume released graph products, not floating plugin behavior.	Keep graph outputs behind manifests and governed APIs; do not let rendered network overlays depend on unpinned GDS procedures.	Driver/GDS compatibility matrix; rollback rehearsal; graph-projection fixture diff.
ML-062-002	Driver-first graph upgrades reduce rollback risk for UI/API paths	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-062 pp.4-6	The source recommends upgrading drivers before Neo4j server upgrades because application rollback is easier than database rollback.	For map APIs that query graph projections, pin client drivers and test older/newer server pairs before public release.	Compatibility tests; rollback card for graph-backed map layers.
ML-062-003	Graph invariant outputs are evidence artifacts, not map truth	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW SRC-062 pp.7-9	Graph invariant snapshots and PROV records are emitted artifacts; they support governance but do not replace EvidenceBundle or release state.	Treat graph delta/invariant overlays as downstream carriers in the MapLibre shell.	No direct graph-to-popup truth test; EvidenceBundle resolution required.
ML-062-004	STAC Collections as stable layer contracts	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.10-14	STAC Collections should use stable IDs, license, provider, and spatial/temporal extent; they are good dataset contracts for layer catalog entries.	LayerManifest should reference stable STAC Collection IDs rather than ad-hoc source URLs.	STAC schema validation and stable-id non-reuse test.
ML-062-005	STAC API extension set supports governed layer search	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW SRC-062 pp.11-17	The source highlights filter, query, fields, sort, and free-text search as practical STAC API capabilities.	Layer catalog search can expose these functions through governed APIs, not direct client scraping.	Search contract tests for bbox, datetime, field selection, sort, and limit behavior.
ML-062-006	STAC search GeoJSON FeatureCollections are layer candidates	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW SRC-062 pp.16-17	STAC API search returns GeoJSON FeatureCollections of Items that can feed candidate lists and map extents.	Use search results as candidates for governed layer selection; avoid treating them as published layer payloads.	GeoJSON FeatureCollection schema and item asset checks.
ML-062-007	Fields extension minimizes map catalog payloads	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW SRC-062 pp.15-16	The Fields extension can constrain returned STAC fields and reduce payload size.	Map catalog APIs should support minimal field payloads for performance.	Payload-size regression test; required fields remain present.
ML-062-008	STAC asset conventions reduce tile/raster/vector friction	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.14-18	The source emphasizes stable asset roles, media types, and links to avoid client confusion.	TileArtifactManifest and LayerManifest should mirror asset roles for COG, GeoParquet, PMTiles, and 3D Tiles.	Asset role/media-type checks; broken-link tests.
ML-062-009	STAC-ish JSON blocks map release	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW SRC-062 pp.18-19	The source names invalid STAC-ish JSON as a common failure mode.	Do not release style or layer entries whose catalog records fail strict validation.	Fail-closed STAC validation fixture.
ML-062-010	3D Tiles Next is conditional Cesium branch	W. 3D / Cesium / Deck.gl / Overlay Interoperability	CONFIRMED evidence / NEW SRC-062 pp.20-26	The 3D Tiles brief describes implicit tiling, subtree availability, and 3D Tiles Next patterns for meshes, point clouds, and DEMs.	Keep KFM 2D-first while allowing a conditional Cesium/3D scene branch when vertical context is evidence-bearing.	3D scene manifest, tile availability, and evidence fallback tests.
ML-062-011	3D tiles require digest-immutable addressing	W. 3D / Cesium / Deck.gl / Overlay Interoperability	CONFIRMED evidence / NEW SRC-062 pp.23-25	The source recommends SHA-256 digests across 3D tile trees and STAC links to immutable tilesets.	MapReleaseManifest should refer to immutable 3D tileset digests, not mutable folders.	Digest-tree verification; no mutable tileset URL in public release.
ML-062-012	3D feature metadata schema links to STAC	W. 3D / Cesium / Deck.gl / Overlay Interoperability	CONFIRMED evidence / NEW SRC-062 pp.22-24	3D feature semantics are carried in EXT_mesh_features/property schemas and referenced from STAC.	Evidence Drawer or Focus Mode should resolve feature properties through schema-backed metadata.	Picked-feature property schema test.
ML-062-013	3D/Cesium QA includes picked-feature property checks	W. 3D / Cesium / Deck.gl / Overlay Interoperability	CONFIRMED evidence / NEW SRC-062 pp.25-26	The source suggests selecting a feature and checking properties against schema.	3D overlays need the same click-to-evidence discipline as MapLibre 2D layers.	Screenshot plus picked-feature property validation.

idea_id	idea title	category	status / source	normalized statement / evidence	implementation consequence	validation / risk
ML-062-014	Web-to-CloudEvents turns external changes into typed deltas	Y. Implementation Backlog and PR Plan	CONFIRMED evidence / NEW SRC-062 pp.27-31	External web/API changes can become typed CloudEvents with source-family schema and license metadata.	Pre-layer watchers can emit deltas before catalog or tile rebuild decisions.	CloudEvent schema, license, source_id, and digest validation.
ML-062-015	Conditional GET heartbeat prevents fake map changes	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW SRC-062 pp.28-31	If ETag or Last-Modified reports no change, the source emits a lightweight heartbeat rather than a new contentful event.	Avoid creating new map releases and cache invalidations for unchanged sources.	304/no-change fixture emits heartbeat only.
ML-062-016	License travels with deltas before map ingestion	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW SRC-062 pp.29-31	Contentful delta emission is blocked or constrained when license status is unknown.	Map layer admission should fail closed when license or use terms are absent.	License-deny policy fixture.
ML-062-017	Source configs require canonical source-family schemas	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.30-31	The pattern maps extracted fields to a canonical JSON Schema per source family.	SourceDescriptor should declare source family, schema version, and downstream layer eligibility.	SourceDescriptor schema validation.
ML-062-018	Hydrology OGC/NWPS intake normalizes station, series, observation	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW SRC-062 pp.32-44	Hydrology intake separates monitoring locations, time series definitions, raw pulls, normalized artifacts, and catalog entries.	Map time sliders should consume normalized, versioned time-series artifacts only.	Station/series schema tests and stale/late telemetry cases.
ML-062-019	Hydrology station and series IDs use authoritative provider IDs	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.35-36	The source defines canonical keys using provider and station_id/source series identifiers.	Layer feature IDs should be deterministic and provider-based for click resolution.	promoteId/feature ID uniqueness tests.
ML-062-020	No-change hydrology polls do not emit new catalog entities	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW SRC-062 pp.38-39	Confirmed no-change responses should not create new STAC/DCAT/PROV entities.	Avoid unnecessary tile/cache invalidation and false stale-source churn.	No-change poll regression.
ML-062-021	Hydrology drift checks drive stale badges and warnings	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW SRC-062 pp.39-43	Drift checks and hydrology sanity checks can be warnings or blockers depending on severity.	UI should show stale/degraded state separately from denied release.	Stale badge/abstain tests.
ML-062-022	Hydrology telemetry supports polling and latency budgets	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW SRC-062 pp.42-43	The source names poll counts, latency, partial payloads, and throttling as operational metrics.	Map runtime should expose source freshness and avoid uncontrolled polling.	Resource timing and poll-budget tests.
ML-062-023	Hydrology Definition of Done includes STAC/DCAT/PROV raw and normalized artifacts	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-062 pp.44	The intake is done only when raw and normalized artifacts have catalog/provenance records and validation.	No layer release until raw/normalized lineage is complete.	Catalog closure fixture and missing-PROV deny case.
ML-062-024	Freight/logistics intake splits flows, networks, safety, and derived products	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW SRC-062 pp.45-49	The logistics section separates modeled flows, highway/rail networks, carrier/safety registries, incidents, and KFM-only derived products.	Layer catalog should not collapse observed incidents, modeled flows, and derived accessibility surfaces.	Layer type/source-role validation.
ML-062-025	Deterministic network-edge ID uses source, segment, and geometry fingerprint	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW SRC-062 pp.50	Network edge identity is based on source_id, segment_id, and geometry fingerprint.	Use stable edge IDs in vector tiles and graph projections.	Geometry-fingerprint and promoteId tests.
ML-062-026	STAC Items for logistics layers and incidents	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.51	Spatial logistics layers, incidents as points, and datasets get STAC/DCAT/PROV records.	Transport layers must resolve to catalog and provenance before public use.	STAC Item per layer/incident fixture.
ML-062-027	Routable graph exports are derived products	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW SRC-062 pp.52	Routable graph exports are KFM value-adds after authoritative layers are stable.	Graph overlays and routing aids do not replace canonical source records.	No route claim without evidence bundle.
ML-062-028	Soils/surficial geology deterministic ingestion feeds GeoParquet, COG and map layers	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW SRC-062 pp.64-76	The source normalizes SSURGO, gNATSGO, SoilGrids, and KGS/Kansas Geoportal inputs into standard outputs including GeoParquet and COG.	Soil/geology layers should derive tiles from processed artifacts, not raw source downloads.	COG/GeoParquet schema and geometry validity gates.

idea_id	idea title	category	status / source	normalized statement / evidence	implementation consequence	validation / risk
ML-062-029	Store source geometry as delivered plus normalized outputs	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.71	The source requires storing source geometry as delivered and normalized outputs.	Layer manifests should point to both source lineage and normalized tile-ready artifacts.	As-delivered vs normalized lineage check.
ML-062-030	STAC assets include COG and vector roles for soil/geology	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.72	The STAC asset pattern lists COG and vector artifacts as explicit roles.	TileArtifactManifest should align with STAC asset roles.	Asset role/media type validation.
ML-062-031	COG compliance and metadata validation gate raster release	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW SRC-062 pp.72-74	COG compliance and STAC metadata validation are required checks.	Raster layers should not load if COG layout or metadata fails.	COG validate and raster metadata fixture.
ML-062-032	Knowledge graph nodes for STAC/DCAT/PROV are catalog projections	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW SRC-062 pp.75	The knowledge graph includes StacItem, DcatDataset, and ProvActivity nodes.	Graph views support evidence navigation but do not replace the catalog.	Graph node-to-catalog crosswalk test.
ML-062-033	CARE-bound STAC/DCAT metadata adds consent and authority	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW SRC-062 pp.77-83	CARE fields for consent, authority to control, use conditions, locality restrictions, steward contact, and review expiry are embedded in STAC/DCAT.	Sensitive map layers require policy-readable metadata before publication.	CARE schema and review-expiry denial tests.
ML-062-034	Consent attestation binds to exact metadata digest	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW SRC-062 pp.79-82	in-toto/Cosign attestation references the exact STAC/DCAT digest and steward role.	A public-safe layer cannot rely on generic consent detached from the actual metadata artifact.	Attestation digest-match test.
ML-062-035	Locality restrictions imply spatial masking or restricted tier	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW SRC-062 pp.80-83	OPA soft-deny/remediation says locality restrictions require masking or restricted-tier publication.	MapLibre should receive generalized or denied geometry, not exact restricted locations hidden by style.	Restricted-locality deny/generalization fixture.
ML-062-036	CARE gate outputs human-readable remediation	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW SRC-062 pp.80-83	Gate failures include plain-language remediation such as renewing consent or adding terms.	UI can surface deny reasons without revealing sensitive details.	Deny-reason accessibility and no-sensitive-leak checks.
ML-062-037	Reproducible builds make map artifacts reconstructable	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-062 pp.84-91	The source says unreconstructable artifacts cannot defend truth claims.	Tile, COG, GeoParquet, and style builds need repeatable environment/material/process receipts.	Clean rebuild hash comparison.
ML-062-038	Nix, Python lockfiles, containers, SBOMs, and provenance cover distinct planes	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW SRC-062 pp.85-90	No single reproducibility track is sufficient; environment, materials, process, contents, provenance, and signatures interlock.	Use layered reproducibility for map/tile pipelines.	SBOM, lockfile, container digest, and provenance checks.
ML-062-039	Unsigned artifacts are invalid data	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW SRC-062 pp.88-91	The source recommends treating unsigned artifacts as invalid.	Do not publish unsigned tiles, rasters, style bundles, or catalog outputs.	Unsigned artifact deny fixture.
ML-062-040	Zero-trust ingest uses sidecar fetch and content-addressed staging	Y. Implementation Backlog and PR Plan	CONFIRMED evidence / NEW SRC-062 pp.91-96	External intake uses sidecar isolation, short-lived OIDC, signed logs, digest staging, and license-first checks.	Map source ingestion should go through staging and gates before reaching processed/catalog.	Egress/host allowlist, digest, signed-log tests.
ML-062-041	ArcGIS/S3/OCI intake records URL, version, license, checksum	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-062 pp.95-96	The source lists metadata to record for ArcGIS, S3, and OCI sources.	SourceDescriptor and RunReceipt should preserve endpoint, item ID, modified timestamp, ETag, license, digest, and version.	Source-specific intake fixtures.
ML-062-042	PR-first change detectors update artifacts only after gates	Y. Implementation Backlog and PR Plan	CONFIRMED evidence / NEW SRC-062 pp.97-101	Each detected change becomes a branch/PR with machine-readable manifests and required checks.	Tile/style/catalog updates should be PR-first unless an approved ADR says otherwise.	CI required-checks gate and draft PR artifact review.
ML-062-043	Review jobs render Story Nodes and Focus Mode diffs	N. Evidence Drawer Payloads and Click Resolution	CONFIRMED evidence / NEW SRC-062 pp.98-101	Required checks include review render and post-merge Story Node/Focus Mode diffs.	Evidence Drawer and Focus surfaces need review artifacts before publication.	Visual/regression test for drawer/story/focus diffs.

idea_id	idea title	category	status / source	normalized statement / evidence	implementation consequence	validation / risk
ML-062-044	Post-merge promotion updates catalogs, graph, APIs, and UI diffs	Y. Implementation Backlog and PR Plan	CONFIRMED evidence / NEW SRC-062 pp.101	Promotion jobs update STAC/DCAT/PROV, graph deltas, API indices, Story Nodes, and Focus diffs after checks.	Keep API/UI updates downstream of the release manifest.	Release manifest update and rollback replay test.
ML-063-001	GitHub App orchestration for cross-repo map and tile changes	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.1-5	A GitHub App issues short-lived installation tokens, triggers repository_dispatch, opens PRs, and writes check runs.	Use app-based orchestration for governed tile/source PRs instead of PAT shortcuts.	Auth scope, repository_dispatch, and Checks API tests.
ML-063-002	OPA/Rego fail-closed gates on run manifests	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.2-5	Conftest/OPA evaluates run_manifest.json and configs; violations deny promotion.	Run manifests must gate layer/tile artifacts before release.	Negative policy fixtures for missing fields, rights, sensitivity, and release state.
ML-063-003	Cosign/Rekor attestation UUID recorded in run manifest	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.3-5	The source attests run manifests and records Rekor UUIDs back into manifests.	MapReleaseManifest should link to attested RunReceipt/AIRceipt/TileArtifactManifest records.	Attestation lookup and UUID presence checks.
ML-063-004	Checks API rollup blocks merges until receipts verify	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.3-5	A Check Run summarizes OPA, Cosign/Rekor, materiality, and annotations; branch protection blocks merge.	Map/tile PRs should expose receipt rollup checks before publication.	Branch protection/check-name contract test.
ML-063-005	Kill-switch blocks map/data promotion	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW SRC-063 pp.4-5	A kill-switch file causes the aggregator to fail checks and block merge.	Layer release and cache invalidation paths should honor a fail-closed kill switch.	Kill-switch fixture makes publish fail.
ML-063-006	Rollback action emits rollback attestation	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.4-5	Rollback reverts the last publish and emits a rollback attestation.	Every map release needs a rollback target and receipt.	Rollback manifest and restored-style/layer test.
ML-063-007	Acceptance harness validates STAC/DCAT/PROV before release	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.6-9	The acceptance harness validates metadata, policy, signatures, and spec_hash before promotion.	Map release CI should include catalog and provenance validation before tile serving.	STAC/DCAT/PROV and signature validation jobs.
ML-063-008	spec_hash reproducibility gate blocks drift	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.7-8	Computed spec_hash must match the run receipt.	Layer and tile builds should pin build specs and compare hashes.	Spec hash mismatch denial test.
ML-063-009	Earth-observation STAC harvest supports layer discovery	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.10-11	Standardized STAC/collection endpoints make cross-platform EO discovery and harvesting possible.	Layer catalog can discover EO assets via governed STAC harvesting.	STAC endpoint smoke and auth handling tests.
ML-063-010	NASA/CMR auth tokens must not be public map dependencies	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW SRC-063 pp.10-11	Auth tokens are required for some Earthdata metadata and quota tracking.	Clients should not fetch protected catalogs directly; governed API resolves public-safe assets.	No browser token exposure test.
ML-063-011	NAIP/3DEP/3DNTM baselines strengthen terrain and raster layers	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW SRC-063 pp.12	Terrain/elevation baselines are high-value public-domain sources for map products.	Add source descriptors and terrain artifacts only after rights and vintage verification.	3DEP/NAIP source descriptor and COG/DEM fixture tests.
ML-063-012	Soil watcher emits canonical GeoParquet and COG outputs	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW SRC-063 pp.13-18	The soil watcher normalizes SSURGO/gNATSGO into GeoParquet vectors and tiled COG rasters.	Map layers should derive from processed GeoParquet/COG, not raw tables/FGDB.	GeoParquet/COG schema, CRS, and geometry validity tests.
ML-063-013	GeoParquet is vector truth for derived tiles	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW SRC-063 pp.16	The source says store GeoParquet as the single source of truth and derive tippecanoe/PMTiles from it.	PMTiles should be downstream from canonical processed GeoParquet.	Tile artifact digest linked to GeoParquet digest.
ML-063-014	Soil materiality gates centroid, area, and median changes	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.15-18	Material release is triggered by source version change, centroid shift, polygon area delta, or numeric median change.	Cache invalidation and release kind should be data-driven.	Materiality threshold fixtures.
ML-063-015	Soil COGs require overviews/internal tiling and range tuning	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW SRC-063 pp.14-18	gNATSGO COGs require internal tiling, overviews, and block size tuned for HTTP range reads.	Raster layer hosting should be Range-capable and manifest-checked.	COG layout and Range/CORS probe.

idea_id	idea title	category	status / source	normalized statement / evidence	implementation consequence	validation / risk
ML-063-016	CI job chain fails closed before processed/catalog publication	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.17-18	Acquire, QA, change detection, packaging, signing, policy, and publish jobs run in sequence and fail closed.	Tile/style releases need ordered gate stages.	Workflow fixture with intentional failure at each stage.
ML-063-017	Realtime feeds require materiality, SLO, license, and schema gates	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW SRC-063 pp.20-25	Realtime publication checks schema, license, SLOs, material change, and provenance before publishing.	Live-ish map layers must display freshness and materiality, not flood UI with noise.	SLO stale badge and material-change tests.
ML-063-018	COG vs Zarr canonical choices separate raster and datacube needs	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW SRC-063 pp.26-29	COG is recommended for single-file rasters; Zarr v2 consolidated for N-D datacubes.	Use COG for map rasters and Zarr for analytic datacubes, with separate manifests.	COG/Zarr packaging validation.
ML-063-019	Avoid in-place COG updates	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW SRC-063 pp.27	Updating a COG in place can break COG-safe layout.	Treat raster updates as new immutable artifacts, not mutable overwrites.	COG layout regression after update attempts.
ML-063-020	STAC Raster/Projection/Datacube extensions lock raster metadata	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW SRC-063 pp.27-29	STAC should include Raster, Projection, and Datacube where applicable.	Raster/terrain layers need explicit CRS, nodata, bands, stats, and datetime metadata.	STAC extension schema validation.
ML-063-021	Deterministic packaging pins geospatial toolchain	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW SRC-063 pp.27-29	Pin GDAL/PROJ, xarray, zarr, numcodecs, PySTAC; avoid auto chunks and implicit dtype.	Tile and raster build tools need version pins and SBOMs.	Tool version pin and SBOM check.
ML-063-022	Repro gate reruns clean container and compares hashes	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.28-29	Clean rebuild hash mismatch fails promotion.	Map artifacts should be reproducible before release.	Clean-room rebuild test.
ML-063-023	Quick geospatial QC checks CRS, bounds, size, and sample render	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.30-33	Ten-minute QC checks file size, geotransform, bounds, STAC/DCAT, and sample tiles.	Every map artifact gets a lightweight preflight before deeper review.	CRS/bounds/sample-render fixtures.
ML-063-024	XYZ/WMTS planned outputs need sample tile proof	L. WMS / WMTS / External Map Services	CONFIRMED evidence / NEW SRC-063 pp.32	If XYZ/WMTS is planned, render a sample z/x/y tile and save it as proof.	External/derived tile services need a proof sample.	Tile sample render artifact and visual regression test.
ML-063-025	Invalid STAC/DCAT/PROV/checksum/bounds deny publish	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.33	CI denies merge on missing CRS, failed checksum, invalid catalog metadata, insane bounds, or failed sample tile.	MapReleaseManifest requires validation reports.	Negative fixture for each deny condition.
ML-063-026	LLM/chat activity can emit STAC/DCAT/PROV but remains interpretive	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW SRC-063 pp.34-37	Conversation and backfill flows can emit provenance/catalog records but remain process evidence, not truth.	Focus Mode outputs need AIReceipt and citation validation.	AIReceipt and citation pass/fail tests.
ML-063-027	Pipeline orchestrators sign artifacts before merge	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.37-43	Dagster/Prefect/GitHub Actions patterns produce run receipts, sign artifacts, and gate merge.	Map/tile jobs should emit receipts before release.	Signed receipt fixture and merge block test.
ML-063-028	OCI registry can act as provenance hub for geospatial artifacts	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.44-50	OCI/ORAS stores datasets, PROV, SBOMs, and attestations as digest-addressed artifacts.	TileArtifactManifest may include OCI digest and referrer links.	OCI digest/referrer support test.
ML-063-029	Catalogs include contentDigest and distribution.integrity	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.44-50	STAC/DCAT should include contentDigest, distribution integrity, and links to receipt/provenance.	Catalog records must bind to exact PMTiles/COG/GeoParquet bytes.	Digest match and broken referrer checks.
ML-063-030	OCI tags are pointers; manifests reference immutable digests	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW SRC-063 pp.127	Symbolic tags can move; release metadata must reference immutable digests.	Do not put mutable registry tags into MapReleaseManifest as sole identity.	Mutable-tag deny fixture.
ML-063-031	Catalog checklist enforces license, contacts, extents, and links	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.129-130	Dataset PR template/checklist requires STAC/DCAT/PROV, distribution integrity, license, owners, and contact fields.	Layer catalog entries should fail if dataset metadata is incomplete.	Catalog completeness test.

idea_id	idea title	category	status / source	normalized statement / evidence	implementation consequence	validation / risk
ML-063-032	kfm_prov_id and kfm_run_id link logs, catalogs, and Focus grounding	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW SRC-063 pp.131-132	The source recommends log keys tying run ID, provenance bundle, catalog, and gate state.	Focus Mode and Evidence Drawer can show run/provenance context without raw logs.	Drawer payload includes prov/run IDs and hides secrets.
ML-063-033	OpenLineage facets carry KFM receipt, SLO, job, and dataset identity	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.137-140	Custom OpenLineage facets can include KFM receipts and dataset context.	Tile build observability can be standardized across pipelines.	Facet schema validation.
ML-063-034	PM2.5/AQI materiality gates prevent noisy layer churn	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW SRC-063 pp.99-102, pp.108-111	High-frequency air feeds use thresholds such as AQI category change and PM2.5 deltas.	Air-quality MapLibre layers should update on material changes and show stale/provisional state.	Materiality threshold and stale source tests.
ML-063-035	NPMRDS/freight ingest carries rights text in STAC/DCAT	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW SRC-063 pp.112-116	Transportation data ingest must carry provider, portal account, rights text, and PROV.	Restricted freight layers require rights-aware publication.	Rights text missing deny test.
ML-063-036	GTFS diffs produce detour GeoJSON and STAC/DCAT/PROV artifacts	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW SRC-063 pp.117-120	Transit changes produce canonical files, detour GeoJSON, catalog/provenance, and policy gates.	Small GeoJSON overlays are acceptable after normalization and catalog closure.	GeoJSON schema, sort/order, and catalog closure tests.
ML-063-037	Datasets-as-OCI mental model includes GeoParquet, COG, and PMTiles subjects	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.120-125	GeoParquet, COG, and PMTiles can be pushed as OCI artifacts with metadata and attestations.	Artifact manifests should identify media type, digest, and OCI reference.	Artifact type and digest tests.
ML-063-038	ORAS referrers attach provenance, SBOM, and attestations	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.121-128	ORAS/cosign attach PROV, SBOM, and in-toto/SLSA predicates to immutable dataset digests.	Map release proof packs can use OCI referrers where registry support is verified.	Referrer discovery and verification test.
ML-063-039	Immutable release policy blocks tag mutation	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW SRC-063 pp.127-128	Release tags are pointers but immutable release references must point to digests; tag mutation is blocked.	No mutable release labels as authoritative map artifacts.	Tag mutation deny test.
ML-063-040	pgSTAC to GeoParquet to PMTiles to OCI pipeline supports tile publication	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW SRC-063 pp.150-153	The source gives a version-pinned pipeline from STAC items to pgSTAC, GeoParquet, PMTiles, OCI, and cosign.	Create a governed tile publishing pipeline as a future PR, not a direct browser path.	End-to-end fixture with digest, PMTiles inspection, and attestation.
ML-063-041	pgSTAC ingest is transactional and role-scoped	J. Martin and Server-Mediated Tile Serving	CONFIRMED evidence / NEW SRC-063 pp.151	pyPgSTAC migrations and item loads should be transactional with pgstac_admin/ingest/read roles.	Dynamic/catalog-backed tile services need least-privilege roles.	Role and transaction rollback tests.
ML-063-042	GeoParquet v1.1 export requires geometry, CRS, and bbox metadata	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW SRC-063 pp.151-153	GeoParquet export should preserve geometry encoding, bbox, CRS, and version metadata.	GeoParquet can be canonical processed vector artifact feeding tiles.	GeoParquet metadata validation.
ML-063-043	Tippecanoe direct PMTiles or MBTiles-convert path	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW SRC-063 pp.152-153	Tippecanoe can emit PMTiles directly or MBTiles can be converted to PMTiles.	Support both conversion paths but record tool versions and digests.	PMTiles inspect and source-layer tests.
ML-063-044	ORAS push stores GeoParquet and PMTiles as portable blobs	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW SRC-063 pp.152-153	ORAS stores each file as a blob under an OCI manifest.	Published PMTiles can be packaged with metadata and linked to release manifests.	OCI pull/verify fixture.
ML-063-045	Consumers verify cosign attestation on immutable OCI digest	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.152-153	cosign attest and verify-attestation operate on the immutable OCI digest.	Tile consumers or deployment scripts should verify before serving.	Attestation failure denies deployment.
ML-063-046	Ollama context-window changes can destabilize local Focus Mode	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW SRC-063 pp.153-154	The source warns that default context window behavior can change memory use and performance.	Focus Mode runtime configuration must be explicit and logged.	num_ctx/context-length fixture and runtime budget test.
ML-063-047	Focus Mode local runtime needs explicit context-length control	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW SRC-063 pp.154	KFM should verify num_ctx or environment settings after runtime updates.	AIRceipt should record model runtime settings relevant to answer reproducibility.	Runtime config capture and fail-safe fallback test.
ML-063-048	MLT as successor to MVT is a pilot candidate	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW SRC-063 pp.154	The source describes MLT as a ground-up successor to MVT with smaller tiles and faster decoding.	Treat MLT as PROPOSED/PILOT for KFM until validated.	MVT vs MLT benchmark, schema, and rendering parity tests.

idea_id	idea title	category	status / source	normalized statement / evidence	implementation consequence	validation / risk
ML-063-049	MLT column-oriented layout targets GPU/browser performance	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW SRC-063 pp.154	MLT uses a column-oriented layout and GPU-friendly in-memory formats according to the source.	Performance budgets can include MLT experiments for heavy layers.	Tile load/decoding benchmark and browser compatibility tests.
ML-063-050	MLT support in GL JS/Native and Planetiler is experiment-ready	C. MapLibre Native / Mobile / Platform Parity	CONFIRMED evidence / NEW SRC-063 pp.154	The source reports support in MapLibre GL JS, MapLibre Native, and Planetiler, but this remains version-sensitive.	Do not pin versions from this PDF without verification.	Version check and native/web parity test.
ML-063-051	MLT is not production default before KFM validation	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW SRC-063 pp.154	Despite promising claims, MLT should remain pilot until KFM validates tooling, compatibility, and governance manifests.	Keep MVT/PMTiles default unless release gates prove MLT safe.	Pilot gate and rollback comparison.
ML-063-052	Conditional fetch, canonical hash, receipt, and policy gate pattern	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.155-158	The source gives a pattern of conditional fetch, JCS canonicalization, run receipt, and Contest/OPA.	Map source refreshers should emit receipts before tile/catalog updates.	Fetch/no-change/hash/policy fixtures.
ML-063-053	JCS canonicalization makes stable diffs for map source specs	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.155-156	RFC 8785/JCS canonical form produces stable hashes for semantically identical JSON.	SourceDescriptor, LayerManifest, and style specs can use canonical hashes.	Canonical hash golden tests.
ML-063-054	RunReceipt carries source URL, ETag, spec_hash, artifact digests	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.156-158	RunReceipt fields include URL, fetched time, ETag, last_modified, spec_hash, artifacts, provider, and provenance.	RunReceipt is first-class input to release and cache invalidation decisions.	Receipt schema validation.
ML-063-055	Rego denies unapproved provider or missing spec_hash	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.157-158	Policy denies promotion if provider is absent/unapproved or spec_hash is missing.	Map/tile promotion should fail closed on unknown provider.	Unapproved-provider deny fixture.
ML-063-056	Provenance-policy workflow opens PR with STAC/DCAT/PROV after gates	Y. Implementation Backlog and PR Plan	CONFIRMED evidence / NEW SRC-063 pp.159-163	The workflow runs a watcher, evaluates policy, comments on failure, commits catalog/provenance on success, and opens a draft PR.	Use for source-driven layer updates.	Workflow dry-run fixture and PR body receipt checks.
ML-063-057	Watcher dry-run receipts support no-network CI	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.164-170	A dry-run stack emits receipts and validates structure without live ports or network side effects.	First implementation can be no-network and deterministic.	No-network fixture and dry-run receipt validation.
ML-063-058	Watcher schema types include stac, gtfs, tile, file, and api	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.171-172	Watcher schema includes a type enum for stac, gtfs, tile, file, and api.	SourceDescriptor/WatcherDescriptor should classify tile-specific watchers.	Schema enum validation.
ML-063-059	Signature references point to OCI signatures for watchers	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.173-175	Watcher entries include signature_ref values pointing at OCI signature artifacts.	Watcher registry should require signed descriptors before source activation.	Signature ref resolution test.
ML-063-060	Watcher registry links outputs to STAC catalogs	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW SRC-063 pp.173-175	Watcher output records can point to STAC catalog URLs.	Layer catalog can trace from source watcher to catalog to tile artifact.	Watcher-output link validation.
ML-063-061	Ops acceptance gates for stations and artifacts guard map layers	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW SRC-063 pp.176-178	Station and artifact gates test source approval, material probes, and artifact policy.	High-risk layers need operations acceptance before being visible.	Ops acceptance fixture.
ML-063-062	Stable geo-normalize IDs for road events	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW SRC-063 pp.182-184	Road event IDs combine publisher, source ID, and canonical payload hash.	Road event vector tiles should use stable IDs, not ephemeral feed IDs.	Stable ID golden tests.
ML-063-063	Material change policies measure centroid, status, and area changes	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW SRC-063 pp.183-187	Diff policy evaluates centroid shift, status change, and local patch area.	Map release and cache invalidation should be material-change aware.	Material-diff policy tests.
ML-063-064	Policy gate PR comments surface reasons and changed assets	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW SRC-063 pp.160-163	Failure comments show run ID, status, spec_hash, changed assets, centroid shifts, and reasons.	Review UI and PR comments should show trust state and remediation.	Human-readable failure output and accessibility checks.

6. Category Atlas

The atlas below shows which categories changed in v1.8. Categories with no new records remain **RETAINED** through SRC-MASTER-PREV.

category	category title	v1.8 status	delta idea IDs	summary
A	Renderer Boundary and KFM Trust Law	EXPANDED	ML-062-003, ML-062-027, ML-062-032	Current run reinforces the core rule: graph projections, routable exports, STAC graphs, and rendered layers are downstream carriers. They support evidence navigation but do not replace EvidenceBundle or release authority.
B	MapLibre GL JS Web Shell	RETAINED	No new delta records; see Appendix A.	No new GL JS shell mechanics were verified beyond prior master; current sources strengthen the need for the shell to consume cataloged, signed, released artifacts only.
C	MapLibre Native / Mobile / Platform Parity	EXPANDED	ML-063-050	Current source raises MLT web/native parity as version-sensitive. Treat GL JS/Native MLT statements as NEEDS VERIFICATION until official package targets and KFM parity tests are run.
D	MapLibre RS / Experimental Renderers	RETAINED	No new delta records; see Appendix A.	No new MapLibre RS evidence was found; retain prior master posture that experimental renderers remain pilot-only until contracts, tests, and parity are proven.
E	Style Specification, Sources, Layers, Expressions	RETAINED	No new delta records; see Appendix A.	No direct new style-spec record was extracted; current-run geometry normalization and STAC/asset conventions indirectly reinforce checked source/layer definitions tied to stable IDs and allowed attributes.
F	Sprites, Glyphs, Fonts, Design Tokens	RETAINED	No new delta records; see Appendix A.	No new sprites/glyphs/font content was found; carry forward prior master requirements for hosted glyph/sprite manifests and design token governance.
G	GeoJSON and Runtime Data	EXPANDED	ML-062-005, ML-062-006, ML-062-025, ML-063-036, ML-063-042, ML-063-062	GeoJSON appears as STAC search output, GTFS detour output, road-event payloads, and runtime candidate data. The pattern is to normalize, validate, and bound size before the browser sees it.
H	Vector Tiles: MVT and MLT	EXPANDED	ML-063-013, ML-063-043, ML-063-048, ML-063-049	Current run adds a strong MVT/MLT and PMTiles production update: GeoParquet as source of truth, Tippecanoe/PMTiles as derived, and MLT as a pilot candidate only.
I	PMTiles, MBTiles, Static Tiles, Serverless Delivery	EXPANDED	ML-063-040, ML-063-044	PMTiles and serverless delivery are expanded through pgSTAC -> GeoParquet -> PMTiles -> OCI and ORAS packaging patterns. Static archives remain immutable downstream artifacts.
J	Martin and Server-Mediated Tile Serving	EXPANDED	ML-063-041	pgSTAC and transactional/role-scoped PostGIS-backed catalog serving are strengthened. Martin remains retained from prior master; current sources add database role and transaction gates.
K	Raster, COG, DEM, Terrain, Hillshade	EXPANDED	ML-062-028, ML-062-031, ML-063-011, ML-063-012, ML-063-015, ML-063-018, ML-063-020	COG, Zarr, DEM/terrain, NAIP/3DEP, soils, and rasters receive major validation upgrades: COG-safe layout, overviews, range tuning, Raster/Projection/Datacube STAC, and reproducible packaging.
L	WMS / WMTS / External Map Services	EXPANDED	ML-063-024	WMTS/XYZ sample tile rendering appears in the quick QC list; external services remain visualization or source interfaces unless governed analytics records exist.
M	LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	EXPANDED	ML-062-004, ML-062-008, ML-062-017, ML-062-019, ML-062-026, ML-062-029, ML-062-030, ML-062-041, ML-063-009, ML-063-028, ML-063-029, ML-063-031, ML-063-037, ML-063-054, ML-063-058, ML-063-060	Current run strongly expands manifest thinking: STAC/DCAT/PROV RunReceipt, OCI digests, contentDigest, watcher descriptors, and TileArtifactManifest fields must bind exact bytes to provenance.
N	Evidence Drawer Payloads and Click Resolution	EXPANDED	ML-062-043	Evidence Drawer implications are mostly indirect: Story Node and Focus diffs, picked 3D features, and drawer payloads must be reviewable and bound to released evidence.
O	Focus Mode and Governed AI Map Context	EXPANDED	ML-063-026, ML-063-032, ML-063-046, ML-063-047	Focus Mode gains runtime-governance pressure: local model settings such as context length must be explicit, and AI/chat provenance can be captured without becoming truth.
P	Time-Aware Map Interaction and Timeline	EXPANDED	ML-062-018, ML-062-024, ML-063-017, ML-063-034, ML-063-063	Time-aware interaction is expanded by station series, realtime SLO gates, materiality thresholds, and source change windows that determine stale badges and cache invalidation.
Q	Sensitive Geometry, Geoprivacy, Rights, and Policy	EXPANDED	ML-062-016, ML-062-033, ML-062-034, ML-062-035, ML-063-010, ML-063-035	CARE, rights, locality restrictions, NASA tokens, freight rights text, and license-first deltas expand geoprivacy and access control rules.
R	Plugin, Wrapper, and Dependency Governance	EXPANDED	ML-062-038, ML-063-021	Dependency governance gains stronger reproducible-build coverage across Nix, lockfiles, containers, SBOMs, Cosign, pinned geospatial toolchains, and binary/cache trust.

category	category title	v1.8 status	delta idea IDs	summary
S	Accessibility, UX, and Trust-Visible States	EXPANDED	ML-062-021, ML-062-036, ML-063-064	Trust-visible states expand through human-readable gate remediation, stale/degraded badges, policy failure comments, and accessibility expectations for PR and UI review surfaces.
T	Performance, Caching, CDN, Range Requests, and Resource Timing	EXPANDED	ML-062-007, ML-062-015, ML-062-020, ML-062-022	Performance and caching are expanded by conditional GET, heartbeat events, field-minimized STAC payloads, polling budgets, COG range reads, and no-change cache preservation.
U	Testing, CI, Validation, Rollback, and Proof Objects	EXPANDED	ML-062-001, ML-062-002, ML-062-023, ML-062-037, ML-063-001, ML-063-002, ML-063-003, ML-063-004, ML-063-006, ML-063-007, ML-063-008, ML-063-014, ML-063-016, ML-063-022, ML-063-023, ML-063-025, ML-063-027, ML-063-033, ML-063-038, ML-063-045, ML-063-052, ML-063-053, ML-063-055, ML-063-057, ML-063-059, ML-063-061	The largest current-run expansion: CI, OPA/Conftest, Sigstore/Rekor, acceptance harnesses, spec_hash, clean rebuilds, OpenLineage facets, rollback attestations, and no-network fixtures
V	Exports, Screenshots, Reports, and Citation Preservation	RETAINED	No new delta records; see Appendix A.	Exports/screenshots are retained from prior master; current sources strengthen report/PR output through human-readable proof summaries and sample tile proof artifacts.
W	3D / Cesium / Deck.gl / Overlay Interoperability	EXPANDED	ML-062-010, ML-062-011, ML-062-012, ML-062-013	3D/Cesium receives an explicit 3D Tiles Next implicit-tiling pattern with digest-immutable tilesets, feature metadata schemas, and picked-feature QA.
X	Anti-Patterns and Failure Modes	EXPANDED	ML-062-009, ML-062-039, ML-063-005, ML-063-019, ML-063-030, ML-063-039, ML-063-051	New anti-patterns include unsigned artifacts, mutable OCI tags, in-place COG updates, invalid STAC-ish JSON, MLT as default too early, and PR bypasses.
Y	Implementation Backlog and PR Plan	EXPANDED	ML-062-014, ML-062-040, ML-062-042, ML-062-044, ML-063-056	Implementation backlog expands toward PR-first change detectors, source watchers, pgSTAC-to-PMTiles pipelines, zero-trust ingest, and no-network dry-run receipts.
Z	Open Questions and Verification Backlog	RETAINED	No new delta records; see Appendix A.	Open questions now include MLT official support/version target, registry referrer support, COG/Zarr/tool pins, source rights, context length limits, and repository homes.

7. Detailed Category Chapters

Each chapter distinguishes CONFIRMED source evidence from PROPOSED implementation consequences. Objects, file paths, routes, and tests remain PROPOSED unless repo evidence later verifies them.

A. Renderer Boundary and KFM Trust Law

Current run reinforces the core rule: graph projections, routable exports, STAC graphs, and rendered layers are downstream carriers. They support evidence navigation but do not replace EvidenceBundle or release authority.

Relevant v1.8 ideas: ML-062-003, ML-062-027, ML-062-032

idea	source pages	what changed	policy / validation implication
ML-062-003	SRC-062 pp.7-9	Graph invariant snapshots and PROV records are emitted artifacts; they support governance but do not replace EvidenceBundle or release state.	No direct graph-to-popup truth test; EvidenceBundle resolution required.
ML-062-027	SRC-062 pp.52	Routable graph exports are KFM value-adds after authoritative layers are stable.	No route claim without evidence bundle.
ML-062-032	SRC-062 pp.75	The knowledge graph includes StacItem, DcatDataset, and ProvActivity nodes.	Graph node-to-catalog crosswalk test.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

B. MapLibre GL JS Web Shell

No new GL JS shell mechanics were verified beyond prior master; current sources strengthen the need for the shell to consume cataloged, signed, released artifacts only.

No new v1.8 delta record was extracted for this category. Prior master entries are retained in Appendix A unchanged.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: retained without current-run expansion.

C. MapLibre Native / Mobile / Platform Parity

Current source raises MLT web/native parity as version-sensitive. Treat GL JS/Native MLT statements as NEEDS VERIFICATION until official package targets and KFM parity tests are run.

Relevant v1.8 ideas: ML-063-050

idea	source pages	what changed	policy / validation implication
ML-063-050	SRC-063 pp.154	The source reports support in MapLibre GL JS, MapLibre Native, and Planetiler, but this remains version-sensitive.	Version check and native/web parity test.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

D. MapLibre RS / Experimental Renderers

No new MapLibre RS evidence was found; retain prior master posture that experimental renderers remain pilot-only until contracts, tests, and parity are proven.

No new v1.8 delta record was extracted for this category. Prior master entries are retained in Appendix A unchanged.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: retained without current-run expansion.

E. Style Specification, Sources, Layers, Expressions

No direct new style-spec record was extracted; current-run geometry normalization and STAC/asset conventions indirectly reinforce checked source/layer definitions tied to stable IDs and allowed attributes.

No new v1.8 delta record was extracted for this category. Prior master entries are retained in Appendix A unchanged.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: retained without current-run expansion.

F. Sprites, Glyphs, Fonts, Design Tokens

No new sprites/glyphs/font content was found; carry forward prior master requirements for hosted glyph/sprite manifests and design token governance.

No new v1.8 delta record was extracted for this category. Prior master entries are retained in Appendix A unchanged.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: retained without current-run expansion.

G. GeoJSON and Runtime Data

GeoJSON appears as STAC search output, GTFS detour output, road-event payloads, and runtime candidate data. The pattern is to normalize, validate, and bound size before the browser sees it.

Relevant v1.8 ideas: ML-062-005, ML-062-006, ML-062-025, ML-063-036, ML-063-042, ML-063-062

idea	source pages	what changed	policy / validation implication
ML-062-005	SRC-062 pp.11-17	The source highlights filter, query, fields, sort, and free-text search as practical STAC API capabilities.	Search contract tests for bbox, datetime, field selection, sort, and limit behavior.
ML-062-006	SRC-062 pp.16-17	STAC API search returns GeoJSON FeatureCollections of Items that can feed candidate lists and map extents.	GeoJSON FeatureCollection schema and item asset checks.
ML-062-025	SRC-062 pp.50	Network edge identity is based on source_id, segment_id, and geometry fingerprint.	Geometry-fingerprint and promoteId tests.
ML-063-036	SRC-063 pp.117-120	Transit changes produce canonical files, detour GeoJSON, catalog/provenance, and policy gates.	GeoJSON schema, sort/order, and catalog closure tests.
ML-063-042	SRC-063 pp.151-153	GeoParquet export should preserve geometry encoding, bbox, CRS, and version metadata.	GeoParquet metadata validation.
ML-063-062	SRC-063 pp.182-184	Road event IDs combine publisher, source ID, and canonical payload hash.	Stable ID golden tests.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

H. Vector Tiles: MVT and MLT

Current run adds a strong MVT/MLT and PMTiles production update: GeoParquet as source of truth, Tippecanoe/PMTiles as derived, and MLT as a pilot candidate only.

Relevant v1.8 ideas: ML-063-013, ML-063-043, ML-063-048, ML-063-049

idea	source pages	what changed	policy / validation implication
ML-063-013	SRC-063 pp.16	The source says store GeoParquet as the single source of truth and derive tippecanoe/PMTiles from it.	Tile artifact digest linked to GeoParquet digest.
ML-063-043	SRC-063 pp.152-153	Tippecanoe can emit PMTiles directly or MBTiles can be converted to PMTiles.	PMTiles inspect and source-layer tests.
ML-063-048	SRC-063 pp.154	The source describes MLT as a ground-up successor to MVT with smaller tiles and faster decoding.	MVT vs MLT benchmark, schema, and rendering parity tests.
ML-063-049	SRC-063 pp.154	MLT uses a column-oriented layout and GPU-friendly in-memory formats according to the source.	Tile load/decoding benchmark and browser compatibility tests.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

I. PMTiles, MBTiles, Static Tiles, Serverless Delivery

PMTiles and serverless delivery are expanded through pgSTAC -> GeoParquet -> PMTiles -> OCI and ORAS packaging patterns. Static archives remain immutable downstream artifacts.

Relevant v1.8 ideas: ML-063-040, ML-063-044

idea	source pages	what changed	policy / validation implication
ML-063-040	SRC-063 pp.150-153	The source gives a version-pinned pipeline from STAC items to pgSTAC, GeoParquet, PMTiles, OCI, and cosign.	End-to-end fixture with digest, PMTiles inspection, and attestation.
ML-063-044	SRC-063 pp.152-153	ORAS stores each file as a blob under an OCI manifest.	OCI pull/verify fixture.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

J. Martin and Server-Mediated Tile Serving

pgSTAC and transactional/role-scoped PostGIS-backed catalog serving are strengthened. Martin remains retained from prior master; current sources add database role and transaction gates.

Relevant v1.8 ideas: ML-063-041

idea	source pages	what changed	policy / validation implication
ML-063-041	SRC-063 pp.151	pyPgSTAC migrations and item loads should be transactional with pgstac_admin/ingest/read roles.	Role and transaction rollback tests.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

K. Raster, COG, DEM, Terrain, Hillshade

COG, Zarr, DEM/terrain, NAIP/3DEP, soils, and rasters receive major validation upgrades: COG-safe layout, overviews, range tuning, Raster/Projection/Datacube STAC, and reproducible packaging.

Relevant v1.8 ideas: ML-062-028, ML-062-031, ML-063-011, ML-063-012, ML-063-015, ML-063-018, ML-063-020

idea	source pages	what changed	policy / validation implication
ML-062-028	SRC-062 pp.64-76	The source normalizes SSURGO, gNATSGO, SoilGrids, and KGS/Kansas Geoportal inputs into standard outputs including GeoParquet and COG.	COG/GeoParquet schema and geometry validity gates.
ML-062-031	SRC-062 pp.72-74	COG compliance and STAC metadata validation are required checks.	COG validate and raster metadata fixture.
ML-063-011	SRC-063 pp.12	Terrain/elevation baselines are high-value public-domain sources for map products.	3DEP/NAIP source descriptor and COG/DEM fixture tests.
ML-063-012	SRC-063 pp.13-18	The soil watcher normalizes SSURGO/gNATSGO into GeoParquet vectors and tiled COG rasters.	GeoParquet/COG schema, CRS, and geometry validity tests.
ML-063-015	SRC-063 pp.14-18	gNATSGO COGs require internal tiling, overviews, and block size tuned for HTTP range reads.	COG layout and Range/CORS probe.
ML-063-018	SRC-063 pp.26-29	COG is recommended for single-file rasters; Zarr v2 consolidated for N-D datacubes.	COG/Zarr packaging validation.
ML-063-020	SRC-063 pp.27-29	STAC should include Raster, Projection, and Datacube where applicable.	STAC extension schema validation.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

L. WMS / WMTS / External Map Services

WMTS/XYZ sample tile rendering appears in the quick QC list; external services remain visualization or source interfaces unless governed analytics records exist.

Relevant v1.8 ideas: ML-063-024

idea	source pages	what changed	policy / validation implication
ML-063-024	SRC-063 pp.32	If XYZ/WMTS is planned, render a sample z/x/y tile and save it as proof.	Tile sample render artifact and visual regression test.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest

Current run strongly expands manifest thinking: STAC/DCAT/PROV, RunReceipt, OCI digests, contentDigest, watcher descriptors, and TileArtifactManifest fields must bind exact bytes to provenance.

Relevant v1.8 ideas: ML-062-004, ML-062-008, ML-062-017, ML-062-019, ML-062-026, ML-062-029, ML-062-030, ML-062-041, ML-063-009, ML-063-028, ML-063-029, ML-063-031, ML-063-037, ML-063-054, ML-063-058, ML-063-060

idea	source pages	what changed	policy / validation implication
ML-062-004	SRC-062 pp.10-14	STAC Collections should use stable IDs, license, provider, and spatial/temporal extent; they are good dataset contracts for layer catalog entries.	STAC schema validation and stable-id non-reuse test.
ML-062-008	SRC-062 pp.14-18	The source emphasizes stable asset roles, media types, and links to avoid client confusion.	Asset role/media-type checks; broken-link tests.

idea	source pages	what changed	policy / validation implication
ML-062-017	SRC-062 pp.30-31	The pattern maps extracted fields to a canonical JSON Schema per source family.	SourceDescriptor schema validation.
ML-062-019	SRC-062 pp.35-36	The source defines canonical keys using provider and station_id/source series identifiers.	promoteId/feature ID uniqueness tests.
ML-062-026	SRC-062 pp.51	Spatial logistics layers, incidents as points, and datasets get STAC/DCAT/PROV records.	STAC Item per layer/incident fixture.
ML-062-029	SRC-062 pp.71	The source requires storing source geometry as delivered and normalized outputs.	As-delivered vs normalized lineage check.
ML-062-030	SRC-062 pp.72	The STAC asset pattern lists COG and vector artifacts as explicit roles.	Asset role/media type validation.
ML-062-041	SRC-062 pp.95-96	The source lists metadata to record for ArcGIS, S3, and OCI sources.	Source-specific intake fixtures.
additional ideas		ML-063-009, ML-063-028, ML-063-029, ML-063-031, ML-063-037, ML-063-054, ML-063-058, ML-063-060	See Master Idea Index for full rows.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

N. Evidence Drawer Payloads and Click Resolution

Evidence Drawer implications are mostly indirect: Story Node and Focus diffs, picked 3D features, and drawer payloads must be reviewable and bound to released evidence.

Relevant v1.8 ideas: ML-062-043

idea	source pages	what changed	policy / validation implication
ML-062-043	SRC-062 pp.98-101	Required checks include review render and post-merge Story Node/Focus Mode diffs.	Visual/regression test for drawer/story/focus diffs.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

O. Focus Mode and Governed AI Map Context

Focus Mode gains runtime-governance pressure: local model settings such as context length must be explicit, and AI/chat provenance can be captured without becoming truth.

Relevant v1.8 ideas: ML-063-026, ML-063-032, ML-063-046, ML-063-047

idea	source pages	what changed	policy / validation implication
ML-063-026	SRC-063 pp.34-37	Conversation and backfill flows can emit provenance/catalog records but remain process evidence, not truth.	AIReceipt and citation pass/fail tests.
ML-063-032	SRC-063 pp.131-132	The source recommends log keys tying run ID, provenance bundle, catalog, and gate state.	Drawer payload includes prov/run IDs and hides secrets.
ML-063-046	SRC-063 pp.153-154	The source warns that default context window behavior can change memory use and performance.	num_ctx/context-length fixture and runtime budget test.
ML-063-047	SRC-063 pp.154	KFM should verify num_ctx or environment settings after runtime updates.	Runtime config capture and fail-safe fallback test.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

P. Time-Aware Map Interaction and Timeline

Time-aware interaction is expanded by station series, realtime SLO gates, materiality thresholds, and source change windows that determine stale badges and cache invalidation.

Relevant v1.8 ideas: ML-062-018, ML-062-024, ML-063-017, ML-063-034, ML-063-063

idea	source pages	what changed	policy / validation implication
ML-062-018	SRC-062 pp.32-44	Hydrology intake separates monitoring locations, time series definitions, raw pulls, normalized artifacts, and catalog entries.	Station/series schema tests and stale/late telemetry cases.
ML-062-024	SRC-062 pp.45-49	The logistics section separates modeled flows, highway/rail networks, carrier/safety registries, incidents, and KFM-only derived products.	Layer type/source-role validation.
ML-063-017	SRC-063 pp.20-25	Realtime publication checks schema, license, SLOs, material change, and provenance before publishing.	SLO stale badge and material-change tests.
ML-063-034	SRC-063 pp.99-102, pp.108-111	High-frequency air feeds use thresholds such as AQI category change and PM2.5 deltas.	Materiality threshold and stale source tests.
ML-063-063	SRC-063 pp.183-187	Diff policy evaluates centroid shift, status change, and local patch area.	Material-diff policy tests.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

Q. Sensitive Geometry, Geoprivacy, Rights, and Policy

CARE, rights, locality restrictions, NASA tokens, freight rights text, and license-first deltas expand geoprivacy and access control rules.

Relevant v1.8 ideas: ML-062-016, ML-062-033, ML-062-034, ML-062-035, ML-063-010, ML-063-035

idea	source pages	what changed	policy / validation implication
ML-062-016	SRC-062 pp.29-31	Contentful delta emission is blocked or constrained when license status is unknown.	License-deny policy fixture.
ML-062-033	SRC-062 pp.77-83	CARE fields for consent, authority to control, use conditions, locality restrictions, steward contact, and review expiry are embedded in STAC/DCAT.	CARE schema and review-expiry denial tests.
ML-062-034	SRC-062 pp.79-82	in-toto/Cosign attestation references the exact STAC/DCAT digest and steward role.	Attestation digest-match test.
ML-062-035	SRC-062 pp.80-83	OPA soft-deny/remediation says locality restrictions require masking or restricted-tier publication.	Restricted-locality deny/generalization fixture.
ML-063-010	SRC-063 pp.10-11	Auth tokens are required for some Earthdata metadata and quota tracking.	No browser token exposure test.
ML-063-035	SRC-063 pp.112-116	Transportation data ingest must carry provider, portal account, rights text, and PROV.	Rights text missing deny test.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

R. Plugin, Wrapper, and Dependency Governance

Dependency governance gains stronger reproducible-build coverage across Nix, lockfiles, containers, SBOMs, Cosign, pinned geospatial toolchains, and binary/cache trust.

Relevant v1.8 ideas: ML-062-038, ML-063-021

idea	source pages	what changed	policy / validation implication
ML-062-038	SRC-062 pp.85-90	No single reproducibility track is sufficient; environment, materials, process, contents, provenance, and signatures interlock.	SBOM, lockfile, container digest, and provenance checks.
ML-063-021	SRC-063 pp.27-29	Pin GDAL/PROJ, xarray, zarr, numcodecs, PySTAC; avoid auto chunks and implicit dtype.	Tool version pin and SBOM check.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

S. Accessibility, UX, and Trust-Visible States

Trust-visible states expand through human-readable gate remediation, stale/degraded badges, policy failure comments, and accessibility expectations for PR and UI review surfaces.

Relevant v1.8 ideas: ML-062-021, ML-062-036, ML-063-064

idea	source pages	what changed	policy / validation implication
ML-062-021	SRC-062 pp.39-43	Drift checks and hydrology sanity checks can be warnings or blockers depending on severity.	Stale badge/abstain tests.
ML-062-036	SRC-062 pp.80-83	Gate failures include plain-language remediation such as renewing consent or adding terms.	Deny-reason accessibility and no-sensitive-leak checks.
ML-063-064	SRC-063 pp.160-163	Failure comments show run ID, status, spec_hash, changed assets, centroid shifts, and reasons.	Human-readable failure output and accessibility checks.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

T. Performance, Caching, CDN, Range Requests, and Resource Timing

Performance and caching are expanded by conditional GET, heartbeat events, field-minimized STAC payloads, polling budgets, COG range reads, and no-change cache preservation.

Relevant v1.8 ideas: ML-062-007, ML-062-015, ML-062-020, ML-062-022

idea	source pages	what changed	policy / validation implication
ML-062-007	SRC-062 pp.15-16	The Fields extension can constrain returned STAC fields and reduce payload size.	Payload-size regression test; required fields remain present.
ML-062-015	SRC-062 pp.28-31	If ETag or Last-Modified reports no change, the source emits a lightweight heartbeat rather than a new contentful event.	304/no-change fixture emits heartbeat only.
ML-062-020	SRC-062 pp.38-39	Confirmed no-change responses should not create new STAC/DCAT/PROV entities.	No-change poll regression.
ML-062-022	SRC-062 pp.42-43	The source names poll counts, latency, partial payloads, and throttling as operational metrics.	Resource timing and poll-budget tests.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

U. Testing, CI, Validation, Rollback, and Proof Objects

The largest current-run expansion: CI, OPA/Conftest, Sigstore/Rekor, acceptance harnesses, spec_hash, clean rebuilds, OpenLineage facets, rollback attestations, and no-network fixtures.

Relevant v1.8 ideas: ML-062-001, ML-062-002, ML-062-023, ML-062-037, ML-063-001, ML-063-002, ML-063-003, ML-063-004, ML-063-006, ML-063-007, ML-063-008, ML-063-014, ML-063-016, ML-063-022, ML-063-023, ML-063-025, ML-063-027, ML-063-033, ML-063-038, ML-063-045, ML-063-052, ML-063-053, ML-063-055, ML-063-057, ML-063-059, ML-063-061

idea	source pages	what changed	policy / validation implication
ML-062-001	SRC-062 pp.1-8	Neo4j/GDS migration content treats graph analytics as a coupled, pinned operational component; MapLibre graph overlays should consume released graph products, not floating plugin behavior.	Driver/GDS compatibility matrix; rollback rehearsal; graph-projection fixture diff.
ML-062-002	SRC-062 pp.4-6	The source recommends upgrading drivers before Neo4j server upgrades because application rollback is easier than database rollback.	Compatibility tests; rollback card for graph-backed map layers.
ML-062-023	SRC-062 pp.44	The intake is done only when raw and normalized artifacts have catalog/provenance records and validation.	Catalog closure fixture and missing-PROV deny case.
ML-062-037	SRC-062 pp.84-91	The source says unreconstructable artifacts cannot defend truth claims.	Clean rebuild hash comparison.
ML-063-001	SRC-063 pp.1-5	A GitHub App issues short-lived installation tokens, triggers repository_dispatch, opens PRs, and writes check runs.	Auth scope, repository_dispatch, and Checks API tests.
ML-063-002	SRC-063 pp.2-5	Conftest/OPA evaluates run_manifest.json and configs; violations deny promotion.	Negative policy fixtures for missing fields, rights, sensitivity, and release state.
ML-063-003	SRC-063 pp.3-5	The source attests run manifests and records Rekor UUIDs back into manifests.	Attestation lookup and UUID presence checks.
ML-063-004	SRC-063 pp.3-5	A Check Run summarizes OPA, Cosign/Rekor, materiality, and annotations; branch protection blocks merge.	Branch protection/check-name contract test.
additional ideas		ML-063-006, ML-063-007, ML-063-008, ML-063-014, ML-063-016, ML-063-022, ML-063-023, ML-063-025, ML-063-027, ML-063-033, ML-063-038, ML-063-045, ML-063-052, ML-063-053, ML-063-055, ML-063-057, ML-063-059, ML-063-061	See Master Idea Index for full rows.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

V. Exports, Screenshots, Reports, and Citation Preservation

Exports/screenshots are retained from prior master; current sources strengthen report/PR output through human-readable proof summaries and sample tile proof artifacts.

No new v1.8 delta record was extracted for this category. Prior master entries are retained in Appendix A unchanged.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: retained without current-run expansion.

W. 3D / Cesium / Deck.gl / Overlay Interoperability

3D/Cesium receives an explicit 3D Tiles Next implicit-tiling pattern with digest-immutable tilesets, feature metadata schemas, and picked-feature QA.

Relevant v1.8 ideas: ML-062-010, ML-062-011, ML-062-012, ML-062-013

idea	source pages	what changed	policy / validation implication
ML-062-010	SRC-062 pp.20-26	The 3D Tiles brief describes implicit tiling, subtree availability, and 3D Tiles Next patterns for meshes, point clouds, and DEMs.	3D scene manifest, tile availability, and evidence fallback tests.
ML-062-011	SRC-062 pp.23-25	The source recommends SHA-256 digests across 3D tile trees and STAC links to immutable tilesets.	Digest-tree verification; no mutable tileset URL in public release.

idea	source pages	what changed	policy / validation implication
ML-062-012	SRC-062 pp.22-24	3D feature semantics are carried in EXT_mesh_features/property schemas and referenced from STAC.	Picked-feature property schema test.
ML-062-013	SRC-062 pp.25-26	The source suggests selecting a feature and checking properties against schema.	Screenshot plus picked-feature property validation.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

X. Anti-Patterns and Failure Modes

New anti-patterns include unsigned artifacts, mutable OCI tags, in-place COG updates, invalid STAC-ish JSON, MLT as default too early, and PR bypasses.

Relevant v1.8 ideas: ML-062-009, ML-062-039, ML-063-005, ML-063-019, ML-063-030, ML-063-039, ML-063-051

idea	source pages	what changed	policy / validation implication
ML-062-009	SRC-062 pp.18-19	The source names invalid STAC-ish JSON as a common failure mode.	Fail-closed STAC validation fixture.
ML-062-039	SRC-062 pp.88-91	The source recommends treating unsigned artifacts as invalid.	Unsigned artifact deny fixture.
ML-063-005	SRC-063 pp.4-5	A kill-switch file causes the aggregator to fail checks and block merge.	Kill-switch fixture makes publish fail.
ML-063-019	SRC-063 pp.27	Updating a COG in place can break COG-safe layout.	COG layout regression after update attempts.
ML-063-030	SRC-063 pp.127	Symbolic tags can move; release metadata must reference immutable digests.	Mutable-tag deny fixture.
ML-063-039	SRC-063 pp.127-128	Release tags are pointers but immutable release references must point to digests; tag mutation is blocked.	Tag mutation deny test.
ML-063-051	SRC-063 pp.154	Despite promising claims, MLT should remain pilot until KFM validates tooling, compatibility, and governance manifests.	Pilot gate and rollback comparison.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

Y. Implementation Backlog and PR Plan

Implementation backlog expands toward PR-first change detectors, source watchers, pgSTAC-to-PMTiles pipelines, zero-trust ingest, and no-network dry-run receipts.

Relevant v1.8 ideas: ML-062-014, ML-062-040, ML-062-042, ML-062-044, ML-063-056

idea	source pages	what changed	policy / validation implication
ML-062-014	SRC-062 pp.27-31	External web/API changes can become typed CloudEvents with source-family schema and license metadata.	CloudEvent schema, license, source_id, and digest validation.
ML-062-040	SRC-062 pp.91-96	External intake uses sidecar isolation, short-lived OIDC, signed logs, digest staging, and license-first checks.	Egress/host allowlist, digest, signed-log tests.
ML-062-042	SRC-062 pp.97-101	Each detected change becomes a branch/PR with machine-readable manifests and required checks.	CI required-checks gate and draft PR artifact review.
ML-062-044	SRC-062 pp.101	Promotion jobs update STAC/DCAT/PROV, graph deltas, API indices, Story Nodes, and Focus diffs after checks.	Release manifest update and rollback replay test.
ML-063-056	SRC-063 pp.159-163	The workflow runs a watcher, evaluates policy, comments on failure, commits catalog/provenance on success, and opens a draft PR.	Workflow dry-run fixture and PR body receipt checks.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: expanded by current-run source evidence.

Z. Open Questions and Verification Backlog

Open questions now include MLT official support/version target, registry referrer support, COG/Zarr/tool pins, source rights, context length limits, and repository homes.

No new v1.8 delta record was extracted for this category. Prior master entries are retained in Appendix A unchanged.

Required objects / contracts: SourceDescriptor, LayerManifest, TileArtifactManifest, StyleManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, RunReceipt, and rollback target are used where applicable. None is claimed to exist in a repository from this run.

Public UI implication: the MapLibre shell should show released layer state, stale/degraded badges, evidence resolution, denial/abstain reasons, and correction/rollback visibility without exposing raw or restricted data.

Changed since prior master: retained without current-run expansion.

8. Components-Functions-Features Matrix

Matrix entries are implementation guidance. They are not claims of current repo existence. The matrix is intentionally split into compact rows to avoid cut-off text.

component	function	feature	input artifacts	output artifacts	governed dependencies	runtime/UI behavior	tests required
MapLibre GL JS	2D web renderer	Render released vector/raster/style artifacts and expose interaction state.	StyleManifest, TileArtifactManifest, MapReleaseManifest.	Camera/click events and rendered map.	Governed API, released catalog, EvidenceBundle resolver.	No RAW/canonical fetch; click opens Evidence Drawer.	Style/load/click/evidence tests.
MapLibre Native	Mobile parity renderer	Potential mobile runtime for released layers.	Same manifest set plus mobile budget.	Mobile map view.	Platform parity and policy state.	Only after drawer/focus parity.	Native/web parity and MLT pilot tests.
MapLibre RS	Experimental renderer	Potential future renderer only.	Renderer benchmark fixtures.	Prototype render.	Plugin/renderer allowlist.	Not public default.	Parity and security tests.
Style JSON	Cartographic contract	Defines sources, layers, filters, expressions.	StyleManifest, sprite/glyph refs.	Map style.	Style validation and source allowlist.	Style loads only released sources.	Style spec/schema/source-layer tests.
source definition	Data source binding	Connect style to PMTiles/MVT/COG/WMS/GeoJSON through governed URLs.	SourceDescriptor, TileArtifactManifest.	Map source block.	Catalog and policy release checks.	Blocked if unreleased or stale beyond policy.	No unreleased tile load test.
layer definition	Visual layer binding	Layer/source-layer/paint/layout/filter.	Style JSON and LayerManifest.	Rendered layer.	Policy and sensitivity class.	Shows trust badge/stale state.	Layer/source-layer/attribute whitelist.
expressions	Conditional styling	Encode thematic style rules without hiding policy-sensitive facts.	Style JSON.	Paint/filter results.	Allowed attributes.	No sensitive geometry hidden only by expression.	Expression whitelist and visual regression.
filters	Feature subset	Filter features by released attributes.	Style JSON, LayerManifest.	Visible subset.	Policy must happen before filter.	Filters cannot be publication control.	Sensitive geometry deny test.
sprites	Icon atlas	Governed icon assets.	Sprite manifest.	Icons.	Style asset integrity.	No missing icon fallback for trust badges.	Sprite URL/hash tests.
glyphs	Font glyphs	Hosted glyph resources.	Glyph manifest.	Labels.	Glyph hosting and CORS.	Readable labels and accessibility.	Glyph URL/CORS tests.
MVT	Vector tile format	Current production baseline for vector tiles.	GeoParquet/processed vectors.	MVT/PMTiles/XYZ.	TileArtifactManifest.	Rendered vector layers.	Tile smoke/source-layer/attribute tests.
MLT	Experimental vector tile successor	Potential smaller/faster tiles with column layout.	Processed vectors, Planetiler/MLT tooling.	MLT tiles.	Version/toolchain gate.	Pilot only; no default before validation.	MVT parity, decode, performance tests.
PMTiles	Static tile archive	Single-file serverless vector/raster tile carrier.	MVT/Tippecanoe, metadata.	PMTiles artifact.	Range/CORS, manifest, signature.	Loaded via governed source URL only.	pmtiles show, Range/CORS, rollback.
MBTiles	Local/intermediate tile store	Optional intermediate before PMTiles conversion.	Tippecanoe output.	MBTiles or converted PMTiles.	Tool version pins.	Not public default.	Convert/inspect tests.
COG	Cloud raster artifact	Single-file raster with overviews/range reads.	Processed raster.	COG.	STAC Raster/Projection, COG validation.	Rendered via raster source or protocol.	COG layout, nodata, bounds, Range.
Martin	Server-mediated tile serving	Dynamic PostGIS tile path when mediation needed.	PostGIS tables, policy context.	MVT tiles.	Access control and source roles.	Use when dynamic/stewarded access is required.	Auth/policy/tile tests.
Maputnik	Style editing tool	Style authoring/review, not truth.	Style JSON.	Reviewed style.	Style review and export checks.	No uncontrolled style drift.	Style diff and visual regression.
layer catalog	Discoverable layer registry	Expose public-safe layers and status.	STAC/DCAT/LayerManifest.	Layer list and metadata.	Governed API.	Search, badges, stale states.	STAC search and payload minimization tests.
Evidence Drawer	Trust payload UI	Resolve clicked feature to evidence.	Feature ID, EvidenceRef, EvidenceBundle.	EvidenceDrawerPayload.	EvidenceBundle resolver and PolicyDecision.	Shows citations, source, status, denial.	Click-to-drawer and citation tests.

component	function	feature	input artifacts	output artifacts	governed dependencies	runtime/UI behavior	tests required
Focus Mode	Evidence-bounded AI	Answer/abstain/deny/error over released context.	MapContextEnvelope, EvidenceBundle, AIReceipt.	FocusModeResponse.	CitationValidationReport and policy.	No direct model client.	ANSWER/ABSTAIN/DENY/ERROR tests.
timeline / time slider	Time-aware interaction	Filter by valid/source/release times.	Time-indexed LayerManifest.	Map state.	Temporal metadata.	Stale badges and time locks.	Temporal fixture tests.
MapContextEnvelope	Context object	Bounded map state for Focus Mode.	Camera, layers, clicked feature, time.	Context envelope.	Policy and evidence closure.	Provider receives only allowed context.	Schema and deny missing-evidence tests.
LayerManifest	Layer contract	Defines layer, source, evidence, policy, release.	SourceDescriptor, catalog IDs.	Layer manifest.	Policy/release state.	Layer catalog visibility.	Schema, source, policy tests.
StyleManifest	Style release contract	Binds style JSON, sprites, glyphs, source refs.	Style JSON, asset digests.	Style release record.	Style validation.	Public style load.	Style hash and visual regression.
TileArtifactManifest	Tile artifact contract	Binds PMTiles/MVT/MLT/COG/3D tiles to bytes and proof.	Tile/COG/3D artifacts, digests.	Tile manifest.	Receipt, signature, rollback.	Only released artifacts load.	Digest, attestation, Range tests.
MapReleaseManifest	Release state contract	Coordinates layer/style/tile release and rollback.	Layer/Style/Tile manifests, PolicyDecision.	Release manifest.	PromotionDecision.	Map publication.	Release/rollback/cache invalidation.
PolicyDecision	Access decision	Allow/deny/abstain/error before exposure.	Rights, sensitivity, release state.	Policy decision.	OPA/Rego and policy service.	Badges and denial messages.	Policy negative fixtures.
CitationValidationReport	Citation gate	Checks evidence citations before answers/exports.	EvidenceBundle, response.	Citation report.	Citation validator.	Abstain on fail.	Pass/fail citation tests.
AIReceipt	AI process receipt	Records model/provider/config/context/citations.	FocusModeRequest/Response.	AIReceipt.	Model adapter, policy, citation.	Displayed or logged for audit.	No direct model, config capture tests.
rollback target	Reversal anchor	Restores prior manifest and invalidates caches.	Previous MapReleaseManifest.	Rollback receipt.	Release system.	Revert layer/tile/style.	Rollback replay test.
cache invalidation record	Cache governance	Records what was invalidated and why.	Materiality decision, manifest diff.	Cache invalidation receipt.	CDN/tile host.	No no-change invalidation.	304/materiality/cdn tests.
GeoParquet	Canonical vector artifact	Processed vector source of truth for derived tiles.	Normalized vectors.	GeoParquet v1.1.	Catalog/manifest.	Not directly authoritative if unpromoted.	GeoParquet metadata and CRS tests.
pgSTAC	STAC database/search	Transactional STAC storage and query backend.	STAC collections/items.	Search API/query results.	DB roles and catalog API.	Layer discovery via API.	Role/transaction/search tests.
ORAS / OCI	Artifact distribution	Digest-addressed dataset/artifact store with referrers.	PMTiles, COG, GeoParquet, SBOM, PROV.	OCI artifact refs.	Cosign/referrer verification.	Serve only verified artifacts.	OCI digest/referrer tests.
RunReceipt	Run evidence	Records source URL, ETag, spec_hash, artifacts, provider.	Watcher/fetch outputs.	RunReceipt.	Policy and manifest gates.	Supports drawer/focus provenance.	Schema and spec_hash tests.
Watcher schema	Source refresh contract	Classifies stac/gtfs/tile/file/api watchers.	SourceDescriptor, watcher spec.	Watcher record.	Signature and registry.	No source activation if unsigned.	Watcher schema/signature tests.
3D Tiles / Cesium	Conditional 3D overlay	Implicit tiling and picked-feature metadata.	3D tileset, STAC, metadata schema.	3D scene artifact.	3D release manifest.	Only when 3D carries evidence burden.	3D digest, picked-feature, screenshot tests.
WMS / WMTS	External/rasterized map service	Visualization path or sample tile proof.	External service descriptor.	Rendered image tiles.	Source terms and metadata.	Not analytics unless source supports it.	Service metadata/sample tile tests.

9. Tile Strategy Chapter

This chapter compares delivery choices using the KFM trust membrane. A tile strategy wins only when it preserves evidence, policy, release state, correction path, and rollback.

comparison	when it wins	when it fails	KFM trust consequence	release / rollback consequence	validation needed
GeoJSON vs vector tiles	GeoJSON wins for small candidate sets, STAC item search results, GTFS detour snippets, and debugging.	Fails as production default for large dense layers or sensitive geometries.	GeoJSON is runtime data, not proof; must be normalized and policy checked.	GeoJSON overlays can be ephemeral; production layers should use manifests and rollback targets.	Schema, size, bounds, CRS, stale-source, no-sensitive-geometry tests.
MVT vs MLT	MVT wins as mature baseline; MLT is promising for large tiles and GPU/browser performance.	MLT fails as default until KFM validates toolchain, spec support, and GL JS/Native parity.	MLT is PILOT/NEEDS VERIFICATION; do not let hype become release authority.	Release MLT only as separate experimental artifact with rollback to MVT.	MVT/MLT parity, decode, load budget, source-layer, attribute whitelist tests.
PMTiles vs MBTiles	PMTiles wins for immutable serverless hosting and simple CDN/Range distribution.	MBTiles is useful as local/intermediate but not ideal for public static hosting.	Both are downstream carriers; manifest and proof decide release.	PMTiles should be checksum-addressed and cache-invalidation aware.	pmtiles show, MBTiles conversion, Range/CORS, digest, rollback tests.
static tiles vs dynamic tile server	Static wins for stable public-safe releases and reproducibility.	Fails when access must be steward-mediated, dynamic, or per-user.	Dynamic serving must enforce policy before tiles leave the server.	Static releases rollback by manifest; dynamic needs config/version rollback.	No unreleased tile load, policy auth, tile cache and materiality tests.
PMTiles serverless vs Martin server-mediated	PMTiles wins for public-safe immutable bundles; Martin/PostGIS wins for dynamic server-side filtering.	PMTiles fails for restricted per-user access; Martin fails if it bypasses release manifests.	Choose by access model, not convenience.	PMTiles rollback via artifact digest; Martin via service/version config.	Range/CORS vs auth/policy/tile integration tests.
vector vs raster	Vector wins for interactive attributes and expressions; raster wins for continuous surfaces/imagery/DEM.	Vector fails for dense rasters; raster fails for feature-level evidence unless linked to metadata.	Both need EvidenceBundle; pixel values alone are not citations.	Different artifact manifests and validation gates.	Tile smoke/attribute tests vs COG/nodata/band/stats tests.
COG vs raster tiles	COG wins for source-like cloud raster with overviews and range reads.	Raster tiles win for simple basemap-style rendering but lose analytic metadata.	COG is closer to evidence carrier but still not root truth.	COG releases must avoid in-place mutation and keep digests.	COG layout, overviews, Range/CORS, Raster/Projection STAC tests.
terrain-enabled vs non-terrain	Terrain wins when vertical context carries real evidence burden.	Fails when it adds visual drama without policy/evidence value.	3D/terrain is conditional and must retain click/evidence governance.	3D tilesets need their own manifest and rollback path.	3D digest-tree, vertical datum, picked-feature and screenshot tests.
2D default vs 3D conditional	2D wins for clarity, accessibility, and KFM map-first shell.	3D fails if it weakens Evidence Drawer or sensitive geometry controls.	MapLibre GL JS remains default; Cesium is a governed overlay branch.	3D release cannot bypass 2D release gates.	Accessibility, keyboard, 2D fallback and scene parity tests.
public-safe tiles vs restricted/internal tiles	Public-safe wins after rights, sensitivity, provenance, and release gates pass.	Restricted/internal tiles fail if exposed through style filters or public URLs.	Sensitive geometry must be transformed before public tile generation.	Restricted artifacts need access tier and rollback/withdrawal path.	Sensitive deny, CARE, rights, generalization, no-style-hiding tests.

10. Governance and Trust-Membrane Chapter

KFM map flow remains: released layer -> user click -> governed API -> EvidenceBundle -> Evidence Drawer -> Focus Mode answer / abstain / deny / error. Current-run sources strengthen the upstream side of this flow by requiring run receipts, source descriptors, catalog closure, signatures, STAC/DCAT/PROV, and release decisions before the renderer receives any public artifact.

- No public RAW path. RAW, WORK, QUARANTINE, unpublished candidate data, and canonical/internal stores remain behind governed interfaces.
- No direct model client. Focus Mode is an evidence-bounded adapter behind the governed API; local runtime settings such as context length must be explicit and auditable.
- No canonical/internal client fetch. MapLibre consumes released artifacts and governed APIs, not source systems or internal data stores.
- No unreleased tile load. PMTiles, MVT, MLT, COG, 3D Tiles, style JSON, sprites, and glyphs must be released and manifest-bound.
- No sensitive geometry hidden only by style. CARE/locality restrictions require masking, generalization, restricted tier, or denial before public tile generation.
- No popup as Evidence Drawer substitute. Popups can preview; material claims need EvidenceDrawerPayload and EvidenceBundle resolution.
- No uncited export. Screenshots, reports, Story Nodes, and Focus answers retain citations and manifest/version references.

10A. Current-run trust membrane additions

addition	source	trust consequence
GitHub App + Checks API	SRC-063 pp.1-5	Promotion checks become visible, machine-verifiable and merge-blocking.
Acceptance harness	SRC-063 pp.6-9	STAC/DCAT/PROV and spec_hash validation becomes a release gate.
CARE-bound metadata	SRC-062 pp.77-83	Consent, authority, locality restrictions and review expiry enter policy decisions.
Zero-trust ingest	SRC-062 pp.91-96	External source blobs are staged by digest with signed logs before transforms.
pgSTAC -> GeoParquet -> PMTiles -> OCI	SRC-063 pp.150-153	Tile publication gets exact byte identity and attestable distribution.
MLT pilot	SRC-063 p.154	Potential performance gains are not production authority; parity validation required.

11. Implementation-Ready Object Map

Object families below are PROPOSED field-intent maps unless repo evidence later proves they exist. Directory Rules remain binding for path placement; no path is asserted as implemented in this run.

object family	field intent (PROPOSED unless verified)	current-run expansion	validation / policy consequence
SourceDescriptor	source_id, source_role, provider, endpoint, rights, sensitivity, cadence, authority class, retrieval method.	Adds ArcGIS/S3/OCI fields, watcher type, license-first delta handling.	Deny unknown rights, source-role mismatch, unapproved provider.
LayerManifest	layer_id, title, source refs, source-layer, geometry class, evidence dependency, policy class, stale rules.	Tie layer to STAC Collection, RunReceipt, and materiality thresholds.	Layer cannot load without release and policy allow.
StyleManifest	style_id, style JSON digest, sprite/glyph refs, source allowlist, visual regression baseline.	No major new fields; strengthened by reproducible build and artifact digest requirements.	Style drift and unreleased source tests.
TileArtifactManifest	artifact_id, format, digest, byte size, tilejson, min/max zoom, CRS, source artifact, toolchain, attestation.	Adds PMTiles/COG/GeoParquet/OCI digest, COG layout, MLT pilot, 3D tileset digest-tree.	Checksum, attestation, Range/CORS, source-layer tests.
MapReleaseManifest	release_id, version, layer/style/tile refs, PromotionDecision, rollback target, cache invalidation record.	Adds OCI digest refs, kill-switch behavior, PR-first release evidence.	Release/rollback replay and no-file-move promotion tests.
EvidenceBundle	evidence_id, claims, source role, citations, rights, policy, temporal/spatial scope.	CARE fields and exact metadata digest link strengthen sensitive releases.	Citation validation and deny/abstain tests.
EvidenceDrawerPayload	clicked feature, feature_id, evidence refs, citations, source state, policy messages, correction links.	Adds run/prov IDs and human-readable policy remediation.	Click-to-drawer latency and data minimization tests.
MapContextEnvelope	camera, bounds, time, visible layers, clicked feature, released context only.	Focus Mode runtime config and source provenance can be captured.	No direct model/raw path; missing evidence abstain.
FocusModeRequest	question, MapContextEnvelope, EvidenceBundle refs, policy context, model adapter config.	Explicit context-length/model config from SRC-063.	Runtime budget and no-public-model-client tests.
FocusModeResponse	finite outcome, answer text, citations, denial/abstain reasons, report refs.	AI/chat activities can have PROV without becoming truth.	Citation pass/fail, ABSTAIN, DENY, ERROR tests.
AIReceipt	provider, model, context, config, prompt class, evidence refs, output digest, citation report.	Adds local Ollama context length / num_ctx capture.	Reject uncited or config-unrecorded answers.
CitationValidationReport	citation targets, resolved EvidenceBundle IDs, pass/fail, missing refs, source ledger coverage.	No direct expansion; now tied to Focus Mode and exports.	Citation failure abstains.
PolicyDecision	input facts, policy version, allow/deny/abstain/error, reasons, remediation.	CARE, license, provider, source approval, kill-switch, public/restricted tier.	OPA/Rego fixtures and plain-language failure surface.
PromotionDecision	gate results, materiality, review state, attestations, release/rollback.	Adds GitHub App checks, OCI referrers, Clean rebuild hash.	Merge blocked unless gates pass.
RunReceipt	run id, source URL, ETag/Last-Modified, spec_hash, artifacts, provider, provenance.	Expanded by conditional fetch and watcher schemas.	Spec_hash and provider approval tests.
rollback target	previous release manifest, previous artifact digests, cache state, rollback receipt.	Rollback action emits attestation.	Rollback restores prior manifest and invalidates cache.
stale-source fixture	source last seen, expected cadence, latest retrieved, stale reason.	Hydrology/realtime SLOs and heartbeat events.	Stale badge, abstain or deny tests.
sensitive-geometry deny fixture	exact geometry, sensitivity reason, transform/generalization result, policy decision.	CARE locality restrictions and rights controls.	No public exact sensitive geometry; style filters insufficient.

12. Validation and Test Plan

test family	minimum test	current-run expansion	status
schema validation	Validate SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, RunReceipt, Watcher schema, EvidenceDrawerPayload.	Adds watcher enum, RunReceipt fields, STAC/CARE and OCI digest/referrer fields.	PROPOSED
source ledger completeness	Every source has source_id, role, limits, MapLibre/tile relevance.	Adds SRC-062/SRC-063 and preserved baseline.	CONFIRMED in PDF
style source/layer validity	Style JSON sources/layers source-layer/promoteId/filters/expressions validated.	Geometry normalization and stable IDs increase importance.	PROPOSED
PMTiles/MVT/MLT manifest checks	pmtiles show, source-layer whitelist, MVT/MLT parity, tile stats.	MLT pilot and pgSTAC->PMTiles pipeline.	NEEDS VERIFICATION
COG/range request requirements	COG layout, overviews, nodata, Raster/Projection STAC, HTTP Range/CORS.	COG/Zarr deterministic packaging and no in-place COG updates.	PROPOSED
no public raw path	Browser cannot load RAW/WORK/QUARANTINE/candidate/internal store.	Zero-trust ingest reinforces staging.	PROPOSED
no direct model client	Client cannot call Ollama/model runtime directly.	Context-length and model config captured in AIReceipt.	PROPOSED
no unreleased tile load	Layer source URL must appear in released manifest.	OCI digest and attestation required for published PMTiles/COG.	PROPOSED
sensitive geometry deny	CARE/locality restrictions deny public exact geometry.	Spatial masking or restricted tier required.	PROPOSED
source rights unknown blocks release	Rights/license unknown leads to quarantine/deny.	License-first deltas and freight rights text.	PROPOSED
stale source badge / abstain	Source cadence and ETag/no-change handling produce stale/heartbeat states.	Hydrology/realtime SLO and no-change heartbeat.	PROPOSED
clicked feature resolves EvidenceBundle	Feature ID and source-layer map to EvidenceBundle.	3D picked-feature property checks extend to Cesium branch.	PROPOSED
citation validation pass/fail	Focus answers and exports cite evidence or abstain.	AI/chat PROV exists but does not replace citations.	PROPOSED
Focus Mode answer with citations	Finite ANSWER outcome only with citations and policy allow.	Context length/runtime config captured.	PROPOSED
Focus Mode abstain on missing evidence	ABSTAIN on missing EvidenceBundle/citation.	No rendered-feature-only answers.	PROPOSED
rollback restores prior manifest	Rollback action reverts release and emits attestation.	GitHub App rollback and cache invalidation.	PROPOSED
cache invalidation recorded	Material change drives invalidation; no-change does not.	Centroid/area/status/materiality thresholds.	PROPOSED
visual regression for styles	Render sample pages/tiles/screenshots.	Review render for Story Nodes/Focus diffs.	PROPOSED
keyboard navigation / accessibility	Layer catalog, drawer, Focus states keyboard accessible.	Plain-language policy failure output.	PROPOSED
tile load budget	Measure tile count, size, decode, first render.	MLT pilot and PMTiles/COG range budgets.	NEEDS VERIFICATION
interaction-to-drawer latency	Measure click to Evidence Drawer p95.	Run/prov IDs and evidence resolver path.	NEEDS VERIFICATION

13. Anti-Patterns Register

The register preserves prior master anti-patterns and adds v1.8 failures inferred from current sources. Anti-patterns are not deleted; they become regression fixtures.

anti-pattern	why it is dangerous	current-run evidence / consequence	required countermeasure
Treating MapLibre as truth authority	Renderer output is not evidence.	All new sources assume catalog/proof/manifest before UI.	EvidenceBundle and release manifest required.
Treating tiles as proof	Tiles are derived carriers.	PMTiles/COG/GeoParquet require digest and attestation.	TileArtifactManifest + EvidenceBundle.
Layer toggle as publication	Visibility is not release state.	PR-first and promotion gates control release.	MapReleaseManifest required.
Hiding sensitive geometry with style filters	Data still reaches browser.	CARE locality restriction requires masking/restricted tier.	Pre-tile transform or deny.
Dense GeoJSON as production default	Browser performance and leakage risks.	GeoJSON is limited to small candidates/detours.	Use tiles or bounded APIs.
Uncontrolled style drift	Style can change claims visually.	Reproducible builds and style manifests.	Style hash and visual regression.
Plugin adoption without review	Plugins can bypass trust membrane.	Version-sensitive MLT/toolchain claims.	Allowlist, SBOM, pinning.
Direct browser access to model runtime	Bypasses policy and citation.	Ollama Focus Mode config needs governed adapter.	No direct model client test.
AI answers based only on rendered features	Rendered state lacks evidence closure.	Focus Mode must use EvidenceBundle and citations.	MapContextEnvelope + EvidenceBundle.
Uncited popups or exports	Users cannot inspect support.	PR/check summaries and drawer payloads require citations.	CitationValidationReport.
MLT as default before validation	New format may not meet parity/tooling needs.	SRC-063 reports MLT support but needs verification.	Pilot gate and rollback.
3D/globe as default	Can obscure evidence and accessibility.	3D Tiles/Cesium is conditional.	2D default and 3D fallback.
Mapbox-era backports without review	Legal/security/API drift risk.	Retained prior anti-pattern.	Legal/security review.
Unsigned artifact release	Origin/process unprovable.	SRC-062 says unsigned artifacts invalid.	Cosign/Rekor or approved signature gate.
Mutable OCI tag as release identity	Tags can move.	SRC-063 says metadata must reference digests.	Use immutable digest.
In-place COG update	Can break COG layout.	SRC-063 warns against post-writes.	Create new artifact/digest.
Invalid STAC-ish JSON	Catalog clients and layer manifests break.	SRC-062 and SRC-063 flag invalid STAC.	Strict STAC validation.
No-change source causes new release	Creates false churn and cache invalidations.	Heartbeat/304 pattern prevents.	No-change fixture.
Raw feed exposed as public layer	Bypasses lifecycle and policy.	Raw -> processed -> catalog -> published invariant.	No public RAW path test.

14. Open Questions and Verification Backlog

Unresolved items remain visible. The next run should not convert them into facts unless verified from repo evidence, generated artifacts, official version checks, tests, logs, or release manifests.

open item	status	why it matters	next verification
MapLibre GL JS target version	NEEDS VERIFICATION	Prior version and MLT support claims are version-sensitive.	Check official MapLibre release docs before pinning.
MLT support status in GL JS/Native/Planetiler	NEEDS VERIFICATION	SRC-063 reports support; KFM needs parity and toolchain proof.	Official docs + local MVT/MLT benchmark.
Schema home and object path placement	NEEDS VERIFICATION	Directory Rules default schemas/contracts/v1, but repo must be inspected.	Mount repo; check ADR-0001 and current roots.
PMTiles hosting Range/CORS behavior	NEEDS VERIFICATION	PMTiles and COG depend on byte-range capable hosting.	Probe headers for target host/CDN.
COG range/CORS and layout validation	NEEDS VERIFICATION	COG rendering and analytics require correct layout/headers.	Run gdalinfo/validate and HTTP range probe.
OCI registry referrer support	NEEDS VERIFICATION	ORAS/Cosign referrers require registry support.	Test GHCR/target registry referrers and discover.
Cosign/Rekor availability and offline bundle policy	NEEDS VERIFICATION	Release proof may need online or offline verification.	Define verify mode and CI environment.
pgSTAC/PostGIS role model	NEEDS VERIFICATION	Dynamic/catalog-backed services need least privilege.	Inspect deployment and migrations.
Martin deployment/access model	NEEDS VERIFICATION	Dynamic tiles need server-mediated policy.	Decide static vs dynamic per lane.
Plugin/tool allowlist	NEEDS VERIFICATION	Tippecanoe, pmtiles, GDAL, Planetiler, MapLibre plugins need review.	Create SBOM/tool registry.
Style/sprite/glyph hosting	NEEDS VERIFICATION	Public maps require stable assets and CORS.	Probe asset hosting and digests.
Attribution and rights	NEEDS VERIFICATION	Source rights block publication.	Verify source-specific licenses/terms.
CARE/locality restriction transform rules	NEEDS VERIFICATION	Sensitive/cultural/locality data must not leak.	Define generalization/masking rules and fixtures.
Release manifest location	NEEDS VERIFICATION	Publication needs a canonical home.	Check repo; avoid parallel release homes.
CI workflow and branch protection	NEEDS VERIFICATION	Checks API/promotion gates need enforced status.	Inspect GitHub settings and workflow names.
Accessibility acceptance	NEEDS VERIFICATION	Trust-visible UI must be usable by keyboard/screen readers.	Define a11y acceptance matrix.
Performance budgets	NEEDS VERIFICATION	MLT/PMTiles/COG changes need budgets.	Tile load/resource timing benchmarks.
Interaction-to-drawer latency budget	NEEDS VERIFICATION	Evidence Drawer must feel trustworthy and responsive.	Define p95 click-to-drawer target.
Ollama/Focus Mode context length controls	NEEDS VERIFICATION	SRC-063 warns of memory/performance changes.	Pin model runtime settings and log in AIReceipt.
Watcher dry-run fixture home	NEEDS VERIFICATION	No-network CI should be first PR.	Verify repo test conventions and add fixtures.

15. How to Continue This Master PDF Next Run

Attach this PDF again with any new documents.

The next run should read this PDF first as SRC-MASTER-PREV.

Preserve all idea IDs unless merged or superseded.

Update the delta table.

Add new sources to the source ledger.

Add new discoveries to the idea index.

Recompute counts.

Keep uncertainty labels visible.

Carry Appendix A forward or replace it with a new appended baseline only after preserving all prior ideas, source ledgers, category chapters, open questions, and backlog entries. If a prior idea is merged, retain its older idea_id as an alias and document the canonical successor.

Final PDF Self-Check

check	result
No obvious cut-off text	PASS - update packet generated with wrapped paragraphs/tables; sample pages rendered for QA.
No missing source ledger	PASS - Section 4 includes current-run ledger and Appendix A retains prior ledger.
No missing delta section	PASS - Section 3 includes continuity and prior-cluster comparison.
No missing open questions	PASS - Section 14 lists carried-forward and new verification items.
No unsupported implementation claims	PASS - repository paths and object families are labeled PROPOSED/NEEDS VERIFICATION unless evidenced by attached PDFs.
No forgotten prior master ideas	PASS - SRC-MASTER-PREV is appended page-for-page as Appendix A and all prior 2550 idea records are retained.
PDF-only output	PASS - one final user-facing PDF is produced; no archive, scaffold, patch, or loose source tree.

Appendix A begins on the next page and contains the retained v1.7 baseline page-for-page.

Master MapLibre

Components-Functions-Features.pdf

Cumulative KFM MapLibre, Tile, Renderer, Style, UI, Evidence Drawer, and Release Architecture
Atlas

Field	Value
Version	v1.7 cumulative update
Run date	2026-05-10
Attached source count	42 PDFs visible in /mnt/data; prior master detected; one new source packet analyzed and one reattached source recognized.
Prior master PDF detected	Yes - v1.6 copied as SRC-MASTER-PREV before overwrite; prior pages 425; sha256 prefix ac6079aee934601e.
Output mode	PDF only. No ZIP, tarball, scaffold folder, repository patch, loose source tree, or extra user-facing artifact.
Truth posture	CONFIRMED doctrine/source scans; PROPOSED implementation guidance; UNKNOWN current repo/runtime depth; NEEDS VERIFICATION for versions, hosting, rights, source endpoints, policy enforcement, performance budgets, mobile/native parity, and path placement.
Evidence limits	No mounted KFM repo, tests, workflows, dashboards, logs, release manifests, or runtime traces were available. Paths in sources are source-reported and PROPOSED unless repo evidence later verifies them.
Source ledger count	43 current-run records: SRC-RUN-PROMPT, SRC-MASTER-PREV, 40 retained or reattached source records, and NEW SRC-061.
Idea count	2550 cumulative idea records: 2383 prior + 167 v1.7 delta records.
New discoveries count	167 delta records from SRC-061.
Expanded category clusters	22 category clusters expanded by current-run source evidence.
Superseded ideas	0 - no attached evidence invalidated a prior idea.
Open verification items	393 cumulative estimate: prior 329 + 64 new or sharpened items.

Core determination: MapLibre remains a downstream renderer and interaction runtime. Tiles, PMTiles, MVT, MLT, COGs, style JSON, popups, screenshots, Story Nodes, 3D scenes, graph projections, and AI answers remain downstream carriers, not sovereign truth.

1. Table of Contents

Navigation note. This update packet is current-run specific. The full v1.6 baseline follows as Appendix A so prior source ledger entries, category chapters, matrices, idea IDs, and open questions are not silently deleted.

- 1. Cover Page and run metadata
- 2. Executive Determination
- 3. Continuity and Delta Since Last Run
- 4. Source Ledger
- 5. Master Idea Index - v1.7 Delta
- 6. Category Atlas
- 7. Detailed Category Chapters
- 8. Components-Functions-Features Matrix
- 9. Tile Strategy Chapter
- 10. Governance and Trust-Membrane Chapter
- 11. Implementation-Ready Object Map
- 12. Validation and Test Plan
- 13. Anti-Patterns Register
- 14. Open Questions and Verification Backlog
- 15. How to Continue This Master PDF Next Run
- Final PDF Self-Check
- Appendix A. Full retained v1.6 Master PDF baseline (SRC-MASTER-PREV, appended unchanged)

2. Executive Determination

CONFIRMED from SRC-MASTER-PREV and retained doctrine: MapLibre is a disciplined 2D renderer and interaction runtime inside a governed shell. It is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority, or AI authority. Public surfaces use governed APIs, released artifacts, catalog records, tile services, EvidenceBundle resolution, and release manifests.

CONFIRMED from this run: the prior v1.6 master baseline was detected, copied before overwrite, and appended unchanged. The file New Ideas 12-20-25.docx.pdf has the same SHA-256 as prior SRC-060 and is treated as reattached/retained rather than a duplicate discovery source. New source packet SRC-061 adds the strongest v1.7 evidence: SLSA/Sigstore gating for COG/GeoParquet/PMTiles, geometry repair before MVT, H3 GeoParquet quickmaps, OCI/ORAS distribution for PMTiles/COGs, COG + STAC raster requirements, production MVT pipeline guidance, MapLibre PMTiles source patterns, dynamic Tegola/t_rex gates, Cesium/3D fallback guidance, sensitive archaeology generalization, and remote-sensing QA.

PROPOSED synthesis: v1.7 should be treated as an artifact-integrity, vector-tile-production, COG/raster-validation, dynamic-vs-static tile-serving, sensor/air-layer, and sensitive-archaeology map-governance update. It does not reorganize the prior master. It adds one new source, preserves all prior records, and sharpens the implementation-ready object and validation backlogs.

UNKNOWN: actual repo topology, package manager, schema home, UI component homes, MapLibre GL JS target version, PMTiles tooling, PMTiles/COG/Zarr/3D-Tiles hosting headers, browser/device performance, mobile/native parity, plugin allowlist, source rights, actual policy enforcement, and release workflow remain unverified.

NEEDS VERIFICATION: every package version, tool version, source URL, feed schema, hosted header, release manifest, SBOM, telemetry file, artifact digest, Story Node path, sensor calibration source, WZDx/TDx registry behavior, ArcGIS/NFHL endpoint behavior, PMTiles protocol handling, range/CORS behavior, and archaeology/cultural-steward rule before operational use.

Highest-risk anti-pattern: loading a public map source, PMTiles archive, COG, NFHL layer, air-quality surface, incident layer, archaeology heatmap, Story Node, badge, popup, or Focus Mode answer before EvidenceBundle, manifest, PolicyDecision, PromotionDecision, citation validation, proof object, and rollback target exist.

Metric	Value
Previous version	v1.6 cumulative update
Current version	v1.7 cumulative update
Prior ideas carried forward	2383
Reattached source recognized	SRC-060, exact SHA match; no duplicate ideas created
New SRC-061 delta ideas	167
Total cumulative idea count	2550
Superseded ideas	0
Open verification items	393

3. Continuity and Delta Since Last Run

SRC-MASTER-PREV is the prior v1.6 Master MapLibre Components-Functions-Features.pdf. It remains the cumulative memory baseline and is appended unchanged. It does not outrank original evidence. Current-run source evidence can refine, expand, conflict with, or supersede its entries. No supersession occurred in this run.

Delta item	Result
Prior version detected	v1.6, 425 pages, sha256 prefix ac6079aee934601e
Version rule applied	+0.1 ordinary cumulative update. No fundamental structural reorganization.
New sources added	SRC-061 New Ideas 12-24_25_26.docx.pdf.
Reattached source	SRC-060 New Ideas 12-20-25.docx.pdf matches prior SHA; retained, not duplicated.
Ideas retained	All 2383 prior idea records preserved by appending SRC-MASTER-PREV unchanged.
New records	167 ML-061-* delta records.
Merged / superseded	0 merged, 0 superseded.
Conflicts discovered	No direct source conflict. Main tension remains implementation maturity: source packets include implementation-style paths and code snippets, but no mounted repo/runtime proof was available.
Open questions carried forward	Package versions, schema home, source rights, MLT status, mobile parity, hosting headers, sensitive geometry, release manifest home, CI workflow, accessibility and performance budgets.
New open questions	OCI registry/referrer support, artifact SBOM home, geometry repair thresholds, COG range/CORS behavior, dynamic tile serving access model, TDx/NFHL feed behavior, archaeology H3 floor, air-sensor calibration authority, and provenance badge semantics.

3A. Prior idea comparison summary

The full prior idea comparison table cannot be repeated without duplicating 2383 rows. Instead, Appendix A retains the complete prior v1.6 master unchanged. The table below records current status at the prior idea-range and category-cluster level. Each prior idea_id remains preserved in Appendix A unless a future run explicitly merges or supersedes it.

Prior idea range / cluster	Prior title scope	Current status	Source that changed status	Note
All prior idea IDs through v1.6	Full retained v1.6 idea index and category chapters	RETAINED	SRC-MASTER-PREV	No prior idea was deleted or superseded.
ML-060-*	PM2.5 calibration, Zarr/STAC arrays, incident fusion, deterministic run gates	RETAINED / REATTACHED	SRC-060	Same SHA as prior SRC-060; no duplicate delta.
MVT / PMTiles clusters	Vector tile, PMTiles, runtime verification, artifact integrity	EXPANDED	SRC-061 pp.171-178, pp.84-88	Adds production MVT pipeline, static/dynamic decision gates, OCI artifact publication.
COG / raster clusters	COG, DEM, raster, remote-sensing time series	EXPANDED	SRC-061 pp.123-127, pp.520-528	Adds COG/STAC validation, range/cache guidance, remote-sensing QA.
Sensitive archaeology clusters	Generalized sensitive geometry, Story Nodes, Focus Mode	EXPANDED	SRC-061 pp.224-229, pp.340-344	Adds H3 generalization floor, CARE chips, and Focus Mode cautions.

4. Source Ledger

This ledger is a control surface, not a bibliography. It states what each source can support and what it cannot prove. Current external facts embedded in source PDFs remain source-supported but NEEDS VERIFICATION before operational package pins, access decisions, source activation, or release claims.

source_id	file	scan result
SRC-061	New Ideas 12-24_25_26.docx.pdf	550 pages; high signals: STAC/DCAT/PROV, PMTiles, MVT, COG, MapLibre, Cesium, Focus Mode, H3, GeoParquet, NFHL, TDx, LiDAR, remote sensing, sensitive archaeology. sha256 prefix df8a2f1a5069db16
SRC-060	New Ideas 12-20-25.docx.pdf	37 pages; exact SHA prefix 89b0231a26c34e8a matches prior v1.6 SRC-060. Reattached/retained.
SRC-MASTER-PREV	Master MapLibre Components-Functions-Features.pdf v1.6	425 pages; retained baseline; sha256 prefix ac6079aee934601e.

source_id	file / document title	source type	status	role in this PDF	relevance / evidence strength	limitations
SRC-RUN-PR OMPT	Current user prompt	Prompt	CONFIRMED	Controls cumulative PDF-only output, required structure, truth labels, source handling, and KFM invariants.	High requirements source	Instruction source, not implementation proof.
SRC-MASTER -PREV	Master MapLibre Components-Functions-Features.pdf v1.6	Prior master PDF	CONFIRMED / RETAINED	Cumulative baseline. Appended unchanged as Appendix A.	High continuity source Pages 425; sha ac6079aee934601e.	Does not outrank original evidence; current-run evidence can expand or conflict.
SRC-003	KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf	Map/UI doctrine	RETAINED	Renderer boundary, governed UI, Evidence Drawer, Focus Mode, tile/style strategy.	High MapLibre relevance	No mounted repo proof.
SRC-004	Directory Rules.pdf	Directory doctrine	RETAINED	Responsibility roots, schema-home default, lifecycle boundaries, ADR/drift rules.	High for proposed path discipline	Specific paths remain PROPOSED until repo evidence.
SRC-005	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf	Pipeline manual	RETAINED	Lifecycle law, receipts, promotion, query-save-validate-compile loop.	High governance relevance	No runtime proof.
SRC-006	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan_v1_1.pdf	Build plan	RETAINED	Trust spine, pre-RAW event family, public-safe defaults, map architecture.	High doctrine	Greenfield/proposed-to-create posture.
SRC-007	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf	UI + AI plan	RETAINED	Whole UI, governed API, MapLibre adapter boundary, Evidence Drawer, Focus Mode.	High lineage	No mounted repo proof.
SRC-008	KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf	Governed AI plan	RETAINED	Provider-neutral AI, finite outcomes, citation validation, AIReceipt.	High Focus Mode relevance	No runtime proof.
SRC-009	Ollama & Ubuntu Information.pdf	Runtime guide	RETAINED	Local model runtime behind governed API, no direct public model path.	Medium Focus Mode relevance	Version/runtime facts need recheck.
SRC-010	kfm_encyclopedia.pdf	Capability encyclopedia	RETAINED	System-wide KFM domain and capability atlas; maps/tiles/AI as downstream carriers.	High doctrine	Implementation depth unknown.
SRC-011	# Kansas Frontier Matrix Implementation Reference.pdf	Implementation reference	RETAINED	Prior public-repo summary and MapLibre/PMTiles stack signals.	Lineage signal	Prior connector claims not reverified here.
SRC-012	a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf	GIS/cartography reference	RETAINED	Representation, projections, map communication concepts.	Conceptual support	Not KFM implementation evidence.
SRC-013	Advanced-SQL-Concepts.pdf	SQL reference	RETAINED	Analytic SQL concepts.	Low direct map relevance	Not MapLibre-specific.
SRC-014	AI_Concepts_Using_Python.pdf	AI reference	RETAINED	AI/data workflow background.	Low direct map relevance	Not KFM-specific proof.
SRC-015	Archaeological 3D GIS.pdf	3D GIS reference	RETAINED	3D GIS and archaeological visualization context.	Supports conditional 3D	External examples only.

source_id	file / document title	source type	status	role in this PDF	relevance / evidence strength	limitations
SRC-016	CICD_with_Docker_Kubernetes_Semaphore.pdf	CI/CD reference	RETAINED	CI/CD, release, rollback, deployment controls.	Supports validation/rollback	Not KFM proof.
SRC-017	Designing Great Web APIs.pdf	API reference	RETAINED	API design concepts and contracts.	Supports governed API dependency	Not KFM proof.
SRC-018	developing-time-oriented-database-applications-in-sql.pdf	Temporal DB reference	RETAINED	Valid time, transaction time, bitemporal modeling.	Supports timeline maps	Not KFM proof.
SRC-019	Domain-Driven Design Reference.pdf	DDD reference	RETAINED	Bounded context and published language concepts.	Supports responsibility boundaries	Not KFM proof.
SRC-020	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf	GIS lab/textbook	RETAINED	Environmental GIS and hazards/climate layer context.	Conceptual support	ArcGIS-centric.
SRC-022	GIS Succinctly.pdf	GIS reference	RETAINED	Spatial database, geometry, and GIS basics.	Conceptual support	Not KFM proof.
SRC-023	GIS in Sustainable Urban Planning and Management.pdf	Urban GIS reference	RETAINED	Planning support, spatial indicators, resilience cases.	Context for planning layers	Not KFM proof.
SRC-024	instant-web-scraping-with-java.pdf	Scraping reference	RETAINED	Web acquisition and error handling background.	Low direct map relevance	Not KFM proof.
SRC-025	KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Run receipts, deterministic identity, promotion gates.	High governance lineage	Prior synthesis.
SRC-026	KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Source-aware ingest, PMTiles, MLT, map stack guidance.	High map lineage	Prior synthesis.
SRC-027	KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Pre-RAW events, watchers, catalog integrity.	High map/data lineage	Prior synthesis.
SRC-028	KFM_Agriculture_Domain_Implementation_Dossier_REVISÉD_2026-04-21.pdf	Domain blueprint	RETAINED	Agriculture layers, catalog closure, MapLibre/Evidence Drawer.	Domain layer examples	Paths PROPOSED.
SRC-029	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	Domain blueprint	RETAINED	Archaeology exact-location denial and generalized map outputs.	Strong sensitivity relevance	Paths PROPOSED.
SRC-030	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Air/smoke/EO layers and knowledge-character labels.	Air layer relevance	Paths PROPOSED.
SRC-031	KFM_Fauna_Architecture_PDF_Only_Report.pdf	Domain blueprint	RETAINED	Fauna occurrence geoprivacy and public-safe products.	Sensitive occurrence relevance	Paths PROPOSED.
SRC-032	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	Domain blueprint	RETAINED	Rare plant controls and occurrence redaction.	Sensitive flora relevance	Paths PROPOSED.
SRC-033	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Geology/resource layers and public-safe geometry.	Geology tiles relevance	Paths PROPOSED.
SRC-034	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	Domain blueprint	RETAINED	Habitat assignment proof slice and drawer payloads.	Habitat/fauna layer relevance	Paths PROPOSED.
SRC-035	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	Domain blueprint	RETAINED	Habitat patches, suitability, connectivity, public-safe products.	Domain layer examples	Paths PROPOSED.
SRC-036	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	Domain blueprint	RETAINED	Hazard layer boundary and no emergency alerting.	Hazard map relevance	Paths PROPOSED.
SRC-037	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Hydrology proof lane, HUC/NHD/USGS/NFHL layers.	Hydrology map foundation	Paths PROPOSED.

source_id	file / document title	source type	status	role in this PDF	relevance / evidence strength	limitations
SRC-038	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	Domain blueprint	RETAINED	Living-person/DNA restrictions and temporal land assertions.	Sensitivity/parcel relevance	Paths PROPOSED.
SRC-039	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	Domain blueprint	RETAINED	Transport/corridor layers, graph projections, drawer payloads.	Transport map relevance	Paths PROPOSED.
SRC-040	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	Domain blueprint	RETAINED	Settlement/infrastructure layer controls and sensitive assets.	Infrastructure layer relevance	Paths PROPOSED.
SRC-041	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	Domain blueprint	RETAINED	Soil maps, SSURGO/gSSURGO, PMTiles/COG lineage.	Soil tile/raster relevance	Paths PROPOSED.
SRC-056	KFM_Pass_15_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Runtime proof lanes, map-serving proof parity and lane gaps.	High prior master relevance	Already incorporated in prior master.
SRC-060	New Ideas 12-20-25.docx.pdf	New Ideas packet	REATTACHED / RETAINED	Exact SHA matches prior SRC-060; prior v1.6 ideas retained, no duplicate current delta created.	PM2.5/Zarr/incident lineage Pages 37; sha 89b0231a26c34e8a.	Already included in SRC-MASTER-PREV.
SRC-061	New Ideas 12-24_25_26.docx.pdf	New Ideas packet	NEW	Adds SLSA/OCI artifact publication, geometry repair, H3 GeoParquet, COG/STAC, PMTiles/MVT pipeline, sensitive archaeology layers, and remote-sensing validation.	Very high v1.7 delta relevance Pages 550; sha df8a2f1a5069db16.	Implementation-style paths remain PROPOSED; external claims not revalidated.

5. Master Idea Index - v1.7 Delta

The full cumulative v1.6 idea index of 2383 prior ideas is retained in Appendix A via SRC-MASTER-PREV. The records below add the v1.7 delta from SRC-061. Implementation and validation consequences are PROPOSED unless a future mounted-repo inspection verifies them.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-001	SLSA-grade attestations for PMTiles, COG, and GeoParquet	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.1-8	The source frames COGs, GeoParquet, and PMTiles as non-container artifacts that require signatures, attestations, provenance predicates, and promotion policy before trusted release.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-002	Checksums do not prove origin or process	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW idea SRC-061 pp.1-2	Checksums answer byte integrity but not who produced an artifact or which pipeline and inputs produced it.	Keep renderer and tiles downstream of released evidence, policy, and proof objects.	Trust-membrane test: no public raw/canonical/model-only path. Treating SHA-256 alone as publication proof.
ML-061-003	Artifact is evidence; metadata is sovereign	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.1-2	The source explicitly separates the bytes from the trust metadata and says artifact metadata controls origin, process, and promotion.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-004	Blob signature plus DSSE attestation are distinct checks	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.3-4	A blob signature verifies exact bytes and signer identity, while a DSSE/in-toto style attestation records the process that created the artifact.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-005	Keyless CI signing binds workflow identity to artifact	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW idea SRC-061 pp.3-4	The source describes OIDC/Fulcio/Cosign/Rekor style keyless signing as preferable to long-lived keys for CI-built geospatial artifacts.	Pin, review, and allowlist dependencies and tools.	SBOM, tool hash, plugin allowlist, OIDC, sandbox tests. Long-lived keys or manual signing can weaken auditability.
ML-061-006	Offline verification bundles support forensic review	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.4-5	Verification bundles carry certificate, signature, and transparency proof so artifact verification can work even when external services are not reachable.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-007	CUE validates structure; OPA/Rego validates trust logic	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.5-6	The source separates structural validation from trust decisions: schema/predicate shape must pass before policy rules decide whether publication is allowed.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-008	Fail-closed promotion rejects invalid signature, identity, predicate, or policy	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW idea SRC-061 pp.6-7	The source says promotion fails if signature, identity, predicate shape, or policy checks fail; no manual "looks fine" path is acceptable.	Record failure mode and deny or abstain path.	Regression fixture proves failure is denied, abstained, or quarantined. Manual override that promotes unverified tiles or COGs.
ML-061-009	LAZ remains raw elevation source while EPT/COPC serve visualization and analytics	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.8-11	The LiDAR section distinguishes LAZ as archival/scientific source material from EPT/COPC as range-addressable analytics and visualization carriers.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-010	USGS 1 m DEM is a canonical raster surface but not a complete point-cloud substitute	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.11-12	The source treats DEMs as derived surface products useful for hydrology and terrain analysis while noting that non-ground returns require point-cloud fallback.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Using a DEM as proof of buildings, vegetation, or powerlines.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-011	Point clouds and terrain derivatives require STAC pointcloud/catalog records	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.10-11	The source recommends indexing EPT/COPC through STAC pointcloud-style catalog records.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-012	LiDAR acquisition year is a first-class temporal attribute	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.12-13	Kansas LiDAR spans multiple acquisition years; the source calls for acquisition year to support time-slice DEMs, change detection, and vertical bias audits.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-013	State LiDAR mirrors require explicit license assertions	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.11-12	Although many Kansas LiDAR sources are open, the source says project-level licensing and attribution language must be checked before publication.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Assuming all mirrors inherit federal public domain rights.
ML-061-014	Terrain narratives can support Story Nodes but do not replace evidence	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW idea SRC-061 pp.13-14	The source connects terrain products to story-node terrain narratives while keeping provenance and source identity explicit.	Make Focus Mode evidence-bounded and citation-validated.	ANSWER/ABSTAIN/DENY/ERROR with citation validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-015	Vertical datum and CRS belong in terrain artifact manifests	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.9-13	The LiDAR section names horizontal reference, vertical datum, and acquisition quality; KFM terrain/DEM manifests should carry these fields.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-016	Terrain change products need bias and vintage validation	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.12-14	Temporal terrain analysis should validate acquisition vintage, datum, and vertical bias before comparison or publication.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Comparing vintages without datum or QA controls.
ML-061-017	NFHL has analytic vector endpoints and visualization WMS paths	L. WMS / WMTS / External Map Services	CONFIRMED evidence / NEW idea SRC-061 pp.14-20	The source distinguishes FEMA NFHL ArcGIS REST/FeatureServer style access for queries from WMS map rendering for visualization.	Use external services through source descriptors and access-role rules.	Service metadata, attribution, freshness, and analytics-not-visualization test. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-018	NFHL is a regulatory baseline, not a predictive flood model	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW idea SRC-061 pp.14-20	The NFHL section states that it is legally effective flood hazard data, not real-time inundation, climate projection, or hydraulic model output.	Keep renderer and tiles downstream of released evidence, policy, and proof objects.	Trust-membrane test: no public raw/canonical/model-only path. Publishing NFHL as observed flood extent or forecast.
ML-061-019	NFHL updates are localized and event-driven, not one national release date	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.16-19	The source says NFHL changes are tracked through VERSION_ID, EFFECTIVE_DATE, DFIRM_ID, and service metadata rather than a single release date.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-020	Flood hazard features carry regulatory attributes that must be preserved verbatim	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.15-19	Fields such as DFIRM_ID, VERSION_ID, EFFECTIVE_DATE, flood zones, and study references are material attributes for any public layer.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-021	NFHL WMS is visualization-only for KFM analytics purposes	L. WMS / WMTS / External Map Services	CONFIRMED evidence / NEW idea SRC-061 pp.17-18	The source says WMS is useful for rendering overlays but not suitable for analytics; analytical joins need vector/service or archived source data.	Use external services through source descriptors and access-role rules.	Service metadata, attribution, freshness, and analytics-not-visualization test. Using rasterized WMS pixels for regulatory analysis.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-022	NFHL vertical datum and units must be checked before engineering claims	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.18-19	The source highlights vertical datum mismatch as a major error source for BFE and compliance interpretation.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-023	NFHL automation should poll metadata and ingest changed geometries with provenance	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.19-20	The proposed workflow is to poll REST metadata, detect version changes, query changed geometries, preserve original attributes, and maintain audit lineage.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-024	NFHL export language must cite FEMA/reference status and avoid resale determinations	V. Exports, Screenshots, Reports, and Citation Preservation	CONFIRMED evidence / NEW idea SRC-061 pp.18-20	The source warns that regulatory decisions should reference FEMA directly and derivative products should be careful about resale-like determinations.	Keep exports/screenshots attached to citations, manifest, and version.	Export citation, watermark/version, and no-uncited-output tests. Uncited flood determination export.
ML-061-025	TDx road events are schema-validated GeoJSON FeatureCollections	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.20-26	The source describes TDx feeds as GeoJSON-native and schema-driven, with feature collections representing roadway incidents and restrictions.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-026	TDx road impact geometry should represent the affected road, not only a centroid	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.23-24	Allowed geometries are points, lines, and multilines according to impact scope; polygons are described as disallowed.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Centroid-only incident layers misleading users.
ML-061-027	TDx incidents require temporal scope and status transitions	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.22-24	Road incidents carry start/end time, expected end time, and active/planned/completed style state that map timelines must preserve.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-028	TDx restrictions and incidents need separate source and layer manifests	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.21-24	RoadRestrictionFeed and RoadIncidentFeed represent different event classes and should not be collapsed into one map truth layer.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-029	TDx validation belongs in CI before cataloging or map release	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.24-26	The source lists JSON syntax validation, TDx schema validation, enum correctness, time logic, and geometry rules as typical CI checks.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-030	TDx layers can become published transport PMTiles only after event normalization	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW idea SRC-061 pp.24-26	The feed is a data contract, not a transport mechanism; KFM can publish tiles only after normalized, governed event objects exist.	Publish static tiles as immutable, checksum-addressed artifacts.	Range/CORS/cache headers, checksum, and rollback validation. Treating external feed publication as KFM release.
ML-061-031	TDx can map to STAC, DCAT, PROV, and graph edges	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.25-26	The source identifies STAC Items, DCAT datasets, PROV lineage, and knowledge graph relations as natural integrations for TDx feeds.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-032	TDx/WZDx incident layers remain traveler context, not canonical emergency truth	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW idea SRC-061 pp.20-26	The source frames road-event data as interoperable transportation data; KFM should still avoid life-safety claims and use official source references.	Keep renderer and tiles downstream of released evidence, policy, and proof objects.	Trust-membrane test: no public raw/canonical/model-only path. Rendering incidents as definitive operational guidance.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-033	Geometry repair is a prerequisite for vector-tile generation	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW idea SRC-061 pp.26-32	The KGS surficial polygon playbook states that self-intersections, dangling rings, and bowties can cause area or attribute loss in MapLibre/Mapbox vector tiles.	Build MVT/MLT artifacts from validated processed data with manifest records.	Tile smoke test plus source-layer/attribute whitelist checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-034	Run deterministic make-valid before simplification or tiling	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.27-32	The source says topology should be repaired in source CRS and simplification should occur after validation, with heuristics used last.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Simplifying invalid geometry and then hiding defects in tiles.
ML-061-035	Never silently drop geometry during tile preparation	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW idea SRC-061 pp.27-32	The geometry playbook explicitly says to log, count, and fail CI on geometry changes beyond thresholds.	Record failure mode and deny or abstain path.	Regression fixture proves failure is denied, abstained, or quarantined. Silent feature loss in published PMTiles.
ML-061-036	Area-drift thresholds become a CI gate for polygon tiles	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.28-32	The SQL examples compute area drift before and after repair and fail if drift exceeds threshold.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-037	Force MultiPolygon, 2D, right-hand-rule geometry before style/tile handoff	E. Style Specification, Sources, Layers, Expressions	CONFIRMED evidence / NEW idea SRC-061 pp.28-31	The source normalizes types, forces 2D, and enforces polygon orientation to stabilize downstream tiling and styling.	Bind style/source/layer choices to checked style JSON and allowed attributes.	Style JSON, source-layer, filter, expression, and promoteld validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-038	Dissolve by intended tileset attribute before simplification	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW idea SRC-061 pp.31-32	The source advises dissolving by formation/class or equivalent tileset attribute before simplification to avoid micro-gaps.	Build MVT/MLT artifacts from validated processed data with manifest records.	Tile smoke test plus source-layer/attribute whitelist checks. Over-merging units when dissolve keys are wrong.
ML-061-039	Tippecanoe smoke tests can catch tiler surprises before release	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.31-32	The CI sketch runs Tippecanoe to /dev/null as a tile smoke test and treats warnings as potential blockers.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-040	Pin GEOS/GDAL/PostGIS tool versions for deterministic geometry repair	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW idea SRC-061 pp.31-32	The source says determinism requires pinned geometry tooling versions and conservative fallback heuristics.	Pin, review, and allowlist dependencies and tools.	SBOM, tool hash, plugin allowlist, OIDC, sandbox tests. Rebuilding the same tile with different geometry-tool behavior.
ML-061-041	H3 GeoParquet summaries use cell boundaries, not original soil polygons	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.33-38	The source defines the H3 geometry as the H3 cell boundary and stores soil aggregates as normalized cell-level attributes.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-042	Optional PMTiles/XYZ quickmaps are previews for H3 soil summaries	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW idea SRC-061 pp.33-34	The source allows optional PMTiles/XYZ tiles derived from centroids for quick maps while preserving GeoParquet as the data artifact.	Publish static tiles as immutable, checksum-addressed artifacts.	Range/CORS/cache headers, checksum, and rollback validation. Treating quickmap centroids as canonical geometry.
ML-061-043	H3 resolution and aggregation strategy must be versioned	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.33-39	The source recommends H3 resolutions and requires the chosen aggregation strategy to be recorded in provenance.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-044	Area-intersection H3 products can be canonical while centroid products are preview	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.35-39	The source says if both strategies are run, publish area_intersection as canonical and mark centroid outputs as role=preview.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Promoting faster preview as authoritative.
ML-061-045	Unit suffixes and null policy make GeoParquet fields validator-friendly	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.34-36	The schema encodes units in field names and forbids NaNs, using nulls instead.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-046	weights_checksum gives deterministic proof for H3 aggregation	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.35-38	The source uses a checksum over sorted contributing mapunit weights to support reproducible cell aggregation.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-047	H3 GeoParquet outputs require STAC, DCAT, and PROV records	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.36-37	The workflow emits STAC Item, DCAT Dataset, and PROV-O JSON-LD for each aggregated product.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-048	GeoParquet should carry CRS WKT2 and stable row grouping	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.36-37	The source specifies embedded CRS and target row group sizing for the GeoParquet materialization step.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Losing CRS or making non-reproducible partition choices.
ML-061-049	PurpleAir API keys must stay out of STAC/DCAT and source-visible outputs	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW idea SRC-061 pp.39-40	The source requires API keys to be passed by header and kept in secrets/environment variables, never committed or emitted in catalog metadata.	Pin, review, and allowlist dependencies and tools.	SBOM, tool hash, plugin allowlist, OIDC, sandbox tests. Leaking API keys through catalog or tile metadata.
ML-061-050	Historical sensor pulls need deterministic chunking and resumable checkpoints	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.40-45	Historical API ranges should be split into stable chunks with checkpoint state keyed by sensor, average, fields hash, and chunk end time.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-051	PM2.5 sensor fields have channel and variant semantics that must be preserved	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.43-65	The source warns that PurpleAir exposes multiple PM2.5 variants and channel behaviors; KFM must store which field and algorithm variant was used.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-052	Sensor sanity checks should detect channel divergence and impossible PM values	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.43-65	The source describes sanity checks for channel consistency and ranges before corrected PM2.5 is published.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-053	Air-source licensing and redistribution rules gate public layer publication	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.44-65	The source includes licensing/redistribution constraints and recommends source terms be designed into KFM publication rules.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Publishing proprietary sensor data without allowed use.
ML-061-054	Air quality products split realtime and historical STAC collections	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.44-60	The source proposes collections for realtime and history, with items by hourly partition or sensor chunk window.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-055	AQS/PurpleAir station-year or sensor-window products map to STAC, DCAT, and PROV	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.55-61	Air products can be cataloged with STAC Items, DCAT datasets, and PROV activities for fetch and normalization.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-056	PM2.5 calibration should expose trust and correction context, not raw sensor certainty	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW idea SRC-061 pp.39-65	The sensor calibration material supports showing correction/trust context rather than treating low-cost sensor output as direct truth.	Make Focus Mode evidence-bounded and citation-validated.	ANSWER/ABSTAIN/DENY/ERROR with citation validation. Raw sensor popup masquerading as calibrated evidence.
ML-061-057	Air observations can be map tooltips or aggregated daily surfaces	B. MapLibre GL JS Web Shell	CONFIRMED evidence / NEW idea SRC-061 pp.39-65	The source supports sensor-location tooltips and aggregated surfaces, but they must be source-aware and governed.	Expose only through governed shell and declarative layer registry.	Layer loads only from MapReleaseManifest and public-safe TileArtifactManifest. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-058	Stale air sources need visible UI state	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.39-65	Source cadence, retrieval windows, and sensor last_seen values imply visible stale-state chips or badges on air layers.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-059	Reference monitors outrank low-cost sensors for calibration claims	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW idea SRC-061 pp.55-65	The source frames AQS/regulatory-grade data as the backbone and lower-cost sensor networks as layered observations requiring calibration and context.	Keep renderer and tiles downstream of released evidence, policy, and proof objects.	Trust-membrane test: no public raw/canonical/model-only path. Equal-weighting all air sources in public claims.
ML-061-060	Focus Mode should abstain when air evidence lacks calibration or citation support	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW idea SRC-061 pp.39-65	Air-quality narrative must not answer beyond the available EvidenceBundle, calibration receipt, and citation validation.	Make Focus Mode evidence-bounded and citation-validated.	ANSWER/ABSTAIN/DENY/ERROR with citation validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-061	Idempotency keys prevent duplicate ingest side effects	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.71-83	The CI/idempotency section recommends checking a deterministic idempotency key before side effects and short-circuiting duplicate triggers.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-062	WAL intents and checkpoints make ingest recovery auditable	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.71-83	The source uses write-ahead intent records and checkpoint artifacts so crashes can resume without silent mutation.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-063	Kill switches and environment approval halt bad promotions	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.71-83	A bad rollout can be stopped by a kill switch or removing environment approval, keeping promotions from proceeding.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-064	CI tools need short-lived least-privilege OIDC credentials	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW idea SRC-061 pp.128-129	The agent tooling blueprint calls for OIDC-minted credentials bound to repo, ref, job, and audience with short TTL.	Pin, review, and allowlist dependencies and tools.	SBOM, tool hash, plugin allowlist, OIDC, sandbox tests. Broad long-lived credentials in map/tile workflows.
ML-061-065	Tool allowlists must pin command, args, image, and digest	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW idea SRC-061 pp.128-129	External tools run by CI agents should be explicitly allowlisted with hash/image pins and denied by default.	Pin, review, and allowlist dependencies and tools.	SBOM, tool hash, plugin allowlist, OIDC, sandbox tests. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-066	Sandboxed CI tool calls protect the map release pipeline	R. Plugin, Wrapper, and Dependency Governance	CONFIRMED evidence / NEW idea SRC-061 pp.128-129	The source proposes rootless OCI wrappers, read-only filesystems, capability drops, and outbound egress control for agent tool calls.	Pin, review, and allowlist dependencies and tools.	SBOM, tool hash, plugin allowlist, OIDC, sandbox tests. Tool call can mutate or exfiltrate without review.
ML-061-067	Canary runs compare SLIs before broad rollout	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.128-129	The source requires canary SLI comparison to baselines before allowing tool-driven changes to proceed.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-068	Fail-closed rollback follows missing attestations or SLI breaches	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.128-129	If SLIs breach or attestations are absent, the proposed flow aborts and auto-rolls back.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-069	PMTiles, COG, and GeoParquet can be distributed as OCI artifacts	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW idea SRC-061 pp.84-88	The source proposes using ORAS to ship single-file geospatial artifacts as digest-addressed OCI blobs.	Publish static tiles as immutable, checksum-addressed artifacts.	Range/CORS/cache headers, checksum, and rollback validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-070	OCI digest is canonical for catalog references	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW idea SRC-061 pp.84-87	The source says the digest, not the mutable tag, should be used everywhere in catalogs.	Publish static tiles as immutable, checksum-addressed artifacts.	Range/CORS/cache headers, checksum, and rollback validation. Cataloging tag URLs that can drift.
ML-061-071	Data-aware SBOMs describe artifact type, generator, CRS, bbox, schema, and checksums	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.86-87	The source extends SBOMs with a KFM dataProfile for PMTiles, COG, and GeoParquet artifacts.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-072	Cosign should sign artifact digests, not tags	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.86-88	The signing recipe explicitly prefers digest-pinned signing and verification.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-073	SBOMs attach as OCI referrers next to spatial artifacts	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.86-87	The source attaches SPDX and KFM SBOM files as referrers to the artifact digest.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-074	SLSA/Cosign attestations capture pipeline inputs, git commit, seeds, and telemetry	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.86-87	The source attaches a provenance predicate to the spatial artifact digest.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-075	Policy should require KFM signature and SBOM referrer before use	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.86-87	The verification path checks both cosign verification and ORAS referrer discovery before trusting the artifact.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Loading unsigned PMTiles in public UI.
ML-061-076	STAC PMTiles asset href can be an OCI digest URI	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.87-88	The catalog integration uses assets.pmtiles.href as an oci:// digest reference with checksum and provenance references.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-077	DCAT distribution accessURL can point to the OCI digest	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.87-88	The source maps OCI digest URLs into DCAT distribution access URLs with checksums and conformsTo profiles.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-078	PROV links signing identity to generation of the artifact	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.87-88	The source maps Cosign attestation to prov:wasAttributedTo and prov:qualifiedGeneration.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-079	OCI digest pins support registry and CDN pull-through caching	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.87-88	The source describes digest-pinned spatial artifacts as cache-friendly and globally distributable.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-080	SBOM dataProfile should include license, ethics, sovereignty, and sensitivity	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.87-88	The source requires license, ethics, sovereignty notes, and data sensitivity to travel with the artifact metadata.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-081	OpenLineage events should be translated to stable PROV-O semantics	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.88-91	The source describes OpenLineage as operational event data and PROV-O as the permanent semantic layer.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-082	Neo4j provenance graph stores lineage but does not replace source truth	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.88-91	The source frames Neo4j as persistence and traversal for provenance, not the artifact or source itself.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Graph projection becoming canonical truth.
ML-061-083	W3C PROV predicates should not be renamed	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.89-90	The mapping requires prov:used, prov:wasGeneratedBy, prov:wasDerivedFrom, prov:wasAssociatedWith, and prov:wasAttributedTo to retain their semantic meaning.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-084	URN canonical forms are required for lineage identity	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.90-106	The source defines canonical URN patterns for runs, plans, datasets, and artifact snapshots.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-085	Bitemporal properties belong on provenance entities and activities	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.90-106	The source references validFrom/validTo and activity start/end time as part of lineage sanity checks.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-086	Lineage sanity checks catch dangling entities without generator activities	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.178-179	The Neo4j checks include queries for entities with no producing Activity.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Downstream tiles unable to prove provenance origin.
ML-061-087	Derived entities whose source lacks a generator should fail audit checks	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.178-179	The source detects derived-from chains that lack a concrete producing step.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-088	Digest mismatches along derivation chains require review	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.179-180	The source checks same logical-key derivations where source and derived digests differ unexpectedly.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Silent artifact substitution.
ML-061-089	Evidence Drawer can surface lineage breaks rather than hiding them	N. Evidence Drawer Payloads and Click Resolution	CONFIRMED evidence / NEW idea SRC-061 pp.178-180	Lineage sanity results should become drawer-visible ABSTAIN/DENY reasons when a clicked feature lacks a valid provenance chain.	Resolve feature clicks to EvidenceBundle or abstain.	Click-to-EvidenceBundle positive/negative tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-090	Attestation badges should be backed by receipts and not visual trust theater	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.181-183	The attestation badge UI pattern should link to verification proof and expose unknown/state states rather than implying trust by decoration alone.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Badge as proof substitute.
ML-061-091	Timestamp normalization must be deterministic across AQS, OpenAQ, and PurpleAir	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.107-112	The timestamp guide requires stable parsing, timezone handling, and normalized UTC fields for air observations.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-092	Map timelines must distinguish source time, retrieval time, and release time	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.107-112	The source supports separate temporal attributes so UI time sliders do not collapse observation, ingest, and publication moments.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-093	Ambiguous or malformed timestamps should fail validation or enter quarantine	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.107-112	Time parsing ambiguity is a checkable data quality problem before public air or sensor layers are released.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-094	Stale-source UI state derives from normalized timing fields	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.107-112	The normalized timing model enables stale badges and Focus Mode abstention when observations are old or windows are missing.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-095	COGs make soil rasters web-fast but remain derived artifacts	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.123-127	The soil raster guide converts soil property rasters to Cloud Optimized GeoTIFFs for web map streaming and CI validation.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-096	Keep analysis CRS separate from web delivery CRS	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.123-124	The source recommends analysis copies in EPSG:5070 when needed while using EPSG:4326 or EPSG:3857 for web delivery.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Mixing analysis and display projections without manifest fields.
ML-061-097	COG block size, overviews, and compression are explicit performance controls	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.123-124	The guide uses internal tiling, overviews, ZSTD compression, and 512x512 blocks for predictable delivery.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-098	Nodata must stay consistent through overviews	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.124-127	The source warns to keep nodata consistent and propagate it to overviews.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. False raster values at overview zooms.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-099	STAC Raster and Projection extensions belong on COG Items	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.124-126	The guide names raster:bands, proj:epsg, proj:shape, transform, statistics, and asset roles for soil COG STAC Items.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-100	COG products can include PMTiles or JPEG quicklooks for first paint	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW idea SRC-061 pp.126-127	The source includes thumbnail and quicklook assets and recommends a PMTiles overview or tiny PNG/JPEG for tiled clients.	Publish static tiles as immutable, checksum-addressed artifacts.	Range/CORS/cache headers, checksum, and rollback validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-101	Object storage must support range requests for COG delivery	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.126-127	The guide recommends object storage with range requests enabled and immutable cache headers for versioned paths.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-102	COG validation uses gdalinfo, rio-cogeo, and stac-validator	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.124-127	The source provides validation snippets for block metadata, COG compliance, and STAC schema checks.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-103	COGs and STAC Items should be hashed and cosign-attested	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.126-127	The source signs COG outputs and attaches SLSA provenance tied to PROV run records.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-104	COG directory layout separates manifests, artifacts, STAC, and PROV	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.126-127	The proposed layout keeps source manifests, COG artifacts, STAC item, and PROV run file in distinct homes.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-105	COG map layers cannot stand in for soil source authority	A. Renderer Boundary and KFM Trust Law	CONFIRMED evidence / NEW idea SRC-061 pp.123-127	The source treats gSSURGO/gNATSGO and SoilGrids as ingest sources, with COGs as downstream streamable artifacts.	Keep renderer and tiles downstream of released evidence, policy, and proof objects.	Trust-membrane test: no public raw/canonical/model-only path. COG visual layer as proof of soil claim.
ML-061-106	Byte-for-byte or hash-for-hash deterministic COG re-runs are validation targets	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.124-127	The guide says identical parameters should rerun deterministically, with metadata timestamps handled explicitly.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-107	PM2.5 fusion products may use Zarr arrays before raster/COG publication	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	The fusion section references prior, ensemble, and smoothed Zarr stores for PM2.5 analysis before map products.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-108	Sensor fusion narratives must separate gridded priors from observations	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	The source frames PM2.5 fusion as combining reference monitors, low-cost sensors, and gridded priors; Focus Mode must identify those roles.	Make Focus Mode evidence-bounded and citation-validated.	ANSWER/ABSTAIN/DENY/ERROR with citation validation. Merged map surface hides source roles.
ML-061-109	Corrected PM2.5 surfaces require uncertainty outputs	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	Fusion products should carry ensemble size, smooth outputs, or uncertainty summaries rather than only a single corrected surface.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-110	Zarr outputs need STAC/PROV assets and lineage	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	The section uses Zarr entities in provenance and output paths, requiring cataloged array metadata before map use.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-111	Public air maps should show calibrated or aggregated products, not raw private sensor traces	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	The fusion flow implies public products should be corrected and policy-filtered, with sensitive or uncertain raw traces withheld.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Raw sensor trail exposes locations or false precision.
ML-061-112	PM2.5 fusion validation must compare against reference monitors and holdout windows	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	Fusion products need validation windows and reference comparisons before publication.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-113	Array chunking choices affect map-read performance	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	Zarr chunking and layout choices affect how quickly tiled or rasterized air surfaces can be generated.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-114	Hourly PM2.5 fusion requires time-window manifests	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.144-154	A time-aware air layer should record which hourly or daily window each fused surface represents.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-115	Sensor devices can be modeled as STAC-style Items with stable IDs	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.159-166	The device-items section describes stable device records that connect sensor identity, spatial location, and provenance maintenance.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-116	Device status changes create temporal evidence, not overwritten metadata	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.159-166	Sensor deployment, maintenance, and decommissioning should be represented as time-aware provenance state.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-117	Clicked sensor features should resolve to device item plus observation evidence	N. Evidence Drawer Payloads and Click Resolution	CONFIRMED evidence / NEW idea SRC-061 pp.159-166	Evidence Drawer payloads for sensor points need both the device metadata and the observation/calibration evidence.	Resolve feature clicks to EvidenceBundle or abstain.	Click-to-EvidenceBundle positive/negative tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-118	Decommissioned or offline sensors should render as stale or inactive	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.159-166	The device maintenance model implies visible inactive/stale states for map markers.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-119	Device lineage tests verify observation-to-device-to-source chains	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.159-166	Validation should prove each sensor observation can resolve to a valid device and source descriptor.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-120	FlatGeoBuf can be an efficient handoff format before MBTiles	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.168-171	The geology cleanup material suggests FlatGeoBuf as an output for speed before tile conversion.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-121	Production MVT pipeline standardizes KFM vector tile builds	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW idea SRC-061 pp.171-178	The MVT pattern covers Tippecanoe, Tegola/t_rex, MBTiles-to-PMTiles, incremental updates, and 3D fallbacks.	Build MVT/MLT artifacts from validated processed data with manifest records.	Tile smoke test plus source-layer/attribute whitelist checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-122	Tippecanoe pre-generated tiles are preferred for stable daily/weekly/monthly layers	H. Vector Tiles: MVT and MLT	CONFIRMED evidence / NEW idea SRC-061 pp.171-173	The source recommends pre-generating tiles when cadence is not minute-level and fixed cost/CDN throughput matter.	Build MVT/MLT artifacts from validated processed data with manifest records.	Tile smoke test plus source-layer/attribute whitelist checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-123	Dynamic MVT from PostGIS fits sub-hour updates or per-request filtering	J. Martin and Server-Mediated Tile Serving	CONFIRMED evidence / NEW idea SRC-061 pp.173-175	The source recommends Tegola/t_rex dynamic serving for high-frequency updates or request-specific filtering.	Use server-mediated tiles only when cadence/filtering/access justifies it.	Access-control, latency, cache-seed, and SQL/provider tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-124	Each tileset needs STAC Item, DCAT Dataset, and PROV lineage	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.172-173	The MVT pipeline requires catalog records and provenance files for each tileset artifact.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-125	GeoJSONL is preferred as streaming input to Tippecanoe	G. GeoJSON and Runtime Data	CONFIRMED evidence / NEW idea SRC-061 pp.172-173	The source says export GeoJSONL, one feature per line, for streaming-friendly tile builds.	Normalize runtime data before client exposure; avoid dense browser GeoJSON by default.	Schema, size, bounds, CRS, and stale-source validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-126	MBTiles should be converted to PMTiles for single-file CDN-friendly hosting	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	CONFIRMED evidence / NEW idea SRC-061 pp.173-174	The source converts MBTiles to PMTiles and hosts PMTiles on object storage plus CDN.	Publish static tiles as immutable, checksum-addressed artifacts.	Range/CORS/cache headers, checksum, and rollback validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-127	MapLibre uses a PMTiles protocol handler for static archive sources	B. MapLibre GL JS Web Shell	CONFIRMED evidence / NEW idea SRC-061 pp.173-177	The source says to register the PMTiles protocol and add vector sources/layers declaratively.	Expose only through governed shell and declarative layer registry.	Layer loads only from MapReleaseManifest and public-safe TileArtifactManifest. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-128	Governed API may expose signed tile URLs when needed	J. Martin and Server-Mediated Tile Serving	CONFIRMED evidence / NEW idea SRC-061 pp.173-175	The source notes that Node/API can expose signed URLs if direct PMTiles URLs require access mediation.	Use server-mediated tiles only when cadence/filtering/access justifies it.	Access-control, latency, cache-seed, and SQL/provider tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-129	tile-join supports small delta updates to existing MBTiles/PMTiles workflows	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.175-178	The MVT guide uses tile-join to merge delta tiles and preserve a tiny PROV delta record.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-130	Style JSON should keep labels sparse at low zooms	E. Style Specification, Sources, Layers, Expressions	CONFIRMED evidence / NEW idea SRC-061 pp.175-176	Label density rules keep low-zoom maps readable and avoid bloated vector attributes.	Bind style/source/layer choices to checked style JSON and allowed attributes.	Style JSON, source-layer, filter, expression, and promoteld validation. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-131	Centroid labels belong in style logic instead of duplicating polygon feature labels	E. Style Specification, Sources, Layers, Expressions	CONFIRMED evidence / NEW idea SRC-061 pp.175-176	The source recommends centroid labels for geology polygons via style JSON rather than per-feature tile duplication.	Bind style/source/layer choices to checked style JSON and allowed attributes.	Style JSON, source-layer, filter, expression, and promoted validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-132	Vector-tile attributes should be whitelisted to prevent PII leakage	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.176-177	The pipeline checklist says to ensure no PII lands in vector attributes and to whitelist attributes with Tippecanoe -y.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Sensitive attributes embedded in public PMTiles.
ML-061-133	Tile SLO target p95 fetch under 150 ms from CDN	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.176-177	The source gives a concrete SLO for tile fetch latency and mentions error-budget tracking.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-134	Tile build telemetry can include energy and carbon	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.176-177	The MVT guide logs build energy in kfm.telemetry.v4.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-135	PMTiles STAC asset carries digest, signed_by, and energy fields	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.176-177	The source provides a minimal STAC tileset asset with sha256, signed_by, and energy_kwh fields.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-136	Directory Rules route proposed tile files to data, web, docs, catalog, and prov roots	Y. Implementation Backlog and PR Plan	CONFIRMED evidence / NEW idea SRC-061 pp.177-178	The source proposed paths must be treated as PROPOSED and checked against Directory Rules responsibility roots before implementation.	Route proposed files through Directory Rules responsibility roots.	ADR/drift-register check for any proposed path. Creating a parallel tiles root without ADR.
ML-061-137	3D scenes can use 3D Tiles while MapLibre overlays labels and 2D vector UI	W. 3D / Cesium / Deck.gl / Overlay Interoperability	CONFIRMED evidence / NEW idea SRC-061 pp.175-178	The source names MapLibre overlay on Cesium canvas, extrusion to 3D Tiles, and raster/label fallbacks for Cesium.	Keep 3D conditional; maintain 2D MapLibre/evidence parity.	3D smoke, 2D fallback, synchronized camera/time, label fallback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-138	Attestation badge UI can expose artifact verification state	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.181-183	The source introduces a UI badge pattern for attested artifacts so users can see whether a layer or output has proof.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-139	Badge clicks should open proof details rather than replacing the Evidence Drawer	N. Evidence Drawer Payloads and Click Resolution	CONFIRMED evidence / NEW idea SRC-061 pp.181-183	Verification badges should link users to receipts, attestations, and drawer-level evidence rather than acting as a full citation surface.	Resolve feature clicks to EvidenceBundle or abstain.	Click-to-EvidenceBundle positive/negative tests. Badge used as evidence drawer substitute.
ML-061-140	Unknown, stale, or failed verification states need distinct visual treatment	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.181-183	The badge pattern implies finite trust-visible states for verified, stale, unknown, or failed verification.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-141	Exports and screenshots should preserve verification badge state and manifest ID	V. Exports, Screenshots, Reports, and Citation Preservation	CONFIRMED evidence / NEW idea SRC-061 pp.181-183	Any exported map using badges should preserve the linked manifest/proof ID and verification state.	Keep exports/screenshots attached to citations, manifest, and version.	Export citation, watermark/version, and no-uncited-output tests. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-142	SDM and ecosystem-service outputs are raster/NetCDF/COG-like derived layers	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.187-205	The ecological modeling sections repeatedly describe habitat suitability, ecosystem service, and biodiversity outputs as rasters or NetCDF products that need provenance.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-143	Remote sensing time-series artifacts require STAC Items, DCAT datasets, and PROV bundles	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.520-524	The remote-sensing section names STAC, DCAT, and PROV alignment as required for composites, trends, anomalies, and optional forecasts.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-144	COG is the primary raster asset class for remote-sensing time-series outputs	K. Raster, COG, DEM, Terrain, Hillshade	CONFIRMED evidence / NEW idea SRC-061 pp.520-524	The source lists cog as a primary raster asset along with QA summaries, config snapshots, and PROV bundle references.	Treat rasters/terrain as derived, versioned, cataloged artifacts.	COG/DEM/terrain validation plus STAC Raster/Projection checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-145	Config snapshots and QA summaries travel with remote-sensing artifacts	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.520-524	Time-series outputs must publish immutable config snapshots and QA summaries alongside data assets.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-146	Focus Mode may summarize remote-sensing products but cannot change governance posture	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW idea SRC-061 pp.523-524	The source states Focus Mode can provide summarization and navigation aids, but transforms cannot alter governance posture or invent provenance.	Make Focus Mode evidence-bounded and citation-validated.	ANSWER/ABSTAIN/DENY/ERR OR with citation validation. AI summary fabricates dataset relationships.
ML-061-147	Sovereignty and sensitivity labels apply before deriving or publishing outputs	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.524-525	The source says sensitivity labels must be applied before derivation/publication, and restricted data should publish only allowed aggregates.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-148	Restricted remote-sensing products require generalized spatial footprints	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.524-525	Restricted datasets should use coarse region/coarse H3 footprints and avoid coordinate summaries or how-to-locate instructions.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Remote-sensing layer reveals sensitive location.
ML-061-149	Remote-sensing validation includes checksums, STAC validation, CRS metadata, and provenance	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.527-528	The QA module validates checksum, STAC schema, projection metadata, license, and PROV annotations before release.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-150	Remote-sensing accuracy tests include classification, regression, temporal holdouts, and sensor consistency	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.527-528	The source lists confusion matrices, RMSE/R2, temporal holdouts, and sensor transition consistency metrics.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-151	Change-detection publication requires thresholded QA metrics	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.527-528	The source gives example acceptance criteria for land cover, NDVI/NDWI, change detection, and forecast products.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-152	Remote-sensing STAC Items are by spatiotemporal tile/window or aggregate region/window	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.520-524	The source frames time-series items as tile-by-time-window or region-window artifacts depending on policy.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-153	Narrative outputs must anchor to cataloged time-series artifacts and lineage	V. Exports, Screenshots, Reports, and Citation Preservation	CONFIRMED evidence / NEW idea SRC-061 pp.523-524	The source diagram feeds Focus Mode summaries from cataloged artifacts, PROV activity, source imagery, and graph nodes.	Keep exports/screenshots attached to citations, manifest, and version.	Export citation, watermark/version, and no-uncited-output tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-154	Overviews and chunking are performance controls for time-series raster products	T. Performance, Caching, CDN, Range Requests, and Resource Timing	CONFIRMED evidence / NEW idea SRC-061 pp.520-528	Remote-sensing outputs need chunking/overviews and performance-aware asset design for map-scale access.	Set measurable cache, range, CDN, and latency expectations.	Tile load budget, resource timing, cache invalidation record. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-155	Forecast products must be labeled as modeled outputs with uncertainty	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW idea SRC-061 pp.524-525	The source requires explicit modeled-output labeling and uncertainty summaries for optional forecast products.	Record failure mode and deny or abstain path.	Regression fixture proves failure is denied, abstained, or quarantined. Forecast map shown as observed fact.
ML-061-156	Archaeology/cultural landscape map layers use H3-generalized footprints	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.224-229	The source describes archaeological/cultural layers as spatially generalized, including H3 r7-r9 footprints.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-157	MapLibre and Cesium layers can show cultural corridors, geophysics overlays, and historical alignments	W. 3D / Cesium / Deck.gl / Overlay Interoperability	CONFIRMED evidence / NEW idea SRC-061 pp.228-229	The source lists MapLibre/Cesium layers for H3-generalized footprints, cultural corridors, geophysics anomalies, and historical alignments.	Keep 3D conditional; maintain 2D MapLibre/evidence parity.	3D smoke, 2D fallback, synchronized camera/time, label fallback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-158	Exact coordinates of sacred sites, burials, and restricted archives are prohibited without review	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.228-229	The source explicitly prohibits exact coordinates and restricted cultural archive exposure without review.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Precise archaeology location leakage.
ML-061-159	Any geometry below H3 r7 is prohibited for sensitive archaeology products	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	CONFIRMED evidence / NEW idea SRC-061 pp.228-229	The source lists any geometry below H3 r7 among content prohibited without review.	Fail closed on unclear rights, sensitive geometry, or cultural sovereignty.	Sensitive geometry deny and generalized-geometry allow tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-160	CARE labels and sovereignty notice chips are required in UI	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.228-229	The source requires CARE annotation, sovereignty notice chips, generalization logs, and consent or custodial review.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-161	Generalization logs are validation evidence for sensitive map products	U. Testing, CI, Validation, Rollback, and Proof Objects	CONFIRMED evidence / NEW idea SRC-061 pp.228-229	The source requires logs describing how geometry was generalized before publication.	Add fixture-driven tests and negative gates before release.	CI schema/policy/fixture tests including rollback replay. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-162	Focus Mode v3 must be sovereignty-aware and explain which evidence influenced insights	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW idea SRC-061 pp.228-229	Focus Mode for archaeology includes AI summaries with cultural sensitivity gating and explainability overlays.	Make Focus Mode evidence-bounded and citation-validated.	ANSWER/ABSTAIN/DENY/ERR OR with citation validation. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-163	Late Prehistoric cluster summaries must say zones are generalized, not precise sites	O. Focus Mode and Governed AI Map Context	CONFIRMED evidence / NEW idea SRC-061 pp.340-344	The source's Focus Summary example explains clusters as generalized cultural activity zones, not exact archaeological locations.	Make Focus Mode evidence-bounded and citation-validated.	ANSWER/ABSTAIN/DENY/ERR OR with citation validation. Narrative overprecision.

idea_id	idea title	category	status / source	normalized statement	implementation consequence	validation / risk
ML-061-164	Focus Mode archaeology panels show CARE labels, provenance badges, and generalization/uncertainty explanations	S. Accessibility, UX, and Trust-Visible States	CONFIRMED evidence / NEW idea SRC-061 pp.340-344	The source requires CARE labels, provenance-chain badges, and generalization/uncertainty explanations in Focus Mode.	Expose trust state visibly without substituting UI badges for evidence.	Keyboard, contrast, badge-state, and screen-reader checks. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-165	Archaeology map outputs support temporal-layer comparatives	P. Time-Aware Map Interaction and Timeline	CONFIRMED evidence / NEW idea SRC-061 pp.340-344	Late Prehistoric clusters contribute to temporal-layer comparatives across periods, requiring time-aware layer state.	Carry source time, valid time, retrieval time, release time, and version lock.	Timeline/version-lock fixture with stale and missing-time cases. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-166	Archaeology cluster outputs require PROV activity, input datasets, transform logs, WAL metadata, and version pinning	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	CONFIRMED evidence / NEW idea SRC-061 pp.340-344	The source lists provenance and lineage requirements for cluster outputs.	Extend manifest families without claiming repo presence.	Manifest schema, checksum, citation, release-state, and rollback tests. Risk: overclaiming implementation or treating a derivative artifact as proof.
ML-061-167	Archaeological anomaly or cluster layers must not be interpreted as precise site evidence	X. Anti-Patterns and Failure Modes	CONFIRMED evidence / NEW idea SRC-061 pp.224-229, 340-344	The source repeatedly distinguishes generalized outputs and cultural activity zones from exact sites.	Record failure mode and deny or abstain path.	Regression fixture proves failure is denied, abstained, or quarantined. User treats heatmap or cluster tile as a site locator.

6. Category Atlas

The table records whether each master category was expanded by SRC-061 or retained from SRC-MASTER-PREV. Counts are current-run delta records only.

Category	v1.7 delta count	Status / note
A. Renderer Boundary and KFM Trust Law	5	EXPANDED by SRC-061
B. MapLibre GL JS Web Shell	2	EXPANDED by SRC-061
C. MapLibre Native / Mobile / Platform Parity	0	RETAINED; no new current-run native/RS evidence.
D. MapLibre RS / Experimental Renderers	0	RETAINED; no new current-run native/RS evidence.
E. Style Specification, Sources, Layers, Expressions	3	EXPANDED by SRC-061
F. Sprites, Glyphs, Fonts, Design Tokens	0	RETAINED from SRC-MASTER-PREV
G. GeoJSON and Runtime Data	9	EXPANDED by SRC-061
H. Vector Tiles: MVT and MLT	4	EXPANDED by SRC-061
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	6	EXPANDED by SRC-061
J. Martin and Server-Mediated Tile Serving	2	EXPANDED by SRC-061
K. Raster, COG, DEM, Terrain, Hillshade	11	EXPANDED by SRC-061
L. WMS / WMTS / External Map Services	2	EXPANDED by SRC-061
M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	25	EXPANDED by SRC-061
N. Evidence Drawer Payloads and Click Resolution	3	EXPANDED by SRC-061
O. Focus Mode and Governed AI Map Context	7	EXPANDED by SRC-061
P. Time-Aware Map Interaction and Timeline	12	EXPANDED by SRC-061
Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	11	EXPANDED by SRC-061
R. Plugin, Wrapper, and Dependency Governance	6	EXPANDED by SRC-061
S. Accessibility, UX, and Trust-Visible States	8	EXPANDED by SRC-061
T. Performance, Caching, CDN, Range Requests, and Resource Timing	8	EXPANDED by SRC-061
U. Testing, CI, Validation, Rollback, and Proof Objects	33	EXPANDED by SRC-061
V. Exports, Screenshots, Reports, and Citation Preservation	3	EXPANDED by SRC-061
W. 3D / Cesium / Deck.gl / Overlay Interoperability	2	EXPANDED by SRC-061
X. Anti-Patterns and Failure Modes	4	EXPANDED by SRC-061
Y. Implementation Backlog and PR Plan	1	EXPANDED by SRC-061
Z. Open Questions and Verification Backlog	0	RETAINED from SRC-MASTER-PREV

7. Detailed Category Chapters

Each chapter distinguishes CONFIRMED evidence from SRC-061 and PROPOSED implementation consequences. Directory/path references remain PROPOSED unless verified later.

A. Renderer Boundary and KFM Trust Law

What changed since prior master: 5 new records expanded this category. Evidence basis includes pp.1-2, pp.123-127, pp.14-20, pp.20-26, pp.55-65.

Relevant ideas: ML-061-002 - Checksums do not prove origin or process; ML-061-018 - NFHL is a regulatory baseline, not a predictive flood model; ML-061-032 - TDx/WZDx incident layers remain traveler context, not canonical emergency truth; ML-061-059 - Reference monitors outrank low-cost sensors for calibration claims; ML-061-105 - COG map layers cannot stand in for soil source authority.

CONFIRMED: SRC-061 repeatedly separates source or artifact from authority: NFHL is regulatory baseline, not real-time flood model; COG/PMTiles are downstream artifacts; TDx is a data contract, not emergency truth; and reference-grade air data outranks raw low-cost sensor readings. PROPOSED: public map claims should flow through EvidenceBundle, PolicyDecision, PromotionDecision, release manifest, and citation validation before MapLibre renders them.

Required objects/contracts: use existing KFM manifest, evidence, policy, and validation families.

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Trust-membrane test: no public raw/canonical/model-only path.

Anti-patterns: COG visual layer as proof of soil claim., Equal-weighting all air sources in public claims., Publishing NFHL as observed flood extent or forecast., Rendering incidents as definitive operational guidance., Treating SHA-256 alone as publication proof..

B. MapLibre GL JS Web Shell

What changed since prior master: 2 new records expanded this category. Evidence basis includes pp.173-177, pp.39-65.

Relevant ideas: ML-061-057 - Air observations can be map tooltips or aggregated daily surfaces; ML-061-127 - MapLibre uses a PMTiles protocol handler for static archive sources.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: MapLibre GL JS, MapLibre popup governance

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Layer loads only from MapReleaseManifest and public-safe TileArtifactManifest.

Anti-patterns: overclaiming implementation or treating a derivative artifact as proof..

C. MapLibre Native / Mobile / Platform Parity

Status: RETAINED. No new SRC-061 extraction materially expanded this category in v1.7. Prior records remain preserved in Appendix A.

D. MapLibre RS / Experimental Renderers

Status: RETAINED. No new SRC-061 extraction materially expanded this category in v1.7. Prior records remain preserved in Appendix A.

E. Style Specification, Sources, Layers, Expressions

What changed since prior master: 3 new records expanded this category. Evidence basis includes pp.175-176, pp.28-31.

Relevant ideas: ML-061-037 - Force MultiPolygon, 2D, right-hand-rule geometry before style/tile handoff; ML-061-130 - Style JSON should keep labels sparse at low zooms; ML-061-131 - Centroid labels belong in style logic instead of duplicating polygon feature labels.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: LayerDefinition, StyleManifest

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Style JSON, source-layer, filter, expression, and promoted validation.

Anti-patterns: overclaiming implementation or treating a derivative artifact as proof..

F. Sprites, Glyphs, Fonts, Design Tokens

Status: RETAINED. No new SRC-061 extraction materially expanded this category in v1.7. Prior records remain preserved in Appendix A.

G. GeoJSON and Runtime Data

What changed since prior master: 9 new records expanded this category. Evidence basis includes pp.15-19, pp.168-171, pp.172-173, pp.20-26, pp.23-24, pp.33-38, pp.34-36, pp.36-37.

Relevant ideas: ML-061-020 - Flood hazard features carry regulatory attributes that must be preserved verbatim; ML-061-025 - TDx road events are schema-validated GeoJSON FeatureCollections; ML-061-026 - TDx road impact geometry should represent the affected road, not only a centroid; ML-061-041 - H3 GeoParquet summaries use cell boundaries, not original soil polygons; ML-061-045 - Unit suffixes and null policy make GeoParquet fields validator-friendly; ML-061-048 - GeoParquet should carry CRS WKT2 and stable row grouping; ML-061-051 - PM2.5 sensor fields have channel and variant semantics that must be preserved; ML-061-120 - FlatGeoBuf can be an efficient handoff format before MBTiles.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: GeoParquetAssetManifest, IncidentLayerManifest, LayerManifest, ProcessedArtifact, SchemaValidationReport, SensorCalibrationReceipt, SourceDescriptor

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Schema, size, bounds, CRS, and stale-source validation.

Anti-patterns: Centroid-only incident layers misleading users., overclaiming implementation or treating a derivative artifact as proof..

H. Vector Tiles: MVT and MLT

What changed since prior master: 4 new records expanded this category. Evidence basis includes pp.171-173, pp.171-178, pp.26-32, pp.31-32.

Relevant ideas: ML-061-033 - Geometry repair is a prerequisite for vector-tile generation; ML-061-038 - Dissolve by intended tileset attribute before simplification; ML-061-121 - Production MVT pipeline standardizes KFM vector tile builds; ML-061-122 - Tippecanoe pre-generated tiles are preferred for stable daily/weekly/monthly layers.

CONFIRMED: SRC-061 pp.171-178 supplies a production-aware MVT pattern: Tippecanoe for static pre-generation, Tegola/t_rex for high-frequency dynamic use, MBTiles-to-PMTiles conversion, tile-join deltas, STAC/DCAT/PROV per tileset, and MapLibre PMTiles source wiring. PROPOSED: the MVT pipeline should be encoded in TileArtifactManifest and MapReleaseManifest fixtures before public use.

Required objects/contracts: GeometryRepairReport, PMTiles, TileArtifactManifest

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Tile smoke test plus source-layer/attribute whitelist checks.

Anti-patterns: Over-merging units when dissolve keys are wrong., overclaiming implementation or treating a derivative artifact as proof..

I. PMTiles, MBTiles, Static Tiles, Serverless Delivery

What changed since prior master: 6 new records expanded this category. Evidence basis includes pp.126-127, pp.173-174, pp.24-26, pp.33-34, pp.84-87, pp.84-88.

Relevant ideas: ML-061-030 - TDx layers can become published transport PMTiles only after event normalization; ML-061-042 - Optional PMTiles/XYZ quickmaps are previews for H3 soil summaries; ML-061-069 - PMTiles, COG, and GeoParquet can be distributed as OCI artifacts; ML-061-070 - OCI digest is canonical for catalog references; ML-061-100 - COG products can include PMTiles or JPEG quicklooks for first paint; ML-061-126 - MBTiles should be converted to PMTiles for single-file CDN-friendly hosting.

CONFIRMED: SRC-061 pp.84-88 proposes PMTiles, COG, and GeoParquet as OCI artifacts with ORAS, Cosign, SBOM referencers, and digest-pinned catalog references. PROPOSED: KFM tile/style consumers should verify artifact digest, signature, SBOM, and release state before addSource or public delivery.

Required objects/contracts: OCIArtifactRef, PMTiles, QuicklookAsset

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Range/CORS/cache headers, checksum, and rollback validation.

Anti-patterns: Cataloging tag URLs that can drift., Treating external feed publication as KFM release., Treating quickmap centroids as canonical geometry., overclaiming implementation or treating a derivative artifact as proof..

J. Martin and Server-Mediated Tile Serving

What changed since prior master: 2 new records expanded this category. Evidence basis includes pp.173-175.

Relevant ideas: ML-061-123 - Dynamic MVT from PostGIS fits sub-hour updates or per-request filtering; ML-061-128 - Governed API may expose signed tile URLs when needed.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: Layer registry endpoint, Martin and server-mediated serving

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Access-control, latency, cache-seed, and SQL/provider tests.

Anti-patterns: overclaiming implementation or treating a derivative artifact as proof..

K. Raster, COG, DEM, Terrain, Hillshade

What changed since prior master: 11 new records expanded this category. Evidence basis includes pp.11-12, pp.123-124, pp.123-127, pp.124-127, pp.144-154, pp.18-19, pp.187-205, pp.520-524.

Relevant ideas: ML-061-009 - LAZ remains raw elevation source while EPT/COPC serve visualization and analytics; ML-061-010 - USGS 1 m DEM is a canonical raster surface but not a complete point-cloud substitute; ML-061-015 - Vertical datum and CRS belong in terrain artifact manifests; ML-061-022 - NFHL vertical datum and units must be checked before engineering claims; ML-061-095 - COGs make soil rasters web-fast but remain derived artifacts; ML-061-096 - Keep analysis CRS separate from web delivery CRS; ML-061-098 - Nodata must stay consistent through overviews; ML-061-107 - PM2.5 fusion products may use Zarr arrays before raster/COG publication.

CONFIRMED: SRC-061 pp.123-127 adds COG/STAC raster rules, while pp.8-14 adds LiDAR/DEM/EPT/COPC terrain distinctions and pp.520-528 adds remote-sensing QA. PROPOSED: raster and terrain releases should carry COGValidationReport, TerrainTilesetManifest, RasterAssetManifest, and time-window metadata.

Required objects/contracts: ArrayAssetManifest, COGArtifactManifest, Elevation/DEM manifests, RasterAssetManifest, TerrainTilesetManifest, UncertaintyRaster

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: COG/DEM/terrain validation plus STAC Raster/Projection checks.

Anti-patterns: Using a DEM as proof of buildings, vegetation, or powerlines., overclaiming implementation or treating a derivative artifact as proof..

L. WMS / WMTS / External Map Services

What changed since prior master: 2 new records expanded this category. Evidence basis includes pp.14-20, pp.17-18.

Relevant ideas: ML-061-017 - NFHL has analytic vector endpoints and visualization WMS paths; ML-061-021 - NFHL WMS is visualization-only for KFM analytics purposes.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: ExternalServiceDescriptor

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Service metadata, attribution, freshness, and analytics-not-visualization test.

Anti-patterns: Using rasterized WMS pixels for regulatory analysis., overclaiming implementation or treating a derivative artifact as proof..

M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest

What changed since prior master: 25 new records expanded this category. Evidence basis includes pp.1-2, pp.1-8, pp.10-11, pp.124-126, pp.126-127, pp.144-154, pp.159-166, pp.172-173.

Relevant ideas: ML-061-001 - SLSA-grade attestations for PMTiles, COG, and GeoParquet; ML-061-003 - Artifact is evidence; metadata is sovereign; ML-061-011 - Point clouds and terrain derivatives require STAC pointcloud/catalog records; ML-061-028 - TDx restrictions and incidents need separate source and layer manifests; ML-061-031 - TDx can map to STAC, DCAT, PROV, and graph edges; ML-061-047 - H3 GeoParquet outputs require STAC, DCAT, and PROV records; ML-061-054 - Air quality products split realtime and historical STAC collections; ML-061-055 - AQS/PurpleAir station-year or sensor-window products map to STAC, DCAT, and PROV.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: ArrayAssetManifest, DCAT Dataset, DeviceItemManifest, Directory Rules, EvidenceBundle, EvidenceBundle, SourceDescriptor, IncidentEvent, EvidenceBundle, LayerManifest, MapReleaseManifest, PROV-O, RemoteSensingTimeSeriesItem, RunReceipt

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Manifest schema, checksum, citation, release-state, and rollback tests.

Anti-patterns: overclaiming implementation or treating a derivative artifact as proof..

N. Evidence Drawer Payloads and Click Resolution

What changed since prior master: 3 new records expanded this category. Evidence basis includes pp.159-166, pp.178-180, pp.181-183.

Relevant ideas: ML-061-089 - Evidence Drawer can surface lineage breaks rather than hiding them; ML-061-117 - Clicked sensor features should resolve to device item plus observation evidence; ML-061-139 - Badge clicks should open proof details rather than replacing the Evidence Drawer.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: EvidenceDrawerPayload

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Click-to-EvidenceBundle positive/negative tests.

Anti-patterns: Badge used as evidence drawer substitute., overclaiming implementation or treating a derivative artifact as proof..

O. Focus Mode and Governed AI Map Context

What changed since prior master: 7 new records expanded this category. Evidence basis includes pp.13-14, pp.144-154, pp.228-229, pp.340-344, pp.39-65, pp.523-524.

Relevant ideas: ML-061-014 - Terrain narratives can support Story Nodes but do not replace evidence; ML-061-056 - PM2.5 calibration should expose trust and correction context, not raw sensor certainty; ML-061-060 - Focus Mode should abstain when air evidence lacks calibration or citation support; ML-061-108 - Sensor fusion narratives must separate gridded priors from observations; ML-061-146 - Focus Mode may summarize remote-sensing products but cannot change governance posture; ML-061-162 - Focus Mode v3 must be sovereignty-aware and explain which evidence influenced insights; ML-061-163 - Late Prehistoric cluster summaries must say zones are generalized, not precise sites.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: FocusModeRequest, FocusModeResponse, StoryNodeMetadata

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: ANSWER/ABSTAIN/DENY/ERROR with citation validation.

Anti-patterns: AI summary fabricates dataset relationships., Merged map surface hides source roles., Raw sensor popup masquerading as calibrated evidence., overclaiming implementation or treating a derivative artifact as proof..

P. Time-Aware Map Interaction and Timeline

What changed since prior master: 12 new records expanded this category. Evidence basis includes pp.107-112, pp.12-13, pp.144-154, pp.159-166, pp.16-19, pp.22-24, pp.33-39, pp.340-344.

Relevant ideas: ML-061-012 - LiDAR acquisition year is a first-class temporal attribute; ML-061-019 - NFHL updates are localized and event-driven, not one national release date; ML-061-027 - TDx incidents require temporal scope and status transitions; ML-061-043 - H3 resolution and aggregation strategy must be versioned; ML-061-050 - Historical sensor pulls need deterministic chunking and resumable checkpoints; ML-061-085 - Bitemporal properties belong on provenance entities and activities; ML-061-091 - Timestamp normalization must be deterministic across AQS, OpenAQ, and PurpleAir; ML-061-092 - Map timelines must distinguish source time, retrieval time, and release time.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: MapContextEnvelope, MapReleaseManifest, RunReceipt, Temporal database doctrine, TemporalNormalizationReport, Timeline, Timeline, MapContextEnvelope

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Timeline/version-lock fixture with stale and missing-time cases.

Anti-patterns: overclaiming implementation or treating a derivative artifact as proof..

Q. Sensitive Geometry, Geoprivacy, Rights, and Policy

What changed since prior master: 11 new records expanded this category. Evidence basis includes pp.11-12, pp.144-154, pp.176-177, pp.224-229, pp.228-229, pp.35-39, pp.44-65, pp.524-525.

Relevant ideas: ML-061-013 - State LiDAR mirrors require explicit license assertions; ML-061-044 - Area-intersection H3 products can be canonical while centroid products are preview; ML-061-053 - Air-source licensing and redistribution rules gate public layer publication; ML-061-080 - SBOM dataProfile should include license, ethics, sovereignty, and sensitivity; ML-061-111 - Public air maps should show calibrated or aggregated products, not raw private sensor traces; ML-061-132 - Vector-tile attributes should be whitelisted to prevent PII leakage; ML-061-147 - Sovereignty and sensitivity labels apply before deriving or publishing outputs; ML-061-148 - Restricted remote-sensing products require generalized spatial footprints.

CONFIRMED: SRC-061 pp.224-229 and pp.340-344 require H3-generalized footprints, CARE labels, sovereignty chips, generalization logs, and prohibitions on exact sacred/burial-site coordinates. PROPOSED: sensitive geometry deny fixtures should be required before any archaeology or culturally sensitive layer appears in MapLibre, Cesium, Focus Mode, exports, or story nodes.

Required objects/contracts: PolicyDecision, SensitiveGeometryTransform

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Sensitive geometry deny and generalized-geometry allow tests.

Anti-patterns: Assuming all mirrors inherit federal public domain rights., Promoting faster preview as authoritative., Publishing proprietary sensor data without allowed use., Raw sensor trail exposes locations or false precision., overclaiming implementation or treating a derivative artifact as proof..

R. Plugin, Wrapper, and Dependency Governance

What changed since prior master: 6 new records expanded this category. Evidence basis includes pp.128-129, pp.3-4, pp.31-32, pp.39-40.

Relevant ideas: ML-061-005 - Keyless CI signing binds workflow identity to artifact; ML-061-040 - Pin GEOS/GDAL/PostGIS tool versions for deterministic geometry repair; ML-061-049 - PurpleAir API keys must stay out of STAC/DCAT and source-visible outputs; ML-061-064 - CI tools need short-lived least-privilege OIDC credentials; ML-061-065 - Tool allowlists must pin command, args, image, and digest; ML-061-066 - Sandboxed CI tool calls protect the map release pipeline.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: Plugin allowlist

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: SBOM, tool hash, plugin allowlist, OIDC, sandbox tests.

Anti-patterns: Broad long-lived credentials in map/tile workflows., Leaking API keys through catalog or tile metadata., Long-lived keys or manual signing can weaken auditability., Rebuilding the same tile with different geometry-tool behavior., overclaiming implementation or treating a derivative artifact as proof..

S. Accessibility, UX, and Trust-Visible States

What changed since prior master: 8 new records expanded this category. Evidence basis includes pp.107-112, pp.159-166, pp.181-183, pp.228-229, pp.340-344, pp.39-65.

Relevant ideas: ML-061-058 - Stale air sources need visible UI state; ML-061-090 - Attestation badges should be backed by receipts and not visual trust theater; ML-061-094 - Stale-source UI state derives from normalized timing fields; ML-061-118 - Decommissioned or offline sensors should render as stale or inactive; ML-061-138 - Attestation badge UI can expose artifact verification state; ML-061-140 - Unknown, stale, or failed verification states need distinct visual treatment; ML-061-160 - CARE labels and sovereignty notice chips are required in UI; ML-061-164 - Focus Mode archaeology panels show CARE labels, provenance badges, and generalization/uncertainty explanations.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: EvidenceDrawerPayload, StaleSourceFixture, TrustVisibleState, VerifyReceipt

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Keyboard, contrast, badge-state, and screen-reader checks.

Anti-patterns: Badge as proof substitute., overclaiming implementation or treating a derivative artifact as proof..

T. Performance, Caching, CDN, Range Requests, and Resource Timing

What changed since prior master: 8 new records expanded this category. Evidence basis includes pp.123-124, pp.126-127, pp.128-129, pp.144-154, pp.176-177, pp.520-528, pp.87-88.

Relevant ideas: ML-061-067 - Canary runs compare SLIs before broad rollout; ML-061-079 - OCI digest pins support registry and CDN pull-through caching; ML-061-097 - COG block size, overviews, and compression are explicit performance controls; ML-061-101 - Object storage must support range requests for COG delivery; ML-061-113 - Array chunking choices affect map-read performance; ML-061-133 - Tile SLO target p95 fetch under 150 ms from CDN; ML-061-134 - Tile build telemetry can include energy and carbon; ML-061-154 - Overviews and chunking are performance controls for time-series raster products.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: CacheInvalidationRecord, EnergyReceipt, HTTP Range, CDN, PerformanceBudget, ResourceTimingReport

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Tile load budget, resource timing, cache invalidation record.

Anti-patterns: overclaiming implementation or treating a derivative artifact as proof..

U. Testing, CI, Validation, Rollback, and Proof Objects

What changed since prior master: 33 new records expanded this category. Evidence basis includes pp.107-112, pp.12-14, pp.124-127, pp.126-127, pp.128-129, pp.144-154, pp.159-166, pp.175-178.

Relevant ideas: ML-061-004 - Blob signature plus DSSE attestation are distinct checks; ML-061-006 - Offline verification bundles support forensic review; ML-061-007 - CUE validates structure; OPA/Rego validates trust logic; ML-061-016 - Terrain change products need bias and vintage validation; ML-061-023 - NFHL automation should poll metadata and ingest changed geometries with provenance; ML-061-029 - TDx validation belongs in CI before cataloging or map release; ML-061-034 - Run deterministic make-valid before simplification or tiling; ML-061-036 - Area-drift thresholds become a CI gate for polygon tiles.

CONFIRMED: SRC-061 supplies multiple validation gates: geometry repair, area drift, Tippecanoe smoke tests, COG validation, STAC/DCAT/PROV checks, lineage sanity queries, timestamp normalization, remote-sensing QA, idempotent WAL/checkpoint recovery, and fail-closed rollback. PROPOSED: v1.7 backlog should prioritize no-network fixtures for these gates.

Required objects/contracts: AIReceipt, RunReceipt, COGValidationReport, CitationValidationReport, PolicyDecision, DeterministicIdentity, GeometryRepairReport, LineageValidationReport, PromotionDecision, ProofPack, ProofPack, ReleaseManifest, QuarantineReceipt, RedactionReceipt, RemoteSensingValidationReport

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: CI schema/policy/fixture tests including rollback replay.

Anti-patterns: Comparing vintages without datum or QA controls., overclaiming implementation or treating a derivative artifact as proof..

V. Exports, Screenshots, Reports, and Citation Preservation

What changed since prior master: 3 new records expanded this category. Evidence basis includes pp.18-20, pp.181-183, pp.523-524.

Relevant ideas: ML-061-024 - NFHL export language must cite FEMA/reference status and avoid resale determinations; ML-061-141 - Exports and screenshots should preserve verification badge state and manifest ID; ML-061-153 - Narrative outputs must anchor to cataloged time-series artifacts and lineage.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: ExportCitationManifest, StoryNode

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Export citation, watermark/version, and no-uncited-output tests.

Anti-patterns: Uncited flood determination export., overclaiming implementation or treating a derivative artifact as proof..

W. 3D / Cesium / Deck.gl / Overlay Interoperability

What changed since prior master: 2 new records expanded this category. Evidence basis includes pp.175-178, pp.228-229.

Relevant ideas: ML-061-137 - 3D scenes can use 3D Tiles while MapLibre overlays labels and 2D vector UI; ML-061-157 - MapLibre and Cesium layers can show cultural corridors, geophysics overlays, and historical alignments.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: 3DSceneManifest

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: 3D smoke, 2D fallback, synchronized camera/time, label fallback tests.

Anti-patterns: overclaiming implementation or treating a derivative artifact as proof..

X. Anti-Patterns and Failure Modes

What changed since prior master: 4 new records expanded this category. Evidence basis includes pp.224-229, 340-344, pp.27-32, pp.524-525, pp.6-7.

Relevant ideas: ML-061-008 - Fail-closed promotion rejects invalid signature, identity, predicate, or policy; ML-061-035 - Never silently drop geometry during tile preparation; ML-061-155 - Forecast products must be labeled as modeled outputs with uncertainty; ML-061-167 - Archaeological anomaly or cluster layers must not be interpreted as precise site evidence.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: use existing KFM manifest, evidence, policy, and validation families.

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: Regression fixture proves failure is denied, abstained, or quarantined.

Anti-patterns: Forecast map shown as observed fact., Manual override that promotes unverified tiles or COGs., Silent feature loss in published PMTiles., User treats heatmap or cluster tile as a site locator..

Y. Implementation Backlog and PR Plan

What changed since prior master: 1 new records expanded this category. Evidence basis includes pp.177-178.

Relevant ideas: ML-061-136 - Directory Rules route proposed tile files to data, web, docs, catalog, and prov roots.

CONFIRMED: The source material supports the listed ideas at the cited pages. PROPOSED: implementation should bind these ideas to existing KFM object families where possible and avoid creating new parallel homes without ADR or drift-register review.

Required objects/contracts: use existing KFM manifest, evidence, policy, and validation families.

Public UI implications: MapLibre and related UI surfaces must show released, policy-safe, versioned, and citation-capable state; popups and badges cannot substitute for the Evidence Drawer.

Policy implications: fail closed for rights uncertainty, unverified signatures, missing provenance, stale source state, sensitive geometry, and incomplete citation support.

Validation tests: ADR/drift-register check for any proposed path.

Anti-patterns: Creating a parallel tiles root without ADR..

Z. Open Questions and Verification Backlog

Status: RETAINED. No new SRC-061 extraction materially expanded this category in v1.7. Prior records remain preserved in Appendix A.

8. Components-Functions-Features Matrix

component	function	feature	input artifacts	output artifacts	governed API / evidence / policy dependency	tile/style dependency	runtime/UI behavior	tests required	status
MapLibre GL JS	2D renderer and interaction runtime	Render released sources/layers, PMTiles protocol sources, click candidates	Style JSON, MapReleaseManifest, TileArtifactManifest	Map viewport, feature candidate events	Governed API, EvidenceBundle, PolicyDecision	PMTiles/MVT/COG/style/glyphs	addSource/addLayer only after manifest and verification	style validation, no unreleased source, click-to-drawer	EXPANDED
MapLibre Native	Mobile/native renderer	Potential mobile parity for released layers	Same as GL JS plus platform adapter	Native map views	Evidence Drawer parity required	MVT/PMTiles/COG support NEEDS VERIFICATION	Do not release until parity proof	mobile parity tests	RETAINED
MapLibre RS	Experimental renderer	Future renderer experiment	Experimental build artifacts	Rendered map output	Same trust membrane	Unknown tile support	Experimental only	fallback proof	RETAINED
Style JSON	Declarative map style	Source/layer references, label density, filters	LayerManifest, symbols, glyphs	Style artifact	StyleManifest and policy allowlist	source-layer, expressions, filters	Sparse labels, no sensitive hidden-only-by-style	style lint, source-layer tests	EXPANDED
source definition	Declare data source	vector, raster, geojson, raster-dem, PMTiles protocol	TileArtifactManifest, source descriptor	Map source object	ReleaseManifest and evidence dependencies	MVT/PMTiles/COG/GeoJSON	No RAW/candidate URL	source validation	EXPANDED
layer definition	Render source data	Fill/line/symbol/raster layers	Style JSON, LayerManifest	Layer visualization	PolicyDecision and review state	source-layer, paint/layout	Trust badges and stale state	layer manifest tests	EXPANDED
expressions	Data-driven styling	Classify features, labels, filters	Style JSON, attributes	Rendered classes	Attribute whitelist and citation support	style spec	No source role hidden in color only	style expression tests	EXPANDED
filters	Feature selection	Public-safe filtering	LayerManifest, PolicyDecision	Filtered layer output	PolicyDecision required	style filter	Style filters cannot hide exact sensitive geometry alone	deny fixture	EXPANDED
sprites	Symbol sheet	Icons and UI symbols	Sprite manifest	Sprite images	Design token review	sprite URL	Missing icons visible as error	sprite path tests	RETAINED
glyphs	Font glyphs	Text rendering and labels	Glyph manifest, font allowlist	Glyph PBF/assets	Accessibility and license review	glyph URL	Consistent glyphs; fallback fonts	glyph availability tests	EXPANDED
MVT	Vector tile encoding	Scalable vector rendering	Processed GeoJSONL /FlatGeoBuf/PostGIS	MBTiles/PMTiles/MVT endpoints	TileArtifactManifest, STAC/DCAT/PROV	source-layer, attributes	Feature candidates only, not proof	tile smoke, attribute whitelist	EXPANDED
MLT	MapLibre Tile format	Potential next-gen vector tiles	MLT tiles	MLT endpoints	NEEDS VERIFICATION	encoding=mlt	Pilot only	end-to-end toolchain tests	RETAINED
PMTiles	Single-file vector/raster tile archive	Static CDN-friendly tiles	MBTiles, tile build outputs	pmtiles artifact	ReleaseManifest, OCI/SBOM/signature	PMTiles protocol	addSource after verification	range/CORS/cache/checksum	EXPANDED
MBTiles	Intermediate tile package	Build and tile-join deltas	Tippecanoe output	PMTiles conversion	PROV delta record	MVT tiles	Not public default unless justified	conversion tests	EXPANDED
COG	Cloud-optimized raster	Web-streamed rasters/terrain/soil	GeoTIFF/raster outputs	COG + STAC item	COGValidationReport, STAC Raster/Projection	raster source/COG protocol	Range-enabled versioned delivery	rio cogeotools, stac-validator	EXPANDED
Martin	Server-mediated tile serving	Dynamic tile or PMTiles/MBTiles serving	PostGIS/PMTiles	Tile endpoints	Access and policy gates	vector/raster source	Use when dynamic/steward access needed	latency/access tests	RETAINED
Tegola/t_rex	Dynamic MVT service	High-frequency or per-request MVT	PostGIS tables	MVT endpoints	Policy + access + SQL bounds	vector source	Dynamic only when justified	provider SQL tests	NEW/EXPANDED

component	function	feature	input artifacts	output artifacts	governed API / evidence / policy dependency	tile/style dependency	runtime/UI behavior	tests required	status
Maputnik	Style editor	Style authoring/review	Style JSON	Edited style JSON	Plugin/tool review	style spec	Not production authority	style regression tests	RETAINED
layer catalog	Published layer listing	Discover released layers	LayerManifest, MapReleaseManifest	Layer list	Governed API	source/style deps	Shows stale/review/release state	catalog closure tests	EXPANDED
Evidence Drawer	Trust details surface	Resolve clicked feature to evidence	Feature candidate, EvidenceBundle	Drawer payload	EvidenceBundle, CitationValidationReport	Rendered feature candidate	ANSWER/ABSTAIN/DENY/ERROR visible	click-to-drawer tests	EXPANDED
Focus Mode	Evidence-bounded synthesis	Map context, Story Nodes, AI answer/abstain	MapContextEnvelope, EvidenceBundle	Focus answer/envelope	Policy/citation validation, AIReceipt	Layer/time/version context	No direct model client	answer/abstain tests	EXPANDED
timeline / time slider	Temporal map control	Version/time filtering	source/valid/retrieval/release time	Temporal viewport state	MapContextEnvelope	time-indexed layers	Version lock/stale state	timeline fixtures	EXPANDED
MapContextEnvelope	Bound map state	Camera, layers, time, selection, release	UI state + manifests	Focus request context	Evidence and policy dependencies	layer/style/tile refs	Carries bounded context to API	schema tests	RETAINED
LayerManifest	Layer contract	Source, style, evidence, sensitivity	SourceDescriptor, TileArtifact	Layer definition	Policy/evidence/release deps	source-layer/style	Public-safe layer controls	manifest tests	EXPANDED
StyleManifest	Style contract	Style JSON, sprites, glyphs, design tokens	Style assets	Versioned style release	Design/policy review	style JSON	Prevents uncontrolled style drift	visual regression	EXPANDED
TileArtifactManifest	Tile artifact contract	Digest, source, tileset, bounds, schema, signatures	PMTiles/COG/MVT/MBTiles	Tile proof record	Evidence/proof/policy deps	MVT/PMTiles/COG	Verify before render	checksum/signature tests	EXPANDED
MapReleaseManifest	Map release state	Published layer bundle and rollback target	Layer/style/tile manifests	Release bundle	PromotionDecision	all tile/style deps	Only released artifacts render	rollback tests	EXPANDED
PolicyDecision	Allow/deny/restrict/abstain	Record policy outcome	source rights/sensitivity context	Decision record	Policy engine	all exposed layers	Trust-visible outcome	policy fixtures	EXPANDED
CitationValidationReport	Citation gate	Validate evidence/citation support	EvidenceBundle, claims	Pass/fail report	Evidence dependency	popups/exports/focus	Uncited output abstains	citation tests	RETAINED
AIReceipt	AI runtime receipt	Model/provider/prompt/context audit	Focus request/response	Receipt	Policy/citation deps	map context	No chain-of-thought as truth	receipt tests	RETAINED
rollback target	Recover prior release	Prior manifest and cache state	ReleaseManifest	Rollback card	PromotionDecision	all artifacts	Restore previous map release	rollback replay	EXPANDED
cache invalidation record	Control cache freshness	Invalidate CDN/tiles after release/rollback	Artifact digests, cache keys	Cache receipt	Release and policy	PMTiles/COG/CDN	Avoid stale public artifacts	cache invalidation tests	EXPANDED
OCI spatial artifact	Digest-pinned artifact channel	PMTiles/COG/GeoParquet in registry	ORAS push, SBOM, attestation	oci:// digest URI	Proof/policy required	PMTiles/COG	Verify before pull/use	cosign/oras tests	NEW
GeoParquet H3 summary	Analytic/vector table	H3 cell summaries with deterministic weights	SSURGO/gNATSGO inputs	GeoParquet, optional PMTiles preview	STAC/DCAT/PROV	quickmap PMTiles	Public preview vs canonical data	schema/determinism tests	NEW
IncidentLayerManifest	Road event layer control	TDx/WZDx/ArcGIS/GTFS incident fusion	GeoJSON feeds	Incident layer	source role/policy	MVT/PMTiles	Context, not emergency truth	time/schema/conflict tests	EXPANDED

9. Tile Strategy Chapter

The comparisons below combine retained v1.6 doctrine with v1.7 evidence from SRC-061. Each choice remains subordinate to EvidenceBundle resolution, policy decisions, release manifests, and rollback.

Comparison	When it wins	When it fails	KFM trust consequence	Release / rollback consequence	Validation needed
GeoJSON vs vector tiles	GeoJSON wins for small dynamic overlays, editing, and raw event previews. Vector tiles win for large layers and stable public products.	Dense GeoJSON fails in browser performance; vector tiles fail when attributes are not whitelisted or provenance is absent.	GeoJSON runtime data and MVT are both carriers, not proof.	GeoJSON preview may stay WORK; vector tiles require TileArtifactManifest and release state.	CRS, size, schema, source role, tile smoke.
MVT vs MLT	MVT wins now due mature toolchain and SRC-061 production pipeline; MLT remains pilot/NEEDS VERIFICATION.	MVT can be larger/slower; MLT fails without validated end-to-end tooling.	Do not default MLT until evidence and toolchain proof exist.	MVT releases use PMTiles/MBTiles manifests; MLT requires separate pilot manifests.	Encoding, renderer support, source-layer, performance tests.
PMTiles vs MBTiles	PMTiles wins for single-file CDN-friendly static public release; MBTiles is useful as Tippecanoe build intermediate and tile-join target.	PMTiles fails if Range/CORS/cache headers or protocol adapter fail; MBTiles is not ideal browser delivery.	Neither proves source claims without EvidenceBundle and manifest.	MBTiles -> PMTiles conversion must create digest and rollback target.	PMTiles header/range, checksum, signature, style source.
Static tiles vs dynamic tile server	Static pre-generated PMTiles wins for weekly/daily/monthly static layers; dynamic Tegola/T_rex wins for sub-hour updates or per-request filters.	Static fails with high churn; dynamic fails when ops/access/SQL/cache are not governed.	Dynamic servers must not expose canonical stores directly to public clients.	Static release has immutable artifact rollback; dynamic needs versioned config and cache rollback.	Latency, SQL bounds, auth, cache, provider config tests.
PMTiles serverless vs Martin/Tegola server-mediated	Serverless PMTiles wins for public-safe static bundles and predictable cost; server-mediated wins for steward access or dynamic PostGIS.	Serverless fails for restricted or frequently changing data; server-mediated fails if it bypasses governance.	Public clients still use governed layer registry, not raw canonical stores.	PMTiles rollback swaps manifest/digest; server-mediated rollback restores config/schema and cache state.	Range/CORS/CDN for PMTiles; access/SLO/cache tests for server.
Vector vs raster	Vector wins for interactive attributes and style-driven maps; raster wins for continuous surfaces, COGs, remote sensing, and labels fallback in 3D.	Vector fails for massive sensitive attributes; raster fails for feature-level evidence unless linked.	Both require source/evidence/citation resolution before claims.	Vector manifests track source-layer and attrs; raster manifests track bands, CRS, nodata, stats.	MVT smoke/attr whitelist; COG/raster/proj validation.
COG vs raster tiles	COG wins for versioned range-readable rasters with STAC Raster/Projection metadata; raster tiles win for simple basemap-like rendering.	COG fails without Range/cache headers; raster tiles fail as analytic source.	COG is a derived carrier with lineage, not soil/remote-sensing truth.	COG rollback by digest; raster tile rollback by manifest and cache invalidation.	rio-cogeo, gdalinfo, stac-validator, cache tests.
Terrain-enabled vs non-terrain	Terrain wins for DEM, LiDAR, hydrology, archaeology viewshed/affordance stories.	Terrain fails if datum/vintage/accuracy are missing or 3D distracts from evidence.	Terrain narratives must cite terrain evidence and uncertainty.	Terrain release needs TerrainTilesetManifest and fallback.	Datum, CRS, vertical accuracy, 2D fallback, visual regression.
2D default vs 3D conditional	2D MapLibre wins by clarity and lower trust complexity; 3D wins only when vertical/globe context bears evidence burden.	3D fails when labels/styles/provenance do not survive or when Cesium cannot match MVT style parity.	3D must not weaken Evidence Drawer, Focus Mode, or policy controls.	3D release needs 2D fallback and scene manifest rollback.	Camera/time sync, label fallback, metadata checksum.
Public-safe tiles vs restricted/internal tiles	Public-safe tiles win for normal UI. Restricted/internal tiles require role-based governed access or quarantine.	Public tiles fail if sensitive geometry is hidden only by style; restricted tiles fail if served by public CDN.	Sensitive geometry must be generalized/redacted before public release.	Release manifest records access class; rollback withdraws cache/digest.	Sensitive geometry deny, H3 generalization, rights check.

10. Governance and Trust-Membrane Chapter

KFM governed flow for map interactions: released layer -> user click -> governed API -> EvidenceBundle -> Evidence Drawer -> Focus Mode answer / abstain / deny / error. MapLibre supplies the renderer and interaction event; KFM services decide evidence, policy, citations, and release state.

No public RAW path: public clients and normal UI surfaces use governed APIs, released artifacts, catalog records, public-safe tile services, and EvidenceBundle resolution only.

No direct model client: Focus Mode and AI runtime remain behind governed API boundaries and finite outcomes: ANSWER, ABSTAIN, DENY, ERROR.

No canonical/internal client fetch: vector tiles, COGs, PMTiles, GeoParquet, Zarr, graph projections, and scenes are downstream artifacts; clients do not fetch canonical stores or unpublished candidates.

No unreleased tile load: addSource/addLayer should be blocked unless LayerManifest, TileArtifactManifest, MapReleaseManifest, PolicyDecision, and release status allow it.

No sensitive geometry hidden only by style: archaeology, species, living-person, infrastructure, and culturally sensitive geometry must be generalized, redacted, delayed, or denied before public tile production.

No popup as Evidence Drawer substitute: popup text is a convenience surface; consequential claims require drawer payloads and citation validation.

No uncited export: screenshots, PDFs, Story Nodes, Focus Mode summaries, and badges must carry citations, evidence IDs, version locks, and release manifest references where material.

SRC-061 expands this membrane with artifact attestation: PMTiles, COG, and GeoParquet should be digest-pinned, signed, and SBOM-attached before publication; MVT/COG pipelines should emit STAC/DCAT/PROV; and tile/build telemetry can be recorded as proof context.

11. Implementation-Ready Object Map

The object families below are PROPOSED field-intent summaries. They do not claim existing repo files. Directory Rules basis: schemas belong under schemas/contracts/v1/<...> by default; object meaning belongs in contracts/; policy in policy/; data lifecycle and emitted artifacts in data/; release decisions and rollback in release/; and public clients use apps/web or governed API only after repo verification.

Object family	Field intent	Source basis	Implementation consequence / path posture
SourceDescriptor	source_id, source_role, authority, rights, cadence, access, sensitivity, citation rules	SRC-061 pp.14-26, pp.39-65	PROPOSED schema; required before NFHL, TDx, AQS/PurpleAir, LiDAR sources.
LayerManifest	layer_id, source refs, style refs, public_safe, sensitivity, evidence refs, release status	SRC-061 pp.171-178	PROPOSED; blocks addLayer until release and policy allow.
StyleManifest	style_id, style JSON hash, sprites/glyphs, label density, attribute whitelist	SRC-061 pp.175-177	PROPOSED; prevents style drift and label over-density.
TileArtifactManifest	artifact_id, type, href/digest, bounds, source_layer, schema, checksum, signature	SRC-061 pp.84-88, pp.171-178	PROPOSED; central to PMTiles/MVT/COG release.
MapReleaseManifest	release_id, layer list, artifacts, policy decisions, rollback target, cache records	SRC-MASTER-PREV + SRC-061	PROPOSED; normal UI consumes only released manifests.
EvidenceBundle	claim/evidence records, source refs, citations, rights, review state	Retained doctrine	CONFIRMED doctrine / PROPOSED implementation.
EvidenceDrawerPayload	clicked feature, candidate IDs, evidence, policy state, citations, errors	SRC-061 pp.178-180, pp.159-166	PROPOSED; clicked sensor/tile feature resolves or abstains.
MapContextEnvelope	camera, bounds, zoom, selected layer/version/time, map release, feature candidate	SRC-MASTER-PREV + SRC-061 pp.107-112	PROPOSED; passed to Focus Mode.
FocusModeRequest	MapContextEnvelope, evidence refs, user question, policy scope	SRC-061 pp.224-229, pp.340-344	PROPOSED; no direct model or raw source access.
FocusModeResponse	finite outcome, answer/abstain/deny/error, citations, source roles, uncertainty	SRC-061 pp.523-524	PROPOSED; cannot invent provenance or alter governance posture.
AIReceipt	provider/model/runtime, context IDs, citation validation, policy decision, finite outcome	Retained doctrine	PROPOSED; not chain-of-thought.
CitationValidationReport	claim IDs, citation IDs, pass/fail, missing evidence, stale evidence	Retained doctrine + SRC-061	PROPOSED; exports and Focus Mode require it.
PolicyDecision	allow/deny/restrict/abstain, rule IDs, source rights, sensitivity, role	SRC-061 pp.87-88, pp.224-229	PROPOSED; fail closed on sensitive/unsigned/unclear.
PromotionDecision	gate results, proof, review state, target release, rollback target	SRC-061 pp.1-8, pp.71-83	PROPOSED; promotion is state transition.
RunReceipt	pipeline run, inputs, outputs, spec_hash/config hash, timestamp, operator, result	SRC-061 pp.1-8, pp.71-83	PROPOSED; used by ingestion, COG, MVT, OCI, remote sensing.
rollback target	prior release manifest, artifact digests, cache invalidation, replay steps	SRC-061 pp.71-83	PROPOSED; required before publication.
stale-source fixture	source last_seen/retrieved_at, stale threshold, UI label, abstain reason	SRC-061 pp.39-65, pp.107-112	PROPOSED; sensor and incident layers.
sensitive-geometry deny fixture	exact geometry input, policy, redaction/generalization output, denial reason	SRC-061 pp.224-229	PROPOSED; archaeology and other sensitive domains.
OCIArtifactRef	registry, repository, digest, media type, referrers, verification result	SRC-061 pp.84-88	NEW PROPOSED object.
SpatialArtifactSBOM	artifact type, generator, source datasets, CRS/bbox, schema fields, checksums, sensitivity	SRC-061 pp.84-88	NEW PROPOSED object.
GeometryRepairReport	invalid counts, area drift, topology repairs, tool versions, smoke-test result	SRC-061 pp.26-32	NEW PROPOSED object.
COGValidationReport	block size, overview levels, nodata, raster/proj STAC checks, range/cache status	SRC-061 pp.123-127	NEW PROPOSED object.
SensorCalibrationReceipt	sensor IDs, field variant, reference source, correction model, trust score, QA flags	SRC-061 pp.39-65, pp.144-154	EXPANDED PROPOSED object.
IncidentLayerManifest	event source, schema version, time rules, conflict/dedup, status, geometry class	SRC-061 pp.20-26	NEW/EXPANDED PROPOSED object.
RemoteSensingTimeSeriesItem	product family, tile/window, COG, QA summary, config snapshot, PROV bundle	SRC-061 pp.520-528	NEW PROPOSED object.

12. Validation and Test Plan

Test area	Required test	v1.7 expansion / source basis	Expected safe outcome
Schema validation	Validate SourceDescriptor, LayerManifest, TileArtifactManifest, MapReleaseManifest, EvidenceDrawerPayload, FocusModeRequest/Response, PolicyDecision, RunReceipt, and new OCI/COG/geometry objects.	SRC-061 pp.1-8, pp.84-88, pp.123-127	Fail invalid shape before promotion.
Source ledger completeness	Every current source has source_id, role, evidence strength, limitation, and status.	Current run ledger	No unledgured source in claims.
Style source/layer validity	Check source-layer, filters, expressions, promoteld, labels, glyph/sprite references, and attribute whitelist.	SRC-061 pp.171-177	Block invalid or unsafe style.
PMTiles / MVT / MLT manifest checks	Validate MVT source-layer, PMTiles range/CORS/cache, digest, signature, SBOM, STAC/DCAT/PROV. MLT remains pilot.	SRC-061 pp.84-88, pp.171-178	PMTiles load only if released and verified.
COG / range request requirements	Run gdalinfo, rio-cogeo, stac-validator, range-header and cache-header checks.	SRC-061 pp.123-127	COG public only if validated.
No public raw path	Browser map cannot fetch RAW, WORK, QUARANTINE, candidates, canonical stores, or unpublished data.	KFM invariant	DENY or ERROR.
No direct model client	Focus Mode cannot call a model runtime from browser; only governed API.	Retained doctrine	DENY direct runtime path.
No unreleased tile load	MapLibre addSource/addLayer blocked without release manifest and proof.	SRC-061 pp.171-178	ERROR or hidden layer state.
Sensitive geometry deny	Exact archaeology/sensitive coordinates and H3 below allowed threshold denied.	SRC-061 pp.224-229	DENY exact geometry; allow generalized only.
Source rights unknown blocks release	NFHL, LiDAR mirrors, PurpleAir, archaeology and remote sensing assets require rights/source terms.	SRC-061 pp.11-20, pp.39-65, pp.224-229	DENY/QUARANTINE until verified.
Stale source badge / abstain	Sensor last_seen/timestamp normalization produces stale badge and Focus Mode abstain when needed.	SRC-061 pp.39-65, pp.107-112	Visible stale state; no overconfident answer.
Clicked feature resolves EvidenceBundle	Sensor/device/lineage/provenance chain exists for clicked feature.	SRC-061 pp.159-166, pp.178-180	Drawer payload or ABSTAIN.
Citation validation pass/fail	Focus, export, and popup claims with citations pass; missing citations abstain.	KFM invariant	ABSTAIN if unsupported.
Focus Mode answer with citations	Focus Mode can summarize only cataloged evidence, no invented relationships.	SRC-061 pp.523-524	ANSWER with citations.
Focus Mode abstain on missing evidence	Missing EvidenceBundle/provenance/lineage produces ABSTAIN or DENY.	SRC-061 pp.178-180	No fluent unsupported answer.
Rollback restores prior manifest	Rollback target restores old release artifacts and invalidates caches.	SRC-061 pp.71-83	Prior release restored.
Cache invalidation recorded	Tile/COG/CDN cache keys recorded after release/rollback.	SRC-061 pp.84-88, pp.123-127, pp.176-178	No stale artifact leak.
Visual regression for styles	Compare map render, labels, symbols, badges, and 3D fallback.	SRC-061 pp.175-177, pp.181-183	Style regression detected.
Keyboard navigation / accessibility	Layer catalog, version strip, badges, Focus Mode, and drawer keyboard paths.	Retained v1.6 + SRC-061 badges	Accessible trust state.
Tile load budget	Measure p95 tile fetch and resource timing; fail if budget breached.	SRC-061 pp.176-177	Performance gate blocks release.
Interaction-to-drawer latency budget	Measure click-to-drawer resolution including evidence fetch.	KFM UI doctrine	Drawer SLO recorded.

13. Anti-Patterns Register

Anti-pattern	Why it is dangerous	Detection / mitigation
Treating MapLibre as truth authority	Renderer output is downstream and can omit, simplify, or style facts.	Require EvidenceBundle and release manifest before claims.
Treating tiles as proof	Tiles simplify and carry selected attributes, not full source authority.	TileArtifactManifest links to source/evidence/proof.
Treating layer toggle as publication	A visible layer is not a governed state transition.	MapReleaseManifest and PromotionDecision required.
Hiding exact sensitive geometry with style filters	Public bytes can still expose exact geometry.	Redact/generalize before tile build; deny exact geometry.
Dense GeoJSON as production default for large datasets	Browser performance and memory fail; stale features can linger.	Use vector tiles/COGs where appropriate.
Uncontrolled style drift	Colors/labels can alter meaning and trust state.	StyleManifest, visual regression, style lint.
Plugin adoption without review	Protocol handlers, wrappers, and servers can alter trust boundary.	Tool/plugin allowlist, SBOM, sandbox.
Direct browser access to model runtime	Bypasses evidence, policy, citation validation, and AIReceipt.	No direct model client test.
AI answers based only on rendered map features	Rendered features are incomplete carriers.	Focus Mode resolves EvidenceBundle or abstains.
Uncited popups or exports	Users cannot inspect evidence or correction lineage.	Citation validation and export manifest.
MLT as default before KFM validation	Toolchain maturity and format support remain verification gaps.	Pilot with end-to-end validation only.
3D/globe as default when 2D is clearer	3D adds label/provenance/fallback complexity.	2D default; 3D conditional proof.
Mapbox-era backports without legal/security review	Licensing/security divergence can affect dependencies.	Dependency and license review.
Unsigned PMTiles, COG, or GeoParquet in public UI	Artifact origin/process cannot be proven.	Cosign/ORAS/SBOM/referrer checks.
Tag-based artifact catalogs	Tags can drift and undermine reproducibility.	Use digest-pinned OCI and catalog URLs.
Geometry repair that silently drops features	Public tiles lose features without visible correction record.	Area drift and feature count gates.
Aggressive gap-fill or dissolve before tiling	Can erase meaningful unmapped areas or units.	Thresholds, source review, repair report.
Dynamic tile server as raw PostGIS public path	Can bypass governed API and source rights.	Server-mediated tiles only behind release/access gates.
WMS pixels used for analytics	Rendered WMS is visualization, not authoritative feature data.	Use vector/API/bulk data for analytics.
NFHL treated as real-time flood or climate forecast	NFHL is regulatory effective baseline.	Knowledge-character label and citation.
Raw low-cost PM2.5 sensor values as public truth	Sensors require calibration, trust, context, and source role.	SensorCalibrationReceipt and reference validation.
Badge as Evidence Drawer substitute	Visual badge can become trust theater.	Badge links to receipt; drawer required for claims.
Remote-sensing forecast shown as observation	Modeled output can be mistaken for observed fact.	Modeled-output labels and uncertainty.
Archaeology heatmap as site locator	Generalized zones can expose sensitive cultural heritage if overinterpreted.	CARE chips, H3 floor, deny exact coordinates.

14. Open Questions and Verification Backlog

Item	Status	Why it matters	Verification action
Package versions	NEEDS VERIFICATION	MapLibre, PMTiles, GDAL, Tippecanoe, Tegola/t_rex, Cosign, ORAS, stac-validator versions affect compatibility.	Check current repo/package lock and official docs before pins.
Schema home	NEEDS VERIFICATION	Directory Rules default schemas/contracts/v1, but actual repo may differ.	Mount repo; inspect ADR-0001 and schema roots.
Actual repo topology	UNKNOWN	Paths in source packets are implementation-style but not verified.	Mount repo and run directory inventory.
MapLibre GL JS target version	NEEDS VERIFICATION	Renderer feature support and security may change.	Check package lock and official release.
MLT support status	NEEDS VERIFICATION	MLT remains retained/pilot; no new SRC-061 support.	Run MLT toolchain/renderer tests.
Mobile/native parity	UNKNOWN	Public mobile/native support must preserve drawer/policy/release.	Define parity fixtures.
PMTiles hosting Range/CORS behavior	NEEDS VERIFICATION	Static PMTiles requires Range and CORS headers.	Probe headers in staging.
COG hosting Range/cache behavior	NEEDS VERIFICATION	COG streaming depends on range and immutable cache.	Probe object storage/CDN.
OCI registry referrers	NEEDS VERIFICATION	SBOM/attestation referrers require registry support.	Run ORAS discover against target registry.
Martin/Tegola/t_rex deployment/access model	UNKNOWN	Dynamic tiles must not bypass governance.	Design server-mediated access ADR.
Plugin/protocol allowlist	UNKNOWN	PMTiles protocol handler and wrappers must be reviewed.	Create plugin/dependency allowlist.
Style/sprite/glyph hosting	UNKNOWN	Missing glyphs/sprites and license issues affect map rendering.	Create StyleManifest/GlyphManifest fixtures.
Attribution and rights	NEEDS VERIFICATION	NFHL, LiDAR mirrors, PurpleAir, archaeology, remote sensing sources have terms.	SourceDescriptor rights review.
Sensitive geometry transform rules	NEEDS VERIFICATION	H3 floors and generalization policies need steward approval.	Policy fixture with cultural/sensitive reviewers.
Release manifest location	UNKNOWN	Directory Rules constrain but repo conventions unknown.	Mount repo and check release root.
CI workflow	UNKNOWN	No actual workflow inspected.	Implement no-network fixture workflows first.
Accessibility acceptance	NEEDS VERIFICATION	Badges, drawers, layer controls, Focus Mode must be accessible.	Keyboard/screen-reader tests.
Performance budgets	NEEDS VERIFICATION	SRC-061 gives p95 tile fetch <150ms, but target may vary.	Set budgets by device/network class.
Geometry repair thresholds	NEEDS VERIFICATION	Area drift and gap thresholds need domain/steward review.	Create fixture thresholds per domain.
Air calibration authority	NEEDS VERIFICATION	PM2.5 corrections depend on source roles and reference monitors.	Define SensorCalibrationReceipt acceptance.
TDx/WZDx feed behavior	NEEDS VERIFICATION	Road-event feeds may be snapshots or deltas and can be stale.	Prototype ingest with stale/conflict cases.
NFHL effective-date behavior	NEEDS VERIFICATION	Localized updates affect layer releases and rollback.	Probe service metadata and version changes.
LiDAR vertical datum and vintage	NEEDS VERIFICATION	Terrain comparisons can be wrong without datum/vintage.	Terrain manifest tests.
Archaeology H3 minimum and consent rules	NEEDS VERIFICATION	Source says no geometry below H3 r7 without review; must be steward-governed.	Policy review and deny fixtures.
Remote sensing QA thresholds	NEEDS VERIFICATION	Thresholds in source may be illustrative or domain-specific.	Define QA acceptance per product family.

15. How to Continue This Master PDF Next Run

- Attach this PDF again with any new documents.
- The next run should read this PDF first as SRC-MASTER-PREV.
- Preserve all idea IDs unless merged or superseded.
- Update the delta table.
- Add new sources to the source ledger.
- Add new discoveries to the idea index.
- Recompute counts.
- Keep uncertainty labels visible.
- If a newly attached source repeats a prior source by SHA-256, mark it REATTACHED / RETAINED and do not duplicate the old idea records.
- If a mounted repo becomes available, repo evidence can upgrade or conflict with proposed paths, but it must not silently delete prior source-led ideas.

Final PDF Self-Check

Check	Result
No obvious cut-off text	Generated with wrapped paragraphs/tables; render QA performed after build.
No missing source ledger	Section 4 includes current-run source ledger and Appendix A retains prior ledger.
No missing delta section	Section 3 records v1.6 -> v1.7 continuity and delta.
No missing open questions	Section 14 includes verification backlog.
No unsupported implementation claims	All repo paths and object homes are labeled PROPOSED/UNKNOWN unless supported by uploaded source text.
No forgotten prior master ideas	Prior v1.6 master appended unchanged as Appendix A.
Single user-facing artifact	Final output is one PDF named Master MapLibre Components-Functions-Features.pdf.

Appendix A begins after this update packet and contains the full retained SRC-MASTER-PREV baseline unchanged.

Master MapLibre

Components-Functions-Features.pdf

Cumulative KFM MapLibre, Tile, Renderer, Style, UI, Evidence Drawer, and Release Architecture Atlas

Field	Value
Version	v1.6 cumulative update
Run date	2026-05-10
Attached source count	42 PDFs visible in /mnt/data; prior master detected; two new source packets analyzed
Prior master PDF detected	Yes - v1.5 copied as SRC-MASTER-PREV before overwrite; prior pages 375; sha256 prefix 9dddfd6cab61a27a
Output mode	PDF only. No ZIP, tarball, scaffold folder, repository patch or extra user-facing artifact.
Truth posture	CONFIRMED doctrine/source scans; PROPOSED implementation guidance; UNKNOWN current repo/runtime depth; NEEDS VERIFICATION for versions, hosting, rights, source endpoints, policy enforcement, performance budgets, mobile/native parity and path placement.
Evidence limits	No mounted KFM repo, tests, workflows, dashboards, logs, release manifests or runtime traces were available. Paths in source packets are source-reported and PROPOSED unless repo evidence later verifies them.
Source ledger count	43 current-run records: SRC-RUN-PROMPT, SRC-MASTER-PREV, 39 retained attached sources and NEW SRC-059/SRC-060. Prior-only source ledger remains preserved in Appendix A.
Idea count	2383 cumulative idea records: 2257 prior + 126 v1.6 delta records.
New discoveries count	126 delta records from SRC-059 and SRC-060.
Expanded category clusters	23 category clusters expanded by current-run source evidence.
Superseded ideas	0 - no attached evidence invalidated a prior idea.
Open verification items	329 cumulative estimate: prior 295 + 34 new items.

Core determination: MapLibre remains a downstream renderer and interaction runtime. Tiles, PMTiles, COGs, style JSON, popups, screenshots, Story Nodes, 3D scenes and AI answers remain downstream carriers, not sovereign truth.

1. Table of Contents

- 1. Cover Page and run metadata
- 2. Executive Determination
- 3. Continuity and Delta Since Last Run
- 4. Source Ledger
- 5. Master Idea Index - v1.6 Delta
- 6. Category Atlas
- 7. Detailed Category Chapters
- 8. Components-Functions-Features Matrix
- 9. Tile Strategy Chapter
- 10. Governance and Trust-Membrane Chapter
- 11. Implementation-Ready Object Map
- 12. Validation and Test Plan
- 13. Anti-Patterns Register
- 14. Open Questions and Verification Backlog
- 15. How to Continue This Master PDF Next Run
- Final PDF Self-Check
- Appendix A. Full retained v1.5 Master PDF baseline (SRC-MASTER-PREV, appended unchanged)

Navigation note. The v1.6 update packet is current-run specific. The full v1.5 baseline follows as Appendix A so prior source ledger entries, category chapters, matrices, idea IDs and open questions are not silently deleted.

2. Executive Determination

CONFIRMED from SRC-MASTER-PREV and retained doctrine: MapLibre is a disciplined 2D renderer and interaction runtime inside a governed shell. It is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority or AI authority. Public surfaces use governed APIs, released artifacts, catalog records, tile services, EvidenceBundle resolution and release manifests.

CONFIRMED from this run: the prior v1.5 master baseline was detected and copied before overwrite. New source packet SRC-059 adds Focus Mode visualization asset registries, Story Node metadata/versioning, map asset metadata, STAC/DCAT/CIDOC crosswalks, symbol/glyph/legend governance, visual regression baselines, 3D terrain tileset metadata and FAIR+CARE 2D/3D controls. New source packet SRC-060 adds PM2.5 calibration trust patterns, Zarr/STAC array metadata, multiscale overview decisions, EOPF-style assets, Neo4j migration rollback, WZDx/ArcGIS/GTFS incident fusion and deterministic data-run gates.

PROPOSED synthesis: v1.6 should be treated as a Focus Mode asset-governance, Story Node versioning, symbol/legend, 3D terrain, PM2.5 trust-layer and incident-fusion update. It does not reorganize the prior master. The next implementation-ready slice remains schema/fixture/test first: MapAssetMetadata, StoryNodeMetadata, StoryNodeVersionEnvelope, SymbolLegendManifest, 3DSceneManifest, TerrainTilesetManifest, ArrayAssetManifest, IncidentLayerManifest, SensorCalibrationReceipt, ReleaseManifest links and negative-policy fixtures.

UNKNOWN: actual repo topology, package manager, schema home, UI component homes, MapLibre GL JS target version, PMTiles tooling, PMTiles/COG/Zarr/3D-Tiles hosting headers, browser/device performance, mobile/native parity, plugin allowlist, source rights, actual policy enforcement and release workflow remain unverified.

NEEDS VERIFICATION: every package version, tool version, source URL, feed schema, hosted header, release manifest, SBOM, telemetry file, Story Node path, map asset path, sensor calibration source, Zarr convention, WZDx registry, ArcGIS change-tracking behavior, GTFS-RT freshness assumption and 3D terrain source license.

Highest-risk anti-pattern: treating documentation badges, static map assets, Story Node screenshots, 3D tilesets, symbol catalogs, model-corrected PM2.5 surfaces or incident clusters as released evidence before the corresponding EvidenceBundle, manifest, PolicyDecision, PromotionDecision, citation validation and rollback target exist.

Metric	Value
Previous version	v1.5 cumulative update
Current version	v1.6 cumulative update
Prior ideas carried forward	2257
New SRC-059 delta ideas	95
New SRC-060 delta ideas	31
Total cumulative idea count	2383
Superseded ideas	0
Open verification items	329

3. Continuity and Delta Since Last Run

SRC-MASTER-PREV is the prior v1.5 Master MapLibre Components-Functions-Features.pdf. It remains the cumulative memory baseline and is appended unchanged. It does not outrank original evidence. Current-run source evidence can refine, expand, conflict with or supersede its entries. No supersession occurred in this run.

Delta item	Result
Prior version detected	v1.5, 375 pages, sha256 prefix 9dddfd6cab61a27a
Version rule applied	+0.1 ordinary cumulative update. No fundamental structural reorganization.
New sources added	SRC-059 New Ideas 6.pdf; SRC-060 New Ideas 12-20-25.docx (1).pdf.
Ideas retained	All 2257 prior idea records preserved by appending SRC-MASTER-PREV unchanged.
New records	126 ML-059-* and ML-060-* delta records.
Merged / superseded	0 merged, 0 superseded.
Conflicts discovered	No direct source conflict. Main tension is implementation maturity: source packets contain strong implementation-style examples but no mounted repo/runtime proof was available.
Open questions carried forward	Package versions, schema home, source rights, MLT status, mobile parity, hosting headers, sensitive geometry, release manifest home, CI workflow, accessibility and performance budgets.
New open questions	Story Node path placement, map asset registry home, 3D terrain rights, Zarr metadata validation, PM2.5 calibration source authority, WZDx/ArcGIS/GTFS feed behavior and symbol/sprite governance.

3A. Prior category status comparison

Category	v1.6 delta count	Status / note
A. Renderer Boundary and KFM Trust Law	5	EXPANDED by SRC-059/SRC-060
B. MapLibre GL JS Web Shell	2	EXPANDED by SRC-059/SRC-060
C. MapLibre Native / Mobile / Platform Parity	0	RETAINED from SRC-MASTER-PREV
D. MapLibre RS / Experimental Renderers	0	RETAINED from SRC-MASTER-PREV
E. Style Specification, Sources, Layers, Expressions	4	EXPANDED by SRC-059/SRC-060
F. Sprites, Glyphs, Fonts, Design Tokens	7	EXPANDED by SRC-059/SRC-060
G. GeoJSON and Runtime Data	3	EXPANDED by SRC-059/SRC-060
H. Vector Tiles: MVT and MLT	0	RETAINED from SRC-MASTER-PREV
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	2	EXPANDED by SRC-059/SRC-060
J. Martin and Server-Mediated Tile Serving	1	EXPANDED by SRC-059/SRC-060
K. Raster, COG, DEM, Terrain, Hillshade	6	EXPANDED by SRC-059/SRC-060
L. WMS / WMTS / External Map Services	1	EXPANDED by SRC-059/SRC-060
M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	16	EXPANDED by SRC-059/SRC-060
N. Evidence Drawer Payloads and Click Resolution	3	EXPANDED by SRC-059/SRC-060
O. Focus Mode and Governed AI Map Context	10	EXPANDED by SRC-059/SRC-060
P. Time-Aware Map Interaction and Timeline	5	EXPANDED by SRC-059/SRC-060
Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	11	EXPANDED by SRC-059/SRC-060
R. Plugin, Wrapper, and Dependency Governance	2	EXPANDED by SRC-059/SRC-060
S. Accessibility, UX, and Trust-Visible States	7	EXPANDED by SRC-059/SRC-060
T. Performance, Caching, CDN, Range Requests, and Resource Timing	5	EXPANDED by SRC-059/SRC-060
U. Testing, CI, Validation, Rollback, and Proof Objects	15	EXPANDED by SRC-059/SRC-060
V. Exports, Screenshots, Reports, and Citation Preservation	4	EXPANDED by SRC-059/SRC-060
W. 3D / Cesium / Deck.gl / Overlay Interoperability	8	EXPANDED by SRC-059/SRC-060

Category	v1.6 delta count	Status / note
X. Anti-Patterns and Failure Modes	4	EXPANDED by SRC-059/SRC-060
Y. Implementation Backlog and PR Plan	3	EXPANDED by SRC-059/SRC-060
Z. Open Questions and Verification Backlog	2	EXPANDED by SRC-059/SRC-060

4. Source Ledger

This ledger is a control surface, not a bibliography. It states what each source can support and what it cannot prove. Source IDs are stable within this cumulative master. Current external facts embedded in source PDFs remain source-supported but need verification before operational package pins, access decisions, source activation or release claims.

4A. Current-source scan highlights

source_id	file	scan result
SRC-059	New Ideas 6.pdf	435 pages; high signals: Focus Mode 224 mentions, 3D 163, accessibility 103, layers 96, timeline 81, MapLibre 21, COG 10, PMTiles 6, Cesium 23.
SRC-060	New Ideas 12-20-25.docx (1).pdf	37 pages; high signals: PM2.5 trust/calibration, Zarr/STAC arrays, multiscale overviews, Neo4j rollback, WZDx/ArcGIS/GTFS incident fusion, deterministic run gates.
SRC-MASTER-PREV	Master MapLibre Components-Functions-Features.pdf	375 pages; retained v1.5 baseline with 2257 prior idea records.

4B. Complete current-run source ledger

source_id	file name / document title	source type	status	role in this PDF	MapLibre / tile relevance	evidence strength	limitations
SRC-RUN-PRO-MPT	Current user prompt	Prompt	CONFIRMED	Controls output mode, truth labels, source IDs, cumulative rule, required sections, cite-or-abstain posture.	High because it defines the required MapLibre/tile search scope.	High for requirements	Instruction source, not implementation proof.
SRC-MASTER-PREV	Master MapLibre Components-Functions-Features.pdf v1.5 (375 pages; sha 9ddd6c61a27a)	Prior master PDF	CONFIRMED / RETAINED	Cumulative baseline. Appended unchanged as Appendix A after this v1.6 update packet.	Preserves prior 2257 idea records, prior source ledger, category chapters, matrices and open questions.	High for continuity	Does not outrank original source evidence.
SRC-003	KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf	KFM UI / map doctrine	RETAINED	Renderer boundary, governed shell, Evidence Drawer, Focus Mode, tile/style strategy.	Primary doctrine for MapLibre as renderer only.	High doctrine	Implementation depth UNKNOWN.
SRC-004	Directory Rules.pdf	Directory governance doctrine	RETAINED	Responsibility-root placement law, schema-home default, lifecycle boundaries.	Constrains any proposed file homes or schema paths.	High doctrine	Quoted paths remain PROPOSED until mounted repo verification.
SRC-005	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf	Pipeline manual	RETAINED	Lifecycle law and query-save-validate-compile-review-promote-recompile loop.	Supports release/correction/rollback discipline for map artifacts.	High doctrine	No repo/runtime proof.
SRC-006	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan_v1_1.pdf	Build plan	RETAINED	Trust spine, pre-RAW event family, public-safe defaults and map architecture.	Defines map architecture as downstream of evidence and release.	High doctrine	Greenfield/proposed-to-create posture.
SRC-007	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf	UI + AI plan	RETAINED	Whole UI, MapLibre shell, governed API, Evidence Drawer, Focus Mode, review console.	Supports UI trust membrane and finite outcomes.	High lineage	No mounted repo proof.
SRC-008	KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf	Governed AI plan	RETAINED	Provider-neutral AI, finite outcomes, citation validation, AIReceipt.	Supports Focus Mode boundaries.	High lineage	No runtime proof.

source_id	file name / document title	source type	status	role in this PDF	MapLibre / tile relevance	evidence strength	limitations
SRC-009	Ollama & Ubuntu Information.pdf	Runtime guide	RETAINED	Local model behind governed API; no direct public model traffic.	Supports no-direct-model-client anti-pattern and Focus Mode flow.	High lineage	Version/runtime facts need recheck.
SRC-010	kfm_encyclopedia.pdf	Capability encyclopedia	RETAINED	Cross-domain atlas: maps, tiles, graphs and AI answers as downstream carriers.	Good general KFM operating law for map surfaces.	High doctrine	Implementation depth UNKNOWN.
SRC-011	# Kansas Frontier Matrix Implementation Reference.pdf	Implementation reference	RETAINED	Prior public-repo summary and MapLibre/PMTiles stack signals.	Useful repo-lineage signal for app stack and schema-home ambiguity.	Lineage / needs verification	Prior connector claims not reverified here.
SRC-012	a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf	GIS/cartography reference	RETAINED	Representation, projections and cartographic communication.	Conceptual background for map-as-representation.	Conceptual	Not KFM implementation evidence.
SRC-013	Advanced-SQL-Concepts.pdf	SQL reference	RETAINED	Query and analytics background.	Low direct MapLibre relevance.	Conceptual	Not MapLibre-specific.
SRC-014	AI_Concepts_Using_Python.pdf	AI reference	RETAINED	AI/data workflow background.	Supports bounded AI context only.	Conceptual	Not MapLibre-specific.
SRC-015	Archaeological 3D GIS.pdf	3D GIS reference	RETAINED	3D GIS, 3D models and archaeological visualization.	Supports conditional 3D/archaeology visualization thinking.	Conceptual	External examples, not KFM implementation.
SRC-016	CICD_with_Docker_Kubernetes_Semaphore.pdf	CI/CD reference	RETAINED	CI, release and rollback background.	Supports validation/rollback chapter.	Conceptual	Not KFM proof.
SRC-017	Designing Great Web APIs.pdf	API reference	RETAINED	API contract and client/developer experience.	Supports governed API dependency.	Conceptual	Not KFM proof.
SRC-018	developing-time-oriented-database-applications-in-sql.pdf	Temporal DB reference	RETAINED	Valid time, transaction time and bitemporal concepts.	Supports timeline/time-aware map design.	Conceptual	Not MapLibre proof.
SRC-019	Domain-Driven Design Reference.pdf	DDD reference	RETAINED	Bounded contexts, published language, responsibility layers.	Supports responsibility root/domain boundary discipline.	Conceptual	Not KFM proof.
SRC-020	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf	GIS lab/textbook	RETAINED	Environmental GIS and hazard/climate layer context.	Conceptual source for environmental layers.	Conceptual	ArcGIS-centric.
SRC-022	GIS Succinctly.pdf	GIS reference	RETAINED	Spatial database, geometry and GIS application basics.	Conceptual support for spatial data delivery.	Conceptual	Not KFM proof.
SRC-023	GIS in Sustainable Urban Planning and Management.pdf	Urban GIS reference	RETAINED	Planning support, spatial indicators and resilience cases.	Context for planning/transport/resilience layers.	Conceptual	Not KFM proof.
SRC-024	instant-web-scraping-with-java.pdf	Scraping reference	RETAINED	Source acquisition and robust scraping background.	Low direct map relevance; source intake caution.	Conceptual	Not KFM proof.
SRC-025	KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Run receipts, deterministic identity, promotion gates.	Governance backbone for map artifacts.	High lineage	Prior synthesis, not repo proof.
SRC-026	KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Source-aware ingest, PMTiles, MLT and map stack guidance.	Map/tile stack lineage.	High lineage	Not current repo proof.
SRC-027	KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Pre-RAW event family, watchers, catalog integrity, environmental tiles.	Supports new event/watchers in map layer flows.	High lineage	Not current repo proof.
SRC-028	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf	Domain blueprint	RETAINED	Agriculture layers, catalog closure, MapLibre/Evidence Drawer posture.	Domain layer examples.	Domain lineage	Paths PROPOSED.

source_id	file name / document title	source type	status	role in this PDF	MapLibre / tile relevance	evidence strength	limitations
SRC-029	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	Domain blueprint	RETAINED	Exact site-location denial and remote-sensing candidate caution.	Strong sensitivity controls for map layers.	Domain lineage	Paths PROPOSED.
SRC-030	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Air/smoke/EO layers and knowledge-character labels.	Supports PM2.5 and air layer updates.	Domain lineage	Paths PROPOSED.
SRC-031	KFM_Fauna_Architecture_PDF_Only_Report.pdf	Domain blueprint	RETAINED	Occurrence geoprivacy and public-safe derived products.	Sensitive occurrence tile controls.	Domain lineage	Paths PROPOSED.
SRC-032	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	Domain blueprint	RETAINED	Rare-plant controls and occurrence redaction.	Sensitive flora layer controls.	Domain lineage	Paths PROPOSED.
SRC-033	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Geology/resource map layers and public-safe geometry.	Domain layers and evidence drawer.	Domain lineage	Paths PROPOSED.
SRC-034	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	Domain blueprint	RETAINED	Habitat assignment proof slice and Evidence Drawer.	Public-safe habitat/fauna map proof pattern.	Domain lineage	Paths PROPOSED.
SRC-035	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	Domain blueprint	RETAINED	Habitat patches, suitability, connectivity, public-safe products.	Domain layer examples.	Domain lineage	Paths PROPOSED.
SRC-036	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	Domain blueprint	RETAINED	Hazard layers, no emergency alerting, evidence-backed UI.	Hazard map layer boundary.	Domain lineage	Paths PROPOSED.
SRC-037	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf	Domain blueprint	RETAINED	Hydrology proof lane and HUC/NHD/USGS/NFHL layers.	Hydrology map/tile foundation.	Domain lineage	Paths PROPOSED.
SRC-038	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	Domain blueprint	RETAINED	Living-person/DNA restrictions and temporal land assertions.	Sensitivity and land/parcel map caution.	Domain lineage	Paths PROPOSED.
SRC-039	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	Domain blueprint	RETAINED	Transport/corridor layers, graph projections, Evidence Drawer.	Transport map layer lineage.	Domain lineage	Paths PROPOSED.
SRC-040	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	Domain blueprint	RETAINED	Settlement/infrastructure layer manifests and sensitive assets.	Infrastructure layer controls.	Domain lineage	Paths PROPOSED.
SRC-041	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	Domain blueprint	RETAINED	Soil maps, SSURGO/gSSURGO, PMTiles stub, Evidence Drawer.	Soil PMTiles/COG lineage.	Domain lineage	Paths PROPOSED.
SRC-056	KFM_Pass_15_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Idea atlas	RETAINED	Map-serving runtime proof parity and lane gap.	Map serving proof requirements retained from prior master.	High lineage	Already incorporated in prior master.
SRC-059	New Ideas 6.pdf (435 pages; sha723ad2a705caec0f)	New source packet	NEW	Adds Focus Mode visualization, Story Node, map asset metadata, STAC versioning, symbol/glyph/legend governance, visual regression and 3D terrain tileset material.	High: MapLibre pages, Focus Mode, 3D Tiles, COG/PMTiles metadata, glyph/sprite/style and governance.	High for v1.6 delta	Path claims are source text only; implementation/version facts need verification.

source_id	file name / document title	source type	status	role in this PDF	MapLibre / tile relevance	evidence strength	limitations
SRC-060	New Ideas 12-20-25.docx (1).pdf (37 pages; sha 89b0231a26c34e8a)	New source packet	NEW	Adds PM2.5 calibration/trust ideas, Zarr/STAC array metadata, multiscale overviews, Neo4j migration rollback, WZDx/ArcGIS/GTFS incident fusion and deterministic data-run gates.	Medium-high: indirectly strengthens air layers, array/raster assets, incident layers, rollback and governance.	High for v1.6 delta	External study/tool claims are not revalidated; no repo proof.

5. Master Idea Index - v1.6 Delta

The full cumulative v1.5 idea index of 2257 prior ideas is retained in Appendix A via SRC-MASTER-PREV. The records below add the v1.6 delta. Implementation and validation consequences are PROPOSED unless a future mounted-repo inspection verifies them.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-001	Hydrology COG/Parquet assets feed released map layers	K. Raster, COG, DEM, Terrain, Hillshade	NEW SRC-059 pp.1-2	Hydrology outputs are described as COG rasters or Parquet/GeoParquet/GeoJSON products registered in STAC and visualized as timeline map layers, including MapLibre frontend examples.	Represent raster/DEM/COG/terrain assets as released, checksum-addressed derivatives.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-002	Hillshade and flow accumulation MapLibre toggles	B. MapLibre GL JS Web Shell	NEW SRC-059 p.2	The source explicitly describes hillshade or flow accumulation layers being toggled in a MapLibre-based frontend, with river-discharge time series plotted over the historical timeline.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-003	Hazard event layers are analytic context, not emergency alerting	A. Renderer Boundary and KFM Trust Law	NEW SRC-059 pp.2-3	Hazard warnings, declarations, earthquakes and wildfire detections can appear on map/timeline surfaces, but KFM is not an operational alert system.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-004	Flood-resilience overlays combine hazards with roads and parcels	P. Time-Aware Map Interaction and Timeline	NEW SRC-059 p.3	The corpus describes overlaying flood simulation outputs with roads, farmland parcels and infrastructure to examine cutoffs and inundation impacts as cross-layer analysis.	Carry valid time, source time, release time and version lock in map context.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-005	Climate anomaly layers become hazard overlays	K. Raster, COG, DEM, Terrain, Hillshade	NEW SRC-059 p.3	Climate anomaly outputs can serve as drought, heat or precipitation hazard overlays, with AI detection context kept as an overlay rather than root truth.	Represent raster/DEM/COG/terrain assets as released, checksum-addressed derivatives.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-006	Air observations become map/tooltips or aggregated daily surfaces	B. MapLibre GL JS Web Shell	NEW SRC-059 p.4	Air quality outputs can appear as sensor-location tooltips or aggregated pollution surfaces by day and can be incorporated into Focus Mode narratives.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-007	Modeled air fields must stay distinct from observations	A. Renderer Boundary and KFM Trust Law	NEW SRC-059 p.4	CAMS model fields are described as separate assets or graph nodes linked by DERIVED_FROM instead of being mixed with raw observations.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-008	Vector ETL standardizes GeoParquet/GeoJSON WGS84 outputs	G. GeoJSON and Runtime Data	NEW SRC-059 pp.5-6	Vector datasets such as parcels and boundaries are harmonized, reprojected to WGS84, and exported to GeoParquet or GeoJSON before catalog/API/map use.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-009	Temporal land ownership animation remains evidence-bound	P. Time-Aware Map Interaction and Timeline	NEW SRC-059 pp.5-6	Historical land patents and plat boundaries are positioned as time-aware map layers that can animate settlement or homesteading waves, but detailed temporal cadastre remains a future expansion.	Carry valid time, source time, release time and version lock in map context.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-010	Focus Mode visualization assets are governed outputs	O. Focus Mode and Governed AI Map Context	NEW SRC-059 pp.80-82	Focus Mode stores rendered outputs, UI captures, 3D perspectives and layered overlays with FAIR+CARE, ISO metadata and reproducibility requirements.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-011	Focus Mode asset classes include overlays, 3D scenes, story nodes and analytics snapshots	S. Accessibility, UX, and Trust-Visible States	NEW SRC-059 pp.80-81	The source distinguishes core panels, interactive overlays, Cesium/MapLibre-integrated 3D scenes, Story Nodes and analytics snapshots as different asset classes.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-012	Visualization assets require metadata sidecars	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.81-82	Focus Mode visualizations require associated metadata records with IDs, data sources, FAIR/CARE status, asset files, checksums and generation timestamps.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-013	Story Nodes unify narrative, spatial, temporal, provenance and CARE metadata	O. Focus Mode and Governed AI Map Context	NEW SRC-059 pp.83-86	Story Nodes are described as explainable links between datasets, scenes, maps, timeline frames and AI-generated insights, with spatial footprints, temporal coverage, provenance, citations and CARE flags.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-059-014	Story Nodes link directly to 2D maps, 3D scenes and timeline frames	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 pp.84-86	Story Node structures include links to map_2d, scene_3d and timeline_frame assets; Focus Mode can use them as narrative chapters and AI reference units.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-015	Story Nodes enforce context-only spatial disclosure for sensitive narratives	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-059 pp.86-87	The ethics table says provenance, citations and lineage are required, while precise site disclosure is avoided through context-only spatial footprints.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-059-016	STAC versioning becomes a Focus Mode reproducibility primitive	P. Time-Aware Map Interaction and Timeline	NEW SRC-059 pp.88-93	STAC version fields, predecessor/successor/latest links, lineage APIs, diffs, version locks and telemetry are wired into Story Node and Focus Mode flows.	Carry valid time, source time, release time and version lock in map context.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-017	Story Node schema needs version and lineage fields	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.89-90	Story Node schema additions include version, deprecated flag and predecessor/successor/latest lineage references.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-018	Lineage and diff endpoints support Story Node version inspection	N. Evidence Drawer Payloads and Click Resolution	NEW SRC-059 pp.90-91	The source proposes endpoints for /lineage and /diff that return predecessor/current/successor/latest objects and metadata/asset/geometry diffs.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-059-019	MapLibre UI version strip supports keyboard navigation	S. Accessibility, UX, and Trust-Visible States	NEW SRC-059 p.91	The Story Node version strip displays predecessor/current/successor/latest, deprecated flags, hover previews and click-to-diff behavior with keyboard arrow navigation.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-020	Focus Mode version lock freezes map layers and panels	O. Focus Mode and Governed AI Map Context	NEW SRC-059 pp.91-92	When a version is selected, Focus Mode freezes map layers to that version, disables auto-updating datasets, locks panels and emits a version_locked telemetry event.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-059-021	Diff engine compares metadata, assets and geometry/raster changes	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-059 p.92	The diff engine must compare metadata fields, asset inventories, geometry or raster diffs and semantic score changes.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-059-022	Every Focus Mode output embeds version lineage and diff hash	V. Exports, Screenshots, Reports, and Citation Preservation	NEW SRC-059 pp.92-93	Version stamping requires STAC Item ID, version string, lineage chain, diff manifest hash, timestamp and session fingerprint in Focus Mode outputs.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-023	Story Node metadata model includes spatial, temporal, entity, provenance and governance fields	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.94-97	The metadata specification defines Story Nodes as atomic narrative units with geometry/bbox/CRS, temporal intervals, entities, provenance, citations, lineage, CARE and visualization links.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-024	Story Node examples carry citations and dataset lineage	V. Exports, Screenshots, Reports, and Citation Preservation	NEW SRC-059 pp.96-97	Examples show Story Nodes including provenance datasets and citations, reinforcing cite-or-abstain behavior for Focus Mode narrative cards.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-025	Story Node assets root requires indexed maps, legends, overlays, 3D scenes and thumbnails	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.112-114	Every asset referenced by a Story Node must have an assets_index record with path, checksum, provenance, citations, CARE status and update timestamp.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-026	Story Node asset workflows include docs, STAC/DCAT, FAIR+CARE and telemetry validation	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-059 p.114	The source names docs-lint, stac-validate, faircare-validate and telemetry-export workflows for visual asset governance.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-059-027	Story Node map asset registry covers static maps, renderings, overlays, grid surfaces and thematic layers	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.115-118	Map assets include basemaps, archaeological overlays, hydrology layers, climate anomaly surfaces, landcover transitions and generalized/restricted products, each with provenance and CARE classification.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-028	Map asset standards define allowed formats and projection defaults	E. Style Specification, Sources, Layers, Expressions	NEW SRC-059 pp.118-121	Visualization standards list PNG, SVG, GeoJSON, TopoJSON and metadata expectations; metadata fields include projection, spatial/temporal extent, checksum and format values like GeoTIFF, COG, PMTiles, CZML and GLB.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-029	Map asset metadata carries CARE status public/generalized/restricted	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-059 pp.120-121	Map assets require CARE blocks identifying public, generalized or restricted status, reviewers, review dates and handling notes.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-059-030	Map asset metadata must identify upstream datasets, tools, commit, STAC/DCAT linkages and generalization	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.122-127	Provenance and lineage requirements include upstream sources, processing tools, workflow commit SHA, STAC/DCAT links, checksum ledger entries and generalization methods.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-031	Map asset examples include STAC Item and COG/GeoTIFF asset references	K. Raster, COG, DEM, Terrain, Hillshade	NEW SRC-059 pp.122-128	The source provides STAC-like examples for flood maps and generalized archaeology density maps, including COG/GeoTIFF-style asset fields and checksum metadata.	Represent raster/DEM/COG/terrain assets as released, checksum-addressed derivatives.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-032	Map asset crosswalk binds KFM fields to STAC, DCAT and CIDOC CRM	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.122-127	The source introduces cross-standard mapping from KFM fields to STAC, DCAT spatial/temporal fields and CIDOC classes for heritage contexts.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-033	Path claims from source packets remain Directory-Rules pending	Y. Implementation Backlog and PR Plan	NEW SRC-059 pp.80-128	The many docs/reports/visualization paths are source-reported organization patterns; actual repo placement needs Directory Rules and mounted-repo verification before implementation.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-034	Landcover symbol system targets Focus Mode, Story Nodes, timelines and MapLibre layers	F. Sprites, Glyphs, Fonts, Design Tokens	NEW SRC-059 pp.208-224	Landcover symbols are governed assets for Focus Mode ecosystem overlays and MapLibre landcover symbol layers, with SVG glyphs, previews and accessibility rules.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-035	Landcover symbols require visual regression tests	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-059 pp.209-217	The source names validation and visual-regression tests, renderer preview generation and golden baseline images for landcover symbols.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-059-036	Symbol catalogs should use metadata keys, not filenames	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 p.224	Story Nodes and Focus Mode do not reference raw filenames directly; they use metadata keys for symbols and layers.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-037	Hydrology symbol layers need glyphs, accessibility and telemetry	F. Sprites, Glyphs, Fonts, Design Tokens	NEW SRC-059 pp.234-249	Hydrology symbols include river/watershed overlays, streamflow/gauge glyphs, accessibility rules, renderer responsibilities and telemetry logging.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-038	Hydrology symbol visual regression protects map legend consistency	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-059 pp.235-249	Hydrology symbol directories include tests and snapshot visual regression comparing preview renderings with baselines.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-059-039	Event symbols feed Focus Mode, Story Nodes, timelines and MapLibre symbol layers	F. Sprites, Glyphs, Fonts, Design Tokens	NEW SRC-059 pp.262-278	Event symbols include SVG glyphs, category rules, neutral respectful glyphs and derivatives used in Focus Mode, Story Nodes and MapLibre layers.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-040	Event symbol sprites and baselines must be stable	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-059 pp.272-278	Event symbol QA uses golden master baselines and composite sprite-sheet baselines to ensure consistent rendering across engines.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-059-041	Climate symbols support Focus Mode and MapLibre climate overlays	F. Sprites, Glyphs, Fonts, Design Tokens	NEW SRC-059 pp.292-308	Climate glyphs use preferred SVG sources, derivative sprites and QA/visual-regression gates for Focus Mode overlays and MapLibre-based interactive maps.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-042	Climate symbol assets include accessibility and recognition constraints	S. Accessibility, UX, and Trust-Visible States	NEW SRC-059 pp.293-308	Climate symbols should remain recognizable at common UI sizes, with accessibility and symbol consistency baked into the source assets.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-043	Archaeological symbols require FAIR+CARE and generalization governance	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-059 pp.320-332	Archaeological legends for Focus Mode, 2D MapLibre and 3D Cesium scenes must respect FAIR+CARE cultural governance and generalization guidelines for sensitive heritage locations.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-059-044	Archaeological symbol metadata maps to MapLibre styles, STAC/DCAT and graph nodes	F. Sprites, Glyphs, Fonts, Design Tokens	NEW SRC-059 pp.328-330	All downstream symbol catalogs, map styles, STAC/DCAT legend assets and graph nodes must map back to canonical archaeological symbol fields.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-045	Archaeological symbol schema encodes OWL-Time, culture, certainty and sensitivity	E. Style Specification, Sources, Layers, Expressions	NEW SRC-059 pp.330-331	The field table includes geometry type, feature class, period, culture, certainty, sensitivity, access tier, min/max scale and cartographic colors.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-046	Sensitive/sacred symbols must not default to full public display	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-059 pp.331-332	For sensitive or sacred archaeological symbols, public map styles must not default to full exposure; generalized or hidden tiers are required.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-047	Core Focus Mode panels need sidecar metadata and zoom-legibility checks	S. Accessibility, UX, and Trust-Visible States	NEW SRC-059 pp.349-350	Timeline context panels, entity summaries, hydrology overlays and metadata panels need sidecar metadata, checksums, provenance and legibility at 200 percent zoom.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-048	3D Focus Mode Views become governed asset family	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 pp.351-354	3D Views include Cesium terrain layers, archaeological reconstructions, hydrology elevation/anomaly surfaces and time-aware volumetric or overlay scenes, governed by FAIR+CARE, ISO, STAC/DCAT and accessibility metadata.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-049	3D asset standards include 3D Tiles, CZML, GLB/GLTF and camera path JSON	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 pp.351-352	The 3D asset standard separates terrain tilesets, geometry models, volumetric layers, camera paths and metadata assets.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-050	3D assets require generalization, alt text, checksums, STAC extensions and temporal anchoring	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 pp.352-353	3D previews and captures require alt text; 3D assets require SHA-256 checksums, STAC extensions and start/end temporal anchors.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-051	3D layers support AI explainability and 2D/3D comparison	O. Focus Mode and Governed AI Map Context	NEW SRC-059 pp.353-354	3D visual layers are used for contextual terrain orientation, historical landform change, AI explainability, side-by-side 2D/3D validation and guided public education.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-059-052	Global Terrain Tileset metadata defines STAC/DCAT-aligned 3D tile asset	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 pp.355-356	The terrain tileset example includes type=3d-tiles, checksum, CRS, provenance, sources, processing pipeline, bbox, temporal extent and license.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-053	Terrain 3D Tiles support hydrology volumes and time-aware overlays	K. Raster, COG, DEM, Terrain, Hillshade	NEW SRC-059 pp.355-357	The global terrain tileset supports base terrain for 3D scenes, geological/environmental overlays, prairie reconstruction and hydrology drought/flood anomaly volumes.	Represent raster/DEM/COG/terrain assets as released, checksum-addressed derivatives.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-054	3D terrain integrates with Time-Encoded MapLibre plus Cesium layers	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 p.357	The terrain tileset is listed as integrated with Focus Mode Story Nodes, time-encoded MapLibre + Cesium layers, hydrology and archaeology pipelines.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-055	Terrain controls require CARE masking and alt-text	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-059 p.357	When terrain is linked to archaeological locations, coordinate generalization of at least 5 km is required; all captures need alt text and metadata.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-059-056	3D terrain build pipeline needs QA before Focus Mode publication	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-059 pp.357-358	The terrain pipeline moves from raw DEM inputs through normalization, 3D Tiles conversion, LOD optimization and mesh decimation to QA/FAIR+CARE audit and Focus Mode publication.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-059-057	3D render telemetry becomes a release evidence sidecar	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW SRC-059 p.358	The source gives telemetry fields for render tests, energy, carbon and FAIR+CARE status, making performance/sustainability logging part of release evidence.	Define budgets, headers, cache rules or performance checks as release gates.	Resource timing, Range/CORS/cache, tile-load and latency tests.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-058	Cultural symbol standards require neutral accessible vector symbols	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-059 p.382	Archaeological/cultural symbols must avoid sacred symbols or tribal insignia, remain WCAG accessible, use generalized geometry and carry CARE metadata.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-059-059	Symbol metadata records cultural review and telemetry references	F. Sprites, Glyphs, Fonts, Design Tokens	NEW SRC-059 p.382	Symbol metadata includes symbol_id, category, intended use, sensitivity class, FAIR+CARE review, reviewers, license, commit SHA and telemetry reference.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-060	External Kansas source URL list is intake backlog, not verified source registry	Z. Open Questions and Verification Backlog	NEW SRC-059 pp.433-434	The pages list Kansas GIS, historic map, FEMA, drought, storm, LiDAR, wetlands, wildfire, geology and water links, but those URLs require source descriptors, rights checks and freshness verification before map use.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-061	Map popups must not replace Evidence Drawer	N. Evidence Drawer Payloads and Click Resolution	NEW SRC-059 pp.2,4,84-87	Several pages mention contextual pop-ups or narrative panels; under KFM trust law they remain cues that must resolve to EvidenceBundle/Evidence Drawer before consequential claims.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-059-062	Release manifests and SBOM refs in visualization README blocks need proof verification	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.83,88,114,119,350	Many source snippets include sbom_ref, manifest_ref, telemetry_ref and commit_sha; these are useful object references but need actual release-manifest proof before they can be treated as CONFIRMED implementation.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-063	FAIR+CARE badges are not release authority	X. Anti-Patterns and Failure Modes	NEW SRC-059 pp.80-128,351-358	The new docs use FAIR+CARE/status badges heavily; KFM must not treat badges as policy decisions or release proof without EvidenceBundle, PolicyDecision and PromotionDecision records.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-064	Static maps and screenshots remain downstream carriers	V. Exports, Screenshots, Reports, and Citation Preservation	NEW SRC-059 pp.80-82,112-118,349-350	Focus Mode screenshots, static maps and previews require checksums, provenance, alt text and citations; they do not substitute for released evidence or citation-preserving exports.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-065	3D assets are not sovereign truth	X. Anti-Patterns and Failure Modes	NEW SRC-059 pp.351-358	3D terrain, GLB/CZML scenes, camera paths and volumetric layers must remain derived visualization carriers, not canonical truth or policy authority.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-066	AI-narrated Story Nodes require evidence-bounded citations	O. Focus Mode and Governed AI Map Context	NEW SRC-059 pp.80-87,94-97	Story Nodes are AI-narrated or AI-summarizable, but their metadata structure requires provenance, citations and source datasets, reinforcing bounded synthesis.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-059-067	Map asset registry should become a first-class manifest family	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.115-128	The map asset registry functions like a TileArtifactManifest/StyleManifest extension for static maps, overlays, temporal layers, COGs, PMTiles, CZML and GLB assets.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-068	CARE generalization >=5 km recurs across map and 3D assets	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-059 pp.82,118,121,357	The source repeatedly requires generalization or masking for sensitive archaeological or cultural content, including 5 km generalization in several examples.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-069	MapAsset schema should not create parallel schema home without ADR	Y. Implementation Backlog and PR Plan	NEW SRC-059 pp.120-128	The source places schemas under docs/reports/.../schemas; Directory Rules require schema-home verification, so the object should be migrated or aliased under schemas/contracts/v1 only through ADR-compatible placement.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-070	Legend/colorbar metadata drives MapLibre and Cesium rendering consistency	E. Style Specification, Sources, Layers, Expressions	NEW SRC-059 pp.54,62	Flood legend and colorbar metadata are used by MapLibre/Leaflet flood visualization layers, Cesium 3D flood depth renderers, Focus Mode and dashboards.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-071	Flood colorbars require WCAG, STAC/DCAT and provenance metadata	S. Accessibility, UX, and Trust-Visible States	NEW SRC-059 pp.54,62	Colorbars for depth, velocity, risk and uncertainty require visual rules, accessibility compliance, contrast validation and provenance.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-072	MapLibre layer catalogs need legend metadata bindings	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.54,62	Layer manifests should reference legend/colorbar metadata so renderer styles, Evidence Drawer explanations and exports use consistent symbology.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-073	Story Node metadata must track deprecated assets	P. Time-Aware Map Interaction and Timeline	NEW SRC-059 pp.88-93	STAC versioning indicators include deprecated flags and successor/latest relationships; timeline UI should expose deprecated warnings.	Carry valid time, source time, release time and version lock in map context.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-074	Map assets may include PMTiles as a format but require hosting validation	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW SRC-059 pp.121,126,129	The map asset metadata examples list PMTiles among allowed formats; actual PMTiles delivery still requires Range/CORS, manifest and checksum tests.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-075	COG appears as allowed static/raster map asset format	K. Raster, COG, DEM, Terrain, Hillshade	NEW SRC-059 pp.121,126	COG is included as a map asset format for raster assets, with checksum, projection and provenance metadata required.	Represent raster/DEM/COG/terrain assets as released, checksum-addressed derivatives.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-076	GeoJSON/TopoJSON overlay formats must be metadata-bound	G. GeoJSON and Runtime Data	NEW SRC-059 pp.112,118	Story Node overlays can be GeoJSON or TopoJSON, but each asset still requires index metadata, checksum, provenance and CARE status.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-077	Visual regression should cover renderer parity across MapLibre, rasterizer and browser	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-059 pp.209-224,272-278	Symbol and event baselines require renderer outputs to remain consistent across MapLibre, pipeline rasterizer and browser contexts.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-059-078	Sprites and thumbnails are derivatives of authoritative SVG glyphs	F. Sprites, Glyphs, Fonts, Design Tokens	NEW SRC-059 pp.276-277,292-307	Event and climate symbol docs treat source vectors as authoritative and PNG/sprites/thumbnails as derivatives that must preserve alignment and padding.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-079	Scale-dependent archaeological symbol rendering is policy-significant	E. Style Specification, Sources, Layers, Expressions	NEW SRC-059 pp.331-332	Archaeological symbol fields include min_scale and max_scale, and sensitive/sacred tiers constrain public style defaults.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-080	Focus Mode metadata panels serve AI explainability but not proof alone	O. Focus Mode and Governed AI Map Context	NEW SRC-059 pp.349-350	Metadata panels provide provenance and dataset lineage used for AI explainability; they must still resolve to source evidence and catalog records.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-081	Terrain tileset CRS EPSG:4979 implies 3D/geodetic CRS validation	K. Raster, COG, DEM, Terrain, Hillshade	NEW SRC-059 pp.353,356	3D terrain metadata examples use EPSG:4979; validation must ensure CRS and vertical context are represented correctly before rendering.	Represent raster/DEM/COG/terrain assets as released, checksum-addressed derivatives.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-082	Camera paths are governed 3D assets	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 pp.352,357	Camera path JSON is listed as an asset type for Focus Mode cinematic navigation and must be metadata-governed like other scene assets.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-083	Volumetric hydrology/climate layers use CZML/GLB and need conditional 3D gates	W. 3D / Cesium / Deck.gl / Overlay Interoperability	NEW SRC-059 pp.352,356	Volumetric layers for hydrology and climate anomalies are proposed as CZML/GLB assets that should not become default unless evidence burden justifies 3D.	Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.	3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
ML-059-084	Temporal extent is required across map assets, Story Nodes and 3D assets	P. Time-Aware Map Interaction and Timeline	NEW SRC-059 pp.85,121,126,353,356	The source consistently attaches start/end times, named periods or temporal extents to Story Nodes, map assets and 3D tilesets.	Carry valid time, source time, release time and version lock in map context.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-085	Map asset and Story Node telemetry are audit inputs	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW SRC-059 pp.82,87,92-93,358	Telemetry records include validated node counts, render tests, energy/carbon and version events; KFM should treat them as receipts/observability, not truth.	Define budgets, headers, cache rules or performance checks as release gates.	Resource timing, Range/CORS/cache, tile-load and latency tests.
ML-059-086	Static map resolution minimum is an export quality gate	V. Exports, Screenshots, Reports, and Citation Preservation	NEW SRC-059 p.118	Story Node map assets require a minimum 2048 px width and sibling STAC metadata, strengthening export/report quality checks.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-087	Motion/animation alternatives are accessibility obligations	S. Accessibility, UX, and Trust-Visible States	NEW SRC-059 p.82	Motion or animation assets require alternatives, and text overlays/colorbars need WCAG contrast, supporting map animation accessibility gates.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-059-088	Archaeological legend fields include source_dataset_id and source_feature_codes	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-059 pp.331-332	Symbol metadata fields link symbols back to source datasets and upstream feature codes, preventing style catalogs from becoming detached from evidence.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-059-089	Map layers auto-published via STAC/DCAT still need promotion state	X. Anti-Patterns and Failure Modes	NEW SRC-059 pp.2,120-128	Source language sometimes says pipelines automatically publish map layers; KFM must interpret that as candidate release only unless PromotionDecision and release manifest are proven.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-090	Neo4j graph context for popups is downstream projection	A. Renderer Boundary and KFM Trust Law	NEW SRC-059 pp.1-4,90	Graph links and version relationships help contextual map features, but the graph projection does not replace canonical evidence or policy state.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-091	Global source URLs require SourceDescriptor onboarding	Y. Implementation Backlog and PR Plan	NEW SRC-059 pp.433-434	The URL list should be converted into SourceDescriptor records with source role, rights, cadence, expected geometry and public-release limits before map ingestion.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-092	Renderer symbols should log telemetry for traceability	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW SRC-059 pp.212,237,278	Renderer responsibilities include logging symbol usage in Focus Mode telemetry, tying style usage to observability.	Define budgets, headers, cache rules or performance checks as release gates.	Resource timing, Range/CORS/cache, tile-load and latency tests.
ML-059-093	MapLibre/Leaflet parity references need wrapper governance	R. Plugin, Wrapper, and Dependency Governance	NEW SRC-059 pp.54,62	The source mentions MapLibre/Leaflet flood visualization layers; KFM should avoid mixed renderer drift unless wrapper/adaptor choices are reviewed and tested.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-059-094	QGIS2ThreeJS and Cesium tooling are toolchain candidates, not defaults	R. Plugin, Wrapper, and Dependency Governance	NEW SRC-059 p.352	3D asset tools such as Cesium ion, GDAL, Entwine, Blender and QGIS2ThreeJS need license/security/toolchain review before admission.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-059-095	Source text contains strong implementation claims that remain NEEDS VERIFICATION	Z. Open Questions and Verification Backlog	NEW SRC-059 pp.1-6,80-128,351-358	Several passages describe active or implemented KFM behavior; without mounted repo, tests, logs or release artifacts, current implementation maturity stays UNKNOWN or NEEDS VERIFICATION.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-060-001	Low-cost PM2.5 layer calibration cannot be static	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-060 pp.1-6	Static calibration fails under humidity, temperature, aging, drift, unit variability and context shift; public air layers need calibration status and uncertainty visible.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-060-002	Sensor trust scores can drive map reliability badges	S. Accessibility, UX, and Trust-Visible States	NEW SRC-060 pp.1-2	Sensor trust scores integrate accuracy, stability, responsiveness and consensus alignment; map UI can display trust-visible sensor states without making them proof.	Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.	Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
ML-060-003	High-trust versus low-trust sensor correction complexity	O. Focus Mode and Governed AI Map Context	NEW SRC-060 pp.1-2	High-trust sensors get minimal correction while low-trust sensors require stronger correction; Focus Mode explanations should expose correction logic and avoid overconfident answers.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-060-004	Meteorology-aware ML features strengthen air surface manifests	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-060 pp.2-5	Relative humidity, temperature, wind, pressure, time features and lags are high-value predictors that should be reflected in air-layer manifests and model provenance.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-060-005	Co-location windows become validation metadata	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 pp.3-6	The source emphasizes 2-6 week co-location windows with environmental diversity; air-source validators should require co-location evidence or mark NEEDS VERIFICATION.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-060-006	Kansas/Midwest humidity transferability is a map-layer caveat	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-060 p.4	Humidity and aerosol-regime transferability constraints should appear as caveats in air layer metadata, tooltips and Evidence Drawer payloads.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-060-007	Deep-learning calibration has interpretability risks	O. Focus Mode and Governed AI Map Context	NEW SRC-060 pp.4-5	Large-scale DL calibration can help with huge datasets but may be risky under out-of-distribution aerosols or when interpretability is required.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-060-008	Hybrid PM2.5 calibration combines meteorology-aware ML with adaptive trust layer	O. Focus Mode and Governed AI Map Context	NEW SRC-060 pp.5-6	The most stable pattern is a reference-anchored meteorology-aware correction followed by trust-aware adaptive recalibration for dense sensor networks.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-060-009	Air map artifacts should separate observed, corrected and modeled values	A. Renderer Boundary and KFM Trust Law	NEW SRC-060 pp.1-6	The calibration material reinforces that raw sensor readings, corrected PM2.5 estimates and modeled fields are distinct knowledge characters that map surfaces must not collapse.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-060-010	kfm:array metadata block can reduce probing requests for array/raster clients	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-060 pp.8-11	A KFM-namespaced STAC Asset array metadata block can advertise chunk_shape, shape, dtype, dimension names, nodata and storage hints for Zarr/reference assets.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-060-011	Multiscale Zarr/GeoZarr overviews affect map zoom strategy	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW SRC-060 pp.8-11	Multiscale stores can expose zoom levels as child groups with tile matrix set and resampling metadata, reducing client probing and improving map IO planning.	Define budgets, headers, cache rules or performance checks as release gates.	Resource timing, Range/CORS/cache, tile-load and latency tests.
ML-060-012	One-store vs many-store vs virtual-reference assets is a release strategy decision	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	NEW SRC-060 pp.9-11	Catalog-visible shapes for multidimensional raster/array data include one multiscale store, multiple resolution stores or virtual dataset reference assets; each has release/rollback consequences.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-060-013	EOPF-style asset layout supports complex scene products	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-060 pp.10-11	Complex products should expose product assets, metadata JSON assets and per-measurement-group assets, providing a pattern for KFM scene or EO products.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-060-014	STAC-in-Zarr reduces bespoke mapping but creates staleness risk	T. Performance, Caching, CDN, Range Requests, and Resource Timing	NEW SRC-060 pp.10-11	Embedding STAC-like metadata in Zarr helps self-describing stores but can go stale if stores mutate independently; versioning/attestation mitigates the risk.	Define budgets, headers, cache rules or performance checks as release gates.	Resource timing, Range/CORS/cache, tile-load and latency tests.
ML-060-015	Zarr array metadata requires CI schema and source-metadata matching checks	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 p.11	The adoption playbook requires schema validation plus checks that kfm:array matches zarr.json/.zarray metadata.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-060-016	Neo4j migration is a governed transformation event	A. Renderer Boundary and KFM Trust Law	NEW SRC-060 pp.11-18	Graph migrations are described as PROV Activities with declared intent, invariants, blast radius, provenance, reversibility and governance constraints, not ordinary DDL.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-060-017	Graph migration rollback artifacts include pre-state checksums and deterministic ID maps	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 pp.16-18	Every migration should generate pre-state checksum, mutation log, deterministic ID map and tested rollback Cypher; this protects Evidence Drawer graph projections.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-060-018	Graph-diff Story Nodes can explain migrations to reviewers	O. Focus Mode and Governed AI Map Context	NEW SRC-060 p.18	The source proposes graph-diff visual summaries and Focus Mode Story Nodes explaining migrations to reviewers, turning changes into inspectable narrative surfaces.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-060-019	WZDx/ArcGIS/GTFS fusion forms transport incident map layer source pattern	J. Martin and Server-Mediated Tile Serving	NEW SRC-060 pp.18-21	A multi-source incident/work-zone pipeline combines WZDx GeoJSON, ArcGIS Feature Service extractChanges and GTFS-Realtime into deduplicated, trust-ranked promotable incident events.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-060-020	Incident observation envelope includes canonical geometry and source lineage	G. GeoJSON and Runtime Data	NEW SRC-060 pp.19-21	The incident event model includes canonical event_id, source IDs, geometry, road reference, times, event type, severity, status, per-source observed_at/publisher/license/raw_ref, trust score and promotion state.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-060-021	WZDx snapshot feeds need hash-derived deltas	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 p.22	WZDx full-snapshot GeoJSON feeds should be validated, normalized, hashed per feature and compared with last snapshot to produce insert/update/delete candidates.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.

idea_id	idea title	category	status / source	short normalized statement	implementation consequence	validation consequence
ML-060-022	ArcGIS extractChanges requires stored watermarks	L. WMS / WMTS / External Map Services	NEW SRC-060 p.22	ArcGIS change tracking uses last generation number, layer ID, since_time and per-feature GlobalID/ObjectID mapping; delete events are high-signal cleared/removed events.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
ML-060-023	GTFS-Realtime staleness gates affect incident publication	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-060 pp.22-24	GTFS-RT alerts/trip updates/vehicle positions can corroborate incidents, but parsers must handle FULL_DATASET/DIFFERENTIAL semantics and staleness gates before map publication.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-060-024	Incident dedup uses deterministic signatures plus spatial-temporal clustering	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 pp.23-24	Deduplication uses normalized event type, roadway reference, H3/geohash geometry, time buckets, spatial/temporal thresholds and semantic equivalence classes.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-060-025	Incident QC gates include GeoJSON validity, timestamps, monotonic updates and source integrity	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 p.24	Geometry validity, timestamp sanity, update monotonicity, WZDx schema validation and ArcGIS change-tracking checks prevent bad incident layers from publishing.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-060-026	Incident trust tiers and promotion states shape public transport layers	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	NEW SRC-060 pp.25-26	Tier A ArcGIS, Tier B WZDx, Tier C GTFS-RT and lower tiers feed promotion states Candidate, Corroborated, Confirmed and Published with unresolved conflicts blocking release.	Apply policy gates before any public layer, style, popup, screenshot or export uses this content.	Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
ML-060-027	Incident conflicts preserve all observations before canonical status selection	N. Evidence Drawer Payloads and Click Resolution	NEW SRC-060 pp.25-26	When sources disagree, both observations are retained; trust tier and recency choose canonical status, and Evidence Drawer should expose lingering/conflicting signals.	Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.	Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
ML-060-028	Incident output streams need lineage records	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	NEW SRC-060 p.26	The pipeline publishes incident_events, event_updates and event_lineage records; map layers should depend on the lineage stream, not just a flattened event table.	Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.	Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
ML-060-029	Deterministic data run uses lakeFS, DVC, LangGraph, OPA and Cosign as proof pattern	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 pp.28-36	The prototype ingests a JSON feed into a lakeFS branch, runs deterministic DVC transform orchestrated by LangGraph, gates with OPA/Rego and Cosign, and rolls back on failure.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-060-030	OPA gate failure triggers rollback to known good data state	U. Testing, CI, Validation, Rollback, and Proof Objects	NEW SRC-060 pp.34-36	The workflow computes result metadata, verifies attestation, evaluates policy and reverts the lakeFS branch to a known good ref if the gate fails.	Add fixture-driven validator or CI check before promotion.	CI fixture must pass/fail deterministically; include rollback and negative-state cases.
ML-060-031	PM2.5 calibration and incident fusion are not direct MapLibre claims unless manifested	X. Anti-Patterns and Failure Modes	NEW SRC-060 pp.1-36	The new source contains high-value calibration and incident patterns, but they become MapLibre-relevant only when connected to layer manifests, Evidence Drawer payloads, public-safe policies and release artifacts.	Bind this idea to governed API, manifest, evidence, policy and test objects before release.	Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.

6. Category Atlas

The category atlas preserves the v1.5 structure and adds v1.6 deltas where the new source packets provide evidence. Categories with zero v1.6 records are retained and remain available in Appendix A.

Category	v1.6 delta	Representative new ideas	Current status
A. Renderer Boundary and KFM Trust Law	5	ML-059-003, ML-059-007, ML-059-090, ML-060-009, ML-060-016	EXPANDED
B. MapLibre GL JS Web Shell	2	ML-059-002, ML-059-006	EXPANDED
C. MapLibre Native / Mobile / Platform Parity	0	No new direct records; see Appendix A.	RETAINED
D. MapLibre RS / Experimental Renderers	0	No new direct records; see Appendix A.	RETAINED
E. Style Specification, Sources, Layers, Expressions	4	ML-059-028, ML-059-045, ML-059-070, ML-059-079	EXPANDED
F. Sprites, Glyphs, Fonts, Design Tokens	7	ML-059-034, ML-059-037, ML-059-039, ML-059-041, ML-059-044, ML-059-059 ...	EXPANDED
G. GeoJSON and Runtime Data	3	ML-059-008, ML-059-076, ML-060-020	EXPANDED
H. Vector Tiles: MVT and MLT	0	No new direct records; see Appendix A.	RETAINED
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	2	ML-059-074, ML-060-012	EXPANDED
J. Martin and Server-Mediated Tile Serving	1	ML-060-019	EXPANDED
K. Raster, COG, DEM, Terrain, Hillshade	6	ML-059-001, ML-059-005, ML-059-031, ML-059-053, ML-059-075, ML-059-081	EXPANDED
L. WMS / WMTS / External Map Services	1	ML-060-022	EXPANDED
M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	16	ML-059-012, ML-059-017, ML-059-023, ML-059-025, ML-059-027, ML-059-030 ...	EXPANDED
N. Evidence Drawer Payloads and Click Resolution	3	ML-059-018, ML-059-061, ML-060-027	EXPANDED
O. Focus Mode and Governed AI Map Context	10	ML-059-010, ML-059-013, ML-059-020, ML-059-051, ML-059-066, ML-059-080 ...	EXPANDED
P. Time-Aware Map Interaction and Timeline	5	ML-059-004, ML-059-009, ML-059-016, ML-059-073, ML-059-084	EXPANDED
Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	11	ML-059-015, ML-059-029, ML-059-043, ML-059-046, ML-059-055, ML-059-058 ...	EXPANDED
R. Plugin, Wrapper, and Dependency Governance	2	ML-059-093, ML-059-094	EXPANDED
S. Accessibility, UX, and Trust-Visible States	7	ML-059-011, ML-059-019, ML-059-042, ML-059-047, ML-059-071, ML-059-087 ...	EXPANDED
T. Performance, Caching, CDN, Range Requests, and Resource Timing	5	ML-059-057, ML-059-085, ML-059-092, ML-060-011, ML-060-014	EXPANDED
U. Testing, CI, Validation, Rollback, and Proof Objects	15	ML-059-021, ML-059-026, ML-059-035, ML-059-038, ML-059-040, ML-059-056 ...	EXPANDED
V. Exports, Screenshots, Reports, and Citation Preservation	4	ML-059-022, ML-059-024, ML-059-064, ML-059-086	EXPANDED
W. 3D / Cesium / Deck.gl / Overlay Interoperability	8	ML-059-014, ML-059-048, ML-059-049, ML-059-050, ML-059-052, ML-059-054 ...	EXPANDED
X. Anti-Patterns and Failure Modes	4	ML-059-063, ML-059-065, ML-059-089, ML-060-031	EXPANDED
Y. Implementation Backlog and PR Plan	3	ML-059-033, ML-059-069, ML-059-091	EXPANDED
Z. Open Questions and Verification Backlog	2	ML-059-060, ML-059-095	EXPANDED

7. Detailed Category Chapters

A. Renderer Boundary and KFM Trust Law

Separates renderer behavior from truth, evidence, policy, source authority, graph projection and publication authority.

New / expanded evidence: ML-059-003 SRC-059 pp.2-3; ML-059-007 SRC-059 p.4; ML-059-090 SRC-059 pp.1-4,90; ML-060-009 SRC-060 pp.1-6; ML-060-016 SRC-060 pp.11-18

idea_id	title	evidence / statement
ML-059-003	Hazard event layers are analytic context, not emergency alerting	SRC-059 pp.2-3: Hazard warnings, declarations, earthquakes and wildfire detections can appear on map/timeline surfaces, but KFM is not an operational alert system.
ML-059-007	Modeled air fields must stay distinct from observations	SRC-059 p.4: CAMS model fields are described as separate assets or graph nodes linked by DERIVED_FROM instead of being mixed with raw observations.
ML-059-090	Neo4j graph context for popups is downstream projection	SRC-059 pp.1-4,90: Graph links and version relationships help contextual map features, but the graph projection does not replace canonical evidence or policy state.
ML-060-009	Air map artifacts should separate observed, corrected and modeled values	SRC-060 pp.1-6: The calibration material reinforces that raw sensor readings, corrected PM2.5 estimates and modeled fields are distinct knowledge characters that map surfaces must not collapse.
ML-060-016	Neo4j migration is a governed transformation event	SRC-060 pp.11-18: Graph migrations are described as PROV Activities with declared intent, invariants, blast radius, provenance, reversibility and governance constraints, not ordinary DDL.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 5 current-run record(s) from SRC-059/SRC-060.

B. MapLibre GL JS Web Shell

Defines what the web map shell renders and how it exposes camera, time, layer and feature-selection context.

New / expanded evidence: ML-059-002 SRC-059 p.2; ML-059-006 SRC-059 p.4

idea_id	title	evidence / statement
ML-059-002	Hillshade and flow accumulation MapLibre toggles	SRC-059 p.2: The source explicitly describes hillshade or flow accumulation layers being toggled in a MapLibre-based frontend, with river-discharge time series plotted over the historical timeline.
ML-059-006	Air observations become map/tooltips or aggregated daily surfaces	SRC-059 p.4: Air quality outputs can appear as sensor-location tooltips or aggregated pollution surfaces by day and can be incorporated into Focus Mode narratives.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 2 current-run record(s) from SRC-059/SRC-060.

C. MapLibre Native / Mobile / Platform Parity

Tracks mobile/native parity. No new direct evidence in this run; v1.5 entries remain retained.

No new direct evidence in SRC-059 or SRC-060. Category retained from SRC-MASTER-PREV; see Appendix A for the full prior chapter, idea records, tests and open items.

D. MapLibre RS / Experimental Renderers

Tracks experimental renderers and non-default rendering paths. No new direct evidence in this run.

No new direct evidence in SRC-059 or SRC-060. Category retained from SRC-MASTER-PREV; see Appendix A for the full prior chapter, idea records, tests and open items.

E. Style Specification, Sources, Layers, Expressions

Governs style, symbol semantics, source-layer references, scale ranges, filters and expressions.

New / expanded evidence: ML-059-028 SRC-059 pp.118-121; ML-059-045 SRC-059 pp.330-331; ML-059-070 SRC-059 pp.54,62; ML-059-079 SRC-059 pp.331-332

idea_id	title	evidence / statement
ML-059-028	Map asset standards define allowed formats and projection defaults	SRC-059 pp.118-121: Visualization standards list PNG, SVG, GeoJSON, TopoJSON and metadata expectations; metadata fields include projection, spatial/temporal extent, checksum and format values like GeoTIFF, COG, PMTiles, CZML and GLB.
ML-059-045	Archaeological symbol schema encodes OWL-Time, culture, certainty and sensitivity	SRC-059 pp.330-331: The field table includes geometry type, feature class, period, culture, certainty, sensitivity, access tier, min/max scale and cartographic colors.
ML-059-070	Legend/colorbar metadata drives MapLibre and Cesium rendering consistency	SRC-059 pp.54,62: Flood legend and colorbar metadata are used by MapLibre/Leaflet flood visualization layers, Cesium 3D flood depth renderers, Focus Mode and dashboards.
ML-059-079	Scale-dependent archaeological symbol rendering is policy-significant	SRC-059 pp.331-332: Archaeological symbol fields include min_scale and max_scale, and sensitive/sacred tiers constrain public style defaults.

- Required objects / contracts: Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
- Changed since v1.5: EXPANDED by 4 current-run record(s) from SRC-059/SRC-060.

F. Sprites, Glyphs, Fonts, Design Tokens

Governs icons, glyphs, sprites, colorbars, symbols and design tokens as metadata-bound artifacts.

New / expanded evidence: ML-059-034 SRC-059 pp.208-224; ML-059-037 SRC-059 pp.234-249; ML-059-039 SRC-059 pp.262-278; ML-059-041 SRC-059 pp.292-308; ML-059-044 SRC-059 pp.328-330; ML-059-059 SRC-059 p.382; ML-059-078 SRC-059 pp.276-277,292-307

idea_id	title	evidence / statement
ML-059-034	Landcover symbol system targets Focus Mode, Story Nodes, timelines and MapLibre layers	SRC-059 pp.208-224: Landcover symbols are governed assets for Focus Mode ecosystem overlays and MapLibre landcover symbol layers, with SVG glyphs, previews and accessibility rules.
ML-059-037	Hydrology symbol layers need glyphs, accessibility and telemetry	SRC-059 pp.234-249: Hydrology symbols include river/watershed overlays, streamflow/gauge glyphs, accessibility rules, renderer responsibilities and telemetry logging.
ML-059-039	Event symbols feed Focus Mode, Story Nodes, timelines and MapLibre symbol layers	SRC-059 pp.262-278: Event symbols include SVG glyphs, category rules, neutral respectful glyphs and derivatives used in Focus Mode, Story Nodes and MapLibre layers.
ML-059-041	Climate symbols support Focus Mode and MapLibre climate overlays	SRC-059 pp.292-308: Climate glyphs use preferred SVG sources, derivative sprites and QA/visual-regression gates for Focus Mode overlays and MapLibre-based interactive maps.
ML-059-044	Archaeological symbol metadata maps to MapLibre styles, STAC/DCAT and graph nodes	SRC-059 pp.328-330: All downstream symbol catalogs, map styles, STAC/DCAT legend assets and graph nodes must map back to canonical archaeological symbol fields.
ML-059-059	Symbol metadata records cultural review and telemetry references	SRC-059 p.382: Symbol metadata includes symbol_id, category, intended use, sensitivity class, FAIR+CARE review, reviewers, license, commit SHA and telemetry reference.
ML-059-078	Sprites and thumbnails are derivatives of authoritative SVG glyphs	SRC-059 pp.276-277,292-307: Event and climate symbol docs treat source vectors as authoritative and PNG/sprites/thumbnails as derivatives that must preserve alignment and padding.

- Required objects / contracts: Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.

- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
- Changed since v1.5: EXPANDED by 7 current-run record(s) from SRC-059/SRC-060.

G. GeoJSON and Runtime Data

Governs runtime vectors, overlays and normalized event/feature data used by map layers.

New / expanded evidence: ML-059-008 SRC-059 pp.5-6; ML-059-076 SRC-059 pp.112,118; ML-060-020 SRC-060 pp.19-21

idea_id	title	evidence / statement
ML-059-008	Vector ETL standardizes GeoParquet/GeoJSON WGS84 outputs	SRC-059 pp.5-6: Vector datasets such as parcels and boundaries are harmonized, reprojected to WGS84, and exported to GeoParquet or GeoJSON before catalog/API/map use.
ML-059-076	GeoJSON/TopoJSON overlay formats must be metadata-bound	SRC-059 pp.112,118: Story Node overlays can be GeoJSON or TopoJSON, but each asset still requires index metadata, checksum, provenance and CARE status.
ML-060-020	Incident observation envelope includes canonical geometry and source lineage	SRC-060 pp.19-21: The incident event model includes canonical event_id, source IDs, geometry, road reference, times, event type, severity, status, per-source observed_at/publisher/license/raw_ref, trust score and promotion state.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 3 current-run record(s) from SRC-059/SRC-060.

H. Vector Tiles: MVT and MLT

Tracks vector tile formats and source-layer requirements. No new direct MVT/MLT records in this run.

No new direct evidence in SRC-059 or SRC-060. Category retained from SRC-MASTER-PREV; see Appendix A for the full prior chapter, idea records, tests and open items.

I. PMTiles, MBTiles, Static Tiles, Serverless Delivery

Governs static tile-like delivery, PMTiles, archive strategy and asset formats.

New / expanded evidence: ML-059-074 SRC-059 pp.121,126,129; ML-060-012 SRC-060 pp.9-11

idea_id	title	evidence / statement
ML-059-074	Map assets may include PMTiles as a format but require hosting validation	SRC-059 pp.121,126,129: The map asset metadata examples list PMTiles among allowed formats; actual PMTiles delivery still requires Range/CORS, manifest and checksum tests.
ML-060-012	One-store vs many-store vs virtual-reference assets is a release strategy decision	SRC-060 pp.9-11: Catalog-visible shapes for multidimensional raster/array data include one multiscale store, multiple resolution stores or virtual dataset reference assets; each has release/rollback consequences.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 2 current-run record(s) from SRC-059/SRC-060.

J. Martin and Server-Mediated Tile Serving

Governs server-mediated or dynamic tile services and source fusion serving patterns.

New / expanded evidence: ML-060-019 SRC-060 pp.18-21

idea_id	title	evidence / statement
ML-060-019	WZDx/ArcGIS/GTFS fusion forms transport incident map layer source pattern	SRC-060 pp.18-21: A multi-source incident/work-zone pipeline combines WZDx GeoJSON, ArcGIS Feature Service extractChanges and GTFS-Realtime into deduplicated, trust-ranked promotable incident events.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 1 current-run record(s) from SRC-059/SRC-060.

K. Raster, COG, DEM, Terrain, Hillshade

Governs COGs, terrain, DEMs, hillshade and raster/array map products.

New / expanded evidence: ML-059-001 SRC-059 pp.1-2; ML-059-005 SRC-059 p.3; ML-059-031 SRC-059 pp.122-128; ML-059-053 SRC-059 pp.355-357; ML-059-075 SRC-059 pp.121,126; ML-059-081 SRC-059 pp.353,356

idea_id	title	evidence / statement
ML-059-001	Hydrology COG/Parquet assets feed released map layers	SRC-059 pp.1-2: Hydrology outputs are described as COG rasters or Parquet/GeoParquet/GeoJSON products registered in STAC and visualized as timeline map layers, including MapLibre frontend examples.
ML-059-005	Climate anomaly layers become hazard overlays	SRC-059 p.3: Climate anomaly outputs can serve as drought, heat or precipitation hazard overlays, with AI detection context kept as an overlay rather than root truth.
ML-059-031	Map asset examples include STAC Item and COG/GeoTIFF asset references	SRC-059 pp.122-128: The source provides STAC-like examples for flood maps and generalized archaeology density maps, including COG/GeoTIFF-style asset fields and checksum metadata.
ML-059-053	Terrain 3D Tiles support hydrology volumes and time-aware overlays	SRC-059 pp.355-357: The global terrain tileset supports base terrain for 3D scenes, geological/environmental overlays, prairie reconstruction and hydrology drought/flood anomaly volumes.
ML-059-075	COG appears as allowed static/raster map asset format	SRC-059 pp.121,126: COG is included as a map asset format for raster assets, with checksum, projection and provenance metadata required.
ML-059-081	Terrain tileset CRS EPSG:4979 implies 3D/geodetic CRS validation	SRC-059 pp.353,356: 3D terrain metadata examples use EPSG:4979; validation must ensure CRS and vertical context are represented correctly before rendering.

- Required objects / contracts: Represent raster/DEM/COG/terrain assets as released, checksum-addressed derivatives.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 6 current-run record(s) from SRC-059/SRC-060.

L. WMS / WMTS / External Map Services

Governs external services, ArcGIS extractChanges and feed/source-service boundaries.

New / expanded evidence: ML-060-022 SRC-060 p.22

idea_id	title	evidence / statement
ML-060-022	ArcGIS extractChanges requires stored watermarks	SRC-060 p.22: ArcGIS change tracking uses last generation number, layer ID, since_time and per-feature GlobalID/ObjectID mapping; delete events are high-signal cleared/removed events.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.

- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 1 current-run record(s) from SRC-059/SRC-060.

M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest

Governs release manifests, sidecars, object fields, source/provenance linkages and rollback references.

New / expanded evidence: ML-059-012 SRC-059 pp.81-82; ML-059-017 SRC-059 pp.89-90; ML-059-023 SRC-059 pp.94-97; ML-059-025 SRC-059 pp.112-114; ML-059-027 SRC-059 pp.115-118; ML-059-030 SRC-059 pp.122-127; ML-059-032 SRC-059 pp.122-127; ML-059-036 SRC-059 p.224; plus 8 more delta records.

idea_id	title	evidence / statement
ML-059-012	Visualization assets require metadata sidecars	SRC-059 pp.81-82: Focus Mode visualizations require associated metadata records with IDs, data sources, FAIR/CARE status, asset files, checksums and generation timestamps.
ML-059-017	Story Node schema needs version and lineage fields	SRC-059 pp.89-90: Story Node schema additions include version, deprecated flag and predecessor/successor/latest lineage references.
ML-059-023	Story Node metadata model includes spatial, temporal, entity, provenance and governance fields	SRC-059 pp.94-97: The metadata specification defines Story Nodes as atomic narrative units with geometry/bbox/CRS, temporal intervals, entities, provenance, citations, lineage, CARE and visualization links.
ML-059-025	Story Node assets root requires indexed maps, legends, overlays, 3D scenes and thumbnails	SRC-059 pp.112-114: Every asset referenced by a Story Node must have an assets_index record with path, checksum, provenance, citations, CARE status and update timestamp.
ML-059-027	Story Node map asset registry covers static maps, renderings, overlays, grid surfaces and thematic layers	SRC-059 pp.115-118: Map assets include basemaps, archaeological overlays, hydrology layers, climate anomaly surfaces, landcover transitions and generalized/restricted products, each with provenance and CARE classification.
ML-059-030	Map asset metadata must identify upstream datasets, tools, commit, STAC/DCAT linkages and generalization	SRC-059 pp.122-127: Provenance and lineage requirements include upstream sources, processing tools, workflow commit SHA, STAC/DCAT links, checksum ledger entries and generalization methods.
ML-059-032	Map asset crosswalk binds KFM fields to STAC, DCAT and CIDOC CRM	SRC-059 pp.122-127: The source introduces cross-standard mapping from KFM fields to STAC, DCAT spatial/temporal fields and CIDOC classes for heritage contexts.
ML-059-036	Symbol catalogs should use metadata keys, not filenames	SRC-059 p.224: Story Nodes and Focus Mode do not reference raw filenames directly; they use metadata keys for symbols and layers.
ML-059-062	Release manifests and SBOM refs in visualization README blocks need proof verification	SRC-059 pp.83,88,114,119,350: Many source snippets include sbom_ref, manifest_ref, telemetry_ref and commit_sha; these are useful object references but need actual release-manifest proof before they can be treated as CONFIRMED implementation.
ML-059-067	Map asset registry should become a first-class manifest family	SRC-059 pp.115-128: The map asset registry functions like a TileArtifactManifest/StyleManifest extension for static maps, overlays, temporal layers, COGs, PMTiles, CZML and GLB assets.

6 additional records are included in the Master Idea Index above.

- Required objects / contracts: Extend manifest/object-family fields; keep paths PROPOSED pending Directory Rules and repo verification.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Schema validate manifest/sidecar; verify checksums, source IDs, release state and rollback target.
- Changed since v1.5: EXPANDED by 16 current-run record(s) from SRC-059/SRC-060.

N. Evidence Drawer Payloads and Click Resolution

Governs how clicked/rendered features resolve into evidence-bearing payloads.

New / expanded evidence: ML-059-018 SRC-059 pp.90-91; ML-059-061 SRC-059 pp.2,4,84-87; ML-060-027 SRC-060 pp.25-26

idea_id	title	evidence / statement
ML-059-018	Lineage and diff endpoints support Story Node version inspection	SRC-059 pp.90-91: The source proposes endpoints for /lineage and /diff that return predecessor/current/successor/latest objects and metadata/asset/geometry diffs.
ML-059-061	Map popups must not replace Evidence Drawer	SRC-059 pp.2,4,84-87: Several pages mention contextual pop-ups or narrative panels; under KFM trust law they remain cues that must resolve to EvidenceBundle/Evidence Drawer before consequential claims.
ML-060-027	Incident conflicts preserve all observations before canonical status selection	SRC-060 pp.25-26: When sources disagree, both observations are retained; trust tier and recency choose canonical status, and Evidence Drawer should expose lingering/conflicting signals.

- Required objects / contracts: Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
- Changed since v1.5: EXPANDED by 3 current-run record(s) from SRC-059/SRC-060.

O. Focus Mode and Governed AI Map Context

Governs evidence-bounded synthesis, Story Nodes, version locks and AI interpretive context.

New / expanded evidence: ML-059-010 SRC-059 pp.80-82; ML-059-013 SRC-059 pp.83-86; ML-059-020 SRC-059 pp.91-92; ML-059-051 SRC-059 pp.353-354; ML-059-066 SRC-059 pp.80-87,94-97; ML-059-080 SRC-059 pp.349-350; ML-060-003 SRC-060 pp.1-2; ML-060-007 SRC-060 pp.4-5; plus 2 more delta records.

idea_id	title	evidence / statement
ML-059-010	Focus Mode visualization assets are governed outputs	SRC-059 pp.80-82: Focus Mode stores rendered outputs, UI captures, 3D perspectives and layered overlays with FAIR+CARE, ISO metadata and reproducibility requirements.
ML-059-013	Story Nodes unify narrative, spatial, temporal, provenance and CARE metadata	SRC-059 pp.83-86: Story Nodes are described as explainable links between datasets, scenes, maps, timeline frames and AI-generated insights, with spatial footprints, temporal coverage, provenance, citations and CARE flags.
ML-059-020	Focus Mode version lock freezes map layers and panels	SRC-059 pp.91-92: When a version is selected, Focus Mode freezes map layers to that version, disables auto-updating datasets, locks panels and emits a version_locked telemetry event.
ML-059-051	3D layers support AI explainability and 2D/3D comparison	SRC-059 pp.353-354: 3D visual layers are used for contextual terrain orientation, historical landform change, AI explainability, side-by-side 2D/3D validation and guided public education.
ML-059-066	AI-narrated Story Nodes require evidence-bounded citations	SRC-059 pp.80-87,94-97: Story Nodes are AI-narrated or AI-summarizable, but their metadata structure requires provenance, citations and source datasets, reinforcing bounded synthesis.
ML-059-080	Focus Mode metadata panels serve AI explainability but not proof alone	SRC-059 pp.349-350: Metadata panels provide provenance and dataset lineage used for AI explainability; they must still resolve to source evidence and catalog records.
ML-060-003	High-trust versus low-trust sensor correction complexity	SRC-060 pp.1-2: High-trust sensors get minimal correction while low-trust sensors require stronger correction; Focus Mode explanations should expose correction logic and avoid overconfident answers.
ML-060-007	Deep-learning calibration has interpretability risks	SRC-060 pp.4-5: Large-scale DL calibration can help with huge datasets but may be risky under out-of-distribution aerosols or when interpretability is required.
ML-060-008	Hybrid PM2.5 calibration combines meteorology-aware ML with adaptive trust layer	SRC-060 pp.5-6: The most stable pattern is a reference-anchored meteorology-aware correction followed by trust-aware adaptive recalibration for dense sensor networks.
ML-060-018	Graph-diff Story Nodes can explain migrations to reviewers	SRC-060 p.18: The source proposes graph-diff visual summaries and Focus Mode Story Nodes explaining migrations to reviewers, turning changes into inspectable narrative surfaces.

- Required objects / contracts: Resolve EvidenceBundle before drawer or Focus Mode output; no rendered-feature-only answer.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Click-to-drawer and Focus Mode answer/abstain tests with citation validation.
- Changed since v1.5: EXPANDED by 10 current-run record(s) from SRC-059/SRC-060.

P. Time-Aware Map Interaction and Timeline

Governs timelines, versioning, temporal extents, time locks and historical navigation.

New / expanded evidence: ML-059-004 SRC-059 p.3; ML-059-009 SRC-059 pp.5-6; ML-059-016 SRC-059 pp.88-93; ML-059-073 SRC-059 pp.88-93; ML-059-084 SRC-059 pp.85,121,126,353,356

idea_id	title	evidence / statement
ML-059-004	Flood-resilience overlays combine hazards with roads and parcels	SRC-059 p.3: The corpus describes overlaying flood simulation outputs with roads, farmland parcels and infrastructure to examine cutoffs and inundation impacts as cross-layer analysis.
ML-059-009	Temporal land ownership animation remains evidence-bound	SRC-059 pp.5-6: Historical land patents and plat boundaries are positioned as time-aware map layers that can animate settlement or homesteading waves, but detailed temporal cadastre remains a future expansion.
ML-059-016	STAC versioning becomes a Focus Mode reproducibility primitive	SRC-059 pp.88-93: STAC version fields, predecessor/successor/latest links, lineage APIs, diffs, version locks and telemetry are wired into Story Node and Focus Mode flows.
ML-059-073	Story Node metadata must track deprecated assets	SRC-059 pp.88-93: STAC versioning indicators include deprecated flags and successor/latest relationships; timeline UI should expose deprecated warnings.
ML-059-084	Temporal extent is required across map assets, Story Nodes and 3D assets	SRC-059 pp.85,121,126,353,356: The source consistently attaches start/end times, named periods or temporal extents to Story Nodes, map assets and 3D tilesets.

- Required objects / contracts: Carry valid time, source time, release time and version lock in map context.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 5 current-run record(s) from SRC-059/SRC-060.

Q. Sensitive Geometry, Geoprivacy, Rights, and Policy

Governs exact geometry denial, CARE, rights, sensor trust, review and public-safe transforms.

New / expanded evidence: ML-059-015 SRC-059 pp.86-87; ML-059-029 SRC-059 pp.120-121; ML-059-043 SRC-059 pp.320-332; ML-059-046 SRC-059 pp.331-332; ML-059-055 SRC-059 p.357; ML-059-058 SRC-059 p.382; ML-059-068 SRC-059 pp.82,118,121,357; ML-060-001 SRC-060 pp.1-6; plus 3 more delta records.

idea_id	title	evidence / statement
ML-059-015	Story Nodes enforce context-only spatial disclosure for sensitive narratives	SRC-059 pp.86-87: The ethics table says provenance, citations and lineage are required, while precise site disclosure is avoided through context-only spatial footprints.
ML-059-029	Map asset metadata carries CARE status public/generalized/restricted	SRC-059 pp.120-121: Map assets require CARE blocks identifying public, generalized or restricted status, reviewers, review dates and handling notes.
ML-059-043	Archaeological symbols require FAIR+CARE and generalization governance	SRC-059 pp.320-332: Archaeological legends for Focus Mode, 2D MapLibre and 3D Cesium scenes must respect FAIR+CARE cultural governance and generalization guidelines for sensitive heritage locations.
ML-059-046	Sensitive/sacred symbols must not default to full public display	SRC-059 pp.331-332: For sensitive or sacred archaeological symbols, public map styles must not default to full exposure; generalized or hidden tiers are required.
ML-059-055	Terrain controls require CARE masking and alt-text	SRC-059 p.357: When terrain is linked to archaeological locations, coordinate generalization of at least 5 km is required; all captures need alt text and metadata.
ML-059-058	Cultural symbol standards require neutral accessible vector symbols	SRC-059 p.382: Archaeological/cultural symbols must avoid sacred symbols or tribal insignia, remain WCAG accessible, use generalized geometry and carry CARE metadata.
ML-059-068	CARE generalization >=5 km recurs across map and 3D assets	SRC-059 pp.82,118,121,357: The source repeatedly requires generalization or masking for sensitive archaeological or cultural content, including 5 km generalization in several examples.
ML-060-001	Low-cost PM2.5 layer calibration cannot be static	SRC-060 pp.1-6: Static calibration fails under humidity, temperature, aging, drift, unit variability and context shift; public air layers need calibration status and uncertainty visible.
ML-060-006	Kansas/Midwest humidity transferability is a map-layer caveat	SRC-060 p.4: Humidity and aerosol-regime transferability constraints should appear as caveats in air layer metadata, tooltips and Evidence Drawer payloads.

idea_id	title	evidence / statement
ML-060-023	GTFS-Realtime staleness gates affect incident publication	SRC-060 pp.22-24: GTFS-RT alerts/trip updates/vehicle positions can corroborate incidents, but parsers must handle FULL_DATASET/DIFFERENTIAL semantics and staleness gates before map publication.

1 additional records are included in the Master Idea Index above.

- Required objects / contracts: Apply policy gates before any public layer, style, popup, screenshot or export uses this content.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Policy test DENY/ABSTAIN cases for rights, sensitivity, CARE status and stale or missing evidence.
- Changed since v1.5: EXPANDED by 11 current-run record(s) from SRC-059/SRC-060.

R. Plugin, Wrapper, and Dependency Governance

Governs renderer wrappers, tooling, plugins and dependency/tool admission.

New / expanded evidence: ML-059-093 SRC-059 pp.54,62; ML-059-094 SRC-059 p.352

idea_id	title	evidence / statement
ML-059-093	MapLibre/Leaflet parity references need wrapper governance	SRC-059 pp.54,62: The source mentions MapLibre/Leaflet flood visualization layers; KFM should avoid mixed renderer drift unless wrapper/adaptor choices are reviewed and tested.
ML-059-094	QGIS2ThreeJS and Cesium tooling are toolchain candidates, not defaults	SRC-059 p.352: 3D asset tools such as Cesium ion, GDAL, Entwine, Blender and QGIS2ThreeJS need license/security/toolchain review before admission.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 2 current-run record(s) from SRC-059/SRC-060.

S. Accessibility, UX, and Trust-Visible States

Governs contrast, alt text, keyboard access, trust badges, negative states and user comprehension.

New / expanded evidence: ML-059-011 SRC-059 pp.80-81; ML-059-019 SRC-059 p.91; ML-059-042 SRC-059 pp.293-308; ML-059-047 SRC-059 pp.349-350; ML-059-071 SRC-059 pp.54,62; ML-059-087 SRC-059 p.82; ML-060-002 SRC-060 pp.1-2

idea_id	title	evidence / statement
ML-059-011	Focus Mode asset classes include overlays, 3D scenes, story nodes and analytics snapshots	SRC-059 pp.80-81: The source distinguishes core panels, interactive overlays, Cesium/MapLibre-integrated 3D scenes, Story Nodes and analytics snapshots as different asset classes.
ML-059-019	MapLibre UI version strip supports keyboard navigation	SRC-059 p.91: The Story Node version strip displays predecessor/current/successor/latest, deprecated flags, hover previews and click-to-diff behavior with keyboard arrow navigation.
ML-059-042	Climate symbol assets include accessibility and recognition constraints	SRC-059 pp.293-308: Climate symbols should remain recognizable at common UI sizes, with accessibility and symbol consistency baked into the source assets.
ML-059-047	Core Focus Mode panels need sidecar metadata and zoom-legibility checks	SRC-059 pp.349-350: Timeline context panels, entity summaries, hydrology overlays and metadata panels need sidecar metadata, checksums, provenance and legibility at 200 percent zoom.
ML-059-071	Flood colorbars require WCAG, STAC/DCAT and provenance metadata	SRC-059 pp.54,62: Colorbars for depth, velocity, risk and uncertainty require visual rules, accessibility compliance, contrast validation and provenance.
ML-059-087	Motion/animation alternatives are accessibility obligations	SRC-059 p.82: Motion or animation assets require alternatives, and text overlays/colorbars need WCAG contrast, supporting map animation accessibility gates.

idea_id	title	evidence / statement
ML-060-002	Sensor trust scores can drive map reliability badges	SRC-060 pp.1-2: Sensor trust scores integrate accuracy, stability, responsiveness and consensus alignment; map UI can display trust-visible sensor states without making them proof.

- Required objects / contracts: Bind style/glyph/legend assets to metadata, checksums, accessibility and review state.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Visual regression, contrast, sprite/glyph path, style reference and source-layer validation.
- Changed since v1.5: EXPANDED by 7 current-run record(s) from SRC-059/SRC-060.

T. Performance, Caching, CDN, Range Requests, and Resource Timing

Governs IO, resource timing, telemetry, cache, multiscale arrays and release budgets.

New / expanded evidence: ML-059-057 SRC-059 p.358; ML-059-085 SRC-059 pp.82,87,92-93,358; ML-059-092 SRC-059 pp.212,237,278; ML-060-011 SRC-060 pp.8-11; ML-060-014 SRC-060 pp.10-11

idea_id	title	evidence / statement
ML-059-057	3D render telemetry becomes a release evidence sidecar	SRC-059 p.358: The source gives telemetry fields for render tests, energy, carbon and FAIR+CARE status, making performance/sustainability logging part of release evidence.
ML-059-085	Map asset and Story Node telemetry are audit inputs	SRC-059 pp.82,87,92-93,358: Telemetry records include validated node counts, render tests, energy/carbon and version events; KFM should treat them as receipts/observability, not truth.
ML-059-092	Renderer symbols should log telemetry for traceability	SRC-059 pp.212,237,278: Renderer responsibilities include logging symbol usage in Focus Mode telemetry, tying style usage to observability.
ML-060-011	Multiscale Zarr/GeoZarr overviews affect map zoom strategy	SRC-060 pp.8-11: Multiscale stores can expose zoom levels as child groups with tile matrix set and resampling metadata, reducing client probing and improving map IO planning.
ML-060-014	STAC-in-Zarr reduces bespoke mapping but creates staleness risk	SRC-060 pp.10-11: Embedding STAC-like metadata in Zarr helps self-describing stores but can go stale if stores mutate independently; versioning/attestation mitigates the risk.

- Required objects / contracts: Define budgets, headers, cache rules or performance checks as release gates.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Resource timing, Range/CORS/cache, tile-load and latency tests.
- Changed since v1.5: EXPANDED by 5 current-run record(s) from SRC-059/SRC-060.

U. Testing, CI, Validation, Rollback, and Proof Objects

Governs schema validation, visual regression, CI gates, rollback and proof objects.

New / expanded evidence: ML-059-021 SRC-059 p.92; ML-059-026 SRC-059 p.114; ML-059-035 SRC-059 pp.209-217; ML-059-038 SRC-059 pp.235-249; ML-059-040 SRC-059 pp.272-278; ML-059-056 SRC-059 pp.357-358; ML-059-077 SRC-059 pp.209-224,272-278; ML-060-005 SRC-060 pp.3-6; plus 7 more delta records.

idea_id	title	evidence / statement
ML-059-021	Diff engine compares metadata, assets and geometry/raster changes	SRC-059 p.92: The diff engine must compare metadata fields, asset inventories, geometry or raster diffs and semantic score changes.
ML-059-026	Story Node asset workflows include docs, STAC/DCAT, FAIR+CARE and telemetry validation	SRC-059 p.114: The source names docs-lint, stac-validate, faircare-validate and telemetry-export workflows for visual asset governance.
ML-059-035	Landcover symbols require visual regression tests	SRC-059 pp.209-217: The source names validation and visual-regression tests, renderer preview generation and golden baseline images for landcover symbols.
ML-059-038	Hydrology symbol visual regression protects map legend consistency	SRC-059 pp.235-249: Hydrology symbol directories include tests and snapshot visual regression comparing preview renderings with baselines.

idea_id	title	evidence / statement
ML-059-040	Event symbol sprites and baselines must be stable	SRC-059 pp.272-278: Event symbol QA uses golden master baselines and composite sprite-sheet baselines to ensure consistent rendering across engines.
ML-059-056	3D terrain build pipeline needs QA before Focus Mode publication	SRC-059 pp.357-358: The terrain pipeline moves from raw DEM inputs through normalization, 3D Tiles conversion, LOD optimization and mesh decimation to QA/FAIR+CARE audit and Focus Mode publication.
ML-059-077	Visual regression should cover renderer parity across MapLibre, rasterizer and browser	SRC-059 pp.209-224,272-278: Symbol and event baselines require renderer outputs to remain consistent across MapLibre, pipeline rasterizer and browser contexts.
ML-060-005	Co-location windows become validation metadata	SRC-060 pp.3-6: The source emphasizes 2-6 week co-location windows with environmental diversity; air-source validators should require co-location evidence or mark NEEDS VERIFICATION.
ML-060-015	Zarr array metadata requires CI schema and source-metadata matching checks	SRC-060 p.11: The adoption playbook requires schema validation plus checks that kfm:array matches zarr.json/.zarray metadata.
ML-060-017	Graph migration rollback artifacts include pre-state checksums and deterministic ID maps	SRC-060 pp.16-18: Every migration should generate pre-state checksum, mutation log, deterministic ID map and tested rollback Cypher; this protects Evidence Drawer graph projections.

5 additional records are included in the Master Idea Index above.

- Required objects / contracts: Add fixture-driven validator or CI check before promotion.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: CI fixture must pass/fail deterministically; include rollback and negative-state cases.
- Changed since v1.5: EXPANDED by 15 current-run record(s) from SRC-059/SRC-060.

V. Exports, Screenshots, Reports, and Citation Preservation

Governs static maps, screenshots, exports and reports as citation-preserving carriers.

New / expanded evidence: ML-059-022 SRC-059 pp.92-93; ML-059-024 SRC-059 pp.96-97; ML-059-064 SRC-059 pp.80-82,112-118,349-350; ML-059-086 SRC-059 p.118

idea_id	title	evidence / statement
ML-059-022	Every Focus Mode output embeds version lineage and diff hash	SRC-059 pp.92-93: Version stamping requires STAC Item ID, version string, lineage chain, diff manifest hash, timestamp and session fingerprint in Focus Mode outputs.
ML-059-024	Story Node examples carry citations and dataset lineage	SRC-059 pp.96-97: Examples show Story Nodes including provenance datasets and citations, reinforcing cite-or-abstain behavior for Focus Mode narrative cards.
ML-059-064	Static maps and screenshots remain downstream carriers	SRC-059 pp.80-82,112-118,349-350: Focus Mode screenshots, static maps and previews require checksums, provenance, alt text and citations; they do not substitute for released evidence or citation-preserving exports.
ML-059-086	Static map resolution minimum is an export quality gate	SRC-059 p.118: Story Node map assets require a minimum 2048 px width and sibling STAC metadata, strengthening export/report quality checks.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 4 current-run record(s) from SRC-059/SRC-060.

W. 3D / Cesium / Deck.gl / Overlay Interoperability

Governs conditional 3D, Cesium, 3D Tiles, GLB/CZML and 2D/3D parity.

New / expanded evidence: ML-059-014 SRC-059 pp.84-86; ML-059-048 SRC-059 pp.351-354; ML-059-049 SRC-059 pp.351-352; ML-059-050 SRC-059 pp.352-353; ML-059-052 SRC-059 pp.355-356; ML-059-054 SRC-059 p.357; ML-059-082 SRC-059 pp.352,357; ML-059-083 SRC-059 pp.352,356

idea_id	title	evidence / statement
ML-059-014	Story Nodes link directly to 2D maps, 3D scenes and timeline frames	SRC-059 pp.84-86: Story Node structures include links to map_2d, scene_3d and timeline_frame assets; Focus Mode can use them as narrative chapters and AI reference units.
ML-059-048	3D Focus Mode Views become governed asset family	SRC-059 pp.351-354: 3D Views include Cesium terrain layers, archaeological reconstructions, hydrology elevation/anomaly surfaces and time-aware volumetric or overlay scenes, governed by FAIR+CARE, ISO, STAC/DCAT and accessibility metadata.
ML-059-049	3D asset standards include 3D Tiles, CZML, GLB/GLTF and camera path JSON	SRC-059 pp.351-352: The 3D asset standard separates terrain tilesets, geometry models, volumetric layers, camera paths and metadata assets.
ML-059-050	3D assets require generalization, alt text, checksums, STAC extensions and temporal anchoring	SRC-059 pp.352-353: 3D previews and captures require alt text; 3D assets require SHA-256 checksums, STAC extensions and start/end temporal anchors.
ML-059-052	Global Terrain Tileset metadata defines STAC/DCAT-aligned 3D tile asset	SRC-059 pp.355-356: The terrain tileset example includes type=3d-tiles, checksum, CRS, provenance, sources, processing pipeline, bbox, temporal extent and license.
ML-059-054	3D terrain integrates with Time-Encoded MapLibre plus Cesium layers	SRC-059 p.357: The terrain tileset is listed as integrated with Focus Mode Story Nodes, time-encoded MapLibre + Cesium layers, hydrology and archaeology pipelines.
ML-059-082	Camera paths are governed 3D assets	SRC-059 pp.352,357: Camera path JSON is listed as an asset type for Focus Mode cinematic navigation and must be metadata-governed like other scene assets.
ML-059-083	Volumetric hydrology/climate layers use CZML/GLB and need conditional 3D gates	SRC-059 pp.352,356: Volumetric layers for hydrology and climate anomalies are proposed as CZML/GLB assets that should not become default unless evidence burden justifies 3D.

- Required objects / contracts: Treat 3D as conditional companion output; preserve MapLibre 2D evidence parity.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: 3D smoke test, 2D fallback, metadata checksum test, alt-text/accessibility test.
- Changed since v1.5: EXPANDED by 8 current-run record(s) from SRC-059/SRC-060.

X. Anti-Patterns and Failure Modes

Records failure modes that KFM must deny, abstain from or mark as needs verification.

New / expanded evidence: ML-059-063 SRC-059 pp.80-128,351-358; ML-059-065 SRC-059 pp.351-358; ML-059-089 SRC-059 pp.2,120-128; ML-060-031 SRC-060 pp.1-36

idea_id	title	evidence / statement
ML-059-063	FAIR+CARE badges are not release authority	SRC-059 pp.80-128,351-358: The new docs use FAIR+CARE/status badges heavily; KFM must not treat badges as policy decisions or release proof without EvidenceBundle, PolicyDecision and PromotionDecision records.
ML-059-065	3D assets are not sovereign truth	SRC-059 pp.351-358: 3D terrain, GLB/CZML scenes, camera paths and volumetric layers must remain derived visualization carriers, not canonical truth or policy authority.
ML-059-089	Map layers auto-published via STAC/DCAT still need promotion state	SRC-059 pp.2,120-128: Source language sometimes says pipelines automatically publish map layers; KFM must interpret that as candidate release only unless PromotionDecision and release manifest are proven.
ML-060-031	PM2.5 calibration and incident fusion are not direct MapLibre claims unless manifested	SRC-060 pp.1-36: The new source contains high-value calibration and incident patterns, but they become MapLibre-relevant only when connected to layer manifests, Evidence Drawer payloads, public-safe policies and release artifacts.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 4 current-run record(s) from SRC-059/SRC-060.

Y. Implementation Backlog and PR Plan

Turns deltas into implementation-ready but repo-unverified work packages.

New / expanded evidence: ML-059-033 SRC-059 pp.80-128; ML-059-069 SRC-059 pp.120-128; ML-059-091 SRC-059 pp.433-434

idea_id	title	evidence / statement
ML-059-033	Path claims from source packets remain Directory-Rules pending	SRC-059 pp.80-128: The many docs/reports/visualization paths are source-reported organization patterns; actual repo placement needs Directory Rules and mounted-repo verification before implementation.
ML-059-069	MapAsset schema should not create parallel schema home without ADR	SRC-059 pp.120-128: The source places schemas under docs/reports/.../schemas; Directory Rules require schema-home verification, so the object should be migrated or aliased under schemas/contracts/v1 only through ADR-compatible placement.
ML-059-091	Global source URLs require SourceDescriptor onboarding	SRC-059 pp.433-434: The URL list should be converted into SourceDescriptor records with source role, rights, cadence, expected geometry and public-release limits before map ingestion.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 3 current-run record(s) from SRC-059/SRC-060.

Z. Open Questions and Verification Backlog

Tracks unresolved and checkable items that remain UNKNOWN or NEEDS VERIFICATION.

New / expanded evidence: ML-059-060 SRC-059 pp.433-434; ML-059-095 SRC-059 pp.1-6,80-128,351-358

idea_id	title	evidence / statement
ML-059-060	External Kansas source URL list is intake backlog, not verified source registry	SRC-059 pp.433-434: The pages list Kansas GIS, historic map, FEMA, drought, storm, LiDAR, wetlands, wildfire, geology and water links, but those URLs require source descriptors, rights checks and freshness verification before map use.
ML-059-095	Source text contains strong implementation claims that remain NEEDS VERIFICATION	SRC-059 pp.1-6,80-128,351-358: Several passages describe active or implemented KFM behavior; without mounted repo, tests, logs or release artifacts, current implementation maturity stays UNKNOWN or NEEDS VERIFICATION.

- Required objects / contracts: Bind this idea to governed API, manifest, evidence, policy and test objects before release.
- Public UI implication: show source, version, evidence, policy, stale state and review state rather than presenting the rendered layer as truth.
- Policy implication: DENY or ABSTAIN where rights, sensitivity, source authority, stale state, calibration confidence or release proof is missing.
- Validation tests: Add or extend fixtures proving source, artifact, evidence, policy and UI behavior.
- Changed since v1.5: EXPANDED by 2 current-run record(s) from SRC-059/SRC-060.

8. Components-Functions-Features Matrix

Matrix entries are cumulative. Status is EXPANDED where SRC-059/SRC-060 add new evidence and RETAINED where v1.5 already carried the component. All repo paths remain PROPOSED/NEEDS VERIFICATION.

8A. Component, function, feature, input and output artifacts

component	function	feature	input artifacts	output artifacts	status
MapLibre GL JS	2D map renderer and interaction runtime	Toggled hydrology, symbols, Story Node version strip, layer selection	released style/layer manifests; source descriptors	rendered map, click candidate, camera/time context	EXPANDED
MapLibre Native	Mobile/native renderer family	Potential mobile/offline renderer	released mobile-safe artifacts	mobile render state	RETAINED
MapLibre RS	Experimental renderer	Experimental rendering path	released test fixtures	experimental visual output	RETAINED
Style JSON	Renderer style definition	Bindings for layers, sprites, glyphs and scale rules	StyleManifest, sprites, glyphs, source definitions	validated style JSON	EXPANDED
Source definition	Connects style to data source	MapLibre source for tiles/GeoJSON/COG/protocol	LayerManifest, TileArtifactManifest	runtime source descriptor	EXPANDED
Layer definition	Map layer rules	Hydrology, air, incident, story-node, symbol layers	LayerManifest	rendered layer	EXPANDED
Expressions	Renderer styling logic	Scale, certainty, sensitivity and symbol rules	StyleManifest, symbol metadata	style expressions	EXPANDED
Filters	Map feature selection	Timeline/version/sensitivity filters	StyleManifest + MapContextEnvelope	filtered render set	EXPANDED
Sprites	Icon atlas	Symbol derivatives from SVG glyphs	SymbolLegendManifest	sprite sheet	EXPANDED
Glyphs	Text/icon glyph resources	Landcover/hydrology/event/climate/archeology glyphs	symbol metadata	glyphs and previews	EXPANDED
MVT	Vector tile carrier	Vector features for high-scale map delivery	TileArtifactManifest	MVT tiles	RETAINED
MLT	Emerging vector tile carrier	Pilot candidate only	TileArtifactManifest pilot	MLT test tiles	RETAINED
PMTiles	Static serverless tile archive	Allowed map asset format and prior tile strategy	TileArtifactManifest, ReleaseManifest	PMTiles archive	EXPANDED
MBTiles	Build intermediate/offline archive	Possible intermediate before PMTiles	TileArtifactManifest	MBTiles/converted archive	RETAINED
COG	Cloud optimized raster carrier	Hydrology, flood, map assets and terrain/raster products	Raster/COG manifest	COG asset	EXPANDED
Martin	Server-mediated tile serving	Dynamic PostGIS/MVT mediation when needed	source registry, TileArtifactManifest	tile endpoints	RETAINED
Maputnik	Style editing tool	Potential style editor	style draft	style proposal	RETAINED
Layer catalog	Public layer registry	Advertises released map assets	LayerManifest records	catalog listing	EXPANDED
Evidence Drawer	Evidence inspection surface	Resolves clicked features, Story Nodes, incidents and sensor layers	EvidenceDrawerPayload	drawer payload	EXPANDED
Focus Mode	Evidence-bounded synthesis mode	Story Nodes, version locks, 3D views, PM2.5 caveats	MapContextEnvelope, AIRceipt	ANSWER/ABSTAIN/DENY/ERROR	EXPANDED
Timeline / time slider	Temporal map control	Story Node versions, temporal extents, incident times	MapContextEnvelope	time-filtered layer state	EXPANDED

component	function	feature	input artifacts	output artifacts	status
MapContextEnvelope	Bounded map state envelope	Camera, filters, version lock, clicked feature context	map state + release IDs	Focus request context	EXPANDED
LayerManifest	Layer release contract	Layer title, source, style, evidence, policy state	source/layer specs	released layer record	EXPANDED
StyleManifest	Style release contract	Styles, sprites, glyphs, legends, colorbars	style + symbol metadata	released style record	EXPANDED
TileArtifactManifest	Tile/raster/array artifact contract	PMTiles/COG/3D/Zarr/incident artifact sidecar	artifact metadata	released artifact record	EXPANDED
MapReleaseManifest	Map release bundle	Links layer, style, tiles, evidence, policy and rollback target	all manifests	release manifest	EXPANDED
PolicyDecision	Policy result	DENY/ABSTAIN/ALLOW for layers/assets/answers	policy input	decision record	EXPANDED
CitationValidationReport	Citation proof	Validates source/citation claims in drawer/exports/AI	evidence citations	validation report	RETAINED
AIReceipt	AI runtime receipt	Records Focus Mode prompt/context/outcome	Focus request	AIReceipt	EXPANDED
rollback target	Reversible release pointer	Returns to prior layer/tile/style/story node version	ReleaseManifest	rollback record	EXPANDED
cache invalidation record	Cache control evidence	Invalidates tiles/styles/assets after release/rollback	release + cache metadata	cache receipt	EXPANDED

8B. Governance, evidence, policy, tile/style, runtime and tests

component	governed API dependency	evidence dependency	policy dependency	tile/style dependency	runtime/UI behavior	tests required	status
MapLibre GL JS	governed API for click/evidence	EvidenceBundle per feature/layer	PolicyDecision for access/release	StyleManifest + LayerManifest	MapLibre shell state and layer toggles	style, source, click, accessibility, visual regression	EXPANDED
MapLibre Native	governed API or cached released payloads	same evidence semantics as web	mobile access policy	TileArtifactManifest, style/glyphs	offline/low-bandwidth map behavior	mobile parity and denied direct raw tests	RETAINED
MapLibre RS	none until approved	evidence parity required	dependency approval	style/tile fixtures	not default public path	experimental renderer tests	RETAINED
Style JSON	style registry	legend evidence links	sensitive style policy	style/source-layer dependency	style-driven rendering	style spec and no hidden geometry tests	EXPANDED
Source definition	governed source metadata API	source EvidenceBundle refs	source rights policy	tile/style dependency	addSource only after release	source validation and no-unreleased-load tests	EXPANDED
Layer definition	layer catalog API	evidence link per layer	visibility and sensitivity policy	style/layer source refs	layer toggle, state, badges	layer manifest and visibility tests	EXPANDED
Expressions	validated style service	legend/evidence refs	no style-only protection	style JSON	conditional rendering	style expression tests	EXPANDED
Filters	governed API for filters	EvidenceBundle constraints	sensitivity gates	source-layer dependency	time/version locked map	filter and timeline tests	EXPANDED
Sprites	style asset service	symbol evidence refs	culture/access review	style sprite refs	icons on map/legend	sprite integrity and visual regression	EXPANDED
Glyphs	asset endpoint	metadata refs	CARE/accessibility gates	style glyph refs	legends, labels, markers	glyph alignment/contrast tests	EXPANDED
MVT	tile service/CDN	feature evidence refs	public-safe geometry	tile/style source-layer	fast vector render	tile schema/source-layer tests	RETAINED
MLT	tile service pilot	same evidence requirements	NEEDS VERIFICATION	tile tooling	not default	MLT parity tests	RETAINED
PMTiles	CDN/range hosting	tile-to-evidence mapping	public-safe release policy	PMTiles + StyleManifest	protocol handler load	Range/CORS/digest/rollback tests	EXPANDED
MBTiles	build pipeline	manifest evidence	release restrictions	tile conversion	not public raw default	conversion/manifest tests	RETAINED
COG	range-capable hosting	raster EvidenceBundle	rights/sensitivity policy	raster style/protocol	raster display	COG validation/gdalinfo/range tests	EXPANDED
Martin	governed tile API	database evidence refs	access/steward policy	tile server config	dynamic tiles	access/no-raw tests	RETAINED
Maputnik	review service	evidence refs in style	style review policy	StyleManifest	edited style	style diff/review tests	RETAINED
Layer catalog	layer catalog API	EvidenceBundle refs	PolicyDecision refs	style/tile dependency	layer toggles, badges	catalog completeness tests	EXPANDED
Evidence Drawer	governed API	EvidenceBundle required	PolicyDecision required	feature/tile/style mapping	trust-visible details	click-to-drawer tests	EXPANDED
Focus Mode	governed AI API	EvidenceBundle collection	policy pre/post checks	map context	answer or abstain	citation/abstain tests	EXPANDED
Timeline / time slider	time API	temporal evidence refs	stale/rights policy	style/source filter	time lock/version strip	time filter/version lock tests	EXPANDED
MapContextEnvelope	governed API	evidence refs	policy refs	style/tile IDs	sent to Focus Mode	schema and no-direct-model tests	EXPANDED
LayerManifest	catalog API	EvidenceBundle refs	PolicyDecision	Tile/Style manifests	layer availability	schema/release tests	EXPANDED
StyleManifest	style API/CDN	legend/evidence refs	style policy	style JSON/sprite/glyph	visual style	style/source/sprite tests	EXPANDED
TileArtifactManifest	tile asset service	artifact evidence refs	rights/sensitivity	tiles/COG/Zarr/3D	artifact load	digest/Range/rollback tests	EXPANDED
MapReleaseManifest	release API	evidence closure	promotion decision	tile/style/asset deps	public release	release closure/rollback tests	EXPANDED

component	governed API dependency	evidence dependency	policy dependency	tile/style dependency	runtime/UI behavior	tests required	status
PolicyDecision	policy API	evidence refs	policy bundle	manifest refs	UI negative states	deny/allow fixture tests	EXPANDED
CitationValidationReport	citation API	EvidenceBundle	cite-or-abstain	export dependencies	rendered citations	citation pass/fail tests	RETAINED
AIReceipt	governed AI API	EvidenceBundle refs	policy checks	map context	answer audit	finite outcome tests	EXPANDED
rollback target	release API	evidence closure	policy closure	cache invalidation	restore previous map	rollback tests	EXPANDED
cache invalidation record	CDN/API	evidence refs	policy refs	tile/style deps	fresh/stale UI	cache/stale tests	EXPANDED

9. Tile Strategy Chapter

The tile strategy remains conservative: use the lightest public-safe carrier that preserves evidence, policy, citation, release and rollback state. SRC-059 expands static map asset metadata and 3D terrain tileset handling; SRC-060 expands array/Zarr and incident-layer delivery concerns.

Comparison	When it wins	When it fails	KFM trust consequence	Release / rollback consequence	Validation needed
GeoJSON vs vector tiles	GeoJSON wins for small, inspectable, low-volume overlays and Story Node examples. Vector tiles win for large or frequently rendered datasets.	GeoJSON fails as production default for dense or large layers; vector tiles fail if source-layer/evidence mapping is missing.	Rendered vectors are selection candidates, not proof.	Vector tiles require TileArtifactManifest and rollback target; GeoJSON requires release manifest too.	GeoJSON schema, bounds, feature IDs; MVT/MLT source-layer and evidence-ref tests.
MVT vs MLT	MVT wins as mature vector tile carrier; MLT remains pilot/needs verification.	MLT fails as default without KFM toolchain and viewer validation.	No format becomes truth.	Pilot MLT only through release manifest with fallback to MVT.	MVT/MLT parity and tooling tests.
PMTiles vs MBTiles	PMTiles wins for serverless immutable/range-capable public-safe bundles; MBTiles can remain build/offline intermediate.	PMTiles fails if hosting lacks Range/CORS or per-archive manifest; MBTiles fails as direct browser artifact.	Archive is not proof; manifest/evidence is proof path.	Digest, version and rollback reference required.	pmtiles inspect, Range/CORS, digest, style source tests.
Static tiles vs dynamic tile server	Static wins for stable public-safe releases; dynamic server wins for steward-mediated or PostGIS-backed policy-sensitive delivery.	Static fails for access-controlled or changing policy cases; dynamic fails if it bypasses governed APIs.	Public clients still use governed release/catalog and not canonical stores.	Static bundles are versioned; dynamic endpoints must bind to release state.	No unreleased tile load, access policy, source freshness tests.
PMTiles serverless vs Martin server-mediated	PMTiles wins where immutable release artifacts can be CDN-hosted; Martin wins when dynamic PostGIS/MVT mediation is needed.	PMTiles fails for restricted/stewarded datasets; Martin fails if it exposes direct canonical access.	Both remain downstream carriers.	PMTiles rollback by manifest pointer; Martin rollback by release binding/config.	Range/CORS and Martin access/no-raw tests.
Vector vs raster	Vector wins for interactive feature selection, labels and style changes; raster/COG wins for continuous fields, surfaces and analytical grids.	Vector fails for dense continuous fields; raster fails if click resolution/citation is needed but not provided.	Feature clicks need EvidenceBundle; raster cells need raster evidence and lineage.	Both need manifests and source digests.	Vector source-layer tests; COG/raster metadata and cell-to-evidence tests.
COG vs raster tiles	COG wins for cloud-native range-readable analysis/raster delivery; raster tiles win for simple visualization where analysis is not needed.	COG fails if headers/overviews are wrong; raster tiles fail if users infer data values without evidence.	COG is derivative but can preserve metadata better.	COG rollback by digest; tile pyramid rollback by manifest.	gdalinfo, overviews, Range, STAC raster/proj tests.
Terrain-enabled vs non-terrain	Terrain/3D wins when vertical context carries evidence burden, e.g. hydrology volumes or landform explanation.	Terrain fails as default spectacle or when exact sensitive locations are exposed.	3D scenes require same evidence and policy closure as 2D.	3D Scene/Terrain manifest and 2D fallback required.	3D smoke, alt-text, checksum, CRS/temporal, sensitive generalization tests.
2D default vs 3D conditional	2D MapLibre shell wins as default because it is simpler, auditable and clearer for most public map tasks.	3D fails if it weakens accessibility, policy, or citation traceability.	2D remains baseline trust surface.	3D release cannot bypass MapReleaseManifest.	2D/3D parity, fallback and visual QA tests.
Public-safe tiles vs restricted/internal tiles	Public-safe tiles win for released generalized content; restricted/internal tiles require controlled access and are not normal UI surfaces.	Style-only hiding fails for sensitive geometry.	Generalize/redact before public release.	Restricted bundles need access policy and rollback/correction.	Sensitive geometry deny/generalize fixtures.

Comparison	When it wins	When it fails	KFM trust consequence	Release / rollback consequence	Validation needed
Zarr/array assets vs map tiles	Zarr/array wins for multidimensional analysis and lazy loading; map tiles win for simple display.	Zarr fails if array metadata is not discoverable; map tiles fail if analytical truth is inferred from render.	Array metadata needs evidence and versioning.	Array stores require kfm:array manifest and attestation.	Schema matching to zarr metadata, multiscale/overview tests.

10. Governance and Trust-Membrane Chapter

released layer -> user click -> governed API -> EvidenceRef resolution -> EvidenceBundle -> Evidence Drawer -> Focus Mode answer / abstain / deny / error -> citation-preserving export or rollback-aware correction.

No public RAW path: public clients and normal UI surfaces must not fetch RAW, WORK, QUARANTINE, unpublished candidate data or direct canonical/internal stores.

No direct model client: Focus Mode consumes governed API envelopes and EvidenceBundles; the browser never talks directly to Ollama or any model runtime.

No unreleased tile load: addSource/addLayer must only load released artifacts or safe preview fixtures explicitly marked non-public.

No sensitive geometry hidden only by style: exact coordinates for archaeological, cultural, rare species, living-person, DNA, infrastructure or sovereign-sensitive content must be transformed, generalized, delayed, redacted or denied before release.

No popup as Evidence Drawer substitute: popups can summarize, but consequential claims require EvidenceDrawerPayload and EvidenceBundle resolution.

No uncited export: screenshots, static maps, reports, Story Node captures, 3D previews and Focus Mode answers must preserve citation/provenance references and release state.

New v1.6 emphasis: Story Nodes, map asset registries, 3D terrain tilesets, PM2.5 corrected surfaces and incident clusters are still downstream carriers. Badges, manifests, readme paths and telemetry refs in source packets are not implementation proof without release artifacts and validation.

11. Implementation-Ready Object Map

Objects below are PROPOSED unless a future repo inspection proves they exist. Directory Rules basis: responsibility roots matter; schemas default to schemas/contracts/v1/<family>; docs/report paths in source packets are source-proposed and need ADR/migration if the real repo differs.

Object family	Field intent / v1.6 additions	Status	Validation consequence
SourceDescriptor	source_id, authority role, rights, cadence, access URL, license, expected geometry/array form, MapLibre/tile relevance, public release class. Add URL-list intake from SRC-059 pp.433-434 and WZDX/ArcGIS/GTFS feed descriptors from SRC-060 pp.18-26.	PROPOSED	Source descriptor completeness and rights tests.
LayerManifest	layer_id, title, domain, source refs, source-layer, style refs, time extent, evidence refs, policy state, stale state, public-safe flag. Add Story Node, incident, air and symbol/legend layer links.	PROPOSED	No unreleased layer load and stale/rights tests.
StyleManifest	style JSON, sprites, glyphs, legends/colorbars, min/max scale, sensitive styling constraints, checksum, release version. Add symbol/legend schema refs from SRC-059.	PROPOSED	Style/sprite/glyph validation and visual regression.
TileArtifactManifest	artifact_id, type, format, URL, media type, digest, size, range headers, bounds, zooms, source_digests, provenance, rollback target. Add COG/PMTiles/Zarr/3D tileset array/terrain fields.	PROPOSED	Digest, Range/CORS, COG, Zarr and 3D Tiles metadata tests.
MapReleaseManifest	links LayerManifest, StyleManifest, TileArtifactManifest, EvidenceBundle, PolicyDecision, PromotionDecision, cache invalidation and rollback target.	PROPOSED	Catalog closure and rollback restore tests.
EvidenceBundle	evidence_id, source refs, citations, source role, spatial/temporal scope, rights, review, checksum, release state.	PROPOSED	EvidenceRef resolution and citation validation.
EvidenceDrawerPayload	clicked feature, layer ID, evidence_refs, policy state, source/citation display, caveats, conflicts, telemetry. Add incident conflicts and sensor-calibration caveats.	PROPOSED	Click-to-drawer and conflict/caveat tests.
MapContextEnvelope	camera, bounds, zoom, pitch/bearing, selected layer/feature, time, version lock, filters, source IDs, release IDs.	PROPOSED	Focus Mode no-direct-model tests.
FocusModeRequest/Response	request context, evidence set, question, policy result, outcome ANSWER/ABSTAIN/DENY/ERROR, citations, AIReceipt. Add Story Node version lock and 3D scene context.	PROPOSED	Answer with citations and abstain-on-missing-evidence tests.
AIReceipt	model/provider, prompt envelope, evidence IDs, policy decisions, output outcome, citation validation, runtime metadata.	PROPOSED	Finite outcome and no public raw model output tests.
CitationValidationReport	citation IDs, source coverage, failures, unsupported claims, export references.	PROPOSED	Pass/fail citation fixtures.
PolicyDecision	subject, resource, purpose, sensitivity, rights, role, decision, reasons, obligations. Add CARE public/generalized/restricted and sensor trust caveats.	PROPOSED	DENY/ABSTAIN fixtures for sensitive or unreviewed content.
PromotionDecision	promotion gate results, release target, proof refs, rollback target, reviewer, timestamp.	PROPOSED	Gate A-G and proof completeness tests.
RunReceipt	input digests, config hash, transform, output artifacts, source roles, timestamp, attestation. Add deterministic data-run/lakeFS/DVC pattern.	PROPOSED	Receipt and attestation verification.
rollback target	previous manifest pointer, prior digests, cache invalidation, revert commands/records. Add graph migration and lakeFS rollback patterns.	PROPOSED	Rollback restores prior manifest/state.
stale-source fixture	feed/asset whose timestamp, version or source headers are stale.	PROPOSED	UI stale badge/abstain tests.
sensitive-geometry deny fixture	archaeological/heritage/sacred/rare/infrastructure exact geometry.	PROPOSED	Style-only hiding DENY; generalized output only.
StoryNodeMetadata	id, title, summary, temporal, spatial, entities, provenance, citations, CARE, visualization links, version lineage.	NEW PROPOSED	Story Node schema, citation and sensitivity tests.
MapAssetMetadata	id, type, path, projection, temporal/spatial extent, checksum, format, provenance, CARE, STAC/DCAT/CIDOC links.	NEW PROPOSED	Map asset metadata and checksum tests.
SymbolLegendManifest	symbol_id, feature_class, geometry_type, period, culture, certainty, sensitivity, access_tier, scale, colors, source_dataset_id, style/sprite refs.	NEW PROPOSED	Glyph/sprite/style/scale and CARE tests.
3DSceneManifest	asset type, 3D Tiles/CZML/GLB/GLTF/camera path, CRS, checksum, temporal anchors, alt text, sensitivity generalization, 2D fallback.	NEW PROPOSED	3D smoke, metadata, accessibility and 2D parity tests.
ArrayAssetManifest	kfm:array chunk_shape, shape, dtype, dimensions, nodata, storage, multiscales, tile matrix set, measurement assets.	NEW PROPOSED	Zarr metadata matching and overview tests.

Object family	Field intent / v1.6 additions	Status	Validation consequence
SensorCalibrationReceipt	sensor ID, raw/observed/corrected/model fields, calibration model, trust score, co-location evidence, meteorological predictors, drift state.	NEW PROPOSED	Calibration source, co-location and trust badge tests.
IncidentLayerManifest	source observations, canonical event geometry, dedupe signature, trust tier, promotion_state, conflict state, event lineage.	NEW PROPOSED	WZDx/ArcGIS/GTFS schema, staleness, clustering and conflict tests.

12. Validation and Test Plan

Test area	Required validation
schema validation	LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, StoryNodeMetadata, MapAssetMetadata, SymbolLegendManifest, 3DSceneManifest, ArrayAssetManifest, SensorCalibrationReceipt and IncidentLayerManifest schemas validate positive and negative fixtures.
source ledger completeness	Every material claim in a released map surface resolves to source_id and page/source descriptor; no orphan source refs.
style source/layer validity	Every style source/layer/source-layer/sprite/glyph/legend reference resolves and has checksum/manifest evidence.
PMTiles / MVT / MLT manifest checks	Artifact digest, bounds, zoom, media type, source_digests, release state and rollback target pass; MLT stays pilot until verified.
COG / range request requirements	COG overview, block size, range readability, raster/proj metadata and checksum tests pass.
Zarr / array metadata requirements	kfm:array fields match zarr.json/.zarray; multiscales and tile matrix set entries validate.
no public raw path	Browser cannot fetch RAW, WORK, QUARANTINE, candidate or direct canonical data.
no direct model client	Focus Mode browser requests only governed API; no model runtime endpoint in client bundle.
no unreleased tile load	MapLibre addSource/addLayer cannot load artifacts without release manifest and allowed policy state.
sensitive geometry deny	Exact archaeological, sacred, rare species, living-person, DNA or infrastructure geometry is DENIED unless properly generalized/released.
source rights unknown blocks release	Unknown license/source terms keep layer/artifact from PUBLISHED.
stale source badge / abstain	Stale feed headers, timestamps or version locks show stale state and may force ABSTAIN.
clicked feature resolves EvidenceBundle	Click on layer feature/incident/story node resolves EvidenceBundle and EvidenceDrawerPayload.
citation validation pass/fail	Focus Mode and exports pass when cited; fail or abstain when unsupported.
Focus Mode answer with citations	Focus Mode returns ANSWER only with resolved evidence and CitationValidationReport pass.
Focus Mode abstain on missing evidence	Focus Mode returns ABSTAIN when Story Node/PM2.5/incident evidence is missing or conflicted.
rollback restores prior manifest	Release rollback restores prior MapReleaseManifest and records cache invalidation.
cache invalidation recorded	Tile/style/glyph/story/3D cache invalidation record is emitted after release/rollback.
visual regression for styles	Landcover, hydrology, event, climate and archaeology symbols compare against baselines.
keyboard navigation / accessibility	Story Node version strip, drawer, legends and panels support keyboard, ARIA, contrast and 200 percent zoom.
tile load budget	Tile/COG/Zarr/3D asset loads meet budgets or fall back/abstain.
interaction-to-drawer latency budget	Click to Evidence Drawer meets latency budget under released fixtures.
3D parity and fallback	3D scene/terrain assets have checksums, alt text, temporal anchors, 2D fallback and sensitive-geometry controls.
PM2.5 calibration trust	Sensor trust/correction metadata appears in drawer and can block/reduce confidence when co-location/source evidence is inadequate.
incident fusion QC	WZDx/ArcGIS/GTFS fixture tests cover schema, staleness, hash delta, dedupe, conflicts, trust tiers and promotion states.
graph migration rollback	Neo4j migration fixtures verify declared intent, allowed labels/relationships, pre-state checksum, deterministic ID map and rollback script.

13. Anti-Patterns Register

Anti-pattern	Response
treating MapLibre as truth authority	DENY: renderer cannot become source registry, policy engine, citation authority or release authority.
treating tiles, COGs, Zarr stores or 3D tilesets as proof	DENY: artifacts carry evidence only when manifest/EvidenceBundle resolves.
treating layer toggle as publication	DENY: publication is a governed state transition.
hiding exact sensitive geometry with style filters	DENY: transform, redact, generalize or deny before release.
dense GeoJSON as production default for large datasets	NEEDS VERIFICATION: use vector tiles or governed API where scale demands.
uncontrolled style drift	DENY release if style/source/layer/sprite/glyph references are unvalidated.
plugin/wrapper/tool adoption without review	DENY until allowlist, security, license and test evidence exist.
direct browser access to model runtime	DENY; Focus Mode uses governed API only.
AI answers based only on rendered map features	ABSTAIN or DENY; rendered features are selection candidates.
uncited popups, screenshots, Story Nodes or exports	DENY or ABSTAIN; citation preservation required.
MLT as default before KFM validation	NEEDS VERIFICATION.
3D/globe as default when 2D is clearer	PROPOSED only when vertical/globe evidence burden exists.
Mapbox-era backports without legal/security review	NEEDS VERIFICATION.
FAIR+CARE badge as release proof	NEW from SRC-059; DENY if badge lacks EvidenceBundle, PolicyDecision, PromotionDecision and release manifest.
Story Node as evidence substitute	NEW from SRC-059; Story Node links evidence but is not itself sovereign proof.
static map asset as live layer proof	NEW from SRC-059; screenshots/static maps are downstream carriers.
symbol catalog detached from source dataset	NEW from SRC-059; symbol records must map back to source_dataset_id/source_feature_codes and style manifests.
full-precision sacred or sensitive symbol display	NEW from SRC-059; default display tier must generalize or hide.
3D terrain or camera path without 2D fallback	NEW from SRC-059; conditional 3D requires 2D parity and accessibility.
sensor-corrected PM2.5 surface without calibration caveat	NEW from SRC-060; corrected/model outputs must expose trust, co-location and predictor metadata.
incident layer publishes unresolved conflict as fact	NEW from SRC-060; preserve observations and display conflict/corroboration state.
Zarr/array metadata embedded but stale	NEW from SRC-060; validate kfm:array against source metadata and attest version.
graph migration changes evidence drawer projection silently	NEW from SRC-060; graph changes need migration declaration, diff, rollback and review.

14. Open Questions and Verification Backlog

- 1. Actual repo topology, package manager and schema home.
- 2. MapLibre GL JS target version and lockfile pin.
- 3. MLT support status and toolchain maturity.
- 4. Mobile/native parity, including drawer, evidence, policy and offline behavior.
- 5. PMTiles hosting Range/CORS behavior and per-tile/per-archive digest strategy.
- 6. Martin deployment/access model and PostGIS mediation boundaries.
- 7. Plugin allowlist and MapLibre wrapper admission records.
- 8. Style/sprite/glyph/colorbar/legend hosting and cache invalidation policy.
- 9. Attribution, source rights, source terms and export obligations.
- 10. Sensitive geometry transform thresholds and transform receipts.
- 11. Release manifest location and rollback record home.
- 12. CI workflow, OPA/Conftest policy enforcement and bundle digest verification.
- 13. Accessibility acceptance criteria, including 200 percent zoom and keyboard version strip.
- 14. Tile load, resource timing, Zarr IO and interaction-to-drawer latency budgets.
- 15. Generic Evidence Drawer schema and domain specialization pattern.
- 16. Map-serving runtime proof parity or explicit out-of-scope decision.
- 17. No-live-calls route test coverage for map-serving and Focus Mode.
- 18. RunReceipt vs RuntimeReceipt vs SensorCalibrationReceipt schema separation.
- 19. Focus Mode Story Node metadata and versioning schema set.
- 20. MapAssetMetadata canonical home under Directory Rules.
- 21. SymbolLegendManifest and visual regression baseline home.
- 22. 3DSceneManifest and TerrainTilesetManifest field set and rights review.
- 23. Cesium terrain/source license and self-hosting/attribution posture.
- 24. CARE review workflow for archaeological/cultural symbols and story nodes.
- 25. Whether the source-reported docs/reports/visualization paths should be migrated to docs/, schemas/contracts/v1/, styles/, viewer_templates/ or artifacts under Directory Rules.
- 26. PM2.5 sensor calibration co-location requirements, trust scoring source authority and validation thresholds.
- 27. WZDx source registry, schema pin, DOT feed authority, and Kansas source availability.
- 28. ArcGIS extractChanges availability, token/access model and change-tracking cleanup behavior.
- 29. GTFS-RT feed freshness thresholds and Kansas transit applicability.
- 30. Zarr/GeoZarr/EOPF convention status and KFM kfm:array namespace home.
- 31. Neo4j migration declaration home, graph-diff viewer and rollback fixtures.
- 32. External Kansas source URL list verification, source descriptors, rights and update cadence.
- 33. Whether any prior v1.5 source not attached this run should be reattached next run for source-hash continuity.
- 34. Final output remains a PDF-only artifact; any implementation PR must rerun Directory Rules and repo inspection.

15. How to Continue This Master PDF Next Run

- Attach this PDF again with any new documents.
- The next run should read this PDF first as SRC-MASTER-PREV.
- Preserve all idea IDs unless merged or superseded.
- Update the delta table and add new sources to the source ledger.
- Add new discoveries to the idea index and recompute counts.
- Keep uncertainty labels visible.
- If repo evidence is mounted in a later run, let repo evidence correct or supersede PDF-only implementation claims, but do not silently delete prior PDF ideas.
- When merging duplicates, keep older idea_id values as aliases and create a canonical idea_id only with an explicit note.
- Recompute source hash and page counts for any reattached master or new packet.

Final PDF Self-Check

Check	Result
No obvious cut-off text	ReportLab tables use wrapped cells, split rows and landscape pages for wide ledger/index/matrix sections. Final render spot-check required.
No missing source ledger	Included current-run ledger and appended prior v1.5 ledger unchanged.
No missing delta section	Continuity and Delta Since Last Run included.
No missing open questions	Open Questions and Verification Backlog included.
No unsupported implementation claims	All repo paths and source-packet implementation examples labeled PROPOSED/NEEDS VERIFICATION unless supported by attached evidence.
No forgotten prior master ideas	SRC-MASTER-PREV appended unchanged as Appendix A.
No extra user-facing artifact	Only final PDF should be linked to user. Temporary build files are not deliverables.

Appendix A begins on the next page. It contains the full retained v1.5 Master MapLibre Components-Functions-Features.pdf baseline, appended unchanged as SRC-MASTER-PREV.

Master MapLibre Components-Functions-Features.pdf

Cumulative KFM MapLibre, Tile, Renderer, Style, UI, Evidence Drawer, and Release Architecture Atlas

v1.5 cumulative update - run date 2026-05-10 - output mode: PDF only

Truth posture	CONFIRMED doctrine and current-session source scans; PROPOSED implementation guidance; UNKNOWN current repo/runtime depth; NEEDS VERIFICATION for versions, hosting headers, rights, source endpoints, policy enforcement, performance budgets and mobile/native parity.
Prior master PDF detected	Yes - v1.4 copied as SRC-MASTER-PREV before overwrite and appended unchanged. Prior pages: 302; sha256 prefix c34e1b7b520b06e7.
Attached PDF count	42 PDFs detected in /mnt/data, including prior master and two new source packets.
Source ledger count	43 current-run records: SRC-RUN-PROMPT, SRC-MASTER-PREV, retained attached sources, and NEW SRC-057 / SRC-058.
Idea count	2257 cumulative idea records: 2162 prior + 95 v1.5 delta records.
New discoveries count	95 delta records from SRC-057 and SRC-058.
Expanded category clusters	21 category clusters expanded by new evidence.
Superseded ideas	0 - no attached evidence invalidated a prior idea.
Open verification items	295 cumulative estimate: prior 271 + 24 new runtime, hosting, consent, probe, and release-manifest verification items.
Evidence limits	No mounted KFM repo, tests, workflows, dashboards, logs, release manifests or runtime traces were available in this generation context. Paths are PROPOSED unless current repo evidence later verifies them.

Core determination: MapLibre remains a downstream renderer and interaction runtime. Tiles, PMTiles, COGs, styles, popups, screenshots, 3D scenes and AI answers remain downstream carriers, not sovereign truth.

1. Table of Contents

1. Cover Page and run metadata
2. Executive Determination
3. Continuity and Delta Since Last Run
4. Source Ledger
5. Master Idea Index - v1.5 Delta
6. Category Atlas
7. Detailed Category Chapters
8. Components-Functions-Features Matrix
9. Tile Strategy Chapter
10. Governance and Trust-Membrane Chapter
11. Implementation-Ready Object Map
12. Validation and Test Plan
13. Anti-Patterns Register
14. Open Questions and Verification Backlog
15. How to Continue This Master PDF Next Run

Final PDF Self-Check

Appendix A. Full retained v1.4 Master PDF baseline (SRC-MASTER-PREV, appended unchanged)

Navigation note. The v1.5 update packet is current-run specific. The full v1.4 baseline follows as Appendix A so prior source ledger entries, category chapters, matrices, idea IDs and open questions are not silently deleted.

2. Executive Determination

CONFIRMED from attached doctrine and SRC-MASTER-PREV: MapLibre is a disciplined 2D renderer and interaction runtime inside a governed shell. It is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority or AI authority. Public surfaces use governed APIs, released artifacts, catalog records, tile services, EvidenceBundle resolution and release manifests.

CONFIRMED from this run: the prior v1.4 master baseline was detected and copied before overwrite. New source packet SRC-057 adds concrete 3D Story Node, H3 tile-diff, soil PMTiles/COG, on-map provenance badge, DSSE/SLSA data-artifact attestation, Valhalla tile, and WZDx PMTiles patterns. New source packet SRC-058 adds the strongest new runtime-governance material: MapLibre vector vs Cesium server-rasterization decision gates, runtime probes, BAO/BLAKE3 verified PMTiles streaming, VerifyReceipt, /caps/issue, ReleaseManifest and consent sidecars.

PROPOSED synthesis: v1.5 should be treated as a runtime-verification and artifact-integrity update. It does not reorganize the prior master. The next implementation-ready slice is schema/fixture/test first: RuntimeProbeResult, ReleaseRuntimeGate, TileArtifactManifest/ReleaseManifest, VerifyReceipt, PMTiles BAO sidecar manifest, MapLibre verify-before-addSource, governed capability issuance, drawer telemetry and rollback replay fixtures.

UNKNOWN: actual repo topology, package manager, schema home, UI component homes, MapLibre GL JS target version, PMTiles tooling, PMTiles/COG hosting headers, browser/device-class performance, mobile/native parity, plugin allowlist, source rights, actual policy enforcement and release workflow remain unverified.

NEEDS VERIFICATION: every package version, external tool version, public-hosting header, OIDC/cosign/Rekor assumption, BAO/BLAKE3 implementation, Valhalla build behavior, WZDx schema pin, token signer latency, mobile device thresholds, and consent-verification mechanism.

Highest-risk anti-pattern: activating a map source, public Story Node, 3D scene, PMTiles archive, COG, badge, popup, route graph or Focus Mode answer before the corresponding ReleaseManifest, EvidenceBundle, policy state, proof object, citation validation and rollback target are available.

Metric	Value
Previous version	v1.4 cumulative update
Current version	v1.5 cumulative update
Prior ideas carried forward	2162
New SRC-057 delta ideas	50
New SRC-058 delta ideas	45
Total cumulative idea count	2257
Superseded ideas	0
Open verification items	295

3. Continuity and Delta Since Last Run

SRC-MASTER-PREV is the prior v1.4 Master MapLibre Components-Functions-Features.pdf. It remains the cumulative memory baseline and is appended unchanged. It does not outrank original evidence. Current-run source evidence can refine, expand, conflict with or supersede its entries. No supersession occurred in this run.

Delta item	Result
Prior version detected	v1.4, 302 pages, sha256 prefix c34e1b7b520b06e7
Version rule applied	+0.1 ordinary cumulative update. No fundamental structural reorganization.
New sources added	SRC-057 New Ideas 5.pdf; SRC-058 New Ideas 5-8-26.pdf.
Retained sources	All current attached sources and SRC-MASTER-PREV are ledgered. Prior-only sources inside v1.4 remain preserved in Appendix A.
Prior ideas preserved	All 2162 prior ML-* records are preserved by appending SRC-MASTER-PREV unchanged.
New records	95 ML-057-* and ML-058-* delta records.
Merged / superseded	0 merged, 0 superseded.
Conflicts discovered	No direct source conflict. The main tension is runtime maturity: source packets contain strong implementation recipes, but no mounted repo/runtime proof was available.
Open questions carried forward	Package versions, schema home, source rights, MLT status, mobile parity, hosting headers, sensitive geometry, release manifest home, CI workflow, accessibility and performance budgets.

3A. Prior category status comparison

Category	v1.5 delta count	Status / note
A. Renderer Boundary and KFM Trust Law	4	EXPANDED by SRC-057/SRC-058
B. MapLibre GL JS Web Shell	5	EXPANDED by SRC-057/SRC-058
C. MapLibre Native / Mobile / Platform Parity	0	RETAINED from SRC-MASTER-PREV
D. MapLibre RS / Experimental Renderers	0	RETAINED from SRC-MASTER-PREV
E. Style Specification, Sources, Layers, Expressions	1	EXPANDED by SRC-057/SRC-058
F. Sprites, Glyphs, Fonts, Design Tokens	0	RETAINED from SRC-MASTER-PREV
G. GeoJSON and Runtime Data	0	RETAINED from SRC-MASTER-PREV
H. Vector Tiles: MVT and MLT	3	EXPANDED by SRC-057/SRC-058
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	20	EXPANDED by SRC-057/SRC-058
J. Martin and Server-Mediated Tile Serving	3	EXPANDED by SRC-057/SRC-058
K. Raster, COG, DEM, Terrain, Hillshade	8	EXPANDED by SRC-057/SRC-058
L. WMS / WMTS / External Map Services	1	EXPANDED by SRC-057/SRC-058
M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	25	EXPANDED by SRC-057/SRC-058
N. Evidence Drawer Payloads and Click Resolution	2	EXPANDED by SRC-057/SRC-058
O. Focus Mode and Governed AI Map Context	4	EXPANDED by SRC-057/SRC-058
P. Time-Aware Map Interaction and Timeline	3	EXPANDED by SRC-057/SRC-058
Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	16	EXPANDED by SRC-057/SRC-058
R. Plugin, Wrapper, and Dependency Governance	5	EXPANDED by SRC-057/SRC-058
S. Accessibility, UX, and Trust-Visible States	9	EXPANDED by SRC-057/SRC-058
T. Performance, Caching, CDN, Range Requests, and Resource Timing	24	EXPANDED by SRC-057/SRC-058
U. Testing, CI, Validation, Rollback, and Proof Objects	35	EXPANDED by SRC-057/SRC-058
V. Exports, Screenshots, Reports, and Citation Preservation	4	EXPANDED by SRC-057/SRC-058
W. 3D / Cesium / Deck.gl / Overlay Interoperability	9	EXPANDED by SRC-057/SRC-058
X. Anti-Patterns and Failure Modes	3	EXPANDED by SRC-057/SRC-058
Y. Implementation Backlog and PR Plan	8	EXPANDED by SRC-057/SRC-058
Z. Open Questions and Verification Backlog	0	RETAINED from SRC-MASTER-PREV

4. Source Ledger

This ledger is a control surface, not a bibliography. It states what each attached source can support and what it cannot prove. Current external facts embedded in source PDFs remain source-supported but need verification before operational package pins, access decisions, source activation or release claims.

4A. Current-source scan highlights

source_id	file	scan result
SRC-057	New Ideas 5.pdf	944 pages; high signals: Tile/tiles 522, PMTiles 73, COG 131, MapLibre 33, Cesium 30, Focus Mode 221, Range 66.
SRC-058	New Ideas 5-8-26.pdf	275 pages; high signals: Tile/tiles 478, PMTiles 138, MapLibre 36, COG 16, Cesium 23, Range 34, PolicyDecision 10.
SRC-MASTER-PREV	Master MapLibre Components-Functions-Features.pdf	302 pages; retained v1.4 baseline with 2162 prior records.

4B. Complete current-run source ledger

source_id	file / document	type	status	role in this PDF	evidence strength	limitations
SRC-RUN-PR OMPT	Current user prompt	prompt	CONFIRMED	Controls output mode, cumulative rule, truth labels, source IDs, required sections, cite-or-abstain posture.	High	Instruction source, not implementation evidence.
SRC-MASTE R-PREV	Master MapLibre Components-Functions-Features.pdf v1.4	prior master PDF	CONFIRMED / RETAINED	Cumulative lineage baseline. Appended unchanged as Appendix A; preserves 2162 prior ideas and prior source ledger.	High for continuity	Does not outrank original evidence.
SRC-003	KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf (22 pp; sha 77f56ec1ab632b76)	KFM doctrine / UI manual	RETAINED	Renderer boundary, governed shell, Evidence Drawer, Focus Mode, tile/style strategy.	High	Implementation depth remains UNKNOWN without repo evidence.
SRC-004	Directory Rules.pdf (22 pp; sha 759de4fcb51cf0f5)	governance doctrine	RETAINED	Responsibility-root placement law; schema-home default; lifecycle boundaries.	High	Quoted paths remain PROPOSED until mounted repo verification.
SRC-005	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf (30 pp; sha 43d0c6fea4cc64ed)	pipeline manual	RETAINED	Lifecycle law; query-save-validate-compile-review-promote-recompile loop; MapLibre UI coupling.	High	No repo/runtime proof in this run.
SRC-006	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan_v1_1.pdf (28 pp; sha b28b0cba512b95c8)	build plan	RETAINED	Trust spine, pre-RAW event family, public-safe defaults, map architecture.	High	Greenfield/proposed-to-create posture.
SRC-007	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf (23 pp; sha b0766ae7c55a7b59)	UI + AI plan	RETAINED	Whole UI, MapLibre shell, governed API, Evidence Drawer, Focus Mode, review console.	High doctrine / proposed realization	No mounted repo proof.
SRC-008	KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf (36 pp; sha 0abbd5fcb54c21d6)	governed AI plan	RETAINED	Finite outcomes, citation validation, AIRceipt, source-ledger discipline.	High doctrine / proposed realization	No runtime implementation proof.
SRC-009	Ollama & Ubuntu Information.pdf (66 pp; sha 378fec4c010b918d)	runtime guide	RETAINED	Local model behind governed API; no direct public model traffic; EvidenceBundle order.	High doctrine / proposed realization	Version/runtime facts need recheck before use.
SRC-010	kfm_encyclopedia.pdf (82 pp; sha cc899a7a57cbadb5)	domain/capability encyclopedia	RETAINED	Cross-domain atlas; maps, tiles, graphs, AI answers are downstream carriers.	High doctrine	Implementation depth UNKNOWN.
SRC-011	# Kansas Frontier Matrix Implementation Reference.pdf (20 pp; sha d948332b6c5bfcd)	implementation reference	RETAINED	Prior public-repo summary and MapLibre/PMTiles stack signals; schema-home ambiguity.	Lineage / needs verification	Prior connector claims not reverified in this run.
SRC-012	a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf (321 pp; sha 0e75fd2bfbe9024b)	GIS/cartography reference	RETAINED	Map/geographic information representation background.	Conceptual	Not KFM implementation evidence.
SRC-013	Advanced-SQL-Concepts.pdf (112 pp; sha 5f489ba2ab4cc154)	SQL reference	RETAINED	Query/catalog/temporal analytics background.	Conceptual	Not MapLibre-specific.
SRC-014	AI_Concepts_Using_Python.pdf (435 pp; sha cb6275b4bf2c44e5)	AI reference	RETAINED	AI/data workflow background for bounded AI context.	Conceptual	Not MapLibre-specific.
SRC-015	Archaeological 3D GIS.pdf (177 pp; sha 27d60483276de20b)	3D GIS reference	RETAINED	3D GIS and model context for conditional 3D/archaeology visualization.	Conceptual	External archaeology examples, not KFM implementation.
SRC-016	CICD_with_Docker_Kubernetes_Semaphore.pdf (92 pp; sha 0fb1b3c2bd7d8811)	CI/CD reference	RETAINED	CI, rollback, release discipline background.	Conceptual	Not KFM implementation evidence.

source_id	file / document	type	status	role in this PDF	evidence strength	limitations
SRC-017	Designing Great Web APIs.pdf (45 pp; sha 160655602b709eef)	API reference	RETAINED	API contract and client/developer experience background.	Conceptual	Not KFM implementation evidence.
SRC-018	developing-time-oriented-database-applications-in-sql.pdf (528 pp; sha 63e7657d379daf3d)	temporal DB reference	RETAINED	Valid-time/transaction-time concepts for time-aware maps.	Conceptual	Not MapLibre implementation evidence.
SRC-019	Domain-Driven Design Reference.pdf (59 pp; sha 4406daa99ff0e3d5)	DDD reference	RETAINED	Bounded context, published language, responsibility layers.	Conceptual	Not KFM implementation evidence.
SRC-020	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf (141 pp; sha 0ef2469bee053ab1)	GIS lab/textbook	RETAINED	Environmental GIS and hazard/climate layer context.	Conceptual	ArcGIS-centric, not KFM implementation.
SRC-022	GIS Succinctly.pdf (110 pp; sha 391fca4fdee9be58)	GIS reference	RETAINED	Spatial database, geometry, GIS application basics.	Conceptual	Not KFM implementation evidence.
SRC-023	GIS in Sustainable Urban Planning and Management.pdf (365 pp; sha 9b4e13779ccf282e)	urban GIS reference	RETAINED	Planning support, spatial indicators, risk and stakeholder map context.	Conceptual	Not KFM implementation evidence.
SRC-024	instant-web-scraping-with-java.pdf (72 pp; sha b48a86014e094268)	scraping reference	RETAINED	Source acquisition caution and robust scraping background.	Conceptual	Not KFM implementation evidence.
SRC-025	KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf (125 pp; sha 8b7cd350451791b6)	idea atlas	RETAINED	Run receipts, deterministic identity, promotion gates, source intake patterns.	High doctrine/lineage	Prior synthesis, not repo proof.
SRC-026	KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf (123 pp; sha b1c6de0c6eb3c80f)	idea atlas	RETAINED	Prior synthesis of New Ideas 4-25/4-30; source-aware ingest, PMTiles, MLT and map stack guidance.	High lineage	Not current repo proof.
SRC-027	KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf (88 pp; sha 71fe46e2d137c8bd)	idea atlas	RETAINED	Pre-RAW event family, watchers, catalog integrity, environmental tiles.	High lineage	Not current repo proof.
SRC-028	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf (48 pp; sha 0bf41c729bc2139d)	domain blueprint	RETAINED	Agriculture layers, catalog closure, MapLibre/Evidence Drawer posture.	Domain lineage	Implementation paths PROPOSED.
SRC-029	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf (51 pp; sha 585d7e927a4accf8)	domain blueprint	RETAINED	Exact site-location denial and candidate remote-sensing caution.	Domain lineage	Implementation paths PROPOSED.
SRC-030	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf (52 pp; sha 69d0d5e9a6988668)	domain blueprint	RETAINED	Air/smoke/EO layers; strict knowledge-character labels.	Domain lineage	Implementation paths PROPOSED.
SRC-031	KFM_Fauna_Architecture_PDF_Only_Report.pdf (37 pp; sha 47309b5112286697)	domain blueprint	RETAINED	Occurrence geoprivacy and public-safe derived products.	Domain lineage	Implementation paths PROPOSED.
SRC-032	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf (36 pp; sha 5507d694e6500df9)	domain blueprint	RETAINED	Rare-plant controls, occurrence redaction, drawer implications.	Domain lineage	Implementation paths PROPOSED.
SRC-033	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf (42 pp; sha d334f43df8d74f1)	domain blueprint	RETAINED	Geology/resource map layers, public-safe geometry, Focus Mode.	Domain lineage	Implementation paths PROPOSED.
SRC-034	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf (20 pp; sha a95594c12d1dd5fb)	domain blueprint	RETAINED	Habitat assignment proof slice and Evidence Drawer.	Domain lineage	Implementation paths PROPOSED.
SRC-035	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf (28 pp; sha 82626239b2f12690)	domain blueprint	RETAINED	Habitat patches, suitability, connectivity, public-safe products.	Domain lineage	Implementation paths PROPOSED.
SRC-036	kfm_hazards_extended_pro_pdf_only_blueprint.pdf (26 pp; sha 52bd2f3e85a0fc12)	domain blueprint	RETAINED	Hazard layers, no emergency alerting, evidence-backed UI.	Domain lineage	Implementation paths PROPOSED.
SRC-037	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf (43 pp; sha 0e7c3531bd8d94d0)	domain blueprint	RETAINED	Hydrology proof lane, HUC12/NHDPlus/USGS/NFHL, map layers.	Domain lineage	Implementation paths PROPOSED.

source_id	file / document	type	status	role in this PDF	evidence strength	limitations
SRC-038	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf (30 pp; sha 6833e9ecf8ba7302)	domain blueprint	RETAINED	Living-person/DNA restrictions, temporal land assertions, map caution.	Domain lineage	Implementation paths PROPOSED.
SRC-039	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf (35 pp; sha 8a0c22ab452ea5b9)	domain blueprint	RETAINED	Transport/corridor layers, graph projections, Evidence Drawer.	Domain lineage	Implementation paths PROPOSED.
SRC-040	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf (43 pp; sha 5b6a6124c91297ed)	domain blueprint	RETAINED	Settlement/infrastructure layer manifests and sensitive assets.	Domain lineage	Implementation paths PROPOSED.
SRC-041	kfm_soil_architecture_extended_pro_pdf_only_report.pdf (25 pp; sha 7c2d498212b9ad56)	domain blueprint	RETAINED	Soil maps, SSURGO/gSSURGO, PMTiles stub, Evidence Drawer.	Domain lineage	Implementation paths PROPOSED.
SRC-056	KFM_Pass_15_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf (48 pp; sha 1c62fdb3d141068d)	idea atlas	RETAINED	Map-serving lane gap: MapLibre time slider + PMTiles + COG sidecars need runtime proof parity.	High lineage	Already incorporated in v1.4.
SRC-057	New Ideas 5.pdf (944 pp; sha 094200ad69f3843d)	new source packet	NEW	Adds 3D Story Node hybrid pattern, on-map provenance badges, H3 tile diffs, soil PMTiles/COG artifacts, DSSE/SLSA artifact attestations, Valhalla tiles, WZDX PMTiles.	High for v1.5 delta	External version claims inside it are not revalidated.
SRC-058	New Ideas 5-8-26.pdf (275 pp; sha 2faec2c08b68eddd)	new source packet	NEW	Adds runtime verification decision matrix, MapLibre vs Cesium raster fallback probes, BAO/BLAKE3 PMTiles verification, /caps/issue, VerifyReceipt, ReleaseManifest and consent sidecars.	High for v1.5 delta	External version claims inside it are not revalidated.

5. Master Idea Index - v1.5 Delta

The full cumulative v1.4 idea index of 2162 prior ideas is retained in Appendix A via SRC-MASTER-PREV. The records below add the v1.5 delta. Each record includes source page references and is labeled NEW or EXPANDED. Implementation consequences are PROPOSED unless a future mounted-repo inspection verifies them.

idea_id	idea title	cat	status	source	normalized statement
ML-057-001	Cesium + MapLibre hybrid Story Node proof artifact	W / A	NEW	SRC-057 pp.1-4	A low-friction 3D Story Node demo can use CesiumJS for globe/terrain/3D Tiles while MapLibre remains the 2D exploration companion.
ML-057-002	MapLibre remains 2D vector, label, and time-slider runtime	B / P	EXPANDED	SRC-057 pp.1-2	The hybrid pattern keeps MapLibre for vector layers, labels, and 2D time-slider interactions.
ML-057-003	Cesium mode returns to MapLibre timeline	W / P	NEW	SRC-057 pp.2-3	The story flow returns from Cesium Scene back to MapLibre, preserving narrative continuity.
ML-057-004	3D Tiles streaming is future path, not first rearchitecture	W	NEW	SRC-057 pp.1-3	The proposed demo is a proof-of-capability with a future-proof path toward 3D Tiles streaming.
ML-057-005	Terrain overlay / extruded GeoJSON is acceptable proof-only 3D input	W / K	NEW	SRC-057 pp.2-3	The fastest demo path is terrain plus one simple KFM GeoJSON layer or tiny 3D Tiles set.
ML-057-006	PM2.5 fusion outputs COG and tile artifacts	K / L / T	NEW	SRC-057 pp.33-39	The air-quality fusion recipe emits daily analysis TIFF/COG surfaces, tiles, quicklooks, STAC, DCAT and PROV.
ML-057-007	PM2.5 catalog closure uses STAC/DCAT/PROV	M / V	NEW	SRC-057 pp.33-39	The fusion output includes STAC Items, DCAT rolling collection, and PROV entities for sources and activities.
ML-057-008	KGS geology layers may publish COGs/tiles only after license review	K / Q	NEW	SRC-057 pp.41-43	The geology pattern rasterizes categorical geology to COGs and publishes public tiles/extracts only after license review.
ML-057-009	Geology layer metadata needs legends, scale badges, and uncertainty notes	E / S	NEW	SRC-057 pp.41-43	The UI/story notes require unit_code legends, county/state scale badges, and uncertainty notes in map toggles.
ML-057-010	SSURGO/gNATSGO conversion to GeoParquet and COG requires CRS/units/provenance	K / M	EXPANDED	SRC-057 pp.76-77	The soil snippet converts vectors to GeoParquet and rasters to COG with consistent CRS/units plus checks and provenance.
ML-057-011	MapLibre QA preview for H3 gaps and overlaps	S / U	NEW	SRC-057 pp.81-82	Reviewers can preview H3 gaps/overlaps in MapLibre with warning/danger styles and density CSVs.
ML-057-012	H3 density and overlap checks become CI gates for tilings	U / T	NEW	SRC-057 pp.81-82	The H3 QA pattern requires no gaps, no overlaps, and acceptable density z-scores.
ML-057-013	Automation/provenance badges can be pinned on MapLibre and Cesium features	S / V	NEW	SRC-057 pp.96-101	The on-map badge component paints pipeline run status and provenance directly on MapLibre or Cesium features/tilesets.
ML-057-014	AutomationBadge DTO links attestations, SBOMs, manifests and logs	M / S	NEW	SRC-057 pp.97-98	Badge payloads include status, last_run_ts, run id and links to attestation, SBOM, manifest and logs.
ML-057-015	MapLibre badge adapter binds sourceId and featureIdProp	B / S	NEW	SRC-057 p.98	The MapLibre integration attaches badges to a GeoJSON source using sourceId and featureId property matching.
ML-057-016	Badge streams degrade from SSE/WebSocket to polling	T / S	NEW	SRC-057 pp.99-100	Automation status can stream via SSE/WebSocket with fallback polling every 15s and redraw debouncing.
ML-057-017	Browser must not fetch untrusted attestations directly	Q / R	NEW	SRC-057 p.99	The security note says remote attestations should be proxied through a server that verifies signatures and strips secrets; CORS allowlists and content types are enforced.
ML-057-018	On-map badges require accessibility and snapshot tests	S / U	NEW	SRC-057 pp.100-101	The component definition of done includes keyboard access, ARIA labels, MapLibre/Cesium demos and Playwright map snapshots.
ML-057-019	Story Node deep link can open drawer and Focus Mode for a feature	N / O	NEW	SRC-057 p.101	The component touchpoints include deep-linking focus=featureId to open a drawer in Focus Mode.
ML-057-020	Watcher-first H3 tile-diff pattern	I / M	NEW	SRC-057 pp.118-122	The tile-diff pattern detects source edits, computes H3-tile checksums, emits tiny GeoParquet/PMTiles diffs, and updates STAC/DCAT/PROV.
ML-057-021	ArcGIS editingInfo, ETag and Last-Modified can trigger tile rebuilds	T / U	NEW	SRC-057 pp.119-120	Watchers can compare ArcGIS editingInfo, ETags, Last-Modified or checksums before recomputing.
ML-057-022	Canonical row hash and tile hash decide changed tiles	I / M	NEW	SRC-057 pp.120-121	The preprocess recipe hashes canonicalized geometry/properties per row and aggregates sorted hashes per H3 tile.
ML-057-023	Only changed GeoParquet/PMTiles tile artifacts should be emitted	I / T	NEW	SRC-057 p.121	The diff routine emits per-tile artifacts only for H3 cells whose hash changed.
ML-057-024	Tile diffs update STAC, DCAT and PROV with content hashes	M / U	NEW	SRC-057 pp.119-122	Each diff gets STAC Item, DCAT Distribution and PROV activity with content hashes.
ML-057-025	Air-quality drift monitor may emit pairing map tiles for Story/Focus	O / P	NEW	SRC-057 pp.161-162	The drift monitor saves diagnostics and optional pairing map tiles for Story Nodes and Focus Mode.

idea_id	idea title	cat	status	source	normalized statement
ML-057-026	Soil vectors to GeoParquet to tippecanoe to PMTiles	I / H	EXPANDED	SRC-057 pp.162-171	The soil runbook converts harmonized soil vectors to GeoParquet then MBTiles/PMTiles with tippecanoe.
ML-057-027	Soil rasters to COG for range-enabled delivery	K / T	EXPANDED	SRC-057 pp.165-166	The soil raster recipe converts SoilGrids-like rasters to COGs with compression, block size and overviews for HTTP range reads.
ML-057-028	Deterministic soil run IDs derive from source hashes	M	NEW	SRC-057 pp.163-164	The runbook stamps soil artifacts with dataset_id/run_id derived from source hashes and date.
ML-057-029	Tippecanoe parameter changes require version bump	H / U	NEW	SRC-057 pp.165,171	The governance checklist says tippecanoe parameters are pinned and changes require a version bump.
ML-057-030	PMTiles STAC asset uses roles tiles	M / I	EXPANDED	SRC-057 pp.166-167	The STAC example declares a PMTiles asset with type application/vnd.pmtiles and roles [tiles].
ML-057-031	PROV tile-generation activity should include tippecanoe and pmtiles software	M / U	NEW	SRC-057 pp.167-168	The PROV fragment records tile-generation activity using source Parquet and software:tippecanoe/software:pmtiles.
ML-057-032	COG STAC assets need raster/proj/eo fields when known	K / M	NEW	SRC-057 p.168	The runbook instructs emitting similar STAC Items for COGs with raster:bands, proj:* and eo:* fields when known.
ML-057-033	QA with pmtiles inspect and gdalinfo	U / I / K	EXPANDED	SRC-057 pp.168-169	Quick QA calls pmtiles inspect for tiles and gdalinfo/cloud-optimized checks for COGs.
ML-057-034	Artifact governance checklist spans immutable inputs, CRS, sidecars, logs and licenses	U / Q	EXPANDED	SRC-057 p.171	The checklist requires immutable inputs, explicit CRS, deterministic names, STAC/PROV sidecars, digest logs, and license/attribution.
ML-057-035	DSSE/SLSA attestations bind data artifacts to config and identity	M / U	EXPANDED	SRC-057 pp.206-213	The attestation pattern binds COG/PMTiles/GeoParquet to artifact digest, config hash, pipeline run and OIDC builder identity.
ML-057-036	STAC, DCAT and PROV should discover attestation bundles	M / V	NEW	SRC-057 pp.211-212	Catalog fragments link STAC assets, DCAT distributions and PROV entities to DSSE/in-toto attestations.
ML-057-037	Consumers verify artifact digest and attestation before loading	T / U	EXPANDED	SRC-057 pp.212-213	The consumer-side verification step checks artifact digest and cosign verify-attestation before trusting/loading data.
ML-057-038	Publish gate requires verify-attestation pass	U / Y	EXPANDED	SRC-057 p.213	The runbook says gate publish by requiring verify-attestation to pass before promotion.
ML-057-039	Valhalla routing graph tiles are versioned derived artifacts	J / I	NEW	SRC-057 pp.213-220	The Valhalla pattern builds OSM-derived graph tiles/tar files plus metadata for downstream routing.
ML-057-040	Valhalla smoke tests precede STAC promotion	U / J	NEW	SRC-057 pp.218-220	The routing template runs API smoke tests before promoting tiles.tar and metadata through STAC/catalog.
ML-057-041	Valhalla tile metadata records OSM sha256	M / J	NEW	SRC-057 pp.218-220	The Valhalla build writes metadata with built_at and OSM sha256.
ML-057-042	WZDx v4.x validation to GeoParquet and optional PMTiles	I / Q	NEW	SRC-057 pp.243-250	The WZDx pattern validates GeoJSON against JSON Schema, normalizes to GeoParquet, and optionally builds PMTiles.
ML-057-043	WZDx ETag/Last-Modified freshness watcher	T / U	NEW	SRC-057 pp.247-248	The WZDx watcher uses conditional GET with ETag and Last-Modified to avoid unnecessary transforms.
ML-057-044	WZDx STAC asset includes PMTiles with roles tiles	M / I	NEW	SRC-057 pp.248-249	The STAC template includes geoparquet and pmtiles assets; PMTiles uses application/vnd.pmtiles and roles tiles.
ML-057-045	WZDx governance includes source license, masking/redaction and sovereignty checks	Q / X	NEW	SRC-057 pp.249-250	Promotion notes require license from feed, ETag/Last-Modified provenance, masking/redaction rules and Indigenous data sovereignty checks.
ML-057-046	Soil differential tile registry uses WAL and checksum lineage	I / T	NEW	SRC-057 pp.297-303	The soil differential update pattern records per-tile WAL entries, tile_id/src_checksum anchors and observability metrics.
ML-057-047	Rollback shifts tile lineage pointers instead of deleting artifacts	U / I	EXPANDED	SRC-057 pp.301-303	Rollback keeps tile variants and moves lineage/catalog references back to prior tile_id/checksum sets.
ML-057-048	Focus Mode can explain tile re-ingestion and rollback behavior	O / I	NEW	SRC-057 p.303	Focus Mode should explain checksum comparisons, new/changed/unchanged branching and rollback by shifting pointers.
ML-057-049	Release artifacts include manifest, SBOM, attestations and signatures	M / U	EXPANDED	SRC-057 pp.479-483	A governed release should inventory artifacts, SBOM, attestations and signing evidence.
ML-057-050	Story Nodes must reference cataloged checksum-inventoried Items	V / W	EXPANDED	SRC-057 pp.481-482	Story Nodes must reference cataloged Items whose artifacts are checksum-inventoried and verifiable under release rules.
ML-058-001	MapLibre vector vs Cesium server-rasterization decision matrix	A / W / T	NEW	SRC-058 pp.26-30	The source compares MapLibre vector tiles with Cesium plus server-side rasterization based on verification burden and mobile stability.
ML-058-002	MapLibre tile decode median threshold is less than 50 ms	T / U	NEW	SRC-058 pp.27,31	CI probes should measure per-tile decode latency with median under 50 ms and preferred 95p under 75 ms.
ML-058-003	Client WASM hash throughput threshold is greater than 100 MB/s	T / U	NEW	SRC-058 pp.27,31-32	Client integrity checks need hash throughput above 100 MB/s; failures disable client hashing or route to raster fallback.
ML-058-004	Native heap growth threshold is less than 10 MB/min	T / U	NEW	SRC-058 pp.27,32	Heavy pan/zoom soak tests should keep sustained native heap growth below 10 MB/min and stabilize after GC.
ML-058-005	Allowlist token handshake p50 threshold is less than 100 ms	T / Q	NEW	SRC-058 pp.27,32-33	Server-signed raster/PMTiles mode needs client-edge-signer token p50 below 100 ms and low error rate.
ML-058-006	Per-tile telemetry envelope is required	T / U	EXPANDED	SRC-058 pp.27,33-34	Telemetry should capture tile_id, zoom, fetch_ms, decode_ms, render_ms and tile_bytes.

idea_id	idea title	cat	status	source	normalized statement
ML-058-007	Failure telemetry must log OOM, decode, throttling, backpressure and signature mismatches	T / U	NEW	SRC-058 p.34	Failure telemetry must log OOM signals, tile decode exceptions, throttling, backpressure, token failures and signature mismatches.
ML-058-008	Unknown device class defaults to safer path	Q / T	NEW	SRC-058 p.34	KFM governance mapping says unknown device class defaults to a safer path.
ML-058-009	Runtime verification stability is publication prerequisite	U / Y	EXPANDED	SRC-058 pp.37-40	The source states runtime verification stability is a publication prerequisite, not a post-release observation.
ML-058-010	tools/probes is proposed home for runtime probe harnesses	U / R	NEW	SRC-058 pp.37-39	The source proposes tools/probes for MapLibre decode/hash/heap probes, Cesium raster/token probes, fixtures and reports.
ML-058-011	Probes measure runtime behavior and do not decide truth	A / U	NEW	SRC-058 p.38	Runtime probes emit reviewable reports for promotion gates; they do not decide truth.
ML-058-012	Threshold changes need steward/security/release review	R / U	NEW	SRC-058 p.40	Probe implementation, threshold and token/signature changes require maintainers, stewards, security or release review.
ML-058-013	Delta PMTiles with BAO sidecar and signed manifest	I / M	NEW	SRC-058 pp.40-42	Mobile delivery pattern emits latest.pmtiles, latest.pmtiles.bao, manifest.json and manifest.sig.
ML-058-014	Mobile worker verifies PMTiles byte ranges before activation	I / T	NEW	SRC-058 pp.41,47-48	A WebWorker streams PMTiles ranges and verifies chunks against root hash before exposing tiles to MapLibre.
ML-058-015	CI A/B/C checks build-sign-verify, streaming and run receipt	U / I	NEW	SRC-058 pp.41-46	The workflow includes build/sign/verify sanity, emulator streaming benchmark and signed RunReceipt.
ML-058-016	KFM-compatible PMTiles ReleaseManifest includes BAO root hash	M / I	NEW	SRC-058 pp.44-46	The minimal manifest records artifact, bao_sidecar, root_hash, spec_hash, tile_stats and policy.
ML-058-017	Browser worker returns DENY on chunk verification failure	Q / T	NEW	SRC-058 pp.47-48	The worker posts status DENY with reason chunk_verification_failed when BAO verification fails.
ML-058-018	PMTiles publication gates include invalid spec, unsigned manifest, unverified chunk and rollback root mismatch	U / X	NEW	SRC-058 pp.48-49	Recommended gates include invalid_spec_hash, unsigned_release_manifest, unverified_tile_chunk, public_unsigned_delta, rollback_root_mismatch and missing_run_receipt.
ML-058-019	PMTiles promotion flow moves from delta build to release only after range verification and receipt	Y / I	NEW	SRC-058 p.49	Flow: Delta PMTiles built -> BAO root -> manifest -> cosign signature -> range verify -> all gates -> publish candidate -> RunReceipt -> released.
ML-058-020	MapLibre app init verifies sidecar before addSource	B / I / U	EXPANDED	SRC-058 pp.190-194,208-211	MapLibre initialization fetches sidecar, verifies DSSE/cosign signature, checks spec_hash/tiling_scheme/tile_format, samples ranges, then addSource.
ML-058-021	Cesium path prefers server-rasterized PMTiles or COG	W / K	EXPANDED	SRC-058 pp.193-194	For Cesium, the source recommends server-side rasterization or raster PMTiles/COG with signed manifests.
ML-058-022	PMTiles sidecar fields include root hash, tile format, tiling scheme and optional byte ranges	M / I	NEW	SRC-058 pp.192,215-217	Sidecar templates include spec_hash, pmtiles_version, root_hash, root_hash_algo, tile_format, tiling_scheme, min/max zoom, byte_ranges_manifest and signature fields.
ML-058-023	Tippecanoe to PMTiles metadata fields are required	H / M	NEW	SRC-058 pp.202-205	The PMTiles production example records tile_format, tiling_scheme, minzoom and maxzoom metadata.
ML-058-024	Source_head capture and canonical spec_hash precede tile publication	M / T	EXPANDED	SRC-058 pp.202-213	The tile pipeline captures source_head, canonicalizes descriptors and computes deterministic spec_hash before generating PMTiles.
ML-058-025	Invalid signature or missing sidecar must DENY	Q / X	NEW	SRC-058 p.214	The failure table maps invalid signature and missing sidecar to DENY.
ML-058-026	VerifyReceipt records digest, bounds and schema verification	M / U	NEW	SRC-058 pp.236-240	VerifyReceipt generation records spec_hash, tileset and verification flags including digest, bounds and schema verified.
ML-058-027	/caps/issue grants tile capability only after receipt verification	Q / R	NEW	SRC-058 pp.238-240	The governed API issues a short-lived tiles:read JWT only if receipt verification passes.
ML-058-028	OPA capability policy denies missing spec_hash or failed digest/bounds/schema checks	Q / U	NEW	SRC-058 pp.239-240	Rego policy denies capability issuance when spec_hash is missing or digest/bounds/schema verification fails.
ML-058-029	Public client only sees released PMTiles, signed manifests, governed caps, evidence receipts and approved layers	A / Q	EXPANDED	SRC-058 pp.240-241	The public-safe invariant explicitly forbids RAW, WORK, unpublished candidate tiles, direct model runtimes and bypassed caps.
ML-058-030	Recommended build order puts verifier/cap gates before MapLibre runtime integration	Y / U	NEW	SRC-058 pp.240-241	The next build order starts with BLAKE3 verifier, VerifyReceipt, OPA cap gate, PMTiles ReleaseManifest, then MapLibre runtime integration.
ML-058-031	ReleaseManifest binds PMTiles to EvidenceBundles, policy, sensitivity, receipts, rollback, correction and DSSE	M / Y	NEW	SRC-058 pp.241-244	ReleaseManifest becomes the canonical publication object binding PMTiles, EvidenceBundles, policy posture, sensitivity, provenance receipts, rollback, correction lineage, spec_hash and attestations.
ML-058-032	Public runtime rule for ReleaseManifest	Q / U	NEW	SRC-058 pp.244-246	Public clients may consume a ReleaseManifest only when PUBLISHED, policy/rights known, sensitivity public, evidence_refs and artifact hashes exist, rollback is supported, and policy allows.

idea_id	idea title	cat	status	source	normalized statement
ML-058-033	Consent metadata may live alongside EvidenceBundle/tile sidecar without raw PII	Q / M	NEW	SRC-058 pp.247-248	Consent-first metadata stores consent scope, retention, credential refs, receipt pointers and status pointers beside EvidenceBundle/tile sidecar; no raw PII goes in tiles/artifacts.
ML-058-034	Render-time consent verification blocks invalid/revoked/expired displays	Q / S	NEW	SRC-058 pp.247-248	Render-time verification checks VC proof, DSSE receipt signature and revocation/validity; missing, invalid, revoked or expired proof blocks render.
ML-058-035	PMTiles Range/CDN delivery pairs with BAO partial verification	T / I	EXPANDED	SRC-058 pp.40-41	PMTiles one-file archives are hosted on range-friendly static/CDN stores and BAO allows partial range verification.
ML-058-036	Server-side rasterization shifts verification burden to signer/hosting duties	W / T	NEW	SRC-058 pp.26-30	The matrix says server-rasterized Cesium/raster path lowers client CPU/WASM while increasing server signing/hosting complexity.
ML-058-037	MapLibre vector path enables styling and interaction but increases client CPU/memory risk	B / T	EXPANDED	SRC-058 pp.26-30	MapLibre vector tiles support offline styling, rich interaction and client filtering but carry higher client verification and heap risk.
ML-058-038	Cesium raster path is safer for public mobile but less flexible	W / S	NEW	SRC-058 pp.29-36	The decision matrix recommends Cesium/server-rasterized PMTiles/rasters for public mobile clients when bounded verification is the priority.
ML-058-039	RuntimeProbeResult and ReleaseRuntimeGate schemas are proposed deliverables	U / M	NEW	SRC-058 p.37	Recommended deliverables include RuntimeProbeResult.schema.json and ReleaseRuntimeGate.schema.json.
ML-058-040	Signed PMTiles publication ADR and raster fallback policy are required	Y / R	NEW	SRC-058 p.37	The source calls for a signed PMTiles publication ADR and raster fallback publication policy.
ML-058-041	Focus Mode runtime telemetry drawer contract	O / N / T	NEW	SRC-058 p.37	The next deliverables include a Focus Mode runtime telemetry drawer contract.
ML-058-042	MapLibre runtime integration follows verifier, cap gate and release manifest	B / Y	EXPANDED	SRC-058 pp.240-241	Build order places MapLibre runtime integration after verifier, receipt, cap gate and ReleaseManifest.
ML-058-043	Correction and rollback replay verification follows delta PMTiles publish	U / Y	NEW	SRC-058 p.241	The build order includes correction/rollback replay verification after delta PMTiles publish.
ML-058-044	ReleaseManifest artifact kinds include pmtiles, stac, geojson, parquet, model, manifest and receipt	M	NEW	SRC-058 pp.242-243	The ReleaseManifest schema artifact kind enum includes pmtiles, stac, geojson, parquet, model, manifest and receipt.
ML-058-045	ReleaseManifest policy blocks unknown rights and non-public sensitivity	Q / U	EXPANDED	SRC-058 pp.244-246	The Rego policy denies unknown policy_label, unknown rights_status, non-public sensitivity, missing evidence_refs/artifacts or unsupported rollback.

5A. Full v1.5 Delta Idea Records

ML-057-001 - Cesium + MapLibre hybrid Story Node proof artifact

idea_id	ML-057-001
title	Cesium + MapLibre hybrid Story Node proof artifact
category	W / A
status	NEW
source evidence	SRC-057 pp.1-4
normalized statement	A low-friction 3D Story Node demo can use CesiumJS for globe/terrain/3D Tiles while MapLibre remains the 2D exploration companion.
implementation consequence	Implement 3D as a Story Node mode, not a replacement for the governed MapLibre shell.
validation consequence	Validate handoff 2D -> 3D -> 2D, release state, and citation visibility.
risk / anti-pattern	Do not let 3D scene become canonical truth or bypass evidence drawer.

ML-057-002 - MapLibre remains 2D vector, label, and time-slider runtime

idea_id	ML-057-002
title	MapLibre remains 2D vector, label, and time-slider runtime
category	B / P
status	EXPANDED
source evidence	SRC-057 pp.1-2
normalized statement	The hybrid pattern keeps MapLibre for vector layers, labels, and 2D time-slider interactions.
implementation consequence	Keep MapLibre in renderer/interaction role; time controls select released snapshots.
validation consequence	Test time-slider only loads released layer snapshots and preserves bounds/camera state.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-003 - Cesium mode returns to MapLibre timeline

idea_id	ML-057-003
title	Cesium mode returns to MapLibre timeline
category	W / P
status	NEW
source evidence	SRC-057 pp.2-3
normalized statement	The story flow returns from Cesium Scene back to MapLibre, preserving narrative continuity.
implementation consequence	Model Story Node state transitions explicitly and keep a rollback/default MapLibre route.
validation consequence	E2E test camera/fade transition and return-to-timeline behavior.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-004 - 3D Tiles streaming is future path, not first rearchitecture

idea_id	ML-057-004
title	3D Tiles streaming is future path, not first rearchitecture
category	W
status	NEW
source evidence	SRC-057 pp.1-3
normalized statement	The proposed demo is a proof-of-capability with a future-proof path toward 3D Tiles streaming.
implementation consequence	Admit 3D Tiles only after release manifests and evidence context survive in 3D mode.
validation consequence	Validate 3D artifact manifests, review state, and public-safe status before loading.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-005 - Terrain overlay / extruded GeoJSON is acceptable proof-only 3D input

idea_id	ML-057-005
title	Terrain overlay / extruded GeoJSON is acceptable proof-only 3D input
category	W / K
status	NEW
source evidence	SRC-057 pp.2-3
normalized statement	The fastest demo path is terrain plus one simple KFM GeoJSON layer or tiny 3D Tiles set.
implementation consequence	Use small synthetic/released fixtures for early 3D proof rather than live sensitive layers.
validation consequence	Visual regression and evidence-link checks on the simple 3D layer.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-006 - PM2.5 fusion outputs COG and tile artifacts

idea_id	ML-057-006
title	PM2.5 fusion outputs COG and tile artifacts
category	K / L / T
status	NEW
source evidence	SRC-057 pp.33-39
normalized statement	The air-quality fusion recipe emits daily analysis TIFF/COG surfaces, tiles, quicklooks, STAC, DCAT and PROV.
implementation consequence	Treat fused PM2.5 maps as derived surfaces with source-role and uncertainty labels.
validation consequence	Validate COG structure, tile generation, catalog closure and freshness.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-007 - PM2.5 catalog closure uses STAC/DCAT/PROV

idea_id	ML-057-007
title	PM2.5 catalog closure uses STAC/DCAT/PROV
category	M / V
status	NEW
source evidence	SRC-057 pp.33-39
normalized statement	The fusion output includes STAC Items, DCAT rolling collection, and PROV entities for sources and activities.
implementation consequence	Map layers should load only from cataloged/released PM2.5 artifacts.
validation consequence	Require STAC/DCAT/PROV checks before map exposure.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-008 - KGS geology layers may publish COGs/tiles only after license review

idea_id	ML-057-008
title	KGS geology layers may publish COGs/tiles only after license review
category	K / Q
status	NEW
source evidence	SRC-057 pp.41-43
normalized statement	The geology pattern rasterizes categorical geology to COGs and publishes public tiles/extracts only after license review.
implementation consequence	Keep geologic map layers under source-rights and attribution gates.
validation consequence	Fail release when license, source series, or attribution is missing.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-009 - Geology layer metadata needs legends, scale badges, and uncertainty notes

idea_id	ML-057-009
title	Geology layer metadata needs legends, scale badges, and uncertainty notes
category	E / S
status	NEW
source evidence	SRC-057 pp.41-43
normalized statement	The UI/story notes require unit_code legends, county/state scale badges, and uncertainty notes in map toggles.
implementation consequence	LayerManifest should include legend, scale, and uncertainty display fields.
validation consequence	Test layer toggle surfaces legend and scale/review caveats.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-010 - SSURGO/gNATSGO conversion to GeoParquet and COG requires CRS/units/provenance

idea_id	ML-057-010
title	SSURGO/gNATSGO conversion to GeoParquet and COG requires CRS/units/provenance
category	K / M
status	EXPANDED
source evidence	SRC-057 pp.76-77
normalized statement	The soil snippet converts vectors to GeoParquet and rasters to COG with consistent CRS/units plus checks and provenance.
implementation consequence	Use GeoParquet as analysis/pre-tile format and COG for raster delivery.
validation consequence	Validate CRS, units, raster metadata and provenance sidecars.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-011 - MapLibre QA preview for H3 gaps and overlaps

idea_id	ML-057-011
title	MapLibre QA preview for H3 gaps and overlaps
category	S / U
status	NEW
source evidence	SRC-057 pp.81-82
normalized statement	Reviewers can preview H3 gaps/overlaps in MapLibre with warning/danger styles and density CSVs.
implementation consequence	Add QA-only map layers outside public publication path.
validation consequence	CI fails on gaps/overlaps unless allowlisted; MapLibre preview must not imply release.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-012 - H3 density and overlap checks become CI gates for tilings

idea_id	ML-057-012
title	H3 density and overlap checks become CI gates for tilings
category	U / T
status	NEW
source evidence	SRC-057 pp.81-82
normalized statement	The H3 QA pattern requires no gaps, no overlaps, and acceptable density z-scores.
implementation consequence	Use H3 QA before publishing tiled inventories.
validation consequence	Fixture tests for gap, overlap and density edge cases.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-013 - Automation/provenance badges can be pinned on MapLibre and Cesium features

idea_id	ML-057-013
title	Automation/provenance badges can be pinned on MapLibre and Cesium features
category	S / V
status	NEW
source evidence	SRC-057 pp.96-101
normalized statement	The on-map badge component paints pipeline run status and provenance directly on MapLibre or Cesium features/tilesets.
implementation consequence	Use badges as trust-visible UI annotations, not proof substitutes.
validation consequence	Bridge tests bind feature IDs to badge DTOs and drawer links.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-014 - AutomationBadge DTO links attestations, SBOMs, manifests and logs

idea_id	ML-057-014
title	AutomationBadge DTO links attestations, SBOMs, manifests and logs
category	M / S
status	NEW
source evidence	SRC-057 pp.97-98
normalized statement	Badge payloads include status, last_run_ts, run id and links to attestation, SBOM, manifest and logs.
implementation consequence	Define an AutomationBadgePayload projection separate from EvidenceBundle.
validation consequence	Validate links, status enum and run ID; block secret-bearing links.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-015 - MapLibre badge adapter binds sourceId and featureIdProp

idea_id	ML-057-015
title	MapLibre badge adapter binds sourceId and featureIdProp
category	B / S
status	NEW
source evidence	SRC-057 p.98
normalized statement	The MapLibre integration attaches badges to a GeoJSON source using sourceId and featureId property matching.
implementation consequence	Keep feature identity deterministic and use promoted/feature IDs where practical.
validation consequence	Test badge joins cannot drift from layer feature IDs.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-016 - Badge streams degrade from SSE/WebSocket to polling

idea_id	ML-057-016
title	Badge streams degrade from SSE/WebSocket to polling
category	T / S
status	NEW
source evidence	SRC-057 pp.99-100
normalized statement	Automation status can stream via SSE/WebSocket with fallback polling every 15s and redraw debouncing.
implementation consequence	UI should show stale/degraded stream status visibly.
validation consequence	Latency and stale-state tests for stream outage and polling fallback.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-017 - Browser must not fetch untrusted attestations directly

idea_id	ML-057-017
title	Browser must not fetch untrusted attestations directly
category	Q / R
status	NEW
source evidence	SRC-057 p.99
normalized statement	The security note says remote attestations should be proxied through a server that verifies signatures and strips secrets; CORS allowlists and content types are enforced.
implementation consequence	Govern attestation fetching behind an API/proxy boundary.
validation consequence	Security tests for origin allowlist, content type and secret stripping.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-018 - On-map badges require accessibility and snapshot tests

idea_id	ML-057-018
title	On-map badges require accessibility and snapshot tests
category	S / U
status	NEW
source evidence	SRC-057 pp.100-101
normalized statement	The component definition of done includes keyboard access, ARIA labels, MapLibre/Cesium demos and Playwright map snapshots.
implementation consequence	Trust badges must be accessible and testable.
validation consequence	A11y, keyboard and visual regression tests for badge overlays.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-019 - Story Node deep link can open drawer and Focus Mode for a feature

idea_id	ML-057-019
title	Story Node deep link can open drawer and Focus Mode for a feature
category	N / O
status	NEW
source evidence	SRC-057 p.101
normalized statement	The component touchpoints include deep-linking focus=featureId to open a drawer in Focus Mode.
implementation consequence	MapContextEnvelope should carry feature and layer IDs for Focus Mode.
validation consequence	Test deep links resolve released feature evidence and abstain when absent.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-020 - Watcher-first H3 tile-diff pattern

idea_id	ML-057-020
title	Watcher-first H3 tile-diff pattern
category	I / M
status	NEW
source evidence	SRC-057 pp.118-122
normalized statement	The tile-diff pattern detects source edits, computes H3-tile checksums, emits tiny GeoParquet/PMTiles diffs, and updates STAC/DCAT/PROV.
implementation consequence	Use watcher output as candidate bundles, not automatic publication.
validation consequence	Validate tile checksum reproducibility and catalog/provenance closure.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-021 - ArcGIS editingInfo, ETag and Last-Modified can trigger tile rebuilds

idea_id	ML-057-021
title	ArcGIS editingInfo, ETag and Last-Modified can trigger tile rebuilds
category	T / U
status	NEW
source evidence	SRC-057 pp.119-120
normalized statement	Watchers can compare ArcGIS editingInfo, ETags, Last-Modified or checksums before recomputing.
implementation consequence	Record source_head in RunReceipt and reject stale/unstable source state.
validation consequence	Fixtures for 304/no-change, changed ETag and missing header behavior.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-022 - Canonical row hash and tile hash decide changed tiles

idea_id	ML-057-022
title	Canonical row hash and tile hash decide changed tiles
category	I / M
status	NEW
source evidence	SRC-057 pp.120-121
normalized statement	The preprocess recipe hashes canonicalized geometry/properties per row and aggregates sorted hashes per H3 tile.
implementation consequence	Use canonical hashing to avoid silent recomputation drift.
validation consequence	Cross-run deterministic hash tests with reordered inputs.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-023 - Only changed GeoParquet/PMTiles tile artifacts should be emitted

idea_id	ML-057-023
title	Only changed GeoParquet/PMTiles tile artifacts should be emitted
category	I / T
status	NEW
source evidence	SRC-057 p.121
normalized statement	The diff routine emits per-tile artifacts only for H3 cells whose hash changed.
implementation consequence	Implement incremental tile publishing with explicit unchanged no-op receipts.
validation consequence	CI compares changed tile set to expected fixture diffs.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-024 - Tile diffs update STAC, DCAT and PROV with content hashes

idea_id	ML-057-024
title	Tile diffs update STAC, DCAT and PROV with content hashes
category	M / U
status	NEW
source evidence	SRC-057 pp.119-122
normalized statement	Each diff gets STAC Item, DCAT Distribution and PROV activity with content hashes.
implementation consequence	Treat tile diffs as cataloged release candidates.
validation consequence	Schema tests ensure each diff has digest, source, lineage and rollback info.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-025 - Air-quality drift monitor may emit pairing map tiles for Story/Focus

idea_id	ML-057-025
title	Air-quality drift monitor may emit pairing map tiles for Story/Focus
category	O / P
status	NEW
source evidence	SRC-057 pp.161-162
normalized statement	The drift monitor saves diagnostics and optional pairing map tiles for Story Nodes and Focus Mode.
implementation consequence	Diagnostics layers require review labels and should not be public facts by default.
validation consequence	Focus Mode test explains diagnostics and source limitations.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-026 - Soil vectors to GeoParquet to tippecanoe to PMTiles

idea_id	ML-057-026
title	Soil vectors to GeoParquet to tippecanoe to PMTiles
category	I / H
status	EXPANDED
source evidence	SRC-057 pp.162-171
normalized statement	The soil runbook converts harmonized soil vectors to GeoParquet then MBTiles/PMTiles with tippecanoe.
implementation consequence	Use GeoParquet as neutral pre-tile artifact and PMTiles as downstream delivery.
validation consequence	Validate tippecanoe parameters, layer name, zooms and feature attributes.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-027 - Soil rasters to COG for range-enabled delivery

idea_id	ML-057-027
title	Soil rasters to COG for range-enabled delivery
category	K / T
status	EXPANDED
source evidence	SRC-057 pp.165-166
normalized statement	The soil raster recipe converts SoilGrids-like rasters to COGs with compression, block size and overviews for HTTP range reads.
implementation consequence	COGs need TileArtifactManifest or raster artifact manifest parity.
validation consequence	Validate COG profile, range/read capability and raster metadata.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-028 - Deterministic soil run IDs derive from source hashes

idea_id	ML-057-028
title	Deterministic soil run IDs derive from source hashes
category	M
status	NEW
source evidence	SRC-057 pp.163-164
normalized statement	The runbook stamps soil artifacts with dataset_id/run_id derived from source hashes and date.
implementation consequence	Record run_id/spec_hash in manifests and STAC properties.
validation consequence	Cross-run determinism test for same inputs.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-029 - Tippecanoe parameter changes require version bump

idea_id	ML-057-029
title	Tippecanoe parameter changes require version bump
category	H / U
status	NEW
source evidence	SRC-057 pp.165,171
normalized statement	The governance checklist says tippecanoe parameters are pinned and changes require a version bump.
implementation consequence	Treat tile-generalization parameters as release-significant.
validation consequence	Regression test tile output and manifest diff when parameters change.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-030 - PMTiles STAC asset uses roles tiles

idea_id	ML-057-030
title	PMTiles STAC asset uses roles tiles
category	M / I
status	EXPANDED
source evidence	SRC-057 pp.166-167
normalized statement	The STAC example declares a PMTiles asset with type application/vnd.pmtiles and roles [tiles].
implementation consequence	Standardize tile asset roles and media types.
validation consequence	Schema test for asset role/type and released artifact digest.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-031 - PROV tile-generation activity should include tippecanoe and pmtiles software

idea_id	ML-057-031
title	PROV tile-generation activity should include tippecanoe and pmtiles software
category	M / U
status	NEW
source evidence	SRC-057 pp.167-168
normalized statement	The PROV fragment records tile-generation activity using source Parquet and software:tippecanoe/software:pmtiles.
implementation consequence	Source tools become provenance materials, not hidden pipeline details.
validation consequence	Validate PROV entity/activity/agent relation exists for tiles.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-032 - COG STAC assets need raster/proj/eo fields when known

idea_id	ML-057-032
title	COG STAC assets need raster/proj/eo fields when known
category	K / M
status	NEW
source evidence	SRC-057 p.168
normalized statement	The runbook instructs emitting similar STAC Items for COGs with raster:bands, proj:* and eo:* fields when known.
implementation consequence	Require CRS/band metadata for raster map surfaces.
validation consequence	STAC extension presence and field-type tests for COG assets.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-033 - QA with pmtiles inspect and gdalinfo

idea_id	ML-057-033
title	QA with pmtiles inspect and gdalinfo
category	U / I / K
status	EXPANDED
source evidence	SRC-057 pp.168-169
normalized statement	Quick QA calls pmtiles inspect for tiles and gdalinfo/cloud-optimized checks for COGs.
implementation consequence	Add these checks to release candidates.
validation consequence	Fail release on invalid PMTiles index or non-COG raster.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-034 - Artifact governance checklist spans immutable inputs, CRS, sidecars, logs and licenses

idea_id	ML-057-034
title	Artifact governance checklist spans immutable inputs, CRS, sidecars, logs and licenses
category	U / Q
status	EXPANDED
source evidence	SRC-057 p.171
normalized statement	The checklist requires immutable inputs, explicit CRS, deterministic names, STAC/PROV sidecars, digest logs, and license/attribution.
implementation consequence	Map layer release gate should include these checks.
validation consequence	Manifest completeness test for each artifact family.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-035 - DSSE/SLSA attestations bind data artifacts to config and identity

idea_id	ML-057-035
title	DSSE/SLSA attestations bind data artifacts to config and identity
category	M / U
status	EXPANDED
source evidence	SRC-057 pp.206-213
normalized statement	The attestation pattern binds COG/PMTiles/GeoParquet to artifact digest, config hash, pipeline run and OIDC builder identity.
implementation consequence	Publish attestations with tile/raster artifacts and link them from catalogs.
validation consequence	Verify attestation identity, artifact digest and config hash before promotion.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-036 - STAC, DCAT and PROV should discover attestation bundles

idea_id	ML-057-036
title	STAC, DCAT and PROV should discover attestation bundles
category	M / V
status	NEW
source evidence	SRC-057 pp.211-212
normalized statement	Catalog fragments link STAC assets, DCAT distributions and PROV entities to DSSE/in-toto attestations.
implementation consequence	Make attestation refs machine-discoverable.
validation consequence	Schema tests for attestation link presence and media type.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-037 - Consumers verify artifact digest and attestation before loading

idea_id	ML-057-037
title	Consumers verify artifact digest and attestation before loading
category	T / U
status	EXPANDED
source evidence	SRC-057 pp.212-213
normalized statement	The consumer-side verification step checks artifact digest and cosign verify-attestation before trusting/loading data.
implementation consequence	Public clients or gateway should verify artifacts before activation.
validation consequence	Negative tests for digest mismatch and failed identity constraints.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-038 - Publish gate requires verify-attestation pass

idea_id	ML-057-038
title	Publish gate requires verify-attestation pass
category	U / Y
status	EXPANDED
source evidence	SRC-057 p.213
normalized statement	The runbook says gate publish by requiring verify-attestation to pass before promotion.
implementation consequence	PromotionDecision should record verification result and failure reason.
validation consequence	Policy fixture blocks unsigned or unverifiable tile/raster artifacts.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-039 - Valhalla routing graph tiles are versioned derived artifacts

idea_id	ML-057-039
title	Valhalla routing graph tiles are versioned derived artifacts
category	J / I
status	NEW
source evidence	SRC-057 pp.213-220
normalized statement	The Valhalla pattern builds OSM-derived graph tiles/tar files plus metadata for downstream routing.
implementation consequence	Treat routing graph tiles as derived artifacts with source/digest/release state.
validation consequence	Build smoke tests and STAC manifest checks before exposure.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-040 - Valhalla smoke tests precede STAC promotion

idea_id	ML-057-040
title	Valhalla smoke tests precede STAC promotion
category	U / J
status	NEW
source evidence	SRC-057 pp.218-220
normalized statement	The routing template runs API smoke tests before promoting tiles.tar and metadata through STAC/catalog.
implementation consequence	Routing graph publication needs route/isochrone smoke tests and catalog closure.
validation consequence	CI tests for route response, metadata JSON and artifact digest.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-041 - Valhalla tile metadata records OSM sha256

idea_id	ML-057-041
title	Valhalla tile metadata records OSM sha256
category	M / J
status	NEW
source evidence	SRC-057 pp.218-220
normalized statement	The Valhalla build writes metadata with built_at and OSM sha256.
implementation consequence	Source snapshot identity must travel with routing graph artifacts.
validation consequence	Test metadata matches PBF digest and build time format.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-042 - WZDx v4.x validation to GeoParquet and optional PMTiles

idea_id	ML-057-042
title	WZDx v4.x validation to GeoParquet and optional PMTiles
category	I / Q
status	NEW
source evidence	SRC-057 pp.243-250
normalized statement	The WZDx pattern validates GeoJSON against JSON Schema, normalizes to GeoParquet, and optionally builds PMTiles.
implementation consequence	Transport work-zone layers need schema pinning and deterministic transforms.
validation consequence	Validate schema, CRS EPSG:4326, sorted rows/columns and optional PMTiles output.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-043 - WZDx ETag/Last-Modified freshness watcher

idea_id	ML-057-043
title	WZDx ETag/Last-Modified freshness watcher
category	T / U
status	NEW
source evidence	SRC-057 pp.247-248
normalized statement	The WZDx watcher uses conditional GET with ETag and Last-Modified to avoid unnecessary transforms.
implementation consequence	Record freshness headers in PROV and RunReceipt.
validation consequence	Tests for 304 response, changed headers and missing header fallback.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-044 - WZDx STAC asset includes PMTiles with roles tiles

idea_id	ML-057-044
title	WZDx STAC asset includes PMTiles with roles tiles
category	M / I
status	NEW
source evidence	SRC-057 pp.248-249
normalized statement	The STAC template includes geoparquet and pmtiles assets; PMTiles uses application/vnd.pmtiles and roles tiles.
implementation consequence	Standardize transport tile layer asset metadata.
validation consequence	Schema validation for source, license, roles and artifact hashes.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-045 - WZDx governance includes source license, masking/redaction and sovereignty checks

idea_id	ML-057-045
title	WZDx governance includes source license, masking/redaction and sovereignty checks
category	Q / X
status	NEW
source evidence	SRC-057 pp.249-250
normalized statement	Promotion notes require license from feed, ETag/Last-Modified provenance, masking/redaction rules and Indigenous data sovereignty checks.
implementation consequence	Treat roadway event map layers as governed, not automatically public.
validation consequence	Policy tests block unknown license or missing redaction decision.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-046 - Soil differential tile registry uses WAL and checksum lineage

idea_id	ML-057-046
title	Soil differential tile registry uses WAL and checksum lineage
category	I / T
status	NEW
source evidence	SRC-057 pp.297-303
normalized statement	The soil differential update pattern records per-tile WAL entries, tile_id/src_checksum anchors and observability metrics.
implementation consequence	Maintain tile lineage registry separate from rendered tiles.
validation consequence	Test restart/reconciliation and OpenLineage correlation.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-047 - Rollback shifts tile lineage pointers instead of deleting artifacts

idea_id	ML-057-047
title	Rollback shifts tile lineage pointers instead of deleting artifacts
category	U / I
status	EXPANDED
source evidence	SRC-057 pp.301-303
normalized statement	Rollback keeps tile variants and moves lineage/catalog references back to prior tile_id/checksum sets.
implementation consequence	MapReleaseManifest rollback target should be pointer-based.
validation consequence	Rollback drill restores previous manifest and verifies tile digest set.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-048 - Focus Mode can explain tile re-ingestion and rollback behavior

idea_id	ML-057-048
title	Focus Mode can explain tile re-ingestion and rollback behavior
category	O / I
status	NEW
source evidence	SRC-057 p.303
normalized statement	Focus Mode should explain checksum comparisons, new/changed/unchanged branching and rollback by shifting pointers.
implementation consequence	MapContextEnvelope should include tile lineage and release context.
validation consequence	Focus Mode answer fixture cites tile registry and manifest evidence.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-049 - Release artifacts include manifest, SBOM, attestations and signatures

idea_id	ML-057-049
title	Release artifacts include manifest, SBOM, attestations and signatures
category	M / U
status	EXPANDED
source evidence	SRC-057 pp.479-483
normalized statement	A governed release should inventory artifacts, SBOM, attestations and signing evidence.
implementation consequence	Map/tile releases need the same supply-chain manifest coverage.
validation consequence	Release gate checks all artifacts are checksum-inventoried and signed as required.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-057-050 - Story Nodes must reference cataloged checksum-inventoried Items

idea_id	ML-057-050
title	Story Nodes must reference cataloged checksum-inventoried Items
category	V / W
status	EXPANDED
source evidence	SRC-057 pp.481-482
normalized statement	Story Nodes must reference cataloged Items whose artifacts are checksum-inventoried and verifiable under release rules.
implementation consequence	No Story Node should cite a tile/scene as evidence without catalog/proof closure.
validation consequence	Story export test verifies citations and artifact checksums.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-001 - MapLibre vector vs Cesium server-rasterization decision matrix

idea_id	ML-058-001
title	MapLibre vector vs Cesium server-rasterization decision matrix
category	A / W / T
status	NEW
source evidence	SRC-058 pp.26-30
normalized statement	The source compares MapLibre vector tiles with Cesium plus server-side rasterization based on verification burden and mobile stability.
implementation consequence	Choose runtime mode based on evidence, interaction needs and verified performance, not preference.
validation consequence	RuntimeProbeResult must drive publication decision.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-002 - MapLibre tile decode median threshold is less than 50 ms

idea_id	ML-058-002
title	MapLibre tile decode median threshold is less than 50 ms
category	T / U
status	NEW
source evidence	SRC-058 pp.27,31
normalized statement	CI probes should measure per-tile decode latency with median under 50 ms and preferred 95p under 75 ms.
implementation consequence	Set performance budgets before public vector publication.
validation consequence	Fail vector release when decode budget is exceeded.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-003 - Client WASM hash throughput threshold is greater than 100 MB/s

idea_id	ML-058-003
title	Client WASM hash throughput threshold is greater than 100 MB/s
category	T / U
status	NEW
source evidence	SRC-058 pp.27,31-32
normalized statement	Client integrity checks need hash throughput above 100 MB/s; failures disable client hashing or route to raster fallback.
implementation consequence	Do not force integrity verification onto devices that cannot sustain it.
validation consequence	Benchmark hash worker on target device classes.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-004 - Native heap growth threshold is less than 10 MB/min

idea_id	ML-058-004
title	Native heap growth threshold is less than 10 MB/min
category	T / U
status	NEW
source evidence	SRC-058 pp.27,32
normalized statement	Heavy pan/zoom soak tests should keep sustained native heap growth below 10 MB/min and stabilize after GC.
implementation consequence	Block mobile vector mode if heap churn is unstable.
validation consequence	Soak test scripted camera paths and layer toggles.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-005 - Allowlist token handshake p50 threshold is less than 100 ms

idea_id	ML-058-005
title	Allowlist token handshake p50 threshold is less than 100 ms
category	T / Q
status	NEW
source evidence	SRC-058 pp.27,32-33
normalized statement	Server-signed raster/PMTiles mode needs client-edge-signer token p50 below 100 ms and low error rate.
implementation consequence	Token issuance becomes part of runtime gate.
validation consequence	Fail signed-raster path on signer latency or elevated errors.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-006 - Per-tile telemetry envelope is required

idea_id	ML-058-006
title	Per-tile telemetry envelope is required
category	T / U
status	EXPANDED
source evidence	SRC-058 pp.27,33-34
normalized statement	Telemetry should capture tile_id, zoom, fetch_ms, decode_ms, render_ms and tile_bytes.
implementation consequence	Record per-tile metrics in probe reports and runtime observations.
validation consequence	Schema test RuntimeProbeResult metrics fields.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-007 - Failure telemetry must log OOM, decode, throttling, backpressure and signature mismatches

idea_id	ML-058-007
title	Failure telemetry must log OOM, decode, throttling, backpressure and signature mismatches
category	T / U
status	NEW
source evidence	SRC-058 p.34
normalized statement	Failure telemetry must log OOM signals, tile decode exceptions, throttling, backpressure, token failures and signature mismatches.
implementation consequence	Expose runtime problems as release-gate evidence.
validation consequence	Negative fixtures for each failure reason.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-008 - Unknown device class defaults to safer path

idea_id	ML-058-008
title	Unknown device class defaults to safer path
category	Q / T
status	NEW
source evidence	SRC-058 p.34
normalized statement	KFM governance mapping says unknown device class defaults to a safer path.
implementation consequence	Deny or rasterize until device class is approved.
validation consequence	Test unknown profile cannot activate vector mode.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-009 - Runtime verification stability is publication prerequisite

idea_id	ML-058-009
title	Runtime verification stability is publication prerequisite
category	U / Y
status	EXPANDED
source evidence	SRC-058 pp.37-40
normalized statement	The source states runtime verification stability is a publication prerequisite, not a post-release observation.
implementation consequence	ReleaseManifest should require runtime gate status for public/mobile layers.
validation consequence	PromotionDecision blocks release if probe report fails.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-010 - tools/probes is proposed home for runtime probe harnesses

idea_id	ML-058-010
title	tools/probes is proposed home for runtime probe harnesses
category	U / R
status	NEW
source evidence	SRC-058 pp.37-39
normalized statement	The source proposes tools/probes for MapLibre decode/hash/heap probes, Cesium raster/token probes, fixtures and reports.
implementation consequence	Treat path as PROPOSED; Directory Rules basis is tools as shared implementation/validation root.
validation consequence	Validate placement against Directory Rules and actual repo before implementation.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-011 - Probes measure runtime behavior and do not decide truth

idea_id	ML-058-011
title	Probes measure runtime behavior and do not decide truth
category	A / U
status	NEW
source evidence	SRC-058 p.38
normalized statement	Runtime probes emit reviewable reports for promotion gates; they do not decide truth.
implementation consequence	Keep runtime probe results separate from EvidenceBundle content.
validation consequence	Schema separates performance status from evidence truth claims.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-012 - Threshold changes need steward/security/release review

idea_id	ML-058-012
title	Threshold changes need steward/security/release review
category	R / U
status	NEW
source evidence	SRC-058 p.40
normalized statement	Probe implementation, threshold and token/signature changes require maintainers, stewards, security or release review.
implementation consequence	Add review requirements to policy/manifest metadata.
validation consequence	CI fails threshold changes without review record.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-013 - Delta PMTiles with BAO sidecar and signed manifest

idea_id	ML-058-013
title	Delta PMTiles with BAO sidecar and signed manifest
category	I / M
status	NEW
source evidence	SRC-058 pp.40-42
normalized statement	Mobile delivery pattern emits latest.pmtiles, latest.pmtiles.bao, manifest.json and manifest.sig.
implementation consequence	Treat PMTiles+BAO+manifest as a release bundle.
validation consequence	Verify BAO root hash and cosign signature before publish.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-014 - Mobile worker verifies PMTiles byte ranges before activation

idea_id	ML-058-014
title	Mobile worker verifies PMTiles byte ranges before activation
category	I / T
status	NEW
source evidence	SRC-058 pp.41,47-48
normalized statement	A WebWorker streams PMTiles ranges and verifies chunks against root hash before exposing tiles to MapLibre.
implementation consequence	Activate tiles only after successful range verification.
validation consequence	Worker tests for verified and corrupted chunk paths.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-015 - CI A/B/C checks build-sign-verify, streaming and run receipt

idea_id	ML-058-015
title	CI A/B/C checks build-sign-verify, streaming and run receipt
category	U / I
status	NEW
source evidence	SRC-058 pp.41-46
normalized statement	The workflow includes build/sign/verify sanity, emulator streaming benchmark and signed RunReceipt.
implementation consequence	Bundle tile integrity checks with promotion CI.
validation consequence	CI fixture exercises all three gates.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-016 - KFM-compatible PMTiles ReleaseManifest includes BAO root hash

idea_id	ML-058-016
title	KFM-compatible PMTiles ReleaseManifest includes BAO root hash
category	M / I
status	NEW
source evidence	SRC-058 pp.44-46
normalized statement	The minimal manifest records artifact, bao_sidecar, root_hash, spec_hash, tile_stats and policy.
implementation consequence	Extend TileArtifactManifest/MapReleaseManifest to include BAO root hash fields.
validation consequence	Schema validation for root_hash, spec_hash, tile_stats and policy block.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-017 - Browser worker returns DENY on chunk verification failure

idea_id	ML-058-017
title	Browser worker returns DENY on chunk verification failure
category	Q / T
status	NEW
source evidence	SRC-058 pp.47-48
normalized statement	The worker posts status DENY with reason chunk_verification_failed when BAO verification fails.
implementation consequence	Surface corrupted tile activation as DENY/ERROR, not silent retry.
validation consequence	Negative test asserts failed chunk never reaches MapLibre source.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-018 - PMTiles publication gates include invalid spec, unsigned manifest, unverified chunk and rollback root mismatch

idea_id	ML-058-018
title	PMTiles publication gates include invalid spec, unsigned manifest, unverified chunk and rollback root mismatch
category	U / X
status	NEW
source evidence	SRC-058 pp.48-49
normalized statement	Recommended gates include invalid_spec_hash, unsigned_release_manifest, unverified_tile_chunk, public_unsigned_delta, rollback_root_mismatch and missing_run_receipt.
implementation consequence	Add these to anti-pattern/gate register.
validation consequence	OPA/validator fixtures for each failure condition.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-019 - PMTiles promotion flow moves from delta build to release only after range verification and receipt

idea_id	ML-058-019
title	PMTiles promotion flow moves from delta build to release only after range verification and receipt
category	Y / I
status	NEW
source evidence	SRC-058 p.49
normalized statement	Flow: Delta PMTiles built -> BAO root -> manifest -> cosign signature -> range verify -> all gates -> publish candidate -> RunReceipt -> released.
implementation consequence	Model promotion as a state transition with explicit proof artifacts.
validation consequence	End-to-end fixture verifies transition order and rollback target.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-020 - MapLibre app init verifies sidecar before addSource

idea_id	ML-058-020
title	MapLibre app init verifies sidecar before addSource
category	B / I / U
status	EXPANDED
source evidence	SRC-058 pp.190-194,208-211
normalized statement	MapLibre initialization fetches sidecar, verifies DSSE/cosign signature, checks spec_hash/tiling_scheme/tile_format, samples ranges, then addSource.
implementation consequence	Do not call addSource on unverified PMTiles.
validation consequence	Unit/E2E test blocks source registration when sidecar invalid.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-021 - Cesium path prefers server-rasterized PMTiles or COG

idea_id	ML-058-021
title	Cesium path prefers server-rasterized PMTiles or COG
category	W / K
status	EXPANDED
source evidence	SRC-058 pp.193-194
normalized statement	For Cesium, the source recommends server-side rasterization or raster PMTiles/COG with signed manifests.
implementation consequence	Use Cesium for public mobile/3D when vector verification cost is too high.
validation consequence	Test raster manifest verification and token flow.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-022 - PMTiles sidecar fields include root hash, tile format, tiling scheme and optional byte ranges

idea_id	ML-058-022
title	PMTiles sidecar fields include root hash, tile format, tiling scheme and optional byte ranges
category	M / I
status	NEW
source evidence	SRC-058 pp.192,215-217
normalized statement	Sidecar templates include spec_hash, pmtiles_version, root_hash, root_hash_algo, tile_format, tiling_scheme, min/max zoom, byte_ranges_manifest and signature fields.
implementation consequence	Define TileArtifactManifest fields for PMTiles integrity.
validation consequence	Schema tests for required sidecar fields.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-023 - Tippecanoe to PMTiles metadata fields are required

idea_id	ML-058-023
title	Tippecanoe to PMTiles metadata fields are required
category	H / M
status	NEW
source evidence	SRC-058 pp.202-205
normalized statement	The PMTiles production example records tile_format, tiling_scheme, minzoom and maxzoom metadata.
implementation consequence	Add metadata completeness to tile build output.
validation consequence	Validate metadata fields match actual PMTiles inspect output.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-024 - Source_head capture and canonical spec_hash precede tile publication

idea_id	ML-058-024
title	Source_head capture and canonical spec_hash precede tile publication
category	M / T
status	EXPANDED
source evidence	SRC-058 pp.202-213
normalized statement	The tile pipeline captures source_head, canonicalizes descriptors and computes deterministic spec_hash before generating PMTiles.
implementation consequence	SourceDescriptor version and source_head become release preconditions.
validation consequence	Cross-run spec_hash reproducibility and source header tests.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-025 - Invalid signature or missing sidecar must DENY

idea_id	ML-058-025
title	Invalid signature or missing sidecar must DENY
category	Q / X
status	NEW
source evidence	SRC-058 p.214
normalized statement	The failure table maps invalid signature and missing sidecar to DENY.
implementation consequence	PolicyDecision should deny map render if sidecar proof is absent/invalid.
validation consequence	Negative runtime fixtures for missing sidecar and bad signature.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-026 - VerifyReceipt records digest, bounds and schema verification

idea_id	ML-058-026
title	VerifyReceipt records digest, bounds and schema verification
category	M / U
status	NEW
source evidence	SRC-058 pp.236-240
normalized statement	VerifyReceipt generation records spec_hash, tileset and verification flags including digest, bounds and schema verified.
implementation consequence	Use VerifyReceipt before capability issuance.
validation consequence	Schema test and signed receipt fixture.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-027 - /caps/issue grants tile capability only after receipt verification

idea_id	ML-058-027
title	/caps/issue grants tile capability only after receipt verification
category	Q / R
status	NEW
source evidence	SRC-058 pp.238-240
normalized statement	The governed API issues a short-lived tiles:read JWT only if receipt verification passes.
implementation consequence	Public clients fetch tiles with governed capabilities, not raw open URLs when access is constrained.
validation consequence	API test rejects failed digest receipts and checks TTL/scope.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-028 - OPA capability policy denies missing spec_hash or failed digest/bounds/schema checks

idea_id	ML-058-028
title	OPA capability policy denies missing spec_hash or failed digest/bounds/schema checks
category	Q / U
status	NEW
source evidence	SRC-058 pp.239-240
normalized statement	Rego policy denies capability issuance when spec_hash is missing or digest/bounds/schema verification fails.
implementation consequence	Policy-as-code mirrors API checks.
validation consequence	Conftest fixtures for each deny reason.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-029 - Public client only sees released PMTiles, signed manifests, governed caps, evidence receipts and approved layers

idea_id	ML-058-029
title	Public client only sees released PMTiles, signed manifests, governed caps, evidence receipts and approved layers
category	A / Q
status	EXPANDED
source evidence	SRC-058 pp.240-241
normalized statement	The public-safe invariant explicitly forbids RAW, WORK, unpublished candidate tiles, direct model runtimes and bypassed caps.
implementation consequence	Enforce trust membrane for map assets.
validation consequence	No-public-raw-path and no-unreleased-tile tests.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-030 - Recommended build order puts verifier/cap gates before MapLibre runtime integration

idea_id	ML-058-030
title	Recommended build order puts verifier/cap gates before MapLibre runtime integration
category	Y / U
status	NEW
source evidence	SRC-058 pp.240-241
normalized statement	The next build order starts with BLAKE3 verifier, VerifyReceipt, OPA cap gate, PMTiles ReleaseManifest, then MapLibre runtime integration.
implementation consequence	Do not start with UI polish or MapLibre addSource code.
validation consequence	Acceptance gate requires verifier and release manifest before UI binding.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-031 - ReleaseManifest binds PMTiles to EvidenceBundles, policy, sensitivity, receipts, rollback, correction and DSSE

idea_id	ML-058-031
title	ReleaseManifest binds PMTiles to EvidenceBundles, policy, sensitivity, receipts, rollback, correction and DSSE
category	M / Y
status	NEW
source evidence	SRC-058 pp.241-244
normalized statement	ReleaseManifest becomes the canonical publication object binding PMTiles, EvidenceBundles, policy posture, sensitivity, provenance receipts, rollback, correction lineage, spec_hash and attestations.
implementation consequence	Use MapReleaseManifest as release authority for map assets.
validation consequence	Schema and policy tests for all binding fields.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-032 - Public runtime rule for ReleaseManifest

idea_id	ML-058-032
title	Public runtime rule for ReleaseManifest
category	Q / U
status	NEW
source evidence	SRC-058 pp.244-246
normalized statement	Public clients may consume a ReleaseManifest only when PUBLISHED, policy/rights known, sensitivity public, evidence_refs and artifact hashes exist, rollback is supported, and policy allows.
implementation consequence	Encode public map-layer gate in policy.
validation consequence	OPA fixture blocks draft, unknown rights, restricted sensitivity and missing rollback.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-033 - Consent metadata may live alongside EvidenceBundle/tile sidecar without raw PII

idea_id	ML-058-033
title	Consent metadata may live alongside EvidenceBundle/tile sidecar without raw PII
category	Q / M
status	NEW
source evidence	SRC-058 pp.247-248
normalized statement	Consent-first metadata stores consent scope, retention, credential refs, receipt pointers and status pointers beside EvidenceBundle/tile sidecar; no raw PII goes in tiles/artifacts.
implementation consequence	Support consent-bound map rendering with pointers only.
validation consequence	Privacy fixture verifies no PII in tile sidecar.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-034 - Render-time consent verification blocks invalid/revoked/expired displays

idea_id	ML-058-034
title	Render-time consent verification blocks invalid/revoked/expired displays
category	Q / S
status	NEW
source evidence	SRC-058 pp.247-248
normalized statement	Render-time verification checks VC proof, DSSE receipt signature and revocation/validity; missing, invalid, revoked or expired proof blocks render.
implementation consequence	Treat consent failure as DENY before map draw.
validation consequence	Test revoked consent prevents layer/drawer rendering.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-035 - PMTiles Range/CDN delivery pairs with BAO partial verification

idea_id	ML-058-035
title	PMTiles Range/CDN delivery pairs with BAO partial verification
category	T / I
status	EXPANDED
source evidence	SRC-058 pp.40-41
normalized statement	PMTiles one-file archives are hosted on range-friendly static/CDN stores and BAO allows partial range verification.
implementation consequence	Hosting must support byte ranges and immutable bundles.
validation consequence	Range/CORS/head tests before publication.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-036 - Server-side rasterization shifts verification burden to signer/hosting duties

idea_id	ML-058-036
title	Server-side rasterization shifts verification burden to signer/hosting duties
category	W / T
status	NEW
source evidence	SRC-058 pp.26-30
normalized statement	The matrix says server-rasterized Cesium/raster path lowers client CPU/WASM while increasing server signing/hosting complexity.
implementation consequence	Document operational burden before choosing raster fallback.
validation consequence	Signer SLA and token path tests replace client decode/hash tests.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-037 - MapLibre vector path enables styling and interaction but increases client CPU/memory risk

idea_id	ML-058-037
title	MapLibre vector path enables styling and interaction but increases client CPU/memory risk
category	B / T
status	EXPANDED
source evidence	SRC-058 pp.26-30
normalized statement	MapLibre vector tiles support offline styling, rich interaction and client filtering but carry higher client verification and heap risk.
implementation consequence	Use only after device-class probe passes.
validation consequence	Decode/hash/heap stress tests before vector release.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-038 - Cesium raster path is safer for public mobile but less flexible

idea_id	ML-058-038
title	Cesium raster path is safer for public mobile but less flexible
category	W / S
status	NEW
source evidence	SRC-058 pp.29-36
normalized statement	The decision matrix recommends Cesium/server-rasterized PMTiles/rasters for public mobile clients when bounded verification is the priority.
implementation consequence	3D/raster should be conditional and public-safe, not default for all layers.
validation consequence	Gate Cesium layers with raster latency and signed-manifest tests.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-039 - RuntimeProbeResult and ReleaseRuntimeGate schemas are proposed deliverables

idea_id	ML-058-039
title	RuntimeProbeResult and ReleaseRuntimeGate schemas are proposed deliverables
category	U / M
status	NEW
source evidence	SRC-058 p.37
normalized statement	Recommended deliverables include RuntimeProbeResult.schema.json and ReleaseRuntimeGate.schema.json.
implementation consequence	Add as PROPOSED object families; verify schema home before implementation.
validation consequence	Schema validation with pass/fail fixtures.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-040 - Signed PMTiles publication ADR and raster fallback policy are required

idea_id	ML-058-040
title	Signed PMTiles publication ADR and raster fallback policy are required
category	Y / R
status	NEW
source evidence	SRC-058 p.37
normalized statement	The source calls for a signed PMTiles publication ADR and raster fallback publication policy.
implementation consequence	Decision records are required before public adoption.
validation consequence	ADR acceptance check before release workflow is enabled.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-041 - Focus Mode runtime telemetry drawer contract

idea_id	ML-058-041
title	Focus Mode runtime telemetry drawer contract
category	O / N / T
status	NEW
source evidence	SRC-058 p.37
normalized statement	The next deliverables include a Focus Mode runtime telemetry drawer contract.
implementation consequence	Focus Mode should explain runtime-probe status, not just evidence content.
validation consequence	Drawer fixture shows probe metrics, threshold, outcome and citation.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-042 - MapLibre runtime integration follows verifier, cap gate and release manifest

idea_id	ML-058-042
title	MapLibre runtime integration follows verifier, cap gate and release manifest
category	B / Y
status	EXPANDED
source evidence	SRC-058 pp.240-241
normalized statement	Build order places MapLibre runtime integration after verifier, receipt, cap gate and ReleaseManifest.
implementation consequence	Do not expose UI to unverified map assets.
validation consequence	Release readiness checklist requires prior gates.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-043 - Correction and rollback replay verification follows delta PMTiles publish

idea_id	ML-058-043
title	Correction and rollback replay verification follows delta PMTiles publish
category	U / Y
status	NEW
source evidence	SRC-058 p.241
normalized statement	The build order includes correction/rollback replay verification after delta PMTiles publish.
implementation consequence	Rollback must be tested as part of map release.
validation consequence	Replay verification fixture restores previous root hash and manifest.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-044 - ReleaseManifest artifact kinds include pmtiles, stac, geojson, parquet, model, manifest and receipt

idea_id	ML-058-044
title	ReleaseManifest artifact kinds include pmtiles, stac, geojson, parquet, model, manifest and receipt
category	M
status	NEW
source evidence	SRC-058 pp.242-243
normalized statement	The ReleaseManifest schema artifact kind enum includes pmtiles, stac, geojson, parquet, model, manifest and receipt.
implementation consequence	Map releases should inventory all supporting artifacts, not only tiles.
validation consequence	Schema tests for artifact kind coverage.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

ML-058-045 - ReleaseManifest policy blocks unknown rights and non-public sensitivity

idea_id	ML-058-045
title	ReleaseManifest policy blocks unknown rights and non-public sensitivity
category	Q / U
status	EXPANDED
source evidence	SRC-058 pp.244-246
normalized statement	The Rego policy denies unknown policy_label, unknown rights_status, non-public sensitivity, missing evidence_refs/artifacts or unsupported rollback.
implementation consequence	Public map releases fail closed under uncertainty.
validation consequence	OPA fixtures for unknown rights, restricted sensitivity and missing evidence.
risk / anti-pattern	Treat as downstream carrier; do not promote without proof/release state.

6. Category Atlas

Categories A-Z are retained from the prior master. v1.5 adds runtime verification, signed tile bundles, on-map trust-visible states, incremental tile diffs, 3D conditionality and consent-bound render controls.

category	delta idea count	representative idea IDs
A. Renderer Boundary and KFM Trust Law	4	ML-057-001, ML-058-001, ML-058-011, ML-058-029
B. MapLibre GL JS Web Shell	5	ML-057-002, ML-057-015, ML-058-020, ML-058-037, ML-058-042
C. MapLibre Native / Mobile / Platform Parity	0	Retained from prior master
D. MapLibre RS / Experimental Renderers	0	Retained from prior master
E. Style Specification, Sources, Layers, Expressions	1	ML-057-009
F. Sprites, Glyphs, Fonts, Design Tokens	0	Retained from prior master
G. GeoJSON and Runtime Data	0	Retained from prior master
H. Vector Tiles: MVT and MLT	3	ML-057-026, ML-057-029, ML-058-023
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	20	ML-057-020, ML-057-022, ML-057-023, ML-057-026, ML-057-030, ML-057-033, ML-057-039, ML-057-042 ... +12 more
J. Martin and Server-Mediated Tile Serving	3	ML-057-039, ML-057-040, ML-057-041
K. Raster, COG, DEM, Terrain, Hillshade	8	ML-057-005, ML-057-006, ML-057-008, ML-057-010, ML-057-027, ML-057-032, ML-057-033, ML-058-021
L. WMS / WMTS / External Map Services	1	ML-057-006
M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	25	ML-057-007, ML-057-010, ML-057-014, ML-057-020, ML-057-022, ML-057-024, ML-057-028, ML-057-030 ... +17 more
N. Evidence Drawer Payloads and Click Resolution	2	ML-057-019, ML-058-041
O. Focus Mode and Governed AI Map Context	4	ML-057-019, ML-057-025, ML-057-048, ML-058-041
P. Time-Aware Map Interaction and Timeline	3	ML-057-002, ML-057-003, ML-057-025
Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	16	ML-057-008, ML-057-017, ML-057-034, ML-057-042, ML-057-045, ML-058-005, ML-058-008, ML-058-017 ... +8 more
R. Plugin, Wrapper, and Dependency Governance	5	ML-057-017, ML-058-010, ML-058-012, ML-058-027, ML-058-040
S. Accessibility, UX, and Trust-Visible States	9	ML-057-009, ML-057-011, ML-057-013, ML-057-014, ML-057-015, ML-057-016, ML-057-018, ML-058-034 ... +1 more
T. Performance, Caching, CDN, Range Requests, and Resource Timing	24	ML-057-006, ML-057-012, ML-057-016, ML-057-021, ML-057-023, ML-057-027, ML-057-037, ML-057-043 ... +16 more
U. Testing, CI, Validation, Rollback, and Proof Objects	35	ML-057-011, ML-057-012, ML-057-018, ML-057-021, ML-057-024, ML-057-029, ML-057-031, ML-057-033 ... +27 more
V. Exports, Screenshots, Reports, and Citation Preservation	4	ML-057-007, ML-057-013, ML-057-036, ML-057-050
W. 3D / Cesium / Deck.gl / Overlay Interoperability	9	ML-057-001, ML-057-003, ML-057-004, ML-057-005, ML-057-050, ML-058-001, ML-058-021, ML-058-036 ... +1 more
X. Anti-Patterns and Failure Modes	3	ML-057-045, ML-058-018, ML-058-025
Y. Implementation Backlog and PR Plan	8	ML-057-038, ML-058-009, ML-058-019, ML-058-030, ML-058-031, ML-058-040, ML-058-042, ML-058-043
Z. Open Questions and Verification Backlog	0	Retained from prior master

7. Detailed Category Chapters

Each category distinguishes attached-source-supported ideas from proposed implementation consequences. For categories without v1.5 deltas, the prior v1.4 chapter is retained in Appendix A.

A. Renderer Boundary and KFM Trust Law

Status: EXPANDED by 4 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 reinforces the renderer boundary by routing map runtime choices through verification gates and release manifests. Visual layers remain carriers.

idea	title	source	implementation consequence	validation consequence
ML-057-001	Cesium + MapLibre hybrid Story Node proof artifact	SRC-057 pp.1-4	Implement 3D as a Story Node mode, not a replacement for the governed MapLibre shell.	Validate handoff 2D -> 3D -> 2D, release state, and citation visibility.
ML-058-001	MapLibre vector vs Cesium server-rasterization decision matrix	SRC-058 pp.26-30	Choose runtime mode based on evidence, interaction needs and verified performance, not preference.	RuntimeProbeResult must drive publication decision.
ML-058-011	Probes measure runtime behavior and do not decide truth	SRC-058 p.38	Keep runtime probe results separate from EvidenceBundle content.	Schema separates performance status from evidence truth claims.
ML-058-029	Public client only sees released PMTiles, signed manifests, governed caps, evidence receipts and approved layers	SRC-058 pp.240-241	Enforce trust membrane for map assets.	No-public-raw-path and no-unreleased-tile tests.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

B. MapLibre GL JS Web Shell

Status: EXPANDED by 5 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds verify-before-addSource and performance budgets for MapLibre vector mode.

idea	title	source	implementation consequence	validation consequence
ML-057-002	MapLibre remains 2D vector, label, and time-slider runtime	SRC-057 pp.1-2	Keep MapLibre in renderer/interaction role; time controls select released snapshots.	Test time-slider only loads released layer snapshots and preserves bounds/camera state.
ML-057-015	MapLibre badge adapter binds sourceld and featureldProp	SRC-057 p.98	Keep feature identity deterministic and use promoteld/feature IDs where practical.	Test badge joins cannot drift from layer feature IDs.
ML-058-020	MapLibre app init verifies sidecar before addSource	SRC-058 pp.190-194,208-211	Do not call addSource on unverified PMTiles.	Unit/E2E test blocks source registration when sidecar invalid.
ML-058-037	MapLibre vector path enables styling and interaction but increases client CPU/memory risk	SRC-058 pp.26-30	Use only after device-class probe passes.	Decode/hash/heap stress tests before vector release.
ML-058-042	MapLibre runtime integration follows verifier, cap gate and release manifest	SRC-058 pp.240-241	Do not expose UI to unverified map assets.	Release readiness checklist requires prior gates.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

C. MapLibre Native / Mobile / Platform Parity

Status: RETAINED. No new v1.5 evidence changed this category; see Appendix A for full prior chapter.

D. MapLibre RS / Experimental Renderers

Status: RETAINED. No new v1.5 evidence changed this category; see Appendix A for full prior chapter.

E. Style Specification, Sources, Layers, Expressions

Status: EXPANDED by 1 v1.5 idea records. Evidence sources: SRC-057.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-009	Geology layer metadata needs legends, scale badges, and uncertainty notes	SRC-057 pp.41-43	LayerManifest should include legend, scale, and uncertainty display fields.	Test layer toggle surfaces legend and scale/review caveats.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

F. Sprites, Glyphs, Fonts, Design Tokens

Status: RETAINED. No new v1.5 evidence changed this category; see Appendix A for full prior chapter.

G. GeoJSON and Runtime Data

Status: RETAINED. No new v1.5 evidence changed this category; see Appendix A for full prior chapter.

H. Vector Tiles: MVT and MLT

Status: EXPANDED by 3 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-026	Soil vectors to GeoParquet to tippecanoe to PMTiles	SRC-057 pp.162-171	Use GeoParquet as neutral pre-tile artifact and PMTiles as downstream delivery.	Validate tippecanoe parameters, layer name, zooms and feature attributes.
ML-057-029	Tippecanoe parameter changes require version bump	SRC-057 pp.165,171	Treat tile-generalization parameters as release-significant.	Regression test tile output and manifest diff when parameters change.
ML-058-023	Tippecanoe to PMTiles metadata fields are required	SRC-058 pp.202-205	Add metadata completeness to tile build output.	Validate metadata fields match actual PMTiles inspect output.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

I. PMTiles, MBTiles, Static Tiles, Serverless Delivery

Status: EXPANDED by 20 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds H3 tile diffs, PMTiles + BAO sidecars, WZDx PMTiles and soil PMTiles details.

idea	title	source	implementation consequence	validation consequence
ML-057-020	Watcher-first H3 tile-diff pattern	SRC-057 pp.118-122	Use watcher output as candidate bundles, not automatic publication.	Validate tile checksum reproducibility and catalog/provenance closure.
ML-057-022	Canonical row hash and tile hash decide changed tiles	SRC-057 pp.120-121	Use canonical hashing to avoid silent recomputation drift.	Cross-run deterministic hash tests with reordered inputs.
ML-057-023	Only changed GeoParquet/PMTiles tile artifacts should be emitted	SRC-057 p.121	Implement incremental tile publishing with explicit unchanged no-op receipts.	CI compares changed tile set to expected fixture diffs.
ML-057-026	Soil vectors to GeoParquet to tippecanoe to PMTiles	SRC-057 pp.162-171	Use GeoParquet as neutral pre-tile artifact and PMTiles as downstream delivery.	Validate tippecanoe parameters, layer name, zooms and feature attributes.
ML-057-030	PMTiles STAC asset uses roles tiles	SRC-057 pp.166-167	Standardize tile asset roles and media types.	Schema test for asset role/type and released artifact digest.
ML-057-033	QA with pmtiles inspect and gdalinfo	SRC-057 pp.168-169	Add these checks to release candidates.	Fail release on invalid PMTiles index or non-COG raster.
ML-057-039	Valhalla routing graph tiles are versioned derived artifacts	SRC-057 pp.213-220	Treat routing graph tiles as derived artifacts with source/digest/release state.	Build smoke tests and STAC manifest checks before exposure.
ML-057-042	WZDx v4.x validation to GeoParquet and optional PMTiles	SRC-057 pp.243-250	Transport work-zone layers need schema pinning and deterministic transforms.	Validate schema, CRS EPSG:4326, sorted rows/columns and optional PMTiles output.
ML-057-044	WZDx STAC asset includes PMTiles with roles tiles	SRC-057 pp.248-249	Standardize transport tile layer asset metadata.	Schema validation for source, license, roles and artifact hashes.
ML-057-046	Soil differential tile registry uses WAL and checksum lineage	SRC-057 pp.297-303	Maintain tile lineage registry separate from rendered tiles.	Test restart/reconciliation and OpenLineage correlation.
ML-057-047	Rollback shifts tile lineage pointers instead of deleting artifacts	SRC-057 pp.301-303	MapReleaseManifest rollback target should be pointer-based.	Rollback drill restores previous manifest and verifies tile digest set.
ML-057-048	Focus Mode can explain tile re-ingestion and rollback behavior	SRC-057 p.303	MapContextEnvelope should include tile lineage and release context.	Focus Mode answer fixture cites tile registry and manifest evidence.

Additional idea IDs in this category: ML-058-013, ML-058-014, ML-058-015, ML-058-016, ML-058-019, ML-058-020, ML-058-022, ML-058-035.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

J. Martin and Server-Mediated Tile Serving

Status: EXPANDED by 3 v1.5 idea records. Evidence sources: SRC-057.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-039	Valhalla routing graph tiles are versioned derived artifacts	SRC-057 pp.213-220	Treat routing graph tiles as derived artifacts with source/digest/release state.	Build smoke tests and STAC manifest checks before exposure.
ML-057-040	Valhalla smoke tests precede STAC promotion	SRC-057 pp.218-220	Routing graph publication needs route/isochrone smoke tests and catalog closure.	CI tests for route response, metadata JSON and artifact digest.
ML-057-041	Valhalla tile metadata records OSM sha256	SRC-057 pp.218-220	Source snapshot identity must travel with routing graph artifacts.	Test metadata matches PBF digest and build time format.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

K. Raster, COG, DEM, Terrain, Hillshade

Status: EXPANDED by 8 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 expands COG delivery and server-rasterized fallback implications.

idea	title	source	implementation consequence	validation consequence
ML-057-005	Terrain overlay / extruded GeoJSON is acceptable proof-only 3D input	SRC-057 pp.2-3	Use small synthetic/released fixtures for early 3D proof rather than live sensitive layers.	Visual regression and evidence-link checks on the simple 3D layer.
ML-057-006	PM2.5 fusion outputs COG and tile artifacts	SRC-057 pp.33-39	Treat fused PM2.5 maps as derived surfaces with source-role and uncertainty labels.	Validate COG structure, tile generation, catalog closure and freshness.
ML-057-008	KGS geology layers may publish COGs/tiles only after license review	SRC-057 pp.41-43	Keep geologic map layers under source-rights and attribution gates.	Fail release when license, source series, or attribution is missing.
ML-057-010	SSURGO/gNATSGO conversion to GeoParquet and COG requires CRS/units/provenance	SRC-057 pp.76-77	Use GeoParquet as analysis/pre-tile format and COG for raster delivery.	Validate CRS, units, raster metadata and provenance sidecars.
ML-057-027	Soil rasters to COG for range-enabled delivery	SRC-057 pp.165-166	COGs need TileArtifactManifest or raster artifact manifest parity.	Validate COG profile, range/read capability and raster metadata.
ML-057-032	COG STAC assets need raster/proj/geo fields when known	SRC-057 p.168	Require CRS/band metadata for raster map surfaces.	STAC extension presence and field-type tests for COG assets.
ML-057-033	QA with pmtiles inspect and gdalinfo	SRC-057 pp.168-169	Add these checks to release candidates.	Fail release on invalid PMTiles index or non-COG raster.
ML-058-021	Cesium path prefers server-rasterized PMTiles or COG	SRC-058 pp.193-194	Use Cesium for public mobile/3D when vector verification cost is too high.	Test raster manifest verification and token flow.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

L. WMS / WMTS / External Map Services

Status: EXPANDED by 1 v1.5 idea records. Evidence sources: SRC-057.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-006	PM2.5 fusion outputs COG and tile artifacts	SRC-057 pp.33-39	Treat fused PM2.5 maps as derived surfaces with source-role and uncertainty labels.	Validate COG structure, tile generation, catalog closure and freshness.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest

Status: EXPANDED by 25 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 strengthens manifest object families: sidecar, VerifyReceipt, ReleaseManifest, AutomationBadge and attestation references.

idea	title	source	implementation consequence	validation consequence
ML-057-007	PM2.5 catalog closure uses STAC/DCAT/PROV	SRC-057 pp.33-39	Map layers should load only from cataloged/released PM2.5 artifacts.	Require STAC/DCAT/PROV checks before map exposure.
ML-057-010	SSURGO/gNATSGO conversion to GeoParquet and COG requires CRS/units/provenance	SRC-057 pp.76-77	Use GeoParquet as analysis/pre-tile format and COG for raster delivery.	Validate CRS, units, raster metadata and provenance sidecars.
ML-057-014	AutomationBadge DTO links attestations, SBOMs, manifests and logs	SRC-057 pp.97-98	Define an AutomationBadgePayload projection separate from EvidenceBundle.	Validate links, status enum and run ID; block secret-bearing links.
ML-057-020	Watcher-first H3 tile-diff pattern	SRC-057 pp.118-122	Use watcher output as candidate bundles, not automatic publication.	Validate tile checksum reproducibility and catalog/provenance closure.
ML-057-022	Canonical row hash and tile hash decide changed tiles	SRC-057 pp.120-121	Use canonical hashing to avoid silent recomputation drift.	Cross-run deterministic hash tests with reordered inputs.
ML-057-024	Tile diffs update STAC, DCAT and PROV with content hashes	SRC-057 pp.119-122	Treat tile diffs as cataloged release candidates.	Schema tests ensure each diff has digest, source, lineage and rollback info.
ML-057-028	Deterministic soil run IDs derive from source hashes	SRC-057 pp.163-164	Record run_id/spec_hash in manifests and STAC properties.	Cross-run determinism test for same inputs.
ML-057-030	PMTiles STAC asset uses roles tiles	SRC-057 pp.166-167	Standardize tile asset roles and media types.	Schema test for asset role/type and released artifact digest.
ML-057-031	PROV tile-generation activity should include tippecanoe and pmtiles software	SRC-057 pp.167-168	Source tools become provenance materials, not hidden pipeline details.	Validate PROV entity/activity/agent relation exists for tiles.
ML-057-032	COG STAC assets need raster/proj/eo fields when known	SRC-057 p.168	Require CRS/band metadata for raster map surfaces.	STAC extension presence and field-type tests for COG assets.
ML-057-035	DSSE/LSA attestations bind data artifacts to config and identity	SRC-057 pp.206-213	Publish attestations with tile/raster artifacts and link them from catalogs.	Verify attestation identity, artifact digest and config hash before promotion.
ML-057-036	STAC, DCAT and PROV should discover attestation bundles	SRC-057 pp.211-212	Make attestation refs machine-discoverable.	Schema tests for attestation link presence and media type.

Additional idea IDs in this category: ML-057-041, ML-057-044, ML-057-049, ML-058-013, ML-058-016, ML-058-022, ML-058-023, ML-058-024, ML-058-026, ML-058-031, ML-058-033, ML-058-039, ML-058-044.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

N. Evidence Drawer Payloads and Click Resolution

Status: EXPANDED by 2 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-019	Story Node deep link can open drawer and Focus Mode for a feature	SRC-057 p.101	MapContextEnvelope should carry feature and layer IDs for Focus Mode.	Test deep links resolve released feature evidence and abstain when absent.
ML-058-041	Focus Mode runtime telemetry drawer contract	SRC-058 p.37	Focus Mode should explain runtime-probe status, not just evidence content.	Drawer fixture shows probe metrics, threshold, outcome and citation.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

O. Focus Mode and Governed AI Map Context

Status: EXPANDED by 4 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-019	Story Node deep link can open drawer and Focus Mode for a feature	SRC-057 p.101	MapContextEnvelope should carry feature and layer IDs for Focus Mode.	Test deep links resolve released feature evidence and abstain when absent.
ML-057-025	Air-quality drift monitor may emit pairing map tiles for Story/Focus	SRC-057 pp.161-162	Diagnostics layers require review labels and should not be public facts by default.	Focus Mode test explains diagnostics and source limitations.

idea	title	source	implementation consequence	validation consequence
ML-057-048	Focus Mode can explain tile re-ingestion and rollback behavior	SRC-057 p.303	MapContextEnvelope should include tile lineage and release context.	Focus Mode answer fixture cites tile registry and manifest evidence.
ML-058-041	Focus Mode runtime telemetry drawer contract	SRC-058 p.37	Focus Mode should explain runtime-probe status, not just evidence content.	Drawer fixture shows probe metrics, threshold, outcome and citation.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

P. Time-Aware Map Interaction and Timeline

Status: EXPANDED by 3 v1.5 idea records. Evidence sources: SRC-057.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-002	MapLibre remains 2D vector, label, and time-slider runtime	SRC-057 pp.1-2	Keep MapLibre in renderer/interaction role; time controls select released snapshots.	Test time-slider only loads released layer snapshots and preserves bounds/camera state.
ML-057-003	Cesium mode returns to MapLibre timeline	SRC-057 pp.2-3	Model Story Node state transitions explicitly and keep a rollback/default MapLibre route.	E2E test camera/fade transition and return-to-timeline behavior.
ML-057-025	Air-quality drift monitor may emit pairing map tiles for Story/Focus	SRC-057 pp.161-162	Diagnostics layers require review labels and should not be public facts by default.	Focus Mode test explains diagnostics and source limitations.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

Q. Sensitive Geometry, Geoprivacy, Rights, and Policy

Status: EXPANDED by 16 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds capability issuance, consent-bound rendering and fail-closed denial rules for unknown rights/sensitivity.

idea	title	source	implementation consequence	validation consequence
ML-057-008	KGS geology layers may publish COGs/tiles only after license review	SRC-057 pp.41-43	Keep geologic map layers under source-rights and attribution gates.	Fail release when license, source series, or attribution is missing.
ML-057-017	Browser must not fetch untrusted attestations directly	SRC-057 p.99	Govern attestation fetching behind an API/proxy boundary.	Security tests for origin allowlist, content type and secret stripping.
ML-057-034	Artifact governance checklist spans immutable inputs, CRS, sidecars, logs and licenses	SRC-057 p.171	Map layer release gate should include these checks.	Manifest completeness test for each artifact family.
ML-057-042	WZDx v4.x validation to GeoParquet and optional PMTiles	SRC-057 pp.243-250	Transport work-zone layers need schema pinning and deterministic transforms.	Validate schema, CRS EPSG:4326, sorted rows/columns and optional PMTiles output.
ML-057-045	WZDx governance includes source license, masking/redaction and sovereignty checks	SRC-057 pp.249-250	Treat roadway event map layers as governed, not automatically public.	Policy tests block unknown license or missing redaction decision.
ML-058-005	Allowlist token handshake p50 threshold is less than 100 ms	SRC-058 pp.27,32-33	Token issuance becomes part of runtime gate.	Fail signed-raster path on signer latency or elevated errors.
ML-058-008	Unknown device class defaults to safer path	SRC-058 p.34	Deny or rasterize until device class is approved.	Test unknown profile cannot activate vector mode.
ML-058-017	Browser worker returns DENY on chunk verification failure	SRC-058 pp.47-48	Surface corrupted tile activation as DENY/ERROR, not silent retry.	Negative test asserts failed chunk never reaches MapLibre source.
ML-058-025	Invalid signature or missing sidecar must DENY	SRC-058 p.214	PolicyDecision should deny map render if sidecar proof is absent/invalid.	Negative runtime fixtures for missing sidecar and bad signature.
ML-058-027	/caps/issue grants tile capability only after receipt verification	SRC-058 pp.238-240	Public clients fetch tiles with governed capabilities, not raw open URLs when access is constrained.	API test rejects failed digest receipts and checks TTL/scope.
ML-058-028	OPA capability policy denies missing spec_hash or failed digest/bounds/schema checks	SRC-058 pp.239-240	Policy-as-code mirrors API checks.	Conftest fixtures for each deny reason.
ML-058-029	Public client only sees released PMTiles, signed manifests, governed caps, evidence receipts and approved layers	SRC-058 pp.240-241	Enforce trust membrane for map assets.	No-public-raw-path and no-unreleased-tile tests.

Additional idea IDs in this category: ML-058-032, ML-058-033, ML-058-034, ML-058-045.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

R. Plugin, Wrapper, and Dependency Governance

Status: EXPANDED by 5 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-017	Browser must not fetch untrusted attestations directly	SRC-057 p.99	Govern attestation fetching behind an API/proxy boundary.	Security tests for origin allowlist, content type and secret stripping.
ML-058-010	tools/probes is proposed home for runtime probe harnesses	SRC-058 pp.37-39	Treat path as PROPOSED; Directory Rules basis is tools as shared implementation/validation root.	Validate placement against Directory Rules and actual repo before implementation.
ML-058-012	Threshold changes need steward/security/release review	SRC-058 p.40	Add review requirements to policy/manifest metadata.	CI fails threshold changes without review record.
ML-058-027	/caps/issue grants tile capability only after receipt verification	SRC-058 pp.238-240	Public clients fetch tiles with governed capabilities, not raw open URLs when access is constrained.	API test rejects failed digest receipts and checks TTL/scope.
ML-058-040	Signed PMTiles publication ADR and raster fallback policy are required	SRC-058 p.37	Decision records are required before public adoption.	ADR acceptance check before release workflow is enabled.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

S. Accessibility, UX, and Trust-Visible States

Status: EXPANDED by 9 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-009	Geology layer metadata needs legends, scale badges, and uncertainty notes	SRC-057 pp.41-43	LayerManifest should include legend, scale, and uncertainty display fields.	Test layer toggle surfaces legend and scale/review caveats.
ML-057-011	MapLibre QA preview for H3 gaps and overlaps	SRC-057 pp.81-82	Add QA-only map layers outside public publication path.	CI fails on gaps/overlaps unless allowlisted; MapLibre preview must not imply release.
ML-057-013	Automation/provenance badges can be pinned on MapLibre and Cesium features	SRC-057 pp.96-101	Use badges as trust-visible UI annotations, not proof substitutes.	Bridge tests bind feature IDs to badge DTOs and drawer links.
ML-057-014	AutomationBadge DTO links attestations, SBOMs, manifests and logs	SRC-057 pp.97-98	Define an AutomationBadgePayload projection separate from EvidenceBundle.	Validate links, status enum and run ID; block secret-bearing links.
ML-057-015	MapLibre badge adapter binds sourced and featureIdProp	SRC-057 p.98	Keep feature identity deterministic and use promoteId/feature IDs where practical.	Test badge joins cannot drift from layer feature IDs.
ML-057-016	Badge streams degrade from SSE/WebSocket to polling	SRC-057 pp.99-100	UI should show stale/degraded stream status visibly.	Latency and stale-state tests for stream outage and polling fallback.
ML-057-018	On-map badges require accessibility and snapshot tests	SRC-057 pp.100-101	Trust badges must be accessible and testable.	A11y, keyboard and visual regression tests for badge overlays.
ML-058-034	Render-time consent verification blocks invalid/revoked/expired displays	SRC-058 pp.247-248	Treat consent failure as DENY before map draw.	Test revoked consent prevents layer/drawer rendering.
ML-058-038	Cesium raster path is safer for public mobile but less flexible	SRC-058 pp.29-36	3D/raster should be conditional and public-safe, not default for all layers.	Gate Cesium layers with raster latency and signed-manifest tests.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

T. Performance, Caching, CDN, Range Requests, and Resource Timing

Status: EXPANDED by 24 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds concrete decode/hash/heap/token thresholds and telemetry envelopes.

idea	title	source	implementation consequence	validation consequence
ML-057-006	PM2.5 fusion outputs COG and tile artifacts	SRC-057 pp.33-39	Treat fused PM2.5 maps as derived surfaces with source-role and uncertainty labels.	Validate COG structure, tile generation, catalog closure and freshness.
ML-057-012	H3 density and overlap checks become CI gates for tilings	SRC-057 pp.81-82	Use H3 QA before publishing tiled inventories.	Fixture tests for gap, overlap and density edge cases.
ML-057-016	Badge streams degrade from SSE/WebSocket to polling	SRC-057 pp.99-100	UI should show stale/degraded stream status visibly.	Latency and stale-state tests for stream outage and polling fallback.
ML-057-021	ArcGIS editingInfo, ETag and Last-Modified can trigger tile rebuilds	SRC-057 pp.119-120	Record source_head in RunReceipt and reject stale/unstable source state.	Fixtures for 304/no-change, changed ETag and missing header behavior.
ML-057-023	Only changed GeoParquet/PMTiles tile artifacts should be emitted	SRC-057 p.121	Implement incremental tile publishing with explicit unchanged no-op receipts.	CI compares changed tile set to expected fixture diffs.
ML-057-027	Soil rasters to COG for range-enabled delivery	SRC-057 pp.165-166	COGs need TileArtifactManifest or raster artifact manifest parity.	Validate COG profile, range/read capability and raster metadata.
ML-057-037	Consumers verify artifact digest and attestation before loading	SRC-057 pp.212-213	Public clients or gateway should verify artifacts before activation.	Negative tests for digest mismatch and failed identity constraints.
ML-057-043	WZDx ETag/Last-Modified freshness watcher	SRC-057 pp.247-248	Record freshness headers in PROV and RunReceipt.	Tests for 304 response, changed headers and missing header fallback.
ML-057-046	Soil differential tile registry uses WAL and checksum lineage	SRC-057 pp.297-303	Maintain tile lineage registry separate from rendered tiles.	Test restart/reconciliation and OpenLineage correlation.
ML-058-001	MapLibre vector vs Cesium server-rasterization decision matrix	SRC-058 pp.26-30	Choose runtime mode based on evidence, interaction needs and verified performance, not preference.	RuntimeProbeResult must drive publication decision.
ML-058-002	MapLibre tile decode median threshold is less than 50 ms	SRC-058 pp.27,31	Set performance budgets before public vector publication.	Fail vector release when decode budget is exceeded.
ML-058-003	Client WASM hash throughput threshold is greater than 100 MB/s	SRC-058 pp.27,31-32	Do not force integrity verification onto devices that cannot sustain it.	Benchmark hash worker on target device classes.

Additional idea IDs in this category: ML-058-004, ML-058-005, ML-058-006, ML-058-007, ML-058-008, ML-058-014, ML-058-017, ML-058-024, ML-058-035, ML-058-036, ML-058-037, ML-058-041.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

U. Testing, CI, Validation, Rollback, and Proof Objects

Status: EXPANDED by 35 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds runtime probes, BAO verification, rollback replay, OPA policy fixtures and attestation gates.

idea	title	source	implementation consequence	validation consequence
ML-057-011	MapLibre QA preview for H3 gaps and overlaps	SRC-057 pp.81-82	Add QA-only map layers outside public publication path.	CI fails on gaps/overlaps unless allowlisted; MapLibre preview must not imply release.
ML-057-012	H3 density and overlap checks become CI gates for tilings	SRC-057 pp.81-82	Use H3 QA before publishing tiled inventories.	Fixture tests for gap, overlap and density edge cases.
ML-057-018	On-map badges require accessibility and snapshot tests	SRC-057 pp.100-101	Trust badges must be accessible and testable.	A11y, keyboard and visual regression tests for badge overlays.
ML-057-021	ArcGIS editingInfo, ETag and Last-Modified can trigger tile rebuilds	SRC-057 pp.119-120	Record source_head in RunReceipt and reject stale/unstable source state.	Fixtures for 304/no-change, changed ETag and missing header behavior.
ML-057-024	Tile diffs update STAC, DCAT and PROV with content hashes	SRC-057 pp.119-122	Treat tile diffs as cataloged release candidates.	Schema tests ensure each diff has digest, source, lineage and rollback info.
ML-057-029	Tippecanoe parameter changes require version bump	SRC-057 pp.165,171	Treat tile-generalization parameters as release-significant.	Regression test tile output and manifest diff when parameters change.
ML-057-031	PROV tile-generation activity should include tippecanoe and pmtiles software	SRC-057 pp.167-168	Source tools become provenance materials, not hidden pipeline details.	Validate PROV entity/activity/agent relation exists for tiles.
ML-057-033	QA with pmtiles inspect and gdalinfo	SRC-057 pp.168-169	Add these checks to release candidates.	Fail release on invalid PMTiles index or non-COG raster.
ML-057-034	Artifact governance checklist spans immutable inputs, CRS, sidecars, logs and licenses	SRC-057 p.171	Map layer release gate should include these checks.	Manifest completeness test for each artifact family.

idea	title	source	implementation consequence	validation consequence
ML-057-035	DSSE/SLSA attestations bind data artifacts to config and identity	SRC-057 pp.206-213	Publish attestations with tile/raster artifacts and link them from catalogs.	Verify attestation identity, artifact digest and config hash before promotion.
ML-057-037	Consumers verify artifact digest and attestation before loading	SRC-057 pp.212-213	Public clients or gateway should verify artifacts before activation.	Negative tests for digest mismatch and failed identity constraints.
ML-057-038	Publish gate requires verify-attestation pass	SRC-057 p.213	PromotionDecision should record verification result and failure reason.	Policy fixture blocks unsigned or unverifiable tile/raster artifacts.

Additional idea IDs in this category: ML-057-040, ML-057-043, ML-057-047, ML-057-049, ML-058-002, ML-058-003, ML-058-004, ML-058-006, ML-058-007, ML-058-009, ML-058-010, ML-058-011, ML-058-012, ML-058-015, ML-058-018, ML-058-020, ML-058-026, ML-058-028, ML-058-030, ML-058-032, ML-058-039, ML-058-043, ML-058-045.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

V. Exports, Screenshots, Reports, and Citation Preservation

Status: EXPANDED by 4 v1.5 idea records. Evidence sources: SRC-057.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-007	PM2.5 catalog closure uses STAC/DCAT/PROV	SRC-057 pp.33-39	Map layers should load only from cataloged/released PM2.5 artifacts.	Require STAC/DCAT/PROV checks before map exposure.
ML-057-013	Automation/provenance badges can be pinned on MapLibre and Cesium features	SRC-057 pp.96-101	Use badges as trust-visible UI annotations, not proof substitutes.	Bridge tests bind feature IDs to badge DTOs and drawer links.
ML-057-036	STAC, DCAT and PROV should discover attestation bundles	SRC-057 pp.211-212	Make attestation refs machine-discoverable.	Schema tests for attestation link presence and media type.
ML-057-050	Story Nodes must reference cataloged checksum-inventoried Items	SRC-057 pp.481-482	No Story Node should cite a tile/scene as evidence without catalog/proof closure.	Story export test verifies citations and artifact checksums.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

W. 3D / Cesium / Deck.gl / Overlay Interoperability

Status: EXPANDED by 9 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 clarifies conditional 3D/Cesium use and server-rasterization fallback.

idea	title	source	implementation consequence	validation consequence
ML-057-001	Cesium + MapLibre hybrid Story Node proof artifact	SRC-057 pp.1-4	Implement 3D as a Story Node mode, not a replacement for the governed MapLibre shell.	Validate handoff 2D -> 3D -> 2D, release state, and citation visibility.
ML-057-003	Cesium mode returns to MapLibre timeline	SRC-057 pp.2-3	Model Story Node state transitions explicitly and keep a rollback/default MapLibre route.	E2E test camera/fade transition and return-to-timeline behavior.
ML-057-004	3D Tiles streaming is future path, not first rearchitecture	SRC-057 pp.1-3	Admit 3D Tiles only after release manifests and evidence context survive in 3D mode.	Validate 3D artifact manifests, review state, and public-safe status before loading.
ML-057-005	Terrain overlay / extruded GeoJSON is acceptable proof-only 3D input	SRC-057 pp.2-3	Use small synthetic/released fixtures for early 3D proof rather than live sensitive layers.	Visual regression and evidence-link checks on the simple 3D layer.
ML-057-050	Story Nodes must reference cataloged checksum-inventoried Items	SRC-057 pp.481-482	No Story Node should cite a tile/scene as evidence without catalog/proof closure.	Story export test verifies citations and artifact checksums.
ML-058-001	MapLibre vector vs Cesium server-rasterization decision matrix	SRC-058 pp.26-30	Choose runtime mode based on evidence, interaction needs and verified performance, not preference.	RuntimeProbeResult must drive publication decision.
ML-058-021	Cesium path prefers server-rasterized PMTiles or COG	SRC-058 pp.193-194	Use Cesium for public mobile/3D when vector verification cost is too high.	Test raster manifest verification and token flow.
ML-058-036	Server-side rasterization shifts verification burden to signer/hosting duties	SRC-058 pp.26-30	Document operational burden before choosing raster fallback.	Signer SLA and token path tests replace client decode/hash tests.
ML-058-038	Cesium raster path is safer for public mobile but less flexible	SRC-058 pp.29-36	3D/raster should be conditional and public-safe, not default for all layers.	Gate Cesium layers with raster latency and signed-manifest tests.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

X. Anti-Patterns and Failure Modes

Status: EXPANDED by 3 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-045	WZDx governance includes source license, masking/redaction and sovereignty checks	SRC-057 pp.249-250	Treat roadway event map layers as governed, not automatically public.	Policy tests block unknown license or missing redaction decision.
ML-058-018	PMTiles publication gates include invalid spec, unsigned manifest, unverified chunk and rollback root mismatch	SRC-058 pp.48-49	Add these to anti-pattern/gate register.	OPA/validator fixtures for each failure condition.
ML-058-025	Invalid signature or missing sidecar must DENY	SRC-058 p.214	PolicyDecision should deny map render if sidecar proof is absent/invalid.	Negative runtime fixtures for missing sidecar and bad signature.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

Y. Implementation Backlog and PR Plan

Status: EXPANDED by 8 v1.5 idea records. Evidence sources: SRC-057, SRC-058.

v1.5 adds one or more source-backed implementation constraints for this category while preserving prior KFM trust law.

idea	title	source	implementation consequence	validation consequence
ML-057-038	Publish gate requires verify-attestation pass	SRC-057 p.213	PromotionDecision should record verification result and failure reason.	Policy fixture blocks unsigned or unverifiable tile/raster artifacts.
ML-058-009	Runtime verification stability is publication prerequisite	SRC-058 pp.37-40	ReleaseManifest should require runtime gate status for public/mobile layers.	PromotionDecision blocks release if probe report fails.
ML-058-019	PMTiles promotion flow moves from delta build to release only after range verification and receipt	SRC-058 p.49	Model promotion as a state transition with explicit proof artifacts.	End-to-end fixture verifies transition order and rollback target.
ML-058-030	Recommended build order puts verifier/cap gates before MapLibre runtime integration	SRC-058 pp.240-241	Do not start with UI polish or MapLibre addSource code.	Acceptance gate requires verifier and release manifest before UI binding.
ML-058-031	ReleaseManifest binds PMTiles to EvidenceBundles, policy, sensitivity, receipts, rollback, correction and DSSE	SRC-058 pp.241-244	Use MapReleaseManifest as release authority for map assets.	Schema and policy tests for all binding fields.
ML-058-040	Signed PMTiles publication ADR and raster fallback policy are required	SRC-058 p.37	Decision records are required before public adoption.	ADR acceptance check before release workflow is enabled.
ML-058-042	MapLibre runtime integration follows verifier, cap gate and release manifest	SRC-058 pp.240-241	Do not expose UI to unverified map assets.	Release readiness checklist requires prior gates.
ML-058-043	Correction and rollback replay verification follows delta PMTiles publish	SRC-058 p.241	Rollback must be tested as part of map release.	Replay verification fixture restores previous root hash and manifest.

Policy implications: public rendering requires released artifacts, evidence refs, policy labels, sensitivity posture, rights status and rollback target. **Anti-patterns:** direct addSource on unverified PMTiles, hidden sensitive geometry, client hashing without budgets, or using badges as proof.

Z. Open Questions and Verification Backlog

Status: RETAINED. No new v1.5 evidence changed this category; see Appendix A for full prior chapter.

8. Components-Functions-Features Matrix

The matrix is rendered as cards to avoid wide-table cutoffs. Each card preserves the requested columns: component, function, feature, input artifacts, output artifacts, governed API dependency, evidence dependency, policy dependency, tile/style dependency, runtime/UI behavior, tests required, and status.

MapLibre GL JS

component	MapLibre GL JS
function	2D web renderer and interaction runtime
feature	Released style/layers; verify-before-addSource for PMTiles
input artifacts	StyleManifest, LayerManifest, ReleaseManifest, TileArtifactManifest
output artifacts	Rendered map state, click candidates, MapContextEnvelope
governed API dependency	Governed API for evidence/click resolution
evidence dependency	EvidenceBundle refs on features/layers
policy dependency	PolicyDecision before public load
tile/style dependency	Style JSON, source/layer definitions, PMTiles/MVT/GeoJSON/COG overlays
runtime/UI behavior	Client verifies sidecar, loads approved sources, sends clicks to governed API
tests required	Decode/hash/heap probes; no-unreleased-tile; visual regression
status	CONFIRMED doctrine / PROPOSED implementation

MapLibre Native

component	MapLibre Native
function	Mobile/native map runtime
feature	Only after parity and runtime probes
input artifacts	Same as GL JS plus device profile
output artifacts	Native rendered map state
governed API dependency	Governed API/cap issuance
evidence dependency	EvidenceBundle refs
policy dependency	Device class and sensitivity policy
tile/style dependency	PMTiles/vector/raster per approved mode
runtime/UI behavior	Default deny until device probes pass
tests required	Mobile probe and parity tests
status	NEEDS VERIFICATION

MapLibre RS

component	MapLibre RS
function	Experimental renderer
feature	Research/possible future renderer
input artifacts	Test fixtures only
output artifacts	Prototype outputs
governed API dependency	None for public path until approved
evidence dependency	Evidence refs only in tests
policy dependency	Plugin/dependency policy
tile/style dependency	TBD
runtime/UI behavior	Not public default
tests required	Sandbox tests
status	PROPOSED / DEFERRED

Style JSON

component	Style JSON
function	Style definition
feature	Released style manifest only
input artifacts	StyleManifest, sprites, glyphs
output artifacts	Map style
governed API dependency	Style registry endpoint
evidence dependency	Layer evidence refs
policy dependency	Style policy/rights
tile/style dependency	Sources/layers/expressions
runtime/UI behavior	No style-hidden sensitive geometry
tests required	Style validation, visual diff
status	RETAINED

Source definition

component	Source definition
function	MapLibre source wiring
feature	Source points to governed/released artifact
input artifacts	LayerManifest, TileArtifactManifest
output artifacts	Map source state
governed API dependency	Artifact manifest fetch
evidence dependency	SourceDescriptor/EvidenceBundle
policy dependency	Release and access policy
tile/style dependency	PMTiles/MVT/GeoJSON/COG/WMS
runtime/UI behavior	No canonical/raw client fetch
tests required	No raw/work path test
status	EXPANDED

Layer definition

component	Layer definition
function	Layer styling/filtering
feature	Public-safe, evidence-aware layer
input artifacts	LayerManifest
output artifacts	Rendered layer
governed API dependency	Layer catalog API
evidence dependency	Evidence refs, review state
policy dependency	Sensitivity and rights policy
tile/style dependency	Style JSON layers and source-layer
runtime/UI behavior	Click candidate only, not proof
tests required	Layer/source validation
status	EXPANDED

Expressions and filters

component	Expressions and filters
function	Visual classification
feature	Display only; not publication control
input artifacts	StyleManifest
output artifacts	Rendered symbology
governed API dependency	None as authority
evidence dependency	Evidence in drawer only
policy dependency	Policy cannot rely only on filter
tile/style dependency	Style spec expressions
runtime/UI behavior	Do not hide sensitive geometry only by style
tests required	Style tests and deny fixtures
status	RETAINED

Sprites

component	Sprites
function	Icon atlas
feature	Themed trust/status icons
input artifacts	Sprite manifest
output artifacts	Rendered icons
governed API dependency	Static released asset endpoint
evidence dependency	None unless icon conveys trust state
policy dependency	Design-token review
tile/style dependency	Style JSON sprite ref
runtime/UI behavior	Accessible labels required
tests required	Asset hash tests
status	RETAINED

Glyphs / fonts

component	Glyphs / fonts
function	Text rendering
feature	Label glyph delivery
input artifacts	Glyph manifest
output artifacts	Rendered labels
governed API dependency	Static released asset endpoint
evidence dependency	None unless label claim
policy dependency	Attribution/licensing
tile/style dependency	Style JSON glyph URL
runtime/UI behavior	Readable, accessible labels
tests required	Glyph URL/rights tests
status	RETAINED

MVT

component	MVT
function	Vector-tile payload
feature	Efficient vector delivery
input artifacts	TileArtifactManifest
output artifacts	MVT tiles
governed API dependency	Tile service/cap issuance if gated
evidence dependency	Feature evidence refs promoted IDs
policy dependency	Sensitive geometry policy
tile/style dependency	Vector source/layers
runtime/UI behavior	Interactive features through governed click API
tests required	MVT schema and tile budget tests
status	RETAINED

MLT

component	MLT
function	Emerging vector-tile format
feature	Pilot only after KFM validation
input artifacts	Experimental manifest
output artifacts	MLT tiles
governed API dependency	None public until approved
evidence dependency	TBD
policy dependency	Toolchain policy
tile/style dependency	Tile decoder
runtime/UI behavior	Not default
tests required	Toolchain validation
status	NEEDS VERIFICATION

PMTiles

component	PMTiles
function	Single-file tile archive
feature	Static/serverless/offline delivery
input artifacts	PMTiles, sidecar, BAO, ReleaseManifest
output artifacts	Released PMTiles URL
governed API dependency	Capability issuance for gated access
evidence dependency	Evidence refs via ReleaseManifest
policy dependency	Rights/sensitivity/release policy
tile/style dependency	pmtiles:// source
runtime/UI behavior	Verify sidecar before addSource
tests required	Range/CORS, signature, root hash, decode tests
status	EXPANDED

MBTiles

component	MBTiles
function	Intermediate/legacy archive
feature	Tippecanoe output/pre-PMTiles conversion
input artifacts	MBTiles file
output artifacts	PMTiles or server tiles
governed API dependency	No public direct path unless approved
evidence dependency	Source provenance
policy dependency	License/release policy
tile/style dependency	Vector tiles
runtime/UI behavior	Build artifact not truth
tests required	Conversion and inspect tests
status	RETAINED

COG

component	COG
function	Cloud Optimized GeoTIFF
feature	Raster/continuous surface delivery
input artifacts	COG, STAC asset, raster metadata
output artifacts	Range-read raster artifact
governed API dependency	Raster API/tiles if served
evidence dependency	Evidence refs in catalog
policy dependency	Source rights/sensitivity
tile/style dependency	raster source or Cesium drape
runtime/UI behavior	Used as derived surface
tests required	COG profile/range tests
status	EXPANDED

Martin

component	Martin
function	Server-mediated PostGIS/MVT tiles
feature	Dynamic server tile serving
input artifacts	PostGIS, Martin config, TileJSON
output artifacts	MVT/TileJSON
governed API dependency	Governed tile API
evidence dependency	Canonical stores not public
policy dependency	Access policy
tile/style dependency	Vector tile endpoints
runtime/UI behavior	Use when stewarded/dynamic tiles needed
tests required	Access/control tests
status	RETAINED

Maputnik

component	Maputnik
function	Style editing
feature	Style authoring/review tool
input artifacts	Style JSON draft
output artifacts	Candidate style
governed API dependency	No public authority
evidence dependency	None directly
policy dependency	Style review policy
tile/style dependency	Style JSON
runtime/UI behavior	Review only
tests required	Style diff/visual tests
status	RETAINED

Layer catalog

component	Layer catalog
function	Layer discovery
feature	Released layer index
input artifacts	LayerManifest, ReleaseManifest
output artifacts	Layer listing
governed API dependency	Catalog API
evidence dependency	Evidence status and citations
policy dependency	Policy labels
tile/style dependency	Tile/style refs
runtime/UI behavior	Trust-visible layer selection
tests required	Catalog closure tests
status	EXPANDED

Evidence Drawer

component	Evidence Drawer
function	Trust surface for clicked feature
feature	Displays evidence, citation, policy, review and release state
input artifacts	EvidenceDrawerPayload
output artifacts	Drawer UI state
governed API dependency	EvidenceBundle resolution API
evidence dependency	EvidenceBundle
policy dependency	PolicyDecision
tile/style dependency	Layer/feature IDs
runtime/UI behavior	Popup is not substitute
tests required	Projection and citation tests
status	EXPANDED

Focus Mode

component	Focus Mode
function	Bounded AI map-context answer
feature	Answer/abstain/deny/error over evidence only
input artifacts	MapContextEnvelope, FocusModeRequest
output artifacts	FocusModeResponse, AIReceipt
governed API dependency	Governed AI API
evidence dependency	EvidenceBundle/citation validation
policy dependency	Policy pre/postcheck
tile/style dependency	Map context only as scope
runtime/UI behavior	No direct model client
tests required	Answer/abstain fixtures
status	EXPANDED

Timeline / time slider

component	Timeline / time slider
function	Temporal navigation
feature	Selects released snapshot
input artifacts	LayerManifest temporal metadata
output artifacts	Snapshot/layer state
governed API dependency	Layer catalog API
evidence dependency	Temporal evidence refs
policy dependency	Stale/release policy
tile/style dependency	Style/layer filters
runtime/UI behavior	Reject unreleased snapshots
tests required	Temporal fixture tests
status	EXPANDED

MapContextEnvelope

component	MapContextEnvelope
function	Map interaction context
feature	Bounded context for Focus Mode/click
input artifacts	Camera, layer, feature, time, release refs
output artifacts	Context payload
governed API dependency	Governed API
evidence dependency	Evidence refs only
policy dependency	Policy label
tile/style dependency	Map state
runtime/UI behavior	Input to Focus Mode, not proof
tests required	Schema tests
status	EXPANDED

LayerManifest

component	LayerManifest
function	Layer contract
feature	Layer source, style, evidence, policy, release metadata
input artifacts	SourceDescriptor, TileArtifactManifest
output artifacts	Layer catalog record
governed API dependency	Catalog API
evidence dependency	Evidence refs
policy dependency	Policy labels/sensitivity
tile/style dependency	Source/layer/style refs
runtime/UI behavior	Trust badges and states
tests required	Schema and release tests
status	EXPANDED

StyleManifest

component	StyleManifest
function	Style contract
feature	Style version, sources, sprites, glyphs
input artifacts	Style JSON, assets
output artifacts	Released style
governed API dependency	Style asset endpoint
evidence dependency	None unless claims
policy dependency	Rights/plugin policy
tile/style dependency	Style spec
runtime/UI behavior	Client style load
tests required	Style spec validation
status	RETAINED

TileArtifactManifest

component	TileArtifactManifest
function	Tile artifact contract
feature	PMTiles/MVT/COG metadata, hashes, root_hash, range/CORS
input artifacts	PMTiles/COG/MVT, sidecars
output artifacts	Verified artifact record
governed API dependency	Artifact API
evidence dependency	Evidence refs via release
policy dependency	Rights/sensitivity
tile/style dependency	Tile URL, media type
runtime/UI behavior	Blocks unverified addSource
tests required	Hash/range/signature tests
status	EXPANDED

MapReleaseManifest

component	MapReleaseManifest
function	Publication envelope
feature	Binds artifacts, evidence, policy, sensitivity, rollback, corrections
input artifacts	ReleaseManifest
output artifacts	Public release record
governed API dependency	Release API
evidence dependency	Evidence refs
policy dependency	PolicyDecision/PromotionDecision
tile/style dependency	Tile/style/layer refs
runtime/UI behavior	Only PUBLISHED public
tests required	OPA release tests
status	EXPANDED

PolicyDecision

component	PolicyDecision
function	Access/release decision
feature	ANSWER/ABSTAIN/DENY/ERROR or allow/deny reasons
input artifacts	Policy input
output artifacts	Decision envelope
governed API dependency	Policy API/PDP
evidence dependency	Evidence refs
policy dependency	Policy bundle
tile/style dependency	Layer/tile sensitivity
runtime/UI behavior	Render/load allowed or blocked
tests required	Rego fixtures
status	EXPANDED

CitationValidationReport

component	CitationValidationReport
function	Citation check
feature	Confirms answer/export citations resolve
input artifacts	FocusModeResponse/export
output artifacts	Pass/fail report
governed API dependency	Citation API
evidence dependency	EvidenceBundle
policy dependency	Cite-or-abstain
tile/style dependency	N/A
runtime/UI behavior	Blocks uncited output
tests required	Citation fixtures
status	RETAINED

AIReceipt

component	AIReceipt
function	AI response receipt
feature	Records model/provider/context/citation validation
input artifacts	FocusModeRequest/Response
output artifacts	Receipt
governed API dependency	Governed AI API
evidence dependency	Evidence refs
policy dependency	Policy pre/postcheck
tile/style dependency	MapContextEnvelope
runtime/UI behavior	Trace without chain-of-thought
tests required	Receipt schema tests
status	RETAINED

Rollback target

component	Rollback target
function	Reversible release pointer
feature	Prior manifest/root hash/tile checksum set
input artifacts	ReleaseManifest, WAL
output artifacts	Rollback record
governed API dependency	Release API
evidence dependency	Previous evidence refs
policy dependency	Policy/correction state
tile/style dependency	Tile/style manifest refs
runtime/UI behavior	Restores prior release
tests required	Rollback drill
status	EXPANDED

Cache invalidation record

component	Cache invalidation record
function	Cache coherency
feature	Invalidate CDN/client caches after release/correction
input artifacts	ReleaseManifest, cache keys
output artifacts	Invalidation receipt
governed API dependency	Release API/CDN
evidence dependency	None
policy dependency	Policy labels
tile/style dependency	Tile URL/version
runtime/UI behavior	Prevents stale source loads
tests required	Cache purge tests
status	RETAINED

VerifyReceipt

component	VerifyReceipt
function	Runtime verification proof
feature	Records digest/bounds/schema checks before capabilities
input artifacts	PMTiles sidecar, probe result
output artifacts	VerificationReceipt
governed API dependency	/caps/issue
evidence dependency	Release/evidence refs
policy dependency	Capability policy
tile/style dependency	PMTiles root hash
runtime/UI behavior	Enables governed tile access
tests required	Receipt and OPA tests
status	NEW

RuntimeProbeResult

component	RuntimeProbeResult
function	Performance proof
feature	Decode/hash/heap/token metrics
input artifacts	Probe fixtures, device profile
output artifacts	Probe report
governed API dependency	Promotion gate
evidence dependency	None as truth
policy dependency	Release gate policy
tile/style dependency	Renderer/tile artifact
runtime/UI behavior	Blocks release on failed budget
tests required	Threshold tests
status	NEW

9. Tile Strategy Chapter

comparison	when it wins	when it fails	KFM trust consequence	release / rollback consequence	validation needed
GeoJSON vs vector tiles	GeoJSON wins for small, review, QA and low-volume runtime data. Vector tiles win for large, repeated, zoom-dependent layers.	GeoJSON fails at scale; vector tiles fail if feature evidence IDs and release manifests are missing.	GeoJSON can be closer to evidence but still downstream; vector tiles need manifests.	Release vector tiles only via TileArtifactManifest and rollback root.	Size/performance, evidence-link and no-unreleased-tile tests.
MVT vs MLT	MVT wins now through mature tooling. MLT remains promising but pilot-only.	MLT fails as default until KFM support/toolchain is verified.	Do not equate new tile format with better truth.	Separate manifest/version and rollback for each format.	Toolchain, decoder and interop tests.
PMTiles vs MBTiles	PMTiles wins for static/range/CDN delivery; MBTiles useful as build intermediate.	PMTiles fails without range/CORS/manifest/signature; MBTiles fails for browser/serverless direct use.	Both are downstream carriers.	PMTiles needs immutable release root and rollback.	pmtiles inspect, range, root_hash and signature tests.
Static tiles vs dynamic tile server	Static wins for stable public-safe derivatives; dynamic server wins for stewarded, access-controlled or frequently changing PostGIS-backed layers.	Static fails when rights/access are dynamic; dynamic fails if server becomes canonical truth.	Server mediation can enforce policy; static requires strong release manifests.	Static rollback shifts manifest; dynamic rollback uses versioned query/source state.	Access control, cache invalidation and rollback tests.
PMTiles serverless vs Martin server-mediated	PMTiles wins low-ops public bundles; Martin wins dynamic MVT/PostGIS and steward mediation.	PMTiles fails for per-user policy; Martin fails if public clients reach internal canonical stores.	Both must sit below governed API/catalog.	Different release manifests and rollback paths.	TileJSON/access/range tests.
Vector vs raster	Vector wins rich interaction and styling; raster wins bounded client cost and sensitive generalization.	Vector fails with mobile heap/CPU risk; raster fails if it hides uncertainty or becomes uncited picture.	Vector claims need feature evidence; raster claims need source/raster metadata.	Vector rollback root vs raster COG/tile manifest.	Decode/hash/heap or raster/COG tests.
COG vs raster tiles	COG wins analysis-friendly range reads and raster cataloging; raster tiles win simple slippy-map delivery.	COG fails without range metadata; raster tiles fail if generated outside release state.	COG is artifact, not proof.	COG manifest and rollback to prior asset.	COG profile/range/gdalinfo tests.
Terrain-enabled vs non-terrain	Terrain wins when vertical context carries evidence burden; non-terrain wins for clarity and low risk.	Terrain fails when it distracts from evidence or leaks sensitive context.	3D is conditional carrier.	3D assets need release manifest.	3D visual regression and evidence drawer parity.
2D default vs 3D conditional	2D MapLibre default wins for inspection and accessibility; 3D wins for Story Node context.	3D default fails when trust controls are weaker.	Keep 2D trust path canonical.	3D rollback returns to 2D shell.	Handoff, a11y and screenshot tests.
Public-safe vs restricted/internal tiles	Public-safe wins for released generalized products; restricted tiles require governed caps and access records.	Public-safe fails if sensitive geometry hidden only by style; restricted fails if static URLs leak.	Policy decides render/load before client.	ReleaseManifest sensitivity and cap issuance.	DENY fixtures and token tests.
MapLibre vector vs Cesium server-rasterized	MapLibre wins for interaction/offline styling when probes pass; Cesium/raster wins for public mobile stability and bounded verification.	MapLibre fails with decode/hash/heap budgets; Cesium fails if signer/token SLA fails.	Runtime stability is part of publication safety.	ReleaseRuntimeGate controls selected path.	Decode/hash/heap/token probes and fallback tests.

10. Governance and Trust-Membrane Chapter

MapLibre fits KFM as a released-artifact viewer and interaction runtime inside the trust membrane. The flow is: released layer -> user click -> governed API -> EvidenceBundle -> Evidence Drawer -> Focus Mode answer / abstain / deny / error. v1.5 adds a parallel runtime-safety gate: released tile artifact -> sidecar/manifest verification -> runtime probes or capability issuance -> governed load -> click/evidence resolution.

rule	v1.5 interpretation
No public RAW path	Public clients must not load RAW, WORK, QUARANTINE, unpublished candidate data or canonical/internal stores.
No direct model client	Focus Mode and AI run behind governed APIs after EvidenceBundle resolution and policy checks.
No unreleased tile load	PMTiles/COG/MVT load only when release_state, policy, rights, sensitivity, evidence_refs, hashes and rollback are valid.
No sensitive geometry hidden only by style	Generalization/redaction happens before public tile release and must record transform reasons.
No popup as Evidence Drawer substitute	Popups can summarize, but consequential claims resolve through EvidenceDrawerPayload and EvidenceBundle.
No uncited export	Screenshots, Story Nodes and reports must preserve citation/proof context or abstain.
Verify before addSource	MapLibre source registration for PMTiles must wait for sidecar/signature/spec_hash/tile_format checks where the layer is governed.
Runtime probes before public/mobile release	Decode latency, hash throughput, heap stability and token SLA are publication prerequisites for relevant clients.
Consent before render	Consent-bound layers store pointers, not raw PII, and block rendering when consent proof is invalid, expired or revoked.

11. Implementation-Ready Object Map

All object families below are PROPOSED unless a future mounted-repo inspection verifies matching schemas/contracts. Directory Rules supports schemas/contracts/v1/<...> as the default schema-home convention, but exact homes still need current repo verification.

object family	field intent / v1.5 role
SourceDescriptor	Describes source identity, role, rights, sensitivity, cadence, source_head and expected shape.
LayerManifest	Binds layer ID, source refs, style refs, evidence refs, policy labels, temporal scope, review state and release state.
StyleManifest	Versioned style JSON, sprites, glyphs, expressions, design tokens and visual-regression metadata.
TileArtifactManifest	Describes PMTiles/MVT/COG/MBTiles artifact, media type, min/max zoom, tile_format, tiling_scheme, sha256/blake3/root_hash, byte ranges, range/CORS requirements, source refs and attestation refs.
MapReleaseManifest / ReleaseManifest	Canonical publication envelope binding artifacts, evidence_refs, rights, sensitivity, release_state, policy result, attestations, correction_lineage and rollback.
EvidenceBundle	Admissible evidence object resolved from EvidenceRef; outranks maps, tiles and generated text.
EvidenceDrawerPayload	Governed UI projection of EvidenceBundle, citations, policy/review/release state, stale state and correction links.
MapContextEnvelope	Bounded context carrying map camera, layer IDs, feature IDs, temporal snapshot, release refs and selected evidence refs.
FocusModeRequest / FocusModeResponse	Evidence-bounded request/response with finite ANSWER, ABSTAIN, DENY, ERROR outcomes.
AIReceipt	Records provider/model adapter, evidence refs, citation validation, policy result and response metadata without exposing private chain-of-thought.
CitationValidationReport	Pass/fail citation closure object for Focus Mode, Story Nodes, popups and exports.
PolicyDecision	Policy output with allow/deny, finite outcome, reasons, obligations and sensitivity/rights posture.
PromotionDecision	Promotion gate result with gate IDs, inputs, proofs, release target and rollback target.
RunReceipt	Build/run receipt for pipeline or tile artifact generation: inputs, config/spec hash, artifact digests, source_head, tool versions and attestations.
VerifyReceipt	Runtime verification receipt for tile activation: digest_verified, bounds_verified, schema_verified, root_hash, tileset, capability_issued.
RuntimeProbeResult	Probe result for renderer/tile runtime budgets: decode latency, hash throughput, heap growth, token latency, device profile and pass/fail.
ReleaseRuntimeGate	Gate object tying RuntimeProbeResult to release eligibility for a layer/story/device class.
AutomationBadgePayload	Trust-visible on-map status projection for run health, attestation, SBOM, manifest, logs and telemetry.
ConsentMetadata / ConsentReceiptPointer	Pointer-only consent proof metadata beside EvidenceBundle/tile sidecar; no raw PII in public artifacts.
rollback target	Pointer to previous release manifest/root hash/tile checksum set with rollback drill and correction lineage.
stale-source fixture	Negative fixture proving stale source headers trigger ABSTAIN/DENY or stale badge.
sensitive-geometry deny fixture	Negative fixture proving precise sensitive geometry cannot publish or render publicly.

12. Validation and Test Plan

test area	required check
Schema validation	Validate SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceDrawerPayload, MapContextEnvelope, AIReceipt, PolicyDecision, PromotionDecision, RunReceipt, VerifyReceipt, RuntimeProbeResult and ReleaseRuntimeGate.
Source ledger completeness	Every attached source has stable source_id; prior-only sources remain via SRC-MASTER-PREV.
Style source/layer validity	Style JSON sources and layers match manifests; no unreviewed style drift.
PMTiles / MVT / MLT manifest checks	Validate PMTiles root_hash, BAO sidecar, media type, tile_format, tiling_scheme, min/max zoom, source digest and attestation refs; MLT remains pilot-only.
COG / range request requirements	Run gdalinfo COG checks; verify Range/CORS/HEAD behavior before public COG/raster delivery.
No public raw path	Browser cannot reach RAW, WORK, QUARANTINE, candidate tiles, canonical stores or direct model runtimes.
No unreleased tile load	MapLibre addSource is blocked until ReleaseManifest and sidecar verification pass.
Sensitive geometry deny	Exact sensitive geometry cannot publish; style filters are not valid redaction.
Source rights unknown blocks release	Unknown policy_label/rights_status/sensitivity blocks public ReleaseManifest.
Stale source badge / abstain	Stale source_head or missing source freshness emits visible stale state and Focus Mode abstains.
Clicked feature resolves EvidenceBundle	query/click candidate passes through governed API and resolves EvidenceBundle before drawer answer.
Citation validation pass/fail	Focus Mode and exports cite evidence or abstain.
Focus Mode answer with citations	ANSWER fixture includes citations, policy pass and AIReceipt.
Focus Mode abstain on missing evidence	Missing EvidenceBundle yields ABSTAIN/ERROR; no fluent guess.
Rollback restores prior manifest	Rollback shifts lineage pointers/root hash to prior release and records receipt.
Cache invalidation recorded	CDN/client cache invalidation receipt produced on release/correction.
Visual regression for styles	MapLibre/Cesium screenshots, badges and Story Nodes have visual snapshots.
Keyboard navigation / accessibility	Badges, drawers, layer toggles and Focus Mode are keyboard accessible with ARIA labels.
Tile load budget	Decode latency median <50 ms where vector path is public/mobile; 95p target <75 ms.
Interaction-to-drawer latency budget	Click -> governed API -> drawer appears within agreed SLO; stale/error states visible.
BAO/BLAKE3 verified streaming	Corrupted PMTiles range returns DENY and never activates source.
Capability issuance	/caps/issue rejects failed VerifyReceipt and issues short-lived scoped token only after OPA allow.
No-live-calls route tests	Runtime proof fixtures do not depend on live external sources.
Valhalla routing graph smoke	Route smoke test and OSM sha256 metadata required before routing graph publication.
WZDx deterministic transform	Schema validation, sorted rows/columns, CRS EPSG:4326 and STAC/PROV outputs validated.

13. Anti-Patterns Register

anti-pattern	failure mode
MapLibre as truth authority	Renderer output is treated as proof or publication authority.
Tiles as proof	PMTiles, COG, MVT, screenshots or 3D scenes are cited as if they were EvidenceBundles.
Layer toggle as publication	Making a layer visible is mistaken for PromotionDecision or release state.
Style-filter geoprivacy	Exact sensitive geometry exists in public tile but is hidden by style/filter.
Dense GeoJSON production default	Large/recurring datasets use browser GeoJSON instead of governed tile artifacts.
Uncontrolled style drift	Style JSON changes without StyleManifest/version/visual regression.
Unreviewed plugin/wrapper adoption	MapLibre plugin or wrapper adds network/runtime behavior without allowlist/security review.
Direct browser model access	Client sends map features directly to local/cloud model runtime.
AI based only on rendered features	Focus Mode answers from visible map state without EvidenceBundle/citations.
Uncited popups or exports	Popup/screenshot/report contains consequential claim without citations.
MLT default before validation	Emerging tile format adopted as production default without toolchain proof.
3D/globe as default	3D used when 2D communicates more clearly or trust controls are weaker.
Mapbox-era backports without review	Legacy Mapbox assumptions/code imported without legal/security compatibility review.
addSource before verification	MapLibre registers PMTiles before sidecar/signature/spec_hash/tile_format checks.
Runtime probes skipped	Public/mobile vector path promoted without decode/hash/heap/token budgets.
Unsigned PMTiles delta	Delta PMTiles published without signed manifest, BAO/root hash, RunReceipt or rollback root.
Untrusted browser attestation fetch	Browser pulls remote attestations from arbitrary origins instead of verified proxy.
Consent-hidden PII	Consent-bound data or identifiers embedded in tiles/artifacts instead of pointer-only sidecar metadata.
Diagnostic badges as proof	On-map pipeline badges replace the Evidence Drawer or release manifest.

14. Open Questions and Verification Backlog

item	verification needed
Package versions	MapLibre GL JS, pmtiles/protomaps, tippecanoe, GDAL, rio-cogeo, Cesium, Valhalla, cosign, bao/blake3, OPA/Conftest versions are NEEDS VERIFICATION.
Schema home	Directory Rules default schemas/contracts/v1/<...>, but actual repo convention must be verified before file creation.
Actual repo topology	No mounted repo; app paths, package manager, routes, CI and storage are UNKNOWN.
MapLibre GL JS target version	Target version and browser support matrix need current official verification.
MLT support status	MLT remains pilot until KFM toolchain and browser/runtime support are proven.
Mobile/native parity	MapLibre Native/mobile parity, memory budgets and device profiles need tests.
PMTiles hosting Range/CORS	Hosting must prove Range, CORS, immutable URLs, cache headers and byte-range behavior.
Martin deployment/access model	When dynamic tile serving is needed, access model, PostGIS policy and TileJSON must be verified.
Plugin allowlist	MapLibre wrapper/plugin review and allowlist status unknown.
Style/sprite/glyph hosting	Asset locations, hashes, rights and CORS headers need verification.
Attribution and rights	KGS, SSURGO/gNATSGO, SoilGrids, WZDx, OSM/Valhalla, air-quality sources and Story Node assets require rights review.
Sensitive geometry transform rules	Exact thresholds and transform receipts by domain remain to be specified.
Release manifest location	ReleaseManifest/MapReleaseManifest home and registry path need repo verification.
CI workflow	No actual CI workflow inspected; all workflows in source packets are PROPOSED.
Accessibility acceptance	Keyboard/ARIA and map overlay acceptance criteria need implementation tests.
Performance budgets	Runtime thresholds from SRC-058 need calibration against KFM target devices.
BAO/BLAKE3 implementation	Verifier library, WASM security, worker design and performance need validation.
Capability issuance	/caps/issue auth model, signing keys, token TTL and audit logging need security design.
Consent verification	VC mechanism, DSSE receipt store and privacy-preserving status list need legal/security review.
Visual regression	MapLibre/Cesium snapshot tools and baselines need repo-native setup.
Valhalla publication	OSM source terms, build determinism, route test fixtures and catalog semantics need verification.
WZDx publication	Feed rights, schema version, freshness headers and masking rules need verification.
Attestation transparency	Rekor/offline transparency assumptions need environment decision.
CDN cache invalidation	Cache purge receipts and rollback behavior need staging tests.

15. How to Continue This Master PDF Next Run

- Attach this PDF again with any new documents.
- The next run should read this PDF first as SRC-MASTER-PREV.
- Preserve all idea IDs unless merged or superseded.
- Update the delta table.
- Add new sources to the source ledger.
- Add new discoveries to the idea index.
- Recompute counts.
- Keep uncertainty labels visible.

Final PDF Self-Check

check	result
No obvious cut-off text	Update packet uses wrapped cards and narrow tables; prior baseline appended unchanged.
No missing source ledger	Section 4 includes current-run source ledger with prompt, prior master, retained sources and new sources.
No missing delta section	Section 3 records prior version, new sources, retained ideas, and status.
No missing open questions	Section 14 lists version, hosting, policy, source rights, accessibility and performance verification items.
No unsupported implementation claims	All paths and implementations are labeled PROPOSED/UNKNOWN unless attached documents support doctrine.
No forgotten prior master ideas	SRC-MASTER-PREV is appended unchanged as Appendix A.
No extra user-facing deliverables	Only the final PDF is intended for the user.

Appendix A begins on the next page: full retained v1.4 baseline

The following pages are the prior Master MapLibre Components-Functions-Features.pdf copied as SRC-MASTER-PREV. They preserve earlier source ledger entries, category chapters, matrices, idea IDs, open questions and prior deltas. This appendix is retained unchanged and does not silently delete prior discoveries.

Master MapLibre Components-Functions-Features.pdf

Cumulative KFM MapLibre, Tile, Renderer, Style, UI, Evidence Drawer, and Release Architecture
Atlas

Version v1.4 cumulative update - run date 2026-05-10 - output mode: PDF only

Truth posture: CONFIRMED doctrine and source scans; PROPOSED implementation guidance; UNKNOWN current repo/runtime depth; NEEDS VERIFICATION for versions, hosting headers, rights, source endpoints, policy enforcement, performance budgets and mobile parity.

Field	Value
Prior master PDF detected	Yes - v1.3 is copied as SRC-MASTER-PREV and appended as the retained cumulative baseline.
Current input PDFs detected	42 PDFs in /mnt/data, including the prior master, 38 retained attached PDFs, New Ideas 4-25-26, New Ideas 4-30-26 and new SRC-056 Pass 15 Part 2.
Source ledger count	43 current-run records: SRC-RUN-PROMPT, SRC-MASTER-PREV, SRC-003 through SRC-041 except absent SRC-021, SRC-054, SRC-055, SRC-056.
Prior-only continuity source	SRC-021 fullstack-react-with-typescript.pdf appears in v1.3 but was not attached in this run; it is preserved only inside SRC-MASTER-PREV.
Idea count	2162 cumulative idea records: 2128 prior + 34 v1.4 delta records from SRC-056.
New discoveries count	34 SRC-056 delta records; SRC-054 and SRC-055 were already incorporated in v1.3 and are rechecked, not double-counted.
Expanded prior categories	12 category clusters expanded by SRC-056.
Superseded ideas	0 - no attached evidence invalidated a prior idea.
Open verification items	271 cumulative: prior 259 + 12 SRC-056-specific verification items.
Evidence limits	No mounted KFM repo, tests, workflows, dashboards, logs, release manifests or runtime traces were available in this generation context.

1. Table of Contents

1. Cover Page and run metadata
 2. Executive Determination
 3. Continuity and Delta Since Last Run
 4. Source Ledger
 5. Master Idea Index - v1.4 Delta plus retained baseline note
 6. Category Atlas
 7. Detailed Category Chapters
 8. Components-Functions-Features Matrix
 9. Tile Strategy Chapter
 10. Governance and Trust-Membrane Chapter
 11. Implementation-Ready Object Map
 12. Validation and Test Plan
 13. Anti-Patterns Register
 14. Open Questions and Verification Backlog
 15. How to Continue This Master PDF Next Run
- Appendix A. Full retained v1.3 Master PDF baseline (SRC-MASTER-PREV, appended unchanged)
- Final PDF Self-Check

Navigation note. The v1.4 update packet is short and current-run specific. The full v1.3 baseline follows it as Appendix A so prior source ledger entries, category chapters, matrices, idea IDs and open questions are not silently deleted.

2. Executive Determination

CONFIRMED from SRC-MASTER-PREV and attached KFM doctrine: MapLibre remains a downstream renderer and interaction runtime. It is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority or AI authority. Tiles, PMTiles, MVT, COGs, styles, sprites, glyphs, popups, screenshots, exports, scenes and AI answers remain downstream carriers.

CONFIRMED from this run: the prior v1.3 master baseline was detected before overwriting and copied to SRC-MASTER-PREV. The current source scan shows that New Ideas 4-25-26 and New Ideas 4-30-26 match the same source-title and high-signal term profile already incorporated as SRC-054 and SRC-055 in v1.3. They are therefore rechecked, retained and not double-counted as new idea families.

NEW from SRC-056: Pass 15 Part 2 adds a compact but important source-backed clarification: the map-serving lane itself is named, but it is explicitly underdeveloped compared with other proof lanes. It names MapLibre, a time slider, PMTiles and COG sidecars, and it calls for runtime proof, envelope mapping, validators, bridge tests, no-live-calls route tests and a parity-or-out-of-scope decision.

PROPOSED synthesis: v1.4 should be treated as a continuity-and-parity update, not a structural rewrite. The next implementation-ready MapLibre slice is still schema/fixture/test first: map-serving runtime proof, time-slider request fixture, TileArtifactManifest and COG/PMTiles digest checks, EvidenceDrawerPayload projection, generic drawer schema, Focus Mode building blocks, no-live-calls route tests and promotion-gate CI fixtures.

UNKNOWN / NEEDS VERIFICATION: actual repo topology, package manager, schema home, UI component home, MapLibre GL JS target version, PMTiles tooling, COG/PMTiles hosting headers, browser-side signature verification, mobile/native parity, MLT support status, plugin allowlist, sensitive geometry thresholds, source rights, accessibility acceptance and performance budgets.

Highest-risk anti-pattern remains unchanged: treating a rendered map, layer toggle, PMTiles archive, COG, popup, trust badge or generated answer as proof. SRC-056 strengthens that risk by showing that the map-serving lane can sound concrete while still lacking runtime proof.

Metric	Value
Previous version	v1.3 cumulative update
Current version	v1.4 cumulative update
Prior ideas carried forward	2128
New SRC-056 delta idea records	34
Total cumulative idea count	2162
Prior categories expanded	12
Superseded ideas	0
Open verification items	271

3. Continuity and Delta Since Last Run

SRC-MASTER-PREV is the prior v1.3 Master MapLibre Components-Functions-Features.pdf. It remains the cumulative memory baseline and is appended unchanged. It does not outrank original evidence. Current-run source evidence can refine, expand, conflict with or supersede its entries. No such supersession occurred in this run.

Delta item	Result
Prior version detected	v1.3, 281 pages, sha256 prefix a3a4fd76f8604ad7
Version rule applied	+0.1 ordinary cumulative update. No fundamental structural reorganization.
New source added	SRC-056 KFM Pass 15 Part 2 Idea Index, Category Atlas, and Expansion Dossier.
Reconfirmed sources	SRC-054 New Ideas 4-25-26 and SRC-055 New Ideas 4-30-26; already included in v1.3, rechecked here.
Prior ideas preserved	All 2128 prior ML-* records are preserved by appending SRC-MASTER-PREV unchanged.
New records	34 ML-056-* delta records.
Merged/superseded	0 merged, 0 superseded.
Conflict discovered	No direct source conflict. The key unresolved status is an implementation maturity gap: map-serving is named but not proof-complete.
Open questions carried forward	Package versions, schema home, source rights, MLT status, mobile parity, hosting headers, sensitive geometry, release manifest home, CI workflow, accessibility and performance budgets.

3A. Prior idea status comparison by category

Category	v1.3 prior count	v1.4 SRC-056 delta	v1.4 status / note
A. Renderer Boundary and KFM Trust Law	75	2	EXPANDED by SRC-056
B. MapLibre GL JS Web Shell	65	1	EXPANDED by SRC-056
C. MapLibre Native / Mobile / Platform Parity	17	0	RETAINED from SRC-MASTER-PREV
D. MapLibre RS / Experimental Renderers	5	0	RETAINED from SRC-MASTER-PREV
E. Style Specification, Sources, Layers, Expressions	66	0	RETAINED from SRC-MASTER-PREV
F. Sprites, Glyphs, Fonts, Design Tokens	11	0	RETAINED from SRC-MASTER-PREV
G. GeoJSON and Runtime Data	68	0	RETAINED from SRC-MASTER-PREV
H. Vector Tiles: MVT and MLT	54	0	RETAINED from SRC-MASTER-PREV
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	212	3	EXPANDED by SRC-056
J. Martin and Server-Mediated Tile Serving	14	0	RETAINED from SRC-MASTER-PREV
K. Raster, COG, DEM, Terrain, Hillshade	94	2	EXPANDED by SRC-056
L. WMS / WMTS / External Map Services	27	0	RETAINED from SRC-MASTER-PREV
M. Manifests	146	4	EXPANDED by SRC-056
N. Evidence Drawer Payloads and Click Resolution	114	4	EXPANDED by SRC-056
O. Focus Mode and Governed AI Map Context	86	3	EXPANDED by SRC-056
P. Time-Aware Map Interaction and Timeline	94	3	EXPANDED by SRC-056
Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	88	0	RETAINED from SRC-MASTER-PREV
R. Plugin, Wrapper, and Dependency Governance	42	0	RETAINED from SRC-MASTER-PREV
S. Accessibility, UX, and Trust-Visible States	64	1	EXPANDED by SRC-056
T. Performance, Caching, CDN, Range Requests, and Resource Timing	90	0	RETAINED from SRC-MASTER-PREV
U. Testing, CI, Validation, Rollback, and Proof Objects	177	5	EXPANDED by SRC-056
V. Exports, Screenshots, Reports, and Citation Preservation	50	0	RETAINED from SRC-MASTER-PREV
W. 3D / Cesium / Deck.gl / Overlay Interoperability	101	0	RETAINED from SRC-MASTER-PREV
X. Anti-Patterns and Failure Modes	104	0	RETAINED from SRC-MASTER-PREV
Y. Implementation Backlog and PR Plan	114	1	EXPANDED by SRC-056

Category	v1.3 prior count	v1.4 SRC-056 delta	v1.4 status / note
Z. Open Questions and Verification Backlog	150	5	EXPANDED by SRC-056

4. Source Ledger

This source ledger is a control surface, not a bibliography. It states what each source can support and what it cannot prove. Current external facts embedded in PDFs remain source-supported but need verification before operational package pins, access decisions, source activation or release claims.

Scan item	Result
SRC-054	337 pages; term scan high signals: Tile 294, MapLibre 70, PMTiles 55, Focus Mode 36, Evidence Drawer 35, Cesium 31, COG 22.
SRC-055	289 pages; term scan high signals: Tile 706, PMTiles 325, MapLibre 200, Cesium 189, style 73, Range 55, MVT 29, CDN 27.
SRC-056	48 pages; term scan high signals: runtime proof 57, Promotion 55, Evidence Drawer 31, EvidenceBundle 22, DecisionEnvelope 15, MapLibre 8, PMTiles 8, COG 6, time slider 5.
Repo/runtime evidence	No mounted repository, source tree, tests, workflows, dashboards, logs or release manifests were available; implementation depth remains UNKNOWN.

source_id	file / document title	run status	MapLibre / tile role
SRC-RUN-PR OMPT	Current user prompt	CONFIRMED request	Controls PDF-only output, cumulative memory, source IDs, truth labels and required atlas sections.
SRC-MASTER -PREV	Master MapLibre Components-Functions-Features.pdf v1.3	Prior cumulative baseline detected	Preserves 2128 prior ML-* ideas; appended unchanged after the v1.4 update packet.
SRC-003	KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf	CONFIRMED attached	Renderer boundary, governed UI, Evidence Drawer, Focus Mode, tile/style strategy.
SRC-004	Directory Rules.pdf	CONFIRMED attached	Responsibility-root placement law, schema-home default, lifecycle boundaries.
SRC-005	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf	CONFIRMED attached	Lifecycle, query-save-validate-compile-review-promote-recompile loop, MapLibre UI.
SRC-006	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan_v1_1.pdf	CONFIRMED attached	Trust spine, public-safe defaults, pre-RAW event family, map architecture.
SRC-007	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf	CONFIRMED attached	Whole UI, MapLibre shell, governed API, Evidence Drawer, Focus Mode.
SRC-008	KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf	CONFIRMED attached	AI boundary, finite outcomes, citation validation, source-ledger discipline.
SRC-009	Ollama & Ubuntu Information.pdf	CONFIRMED attached	Local model runtime behind governed API; no direct client/model path.
SRC-010	kfm_encyclopedia.pdf	CONFIRMED attached	Cross-domain capability atlas; maps/tiles/AI as carriers, not truth.
SRC-011	# Kansas Frontier Matrix Implementation Reference.pdf	CONFIRMED attached	Prior implementation reference; MapLibre/PMTiles stack signals and schema-home ambiguity.
SRC-012	a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf	CONFIRMED attached	GIS/cartographic representation background.
SRC-013	Advanced-SQL-Concepts.pdf	CONFIRMED attached	SQL/query background for validation, catalogs, temporal analytics.
SRC-014	AI_Concepts_Using_Python.pdf	CONFIRMED attached	AI/data workflow background; not MapLibre-specific.
SRC-015	Archaeological 3D GIS.pdf	CONFIRMED attached	3D GIS, visibility and model context for conditional 3D support.
SRC-016	CICD_with_Docker_Kubernetes_Semaphore.pdf	CONFIRMED attached	CI/CD, rollbacks, deployment testing background.
SRC-017	Designing Great Web APIs.pdf	CONFIRMED attached	Governed API contract background.
SRC-018	developing-time-oriented-database-applications-in-sql.pdf	CONFIRMED attached	Valid/transaction/bitemporal time concepts for time-aware maps.
SRC-019	Domain-Driven Design Reference.pdf	CONFIRMED attached	Bounded-context and published-language background.

source_id	file / document title	run status	MapLibre / tile role
SRC-020	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf	CONFIRMED attached	Environmental GIS layer/hazard/climate context.
SRC-022	GIS Succinctly.pdf	CONFIRMED attached	Spatial database, geometry and GIS application basics.
SRC-023	GIS in Sustainable Urban Planning and Management.pdf	CONFIRMED attached	Planning support, spatial indicators and risk map context.
SRC-024	instant-web-scraping-with-java.pdf	CONFIRMED attached	Source acquisition/scrapper caution background.
SRC-025	KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	CONFIRMED attached	Run receipts, deterministic identity, promotion gates, source intake patterns.
SRC-026	KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	CONFIRMED attached	Prior synthesis of New Ideas 4-25 and 4-30; supports continuity.
SRC-027	KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	CONFIRMED attached	Pre-RAW event family, watchers, catalog integrity, environmental tiles.
SRC-028	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf	CONFIRMED attached	Agriculture layer and Evidence Drawer posture.
SRC-029	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	CONFIRMED attached	Exact site-location denial and candidate remote-sensing caution.
SRC-030	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	CONFIRMED attached	Air/smoke/EO layers and knowledge-character labels.
SRC-031	KFM_Fauna_Architecture_PDF_Only_Report.pdf	CONFIRMED attached	Occurrence geoprivacy and public-safe derived products.
SRC-032	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	CONFIRMED attached	Rare-plant controls, occurrence redaction and drawer implications.
SRC-033	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf	CONFIRMED attached	Geology/resource map layers, public-safe geometry, Focus Mode.
SRC-034	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	CONFIRMED attached	Habitat assignment proof slice and Evidence Drawer.
SRC-035	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	CONFIRMED attached	Habitat patches, suitability, connectivity, public-safe products.
SRC-036	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	CONFIRMED attached	Hazard layers, no emergency alerting, evidence-backed UI.
SRC-037	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf	CONFIRMED attached	Hydrology proof lane, HUC12/NHDPlus/USGS/NFHL, map layers.
SRC-038	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	CONFIRMED attached	Living-person/DNA restrictions, temporal land assertions, map caution.
SRC-039	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	CONFIRMED attached	Transport/corridor layers, graph projections, Evidence Drawer.
SRC-040	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	CONFIRMED attached	Settlement/infrastructure layer manifests and sensitive assets.
SRC-041	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	CONFIRMED attached	Soil maps, SSURGO/gSSURGO, PMTiles stub, Evidence Drawer.
SRC-054	New Ideas 4-25-26.pdf	CONFIRMED attached; same source title as v1.3 SRC-054	Already incorporated in v1.3; current run rechecked scan counts and page hits.
SRC-055	New Ideas 4-30-26.pdf	CONFIRMED attached; same source title as v1.3 SRC-055	Already incorporated in v1.3; current run rechecked scan counts and page hits.
SRC-056	KFM_Pass_15_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	NEW attached source in v1.4	Adds explicit map-serving lane finding: MapLibre time slider + PMTiles + COG sidecars need runtime proof parity.

5. Master Idea Index - v1.4 Delta

The full cumulative v1.3 index of 2128 prior ideas is retained in Appendix A via SRC-MASTER-PREV. The table below adds the 34 v1.4 delta records derived from SRC-056. These IDs are stable for future runs. They are source-backed extraction or synthesis records; they do not assert current repo implementation.

ML-056-001 - Map-serving lane is named but underbuilt

Field	Value
Category	Z / Y
Status	NEW
Evidence	SRC-056 p.31, p.35
Normalized statement	The corpus identifies a map-serving lane built around MapLibre time slider, PMTiles and COG sidecars, but says it lacks runtime envelope mapping, validators and runtime proof.
Implementation consequence	Treat map-serving as PROPOSED until brought to proof-lane parity.
Validation consequence	Add a map-serving parity checklist and do not publish as a proven lane until runtime proof exists.

ML-056-002 - MapLibre client-side renderer in map-serving lane

Field	Value
Category	B / A
Status	EXPANDED
Evidence	SRC-056 p.31
Normalized statement	MapLibre is explicitly the client-side renderer in the map-serving lane.
Implementation consequence	Keep MapLibre in the renderer/interaction runtime role, not as registry or truth authority.
Validation consequence	Assert no direct MapLibre-to-canonical-store path.

ML-056-003 - Time slider selects snapshot

Field	Value
Category	P
Status	EXPANDED
Evidence	SRC-056 p.31
Normalized statement	The map-serving lane uses a time slider as the temporal navigation control selecting which snapshot to render.
Implementation consequence	Layer manifests need temporal snapshot IDs and valid/release time semantics.
Validation consequence	Fixture test: time slider chooses a released snapshot and rejects unreleased snapshots.

ML-056-004 - PMTiles as vector/raster delivery format

Field	Value
Category	I / H
Status	EXPANDED
Evidence	SRC-056 p.31
Normalized statement	PMTiles is named as the tile format for vector and raster data in the map-serving lane.
Implementation consequence	PMTiles artifacts need manifests, digests, release state and rollback targets.
Validation consequence	Validate PMTiles sidecar manifests and source digests before release.

ML-056-005 - COG sidecars for raster evidence surfaces

Field	Value
Category	K
Status	EXPANDED
Evidence	SRC-056 p.31
Normalized statement	COG sidecars are named as per-raster sidecars in the map-serving lane.
Implementation consequence	Raster surfaces need range-capable hosting and manifest closure like vector tiles.
Validation consequence	Test COG Range/CORS metadata and record artifact digest.

ML-056-006 - Tiles and rasters live in registry

Field	Value
Category	M / A
Status	EXPANDED
Evidence	SRC-056 p.31
Normalized statement	The corpus says underlying tiles and rasters live in the registry.
Implementation consequence	Do not let ad-hoc static files become release authority.
Validation consequence	Release gate checks registry record and manifest before map load.

ML-056-007 - Map-serving runtime queries go through envelope grammar

Field	Value
Category	U / O
Status	EXPANDED
Evidence	SRC-056 p.31
Normalized statement	Runtime queries against the map-serving lane are expected to go through the envelope grammar.
Implementation consequence	Map tile-related answers should be surfaced as ANSWER, ABSTAIN, DENY or ERROR when evidence/policy requires.
Validation consequence	Runtime proof must assert envelope shape for map-serving requests.

ML-056-008 - Map-serving admission uses standard gates

Field	Value
Category	U / Y
Status	EXPANDED
Evidence	SRC-056 p.31
Normalized statement	Admission for the map-serving lane goes through the standard gates.
Implementation consequence	Use the same gate discipline as other lanes, not a special map shortcut.
Validation consequence	Gate tests cover schema, policy, proof, release decision and manifest checks.

ML-056-009 - Runtime proof for map-serving lane is required

Field	Value
Category	U / Y
Status	NEW
Evidence	SRC-056 p.31, p.39
Normalized statement	The source explicitly calls for a runtime-proof entry for the map-serving lane.
Implementation consequence	Build a proof harness before treating map-serving as operational.
Validation consequence	Add fixture proving time-slider request and envelope response.

ML-056-010 - Time-slider request proof target

Field	Value
Category	P / U
Status	NEW
Evidence	SRC-056 p.31, p.39
Normalized statement	The expansion direction is a runtime proof that exercises a time-slider request.
Implementation consequence	Temporal map controls need testable request/response semantics.
Validation consequence	Test released snapshot, stale snapshot and missing evidence outcomes.

ML-056-011 - Map-serving parity or out-of-scope decision

Field	Value
Category	Y / Z
Status	NEW
Evidence	SRC-056 p.35
Normalized statement	The map-serving lane should either be developed to parity or explicitly designated out-of-scope.
Implementation consequence	Avoid ambiguous half-lane status.
Validation consequence	Decision record must mark parity plan or deferral.

ML-056-012 - Evidence Drawer as governed UI surface

Field	Value
Category	N / S
Status	EXPANDED
Evidence	SRC-056 p.6, p.45
Normalized statement	UIX category gathers Evidence Drawer and side-by-side patterns as governed public surfaces.
Implementation consequence	Drawer is part of the trust interface, not a decorative popup.
Validation consequence	Bridge test drawer payload shape against runtime envelope.

ML-056-013 - ANSWER/ABSTAIN drawer rendering target

Field	Value
Category	N / O
Status	NEW
Evidence	SRC-056 p.8
Normalized statement	The source asks for a worked example where the same query produces ANSWER and ABSTAIN with drawer rendering both outcomes.
Implementation consequence	UX must make negative outcomes legible, not hide them.
Validation consequence	Create paired fixtures for cited answer and evidence-missing abstention.

ML-056-014 - Generic Evidence Drawer schema gap

Field	Value
Category	N / M / Z
Status	NEW
Evidence	SRC-056 p.13
Normalized statement	The source notes per-domain drawer schemas scale linearly and calls for a generic Evidence Drawer schema specialized by domains.
Implementation consequence	Prevent divergent drawer contracts across lanes.
Validation consequence	Design generic schema plus domain specialization fixtures.

ML-056-015 - Drawer projection separates UI from canonical evidence

Field	Value
Category	N / M
Status	EXPANDED
Evidence	SRC-056 p.13
Normalized statement	The drawer schema describes what UI expects and lets UI evolve without renegotiating canonical evidence shape.
Implementation consequence	Keep EvidenceBundle canonical; EvidenceDrawerPayload is a governed projection.
Validation consequence	Test projection does not drop citation, policy, review or release state.

ML-056-016 - Focus Mode Evidence Building Blocks Standard

Field	Value
Category	O / M
Status	NEW
Evidence	SRC-056 p.14
Normalized statement	The Focus Mode Evidence Building Blocks Standard is a proposal for reusable typed contract artifacts.
Implementation consequence	Treat as PROPOSED until actual schemas/tests exist.
Validation consequence	Create no-network fixtures for building block composition.

ML-056-017 - Reusable Focus Mode contracts

Field	Value
Category	O / M
Status	NEW
Evidence	SRC-056 p.14
Normalized statement	Examples include small reusable artifacts such as geometry.surface.v1.yaml and citation.set.v1.yaml.
Implementation consequence	Focus Mode context should compose typed blocks instead of freeform map text.
Validation consequence	Validate blocks before model context assembly.

ML-056-018 - Uniform DecisionEnvelope grammar

Field	Value
Category	O / U
Status	EXPANDED
Evidence	SRC-056 p.2, p.43
Normalized statement	The source highlights uniform ANSWER / ABSTAIN / DENY / ERROR grammar at the trust membrane.
Implementation consequence	Map-serving, drawer and Focus Mode all need finite outcome surfaces.
Validation consequence	Runtime tests assert all four outcomes are representable.

ML-056-019 - Validator severity feeds envelope and drawer

Field	Value
Category	U / N
Status	EXPANDED
Evidence	SRC-056 p.13
Normalized statement	Validation outcomes are reasoned over by route, drawer and review payloads.
Implementation consequence	Do not recompute policy meaning in UI.
Validation consequence	Fixture maps severity to envelope outcome and drawer badge.

ML-056-020 - Side-by-side explorer/review console pattern

Field	Value
Category	S / U
Status	EXPANDED
Evidence	SRC-056 p.3, p.6
Normalized statement	The source names side-by-side explorer/review-console UI and review handoff.
Implementation consequence	Review surfaces need explicit contract boundaries.
Validation consequence	Bridge tests compare review payload to drawer/layout manifest.

ML-056-021 - Bridge tests for drawer and side-by-side layout manifests

Field	Value
Category	U / S
Status	NEW
Evidence	SRC-056 p.39
Normalized statement	The source asks for bridge tests for Evidence Drawer and side-by-side console layout manifests.
Implementation consequence	Prevent UI/payload drift.
Validation consequence	Add tests that fail when drawer/layout schema and UI payload diverge.

ML-056-022 - No-live-calls tests for governed routes

Field	Value
Category	U
Status	NEW
Evidence	SRC-056 p.39
Normalized statement	The source proposes tests/runtime/no_live_calls.py parameterized over all routes.
Implementation consequence	Runtime proofs should not depend on live external calls.
Validation consequence	No-network route fixtures must pass before CI release gate.

ML-056-023 - Run receipts and runtime receipts are distinct

Field	Value
Category	M / U
Status	EXPANDED
Evidence	SRC-056 p.13
Normalized statement	The source distinguishes build-run receipts from live-answer runtime receipts.
Implementation consequence	Do not collapse pipeline provenance and response provenance.
Validation consequence	Schemas and validators for both receipt families.

ML-056-024 - Source descriptors are versioned artifacts

Field	Value
Category	M / A
Status	EXPANDED
Evidence	SRC-056 p.11
Normalized statement	The source says new source shapes require admitting a new descriptor, superseding old descriptor and migrating through gates.
Implementation consequence	No ad-hoc client fallback when source shape changes.
Validation consequence	Migration tests require descriptor version and gate outcome.

ML-056-025 - Source migration through promotion gates

Field	Value
Category	Y / U
Status	EXPANDED
Evidence	SRC-056 p.11
Normalized statement	Descriptor changes should run through promotion gates rather than client-side adaptation.
Implementation consequence	Treat source-shape changes as governed releases.
Validation consequence	Gate fixture for descriptor supersession.

ML-056-026 - OPA governance substrate still needs enforcement proof

Field	Value
Category	U / Z
Status	NEW
Evidence	SRC-056 p.4
Normalized statement	The source says OPA is argued for but policy bundles are not shown enforced.
Implementation consequence	Policy-as-code is PROPOSED until CI/runtime proof exists.
Validation consequence	Add policy bundle digest and test run evidence.

ML-056-027 - Composable promotion gates in CI

Field	Value
Category	U / Y
Status	EXPANDED
Evidence	SRC-056 p.39
Normalized statement	The source prioritizes wiring four composable promotion gates into CI on fixture bundles.
Implementation consequence	Promotion gate is a continuous check, not prose.
Validation consequence	CI workflow exercises success and failure modes.

ML-056-028 - Runtime proof harness as lane parity floor

Field	Value
Category	U
Status	NEW
Evidence	SRC-056 p.39
Normalized statement	Runtime proof is repeatedly named as the parity artifact for lanes.
Implementation consequence	Map-serving cannot be treated as mature without proof harness.
Validation consequence	Runtime proof matrix includes map-serving.

ML-056-029 - Signed sidecars and proof packs are open work in adjacent lanes

Field	Value
Category	U / M
Status	NEW
Evidence	SRC-056 p.35
Normalized statement	The source says structural specification for signed sidecars, EvidenceBundle wrappers and proof packs is complete in another lane but runtime proof and stub removal remain open.
Implementation consequence	For MapLibre assets, proof packs should be treated as necessary but not sufficient.
Validation consequence	Proof pack verifier plus stub-removal plan.

ML-056-030 - Domain drawer schemas need generic specialization strategy

Field	Value
Category	N / M / Z
Status	NEW
Evidence	SRC-056 p.13
Normalized statement	Per-domain drawer schemas are useful but scale linearly; a generic schema is an expansion direction.
Implementation consequence	Avoid schema sprawl across map popups/drawers.
Validation consequence	Add compatibility tests for generic + domain extensions.

ML-056-031 - Watcher boundary tied to promotion and bundle registry

Field	Value
Category	M / U
Status	EXPANDED
Evidence	SRC-056 p.5, p.6
Normalized statement	WCH category connects catalog, watcher, webhook boundary with promotion and bundle registry conventions.
Implementation consequence	Map-serving updates should arrive as candidate bundles, not public auto-loads.
Validation consequence	Watcher output must create candidate/review/promote path.

ML-056-032 - Map tile queries need DecisionEnvelope not raw asset answer

Field	Value
Category	A / I / U
Status	NEW
Evidence	SRC-056 p.31, p.39
Normalized statement	The source calls for runtime envelopes to apply to map-tile queries.
Implementation consequence	A tile load may be static, but a claim about the tile must pass governed API.
Validation consequence	Test clicked feature resolves through envelope to evidence.

ML-056-033 - Open-review interaction contract matters for map review

Field	Value
Category	S / N
Status	NEW
Evidence	SRC-056 p.3
Normalized statement	Open-review interaction contract and side-by-side review payload are named in the fauna runtime context.
Implementation consequence	Map review UI needs typed handoff, not screenshots alone.
Validation consequence	Review payload fixture with drawer and candidate evidence.

ML-056-034 - Map-serving claims remain PROPOSED until repo/runtime verified

Field	Value
Category	A / Z
Status	NEW
Evidence	SRC-056 p.4, p.31, p.35
Normalized statement	The source repeatedly says map-serving notes are practical but compressed and not a built lane.
Implementation consequence	Use truth labels to prevent premature publication authority.
Validation consequence	Self-check must ensure no unsupported implementation claim is made.

6. Category Atlas

The v1.4 category structure is unchanged from v1.3. SRC-056 does not reorganize the atlas; it strengthens the evidence that the map-serving lane needs the same proof discipline as the other lanes. Categories not touched by SRC-056 are retained from SRC-MASTER-PREV.

Category	v1.4 determination
A. Renderer Boundary and KFM Trust Law	EXPANDED: map-serving lane must not become a shortcut around envelopes or gates.
B. MapLibre GL JS Web Shell	EXPANDED: SRC-056 names MapLibre as client-side renderer in the map-serving lane.
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	EXPANDED: PMTiles named for vector and raster delivery; manifests/digests remain required.
K. Raster, COG, DEM, Terrain, Hillshade	EXPANDED: COG sidecars named as raster companion artifacts requiring manifest closure.
M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	EXPANDED: tiles/rasters live in registry; run receipts, runtime receipts and generic drawer schema need separation.
N. Evidence Drawer Payloads and Click Resolution	EXPANDED: Evidence Drawer is governed UI surface; generic schema gap is explicit.
O. Focus Mode and Governed AI Map Context	EXPANDED: Focus Mode Evidence Building Blocks are proposed typed context fragments.
P. Time-Aware Map Interaction and Timeline	EXPANDED: time slider request needs runtime proof and released-snapshot semantics.
S. Accessibility, UX, and Trust-Visible States	EXPANDED: side-by-side review and bridge tests become UI drift controls.
U. Testing, CI, Validation, Rollback, and Proof Objects	EXPANDED: runtime proof, no-live-calls route tests and promotion-gate CI are the parity floor.
Y. Implementation Backlog and PR Plan	EXPANDED: map-serving must be brought to parity or explicitly deferred.
Z. Open Questions and Verification Backlog	EXPANDED: map-serving maturity, policy enforcement proof and generic drawer schema added.
All other categories	RETAINED from SRC-MASTER-PREV with no SRC-056-specific change.

7. Detailed Category Chapters

A/B//K - Map-serving lane and renderer boundary

SRC-056 confirms the map-serving lane names MapLibre, time slider, PMTiles and COG sidecars. The same source also warns that the lane lacks runtime proof, validator stack and envelope mapping. The category change is not that KFM should trust map-serving more; it is that map-serving must be held to the same proof burden as other lanes. Public UI implication: a user may render released tiles, but claims about those tiles must resolve through governed API and EvidenceBundle. Policy implication: no unreleased tile load, no style-only sensitivity hiding and no registry bypass. Tests: runtime-proof fixture, time-slider request fixture, PMTiles/COG manifest checks, no-live-calls route test.

M/N - Manifest and Evidence Drawer contract split

SRC-056 strengthens the distinction between canonical evidence and UI projection. EvidenceBundle remains the canonical evidence object. EvidenceDrawerPayload is a governed projection. The source explicitly notes that per-domain drawer schemas scale linearly and calls for a generic Evidence Drawer schema with domain specialization. Public UI implication: popups cannot replace drawers. Tests: projection preserves citation, source, policy, review and release state; generic schema compatibility tests.

O/P - Focus Mode and time-aware map context

SRC-056 introduces the Focus Mode Evidence Building Blocks Standard as PROPOSED. It frames typed context fragments such as geometry.surface and citation.set as reusable building blocks. For time-aware maps, the time slider needs a runtime proof that exercises a request and receives a finite DecisionEnvelope. Public UI implication: Focus Mode should answer, abstain, deny or error over released, typed context only. Tests: context-block schema validation and time-slider ANSWER/ABSTAIN fixtures.

S/UI/Y/Z - Review surfaces, proofs and backlog

SRC-056 identifies runtime proof, no-live-calls route tests, Evidence Drawer bridge tests, side-by-side layout manifest tests and promotion-gate CI as the mechanisms that turn a map-serving note into a governed lane. It also states that map-serving is currently practical but compressed. Backlog implication: either bring map-serving to parity or mark it out-of-scope for the current implementation phase. Rollback implication: every tile/style release needs a rollback target and cache invalidation record.

8. Components-Functions-Features Matrix

Component	Function	Feature / consequence	Required tests	Status
MapLibre GL JS	Render released map artifacts and expose interaction state	Renderer only; map-serving lane cannot bypass envelope or evidence gates.	No public RAW path; clicked feature resolves EvidenceBundle.	CONFIRMED doctrine / PROPOSED implementation
Time slider	Select temporal snapshot for layer rendering	Snapshot must be released, manifest-backed and valid for selected time.	Released snapshot, stale snapshot, missing evidence and unreleased snapshot fixtures.	EXPANDED by SRC-056
PMTiles	Static/serverless tile bundle delivery	Needs TileArtifactManifest, digest, release state, rollback and cache invalidation.	PMTiles manifest/digest/range tests.	EXPANDED
COG sidecar	Raster companion artifact	Range-capable hosting and source digest required; not proof by itself.	COG Range/CORS and digest checks.	EXPANDED
Evidence Drawer	Resolve clicked feature or runtime answer to evidence-facing payload	Generic schema gap identified; domain specialization required.	Projection and bridge tests.	EXPANDED
Focus Mode	Evidence-bounded synthesis over map context	Typed building blocks proposed; model cannot answer from rendered features alone.	Citation validation and ABSTAIN fixture.	PROPOSED / EXPANDED
DecisionEnvelope	Trust-membrane runtime response	ANSWER / ABSTAIN / DENY / ERROR for map-serving queries.	Runtime proof and all-outcome tests.	EXPANDED
Promotion gates	Govern release transition	Map-serving artifacts must use gates, not file moves.	CI fixture bundle with pass/fail cases.	EXPANDED
Review console	Human review of candidates and payloads	Side-by-side layout and open-review contract prevent UI drift.	Layout manifest bridge tests.	EXPANDED

9. Tile Strategy Chapter

Comparison	When it wins	When it fails	KFM trust consequence	Validation needed
GeoJSON vs vector tiles	GeoJSON wins for small interactive fixtures and runtime examples. Vector tiles win for large public-safe layers.	Dense GeoJSON fails as production default at scale; vector tiles fail if treated as proof.	Rendered features are candidates only.	Feature click resolves to EvidenceBundle.
MVT vs MLT	MVT wins on current ecosystem maturity. MLT remains a pilot/verification target.	MLT fails as default before KFM toolchain validation.	Do not make newer format release authority.	Format-specific fixtures and visual regression.
PMTiles vs MBTiles	PMTiles wins for static, public-safe, serverless/Range-capable delivery. MBTiles can be build-time or local staging.	PMTiles fails when dynamic access control or steward mediation is required.	PMTiles needs release manifest, digest, rollback target.	Range/CORS, per-tile digests, cache invalidation.
PMTiles vs Martin	PMTiles wins for immutable public bundles; Martin wins when PostGIS/dynamic tile serving or server mediation is needed.	Serverless fails for restricted or dynamic data; Martin fails if exposed without governed access.	Choose by release/access model, not taste.	Access tests and layer manifest checks.
COG vs raster tiles	COG wins for range-capable raster evidence and analysis sidecars; raster tiles win for simple display.	Both fail if rendered pixels are treated as proof.	COG sidecar must resolve to source and manifest.	COG range/digest/source-ref tests.
2D default vs 3D conditional	2D wins for clarity and governance; 3D wins only when vertical context carries evidence burden.	3D/globe fails when it obscures uncertainty or bypasses drawer/policy.	Conditional 3D needs same evidence drawer and release controls.	Camera sync, attribution, performance and drawer tests.
Public-safe vs restricted tiles	Public-safe wins for ordinary UI; restricted/internal tiles only after access governance.	Style hiding exact geometry fails as geoprivacy.	Redaction/generalization happens before release.	Sensitive geometry deny fixture and redaction receipt.

10. Governance and Trust-Membrane Chapter

The governed MapLibre flow remains: released layer -> user click or time selection -> governed API -> EvidenceBundle resolution -> Evidence Drawer -> Focus Mode answer, abstain, deny or error. SRC-056 adds that map-serving itself needs a runtime proof entry, not just tile files.

No.	Trust membrane rule	Reason
1	No public RAW, WORK, QUARANTINE, unpublished candidate or canonical/internal client fetch.	Public UI must use governed APIs, released artifacts, catalog records and tile services.
2	No direct model client from browser or map shell.	AI is interpretive and evidence-subordinate.
3	No unreleased tile load.	Layer toggle is not publication.
4	No sensitive geometry hidden only by style filters.	Geoprivacy requires transformation, redaction receipts and policy gates.
5	No popup as Evidence Drawer substitute.	Popup may summarize; drawer resolves evidence and policy state.
6	No Focus Mode answer from rendered features alone.	Rendered features are candidates; evidence bundle carries truth support.
7	No uncited export or screenshot.	Exports must preserve citation and release context.

11. Implementation-Ready Object Map

Object family	Field intent / v1.4 consequence	Status
SourceDescriptor	Versioned source shape; source changes migrate through promotion gates.	PROPOSED / SRC-056 expanded
LayerManifest	Released layer metadata, IDs, sources, tiles, policy, time and rollback.	PROPOSED
StyleManifest	Style JSON version, source/layer references, sprite/glyph refs, visual regression target.	PROPOSED
TileArtifactManifest	PMTiles/MVT/MLT/COG artifact digest, source digests, release state, cache invalidation.	PROPOSED
MapReleaseManifest	Release bundle linking layer/style/tile manifests and rollback target.	PROPOSED
EvidenceBundle	Canonical evidence support resolved from EvidenceRef; outranks UI projection.	CONFIRMED doctrine / PROPOSED schema
EvidenceDrawerPayload	Governed projection of canonical evidence to UI; generic schema gap identified.	EXPANDED
MapContextEnvelope	Typed map context for Focus Mode; should use building blocks, not free text.	PROPOSED
FocusModeRequest/Response	Evidence-bounded synthesis request and finite response.	PROPOSED
AIReceipt	Runtime model/evidence/citation validation receipt.	PROPOSED
CitationValidationReport	Pass/fail citation support report before answer/export.	PROPOSED
PolicyDecision	DENY/ALLOW/ABSTAIN/ERROR reasoning across rights, sensitivity and release.	PROPOSED
PromotionDecision	Release decision and gate results; map-serving must use it.	EXPANDED
RunReceipt / RuntimeReceipt	Build-run receipt and live-answer receipt remain distinct.	EXPANDED
Rollback target	Prior release manifest and cache invalidation reference.	PROPOSED
Stale-source fixture	Fixture proving STALE badge/ABSTAIN behavior.	PROPOSED
Sensitive-geometry deny fixture	Fixture proving style-only hiding is not accepted.	PROPOSED

12. Validation and Test Plan

1. Schema validation for SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceDrawerPayload, MapContextEnvelope, RunReceipt and RuntimeReceipt.
2. Source ledger completeness check: every attached source has source_id, status, role, limitation and relevance.
3. Style source/layer validity, source-layer references, promoteId/feature ID consistency and visual regression.
4. PMTiles / MVT / MLT manifest checks and source_digests verification.
5. COG / Range request requirements with CORS/header fixture.
6. No public RAW/WORK/QUARANTINE/canonical path in UI or route fixtures.
7. No direct model client from browser or map shell.
8. No unreleased tile load from layer catalog or style JSON.
9. Sensitive geometry deny and redaction receipt fixtures.
10. Source rights unknown blocks release.
11. Stale source badge and Focus Mode ABSTAIN fixture.
12. Clicked feature resolves EvidenceBundle and EvidenceDrawerPayload.
13. Citation validation pass/fail and export-citation preservation.
14. Focus Mode answer with citations; Focus Mode abstain on missing evidence.
15. Rollback restores prior manifest and records cache invalidation.
16. Keyboard navigation, drawer accessibility, trust-visible negative states.
17. Tile load budget, interaction-to-drawer latency budget, no-live-calls route tests.
18. Map-serving runtime proof that exercises a time-slider request and receives a DecisionEnvelope.

13. Anti-Patterns Register

Anti-pattern	Response
Treating MapLibre as truth authority	DENY: renderer cannot become source registry, policy engine, citation authority or release authority.
Treating tiles or COGs as proof	DENY: artifacts carry evidence only when manifest/evidence bundle resolves.
Treating layer toggle as publication	DENY: publication is a governed state transition.
Style-only sensitive geometry hiding	DENY: transform/redact/generalize before release.
Dense GeoJSON as large production default	NEEDS VERIFICATION: use vector tiles or governed API where scale demands.
Uncontrolled style drift	DENY release if style source/layer references are unvalidated.
Plugin/wrapper adoption without review	DENY until allowlist, security, license and test evidence exist.
Direct browser access to model runtime	DENY; Focus Mode uses governed API.
AI answer from rendered map features only	ABSTAIN or DENY; rendered features are selection candidates.
Uncited popups, screenshots or exports	DENY or ABSTAIN; citation preservation required.
MLT as default before KFM validation	NEEDS VERIFICATION.
3D/globe as default when 2D is clearer	PROPOSED only when vertical/globe evidence burden exists.
Mapbox-era backports without legal/security review	NEEDS VERIFICATION.
Map-serving lane declared mature without runtime proof	NEW anti-pattern from SRC-056; DENY maturity claim until proof harness passes.

14. Open Questions and Verification Backlog

1. Actual repo topology, package manager and schema home.
2. MapLibre GL JS target version and lockfile pin.
3. MLT support status and toolchain maturity.
4. Mobile/native parity, including drawer, evidence, policy and offline behavior.
5. PMTiles hosting Range/CORS behavior and per-tile digest strategy.
6. Martin deployment/access model and PostGIS mediation boundaries.
7. Plugin allowlist and MapLibre wrapper admission records.
8. Style/sprite/glyph hosting and cache invalidation policy.
9. Attribution, source rights, source terms and export obligations.
10. Sensitive geometry transform thresholds and receipts.
11. Release manifest location and rollback record home.
12. CI workflow, OPA/Conftest policy enforcement and bundle digest verification.
13. Accessibility acceptance criteria and keyboard navigation target.
14. Tile load, resource timing and interaction-to-drawer latency budgets.
15. Generic Evidence Drawer schema and domain specialization pattern.
16. Map-serving runtime proof parity or explicit out-of-scope decision.
17. No-live-calls route test coverage for map-serving and Focus Mode.
18. RunReceipt vs RuntimeReceipt schema separation and validation.
19. Focus Mode Evidence Building Blocks schema set and validation harness.
20. Whether any prior v1.3 source not attached this run should be reattached next run for source-hash continuity.

15. How to Continue This Master PDF Next Run

- Attach this PDF again with any new documents.
- The next run should read this PDF first as SRC-MASTER-PREV.
- Preserve all idea IDs unless merged or superseded.
- Update the delta table and add new sources to the source ledger.
- Add new discoveries to the idea index and recompute counts.
- Keep uncertainty labels visible.
- Reattach any prior-only sources when byte-level continuity matters; this run preserved the prior-only fullstack React source only through SRC-MASTER-PREV.
- If the actual KFM repo is mounted, verify paths against Directory Rules before asserting implementation behavior or proposing file homes.

Appendix notice. The next pages are the full retained v1.3 baseline from SRC-MASTER-PREV. They preserve the earlier source ledger, master idea index, category chapters, matrices, tile strategy, governance chapter, object map, validation plan, anti-pattern register and open questions.

Master MapLibre Components-Functions-Features.pdf

Cumulative KFM MapLibre, Tile, Renderer, Style, UI, Evidence Drawer, and Release Architecture
Atlas

Field	Value
Version	v1.3 cumulative update
Run date	2026-05-10
Attached source count	43 active source records: SRC-MASTER-PREV, SRC-RUN-PROMPT, 39 retained PDFs, and 2 current-run new idea PDFs.
Prior master PDF detected	Yes - v1.2 treated as SRC-MASTER-PREV.
Output mode	PDF only
Truth posture	CONFIRMED / PROPOSED / UNKNOWN / NEEDS VERIFICATION / CONFLICTED / SUPERSEDED / RETAINED / NEW / EXPANDED
Evidence limits	No mounted KFM repository, tests, workflows, dashboards, logs, release manifests, or runtime traces were available. Runtime implementation depth remains UNKNOWN.
Source ledger count	43
Idea count	2128 total: 1802 prior + 326 new current-run extractions/syntheses
New discoveries count	326
Expanded prior ideas count	1797
Retained prior ideas count	5
Superseded ideas count	0
Open verification items count	259

Truth posture summary: CONFIRMED evidence supports the prior cumulative baseline, current-run source scans, MapLibre renderer-boundary doctrine, Directory Rules placement doctrine, and current-run MapLibre/tile/Evidence Drawer findings from SRC-054 and SRC-055. PROPOSED content identifies implementation-ready objects, tests, and paths that remain unverified in a mounted repository. UNKNOWN and NEEDS VERIFICATION remain visible for package versions, source rights, endpoints, sensitivity controls, hosting headers, browser verification, and actual repo topology.

Lineage note: prior v1.2 source IDs SRC-052 and SRC-053 remain preserved inside SRC-MASTER-PREV only. Current-run new source PDFs are assigned SRC-054 and SRC-055 to avoid alias collision and preserve cumulative continuity.

1. Table of Contents

1. Cover Page
2. Executive Determination
3. Continuity and Delta Since Last Run
4. Source Ledger
5. Master Idea Index
6. Category Atlas
7. Detailed Category Chapters
8. Components-Functions-Features Matrix
9. Tile Strategy Chapter
10. Governance and Trust-Membrane Chapter
11. Implementation-Ready Object Map
12. Validation and Test Plan
13. Anti-Patterns Register
14. Open Questions and Verification Backlog
15. How to Continue This Master PDF Next Run
16. Final PDF Self-Check

Navigation note: stable source IDs and idea IDs are the primary navigation anchors. Wide matrices are split, carded, or wrapped instead of squeezed into unreadable columns.

2. Executive Determination

CONFIRMED from SRC-MASTER-PREV, SRC-RUN-PROMPT, SRC-003, SRC-004, SRC-005, SRC-006 and the attached KFM doctrine corpus: MapLibre remains a downstream renderer and interaction runtime. It is not the canonical truth store, source registry, policy engine, citation authority, review authority, publication authority, or AI authority. Tiles, COGs, PMTiles, MVT, style JSON, popups, drawers, exports, screenshots, scenes, and AI answers are downstream carriers, not sovereign truth.

CONFIRMED from this run: the prior v1.2 cumulative update master was detected and preserved as SRC-MASTER-PREV before regeneration. It contained 1802 unique ML-* idea IDs. This v1.3 cumulative update carries all prior IDs forward, updates continuity labels, and adds current-run discoveries from SRC-054 New Ideas 4-25-26.pdf and SRC-055 New Ideas 4-30-26.pdf.

NEW from SRC-054: tile schema design, temporal vector tile properties, Evidence Drawer click resolution, PMTiles/COG placement, CDN/Range caching requirements, 2D/3D ID alignment, human review ergonomics, offline/abstain behavior, and fail-closed layer gates. Key evidence: SRC-054 pp.31-43, pp.49-52, pp.313-320, pp.327-336.

NEW from SRC-055: PMTiles delta manifests, per-tile digests, base/delta archive strategy, mobile MapLibre verifier budgets, CORS/Range hosting checks, MapLibre/Cesium interop choices, HUC12-COMID manifest resolution into the Evidence Drawer, service-worker caching of PMTiles/style/glyphs/sprites, and browser-side signature verification harnesses. Key evidence: SRC-055 pp.29-43, pp.44-46, pp.80-87, pp.268-276.

PROPOSED synthesis: the next implementation-ready MapLibre slice remains schema/fixture/test first. Add TileArtifactManifest and source_digests checks, PMTiles/COG manifest plus signed-bundle fixtures, source-meta/sidecar validators, timeline filter fixtures, Evidence Drawer payload fixtures, Focus Mode composed-claim fixtures, habitat/protection redaction fixtures, and rollback/cache invalidation tests before public publication.

UNKNOWN / NEEDS VERIFICATION: actual repo topology, package manager, schema home, UI component home, MapLibre GL JS target version, PMTiles tooling, COG and PMTiles hosting headers, browser-side signature verification, mobile/native parity, MLT support status, plugin allowlist, sensitive geometry thresholds, source rights, policy tiers for ABSTAIN versus DENY, accessibility acceptance, and performance budgets.

Highest-risk anti-patterns: treating PMTiles, MVT, COG, or source metadata as proof; treating MapLibre, a layer toggle, or a trust badge as release authority; using style filters instead of real sensitive-geometry transformation; auto-publishing provisional detections before receipts, validation, catalog closure, and promotion; letting Focus Mode answer from rendered features alone instead of resolved EvidenceBundles.

Metric	Value
Previous version	v1.2 cumulative update
Current version	v1.3 cumulative update
Prior ideas carried forward	1802
Prior ideas expanded	1797
Prior ideas retained	5
New ideas	326
Total idea count	2128
Superseded ideas	0
Open verification items	259

3. Continuity and Delta Since Last Run

SRC-MASTER-PREV is the prior v1.2 Master MapLibre Components-Functions-Features.pdf. It is the cumulative memory baseline, not an authority above original evidence. No prior idea is deleted. Prior ideas are classified as EXPANDED where SRC-054/SRC-055 strengthen the category and RETAINED where this run added no material category evidence. No idea is marked SUPERSEDED in this run.

Delta item	Result
Previous version	v1.2 cumulative update
Current version	v1.3 cumulative update
New sources added	SRC-054 New Ideas 4-25-26.pdf; SRC-055 New Ideas 4-30-26.pdf
Prior ideas preserved	1802 unique ML-* IDs from SRC-MASTER-PREV
Expanded prior ideas	1797
Retained prior ideas	5
Merged ideas	0 - duplicates normalized into current-run evidence and cross-page synthesis without retiring older IDs
Superseded ideas	0 - no prior idea was invalidated by attached evidence
Conflicts carried forward	Schema/path uncertainty, package/version unknowns, MLT status, plugin allowlist, source rights, sensitive geometry thresholds, browser verification, CORS/Range hosting, performance budgets.
Version change	+0.1 ordinary cumulative update; required structure retained; no structural reorganization.

3A. Prior idea status comparison by category

Category	Prior count	New count	Current count	Current status	Source changing status	Note
A. Renderer Boundary and KFM Trust Law	72	3	75	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
B. MapLibre GL JS Web Shell	61	4	65	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
C. MapLibre Native / Mobile / Platform Parity	14	3	17	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
D. MapLibre RS / Experimental Renderers	5	0	5	RETAINED	SRC-MASTER-PR EV	No new direct current-run extraction; carried forward as prior baseline.
E. Style Specification, Sources, Layers, Expressions	63	3	66	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
F. Sprites, Glyphs, Fonts, Design Tokens	9	2	11	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
G. GeoJSON and Runtime Data	66	2	68	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
H. Vector Tiles: MVT and MLT	51	3	54	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	106	106	212	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
J. Martin and Server-Mediated Tile Serving	12	2	14	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
K. Raster, COG, DEM, Terrain, Hillshade	78	16	94	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
L. WMS / WMTS / External Map Services	25	2	27	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	112	34	146	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
N. Evidence Drawer Payloads and Click Resolution	85	29	114	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
O. Focus Mode and Governed AI Map Context	69	17	86	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.

Category	Prior count	New count	Current count	Current status	Source changing status	Note
P. Time-Aware Map Interaction and Timeline	78	16	94	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	83	5	88	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
R. Plugin, Wrapper, and Dependency Governance	39	3	42	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
S. Accessibility, UX, and Trust-Visible States	62	2	64	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
T. Performance, Caching, CDN, Range Requests, and Resource Timing	82	8	90	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
U. Testing, CI, Validation, Rollback, and Proof Objects	171	6	177	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
V. Exports, Screenshots, Reports, and Citation Preservation	48	2	50	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
W. 3D / Cesium / Deck.gl / Overlay Interoperability	59	42	101	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
X. Anti-Patterns and Failure Modes	100	4	104	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
Y. Implementation Backlog and PR Plan	110	4	114	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.
Z. Open Questions and Verification Backlog	142	8	150	EXPANDED	SRC-054/SRC-055	Current-run evidence adds page-level extraction or cross-page synthesis.

4. Source Ledger

The source ledger is a control surface, not a bibliography. It records what each source can support and what it cannot prove. External facts embedded in PDFs remain source-supported only and NEED VERIFICATION before operational use if version-, rights-, endpoint-, policy-, or security-sensitive.

Automated scan item	Result			
SRC-054 term scan	337 pages; 131 pages with MapLibre/tile/governance hits; top terms: MapLibre=70, PMTiles=55, 3D=40, Focus Mode=36, Evidence Drawer=35, Cesium=31, zoom=29, style=25, COG=22, GeoJSON=16, vector tile=12, PromotionDecision=11			
SRC-055 term scan	289 pages; 151 pages with MapLibre/tile/governance hits; top terms: PMTiles=325, MapLibre=200, Cesium=189, camera=111, zoom=101, style=73, pitch=50, globe=34, maplibre-gl=32, MVT=29, maplibregl=28, CDN=27			
Repo evidence	No mounted KFM repository, tests, workflows, dashboards, logs, release manifests, or runtime traces were available in this generation context; implementation depth remains UNKNOWN.			
source_id	file / document title	role / status	MapLibre / tile relevance	evidence strength / limitations
SRC-RUN-PR OMPT	Current user prompt	Run directive CONFIRMED request	Very high Controls PDF-only output, cumulative memory rule, required categories, truth labels, and extraction scope.	current conversation pages or pointer. Directive evidence, not implementation proof.
SRC-MASTE R-PREV	Master MapLibre Components-Functions-Features.pdf (prior v1.2)	Prior cumulative master CONFIRMED attached PDF	Very high Continuity baseline with 1802 prior ML-* ideas and v1.2 source ledger.	240 pages or pointer. Continuity memory; original source evidence can outrank it.
SRC-003	KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf	MapLibre operating doctrine CONFIRMED attached PDF	Very high MapLibre renderer boundary, tile/style strategy, Evidence Drawer, Focus Mode, governed UI.	22 pages or pointer. Does not prove mounted repo behavior.
SRC-004	Directory Rules.pdf	Directory governance doctrine CONFIRMED attached PDF	High Placement law for responsibility roots, lifecycle, schema-home default, compatibility roots.	22 pages or pointer. Paths remain PROPOSED without repo evidence.
SRC-005	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf	Pipeline manual CONFIRMED attached PDF	High Lifecycle, query-save-validate-compile-review-promote-recompile loop, governed APIs and MapLibre UI.	30 pages or pointer. No implementation proof.
SRC-006	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan_v1_1.pdf	Greenfield plan CONFIRMED attached PDF	High Trust spine, public-safe defaults, pre-RAW event family, map architecture.	28 pages or pointer. Greenfield assumption; not repo proof.
SRC-007	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf	Whole UI / AI plan CONFIRMED attached PDF	High MapLibre shell, Evidence Drawer, Focus Mode, review console, governed API boundaries.	23 pages or pointer. No mounted repo proof.
SRC-008	KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf	Governed AI source-ledger report CONFIRMED attached PDF	High Provider-neutral AI boundary, finite outcomes, citation validation, AIReceipt.	36 pages or pointer. No runtime proof.
SRC-009	Ollama & Ubuntu Information.pdf	Ollama/local AI runtime guide CONFIRMED attached PDF	Medium AI runtime behind governed API; no direct client/model path; evidence-bounded behavior.	66 pages or pointer. Version-sensitive runtime facts not revalidated.
SRC-010	kfm_encyclopedia.pdf	Domain/capability encyclopedia CONFIRMED attached PDF	Medium Cross-domain map features, evidence, tiles as carriers, governance posture.	82 pages or pointer. Not current implementation proof.
SRC-011	# Kansas Frontier Matrix Implementation Reference.pdf	Implementation reference CONFIRMED attached PDF	Medium Prior connector/repo summary, MapLibre/PMTiles stack signals, schema-home ambiguity.	20 pages or pointer. Prior connector claims not reverified.
SRC-012	a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf	GIS/cartography reference CONFIRMED attached PDF	Background Cartographic representation and map-as-communication context.	321 pages or pointer. Not KFM implementation evidence.
SRC-013	Advanced-SQL-Concepts.pdf	SQL reference CONFIRMED attached PDF	Background Analytics/query background for temporal, catalog, or validation logic.	112 pages or pointer. Not MapLibre-specific.
SRC-014	AI_Concepts_Using_Python.pdf	AI/Python reference CONFIRMED attached PDF	Background General AI/data workflow context.	435 pages or pointer. Not KFM implementation evidence.

source_id	file / document title	role / status	MapLibre / tile relevance	evidence strength / limitations
SRC-015	Archaeological 3D GIS.pdf	3D GIS reference CONFIRMED attached PDF	Medium 3D GIS, models, visibility, volumes, archaeological 3D context.	177 pages or pointer. Not MapLibre implementation proof.
SRC-016	CICD_with_Docker_Kubernetes_Semaphore.pdf	CI/CD reference CONFIRMED attached PDF	Background CI/CD, rollback, deployment and testing patterns.	92 pages or pointer. Not KFM-specific.
SRC-017	Designing Great Web APIs.pdf	Web API reference CONFIRMED attached PDF	Background API contracts and client/developer experience.	45 pages or pointer. Not KFM-specific.
SRC-018	developing-time-oriented-database-applications-in-sql.pdf	Temporal database reference CONFIRMED attached PDF	Background Valid time, transaction time, bitemporal concepts for time-aware layers.	528 pages or pointer. Not MapLibre-specific.
SRC-019	Domain-Driven Design Reference.pdf	DDD reference CONFIRMED attached PDF	Background Bounded contexts, ubiquitous language, repositories, services.	59 pages or pointer. Not KFM implementation proof.
SRC-020	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf	Environmental GIS reference CONFIRMED attached PDF	Medium GIS layers, hazards, climate, Mars/earth science map contexts.	141 pages or pointer. ArcGIS-oriented, not MapLibre proof.
SRC-021	fullstack-react-with-typescript.pdf	React/TypeScript reference CONFIRMED attached PDF	Medium Frontend/testing patterns that may support MapLibre UI.	321 pages or pointer. Not KFM-specific.
SRC-022	GIS Succinctly.pdf	GIS reference CONFIRMED attached PDF	Medium Spatial databases, geometry types, GIS application basics.	110 pages or pointer. Not KFM implementation proof.
SRC-023	GIS in Sustainable Urban Planning and Management.pdf	Urban GIS reference CONFIRMED attached PDF	Medium Planning support, collaborative GIS, spatial indicators and risk contexts.	365 pages or pointer. Not MapLibre-specific.
SRC-024	instant-web-scraping-with-java.pdf	Scraping/reference source CONFIRMED attached PDF	Background Source acquisition concepts and scraper caution.	72 pages or pointer. Not KFM implementation proof.
SRC-025	KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Components pass / idea atlas CONFIRMED attached PDF	High Run receipts, deterministic identity, promotion gates, source intake patterns.	125 pages or pointer. Lineage, not repo proof.
SRC-026	KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Components pass / idea atlas CONFIRMED attached PDF	High Synthesizes New Ideas 4-25 and 4-30; strong continuity pressure for this run.	123 pages or pointer. Secondary synthesis; originals still scanned as SRC-054/SRC-055.
SRC-027	KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf	Components pass / idea atlas CONFIRMED attached PDF	High Pre-RAW event family, watchers, catalog integrity, environmental tiles, AI risk alignment.	88 pages or pointer. Lineage, not repo proof.
SRC-028	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf	Agriculture domain report CONFIRMED attached PDF	Medium MapLibre/Evidence Drawer/Focus Mode domain-layer posture.	48 pages or pointer. No repo proof.
SRC-029	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	Archaeology domain report CONFIRMED attached PDF	Medium Exact site location denial, 3D/remote-sensing candidate caution, public-safe map outputs.	51 pages or pointer. No repo proof.
SRC-030	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	Atmosphere/air domain report CONFIRMED attached PDF	Medium AQI, smoke, EO layers, knowledge-character labels, MapLibre UI posture.	52 pages or pointer. No repo proof.
SRC-031	KFM_Fauna_Architecture_PDF_Only_Report.pdf	Fauna domain report CONFIRMED attached PDF	Medium Occurrence geoprivacy, public-safe derived products, MapLibre layers.	37 pages or pointer. No repo proof.
SRC-032	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	Flora domain report CONFIRMED attached PDF	Medium Rare plant controls, specimens, occurrence redaction, Evidence Drawer.	36 pages or pointer. No repo proof.
SRC-033	KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf	Geology/natural resources domain report CONFIRMED attached PDF	Medium Geology/resource layers, public-safe geometry, Evidence Drawer/Focus Mode.	42 pages or pointer. No repo proof.

source_id	file / document title	role / status	MapLibre / tile relevance	evidence strength / limitations
SRC-034	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	Habitat/fauna thin slice report CONFIRMED attached PDF	Medium Habitat assignment proof slice, MapLibre layer, Evidence Drawer.	20 pages or pointer. No repo proof.
SRC-035	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	Habitat domain report CONFIRMED attached PDF	Medium Habitat patches, suitability, connectivity, public-safe products.	28 pages or pointer. No repo proof.
SRC-036	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	Hazards domain report CONFIRMED attached PDF	Medium Hazard layers, operational-context caution, no emergency alerting, Evidence Drawer.	26 pages or pointer. No repo proof.
SRC-037	KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf	Hydrology domain report CONFIRMED attached PDF	Medium Hydrology proof lane, HUC12, NHDPlus, USGS Water Data, NFHL, MapLibre layers.	43 pages or pointer. No repo proof.
SRC-038	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	People/genealogy/DNA/land report CONFIRMED attached PDF	Medium Living-person/DNA restrictions, land evidence, temporal assertions, map caution.	30 pages or pointer. No repo proof.
SRC-039	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	Roads/rail/trade routes report CONFIRMED attached PDF	Medium Transport network layers, corridors, graph projections, MapLibre/Evidence Drawer.	35 pages or pointer. No repo proof.
SRC-040	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	Settlements/infrastructure report CONFIRMED attached PDF	Medium Settlements, infrastructure, sensitive assets, layer manifests, governed APIs.	43 pages or pointer. No repo proof.
SRC-041	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	Soil domain report CONFIRMED attached PDF	Medium Soil maps, SSURGO/gSSURGO, PMTiles stub, Evidence Drawer.	25 pages or pointer. No repo proof.
SRC-054	New Ideas 4-25-26.pdf	Current-run new ideas source: MapLibre/tile/evidence/governance extraction CONFIRMED attached PDF	Very high 131 relevant hit pages by current scan; top terms: MapLibre=70, PMTiles=55, 3D=40, Focus Mode=36, Evidence Drawer=35, Cesium=31, zoom=29, style=25.	337 pages or pointer. Source text is design/source evidence; operational claims remain NEEDS VERIFICATION.
SRC-055	New Ideas 4-30-26.pdf	Current-run new ideas source: MapLibre/tile/evidence/governance extraction CONFIRMED attached PDF	Very high 151 relevant hit pages by current scan; top terms: PMTiles=325, MapLibre=200, Cesium=189, camera=111, zoom=101, style=73, pitch=50, globe=34.	289 pages or pointer. Source text is design/source evidence; operational claims remain NEEDS VERIFICATION.

5. Master Idea Index

This searchable index preserves every prior ML-* idea ID found in SRC-MASTER-PREV and adds current-run new IDs. Count: 2128 ideas = 1802 prior + 326 new. Status labels in this section are line-level continuity labels, not claims of implemented repository behavior.

5.A. A. Renderer Boundary and KFM Trust Law idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-001	Renderer is downstream of trust	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-002	Tiles and styles are downstream carriers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-003	Public-client no-RAW rule	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-004	Cite-or-abstain map posture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-005	Publication is a governed state transition	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-006	Map features are selection candidates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-007	Story maps remain evidence-bound	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-008	Run-receipted geodata artifacts outrank	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-009	Material-change gates replace cron-only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-010	Story Nodes fail closed when	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-011	Run-receipted geodata artifacts outrank	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-012	Catalog closure is a precondition for map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-013	Provisional feeds must remain visibly	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-014	PMTiles deltas are governed release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-015	GeoParquet, COG, and PMTiles outputs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-016	Watchers can trigger map artifact emission	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-017	Tile-level embeddings are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-018	Pre-render verification belongs inside the	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-019	Source-edge map outputs remain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-020	Map delivery requires evidence at	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-021	Client trust widgets must not mint	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-022	External public GIS services are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-023	Source-level metadata is the UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-024	Watcher-Planner-Executor cannot publish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-025	Agent outputs must be PR-ready artifacts with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-026	Deterministic idempotency keys for map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-027	Kill switch for Planner and Executor	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-028	No direct write path to production map stores	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-029	Run receipts are the map-artifact evidence atom	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-030	Promotion remains a governed state transition	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-031	Corrected air-quality surfaces are not regulatory	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-032	Catalog closure precedes UI exposure	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-033	Map readiness is allowed to deny emission	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-034	MapServer fetches are admissible source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-035	Geometry fingerprints outrank metadata churn	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-036	Webhook events are signals, not authoritative	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-037	Client integrity badges are governed trust	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-038	Render engines are untrusted compute	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-039	Runtime security headers are boundaries, not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-040	Catalog closure precedes map release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-041	Receipts, proofs, and policy decisions stay	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-042	Time-series promotion requires	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-043	Receipts and proofs remain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-044	Signed rollback is part of release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-045	Promotion decisions must	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-046	Tiles are auditable artifacts, not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-047	Promotion decision and runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-048	Individually valid sources are not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-049	MapLibre source meta is a hint, not authority	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-050	Digest linkage upgrades tiles to evidence surfaces	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-051	Closed epochs are immutable release artifacts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-052	Promotion-gated PMTiles are downstream carriers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-053	Map artifact lifecycle follows watcher-to-catalog path	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-054	Historical map layers remain precision-labeled claims	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-055	Soils PMTiles and COGs remain downstream carriers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-056	Manifest comparison outranks renderer freshness assumptions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-057	Habitat-fauna joins are derived artifacts rather than canonical truth	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-058	MapLibre may render only after EvidenceBundle and catalog closure	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-059	PMTiles/COG trust badges display verification but do not mint truth	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-060	Hydrology crossing decisions must travel with events and tiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-061	Focus Mode building blocks enforce evidence sufficiency before synthesis	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-062	Public map outputs inherit KFM no-RAW no-WORK no-QUARANTINE law	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-063	Biodiversity matrix tiles are downstream carriers, not source truth	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-064	LANDFIRE-derived COGs/MVTs need signed derivatives and provenance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-065	MapLibre and Cesium consume released artifacts only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-066	Precomputed zoom layers do not replace canonical ecoregion/PLSS/WBD sources	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-067	HLS change candidates are not burn truth until event-supported and promoted	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-068	Run receipts are separate from candidates, proofs, catalogs, and publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-069	No-change and insufficient-evidence outputs must be retained	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-070	Detection maps remain provisional until promotion gate survival	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-071	Evidence Drawer must link rendered artifacts back to sources, licenses, and receipts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-A-072	Public habitat/protection overlays require sensitivity labels before rendering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under A. Renderer Boundary and KFM Trust Law. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-A-073	Renderer must never become evidence authority	NEW	SRC-054 pp.42-43; SRC-055 pp.271-276	Current sources repeat that tiles and PMTiles are derived delivery artifacts; the authoritative claim must resolve through governed APIs and EvidenceBundles. Impl: Keep renderer, tile, and popup behavior downstream of release manifests and evidence resolution. Validation: Test that public clients cannot load RAW, WORK, candidate, or unverifiable artifacts. Evidence: SRC-054 pp.42-43; SRC-055 pp.271-276; cross-page normalized discovery.
ML-A-074	Published artifacts beat operational coupling	NEW	SRC-055 pp.268-276	PMTiles is framed as useful because it encourages distributable release artifacts instead of coupling public clients to internal infrastructure. Impl: Publish immutable artifacts with ReleaseManifest, EvidenceBundle, DecisionEnvelope, and spec_hash pairings. Validation: Verify artifact digests, release state, and rollback targets before rendering. Evidence: SRC-055 pp.268-276; cross-page normalized discovery.
ML-A-075	No API means no claims in offline mode	NEW	SRC-054 p.51	The offline strategy states that when no API is available the UI should not make new claims and should rely only on cached evidence. Impl: Make offline mode visibly evidence-bounded. Validation: Add an offline fixture where layer rendering works but claims abstain without cached evidence. Evidence: SRC-054 p.51; cross-page normalized discovery.

5.B. B. MapLibre GL JS Web Shell idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-001	MapLibre GL JS is the first 2D shell default	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-002	Persistent governed shell	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-003	MapLibre adapter boundary	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-004	MapContextEnvelope captures camera	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-005	Layer catalog is governed surface	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-006	Renderer receipts for important	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-007	StoryNode UI shell uses governed time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-008	Receipt Drawer plus Uncertainty Toggle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-009	MapLibre and Cesium friendly drawer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-010	Feature or Story Node focus opens staged	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-011	MapLibre must defer addSource until	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-012	Map shell consumes STAC, TileJSON,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-013	Digest-cached pre-render verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-014	Map and timeline synchronized selection	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-015	MapLibre layer registration needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-016	EvidenceTooltipController as	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-017	Trust-state expressions for steward	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-018	Public/steward view mode source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-019	Focus filter uses governed trust	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-020	Map shell selection opens receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-021	PMTiles protocol adapter belongs inside	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-022	Map source and layer creation must be	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-023	Watcher-signal chips become trust-visible UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-024	Feature clicks are candidates for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-025	Map shell must support steward and public	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-026	Focus Mode change-surfacing is a UI adjunct,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-027	One PMTiles source can power time filtering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-028	MapLibre PMTiles protocol setup is part of	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-029	setFilter becomes the cheap time-switch	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-030	Raster crossfade can use opacity animation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-031	MapLibre integrity control can implement	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-032	sourcedata hooks can trigger manifest checks	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-033	Map controls can expose manifest and asset	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-034	Trust control output uses finite KFM outcomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-035	MapLibre custom layer can host	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-036	MapLibre epoch switching	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-037	MapLibre fetch/paint should be B. B. MapLibre GL JS Web Shell	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-038	Map triggerRepaint becomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-039	MapLibre trust surfaces belong B. B. MapLibre GL JS Web Shell	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-040	Epoch picker reads style sources	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-041	Source-swap mode for deterministic releases	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-042	Feature-filter mode for streaming partial data	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-043	MapLibre time slider uses numeric feature time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-044	MapLibre addSource/addLayer PMTiles pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-045	Worker-prepared bucket swap messages	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-046	queryRenderedFeatures inherits temporal filter state	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-047	MapLibre unknown-key compatibility needs verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-048	MapLibre setFilter should be applied atomically after worker computation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-049	Time-slider interactions should debounce slider scrubs before changing filters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-050	Stable feature IDs enable faster ID-list filtering in MapLibre layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-051	PMTiles protocol registration is a shell integration point not a truth gate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-052	MapLibre trust badge reads signed manifests before layer confidence display	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-053	MVT decode and time checks belong off the main UI thread	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-054	Overzoom pop-in can justify server-baked epoch sources for selected zooms	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-055	MapLibre GL JS should load released PMTiles directly	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-056	MapLibre color ramps should be tied to released index or facies attributes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-057	MapLibre time slider can filter feature start/end attributes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-058	MapLibre web shell should show threshold-passing vectors only after promotion	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-059	MapLibre public stories can use small Geo/TopoJSON packs plus PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-B-060	MapLibre layer trust state should be driven by manifest and receipt fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-061	Do not start with UI before schema, validator, STAC, and PMTiles proof slice	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under B. MapLibre GL JS Web Shell. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-B-062	MapLibre PMTiles protocol wiring needs verification first	NEW	SRC-055 pp.36-38	The MapLibre sketch installs PMTiles protocol, mounts base and delta archives, and verifies the time index before rendering. Impl: Add a verified PMTiles source adapter behind the governed map shell. Validation: Fail load if index verification, archive digest, or policy check fails. Evidence: SRC-055 pp.36-38; cross-page normalized discovery.
ML-B-063	MapLibre source-layer names are contract-bearing	NEW	SRC-055 pp.40-43	The MapLibre hookup warns that the source-layer must match vector layer metadata. Impl: Bind source-layer names to LayerManifest and TileArtifactManifest. Validation: Validate style JSON against vector layer metadata before release. Evidence: SRC-055 pp.40-43; cross-page normalized discovery.
ML-B-064	MapLibre overlay can synchronize with Cesium only as controlled shell behavior	NEW	SRC-055 pp.40-43; SRC-054 pp.160-168	The current corpus includes overlay and camera synchronization sketches for MapLibre and Cesium. Impl: Keep synchronization code in UI/runtime surfaces, not canonical evidence objects. Validation: Test camera, pitch, bearing, and zoom conversion with bounds and performance budgets. Evidence: SRC-055 pp.40-43; SRC-054 pp.160-168; cross-page normalized discovery.
ML-B-065	MapLibre errors are trust-visible events	NEW	SRC-055 pp.80-86	The mobile harness makes verification errors throw and fail the rendering run. Impl: MapLibre error handling should expose DENY/ERROR states, not silent fallback. Validation: Add map.on(error) tests that assert fail-closed UI copy. Evidence: SRC-055 pp.80-86; cross-page normalized discovery.

5.C. C. MapLibre Native / Mobile / Platform Parity idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-C-001	Native/mobile support is conditional	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-002	Use restricted style profile or variants for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-003	Offline/mobile artifacts require manifests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-C-004	Mobile parity includes negative outcomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-005	Offline packs require manifest and digest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-006	Mobile parity includes drawer confidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-007	Offline and edge PMTiles overlays require	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-008	Mobile and edge privacy overlays publish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-009	On-device field capture should emit	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-010	One-tap receipt drawer supports	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-011	Mobile/native parity remains version- and platform-sensitive for vector and 3D timelines	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-012	Signed COG/MVT derivatives are intended for web and mobile GIS apps	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-C-013	Mobile or low-overhead clients benefit from single-file Range-capable PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-014	Native parity remains conditional on matching drawer, citation, and release behavior	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under C. MapLibre Native / Mobile / Platform Parity. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-C-015	Mobile clients need lower verifier and request budgets	NEW	SRC-055 pp.34-39; SRC-055 pp.80-86	The mobile playbook and harness use one or two workers, capped parallel fetches, and memory ceilings. Impl: Publish mobile acceptance budgets separately from desktop budgets. Validation: Run throttled Puppeteer or equivalent tests for verify-decode-render latency. Evidence: SRC-055 pp.34-39; SRC-055 pp.80-86; cross-page normalized discovery.
ML-C-016	PWA offline bundles need evidence-aware cache groups	NEW	SRC-055 pp.268-276	The PWA stack pairs PMTiles, style JSON, glyphs, sprites, icons, service workers, and offline cache. Impl: Cache map artifacts and trust artifacts together under one release identity. Validation: Test first-run online, partial download, offline boot, stale release, and policy mismatch. Evidence: SRC-055 pp.268-276; cross-page normalized discovery.
ML-C-017	Native/mobile parity remains unverified until drawer and focus behavior match	NEW	SRC-055 pp.268-276; SRC-054 p.51	Mobile/offline delivery is useful only when Evidence Drawer and claim behavior remain governed. Impl: Do not approve native or PWA parity by tile rendering alone. Validation: Acceptance must include Evidence Drawer resolution, abstain paths, and policy mismatch. Evidence: SRC-055 pp.268-276; SRC-054 p.51; cross-page normalized discovery.

5.D. D. MapLibre RS / Experimental Renderers idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-D-001	MapLibre RS remains experimental for	RETAINED	SRC-MASTER-PREV	Prior cumulative idea retained from v1.2 cumulative update: MapLibre RS remains experimental for. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-D-002	Renderer substitution must preserve	RETAINED	SRC-MASTER-PREV	Prior cumulative idea retained from v1.2 cumulative update: Renderer substitution must preserve. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-D-003	WebGPU/Vulkan/globe work needs ADR	RETAINED	SRC-MASTER-PREV	Prior cumulative idea retained from v1.2 cumulative update: WebGPU/Vulkan/globe work needs ADR. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-D-004	Experimental renderers must preserve	RETAINED	SRC-MASTER-PREV	Prior cumulative idea retained from v1.2 cumulative update: Experimental renderers must preserve. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-D-005	Alternative renderers inherit the same	RETAINED	SRC-MASTER-PREV	Prior cumulative idea retained from v1.2 cumulative update: Alternative renderers inherit the same. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

5.E. E. Style Specification, Sources, Layers, Expressions idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-001	StyleManifest binds style JSON to Style	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-002	Sources and layers are validated	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-003	Expressions and filters are not policy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-004	promoteId and feature-state support stable	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-005	Runtime addSource/addLayer only from	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-006	Maputnik is a tool, not release authority E.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-007	Style filters may reflect confidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-008	Source-layer definitions should carry E.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-009	Style edits must be linked to catalog and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-010	TileJSON is a contract input for spec_hash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-011	Runtime source and layer adds are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-012	Source-layer metadata must expose	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-013	Source-level metadata attaches to	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-014	Trust styling via	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-015	Mode and focus filters precompute	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-016	Confidence and temporal	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-017	source-layer name is part of the contract	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-018	Compact attributes reduce leakage and tile	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-019	Trust status should be style-visible but not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-020	Warning and hold flags must survive style	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-021	Layer IDs remain stable across rebuilds	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-022	Style filters are never geoprivacy controls	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-023	Map attributes need provenance columns	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-024	Catalog properties can feed labels and badges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-025	source-layer names are release contract fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-026	Time belongs in attributes, not layer names	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-027	Vector choropleth transitions may use	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-028	Keep live source counts low	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-029	Asset roles drive style/link behavior	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-030	Epoch bundles require E. Style Specification, Sources, Layers, NEW; CONFIRMED/ SRC-046 A tim...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-031	Throttle style changes to E. Style Specification, Sources, Layers, NEW; CONFIRMED/ SRC-046 S...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-032	Single-pass shader blending E. Style Specification, Sources, Layers, NEW; CONFIRMED/ SRC-046...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-033	Layer count minimization is a E. Style Specification, Sources, Layers, NEW; CONFIRMED/ SRC-0...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-034	Proof state should influence E. Style Specification, Sources, Layers, NEW; CONFIRMED/ SRC-04...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-035	Federation outcome can choose	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-036	Minimal source.meta fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-037	KFM source.meta adds proof_ref	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-038	KFM source.meta adds manifest_ref	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-039	Epoch may be range or short ID	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-040	Strict reversed-epoch denial	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-041	License shape is policy input	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-042	Digest must match manifest source_digests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-043	Missing source meta can abstain or deny	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-044	Feature time filters use all expressions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-045	Source-layer names must remain manifest-bound	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-046	Style fixtures model valid/invalid source meta	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-047	Style metadata cannot replace TileJSON/PMTiles metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-048	Temporal start_ts and end_ts properties support style filter expressions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-049	Source metadata sidecars should expose epoch license digest and release refs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-050	Feature properties should retain source_uri and digest for clicked evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-051	MapLibre filters need explicit numeric epoch fields for time comparisons	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-052	Soil PMTiles layer fields require mukey areasymbol hydgrp and drainage class	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-053	Style updates must avoid repeated setFilter thrash during slider movement	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-054	Unknown style/source metadata should produce visible ABSTAIN or DENY states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-055	Layer definitions should separate observed modeled regulatory and derived roles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-056	Vegetation facies layers need minimal style palettes tied to attributes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-057	Low zoom should show broad ecoregions while high zoom reveals detailed Level IV and PLSS lines	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-058	HLS vector features need start, end, confidence, and source_event attributes for filters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-059	Color ramps and output units must be documented in sidecar metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-060	Change-feature tiles should carry confidence, vote_count, onset, and coherence fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-061	Habitat protection PMTiles should trim attributes to explicit include fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-062	CRS should stay 5070/4326 for provenance and reproject to 3857 only for tiling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-E-063	Style sources should separate raw/source catalogs from derived catalogs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under E. Style Specification, Sources, Layers, Expressions. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-E-064	Tile properties should stay flat, typed, and filterable	NEW	SRC-054 pp.36-37	The current corpus warns that vector tile schema mistakes cascade and recommends flat typed properties such as time fields, confidence, source_id, and evidence_ref. Impl: Keep heavy attributes out of tiles and in governed lookups. Validation: Validate tile metadata, field types, source_id, evidence_ref, and temporal fields. Evidence: SRC-054 pp.36-37; cross-page normalized discovery.
ML-E-065	Style expressions should consume simple worker-set flags	NEW	SRC-054 p.37	The corpus recommends a worker-set visible flag and minimal style expressions to avoid render-loop recomputation. Impl: Use workers for temporal pruning and simple style expressions for final display. Validation: Performance tests must distinguish worker cost from style expression cost. Evidence: SRC-054 p.37; cross-page normalized discovery.
ML-E-066	Style versioning must align with layer release versions	NEW	SRC-054 pp.49-50; SRC-055 p.274	The corpus requires layer_id, version, spec_hash, release_time, supersedes, and rollback state. Impl: Include style JSON as a release-scoped artifact. Validation: Check that style digests match ReleaseManifest and cache invalidation records. Evidence: SRC-054 pp.49-50; SRC-055 p.274; cross-page normalized discovery.

5.F. F. Sprites, Glyphs, Fonts, Design Tokens idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-F-001	Sprites and glyphs are release artifacts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-002	Font and glyph licensing must be visible	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-003	Design tokens unify style variants	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-004	Sprite glyph and style assets need license	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-005	Status chips and symbols must encode	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-F-006	Policy badges require text, icon,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-007	Sprites, glyphs, fonts are runtime-loaded assets	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-008	Font/style hosting must avoid unsafe defaults	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-009	Design-token assets need the same release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under F. Sprites, Glyphs, Fonts, Design Tokens. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-F-010	Glyphs and sprites are part of the release bundle	NEW	SRC-055 pp.273-274	Service-worker recommendations cache PMTiles, style.json, glyphs, sprites, and icons together. Impl: Record sprite/glyph versions and hashes with style and tile artifacts. Validation: Offline boot tests must include missing glyph and missing sprite cases. Evidence: SRC-055 pp.273-274; cross-page normalized discovery.
ML-F-011	Fonts and icons need policy-visible fallback	NEW	SRC-055 p.274	The offline cache behavior includes fail-closed graceful handling for missing support assets. Impl: Provide fallback symbols for denied, stale, abstained, and no-data states. Validation: Visual regression should test fallback glyph/sprite behavior. Evidence: SRC-055 p.274; cross-page normalized discovery.

5.G. G. GeoJSON and Runtime Data idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-001	GeoJSON is good for small dynamic	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-002	Dense GeoJSON is a production	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-003	GeoJSON source performance is	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-004	GeoJSON/NDJSON can be tile build input	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-005	Runtime data updates need receipts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-006	GeoJSON-to-tile fixture proves runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-007	WZDx/KanDrive events use GeoJSON	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-008	Event merge keeps provenance arrays	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-009	GeoParquet is the analytical precursor to	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-010	Feed-card JSONL and GeoJSON overlays	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-011	WZDx and KanDrive GeoJSON events are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-012	Runtime provisional points must expose	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-013	GeoJSON build inputs must preserve	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-014	Core layer attributes span GeoJSON,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-015	JSON-LD sidecars preserve semantics	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-016	GeoParquet searchable columns include	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-017	GeoParquet to GeoJSONSeq to PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-018	geometry_hash separates topology	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-019	Runtime overlays from event feeds need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-020	GeoJSON feature shape carries	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-021	Historical route GeoJSON requires	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-022	Genealogy overlay JSON strips	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-023	Station and monitor features need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-024	WZDx RoadEvent GeoJSON becomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-025	ArcGIS FeatureService f=geojson is an ingest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-026	NDVI alert outputs use GeoJSON plus COG	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-027	Soil moisture sidecars may be CSV or	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-028	GNIS and Kansas Geoportal supply GIS-ready	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-029	Hydro REST streams support GeoJSON and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-030	KDOT infrastructure layers are queryable as	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-031	Genealogy timelines can become map-ready	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-032	GeoJSON features remain downstream carriers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-033	WBD f=geojson endpoints are useful source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-034	Viewport bounding boxes can scope Focus	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-035	Single-feature GeoJSON fetch supports	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-036	ArcGIS OpenData feature layers can carry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-037	KGS geology GeoJSON/FGDB layers become	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-038	PBF can be an ingestion optimization, not proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-039	Large GeoJSON should stay off the main thread	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-040	GeoJSON fixtures remain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-041	Runtime receipts should wrap	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-042	Trust federation consumes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-043	People-place-event GeoJSON emission	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-044	Geometry-ready records serve MapLibre and Cesium	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-045	Historical post offices canonical Feature shape	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-046	Santa Fe Trail canonical LineString shape	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-047	Precision tagging is governance-critical	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-048	Event GeoJSON belongs to STAC-like assets	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-049	GeoJSON FeatureCollection remains a candidate artifact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-050	Runtime streamflow events should emit finite ANSWER or ABSTAIN JSON objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-051	Fauna occurrence payloads need public_safe_geometry and precision fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-052	Habitat cell and derived join records should remain separate runtime objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-053	GeoJSON is suitable for controlled fixtures but not bulk production default	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-054	Crosswalk decision JSON should preserve best_comid alternatives and reason	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-055	Occurrence evidence objects require source role rights sensitivity and provenance blocks	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-056	Runtime proof fixtures should contain answer abstain deny and error objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-057	Event content hashes suppress non-material repeated emissions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-058	Species-by-county matrices become map-ready runtime backbones only with provenance sidecars	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-059	GeoJSON/TopoJSON story packs can complement PMTiles full-state exploration	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-060	Topology-cleaned GeoJSON should precede web simplification and PMTiles builds	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-061	Runtime hazard responses should use finite ANSWER/ABSTAIN/DENY/ERROR outcomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-062	ABSTAIN is required for burn questions lacking event support even when vegetation changed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-063	Composed claim runtime data must fail closed when required evidence refs are unresolved	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-064	Evidence drawer runtime examples must preserve unresolved evidence status	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-065	Joinable habitat/protection attributes should keep source_id, license, HUC, sensitivity, and spec_hash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-G-066	Observation point summaries require effort covariates before runtime maps become defensible	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under G. GeoJSON and Runtime Data. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-G-067	GeoJSON is useful for synthetic fixtures and small overlays	NEW	SRC-055 pp.32,88-89	The PMTiles examples convert projected source data to temporary GeoJSON and synthetic grids before tiling. Impl: Keep GeoJSON in work/fixture contexts or small runtime overlays, not large production defaults. Validation: Add size, CRS, and topology checks before using GeoJSON in map runtime. Evidence: SRC-055 pp.32,88-89; cross-page normalized discovery.
ML-G-068	Runtime data must carry lineage fields before catalog promotion	NEW	SRC-055 p.240	The minimal pipeline sketch adds lineage fields such as source_id and source_updated_at before validation and catalog. Impl: Require runtime payloads to carry source identity and temporal support. Validation: Reject payloads missing source_id, temporal fields, or evidence references. Evidence: SRC-055 p.240; cross-page normalized discovery.

5.H. H. Vector Tiles: MVT and MLT idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-H-001	MVT remains production default until MLT	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-002	Deterministic MVT encoding	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-003	Source-layer names are contract-bearing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-004	MLT is a pilot/bench path	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-005	Vector tiles need per-tile hash index	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-006	Vector tiles are not proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-007	source_ref IDs belong in feature/tile	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-H-008	PMTiles can deliver vector tiles or derived	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-009	Hazard and event overlays can derive	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-010	Point-to-tile or point-to-hex redaction is a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-011	MVT outputs need output digests in	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-012	Dedicated tile builders need deterministic	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-013	Vector tile layer definitions are release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-014	Vector tile production needs pinned	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-015	MVT and PMTiles outputs need TileJSON	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-016	Changed TileID lists enable deterministic	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-H-017	Content-addressed tiles support dedupe	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-018	Vector tiles only after proof closure	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-019	MVT can be derived from GeoParquet for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-020	PBF streams are ingest carriers, not public truth	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-021	Vector-tile parity requires shared source artifact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-022	MVT source-layer names need manifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-023	MLT remains a verification backlog item	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-024	MapLibre is the easier 2D vector-tile renderer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-025	Time-sliced vector tiles should carry compact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-H-026	Vector tile release can be one archive with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-027	Vector PMTiles per epoch	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-028	Vector and raster bundles need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-029	MLT remains unpromoted until H. H. Vector Tiles: MVT and MLT	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-030	Historical post office vector tile strategy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-031	Santa Fe Trail vector tile generation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-032	Vector tile outputs need precision gates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-033	Temporal service networks use vector tiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-034	MVT/PMTiles conversion is release-significant	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-H-035	Vector tiles remain subordinate to canonical geometry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-036	PMTiles vector archives carry standard MVT inner tiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-037	MVT payload budgets should be enforced at interactive zooms	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-038	Worker-side MVT prefiltering supports time-aware layer interaction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-039	Vector tile feature IDs should be stable for filter-list strategies	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-040	Soil map unit polygons are strong PMTiles vector tile candidates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-041	Historical and hydrology events need source-role attributes inside tiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-042	MVT generation requires per-feature provenance-retention checks	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-043	MLT remains non-default until KFM toolchain validation is complete	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-H-044	Official raster and vector extracts can be converted to MVT only after governed processing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-045	Vectorized HLS change zones should be published as MVT after promotion	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-046	Change-feature MVT metadata should record generalization tolerances and confidence class	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-047	Vector tile service is an alternative to PMTiles when server mediation is needed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-048	MVT is suitable for snappy panning and zooming of vectorized change features	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-049	MLT remains mentioned as a stack candidate but still needs KFM validation before default use	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-050	Habitat/protection PMTiles carry vector-tile exploration for full-state overlays	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-051	Precompute dominant area keys before tiling to avoid heavy joins at render time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under H. Vector Tiles: MVT and MLT. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-H-052	Vector tile extraction - SRC-054 p.36	NEW	SRC-054 p.36	MVT/MLT/vector tile choices need schema, source-layer, digest, and evidence-ref validation before publication. Evidence page also references: MapLibre, vector tile, filters, Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.36; page-level extraction from terms: MapLibre, vector tile, filters, Evidence Drawer, Focus Mode.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-H-053	Vector tile extraction - SRC-055 p.191	NEW	SRC-055 p.191	MVT/MLT/vector tile choices need schema, source-layer, digest, and evidence-ref validation before publication. Evidence page also references: MVT. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.191; page-level extraction from terms: MVT.
ML-H-054	Vector tile extraction - SRC-055 p.231	NEW	SRC-055 p.231	MVT/MLT/vector tile choices need schema, source-layer, digest, and evidence-ref validation before publication. Evidence page also references: MVT, CORS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.231; page-level extraction from terms: MVT, CORS.

5.I. I. PMTiles, MBTiles, Static Tiles, Serverless Delivery idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-001	PMTiles for immutable public-safe bundles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-002	PMTiles is single-file and read-only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-003	Clustered PMTiles v3 supports efficient	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-004	MBTiles is server/local packaging path I.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-005	Static tiles only for public-safe released	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-006	Serverless PMTiles needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-007	OCI registry can publish PMTiles, COG,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-008	Delta referrers carry tile and dataset	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-009	Canonical geodata media types must be I.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-010	Map UI may fetch verified PMTiles from	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-011	PMTiles/MBTiles/COPC/EP T are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-012	PMTiles packaging from MBTiles should	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-013	PMTiles suits stable public-safe serverless	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-014	PMTiles invalidation is part of redaction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-015	PMTiles artifacts belong beside STAC	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-016	OCI S3 and CDN distribution must not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-017	PMTiles v3 immutability drives overlay,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-018	Overlay PMTiles manifests bind base,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-019	Overlay chains need caps and rebase	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-020	Server-side overlay merge is preferred for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-021	Client-side overlay merge is reserved for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-022	Signed discoverable diffs and reverse	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-023	OCI referrers provide a discoverable	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-024	Registry retention and garbage collection	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-025	Offline pack loaders must verify claim	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-026	PMTiles preview is a derived layer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-027	STAC items can point to GeoJSON	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-028	PMTiles previews support change	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-029	Serverless PMTiles requires	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-030	Bounded collection before full atlas	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-031	GeoParquet plus PMTiles is a canonical	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-032	MBTiles can be a build intermediate only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-033	PMTiles metadata and sample tile fetch are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-034	Single PMTiles archive per unit supports	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-035	PMTiles asset roles must distinguish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-036	PMTiles hashes belong in manifest.yml	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-037	PMTiles public release needs Range-capable	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-038	Tile cache invalidation must be tied to release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-039	PMTiles preview URLs are not evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-040	PMTiles can bundle all vector epochs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-041	PMTiles deep verification should be explicit and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-042	PMTiles manifest entries need size and sha256	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-043	PMTiles loads directly in browser only after	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-044	PMTiles role metadata distinguishes overlay	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-045	PMTiles URL swaps should be infrequent and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-046	PMTiles pairs naturally with STAC asset	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-047	MBTiles remains an intermediate unless release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-048	One PMTiles archive per epoch I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-049	Signed epoch manifest gates I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/ SR...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-050	Digest-only fallback is a I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/ SRC-...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-051	Cache verified manifests and I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/ S...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-052	Hash per-tile payloads only on I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-053	Tile gate provides one source of I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRME...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-054	Proof pack assembly should I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/ SRC...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-055	Runtime worker receives I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/ SRC-04...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-056	PMTiles source changes must I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/ SR...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-057	Static tile delivery can be I. PMTiles, MBTiles, Static Tiles, Serverless NEW; CONFIRMED/ SR...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-058	Station-history PMTiles family	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-059	Static SSURGO PMTiles family	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-060	PMTiles as primary soil-moisture delivery surface	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-061	PMTiles partition path convention	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-062	PMTiles cache index and level-zero tiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-063	Bundle adjacent epochs in PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-064	Closed PMTiles epochs receive immutable cache headers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-065	PMTiles release verify step	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-066	PMTiles signature bundle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-067	PMTiles v3 spec pin note	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-068	PMTiles upload requires remote digest handling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-069	PMTiles file plus sigbundle are proof objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-070	Fail-closed unsigned PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-071	PMTiles interop supports downstream QA	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-072	PMTiles digest check belongs in worker	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-073	County soil polygons can be baked to per-county PMTiles artifacts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-074	PMTiles sidecars should contain dataset version source and provenance metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-075	Object storage plus CDN can serve public-safe immutable tile bundles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-076	PMTiles artifact digests must be recorded in run receipts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-077	PMTiles release should be blocked by missing provenance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-078	PMTiles manifests can be signed and verified offline with bundle material	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-079	PMTiles and COG builds should share a compact sidecar manifest pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-080	PMTiles current/live feeds must not be immutable-cached without freshness rules	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-081	PMTiles protocol handler is a runtime dependency needing allowlist review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-082	PMTiles metadata/header capability remains a verification question	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-083	Source-swap and client-filter modes require separate release and cache strategies	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-084	Historical PMTiles should preserve precision tags and source lineage	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-085	Serverless PMTiles wins for stable public-safe layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-086	Dynamic or steward-mediated layers may not fit static PMTiles release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-087	PMTiles verify/sign dry-runs belong before public publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-088	PMTiles build receipts should include CLI/tool versions and spec_hash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-089	PMTiles delta publication requires rollback target and cache invalidation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-090	PMTiles tile archives must not be treated as catalog closure	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-091	PMTiles assets should link to STAC/DCAT/PROV records	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-092	PMTiles layer badge failures should surface missing invalid or verified states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-093	PMTiles can carry biodiversity or conservation matrices as lightweight map products	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-094	PMTiles can package NDVI, burn severity, and change vectors	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-095	PMTiles is a single-file archive suited to static object-store delivery	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-096	Tippecanoe PMTiles output simplifies vector publish paths	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-097	PMTiles requires sidcar metadata for source licenses and required attribution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-098	PMTiles artifacts must be registered in DCAT/STAC/PROV rather than treated as proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-099	Habitat/protection overlays use PMTiles for full-state exploration	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-100	PMTiles publish should copy receipt alongside published artifact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-101	PMTiles releases should be immutable with cheap rollback by manifest and receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-102	PMTiles with HTTP Range improves request count and latency for cloud-native delivery	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-103	PMTiles must not auto-publish provisional detections	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-104	PMTiles path should be covered by TileArtifactManifest and MapReleaseManifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-105	PMTiles should trim attributes to reduce size and avoid leaking irrelevant fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-106	Static PMTiles is preferred only for stable public-safe derivatives	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under I. PMTiles, MBTiles, Static Tiles, Serverless Delivery. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-I-107	PMTiles/static tile extraction - SRC-054 p.31	NEW	SRC-054 p.31	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, vector tile, HTTP Range, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.31; page-level extraction from terms: MapLibre, PMTiles, vector tile, HTTP Range, 3D.
ML-I-108	PMTiles/static tile extraction - SRC-054 p.32	NEW	SRC-054 p.32	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, vector tile, COG, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.32; page-level extraction from terms: MapLibre, PMTiles, vector tile, COG, Cesium.
ML-I-109	PMTiles/static tile extraction - SRC-054 p.35	NEW	SRC-054 p.35	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, markers, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.35; page-level extraction from terms: MapLibre, PMTiles, markers, Cesium.
ML-I-110	PMTiles/static tile extraction - SRC-054 p.37	NEW	SRC-054 p.37	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, style, expressions, zoom, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.37; page-level extraction from terms: PMTiles, style, expressions, zoom, CDN.
ML-I-111	PMTiles/static tile extraction - SRC-054 p.39	NEW	SRC-054 p.39	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.39; page-level extraction from terms: MapLibre, PMTiles, Cesium.
ML-I-112	PMTiles/static tile extraction - SRC-054 p.40	NEW	SRC-054 p.40	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.40; page-level extraction from terms: PMTiles, COG, style.
ML-I-113	PMTiles/static tile extraction - SRC-054 p.41	NEW	SRC-054 p.41	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, COG, controls, timeline. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.41; page-level extraction from terms: MapLibre, PMTiles, COG, controls, timeline.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-114	PMTiles/static tile extraction - SRC-054 p.42	NEW	SRC-054 p.42	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, Evidence Drawer, Focus Mode, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.42; page-level extraction from terms: MapLibre, PMTiles, Evidence Drawer, Focus Mode, Cesium.
ML-I-115	PMTiles/static tile extraction - SRC-054 p.43	NEW	SRC-054 p.43	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, COG, Cesium, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.43; page-level extraction from terms: MapLibre, PMTiles, COG, Cesium, 3D.
ML-I-116	PMTiles/static tile extraction - SRC-054 p.51	NEW	SRC-054 p.51	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, Cesium, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.51; page-level extraction from terms: MapLibre, PMTiles, Cesium, 3D.
ML-I-117	PMTiles/static tile extraction - SRC-054 p.52	NEW	SRC-054 p.52	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, Evidence Drawer, Focus Mode, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.52; page-level extraction from terms: MapLibre, PMTiles, Evidence Drawer, Focus Mode, Cesium.
ML-I-118	PMTiles/static tile extraction - SRC-054 p.55	NEW	SRC-054 p.55	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG, style, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.55; page-level extraction from terms: PMTiles, COG, style, Evidence Drawer.
ML-I-119	PMTiles/static tile extraction - SRC-054 p.56	NEW	SRC-054 p.56	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.56; page-level extraction from terms: PMTiles, COG, zoom.
ML-I-120	PMTiles/static tile extraction - SRC-054 p.57	NEW	SRC-054 p.57	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.57; page-level extraction from terms: PMTiles, zoom.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-121	PMTiles/static tile extraction - SRC-054 p.58	NEW	SRC-054 p.58	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, MVT, MLT, vector tile. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.58; page-level extraction from terms: MapLibre, PMTiles, MVT, MLT, vector tile.
ML-I-122	PMTiles/static tile extraction - SRC-054 p.66	NEW	SRC-054 p.66	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, vector tile, WMS, controls. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.66; page-level extraction from terms: PMTiles, MVT, vector tile, WMS, controls.
ML-I-123	PMTiles/static tile extraction - SRC-054 p.86	NEW	SRC-054 p.86	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, vector tile, COG, WMS, WMTS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.86; page-level extraction from terms: PMTiles, vector tile, COG, WMS, WMTS.
ML-I-124	PMTiles/static tile extraction - SRC-054 p.106	NEW	SRC-054 p.106	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, vector tile, WMTS, GeoJSON, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.106; page-level extraction from terms: PMTiles, vector tile, WMTS, GeoJSON, zoom.
ML-I-125	PMTiles/static tile extraction - SRC-054 p.109	NEW	SRC-054 p.109	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.109; page-level extraction from terms: PMTiles, MVT, zoom.
ML-I-126	PMTiles/static tile extraction - SRC-054 p.118	NEW	SRC-054 p.118	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG, WMTS, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.118; page-level extraction from terms: PMTiles, COG, WMTS, zoom.
ML-I-127	PMTiles/static tile extraction - SRC-054 p.121	NEW	SRC-054 p.121	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.121; page-level extraction from terms: PMTiles, Evidence Drawer.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-128	PMTiles/static tile extraction - SRC-054 p.153	NEW	SRC-054 p.153	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.153; page-level extraction from terms: PMTiles.
ML-I-129	PMTiles/static tile extraction - SRC-054 p.154	NEW	SRC-054 p.154	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, TileJSON, HTTP Range, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.154; page-level extraction from terms: PMTiles, TileJSON, HTTP Range, CDN.
ML-I-130	PMTiles/static tile extraction - SRC-054 p.253	NEW	SRC-054 p.253	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.253; page-level extraction from terms: PMTiles, COG.
ML-I-131	PMTiles/static tile extraction - SRC-054 p.255	NEW	SRC-054 p.255	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.255; page-level extraction from terms: PMTiles, COG.
ML-I-132	PMTiles/static tile extraction - SRC-054 p.256	NEW	SRC-054 p.256	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.256; page-level extraction from terms: PMTiles, COG.
ML-I-133	PMTiles/static tile extraction - SRC-054 p.258	NEW	SRC-054 p.258	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, COG. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.258; page-level extraction from terms: PMTiles, COG.
ML-I-134	PMTiles/static tile extraction - SRC-054 p.265	NEW	SRC-054 p.265	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, TileJSON, style, bounds, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.265; page-level extraction from terms: PMTiles, TileJSON, style, bounds, zoom.
ML-I-135	PMTiles/static tile extraction - SRC-054 p.313	NEW	SRC-054 p.313	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, TileJSON, timeline. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.313; page-level extraction from terms: PMTiles, TileJSON, timeline.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-136	PMTiles/static tile extraction - SRC-054 p.316	NEW	SRC-054 p.316	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, TileJSON, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.316; page-level extraction from terms: PMTiles, MVT, TileJSON, CDN.
ML-I-137	PMTiles/static tile extraction - SRC-054 p.317	NEW	SRC-054 p.317	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, style, GeoJSON, Focus Mode, static tiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.317; page-level extraction from terms: MapLibre, style, GeoJSON, Focus Mode, static tiles.
ML-I-138	PMTiles/static tile extraction - SRC-055 p.29	NEW	SRC-055 p.29	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.29; page-level extraction from terms: PMTiles, PromotionDecision.
ML-I-139	PMTiles/static tile extraction - SRC-055 p.30	NEW	SRC-055 p.30	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.30; page-level extraction from terms: PMTiles.
ML-I-140	PMTiles/static tile extraction - SRC-055 p.31	NEW	SRC-055 p.31	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.31; page-level extraction from terms: PMTiles.
ML-I-141	PMTiles/static tile extraction - SRC-055 p.32	NEW	SRC-055 p.32	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, GeoJSON. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.32; page-level extraction from terms: PMTiles, GeoJSON.
ML-I-142	PMTiles/static tile extraction - SRC-055 p.33	NEW	SRC-055 p.33	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.33; page-level extraction from terms: MapLibre, PMTiles, zoom.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-143	PMTiles/static tile extraction - SRC-055 p.34	NEW	SRC-055 p.34	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.34; page-level extraction from terms: MapLibre, PMTiles, zoom.
ML-I-144	PMTiles/static tile extraction - SRC-055 p.35	NEW	SRC-055 p.35	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, LayerManifest. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.35; page-level extraction from terms: PMTiles, LayerManifest.
ML-I-145	PMTiles/static tile extraction - SRC-055 p.36	NEW	SRC-055 p.36	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, maplibre-gl, maplibregl, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.36; page-level extraction from terms: MapLibre, PMTiles, maplibre-gl, maplibregl, zoom.
ML-I-146	PMTiles/static tile extraction - SRC-055 p.37	NEW	SRC-055 p.37	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibregl, addSource. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.37; page-level extraction from terms: MapLibre, PMTiles, style, maplibregl, addSource.
ML-I-147	PMTiles/static tile extraction - SRC-055 p.38	NEW	SRC-055 p.38	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, style, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.38; page-level extraction from terms: PMTiles, style, zoom.
ML-I-148	PMTiles/static tile extraction - SRC-055 p.39	NEW	SRC-055 p.39	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, vector tile, style, LayerManifest. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.39; page-level extraction from terms: MapLibre, PMTiles, vector tile, style, LayerManifest.
ML-I-149	PMTiles/static tile extraction - SRC-055 p.40	NEW	SRC-055 p.40	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, MVT, TileJSON, WMTS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.40; page-level extraction from terms: MapLibre, PMTiles, MVT, TileJSON, WMTS.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-150	PMTiles/static tile extraction - SRC-055 p.41	NEW	SRC-055 p.41	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, MVT, WMTS, addLayer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.41; page-level extraction from terms: MapLibre, PMTiles, MVT, WMTS, addLayer.
ML-I-151	PMTiles/static tile extraction - SRC-055 p.42	NEW	SRC-055 p.42	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, WMTS, zoom, CORS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.42; page-level extraction from terms: PMTiles, MVT, WMTS, zoom, CORS.
ML-I-152	PMTiles/static tile extraction - SRC-055 p.43	NEW	SRC-055 p.43	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, MVT, TileJSON, WMTS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.43; page-level extraction from terms: MapLibre, PMTiles, MVT, TileJSON, WMTS.
ML-I-153	PMTiles/static tile extraction - SRC-055 p.80	NEW	SRC-055 p.80	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.80; page-level extraction from terms: MapLibre, PMTiles.
ML-I-154	PMTiles/static tile extraction - SRC-055 p.87	NEW	SRC-055 p.87	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, vector tile, style, CORS, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.87; page-level extraction from terms: PMTiles, vector tile, style, CORS, CDN.
ML-I-155	PMTiles/static tile extraction - SRC-055 p.88	NEW	SRC-055 p.88	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, GeoJSON. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.88; page-level extraction from terms: PMTiles, GeoJSON.
ML-I-156	PMTiles/static tile extraction - SRC-055 p.89	NEW	SRC-055 p.89	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, GeoJSON. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.89; page-level extraction from terms: PMTiles, GeoJSON.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-157	PMTiles/static tile extraction - SRC-055 p.90	NEW	SRC-055 p.90	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.90; page-level extraction from terms: PMTiles.
ML-I-158	PMTiles/static tile extraction - SRC-055 p.91	NEW	SRC-055 p.91	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.91; page-level extraction from terms: PMTiles.
ML-I-159	PMTiles/static tile extraction - SRC-055 p.92	NEW	SRC-055 p.92	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.92; page-level extraction from terms: PMTiles.
ML-I-160	PMTiles/static tile extraction - SRC-055 p.93	NEW	SRC-055 p.93	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.93; page-level extraction from terms: PMTiles.
ML-I-161	PMTiles/static tile extraction - SRC-055 p.94	NEW	SRC-055 p.94	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.94; page-level extraction from terms: PMTiles.
ML-I-162	PMTiles/static tile extraction - SRC-055 p.95	NEW	SRC-055 p.95	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.95; page-level extraction from terms: PMTiles.
ML-I-163	PMTiles/static tile extraction - SRC-055 p.96	NEW	SRC-055 p.96	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.96; page-level extraction from terms: PMTiles.
ML-I-164	PMTiles/static tile extraction - SRC-055 p.97	NEW	SRC-055 p.97	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, GeoJSON, maplibregl, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.97; page-level extraction from terms: MapLibre, PMTiles, GeoJSON, maplibregl, zoom.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-165	PMTiles/static tile extraction - SRC-055 p.101	NEW	SRC-055 p.101	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.101; page-level extraction from terms: PMTiles, style.
ML-I-166	PMTiles/static tile extraction - SRC-055 p.102	NEW	SRC-055 p.102	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.102; page-level extraction from terms: PMTiles.
ML-I-167	PMTiles/static tile extraction - SRC-055 p.103	NEW	SRC-055 p.103	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.103; page-level extraction from terms: PMTiles, MVT.
ML-I-168	PMTiles/static tile extraction - SRC-055 p.147	NEW	SRC-055 p.147	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.147; page-level extraction from terms: MapLibre, PMTiles.
ML-I-169	PMTiles/static tile extraction - SRC-055 p.148	NEW	SRC-055 p.148	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibre-gl, maplibregl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.148; page-level extraction from terms: MapLibre, PMTiles, style, maplibre-gl, maplibregl.
ML-I-170	PMTiles/static tile extraction - SRC-055 p.149	NEW	SRC-055 p.149	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style JSON, style, sprites. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.149; page-level extraction from terms: MapLibre, PMTiles, style JSON, style, sprites.
ML-I-171	PMTiles/static tile extraction - SRC-055 p.150	NEW	SRC-055 p.150	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, CORS, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.150; page-level extraction from terms: MapLibre, PMTiles, style, CORS, CDN.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-172	PMTiles/static tile extraction - SRC-055 p.152	NEW	SRC-055 p.152	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, terrain, camera. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.152; page-level extraction from terms: MapLibre, PMTiles, style, terrain, camera.
ML-I-173	PMTiles/static tile extraction - SRC-055 p.153	NEW	SRC-055 p.153	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, terrain, camera, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.153; page-level extraction from terms: MapLibre, PMTiles, terrain, camera, zoom.
ML-I-174	PMTiles/static tile extraction - SRC-055 p.154	NEW	SRC-055 p.154	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, vector tile, style, terrain. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.154; page-level extraction from terms: MapLibre, PMTiles, vector tile, style, terrain.
ML-I-175	PMTiles/static tile extraction - SRC-055 p.183	NEW	SRC-055 p.183	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, zoom, CDN, tile server. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.183; page-level extraction from terms: MapLibre, PMTiles, zoom, CDN, tile server.
ML-I-176	PMTiles/static tile extraction - SRC-055 p.184	NEW	SRC-055 p.184	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibre-gl, maplibregl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.184; page-level extraction from terms: MapLibre, PMTiles, style, maplibre-gl, maplibregl.
ML-I-177	PMTiles/static tile extraction - SRC-055 p.185	NEW	SRC-055 p.185	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, vector tile, style, maplibregl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.185; page-level extraction from terms: MapLibre, PMTiles, vector tile, style, maplibregl.
ML-I-178	PMTiles/static tile extraction - SRC-055 p.186	NEW	SRC-055 p.186	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, TileJSON. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.186; page-level extraction from terms: PMTiles, MVT, TileJSON.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-179	PMTiles/static tile extraction - SRC-055 p.187	NEW	SRC-055 p.187	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.187; page-level extraction from terms: PMTiles.
ML-I-180	PMTiles/static tile extraction - SRC-055 p.188	NEW	SRC-055 p.188	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.188; page-level extraction from terms: PMTiles.
ML-I-181	PMTiles/static tile extraction - SRC-055 p.189	NEW	SRC-055 p.189	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, TileJSON, CORS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.189; page-level extraction from terms: PMTiles, MVT, TileJSON, CORS.
ML-I-182	PMTiles/static tile extraction - SRC-055 p.192	NEW	SRC-055 p.192	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, TileJSON, zoom, CORS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.192; page-level extraction from terms: PMTiles, TileJSON, zoom, CORS.
ML-I-183	PMTiles/static tile extraction - SRC-055 p.194	NEW	SRC-055 p.194	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, MVT, TileJSON, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.194; page-level extraction from terms: MapLibre, PMTiles, MVT, TileJSON, style.
ML-I-184	PMTiles/static tile extraction - SRC-055 p.195	NEW	SRC-055 p.195	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, filters, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.195; page-level extraction from terms: PMTiles, filters, CDN.
ML-I-185	PMTiles/static tile extraction - SRC-055 p.223	NEW	SRC-055 p.223	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.223; page-level extraction from terms: MapLibre, PMTiles, CDN.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-186	PMTiles/static tile extraction - SRC-055 p.224	NEW	SRC-055 p.224	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibre-gl, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.224; page-level extraction from terms: MapLibre, PMTiles, style, maplibre-gl, zoom.
ML-I-187	PMTiles/static tile extraction - SRC-055 p.225	NEW	SRC-055 p.225	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibregl, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.225; page-level extraction from terms: MapLibre, PMTiles, style, maplibregl, zoom.
ML-I-188	PMTiles/static tile extraction - SRC-055 p.226	NEW	SRC-055 p.226	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, MVT, vector tile, TileJSON. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.226; page-level extraction from terms: MapLibre, PMTiles, MVT, vector tile, TileJSON.
ML-I-189	PMTiles/static tile extraction - SRC-055 p.227	NEW	SRC-055 p.227	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.227; page-level extraction from terms: PMTiles.
ML-I-190	PMTiles/static tile extraction - SRC-055 p.228	NEW	SRC-055 p.228	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.228; page-level extraction from terms: PMTiles.
ML-I-191	PMTiles/static tile extraction - SRC-055 p.229	NEW	SRC-055 p.229	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, TileJSON. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.229; page-level extraction from terms: PMTiles, MVT, TileJSON.
ML-I-192	PMTiles/static tile extraction - SRC-055 p.230	NEW	SRC-055 p.230	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, CORS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.230; page-level extraction from terms: PMTiles, CORS.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-193	PMTiles/static tile extraction - SRC-055 p.232	NEW	SRC-055 p.232	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, MVT, TileJSON, CORS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.232; page-level extraction from terms: PMTiles, MVT, TileJSON, CORS.
ML-I-194	PMTiles/static tile extraction - SRC-055 p.234	NEW	SRC-055 p.234	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, MVT, TileJSON, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.234; page-level extraction from terms: MapLibre, PMTiles, MVT, TileJSON, style.
ML-I-195	PMTiles/static tile extraction - SRC-055 p.235	NEW	SRC-055 p.235	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, filters, CDN, globe. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.235; page-level extraction from terms: PMTiles, filters, CDN, globe.
ML-I-196	PMTiles/static tile extraction - SRC-055 p.260	NEW	SRC-055 p.260	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, raster tile, style, terrain. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.260; page-level extraction from terms: MapLibre, PMTiles, raster tile, style, terrain.
ML-I-197	PMTiles/static tile extraction - SRC-055 p.261	NEW	SRC-055 p.261	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, HTTP Range. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.261; page-level extraction from terms: PMTiles, HTTP Range.
ML-I-198	PMTiles/static tile extraction - SRC-055 p.262	NEW	SRC-055 p.262	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibre-gl, maplibregl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.262; page-level extraction from terms: MapLibre, PMTiles, style, maplibre-gl, maplibregl.
ML-I-199	PMTiles/static tile extraction - SRC-055 p.263	NEW	SRC-055 p.263	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, vector tile, style, sprites. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.263; page-level extraction from terms: MapLibre, PMTiles, vector tile, style, sprites.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-200	PMTiles/static tile extraction - SRC-055 p.264	NEW	SRC-055 p.264	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.264; page-level extraction from terms: PMTiles, style.
ML-I-201	PMTiles/static tile extraction - SRC-055 p.265	NEW	SRC-055 p.265	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, tile server. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.265; page-level extraction from terms: PMTiles, tile server.
ML-I-202	PMTiles/static tile extraction - SRC-055 p.266	NEW	SRC-055 p.266	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, style, HTTP Range, CDN, tile server. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.266; page-level extraction from terms: PMTiles, style, HTTP Range, CDN, tile server.
ML-I-203	PMTiles/static tile extraction - SRC-055 p.267	NEW	SRC-055 p.267	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.267; page-level extraction from terms: PMTiles, CDN.
ML-I-204	PMTiles/static tile extraction - SRC-055 p.268	NEW	SRC-055 p.268	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, tile server. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.268; page-level extraction from terms: PMTiles, tile server.
ML-I-205	PMTiles/static tile extraction - SRC-055 p.269	NEW	SRC-055 p.269	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibre-gl, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.269; page-level extraction from terms: MapLibre, PMTiles, style, maplibre-gl, zoom.
ML-I-206	PMTiles/static tile extraction - SRC-055 p.270	NEW	SRC-055 p.270	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, style, maplibre-gl, maplibregl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.270; page-level extraction from terms: MapLibre, PMTiles, style, maplibre-gl, maplibregl.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-I-207	PMTiles/static tile extraction - SRC-055 p.271	NEW	SRC-055 p.271	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.271; page-level extraction from terms: PMTiles, CDN.
ML-I-208	PMTiles/static tile extraction - SRC-055 p.272	NEW	SRC-055 p.272	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.272; page-level extraction from terms: PMTiles.
ML-I-209	PMTiles/static tile extraction - SRC-055 p.273	NEW	SRC-055 p.273	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, sensitive geometry. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.273; page-level extraction from terms: PMTiles, sensitive geometry.
ML-I-210	PMTiles/static tile extraction - SRC-055 p.274	NEW	SRC-055 p.274	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, style, sprites, glyphs. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.274; page-level extraction from terms: PMTiles, style, sprites, glyphs.
ML-I-211	PMTiles/static tile extraction - SRC-055 p.275	NEW	SRC-055 p.275	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: PMTiles, public-safe layer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.275; page-level extraction from terms: PMTiles, public-safe layer.
ML-I-212	PMTiles/static tile extraction - SRC-055 p.276	NEW	SRC-055 p.276	PMTiles and related static bundles are derived release artifacts that need manifests, digests, and rollback-aware delivery. Evidence page also references: MapLibre, PMTiles, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.276; page-level extraction from terms: MapLibre, PMTiles, Evidence Drawer.

5.J. J. Martin and Server-Mediated Tile Serving idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-J-001	Martin for dynamic/server-mediated	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-J-002	PostGIS-backed tiles require runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-003	Martin style/catalog features must be	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-004	Server-mediated tiles can support	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-005	Tile server purge is a governed revocation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-006	Render-time redaction requires PDP	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-007	Server mediation can hide overlay	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-008	Pull-through registry or tile server should	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-009	PostGIS tile service remains	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-010	PostGIS remains an optional analysis backend,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-J-011	Server-mediated tiles are useful when	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-012	Martin deployment remains	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under J. Martin and Server-Mediated Tile Serving. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-J-013	Server-side conversion remains the production path for Cesium consumers	NEW	SRC-055 pp.40-43	The source recommends MVT to WMTS/TMS imagery or 3D Tiles for production Cesium support. Impl: Use server-mediated conversion when client adapters are not mature or policy-safe. Validation: Validate converted imagery/3D Tiles against the source PMTiles digest and release manifest. Evidence: SRC-055 pp.40-43; cross-page normalized discovery.
ML-J-014	Tile server choices need access-model justification	NEW	SRC-055 p.87	The corpus mentions pmtiles-rs and tileserver-rs as tooling for serving PMTiles or MBTiles through modern backends. Impl: Record why static artifact delivery is insufficient before adopting tile servers. Validation: Test range behavior, access controls, caching, and audit logs. Evidence: SRC-055 p.87; cross-page normalized discovery.

5.K. K. Raster, COG, DEM, Terrain, Hillshade idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-001	COG for large raster and EO overlays	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-002	Terrain and hillshade are contextual	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-003	3D terrain is conditional, not default	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-004	Raster tiles vs COG depends on access	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-005	COG is not a substitute for source registry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-006	SSURGO/gNATSGO thin slice emits	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-007	SoilGrids overlays are continuous	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-008	NLCD and EPA ecoregion baselines can	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-009	COG publication needs deterministic build	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-010	COGs are preferred gridded artifacts for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-011	Soil-moisture map outputs split point time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-012	SSURGO/gNATSGO soil packages	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-013	Raster-to-COG conversion requires CRS,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-014	Soils static raster products should publish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-015	COGs require overviews tiling and HTTP	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-016	Hillshade and DEM derivatives are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-017	COG PMTiles and raster tiles form a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-018	NAIP 3DEP and point-cloud derivatives	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-019	Sensor time series and raster baselines	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-020	Historical map scans become	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-021	Soil Wetness and Access Index splits	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-022	Air quality NowCast and sensor-correction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-023	MRMS GRIB2 hail tiles and SPC/NCEI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-024	BirdCast and NEXRAD migration rasters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-025	Raster or COG mosaics with time sliders	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-026	COG conversion receipts record GDAL	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-027	RAP and HLS COG layers are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-028	HLS/RAP stacks support	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-029	COG-to-preview tile workflow for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-030	NFHL WMS is regulatory context,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-031	Severe weather archives need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-032	Daily readiness tile is a COG plus JSON sidecar	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-033	COG bands can carry mask_health and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-034	COG metadata can carry contributing source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-035	COG internal tiling and overviews are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-036	HLS COG assets are STAC-searchable raster	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-037	NDVI delta COG pairs with GeoJSON alerts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-038	Equal-area grid snapping makes raster statistics	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-039	Valid-pixel footprints are evidence artifacts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-040	COG quick-checks are release gates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-041	Raster outputs must retain STAC ETags and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-042	COGs support raster time slices	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-043	COG internal tiling and overviews are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-044	COG protocol plugins need governance review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-045	Raster opacity crossfade is the default simple	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-046	Sensitive high-resolution frames may need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-047	Terrain and quantized-mesh formats are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-048	COGs belong in the same epoch	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-049	Hydrology should tile derived	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-050	Terrain and imagery tiling needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-051	Surface truth and visualization	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-052	Tileset validation must check	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-053	Tiling receipts need tile count	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-054	HLS COG assets enter catalog via STAC	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-055	Raster tile hashes for derived attributes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-056	COG plus STAC is cloud-native delivery	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-057	Terrain/context omissions need drawer caveats	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-058	Continuous soil properties should publish as COG rasters with overviews	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-059	COGs need internal tiling nodata CRS and resampling rules	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-060	COG sidecars should align with PMTiles manifests and signatures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-061	Raster stacks should separate 10m analysis layers from 30m rollup derivatives	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-062	Categorical rasters need nearest or mode resampling rather than bilinear	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-063	Continuous rasters may use bilinear resampling with declared statistics	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-064	COG range-friendly hosting requires HTTP Range and CORS verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-065	COG raster histogram shifts belong in promotion diff reports	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-066	COG and PMTiles mixed maps require unified trust badge semantics	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-067	LANDFIRE EVT rasters can be converted to COGs with signed derivative provenance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-068	HLS burn slice should emit baseline and current NDVI COGs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-069	HLS burn slice should emit baseline and current NBR COGs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-070	HLS burn slice should emit delta NDVI and dNBR COGs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-071	COG outputs need internal tiling and overviews enabled	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-072	Raster summaries such as occupancy probability and detection rates can be COGs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-073	COGs allow clients or services to pull only needed pixels	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-074	COG metadata must include nodata, band stats, compression, and QA metrics	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-075	COG output should carry QA summary JSON and source STAC refs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-076	COG and MVT should be paired by raster evidence and vectorized promoted zones	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-077	DEM/terrain data should remain evidence-derived and not a 3D default	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-078	Cesium terrain-aware display is conditional on evidence burden, not visual appeal	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under K. Raster, COG, DEM, Terrain, Hillshade. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-K-079	Raster/COG/terrain extraction - SRC-054 p.19	NEW	SRC-054 p.19	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: controls. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.19; page-level extraction from terms: controls.
ML-K-080	Raster/COG/terrain extraction - SRC-054 p.59	NEW	SRC-054 p.59	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: MapLibre, MVT, MLT, style JSON, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.59; page-level extraction from terms: MapLibre, MVT, MLT, style JSON, style.
ML-K-081	Raster/COG/terrain extraction - SRC-054 p.76	NEW	SRC-054 p.76	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: COG, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.76; page-level extraction from terms: COG, style.
ML-K-082	Raster/COG/terrain extraction - SRC-054 p.84	NEW	SRC-054 p.84	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: Cloud Optimized GeoTIFF. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.84; page-level extraction from terms: Cloud Optimized GeoTIFF.
ML-K-083	Raster/COG/terrain extraction - SRC-054 p.110	NEW	SRC-054 p.110	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: COG, Cloud Optimized GeoTIFF, WMTS, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.110; page-level extraction from terms: COG, Cloud Optimized GeoTIFF, WMTS, zoom.
ML-K-084	Raster/COG/terrain extraction - SRC-054 p.150	NEW	SRC-054 p.150	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: MapLibre, style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.150; page-level extraction from terms: MapLibre, style.
ML-K-085	Raster/COG/terrain extraction - SRC-054 p.155	NEW	SRC-054 p.155	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.155; page-level extraction from terms: CDN.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-086	Raster/COG/terrain extraction - SRC-054 p.203	NEW	SRC-054 p.203	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.203; page-level extraction from terms: MapLibre.
ML-K-087	Raster/COG/terrain extraction - SRC-054 p.263	NEW	SRC-054 p.263	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.263; page-level extraction from terms: Evidence Drawer.
ML-K-088	Raster/COG/terrain extraction - SRC-054 p.275	NEW	SRC-054 p.275	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: raster tile. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.275; page-level extraction from terms: raster tile.
ML-K-089	Raster/COG/terrain extraction - SRC-054 p.288	NEW	SRC-054 p.288	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.288; page-level extraction from terms: PromotionDecision.
ML-K-090	Raster/COG/terrain extraction - SRC-054 p.295	NEW	SRC-054 p.295	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.295; page-level extraction from terms: filters.
ML-K-091	Raster/COG/terrain extraction - SRC-055 p.81	NEW	SRC-055 p.81	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: MapLibre, style, maplibre-gl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.81; page-level extraction from terms: MapLibre, style, maplibre-gl.
ML-K-092	Raster/COG/terrain extraction - SRC-055 p.85	NEW	SRC-055 p.85	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.85; page-level extraction from terms: zoom.
ML-K-093	Raster/COG/terrain extraction - SRC-055 p.125	NEW	SRC-055 p.125	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.125; page-level extraction from terms: Focus Mode.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-K-094	Raster/COG/terrain extraction - SRC-055 p.243	NEW	SRC-055 p.243	COGs, rasters, DEMs, terrain, and hillshade layers need range-capable hosting, overviews, source metadata, and caveats. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.243; page-level extraction from terms: MapLibre.

5.L. L. WMS / WMTS / External Map Services idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-L-001	External WMS/WMTS is contextual until	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-002	No unreviewed external service loads	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-003	External service export citations are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-004	WMS/WFS/GeoRSS/map services are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-005	SDA WMS WFS and external map links	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-006	STAC web map links advertise visual	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-007	Live ArcGIS, PBF, and GeoJSON services	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-008	AirNow, AQS, HRRR, and 511 feeds can	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-L-009	NFHL WMS/WFS/REST source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-010	ArcGIS MapServer and FeatureService need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-011	External map services are not citation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-012	WMS/WMTS remain contextual until source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-013	f=json, f=geojson, and f=csv modes are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-014	HTTP validators belong on external service	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-015	WBD MapServer layer 6 is a concrete HUC12	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-016	Trusted source allowlists protect webhook	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-017	External Feature/Map services should emit	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-L-018	KGS ArcGIS geology services are context	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-019	External overlay feeds need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-020	SSURGO WFS/WMS as source path	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-021	NPS feature services feed static trail tiling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-022	Live map services need cataloged source role	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-023	External wetlands and federal map services are context sources until SourceDescriptor rights and access behavior are verified	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-024	External WMS or similar services need source descriptors before use as context layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-025	External services remain contextual unless evidence, rights, and freshness are resolved	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under L. WMS / WMTS / External Map Services. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-L-026	WMTS/TMS is a stable fallback for 2.5D or unverifiable high-res layers	NEW	SRC-055 pp.40-43	The current source proposes WMTS/TMS imagery as server-side conversion for Cesium and as low-res fallback. Impl: Use WMTS/TMS context only with attribution and freshness rules. Validation: Test fallback selection when PMTiles verification fails. Evidence: SRC-055 pp.40-43; cross-page normalized discovery.
ML-L-027	External map services remain contextual unless evidence-linked	NEW	SRC-054 pp.49-50; SRC-055 pp.40-43	The corpus requires catalog completeness and verification before consequential display. Impl: Treat WMS/WMTS as context until it resolves to EvidenceBundle or source metadata. Validation: Validate source URI, attribution, temporal fields, and rights before rendering. Evidence: SRC-054 pp.49-50; SRC-055 pp.40-43; cross-page normalized discovery.

5.M. M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-001	SourceDescriptor as source truth	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-002	LayerManifest links UI layer to artifacts M. LayerManifest, StyleManifest, TileArtifactManifest...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-003	StyleManifest controls style JSON M. M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-004	TileArtifactManifest controls tile artifacts M. LayerManifest, StyleManifest, TileArtifactManifest...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-005	MapReleaseManifest binds public map M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-006	spec_hash and digests make release M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-007	Cache invalidation is part of rollback M. M. LayerManifest, StyleManifest, TileArtifactManifest...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-008	Manifest annotations support zero-byte M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-009	Artifact manifests need registry digest as a M. LayerManifest, StyleManifest, TileArtifactManifest...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-010	STAC/DCAT/PROV minima are required M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-011	STAC-to-DCAT crosswalk preserves M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-012	Release index entries bind dataset, M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-013	Domain freshness windows become M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-014	Feed-card JSONL is a release-note M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-015	RunReceipt is mandatory for tile raster M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-016	STAC Item assets can carry GeoParquet M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-017	PROV-O links raw inputs transforms and M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-018	DCAT license and accessURL drive M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-019	Release manifest includes media type M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-020	RedactionProfile belongs in published item M. LayerManifest, StyleManifest, TileArtifactMani...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-021	AIRceipt records map materialization DQ M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-022	Source freshness validators belong in M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-023	Minimal PMTiles claim JSON contains M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-024	TileArtifactManifest links TileJSON, STAC M. LayerManifest, StyleManifest, TileArtifactManif...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-025	Tile patch RunReceipt captures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-026	MapReleaseManifest lists base+overlay M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-027	Event/Occurrence sidecar is a shared M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-028	ConsentManifest, RevocationEntry, and M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-029	Layer source metadata block M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-030	Historical route STAC Collection	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-031	Historical route entity schema M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-032	TrustStatus manifest for visibility	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-033	Revocation manifest for genealogy M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-034	Run receipt URLs belong in layer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-035	STAC/COG/PMTiles assets require M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-036	Soil/surficial artifacts need one manifest.yml M. LayerManifest, StyleManifest, TileArtifact...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-037	TileArtifactManifest should link GeoParquet, M. LayerManifest, StyleManifest, TileArtifactMa...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-038	MapReleaseManifest carries rollback tag M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-039	STAC/DCAT/PROV are distinct	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-040	RunReceipt resolves from readiness sidecars M. LayerManifest, StyleManifest, TileArtifactMan...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-041	LayerManifest should record source-layer and M. LayerManifest, StyleManifest, TileArtifactMa...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-042	SourceDescriptor must record service update M. LayerManifest, StyleManifest, TileArtifactMan...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-043	Stale-source fixtures are needed for live feeds M. LayerManifest, StyleManifest, TileArtifac...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-044	Catalog closure is a release gate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-045	ReleaseManifest declares assets and hashes M. LayerManifest, StyleManifest, TileArtifactMani...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-046	manifestSpecHash should exclude spec_hash M. LayerManifest, StyleManifest, TileArtifactManif...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-047	STAC/DCAT/PROV catalog matrix is a release M. LayerManifest, StyleManifest, TileArtifactMani...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-048	MetaBlock v2 anchors release artifacts M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-049	EvidenceBundle is proof surface, not receipt M. LayerManifest, StyleManifest, TileArtifactMa...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-050	PolicyDecision should not replace	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-051	QuarantineRecord links blocked map artifacts M. LayerManifest, StyleManifest, TileArtifactMa...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-052	RunReceipt is process memory keyed by run_id M. LayerManifest, StyleManifest, TileArtifactMa...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-053	DecisionEnvelope schema gives finite M. LayerManifest, StyleManifest, TileArtifactManifest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-054	Catalog matrix validator should read embedded M. LayerManifest, StyleManifest, TileArtifactM...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-055	Spec_hash binds manifest, M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-046 p.1-3 Pro...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-056	Minimal STAC metadata M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-046 p.1 Watcher c...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-057	Catalog closure crosses STAC M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-046 Overla...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-058	DecisionEnvelope needs seven	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-059	Promotion summary is M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-046 CI renderer em...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-060	Signed DecisionEnvelope links	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-061	TileVerificationReceipt is a M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-047 Runtime...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-062	Release_ref belongs in runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-063	ProofPack artifact index makes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-064	EvidenceTimeline contract M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-047 Timeline...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-065	TrustFederation schema records	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-066	RuntimeResponseEnvelope can	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-067	Artifact manifest should M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-047 Epoch mani...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-068	TilingReceipt joins tiler run to M. LayerManifest, StyleManifest, NEW; CONFIRMED/ SRC-047 p....	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-069	Source meta maps to release manifest fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-070	source_digests manifest map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-071	Artifact refs include path and sha256	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-072	TileArtifactManifest minimum fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-073	Catalog refs and bounds stay optional first-wave fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-074	Proof refs distinguish receipt/proof/catalog/publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-075	Release manifest connects PMTiles and sigbundle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-076	Lineage manifests tie source to runtime decision	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-077	Geometry digest for historical layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-078	Manifest-driven style validation fixtures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-079	TileArtifactManifest should record PMTiles and COG digest spec_hash and promotion_id	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-080	LayerManifest should link source role policy label release ref and evidence route	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-081	MapReleaseManifest should enumerate tile raster style sprite glyph and rollback refs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-082	StyleManifest should capture MapLibre version and source/layer compatibility	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-083	Manifest digests should join CI receipts with runtime trust badges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-084	Soil PMTiles metadata should include source system retrieved_at digest and version	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-085	COG metadata should include layer resolution statistic derivation and provenance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-086	RunReceipt remains process memory separate from proof pack	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-087	Proof pack should include manifest bundle schema validation and diff summary	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-088	PromotionDecision should block area mismatch coverage shift and missing provenance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-089	CorrectionNotice should reference tile and summary rebuild refs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-090	MapReleaseManifest needs cache invalidation records for replaced tile bundles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-091	TileArtifactManifest must not replace STAC/DCAT/PROV catalog records	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-092	SourceDescriptor activation should fail closed when rights are unknown	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-093	Layer catalog entries should expose stale state and release state	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-094	Manifest schema should include offline verification materials where used	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-095	RunReceipt should include exact inputs, parameters, spec_hash, software digests, and signatures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-096	PromotionReceipt should record gate status and final promotion decision	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-097	EvidenceBundle must link artifacts, inputs, checks, integrity, and signatures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-098	EvidenceRef must resolve or fail closed before runtime claim rendering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-099	HLS burn candidate should carry audit_ref, validator_result_ref, and run_receipt_ref	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-100	Derived STAC items should link COGs, vectors, provenance, and receipt/proof refs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-101	Source STAC and derived STAC should be separate catalog layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-102	DCAT/STAC/PROV closure should be checked, not decorative metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-103	Habitat/protection run receipt should carry source_uris, extraction_date, signer, notes, and spec_hash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-104	TileArtifactManifest should include PMTiles or COG artifact digests and sidecar links	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-105	MapReleaseManifest should bind released layer sources to rollback targets	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-106	LayerManifest should carry sensitivity label, attribution, and source/license obligations	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-107	StyleManifest should document color ramps, output units, uncertainty, and legends	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-108	PolicyDecision should include promote, hold, deny, reviewer, and reason fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-109	CitationValidationReport should verify source/license/receipt links before UI export	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-110	Signed manifests let clients verify assets without fetching huge blobs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-111	Spec_hash must be computed from a pinned build spec before artifact copy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-112	Catalog json should point to artifacts, receipts, licenses, and limitations	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-M-113	Manifest/release object extraction - SRC-054 p.54	NEW	SRC-054 p.54	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: bounds, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.54; page-level extraction from terms: bounds, zoom.
ML-M-114	Manifest/release object extraction - SRC-054 p.71	NEW	SRC-054 p.71	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.71; page-level extraction from terms: filters.
ML-M-115	Manifest/release object extraction - SRC-054 p.74	NEW	SRC-054 p.74	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.74; page-level extraction from terms: PromotionDecision.
ML-M-116	Manifest/release object extraction - SRC-054 p.96	NEW	SRC-054 p.96	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.96; page-level extraction from terms: CDN.
ML-M-117	Manifest/release object extraction - SRC-054 p.105	NEW	SRC-054 p.105	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: PostGIS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.105; page-level extraction from terms: PostGIS.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-118	Manifest/release object extraction - SRC-054 p.117	NEW	SRC-054 p.117	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.117; page-level extraction from terms: style.
ML-M-119	Manifest/release object extraction - SRC-054 p.123	NEW	SRC-054 p.123	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.123; page-level extraction from terms: style.
ML-M-120	Manifest/release object extraction - SRC-054 p.169	NEW	SRC-054 p.169	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style, PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.169; page-level extraction from terms: style, PromotionDecision.
ML-M-121	Manifest/release object extraction - SRC-054 p.218	NEW	SRC-054 p.218	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.218; page-level extraction from terms: MapLibre.
ML-M-122	Manifest/release object extraction - SRC-054 p.269	NEW	SRC-054 p.269	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.269; page-level extraction from terms: PromotionDecision.
ML-M-123	Manifest/release object extraction - SRC-054 p.270	NEW	SRC-054 p.270	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.270; page-level extraction from terms: style.
ML-M-124	Manifest/release object extraction - SRC-054 p.276	NEW	SRC-054 p.276	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.276; page-level extraction from terms: PromotionDecision.
ML-M-125	Manifest/release object extraction - SRC-054 p.278	NEW	SRC-054 p.278	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.278; page-level extraction from terms: PromotionDecision.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-126	Manifest/release object extraction - SRC-054 p.294	NEW	SRC-054 p.294	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: MapLibre, filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.294; page-level extraction from terms: MapLibre, filters.
ML-M-127	Manifest/release object extraction - SRC-054 p.320	NEW	SRC-054 p.320	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.320; page-level extraction from terms: style.
ML-M-128	Manifest/release object extraction - SRC-054 p.327	NEW	SRC-054 p.327	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: sensitive geometry. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.327; page-level extraction from terms: sensitive geometry.
ML-M-129	Manifest/release object extraction - SRC-055 p.3	NEW	SRC-055 p.3	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.3; page-level extraction from terms: filters.
ML-M-130	Manifest/release object extraction - SRC-055 p.14	NEW	SRC-055 p.14	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.14; page-level extraction from terms: filters.
ML-M-131	Manifest/release object extraction - SRC-055 p.16	NEW	SRC-055 p.16	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style, filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.16; page-level extraction from terms: style, filters.
ML-M-132	Manifest/release object extraction - SRC-055 p.19	NEW	SRC-055 p.19	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: filters, controls. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.19; page-level extraction from terms: filters, controls.
ML-M-133	Manifest/release object extraction - SRC-055 p.28	NEW	SRC-055 p.28	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.28; page-level extraction from terms: style.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-134	Manifest/release object extraction - SRC-055 p.50	NEW	SRC-055 p.50	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.50; page-level extraction from terms: style.
ML-M-135	Manifest/release object extraction - SRC-055 p.52	NEW	SRC-055 p.52	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.52; page-level extraction from terms: style.
ML-M-136	Manifest/release object extraction - SRC-055 p.62	NEW	SRC-055 p.62	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.62; page-level extraction from terms: CDN.
ML-M-137	Manifest/release object extraction - SRC-055 p.63	NEW	SRC-055 p.63	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.63; page-level extraction from terms: CDN.
ML-M-138	Manifest/release object extraction - SRC-055 p.71	NEW	SRC-055 p.71	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.71; page-level extraction from terms: PromotionDecision.
ML-M-139	Manifest/release object extraction - SRC-055 p.78	NEW	SRC-055 p.78	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.78; page-level extraction from terms: PromotionDecision.
ML-M-140	Manifest/release object extraction - SRC-055 p.83	NEW	SRC-055 p.83	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: MapLibre, style, maplibregl, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.83; page-level extraction from terms: MapLibre, style, maplibregl, zoom.
ML-M-141	Manifest/release object extraction - SRC-055 p.104	NEW	SRC-055 p.104	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.104; page-level extraction from terms: style.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-M-142	Manifest/release object extraction - SRC-055 p.117	NEW	SRC-055 p.117	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: bounds. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.117; page-level extraction from terms: bounds.
ML-M-143	Manifest/release object extraction - SRC-055 p.132	NEW	SRC-055 p.132	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.132; page-level extraction from terms: style.
ML-M-144	Manifest/release object extraction - SRC-055 p.250	NEW	SRC-055 p.250	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: sensitive geometry. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.250; page-level extraction from terms: sensitive geometry.
ML-M-145	Manifest/release object extraction - SRC-055 p.277	NEW	SRC-055 p.277	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.277; page-level extraction from terms: style.
ML-M-146	Manifest/release object extraction - SRC-055 p.278	NEW	SRC-055 p.278	Layer, style, tile, and release manifests bind artifacts to spec_hash, receipts, catalog records, and rollback targets. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.278; page-level extraction from terms: style.

5.N. N. Evidence Drawer Payloads and Click Resolution idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-001	Click resolution must resolve	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-002	EvidenceDrawerPayload is a contract N.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-003	Popup is not Evidence Drawer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-004	Drawer must show policy and correction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-005	Drawer error cannot fall back to raw	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-006	Role-gated steward fields N. Evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-007	Cite button opens evidence side panel N.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-008	Tooltips can carry source_ref without	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-009	Evidence bundles are the resolution target	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-010	Evidence Drawer should expose trace_id	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-011	Evidence Drawer evolves into Receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-012	Drawer field binding maps STAC DCAT	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-013	Drawer must show catalog missing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-014	Evidence list may expose asset actions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-015	PROV graph should be collapsible and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-016	Evidence Drawer surfaces run_receipt,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-017	Collapsed citation pill expands inline to	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-018	Evidence Drawer fails closed to	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-019	PMTiles verifier rejection yields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-020	Evidence tooltip shows top three	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-021	Receipt Drawer is a governed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-022	Policy badge opens constraints	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-023	Evidence summary validation gates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-024	Internal policy blocks public receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-025	Superseded and correction-pending	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-026	Evidence Drawer can trace	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-027	Pixel-level evidence refs belong in readiness	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-028	RoadEvent drawer payloads should include	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-029	Soil/surficial drawer payloads cite GeoParquet	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-030	NDVI drawer payloads include mask health and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-031	KDOT drawer payloads must distinguish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-032	Evidence Drawer is not a popup substitute	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-033	Drawer denies internal or restricted receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-034	Feature clicks are candidates for governed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-035	Evidence badge advertises traceability state	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-036	Manifest details become drawer inspectables	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-037	Drift state should be visible on hover/card	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-038	Unknown verification must remain visible	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-039	KFM runtime links asset to source and citation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-040	Evidence Drawer should expose deny/error	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-041	Evidence Drawer should display	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-042	Evidence Drawer and trust chip	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-043	Federated Trust Panel lives N. Evidence Drawer Payloads and Click NEW; CONFIRMED/ SRC-047 Cr...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-044	Split-view conflict inspection is N. Evidence Drawer Payloads and Click NEW; CONFIRMED/ SRC-...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-045	Timeline entries enrich click N. Evidence Drawer Payloads and Click NEW; CONFIRMED/ SRC-047...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-046	Promotion summary should be N. Evidence Drawer Payloads and Click NEW; CONFIRMED/ SRC-046 Re...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-047	Proof pack artifact index N. Evidence Drawer Payloads and Click NEW; CONFIRMED/ SRC-047 Proo...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-048	Evidence Drawer is not a popup	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-049	Hover panel displays source meta	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-050	Tile to EvidenceBundle linkage	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-051	Evidence Drawer shows historical confidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-052	Drawer caveats for remote-sensing derived layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-053	Evidence Drawer must not substitute popup proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-054	Evidence Drawer payloads should be downstream of governed evidence resolution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-055	Fauna occurrence drawer payloads need outcome-specific expectations	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-056	Drawer support blocks should distinguish observed modeled regulatory and context roles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-057	Clicked map features should resolve through feature id to EvidenceBundle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-058	Drawer summaries must expose precision served and generalization state	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-059	Drawer payloads should carry policy review freshness and correction fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-060	Negative outcomes must render ABSTAIN DENY and ERROR visibly in the drawer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-061	Soil mukey clicks should resolve to source_uri digest and run receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-062	Hydrology crossing drawers should show contributing COMID and permanence factors	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-063	Trust badges should expand to manifest digest spec_hash and signer details	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-064	Unreleased or restricted feature clicks should deny rather than silently hide evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-065	Drawer fixtures should align request decision envelope and drawer JSON	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-066	Drawer payload contract paths remain proposed until explorer-web conventions are verified	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-067	Popup text must not substitute for the Evidence Drawer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-068	Evidence Drawer is mandatory inspection surface for consequential runtime claims	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-069	Evidence Drawer must not hide abstention from review or stewardship surfaces	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-070	Evidence Drawer should show unresolved evidence status rather than render fact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-071	Evidence Drawer should link tiles to sources, license, and receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-072	Resolved evidence roles should show bundle and promotion identity	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-073	Unresolved evidence roles should show abstention reason	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-074	Composed claims need drawer role groups for each required EvidenceRef	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-075	Map click should resolve feature id to EvidenceBundle before explanation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-076	Hazards Evidence Drawer should disclose validator result and observed window	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-077	Habitat/protection drawer should disclose sensitivity transforms and license status	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-078	Biodiversity drawer should disclose effort, bias, and uncertainty fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-079	Drawer should expose promotion decision and reviewers for published outputs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-080	Drawer should distinguish source catalog, derived catalog, and release artifact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-081	Drawer should preserve negative evidence such as no-change or insufficient evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-082	Drawer should show STAC/DCAT/PROV references for released tiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-083	Drawer should be compatible with Focus Mode and cite-or-abstain answers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-084	Drawer should show stale source or provisional status for unpromoted detections	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-085	Drawer should deny or generalize sensitive geometry and explain why	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under N. Evidence Drawer Payloads and Click Resolution. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-N-086	Evidence Drawer extraction - SRC-054 p.20	NEW	SRC-054 p.20	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.20; page-level extraction from terms: Evidence Drawer.
ML-N-087	Evidence Drawer extraction - SRC-054 p.24	NEW	SRC-054 p.24	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.24; page-level extraction from terms: Evidence Drawer.
ML-N-088	Evidence Drawer extraction - SRC-054 p.30	NEW	SRC-054 p.30	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.30; page-level extraction from terms: Evidence Drawer.
ML-N-089	Evidence Drawer extraction - SRC-054 p.49	NEW	SRC-054 p.49	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.49; page-level extraction from terms: MapLibre, Evidence Drawer, Focus Mode.
ML-N-090	Evidence Drawer extraction - SRC-054 p.94	NEW	SRC-054 p.94	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, Evidence Drawer, Focus Mode, CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.94; page-level extraction from terms: MapLibre, Evidence Drawer, Focus Mode, CDN.
ML-N-091	Evidence Drawer extraction - SRC-054 p.119	NEW	SRC-054 p.119	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.119; page-level extraction from terms: MapLibre, Evidence Drawer, Focus Mode.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-092	Evidence Drawer extraction - SRC-054 p.131	NEW	SRC-054 p.131	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.131; page-level extraction from terms: Evidence Drawer.
ML-N-093	Evidence Drawer extraction - SRC-054 p.205	NEW	SRC-054 p.205	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.205; page-level extraction from terms: Evidence Drawer.
ML-N-094	Evidence Drawer extraction - SRC-054 p.206	NEW	SRC-054 p.206	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.206; page-level extraction from terms: MapLibre, Evidence Drawer.
ML-N-095	Evidence Drawer extraction - SRC-054 p.212	NEW	SRC-054 p.212	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.212; page-level extraction from terms: Evidence Drawer.
ML-N-096	Evidence Drawer extraction - SRC-054 p.214	NEW	SRC-054 p.214	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.214; page-level extraction from terms: Evidence Drawer, Focus Mode.
ML-N-097	Evidence Drawer extraction - SRC-054 p.240	NEW	SRC-054 p.240	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.240; page-level extraction from terms: Evidence Drawer.
ML-N-098	Evidence Drawer extraction - SRC-054 p.250	NEW	SRC-054 p.250	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.250; page-level extraction from terms: MapLibre, Evidence Drawer, Focus Mode.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-099	Evidence Drawer extraction - SRC-054 p.252	NEW	SRC-054 p.252	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.252; page-level extraction from terms: Evidence Drawer, Focus Mode.
ML-N-100	Evidence Drawer extraction - SRC-054 p.274	NEW	SRC-054 p.274	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer, Focus Mode, sensitive geometry. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.274; page-level extraction from terms: Evidence Drawer, Focus Mode, sensitive geometry.
ML-N-101	Evidence Drawer extraction - SRC-054 p.289	NEW	SRC-054 p.289	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.289; page-level extraction from terms: Evidence Drawer.
ML-N-102	Evidence Drawer extraction - SRC-054 p.290	NEW	SRC-054 p.290	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.290; page-level extraction from terms: MapLibre, Evidence Drawer, Focus Mode.
ML-N-103	Evidence Drawer extraction - SRC-054 p.291	NEW	SRC-054 p.291	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, zoom, Evidence Drawer, LayerManifest. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.291; page-level extraction from terms: MapLibre, zoom, Evidence Drawer, LayerManifest.
ML-N-104	Evidence Drawer extraction - SRC-054 p.293	NEW	SRC-054 p.293	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.293; page-level extraction from terms: Evidence Drawer, Focus Mode.
ML-N-105	Evidence Drawer extraction - SRC-054 p.308	NEW	SRC-054 p.308	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: LayerManifest. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.308; page-level extraction from terms: LayerManifest.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-106	Evidence Drawer extraction - SRC-054 p.319	NEW	SRC-054 p.319	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: GeoJSON, Evidence Drawer, Focus Mode, LayerManifest. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.319; page-level extraction from terms: GeoJSON, Evidence Drawer, Focus Mode, LayerManifest.
ML-N-107	Evidence Drawer extraction - SRC-054 p.331	NEW	SRC-054 p.331	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer, Focus Mode, sensitive geometry. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.331; page-level extraction from terms: Evidence Drawer, Focus Mode, sensitive geometry.
ML-N-108	Evidence Drawer extraction - SRC-054 p.336	NEW	SRC-054 p.336	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: markers, Evidence Drawer, PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.336; page-level extraction from terms: markers, Evidence Drawer, PromotionDecision.
ML-N-109	Evidence Drawer extraction - SRC-055 p.4	NEW	SRC-055 p.4	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.4; page-level extraction from terms: Evidence Drawer.
ML-N-110	Evidence Drawer extraction - SRC-055 p.10	NEW	SRC-055 p.10	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.10; page-level extraction from terms: MapLibre, Evidence Drawer.
ML-N-111	Evidence Drawer extraction - SRC-055 p.11	NEW	SRC-055 p.11	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, style JSON, style, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.11; page-level extraction from terms: MapLibre, style JSON, style, Evidence Drawer.
ML-N-112	Evidence Drawer extraction - SRC-055 p.15	NEW	SRC-055 p.15	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: MapLibre, zoom, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.15; page-level extraction from terms: MapLibre, zoom, Evidence Drawer.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-N-113	Evidence Drawer extraction - SRC-055 p.45	NEW	SRC-055 p.45	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: style, Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.45; page-level extraction from terms: style, Evidence Drawer.
ML-N-114	Evidence Drawer extraction - SRC-055 p.46	NEW	SRC-055 p.46	Clicked features must resolve through governed APIs into Evidence Drawer payloads before consequential claims are shown. Evidence page also references: Evidence Drawer. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.46; page-level extraction from terms: Evidence Drawer.

5.O. O. Focus Mode and Governed AI Map Context idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-001	Focus Mode consumes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-002	No raw feature/canonical data to model	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-003	Finite Focus outcomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-004	CitationValidationReport gates answers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-005	AIRceipt preserves runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-006	No chain-of-thought truth object	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-007	Story/Focus transition styles are reversible	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-008	PII-safe map telemetry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-009	Focus Mode Context Loop is bounded,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-010	Focus citation bundle includes feature or	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-011	LLM side effects are gated by attestation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-012	Focus Mode can cite spec_hash to catalog	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-013	Focus/render denies missing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-014	Focus trust badge can link to per-node	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-015	IIIF/LoC map manifests can support exact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-016	Uncertainty threshold and strict mode gate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-017	Focus should prefer validated sources	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-018	Missing confidence degrades disclosure	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-019	Focus labels explain confidence filtering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-020	Focus receives map state plus O. O. Focus Mode and Governed AI Map Context	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-021	Focus Mode uses time and H3 prefilters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-022	Focus Mode deep links include evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-023	AI-generated geospatial patches need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-024	Field-access Focus answers cite decision	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-025	Focus Mode consumes governed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-026	Genealogy Focus layer uses	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-027	Steward focus modes support	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-028	Focus answers abstain on missing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-029	Watcher-signal STAC fields can mirror into	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-030	Focus Mode must abstain when evidence refs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-031	Focus Mode can explain change gates, not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-032	Focus Mode UI for genealogy must respect	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-033	Focus Mode answers must cite run receipts and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-034	Focus Mode viewport pulls require governed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-035	Focus Mode can consume trust badge states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-036	Focus Mode must abstain on missing manifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-037	AIReceipt stays conditional and separate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-038	Model context should include finite negative	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-039	Focus Mode must consume O. Focus Mode and Governed AI Map NEW; CONFIRMED/ SRC-047 p.352-3 Ru...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-040	Cross-source conflict should O. Focus Mode and Governed AI Map NEW; CONFIRMED/ SRC-047 Feder...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-041	Source influence tracking O. Focus Mode and Governed AI Map NEW; CONFIRMED/ SRC-047 p.395 Ea...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-042	Revoked sources must not O. Focus Mode and Governed AI Map NEW; CONFIRMED/ SRC-047 No source...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-043	Focus overlays can use O. Focus Mode and Governed AI Map NEW; CONFIRMED/ SRC-046 p.461 Hydro...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-044	Trust timeline supports narrative	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-045	ABSTAIN is a visible trust O. Focus Mode and Governed AI Map NEW; CONFIRMED/ SRC-047 ABSTAIN...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-046	Runtime federation is a O. Focus Mode and Governed AI Map NEW; CONFIRMED/ SRC-047 Federation...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-047	Worker verification protects Focus Mode responsiveness	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-048	Focus Mode uses lineage manifests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-049	Finite validator outcomes align with Focus outcomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-050	Temporal map context should carry epoch	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-051	Live feeds require different Focus posture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-052	Focus Mode should evaluate declared evidence blocks before answering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-053	Citation sets are required building blocks for accountable Focus Mode responses	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-054	Focus Mode bundle sufficiency should produce ANSWER ABSTAIN DENY or ERROR	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-055	Building block registries allow replaceable AI runtimes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-056	Focus Mode cannot use rendered features alone as evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-057	MapContextEnvelope should include released layer refs and evidence refs only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-058	Policy-incompatible bundles should produce DENY with policy trace	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-059	Missing citation blocks should produce ABSTAIN for insufficient evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-060	Focus Mode should remain downstream of Evidence Drawer resolution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-061	Focus Mode tests should cover missing evidence restricted policy and valid answer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-062	Focus Mode may answer only from validator-backed runtime evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-063	Focus Mode must abstain when burn-specific evidence is missing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-064	Focus Mode composed claims require all required EvidenceRefs resolved	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-065	Focus Mode payload should inherit Evidence Drawer role statuses	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-066	Focus Mode should never infer from rendered map features alone	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-067	Focus Mode should cite STAC/DCAT/PROV and EvidenceBundle refs for map claims	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-068	Focus Mode must surface DENY when sensitivity or rights block release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-O-069	Focus Mode map context should include time window, layer ids, feature ids, and spec_hash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under O. Focus Mode and Governed AI Map Context. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-070	Focus Mode extraction - SRC-054 p.26	NEW	SRC-054 p.26	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.26; page-level extraction from terms: Focus Mode.
ML-O-071	Focus Mode extraction - SRC-054 p.44	NEW	SRC-054 p.44	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.44; page-level extraction from terms: Focus Mode.
ML-O-072	Focus Mode extraction - SRC-054 p.89	NEW	SRC-054 p.89	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: MapLibre, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.89; page-level extraction from terms: MapLibre, Focus Mode.
ML-O-073	Focus Mode extraction - SRC-054 p.111	NEW	SRC-054 p.111	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.111; page-level extraction from terms: Focus Mode.
ML-O-074	Focus Mode extraction - SRC-054 p.120	NEW	SRC-054 p.120	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.120; page-level extraction from terms: Focus Mode.
ML-O-075	Focus Mode extraction - SRC-054 p.168	NEW	SRC-054 p.168	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.168; page-level extraction from terms: Focus Mode.
ML-O-076	Focus Mode extraction - SRC-054 p.183	NEW	SRC-054 p.183	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: MapLibre, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.183; page-level extraction from terms: MapLibre, Focus Mode.
ML-O-077	Focus Mode extraction - SRC-054 p.196	NEW	SRC-054 p.196	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: MapLibre, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.196; page-level extraction from terms: MapLibre, Focus Mode.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-078	Focus Mode extraction - SRC-054 p.197	NEW	SRC-054 p.197	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.197; page-level extraction from terms: Focus Mode.
ML-O-079	Focus Mode extraction - SRC-054 p.200	NEW	SRC-054 p.200	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.200; page-level extraction from terms: Focus Mode.
ML-O-080	Focus Mode extraction - SRC-054 p.202	NEW	SRC-054 p.202	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.202; page-level extraction from terms: Focus Mode.
ML-O-081	Focus Mode extraction - SRC-054 p.204	NEW	SRC-054 p.204	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: style, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.204; page-level extraction from terms: style, Focus Mode.
ML-O-082	Focus Mode extraction - SRC-054 p.260	NEW	SRC-054 p.260	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: MapLibre, Focus Mode, PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.260; page-level extraction from terms: MapLibre, Focus Mode, PromotionDecision.
ML-O-083	Focus Mode extraction - SRC-054 p.285	NEW	SRC-054 p.285	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode, PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.285; page-level extraction from terms: Focus Mode, PromotionDecision.
ML-O-084	Focus Mode extraction - SRC-054 p.318	NEW	SRC-054 p.318	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: style, GeoJSON, Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.318; page-level extraction from terms: style, GeoJSON, Focus Mode.
ML-O-085	Focus Mode extraction - SRC-055 p.111	NEW	SRC-055 p.111	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: AIRceipt. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.111; page-level extraction from terms: AIRceipt.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-O-086	Focus Mode extraction - SRC-055 p.127	NEW	SRC-055 p.127	Focus Mode may synthesize only over resolved, visible, policy-safe evidence and must cite, abstain, deny, or error. Evidence page also references: Focus Mode. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.127; page-level extraction from terms: Focus Mode.

5.P. P. Time-Aware Map Interaction and Timeline idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-001	Time slider controls active support window	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-002	Unsupported time requests abstain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-003	Hidden time slice leaks are denied	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-004	Source, valid, retrieval, release, correction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-005	Timeline performance can use time-sliced	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-006	Two-minute HUC12 story pattern is a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-007	Prebuilt bitemporal spans support timeline	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-008	Event overlays require validity windows,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-009	AirNow NowCast and AQS reconciliation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-010	Real-time events require freshness and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-011	Timeline states distinguish provisional	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-012	Event/Occurrence sidecar provides	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-013	Accessible time scrubber emits typed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-014	Hover-or-focus time-slice preview works	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-015	Interval ribbons and precision glyphs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-016	Historical trail and rail atlas layers use	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-017	Timeline controls expose	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-018	Strict versus exploratory timelines	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-019	Frozen-soil and stale temporal	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-020	HLS/RAP time-series drive change	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-021	Delta-only ingestion reduces silent drift	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-022	HLS time windows must be pinned	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-023	Station health uses activity timestamps	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-024	RoadEvent time fields become temporal map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-025	Release time differs from source and observed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-026	Time slider should mutate filter state on	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-027	Epoch is the stable vector time key	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-028	Raster time is represented as COG slices or	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-029	Cesium uses one Clock for 3D time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-030	3D Tiles availability can reduce time-window	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-031	Observed, retrieval, source, and release time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-032	Consent and vendor-link expiry affect map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-033	Epoch id is the timeline join key	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-034	TimeRange belongs in signed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-035	Prefetch low-res LODs before	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-036	EvidenceTimeline records	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-037	Truth is time-scoped	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-038	No revoked release may	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-039	Correction graph prevents silent	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-040	Rollback is a visible timeline	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-041	Temporal trust supports story	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-042	Runtime receipts become	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-043	Source epoch range controls timelapse	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-044	Timelapse can swap sources	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-045	Timelapse can filter features	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-046	Numeric epoch property t	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-047	Fixed-window scrubber pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-048	Prebucketed epochs for coarse jumps	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-049	Worker PMTiles bucket preparation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-050	CZML availability drives 3D time visibility	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-051	3D Tiles epoch toggle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-052	Post office temporal slicing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-053	Trail period semantics	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-054	Compare mode swaps epochs side-by-side	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-055	MapLibre time sliders can choose between server-baked epoch sources and client filters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-056	Server-baked epoch sources win for tiny draw times and predictable performance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-057	Client-side filters win for ad hoc time ranges and one compact archive	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-058	Hybrid time filtering should offload heavy work to Web Workers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-059	Slider changes should debounce to avoid spamming workers and style updates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-060	Active timeline memory should recycle decoded tiles during long scrubs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-061	Cesium CZML timelines should slice bundles by time windows	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-062	Cesium prefetch should load neighboring slices near interval edges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-063	Time-aware maps must preserve source time retrieval time and release time distinctions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-064	Time-filtered outputs need cache keys tied to epoch or filter mode	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-065	Historical layers should surface precision and temporal uncertainty together	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-066	Timeline views should not hide stale sources behind animation polish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-067	HLS workflow uses baseline and current temporal composites	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-068	Change detection should trigger on material semantic change, not every acquisition	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-069	HLS vector features carry start and end windows for map filtering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-070	Runtime envelope should include observed_window for hazard claims	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-071	Freshness should be evaluated and returned with runtime map results	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-072	Temporal persistence checks prevent single bad scenes from promotion	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-073	STAC items should include start_datetime and end_datetime for time-sliced clients	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-074	Time slider should filter by timestamp or source window rather than mutate evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-075	Event windows distinguish burn claims from generic vegetation disturbance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-076	Source, retrieval, generated, and release dates belong in run receipts and catalogs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-077	Rollbacks are cheap when releases are immutable artifacts plus receipts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-078	Temporal comparison modes need adjacent epoch fixtures before release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under P. Time-Aware Map Interaction and Timeline. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-P-079	Time/timeline extraction - SRC-054 p.68	NEW	SRC-054 p.68	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: controls. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.68; page-level extraction from terms: controls.
ML-P-080	Time/timeline extraction - SRC-054 p.98	NEW	SRC-054 p.98	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: PostGIS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.98; page-level extraction from terms: PostGIS.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-081	Time/timeline extraction - SRC-054 p.104	NEW	SRC-054 p.104	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.104; page-level extraction from terms: PromotionDecision.
ML-P-082	Time/timeline extraction - SRC-054 p.201	NEW	SRC-054 p.201	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.201; page-level extraction from terms: MapLibre.
ML-P-083	Time/timeline extraction - SRC-054 p.264	NEW	SRC-054 p.264	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.264; page-level extraction from terms: MapLibre.
ML-P-084	Time/timeline extraction - SRC-054 p.268	NEW	SRC-054 p.268	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: bounds, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.268; page-level extraction from terms: bounds, zoom.
ML-P-085	Time/timeline extraction - SRC-055 p.72	NEW	SRC-055 p.72	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: zoom, PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.72; page-level extraction from terms: zoom, PromotionDecision.
ML-P-086	Time/timeline extraction - SRC-055 p.77	NEW	SRC-055 p.77	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: zoom, PromotionDecision. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.77; page-level extraction from terms: zoom, PromotionDecision.
ML-P-087	Time/timeline extraction - SRC-055 p.84	NEW	SRC-055 p.84	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: timeline. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.84; page-level extraction from terms: timeline.
ML-P-088	Time/timeline extraction - SRC-055 p.98	NEW	SRC-055 p.98	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: HTTP Range. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.98; page-level extraction from terms: HTTP Range.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-P-089	Time/timeline extraction - SRC-055 p.126	NEW	SRC-055 p.126	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: CitationValidationReport. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.126; page-level extraction from terms: CitationValidationReport.
ML-P-090	Time/timeline extraction - SRC-055 p.155	NEW	SRC-055 p.155	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: timeline. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.155; page-level extraction from terms: timeline.
ML-P-091	Time/timeline extraction - SRC-055 p.156	NEW	SRC-055 p.156	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: timeline. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.156; page-level extraction from terms: timeline.
ML-P-092	Time/timeline extraction - SRC-055 p.237	NEW	SRC-055 p.237	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.237; page-level extraction from terms: filters.
ML-P-093	Time/timeline extraction - SRC-055 p.242	NEW	SRC-055 p.242	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.242; page-level extraction from terms: filters.
ML-P-094	Time/timeline extraction - SRC-055 p.249	NEW	SRC-055 p.249	Timeline and temporal filtering need explicit time windows, versioned slices, and stable time-index manifests. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.249; page-level extraction from terms: MapLibre.

5.Q. Q. Sensitive Geometry, Geoprivacy, Rights, and Policy idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-001	Sensitive geometry cannot be hidden only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-002	Public-safe transforms need receipts Q.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-003	Rights unknown blocks release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-004	Restricted/internal tiles stay off public	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-005	PolicyDecision precedes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-006	Domain-sensitive lanes influence map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-007	Generalized sensitive geometry needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-008	Sensitive geometry transforms should be	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-009	Sensitive occurrence records must carry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-010	GeoPrivacy record shape binds DwC	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-011	Authorized unmasking requires signed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-012	Embargo no-public profiles block public	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-013	Masked geometry assets should be	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-014	Redaction profiles are deterministic	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-015	Differential privacy belongs to aggregates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-016	Sensitivity policy maps to radius grid	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-017	No redaction profile means deny publish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-018	Revocation issues tombstone new	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-019	Public map uses generalized_geometry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-020	Redaction params and sensitivity labels	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-021	Living-person and DNA overlays publish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-022	Revocation disables overlays and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-023	Recent-death and ancestry overlays	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-024	Unknown sensitivity category fails to the	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-025	Protected records collapse to tiles or	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-026	Rare species exact locations	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-027	DNA overlays use HMAC-salted kit	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-028	Geohash precision constrained by	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-029	Consent expiry and revocation Q. Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-030	Sensitive geometry cannot be	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-031	Museum and photo rights separate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-032	NatureServe DPP precision controls are policy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-033	KDWP ERT sensitive records require	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-034	Public layers can be watershed-level	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-035	Genealogy/DNA map overlays are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-036	Living-person data is not map-default	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-037	KDOT infrastructure can have exposure risk	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-038	Readiness products fail closed on unsigned	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-039	Corrected sensor data must be labeled	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-040	Sensitive geometry cannot be hidden only by	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-041	Rights unknown blocks public map release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-042	Genealogy overlays default to coarse place/time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-043	DNA exports remain high-sensitivity short-lived	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-044	eDNA detections require chain-of-custody	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-045	Sensitive time slices may need coarse buckets	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-046	Occurrence/asset licenses must travel with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-047	Signed URLs may be needed for high-resolution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-048	Policy label travels through Q. Sensitive Geometry, Geoprivacy, Rights, NEW; CONFIRMED/ SRC-...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-049	ABSTAIN can permit low-res Q. Sensitive Geometry, Geoprivacy, Rights, NEW; CONFIRMED/ SRC-04...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-050	REVOKED_PRESENT is a Q. Sensitive Geometry, Geoprivacy, Rights, NEW; CONFIRMED/ SRC-047 Fede...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-051	Cross-source tolerance profiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-052	Source influence visibility is a Q. Sensitive Geometry, Geoprivacy, Rights, NEW; CONFIRMED/...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-053	Manifest verification is not a Q. Sensitive Geometry, Geoprivacy, Rights, NEW; CONFIRMED/ SR...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-054	Proof details may require staged	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-055	Public tiles must not rely on style	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-056	License policy hook before rendering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-057	Public-safe default for historical layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-058	Centroid-derived points are approximate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-059	Consent revocation stronger than cache	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-060	source.meta license is passive without policy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-061	Current/live feed cache denial	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-062	Rare species and fine-scale occurrence locations require public-safe geometry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-063	Location sensitivity is evaluated before tile publication not in style filters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-064	Precision tags should persist through tiles drawers and exports	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-065	Restricted license records produce DENY rather than low-friction display	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-066	Unknown rights block release until SourceDescriptor terms are verified	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-067	Generalized public geometry should be distinct from exact internal geometry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-068	Historical and cultural route geometry may require generalized display	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-069	Sensitive geometry cannot be hidden only by MapLibre styling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-070	Public-safe habitat/fauna surfaces should use derived aggregates where needed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-071	Review-required uncertainty should suppress automatic publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-072	NatureServe-style gated rare-species layers require access controls and redaction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-073	Sensitive taxa should not publish precise points	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-074	Restricted taxa can be generalized to county or ecoregion polygons	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-075	Centroid fuzzing and buffer methods need recorded transform metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-076	Public habitat/protection artifacts require sensitivity_label public/restricted/redacted	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-077	Restricted habitat geometry should become coarse footprints or buffered centroids	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-078	Redacted habitat geometry should publish metadata only plus regional envelope	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-079	Exact restricted coordinates stay internal and access-controlled	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-080	Unknown source rights block public promotion	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-081	Consent revocation restricts future access and must be reflected in downstream artifacts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-082	Style filters are not a valid sensitive-geometry protection mechanism	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Q-083	Sensitivity transforms need receipts and EvidenceBundle linkage before publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Q. Sensitive Geometry, Geoprivacy, Rights, and Policy. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Q-084	Sensitive geometry/policy extraction - SRC-054 p.143	NEW	SRC-054 p.143	Sensitive geometry, unknown rights, and unmet obligations must be denied, generalized, quarantined, or abstained. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.143; page-level extraction from terms: style.
ML-Q-085	Sensitive geometry/policy extraction - SRC-054 p.305	NEW	SRC-054 p.305	Sensitive geometry, unknown rights, and unmet obligations must be denied, generalized, quarantined, or abstained. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.305; page-level extraction from terms: MapLibre.
ML-Q-086	Sensitive geometry/policy extraction - SRC-055 p.110	NEW	SRC-055 p.110	Sensitive geometry, unknown rights, and unmet obligations must be denied, generalized, quarantined, or abstained. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.110; page-level extraction from terms: filters.
ML-Q-087	Sensitive geometry/policy extraction - SRC-055 p.280	NEW	SRC-055 p.280	Sensitive geometry, unknown rights, and unmet obligations must be denied, generalized, quarantined, or abstained. Evidence page also references: sensitive geometry. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.280; page-level extraction from terms: sensitive geometry.
ML-Q-088	Sensitive geometry/policy extraction - SRC-055 p.287	NEW	SRC-055 p.287	Sensitive geometry, unknown rights, and unmet obligations must be denied, generalized, quarantined, or abstained. Evidence page also references: sensitive geometry. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.287; page-level extraction from terms: sensitive geometry.

5.R. R. Plugin, Wrapper, and Dependency Governance idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-R-001	Plugin allowlist required R. Plugin,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-002	PMTiles protocol adapter needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-003	React wrappers are implementation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-R-004	Mapbox-era code/tooling requires review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-005	Package versions and SBOMs are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-006	Map and provenance components need R.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-007	GeoParquet PMTiles GDAL Tippecanoe	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-008	MapLibre verifier plugin admission needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-009	PMTiles protocol and verification plugin	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-010	Cloud watcher dependencies are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-011	Evidence tooltip package is	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-012	Mapbox-style concepts are mental	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-R-013	PMTiles plugin admission needs allowlist review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-014	Protocol handlers need audit and rollback plan	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-015	Tile tooling versions must be pinned	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-016	PMTiles protocol plugin belongs on allowlist	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-017	COG protocol plugins need explicit review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-018	Signature verifier abstraction avoids browser	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-019	Verified worker helper is for custom KFM	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-020	Browser verifier implementation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-021	Sigstore JS browser path needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-R-022	Cesium tiler is an adapter, not R. Plugin, Wrapper, and Dependency NEW; CONFIRMED/ SRC-046 U...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-023	three.js / 3d-tiles-renderer R. Plugin, Wrapper, and Dependency NEW; CONFIRMED/ SRC-046 MapL...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-024	PMTiles protocol handler is dependency boundary	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-025	Source-meta TypeScript helper surface	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-026	Worker message interface is an adapter contract	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-027	pmtiles CLI in CI is toolchain dependency	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-028	cosign installer is signing dependency	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-029	PMTiles protocol handler requires dependency and plugin allowlist review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-030	Vector-tile worker libraries require package pinning and security review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-R-031	Cosign and signature tooling need CI pinning before release reliance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-032	MapLibre and Cesium release updates require compatibility tests before adoption	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-033	Wrapper components should not obscure governed API boundaries	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-034	PMTiles protocol adapter use should be allowlisted and version-pinned	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-035	Tippecanoe PMTiles output should be pinned and regression-tested	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-036	Mapshaper simplification and topology steps should be pinned in build specs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-037	Client-side verification plugins need digest and signature checks before trust badges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-038	Heavy filtering or timeline work should move into background workers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-039	MapLibre/Cesium compatibility checks remain required before overlay synchronization	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under R. Plugin, Wrapper, and Dependency Governance. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-R-040	PMTiles protocol handlers require dependency admission	NEW	SRC-055 pp.36-38	The MapLibre client imports PMTiles reader/protocol behavior to mount archives. Impl: Create a dependency admission record for pmtiles JS, protocol adapters, and worker code. Validation: Security, license, version, and fail-closed tests before merge. Evidence: SRC-055 pp.36-38; cross-page normalized discovery.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-R-041	Cesium MVT adapters are prototype-only until hardened	NEW	SRC-055 pp.40-43	The source marks community MVT imagery providers and MapLibre canvas overlays as quick but more complex. Impl: Keep adapters on a reviewed allowlist with maturity labels. Validation: Test CORS, Range, memory, render drift, and evidence identity. Evidence: SRC-055 pp.40-43; cross-page normalized discovery.
ML-R-042	Browser verification libraries need explicit trust roots	NEW	SRC-055 pp.80-86	The source discusses @sigstore/core, sigstore-browser, SubtleCrypto, JWK, and DSSE paths. Impl: Dependency governance must include trust-root configuration and bundle format decisions. Validation: Negative tests for invalid signature, wrong key, stale bundle, and policy mismatch. Evidence: SRC-055 pp.80-86; cross-page normalized discovery.

5.S. S. Accessibility, UX, and Trust-Visible States idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-001	Trust badges are first-class UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-002	Non-color trust cues and contrast	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-003	Keyboard navigation and focus order	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-004	Visible negative states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-005	Progressive heavy-layer loading	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-006	Telemetry informs UX without	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-007	Feed-card/update panel shows S. S. Accessibility, UX, and Trust-Visible States	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-008	Story Node abstain template surfaces	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-009	PR Friday Snapshot makes proof and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-010	Confidence badges and uncertainty sliders	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-011	Keyboard and ARIA requirements are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-012	License chips link to DCAT license and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-013	Telemetry for evidence interactions must	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-014	Status chips need keyboard and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-015	Evidence Drawer is an ARIA dialog with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-016	Time-slice preview is a tooltip with focus	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-017	Deep links round-trip to exact time slice in	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-018	Tooltip hover/focus keyboard	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-019	Drawer dialog focus management	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-020	No color-alone trust styling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-021	Steward bulk actions require	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-022	Trust legend explains map styling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-023	Trust-visible readiness states belong on the	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-024	Validation summaries can become UI chips	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-025	Public/steward mode differences must be visible	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-026	Negative states are first-class UI states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-027	Map UX must make provenance discoverable	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-028	Badge cluster defines user-visible trust states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-029	Full asset verification needs progress UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-030	Degraded isolation must be explicit	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-031	Verification details support reviewer inspection	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-032	Unknown is not failure but prevents authoritative	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-033	Map UI should remain smooth during trust work	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-034	Trust chip maps receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-035	Trust chip should be always	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-036	Evidence drawer toggle belongs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-037	UI state determines map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-038	Federated trust panel uses	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-039	Timeline rail visualizes trust	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-040	Placeholders are acceptable	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-041	Inspectable disagreement is a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-042	Short digest display	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-043	Freshness chip from source metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-044	Epoch selector as trust-visible control	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-045	Compare/review/focus share state	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-046	Historical confidence visible in story surfaces	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-047	Layer trust badges should show verified invalid and missing states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-048	Stale source and provisional source states must be visible in the shell	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-049	Keyboard-accessible drawer and badge controls are required for trust visibility	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-050	Drilldown from density map to atomic observation should preserve support counts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-051	Time slider state should be perceivable and not only visual	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-052	Error and deny states should use consistent finite outcome language	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-053	User-facing layer catalog should distinguish observed modeled and regulatory layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-054	Exports should preserve visible uncertainty and citation state	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-055	Abstention must be visible and not hidden behind a map popup	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-056	Trust badges should reflect receipt, proof, catalog, and release status separately	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-057	Legends should disclose units, uncertainty, and known limitations	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-058	Freshness chips should show fresh/stale/unknown status from source metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-059	Color ramps should be documented and not imply false certainty	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-060	Users should see provisional status for detections that have not survived promotion	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-061	Keyboard-accessible Evidence Drawer and Focus Mode remain required for trust surfaces	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-S-062	Public story maps should use simplified artifacts that keep attribution and sensitivity visible	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under S. Accessibility, UX, and Trust-Visible States. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-S-063	No-data and unverifiable states are UI requirements	NEW	SRC-054 pp.39,49-52; SRC-055 pp.274-276	The sources call for missing-tile no-data states, verification fallback, policy mismatch denial, and stale release notices. Impl: Map shell components need explicit trust-state displays. Validation: Accessibility and visual-regression tests must cover no-data, stale, deny, abstain, and error states. Evidence: SRC-054 pp.39,49-52; SRC-055 pp.274-276; cross-page normalized discovery.
ML-S-064	Reviewer ergonomics protects governance quality	NEW	SRC-054 p.52	The corpus says review must show failure reasons, geometry, metadata, and evidence together. Impl: Design review console around evidence, geometry, policy, and receipt co-inspection. Validation: User tests should measure review task completion and traceability. Evidence: SRC-054 p.52; cross-page normalized discovery.

5.T. T. Performance, Caching, CDN, Range Requests, and Resource Timing idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-001	Tile load budget T. Performance, Caching, T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-002	HTTP Range and CORS gates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-003	CDN cache is release-governed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-004	Low-quality transition tiles for Story Nodes T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-005	Resource timing and interaction latency T. T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-006	Tile diff materiality gate T. Performance, T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-007	Registry/CDN delivery still needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-008	Story layer performance budget is explicit T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-009	Conditional fetch with HEAD/Range and T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-010	trace_id links telemetry, receipts,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-011	Debounce/coalesce windows reduce noisy T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-012	Conditional HEAD ETag Last-Modified T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-013	HTTP Range support is hard gate for COG T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-014	Byte-level output regression protects T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-015	Drawer latency stages load summary T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-016	GeoParquet H3/time partitioning and bbox T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-017	Tile overlay chains have a performance T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-018	Digest-cached verification reduces T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-019	Cache invalidation after revocation is T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-020	CDN cache bust keys on spec_hash, not T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-021	STAC/COG harvest improves cloud T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-022	Spec-hash no-op updates reduce	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-023	Revocation requires cache T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-024	EvidenceRefs include source T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-025	ETag and Last-Modified drive cheap change T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-026	GeoParquet spatial ordering improves scan T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-027	COG overviews support fast rollups T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-028	Skip reprocessing when spec_hash unchanged T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-029	Sample tile fetch is a performance and T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-030	Cache invalidation follows rollback and release T. Performance, Caching, CDN, Range Requests...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-031	Tile load budget remains open but required T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-032	Range/CORS behavior needs preflight tests T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-033	Conditional GETs reduce map artifact churn T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-034	Lean per-tile metadata improves selection T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-035	Worker hashing avoids render-thread jank T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-036	WASM incremental hashing may be needed for T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-037	Same-origin hosting avoids most COEP/CORS T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-038	CDN origins require explicit CORS/CORP T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-039	Low-CPU devices need opt-out or slow-path T. Performance, Caching, CDN, Range Requests, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-040	Tile cache and request throttling matter for time T. Performance, Caching, CDN, Range Reques...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-041	Worker verification keeps T. Performance, Caching, CDN, Range NEW; CONFIRMED/ SRC-047 Use Wo...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-042	Root digest before signature is a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-043	Single-pass blending protects T. Performance, Caching, CDN, Range NEW; CONFIRMED/ SRC-046 p....	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-044	Source/layer minimization T. Performance, Caching, CDN, Range NEW; CONFIRMED/ SRC-046 p.454...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-045	Cache reuse depends on T. Performance, Caching, CDN, Range NEW; CONFIRMED/ SRC-046 Stable ep...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-046	Low-res prefetch improves first T. Performance, Caching, CDN, Range NEW; CONFIRMED/ SRC-046...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-047	GPU memory must be explicitly	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-048	Verified manifest cache skips T. Performance, Caching, CDN, Range NEW; CONFIRMED/ SRC-047 p....	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-049	Per-tile hashing is T. Performance, Caching, CDN, Range NEW; CONFIRMED/ SRC-047 p.329 Hash p...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-050	Coalesced style updates reduce	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-051	PMTiles header root index hot cache	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-052	Prewarm first zoom levels	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-053	Immutable HTTP caching for closed epochs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-054	Off-main-thread digest verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-055	Epoch bundling reduces layer churn	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-056	Prebucketed PMTiles reduces re-styling per frame	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-057	Worker PMTiles index parsing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-058	Hit-testing filter efficiency	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-059	Main-thread map interactions should target less than 16 ms of work	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-060	Interactive vector-tile payloads should be budgeted around 50-150 KB where practical	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-061	Worker decoding reduces UI jank during time filtering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-062	COG and PMTiles hosting require HTTP Range and CORS behavior verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-063	PMTiles and COG manifests should be tiny enough to fetch before layer load	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-064	Prefetching should be bounded for Cesium time slices	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-065	Cache invalidation must be recorded when manifests or tile bundles change	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-066	No-op spec_hash runs reduce unnecessary rebuilds and tile churn	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-067	Range request failures should degrade to ABSTAIN or disabled layer not silent error	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-068	Layer load budget should include manifest verification cost	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-069	Client-filter mode trades smaller release footprint for larger tile payload and CPU	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-070	Server-baked mode trades storage and releases for faster draws	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-071	Precompute dominant area keys before map service or tile generation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-072	Use zoom-specific geometry simplification to preserve performance and avoid false precision	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-073	PMTiles Range delivery can reduce request overhead and latency	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-074	COG overviews reduce raster transfer by serving only needed pixels	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-075	Per-tile decode time and memory footprint should be capped	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-076	Heavy decoding, filtering, and timelines should move to workers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-077	Prune unused attributes and long strings before tiling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-078	TopoJSON story packs need quantization and simplification regimes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-079	PMTiles full-state exploration should use attribute trimming and per-zoom simplification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-080	Chunked raster processing should use fixed windows with overlap for edge effects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-081	Tile load budget should include both vector and raster artifacts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-082	Cache invalidation should be tied to spec_hash/release manifest rather than style edits	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under T. Performance, Caching, CDN, Range Requests, and Resource Timing. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-T-083	Performance/cache extraction - SRC-054 p.50	NEW	SRC-054 p.50	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: markers. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.50; page-level extraction from terms: markers.
ML-T-084	Performance/cache extraction - SRC-054 p.83	NEW	SRC-054 p.83	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: style, markers. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.83; page-level extraction from terms: style, markers.
ML-T-085	Performance/cache extraction - SRC-054 p.87	NEW	SRC-054 p.87	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: CDN. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.87; page-level extraction from terms: CDN.
ML-T-086	Performance/cache extraction - SRC-054 p.222	NEW	SRC-054 p.222	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: MapLibre, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.222; page-level extraction from terms: MapLibre, zoom.
ML-T-087	Performance/cache extraction - SRC-054 p.315	NEW	SRC-054 p.315	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: MapLibre. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.315; page-level extraction from terms: MapLibre.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-T-088	Performance/cache extraction - SRC-055 p.82	NEW	SRC-055 p.82	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: MapLibre, style, maplibre-gl, maplibregl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.82; page-level extraction from terms: MapLibre, style, maplibre-gl, maplibregl.
ML-T-089	Performance/cache extraction - SRC-055 p.190	NEW	SRC-055 p.190	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: CORS. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.190; page-level extraction from terms: CORS.
ML-T-090	Performance/cache extraction - SRC-055 p.233	NEW	SRC-055 p.233	Range requests, CORS, CDN caching, workers, memory caps, and latency budgets are part of release readiness. Evidence page also references: zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.233; page-level extraction from terms: zoom.

5.U. U. Testing, CI, Validation, Rollback, and Proof Objects idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-001	Schema validation for all contracts U.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-002	Style source/layer validity tests U. Testing,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-003	PMTiles/MVT/MLT manifest checks U.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-004	COG range request tests U. Testing, CI,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-005	No public RAW path test U. Testing, CI,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-006	No direct model client test	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-007	No unreleased tile load test	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-008	Sensitive geometry deny test U. Testing,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-009	Stale source badge and abstain test U.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-010	Click resolves EvidenceBundle test U.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-011	Citation validation pass/fail test U. Testing,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-012	Rollback restores prior manifest and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-013	Visual regression for styles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-014	Accessibility smoke tests U. Testing, CI,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-015	Deterministic tile build tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-016	Mini provenance fixtures cover imagery,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-017	OCI artifact gates are release blockers U.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-018	Vector_min fixture proves feature count	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-019	Rego v1/OPA 1.0 migration affects map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-020	Dataset Promotion Gate checklist is a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-021	COG/GeoParquet/STAC validation covers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-022	Story Node Go-Live audit gates every	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-023	Run-receipt schema and in-toto/SLSA	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-024	Sensitive occurrence OPA policy blocks	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-025	OTel trace receipt probe gates trace to	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-026	Auto-revert PRs preserve evidence and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-027	STAC linter extension validator and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-028	GeoParquet schema and COG layout	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-029	PMTiles conversion validator and byte	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-030	ConfTest OPA maps sensitivity to redaction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-031	CI fails closed on catalog proof and rights	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-032	AQS revision reconciliation requires	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-033	Receipt Drawer tests cover STAC-only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-034	Materialization receipts enforce DQ	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-035	PMTiles overlay release tests rehydrate,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-036	Forward plus reverse patch idempotence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-037	STAC-to-PMTiles gates check TileJSON,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-038	Watcher preflight rejects unsigned,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-039	Cloud watcher tests assert no duplicate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-040	Redaction and generalization golden tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-041	PMTiles claim verification tests reject	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-042	Accessibility tests cover chips, drawers,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-043	SWI product tests cover normalization,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-044	All published features have	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-045	GeoJSON coordinate and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-046	Spec hash recomputation tests U. U. Testing, CI, Validation, Rollback, and Proof Objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-047	MapLibre tooltip integration tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-048	Trust styling palette and decision	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-049	Mode/focus filter contract tests U. U. Testing, CI, Validation, Rollback, and Proof Objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-050	Overlay schema and semantic	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-051	Source URI resolve-or-quarantine	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-052	Temporal uncertainty publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-053	Public/steward rendering	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-054	Geometry validity is a release gate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-055	Attribute range and null checks are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-056	mukey and musym uniqueness must be	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-057	COG quick-check validates tiling, overviews,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-058	GeoParquet schema and stats are required	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-059	PMTiles metadata and tile fetch must pass	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-060	STAC/DCAT/PROV schema validation is	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-061	Signatures must be verified in CI before	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-062	Reproducibility rebuild compares hashes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-063	OPA/Confest gates enforce no missing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-064	NDVI promotion gate includes coverage and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-065	Station anomaly artifacts survive gating	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-066	Visual regression remains required for style	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-067	No direct model client is still a validation target	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-068	Rollback restores prior manifest, not just prior	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-069	Source ledger completeness is a map release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-070	Geometry hash diff emits reviewer summaries	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-071	HEAD probes plus idempotency keys support	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-072	OPA gate must fail closed before side effects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-073	Cosign/signature verification gates release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-074	Catalog matrix boolean is not enough	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-075	Manifest spec_hash mismatch must fail closed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-076	Asset sha256 mismatch must fail closed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-077	Missing asset hash blocks deep verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-078	Cross-origin isolation can be tested in	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-079	DecisionEnvelope freshness status supports	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-080	PolicyDecision and QuarantineRecord linkage	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-081	Finite outcome schemas prevent free-text	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-082	No direct browser model client remains a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-083	No public raw path applies to source service	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-084	Policy gate must fail closed U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-0...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-085	Manifest no-overwrite rule U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-046...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-086	Receipt/manifest disagreement	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-087	Catalog closure can HOLD or U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-04...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-088	AI-mediated synthesis requires U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-089	Correction candidate needs U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-046...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-090	Promotion Gate Summary is a U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-04...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-091	Bad checksum must flow to U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-046...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-092	DecisionEnvelope signature U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-046...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-093	Worker digest mismatch returns	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-094	Worker exceptions return U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-047 W...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-095	Tile verification schema is U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-04...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-096	CI must replay browser U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-047 Pro...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-097	Proof pack completeness blocks	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-098	PromotionDecision and U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-047 CI P...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-099	EvidenceTimeline schema U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-047 Ti...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-100	Revoked release must never U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-047...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-101	Trust federation evaluator filters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-102	REVOKED_PRESENT test is U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-047 p....	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-103	Conflict type tests cover U. Testing, CI, Validation, Rollback, and NEW; CONFIRMED/ SRC-047...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-104	kfm-source-meta-verify validator	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-105	Validator exit code contract	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-106	Manifest digest mismatch denial	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-107	Missing proof ref required denial	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-108	No matching style source error	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-109	Epoch shape validation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-110	License shape validation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-111	Validator artifact hashing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-112	Helper verifySourceDigest tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-113	buildFeatureTimeFilter tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-114	PMTiles verify before publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-115	cosign sign-blob bundle validation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-116	GitHub release attaches signed tile assets	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-117	OIDC keyless signing permission	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-118	Schema validation for TileArtifactManifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-119	Style and manifest local/CI parity	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-120	Regression tests for source meta fixtures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-121	Historical tile pipeline validation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-122	Soil artifact validation should check PMTiles COG receipts and schemas	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-123	Soil diff reports should test dominant mukey coverage area mismatch and hydrp shift	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-124	Missing provenance should fail promotion gate	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-125	Runtime proof should validate mukey lookup to EvidenceBundle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-126	Streamflow proof fixtures need ANSWER ABSTAIN DENY and ERROR cases	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-127	Focus Mode runtime proof should test missing citation policy deny and valid answer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-128	PMTiles verify/sign dry-run belongs in CI before publish	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-129	Source ledger completeness must include current-run SRC-050 and SRC-051	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-130	Style source/layer validity tests must cover time filter fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-131	Worker bucket-message contracts need schema validation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-132	Visual regression should cover verified invalid missing trust badges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-133	Accessibility tests should cover evidence drawer and time slider controls	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-134	Tile load budget tests should guard interactive payload size	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-135	Interaction-to-drawer latency budget should include governed API resolution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-136	Rollback tests should restore prior MapReleaseManifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-137	Cache invalidation record tests should accompany tile replacement	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-138	Sensitive geometry deny fixtures should prevent exact public display	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-139	Stale source badge tests should trigger on old retrieval windows	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-140	Citation validation pass/fail fixtures should affect Focus Mode outcomes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-141	No public raw path tests should scan map shell and routes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-142	No direct model client tests should scan browser bundles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-143	COG sidecar schema tests should check nodata CRS overviews and provenance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-144	PMTiles sidecar schema tests should check source version digest and release ref	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-145	Promotion bundle diff summaries should be reviewer-readable	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-146	Offline signature verification should have invalid and missing bundle fixtures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-147	Hard fail if HLS burn candidate schema is invalid	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-148	Hard fail if audit_ref is missing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-149	Hard fail if run_receipt_ref is missing	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-150	Hard fail if spec_hash is missing or mismatched	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-151	Hard fail if valid pixel coverage is below threshold	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-152	Hard fail if baseline or current composite assets are absent	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-153	Hard fail if burn promotion lacks event evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-154	Validate NDVI/NBR/dNBR candidate computation fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-155	Validate event evidence distinguishes burn from vegetation disturbance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-156	Validate source STAC and derived STAC separation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-157	Validate derived STAC includes all COGs and optional vectors	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-158	Validate PMTiles vector publish path works before runtime route	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-159	Validate runtime envelope can answer or abstain from promoted objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-160	Validate topology after every simplify operation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-161	Validate sidecar licenses and source lists before publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-162	Validate effort fields before biodiversity summaries are promoted	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-163	Validate sensitivity rules applied before exact geometry leaves internal storage	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-164	Validate receipt, input resolvability, masks, ensemble votes, metrics, and provenance at promotion	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-165	Validate confidence class and coherence radius in MVT metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-166	Validate COG compression, nodata, stats, and overviews	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-167	Validate PMTiles HTTP Range hosting and CORS where applicable	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-168	Validate signed manifest and artifact digest before trust badge display	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-169	Validate catalog.json points to artifacts, receipts, licenses, and limitations	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-170	Validate rollback restores previous manifest and invalidates derived cache	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-171	Validate no auto-publish path from detector output to public map	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under U. Testing, CI, Validation, Rollback, and Proof Objects. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-U-172	Validation/proof extraction - SRC-054 p.33	NEW	SRC-054 p.33	Schema checks, policy gates, digest verification, CI, rollback, and receipts must fail closed before publication. Evidence page also references: markers. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.33; page-level extraction from terms: markers.
ML-U-173	Validation/proof extraction - SRC-054 p.80	NEW	SRC-054 p.80	Schema checks, policy gates, digest verification, CI, rollback, and receipts must fail closed before publication. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.80; page-level extraction from terms: style.
ML-U-174	Validation/proof extraction - SRC-054 p.149	NEW	SRC-054 p.149	Schema checks, policy gates, digest verification, CI, rollback, and receipts must fail closed before publication. Evidence page also references: style. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.149; page-level extraction from terms: style.
ML-U-175	Validation/proof extraction - SRC-054 p.243	NEW	SRC-054 p.243	Schema checks, policy gates, digest verification, CI, rollback, and receipts must fail closed before publication. Evidence page also references: accessibility. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.243; page-level extraction from terms: accessibility.
ML-U-176	Validation/proof extraction - SRC-055 p.236	NEW	SRC-055 p.236	Schema checks, policy gates, digest verification, CI, rollback, and receipts must fail closed before publication. Evidence page also references: filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.236; page-level extraction from terms: filters.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-U-177	Validation/proof extraction - SRC-055 p.254	NEW	SRC-055 p.254	Schema checks, policy gates, digest verification, CI, rollback, and receipts must fail closed before publication. Evidence page also references: controls. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.254; page-level extraction from terms: controls.

5.V. V. Exports, Screenshots, Reports, and Citation Preservation idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-V-001	Exports preserve citations and release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-002	Screenshot is not proof V. Exports,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-003	Export denial states are first-class	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-004	External attribution travels with export V.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-005	Verified digest badge links exports and UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-006	Renderer/export preflight writes redaction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-007	Reviewer/export snapshots should carry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-008	Exports and screenshots include receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-V-009	Recommended citation travels with STAC	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-010	Historical map exports preserve	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-011	Copy citation and copy receipt are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-012	Uncited timeline cards are blocked or	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-013	Exports retain run receipt links	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-014	STAC severe-weather tiles keep	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-015	Exports must preserve STAC/DCAT/PROV links	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-016	Readiness sidecar is a compact export artifact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-017	Uncited screenshots are denied by policy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-V-018	QA reports should be exportable metadata	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-019	Focus Mode reports cite receipts and evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-020	Exports should preserve manifest and proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-021	STAC/DCAT/PROV provide citation-ready	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-022	Run receipts support rollback notes in reports	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-023	Evidence assets can include photos/specimen	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-024	Promotion summaries support V. Exports, Screenshots, Reports, and NEW; CONFIRMED/ SRC-046 Re...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-025	Trust chip state should appear in	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-026	Evidence Drawer raw receipt V. Exports, Screenshots, Reports, and NEW; CONFIRMED/ SRC-047 Dr...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-V-027	Timeline exports preserve V. Exports, Screenshots, Reports, and NEW; CONFIRMED/ SRC-047 Evid...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-028	Split-view conflict screenshots V. Exports, Screenshots, Reports, and NEW; CONFIRMED/ SRC-04...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-029	STAC/DCAT/PROV after PMTiles build	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-030	Signed PMTiles enable auditable rollbacks	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-031	Geometry digest enters evidence bundle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-032	Lineage manifests cite runtime decisions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-033	Uncataloged screenshots remain unsupported	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-034	Exports should carry manifest digest spec_hash and citation set	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-035	Screenshots should preserve layer release and verification state	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-V-036	Dossier and story rebuild refs should appear in correction notices	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-037	Uncited exports must be blocked or marked ABSTAIN	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-038	Review surfaces should show supersede and withdraw correction notices	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-039	Public notes should accompany withdrawn or superseded map releases	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-040	Export citation governance should follow the same EvidenceBundle as the drawer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-041	Reviewer summary artifact should be generated separately from candidate and receipt	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-042	Exported map products should carry EvidenceBundle and run receipt links	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-043	Screenshots and reports should preserve release manifest and spec_hash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-044	Story-map Geo/TopoJSON artifacts should retain provenance_uri and source_license	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-V-045	Catalog json should be copied with published artifacts for citation continuity	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-046	Exports should include known limitations and uncertainty layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-047	Hazard reports should distinguish burn detection from generic vegetation disturbance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-048	Public exports must not include precise restricted geometry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under V. Exports, Screenshots, Reports, and Citation Preservation. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-V-049	Uncited exports are not acceptable release artifacts	NEW	SRC-054 pp.49-50; SRC-055 p.111	The promotion checklist and catalog completeness rules require EvidenceBundle, run receipts, obligations, and citations before release. Impl: Exports and screenshots must preserve source, release, style, and evidence metadata. Validation: Export tests must fail on missing citations or unresolved EvidenceRefs. Evidence: SRC-054 pp.49-50; SRC-055 p.111; cross-page normalized discovery.
ML-V-050	Catalog closure is mandatory before reports or screenshots	NEW	SRC-054 p.50; SRC-055 p.240	Current sources state missing DCAT/STAC/PROV or packaging data should fail release. Impl: Report generation must depend on catalog closure, not just rendered map state. Validation: Test generated report metadata against catalog and release manifests. Evidence: SRC-054 p.50; SRC-055 p.240; cross-page normalized discovery.

5.W. W. 3D / Cesium / Deck.gl / Overlay Interoperability idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-001	2D-first default W. 3D / Cesium / Deck.gl /	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-002	3D/globe only when evidence-bearing W.	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-003	Cesium synchronization needs manifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-004	deck.gl overlays are visual analytics, not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-005	3D Tiles/glTF/CZML are downstream	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-006	MapLibre/Cesium shared artifacts remain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-007	Cesium 3D tiles for point clouds are	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-008	3D and terrain layers require same	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-009	MapLibre Cesium overlay sync preserves	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-010	MapLibre or Leaflet time-slider imagery	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-011	Cesium, 3D, or alternate visual contexts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-012	Vertical or altitude migration layers need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-013	2D map first; alternate GIS/3D W. W. 3D / Cesium / Deck.gl / Overlay Interoperability	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-014	2D remains the default for governed public	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-015	3D outputs require the same EvidenceBundle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-016	Deck.gl/Cesium overlays remain conditional	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-017	Terrain/hillshade should not imply field truth	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-018	Cesium Clock is conditional time controller for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-019	3D Tiles availability can gate downloads by time	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-020	Cesium asset sprawl requires stricter packaging	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-021	MapLibre and Cesium need separate runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-022	3D terrain remains conditional evidence context	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-023	Overlay synchronization must derive from one	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-024	Unified epoch source works W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-046 One so...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-025	Cesium 3D Tileset remains W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-046 Cesium...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-026	3D Tiles 1.1 can be internal W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-046 Adop...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-027	Cesium CustomShader is a W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-046 CustomSh...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-028	MapLibre custom WebGL layer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-029	Cesium Resource wrapper can	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-030	Cross-engine manifest W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-046;SRC-04 Mani...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-031	Terrain/imaging tiler adapters W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-046 Ce...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-032	3D hydrology surfaces should W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-046 p.46...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-033	Deck.gl remains a candidate W. 3D / Cesium / Deck.gl / Overlay NEW; CONFIRMED/ SRC-MASTER-P...	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-034	Cesium CZML as time-aware sibling output	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-035	3D Tiles extras carry provenance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-036	3D Tiles extras use spec-safe extension area	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-037	Tile to catalog linkage in 3D	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-038	3D tile time semantics	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-039	Cesium availability for vector entities	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-040	3D Tiles style swap by epoch	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-041	Do not reload scene for every time change	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-042	2D default, 3D conditional remains intact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-043	3D tileset epoch naming requires manifest tie	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-044	Cesium CZML time slicing should keep active entities under budget	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-045	3D Tiles can carry provenance hooks for static heavy geometry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-046	Cesium overlays need the same manifest and evidence controls as MapLibre layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-047	3D timelines remain conditional rather than default for KFM	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-048	MapLibre and Cesium compatibility testing is required after upstream releases	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-049	3D scenes should not weaken citation or policy gates	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-050	CZML slices should be independently valid with clock definitions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-051	Deck.gl remains a candidate overlay only after boundary and evidence tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-052	Cesium is appropriate for terrain-aware burn visualization only when evidence needs 3D context	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-053	Cesium can show plume or smoke overlays with temporal playback	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-054	3D terrain and globe views remain conditional and downstream of released evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-055	3D asset compression and texture packing need performance and provenance controls	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-056	MapLibre and Cesium overlay synchronization must preserve the same evidence payloads	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-057	2D shell remains the default when 3D adds visual complexity without evidence gain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-058	Deck.gl or 3D overlays need shared manifest and rollback discipline	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-059	3D scenes are downstream carriers and must not become proof objects	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under W. 3D / Cesium / Deck.gl / Overlay Interoperability. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-W-060	3D/Cesium/deck extraction - SRC-054 p.38	NEW	SRC-054 p.38	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, Evidence Drawer, Focus Mode, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.38; page-level extraction from terms: MapLibre, Evidence Drawer, Focus Mode, 3D.
ML-W-061	3D/Cesium/deck extraction - SRC-054 p.63	NEW	SRC-054 p.63	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.63; page-level extraction from terms: 3D.
ML-W-062	3D/Cesium/deck extraction - SRC-054 p.142	NEW	SRC-054 p.142	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.142; page-level extraction from terms: 3D.
ML-W-063	3D/Cesium/deck extraction - SRC-054 p.151	NEW	SRC-054 p.151	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.151; page-level extraction from terms: 3D.
ML-W-064	3D/Cesium/deck extraction - SRC-054 p.216	NEW	SRC-054 p.216	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, Cesium, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.216; page-level extraction from terms: MapLibre, Cesium, 3D.
ML-W-065	3D/Cesium/deck extraction - SRC-054 p.217	NEW	SRC-054 p.217	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, GeoJSON, camera, zoom, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.217; page-level extraction from terms: MapLibre, GeoJSON, camera, zoom, Cesium.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-066	3D/Cesium/deck extraction - SRC-054 p.219	NEW	SRC-054 p.219	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: GeoJSON, zoom, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.219; page-level extraction from terms: GeoJSON, zoom, Cesium.
ML-W-067	3D/Cesium/deck extraction - SRC-054 p.220	NEW	SRC-054 p.220	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: GeoJSON, camera, zoom, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.220; page-level extraction from terms: GeoJSON, camera, zoom, Cesium.
ML-W-068	3D/Cesium/deck extraction - SRC-054 p.221	NEW	SRC-054 p.221	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, maplibregl, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.221; page-level extraction from terms: MapLibre, maplibregl, Cesium.
ML-W-069	3D/Cesium/deck extraction - SRC-054 p.223	NEW	SRC-054 p.223	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: zoom, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.223; page-level extraction from terms: zoom, Cesium.
ML-W-070	3D/Cesium/deck extraction - SRC-054 p.224	NEW	SRC-054 p.224	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: Focus Mode, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.224; page-level extraction from terms: Focus Mode, 3D.
ML-W-071	3D/Cesium/deck extraction - SRC-054 p.225	NEW	SRC-054 p.225	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, GeoJSON, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.225; page-level extraction from terms: MapLibre, style, GeoJSON, Cesium.
ML-W-072	3D/Cesium/deck extraction - SRC-054 p.226	NEW	SRC-054 p.226	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, Evidence Drawer, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.226; page-level extraction from terms: MapLibre, Evidence Drawer, Cesium.
ML-W-073	3D/Cesium/deck extraction - SRC-054 p.241	NEW	SRC-054 p.241	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: terrain, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.241; page-level extraction from terms: terrain, 3D.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-074	3D/Cesium/deck extraction - SRC-054 p.242	NEW	SRC-054 p.242	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: terrain, hillshade, 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.242; page-level extraction from terms: terrain, hillshade, 3D.
ML-W-075	3D/Cesium/deck extraction - SRC-054 p.312	NEW	SRC-054 p.312	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-054 p.312; page-level extraction from terms: MapLibre, Cesium.
ML-W-076	3D/Cesium/deck extraction - SRC-055 p.151	NEW	SRC-055 p.151	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, terrain, maplibregl, filters. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.151; page-level extraction from terms: MapLibre, style, terrain, maplibregl, filters.
ML-W-077	3D/Cesium/deck extraction - SRC-055 p.157	NEW	SRC-055 p.157	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, maplibregl, camera, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.157; page-level extraction from terms: MapLibre, style, maplibregl, camera, zoom.
ML-W-078	3D/Cesium/deck extraction - SRC-055 p.158	NEW	SRC-055 p.158	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, maplibregl, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.158; page-level extraction from terms: MapLibre, style, maplibregl, Cesium.
ML-W-079	3D/Cesium/deck extraction - SRC-055 p.159	NEW	SRC-055 p.159	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, terrain, camera, pitch, timeline. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.159; page-level extraction from terms: MapLibre, terrain, camera, pitch, timeline.
ML-W-080	3D/Cesium/deck extraction - SRC-055 p.160	NEW	SRC-055 p.160	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, GeoJSON, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.160; page-level extraction from terms: MapLibre, style, GeoJSON, Cesium.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-081	3D/Cesium/deck extraction - SRC-055 p.161	NEW	SRC-055 p.161	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, maplibregl, camera, zoom. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.161; page-level extraction from terms: MapLibre, style, maplibregl, camera, zoom.
ML-W-082	3D/Cesium/deck extraction - SRC-055 p.162	NEW	SRC-055 p.162	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, zoom, pitch, bearing. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.162; page-level extraction from terms: MapLibre, camera, zoom, pitch, bearing.
ML-W-083	3D/Cesium/deck extraction - SRC-055 p.163	NEW	SRC-055 p.163	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: terrain, camera, Cesium, globe. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.163; page-level extraction from terms: terrain, camera, Cesium, globe.
ML-W-084	3D/Cesium/deck extraction - SRC-055 p.164	NEW	SRC-055 p.164	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, zoom, Cesium, globe. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.164; page-level extraction from terms: MapLibre, camera, zoom, Cesium, globe.
ML-W-085	3D/Cesium/deck extraction - SRC-055 p.165	NEW	SRC-055 p.165	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, zoom, pitch, bearing. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.165; page-level extraction from terms: MapLibre, camera, zoom, pitch, bearing.
ML-W-086	3D/Cesium/deck extraction - SRC-055 p.166	NEW	SRC-055 p.166	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.166; page-level extraction from terms: MapLibre, camera, Cesium.
ML-W-087	3D/Cesium/deck extraction - SRC-055 p.167	NEW	SRC-055 p.167	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, terrain, maplibregl, camera, pitch. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.167; page-level extraction from terms: MapLibre, terrain, maplibregl, camera, pitch.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-088	3D/Cesium/deck extraction - SRC-055 p.168	NEW	SRC-055 p.168	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, vector tile, GeoJSON, controls, camera. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.168; page-level extraction from terms: MapLibre, vector tile, GeoJSON, controls, camera.
ML-W-089	3D/Cesium/deck extraction - SRC-055 p.197	NEW	SRC-055 p.197	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, zoom, pitch, bearing. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.197; page-level extraction from terms: MapLibre, camera, zoom, pitch, bearing.
ML-W-090	3D/Cesium/deck extraction - SRC-055 p.198	NEW	SRC-055 p.198	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, maplibre-gl, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.198; page-level extraction from terms: MapLibre, style, maplibre-gl, Cesium.
ML-W-091	3D/Cesium/deck extraction - SRC-055 p.199	NEW	SRC-055 p.199	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, terrain, maplibregl, timeline. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.199; page-level extraction from terms: MapLibre, style, terrain, maplibregl, timeline.
ML-W-092	3D/Cesium/deck extraction - SRC-055 p.200	NEW	SRC-055 p.200	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, GeoJSON, camera, pitch. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.200; page-level extraction from terms: MapLibre, style, GeoJSON, camera, pitch.
ML-W-093	3D/Cesium/deck extraction - SRC-055 p.201	NEW	SRC-055 p.201	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, style, maplibregl, camera, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.201; page-level extraction from terms: MapLibre, style, maplibregl, camera, Cesium.
ML-W-094	3D/Cesium/deck extraction - SRC-055 p.202	NEW	SRC-055 p.202	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, zoom, pitch, bearing. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.202; page-level extraction from terms: MapLibre, camera, zoom, pitch, bearing.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-W-095	3D/Cesium/deck extraction - SRC-055 p.203	NEW	SRC-055 p.203	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: terrain, camera, Cesium, globe. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.203; page-level extraction from terms: terrain, camera, Cesium, globe.
ML-W-096	3D/Cesium/deck extraction - SRC-055 p.204	NEW	SRC-055 p.204	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: camera, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.204; page-level extraction from terms: camera, Cesium.
ML-W-097	3D/Cesium/deck extraction - SRC-055 p.205	NEW	SRC-055 p.205	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, zoom, pitch, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.205; page-level extraction from terms: MapLibre, camera, zoom, pitch, Cesium.
ML-W-098	3D/Cesium/deck extraction - SRC-055 p.206	NEW	SRC-055 p.206	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, zoom, pitch, bearing. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.206; page-level extraction from terms: MapLibre, camera, zoom, pitch, bearing.
ML-W-099	3D/Cesium/deck extraction - SRC-055 p.207	NEW	SRC-055 p.207	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, camera, pitch, Cesium. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.207; page-level extraction from terms: MapLibre, camera, pitch, Cesium.
ML-W-100	3D/Cesium/deck extraction - SRC-055 p.208	NEW	SRC-055 p.208	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: MapLibre, vector tile, terrain, GeoJSON, maplibregl. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.208; page-level extraction from terms: MapLibre, vector tile, terrain, GeoJSON, maplibregl.
ML-W-101	3D/Cesium/deck extraction - SRC-055 p.259	NEW	SRC-055 p.259	3D, Cesium, deck.gl, and globe views are conditional overlays that must preserve evidence identity and performance limits. Evidence page also references: 3D. Impl: Bind this page-level discovery to the relevant manifest, API, UI, policy, or test object before public release. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior. Evidence: SRC-055 p.259; page-level extraction from terms: 3D.

5.X. X. Anti-Patterns and Failure Modes idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-001	MapLibre as truth authority	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-002	Tiles as proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-003	Layer toggle as publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-004	Style-only sensitive geometry hiding	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-005	Dense GeoJSON production default	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-006	Uncontrolled style drift	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-007	Unreviewed plugin adoption	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-008	Direct browser model access	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-009	AI answers from rendered features only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-010	Uncited popup/export	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-011	MLT as default before KFM validation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-012	3D/globe as default	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-013	Mapbox-era backports without review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-014	Publishing raw sensitive points with jitter	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-015	Treating provisional NowCast tiles as	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-016	Catalog-missing map visibility is an	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-017	Cache not invalidated after revocation is	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-018	Ignoring byte drift in tile packaging is an	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-019	Hiding confidence controls is an	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-020	In-place editing a PMTiles monolith	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-021	Client-side overlay merge without manifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-022	Verifying TileJSON but not STAC, digest,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-023	Treating a live ArcGIS/PBF service as a	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-024	Publishing raw identity or DNA tokens as	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-025	Redaction by client style or layer toggle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-026	Untested overlay-chain growth	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-027	Map popup hiding policy or receipt details	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-028	Publicly styling denied features X. X. Anti-Patterns and Failure Modes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-029	Receipt drawer from internal	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-030	Promoting routes before evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-031	Treating COG/PMTiles previews as	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-032	Over-precise genealogy geohashes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-033	Using WMS as citation authority	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-034	Color-only policy badges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-035	AI answers from rendered features	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-036	Treating PMTiles as proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-037	Using style filters as redaction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-038	Cron-only refresh without spec_hash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-039	Publishing corrected AQ as regulatory fact	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-040	Exposing exact rare-species records	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-041	Over-precise genealogy geographies	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-042	Bypassing PR review through agent automation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-043	Dropping STAC checksums or ETags	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-044	Ignoring tile-build warnings	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-045	Using WMS imagery as citation authority	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-046	Focus Mode answering from rendered features	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-047	Publishing COGs without overviews or sidecar	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-048	Per-time layers are a failure mode	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-049	Metadata dates alone are a false change	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-050	Silent verification fallback is prohibited	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-051	Giant GeoJSON on main thread is not	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-052	Runtime headers are not source rights	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-053	Browser signature implementation details	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-054	Style filters are not sensitive-geometry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-055	Catalog closed flag without actual comparison is	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-056	Treating tiles as canonical data X. X. Anti-Patterns and Failure Modes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-057	Blindly adopting vendor tiler as X. X. Anti-Patterns and Failure Modes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-058	Putting catalog provenance only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-059	Skipping deterministic	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-060	Painting before verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-061	Assuming signature verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-062	Using logs instead of signed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-063	Hiding trust state behind UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-064	Letting revoked source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-065	Collapsing all source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-066	Saving proof cache through	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-067	Treating source.meta as proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-068	Immutable caching current/live feeds	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-069	Per-feature filtering as release proof	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-070	License text without policy enforcement	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-071	Digest without proof linkage	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-072	Dropping precision during tiling	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-073	Publishing unsigned PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-074	Assuming external tool latest is safe	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-075	3D scene as evidence substitute	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-076	Direct external map service as public layer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-077	Treating PMTiles source metadata as proof is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-078	Treating MapLibre trust badges as release authority is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-079	Using style filters as sensitive-geometry redaction is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-080	Immutable-caching current feeds without freshness semantics is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-081	Publishing MVT tiles before catalog closure is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-082	Using rendered features alone for AI answers is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-083	Unpinned vector tile or cosign tooling in release CI is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-084	Letting runtime verification and CI promotion diverge is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-085	Using dense GeoJSON as production default for large layers is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-086	Collapsing observed modeled regulatory and derived source roles is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-087	Hiding negative states to make the UI smoother is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-088	Adopting 3D/globe as default when 2D is clearer is an anti-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-089	Treating a vegetation-loss tile as a burn claim without event evidence	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-090	Treating receipts as proof, catalog closure, or publication	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-091	Treating PMTiles as proof of source authority	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-092	Auto-publishing provisional detections	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-093	Hiding abstention in a popup or drawer UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-094	Publishing exact restricted taxa or habitat geometries	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-095	Using style filters as sensitive-geometry redaction	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-096	Running expensive spatial joins at render time instead of precomputing keys	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-097	Ignoring effort bias in biodiversity maps	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-098	Unpinned tool versions in COG/PMTiles build pipelines	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-099	Letting 3D/globe visual appeal outrank evidence clarity	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-X-100	Showing trust badges before digest/signature validation succeeds	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under X. Anti-Patterns and Failure Modes. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-X-101	PMTiles is not ideal for highly transactional GIS	NEW	SRC-055 pp.272-276	The corpus says tiny updates can be expensive and real-time edits or transactional GIS are poor PMTiles-only fits. Impl: Use overlays, regional splits, or server-mediated serving for fast-changing data. Validation: Run update-size and rollback-complexity checks. Evidence: SRC-055 pp.272-276; cross-page normalized discovery.
ML-X-102	Style filters do not protect sensitive geometry	NEW	SRC-054 p.43; SRC-055 p.280	The current sources require public-safe transforms and negative tests for sensitive geometry. Impl: Apply redaction/generalization before tile publication. Validation: Test that exact sensitive geometry is absent from artifacts, not only invisible. Evidence: SRC-054 p.43; SRC-055 p.280; cross-page normalized discovery.
ML-X-103	Verification failures must not degrade into high-resolution display	NEW	SRC-055 pp.40-43,80-86	Sources call for verify-then-fallback and fail-closed rendering. Impl: Block high-res layers until integrity checks pass. Validation: Inject bad signature and digest fixtures. Evidence: SRC-055 pp.40-43,80-86; cross-page normalized discovery.
ML-X-104	3D is not the default when 2D explains the evidence	NEW	SRC-054 p.51; SRC-055 pp.40-43	Current sources allow 3D only when it conveys something 2D cannot and performance budget is respected. Impl: Keep MapLibre 2D default and require a 3D justification record. Validation: Test 3D opt-in, evidence binding, and performance budgets. Evidence: SRC-054 p.51; SRC-055 pp.40-43; cross-page normalized discovery.

5.Y. Y. Implementation Backlog and PR Plan idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-001	Real repo scan and schema-home ADR	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-002	Schema and fixture first PR	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-003	Public-safe hydrology/HUC12 map slice	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-004	Mock governed API resolver	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-005	Layer/style/tile manifests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-006	E2E click/Drawer/Focus test	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-007	Rollback/cache invalidation test	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-008	Dependency admission records	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-009	Source rights and attribution review	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-010	Documentation extraction package	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-011	Standardize media types, annotation keys,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-012	Add STAC/DCAT/PROV crosswalk and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-013	Add feed-card JSONL and governed	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-014	Add WZDx/KanDrive event watcher with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-015	Add occurrence geoprivacy schema,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-016	Add OTel trace/evidence link checker	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-017	Add soils GeoParquet COG PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-018	Add ReceiptDrawer and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-019	Add AQS AirNow reconciliation badges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-020	Add redaction profile library with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-021	Add deterministic PMTiles packaging CI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-022	Add STAC DCAT PROV drawer adapters	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-023	Add PMTiles claim schema and verifier	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-024	Add PMTiles delta ADR, runbook, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-025	Add STAC to GeoParquet to PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-026	Add time-aware layer attribute contract	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-027	Add Evidence Drawer micro-pattern	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-028	Add cloud watcher preflight fixture for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-029	Add SWI map product fixture and Focus	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-030	Add historical map COG/PMTiles pipeline	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-031	Add Evidence Tooltip package	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-032	Add Kansas Frontier Routes v0 thin	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-033	Add PMTiles preview builder with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-034	Add TrustStatus manifest and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-035	Add genealogy overlay schema and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-036	Add COG/HLS/RAP preview	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-037	Add NFHL external-service	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-038	Add severe-weather STAC-to-tile	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-039	Add map source metadata registry	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-040	Add tooltip and drawer accessibility	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-041	Add WPE agent governance docs before	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-042	Build soil/surficial GeoParquet-PMTiles fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-043	Add MapLibre PMTiles smoke page	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-044	Add readiness COG fixture and validator	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-045	Add WZDx RoadEvent ingest fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-046	Add HLS/NDVI HUC12 alert fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-047	Add KDWP ERT source descriptor and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-048	Add KDOT infrastructure descriptor and drawer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-049	Add STAC watcher-signal extension fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-050	Add cache and hosting preflight tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-051	Build WBD HUC12 probe fixture first	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-052	Build time-series MapLibre fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-053	Build KFM integrity control behind manifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-054	Add worker hashing proof-of-concept	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-055	Add explorer-web security headers document	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-056	Add catalog matrix validator to map release	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-057	Add source descriptor validators for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-058	Add sensitive overlay sidecar fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-059	Add PMTiles/COG manifest generator	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-060	Add promotion gate schema to map release CI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-061	Build schema fixture test slice	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-062	Add tile verification contract	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-063	Create stable tile proof verifier	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-064	Add browser/worker runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-065	Add viewer trust chip and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-066	Integrate Promotion Gate with	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-067	Build Evidence Timeline and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-068	Build Trust Federation conflict	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-069	Create governed tiler adapter	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-070	Add mixed epoch manifest	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-071	Add source-meta validator lane	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-072	Add source-meta helper library	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-073	Add style/manifest fixtures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-074	Add PMTiles verify/sign workflow	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-075	Mixed epoch manifest fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-076	Add historical post office tile thin slice	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-077	Add Santa Fe Trail overlay thin slice	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-078	Add 3D Tiles extras validator	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-079	Add worker bucket manifest protocol	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-080	Add policy hook for source license	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-081	Add freshness field proposal	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-082	Add catalog lineage manifests for map decisions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-083	Add TileArtifactManifest schema for PMTiles and COG sidecars	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-084	Add source-meta and manifest validator fixtures	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-085	Add PMTiles verify/sign dry-run workflow step	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-086	Add worker prefilter message contract tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-087	Add timeline source-swap and client-filter helper tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-088	Add fauna Evidence Drawer payload contract note	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-089	Add hydrology crossing decision drawer fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-090	Add soils PMTiles/COG promotion diff summary fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-091	Add public-safe historical tile fixture with precision tags	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-092	Add trust badge verified invalid missing visual regression	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-093	Add MapReleaseManifest rollback and cache invalidation tests	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-094	Add Directory Rules ADR if schema home or UI path differs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-095	Add source rights verification tasks for habitat and historical layers	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-096	Add performance budget checks for time-slider interactions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-097	Build one tile, one event window, one candidate schema, one validator, one STAC item, one PMTiles output	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-098	Implement HLS burn-change candidate schema first	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-099	Implement hls-burn-watch detect_change pipeline after schema	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-100	Implement hazards_burn validator before catalog or UI	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-101	Implement derived STAC item writer before public runtime	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-102	Implement PMTiles export for promoted vectors only	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-103	Implement governed runtime route last in the thin slice	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-104	Add biodiversity PMTiles/COG sidecar schema aligned to EvidenceBundle	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-105	Add habitat/protection buildspec and receipt fixture	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-106	Add topology-clean and simplification fixtures for story maps	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-107	Add signed manifest verification fixture for client trust badges	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-108	Add no-auto-publish promotion gate fixture for provisional detections	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-109	Add Focus Mode composed-claim fixtures for resolved and abstained claims	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-110	Add Directory Rules ADR check before creating any new schema/policy/source homes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Y. Implementation Backlog and PR Plan. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Y-111	Next PR should add verified mobile PMTiles fixtures	NEW	SRC-055 pp.33-39,80-86	The sources provide CI, client, and harness patterns for verified base/delta PMTiles. Impl: Create no-network fixtures for index, archive digests, signatures, and MapLibre boot. Validation: Run mobile-emulated verify-decode-render tests. Evidence: SRC-055 pp.33-39,80-86; cross-page normalized discovery.
ML-Y-112	Next PR should add tile manifest and layer registry binding schemas	NEW	SRC-054 pp.41-43; SRC-055 pp.40-43	Current sources propose map layer binding, layer registry, TileJSON, and manifest fields. Impl: Add PROPOSED schemas under Directory Rules default schema home after repo verification. Validation: Schema tests: required fields, digests, source-layer match, rollback target. Evidence: SRC-054 pp.41-43; SRC-055 pp.40-43; cross-page normalized discovery.
ML-Y-113	Next PR should add Evidence Drawer click-resolution fixture	NEW	SRC-054 pp.38,42-43; SRC-055 pp.44-46	Sources require feature_id/evidence_ref resolution to EvidenceBundle and drawer display of slice metadata. Impl: Implement mock governed API and drawer payload fixture before UI polish. Validation: Test feature click to EvidenceBundle, ABSTAIN on missing evidence, and DENY on policy failure. Evidence: SRC-054 pp.38,42-43; SRC-055 pp.44-46; cross-page normalized discovery.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Y-114	Next PR should add PMTiles service-worker cache tests	NEW	SRC-055 pp.268-276	The PWA/offline section defines cache groups and behavior for first run, partial download, stale release, and policy mismatch. Impl: Implement service-worker fixtures and release-scoped cache keys. Validation: Test offline boot, missing glyphs/sprites, stale release, and deny load. Evidence: SRC-055 pp.268-276; cross-page normalized discovery.

5.Z. Z. Open Questions and Verification Backlog idea index

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-001	Package versions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-002	Schema home	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-003	Actual repo topology	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-004	MapLibre GL JS target version	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-005	MLT support status	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-006	Native/mobile parity	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-007	PMTiles hosting Range/CORS	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-008	Martin deployment/access model	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-009	Plugin allowlist	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-010	Style/sprite/glyph hosting	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-011	Attribution and rights	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-012	Sensitive geometry transform rules	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-013	Release manifest location	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-014	CI workflow	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-015	Accessibility acceptance	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-016	Performance budgets	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-017	OCI registry retention and garbage	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-018	Proposed web/public/tiles path conflicts	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-019	MapLibre package and MLT status remain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-020	Rego v1 and Conftest defaults need	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-021	COG/PMTiles/GeoParquet hosting and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-022	WMS/WFS/KanDrive/WZDx access terms	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-023	IIIF and LoC rights mapping to	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-024	OTel collector, Grafana, evidence viewer,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-025	Exact release, evidence, source registry,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-026	Event-feed freshness and debounce	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-027	GeoParquet GDAL PMTiles tool versions	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-028	USDA NRCS KDHE KanDrive FRA rights	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-029	Range CORS CDN behavior for COG and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-030	AQS flags concurrence and API details	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-031	Redaction radius grid and DP defaults	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-032	Receipt drawer UX and accessibility	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-033	STAC 1.x extension and PySTAC	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-034	GeoParquet/COG/PMTiles media types	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-035	Offline pack artifact type	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-036	Tile visual diff thresholds	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-037	Confidence property source	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-038	Provenance graph renderer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-039	PMTiles delta strategy toolchain support	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-040	Client versus server overlay merge	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-041	OCI referrer retention and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-042	COG and PMTiles media types, Range,	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-043	TileJSON canonicalization and PMTiles	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-044	STAC extension for kfm tile claim fields	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-045	Verifier plugin security, accessibility, and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-046	Historical map rights and control-point	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-047	SWI thresholds and operational disclaimer	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-048	AirNow, AQS, PurpleAir, HRRR terms and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-049	BirdCast and NEXRAD derived-product	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-050	DNA and genealogy overlay revocation	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-051	Map/timeline accessibility acceptance in	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-052	Cloud watcher event ordering and	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-053	PMTiles binary patching operational costs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-054	GeoParquet partitioning and H3 resolution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-055	Actual MapLibre target version	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-056	Plugin package home and wrapper	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-057	PMTiles preview toolchain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-058	COG Range/CORS hosting	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-059	Historical route source rights	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-060	Genealogy revocation manifest Z. Z. Open Questions and Verification Backlog	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-061	Acceptable public geohash	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-062	Stale thresholds by source type	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-063	Steward/public mode	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-064	Evidence Drawer component path	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-065	NFHL WMS rights and attribution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-066	NOAA severe-weather update lag	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-067	AirData/KDHE crosswalk authority	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-068	Mesonet citation policy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-069	Rare species exact-location policy	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-070	Runtime rights registry path	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-071	MapLibre GL JS target version	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-072	PMTiles hosting Range and CORS behavior	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-073	COG Range requests and media types	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-074	KDWP ERT endpoint discovery and terms	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-075	KDOT service terms and attribution	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-076	NatureServe licensed data precision rules	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-077	AQS/AirNow/KDHE cadence and rights	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-078	Schema home and UI component homes	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-079	Sensitive geometry transform thresholds	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-080	Plugin allowlist and SBOM process	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-081	Accessibility acceptance for trust states	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-082	Tile load and interaction latency budgets	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-083	Martin deployment and access model	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-084	MLT support status and toolchain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-085	MapLibre GL JS target version remains	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-086	PMTiles and COG hosting Range/CORS	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-087	Browser Sigstore viability needs verification	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-088	MapLibre worker URL control needs repo	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-089	Source terms for WBD, KGS, FWS, KDOT-like	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-090	Sensitive geometry transform thresholds remain	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-091	Schema home remains proposed until repo	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-092	COG protocol implementation choice remains	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-093	WebCrypto streaming hash status needs	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-094	MLT support status remains unchanged as	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-095	Verify browser Sigstore bundle Z. Z. Open Questions and Verification Backlog	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-096	Verify MapLibre	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-097	Verify OffscreenCanvas support	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-098	Verify 3D Tiles 1.1 and Cesium Z. Z. Open Questions and Verification Backlog	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-099	Verify Range CORS CORP	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-100	Verify schema home in mounted	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-101	Verify policy tiers for ABSTAIN	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-102	Verify performance budgets for Z. Z. Open Questions and Verification Backlog	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-103	Verify source rights for	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-104	Verify trust federation thresholds	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-105	Can MapLibre core accept formal source.meta?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-106	Can PMTiles headers embed KFM metadata?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-107	Should meta move to TileJSON extension?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-108	What outcome for missing source.meta?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-109	What MapLibre GL JS version is target?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-110	How should live feeds express freshness?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-111	What PMTiles CLI version is acceptable?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-112	How to verify browser-side signatures?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-113	Are historical source rights sufficient for public release?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-114	How to preserve precision through all tiles?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-115	What MapLibre GL JS version is target for this release?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-116	How should browser-side signature verification be implemented safely?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-117	Does PMTiles metadata carry enough KFM manifest data or need TileJSON extension?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-118	What exact CORS and HTTP Range behavior will hosting provide?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-119	What PMTiles CLI and protocol handler versions are acceptable?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-120	Where is the actual repo schema home after Directory Rules and ADR review?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-121	What policy tier maps missing source metadata to ABSTAIN versus DENY?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-122	How should mobile/native shells match Evidence Drawer behavior?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-123	What MLT support status is acceptable for pilot use?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-124	What exact sensitive geometry thresholds are domain-specific?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-125	How are stale source windows calculated per source cadence?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-126	What plugin allowlist governs MapLibre protocol adapters?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-127	How should historical rights and attribution be displayed in exports?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-128	What is the acceptable interaction-to-drawer latency budget?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-129	Which MapLibre GL JS version is the target package pin?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-130	Which PMTiles CLI/spec version is accepted for KFM release builds?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-131	What CORS and HTTP Range behavior is guaranteed by the actual hosting path?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-132	What source rights permit redistribution for PAD-US, KDWP, ecoregions, and herbarium-derived artifacts?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-133	What exact sensitive-geometry thresholds apply per domain?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-134	How should effort-standardized biodiversity metrics be validated for public display?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-135	What policy tier separates ABSTAIN from DENY for missing event evidence?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-136	What metadata is mandatory in TileArtifactManifest for COG versus PMTiles?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-137	What browser-side signature verification approach is acceptable for trust badges?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-138	What is the performance budget for worker decoding, filtering, and timeline updates?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-139	What repo path owns MapLibre UI components after Directory Rules verification?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-140	What release-manifest location and rollback process is actually implemented?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-141	How should WMS/WMTS context layers be attributed and freshness-checked?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-142	When does Cesium/3D carry enough evidence burden to justify non-default use?	EXPANDED	SRC-MASTER-PREV ; SRC-054/SRC-055	Prior cumulative idea expanded by current-run evidence under Z. Open Questions and Verification Backlog. Impl: Carry forward prior implementation consequence; do not claim repo implementation without current evidence. Validation: Revalidate only through mounted repo tests, manifests, release receipts, and source-ledger checks. Evidence: SRC-MASTER-PREV, Master Idea Index.
ML-Z-143	Which MapLibre GL JS version should KFM pin after current verification?	NEW	SRC-003 p.3; SRC-055 pp.36-38	The prior MapLibre manual treated version as version-sensitive and the current docs use generic MapLibre examples. Impl: Package pins require a current official check and dependency admission record. Validation: Verify official release, compatibility with PMTiles protocol, and browser support. Evidence: SRC-003 p.3; SRC-055 pp.36-38; cross-page normalized discovery.

idea_id	idea title	status	source_ids	short normalized statement and consequences
ML-Z-144	What browser-side signature verification stack is acceptable?	NEW	SRC-055 pp.80-86	Current docs propose @sigstore/core, sigstore-browser, SubtleCrypto, JWK, and DSSE options but do not settle a KFM implementation. Impl: Choose trust roots, bundle format, worker boundary, and failure behavior. Validation: Test valid, invalid, stale, wrong-key, and missing transparency proof cases. Evidence: SRC-055 pp.80-86; cross-page normalized discovery.
ML-Z-145	What are final mobile tile budgets by device tier?	NEW	SRC-055 pp.34-39,80-86	Current docs propose example budgets but device floor and target acceptance remain unsettled. Impl: Define low-end Android, tablet, desktop, and offline budgets. Validation: Run calibrated traces and store budget receipts. Evidence: SRC-055 pp.34-39,80-86; cross-page normalized discovery.
ML-Z-146	Which PMTiles hosting environments satisfy Range and CORS requirements?	NEW	SRC-055 pp.40-43,271-274	Sources require partial byte reads and exposed Range headers. Impl: Verify S3/R2/CloudFront/static nginx/local PWA behavior. Validation: Automated header tests before release. Evidence: SRC-055 pp.40-43,271-274; cross-page normalized discovery.
ML-Z-147	When should KFM choose PMTiles overlays instead of republishing a base archive?	NEW	SRC-055 pp.29-38,272-276	Current docs describe base/delta PMTiles, regional splits, and overlay archives. Impl: Define update-size, review-churn, rollback, and mobile-size thresholds. Validation: Run delta-manifest and rollback simulations. Evidence: SRC-055 pp.29-38,272-276; cross-page normalized discovery.
ML-Z-148	What is the authoritative schema home for map-layer binding objects in the mounted repo?	NEW	SRC-004 pp.1-3; SRC-054 p.43	Directory Rules default to schemas/contracts/v1, but actual repo topology was not mounted. Impl: Verify repo ADRs and do not create parallel homes. Validation: Open drift/ADR if repo conflicts with Directory Rules. Evidence: SRC-004 pp.1-3; SRC-054 p.43; cross-page normalized discovery.
ML-Z-149	What sensitive-geometry thresholds apply per domain and zoom level?	NEW	SRC-054 p.43; SRC-055 pp.69-78,280	Current docs require generalization/redaction but do not settle domain thresholds. Impl: Define per-domain precision, aggregation, embargo, and steward review rules. Validation: Negative fixtures for exact geometry and missing receipts. Evidence: SRC-054 p.43; SRC-055 pp.69-78,280; cross-page normalized discovery.
ML-Z-150	What release rollback behavior should the map shell perform when a PMTiles delta is rejected?	NEW	SRC-055 pp.29-38,273-276	Current docs say abort, reject, or notify but detailed UX remains open. Impl: Define rollback target selection, cache invalidation, and user messaging. Validation: Test rejected delta, stale cache, and rollback manifest selection. Evidence: SRC-055 pp.29-38,273-276; cross-page normalized discovery.

6. Category Atlas

The atlas organizes all ideas into the requested KFM MapLibre categories. Category counts include prior ideas from SRC-MASTER-PREV plus current-run new extractions from SRC-054 and SRC-055.

Cat	Category	Prior	New	Current	Change	Primary current-run evidence
A	A. Renderer Boundary and KFM Trust Law	72	3	75	EXPANDED	SRC-054/SRC-055
B	B. MapLibre GL JS Web Shell	61	4	65	EXPANDED	SRC-054/SRC-055
C	C. MapLibre Native / Mobile / Platform Parity	14	3	17	EXPANDED	SRC-054/SRC-055
D	D. MapLibre RS / Experimental Renderers	5	0	5	RETAINED	SRC-MASTER-PREV
E	E. Style Specification, Sources, Layers, Expressions	63	3	66	EXPANDED	SRC-054/SRC-055
F	F. Sprites, Glyphs, Fonts, Design Tokens	9	2	11	EXPANDED	SRC-054/SRC-055
G	G. GeoJSON and Runtime Data	66	2	68	EXPANDED	SRC-054/SRC-055
H	H. Vector Tiles: MVT and MLT	51	3	54	EXPANDED	SRC-054/SRC-055
I	I. PMTiles, MBTiles, Static Tiles, Serverless Delivery	106	106	212	EXPANDED	SRC-054/SRC-055
J	J. Martin and Server-Mediated Tile Serving	12	2	14	EXPANDED	SRC-054/SRC-055
K	K. Raster, COG, DEM, Terrain, Hillshade	78	16	94	EXPANDED	SRC-054/SRC-055
L	L. WMS / WMTS / External Map Services	25	2	27	EXPANDED	SRC-054/SRC-055
M	M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest	112	34	146	EXPANDED	SRC-054/SRC-055
N	N. Evidence Drawer Payloads and Click Resolution	85	29	114	EXPANDED	SRC-054/SRC-055
O	O. Focus Mode and Governed AI Map Context	69	17	86	EXPANDED	SRC-054/SRC-055
P	P. Time-Aware Map Interaction and Timeline	78	16	94	EXPANDED	SRC-054/SRC-055
Q	Q. Sensitive Geometry, Geoprivacy, Rights, and Policy	83	5	88	EXPANDED	SRC-054/SRC-055
R	R. Plugin, Wrapper, and Dependency Governance	39	3	42	EXPANDED	SRC-054/SRC-055
S	S. Accessibility, UX, and Trust-Visible States	62	2	64	EXPANDED	SRC-054/SRC-055
T	T. Performance, Caching, CDN, Range Requests, and Resource Timing	82	8	90	EXPANDED	SRC-054/SRC-055
U	U. Testing, CI, Validation, Rollback, and Proof Objects	171	6	177	EXPANDED	SRC-054/SRC-055
V	V. Exports, Screenshots, Reports, and Citation Preservation	48	2	50	EXPANDED	SRC-054/SRC-055
W	W. 3D / Cesium / Deck.gl / Overlay Interoperability	59	42	101	EXPANDED	SRC-054/SRC-055
X	X. Anti-Patterns and Failure Modes	100	4	104	EXPANDED	SRC-054/SRC-055
Y	Y. Implementation Backlog and PR Plan	110	4	114	EXPANDED	SRC-054/SRC-055
Z	Z. Open Questions and Verification Backlog	142	8	150	EXPANDED	SRC-054/SRC-055

7. Detailed Category Chapters

7.A. A. Renderer Boundary and KFM Trust Law

Renderer boundary and KFM trust law explains why MapLibre is a renderer and interaction runtime only. It preserves cite-or-abstain and the public-safe trust membrane.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-A-073	Renderer must never become evidence authority	SRC-054 pp.42-43; SRC-055 pp.271-276	Treating rendered features as proof. Validation: Test that public clients cannot load RAW, WORK, candidate, or unverifiable artifacts.
ML-A-074	Published artifacts beat operational coupling	SRC-055 pp.268-276	Client access to internal tile generation or databases. Validation: Verify artifact digests, release state, and rollback targets before rendering.
ML-A-075	No API means no claims in offline mode	SRC-054 p.51	Offline rendering presented as current truth. Validation: Add an offline fixture where layer rendering works but claims abstain without cached evidence.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.B. B. MapLibre GL JS Web Shell

The MapLibre GL JS shell is the governed web map runtime. It mounts sources, layers, controls, camera state, and interactions only through released artifacts and governed APIs.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-B-062	MapLibre PMTiles protocol wiring needs verification first	SRC-055 pp.36-38	Protocol handler added without trust checks. Validation: Fail load if index verification, archive digest, or policy check fails.
ML-B-063	MapLibre source-layer names are contract-bearing	SRC-055 pp.40-43	Style drift silently hides or mislabels features. Validation: Validate style JSON against vector layer metadata before release.
ML-B-064	MapLibre overlay can synchronize with Cesium only as controlled shell behavior	SRC-055 pp.40-43; SRC-054 pp.160-168	Overlay sync treated as evidence equivalence. Validation: Test camera, pitch, bearing, and zoom conversion with bounds and performance budgets.
ML-B-065	MapLibre errors are trust-visible events	SRC-055 pp.80-86	Silent tile verification failure. Validation: Add map.on(error) tests that assert fail-closed UI copy.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.C. C. MapLibre Native / Mobile / Platform Parity

Mobile, native, and PWA parity requires the same Drawer, Focus, manifest, and policy behavior under tighter memory, request, and offline budgets.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-C-015	Mobile clients need lower verifier and request budgets	SRC-055 pp.34-39; SRC-055 pp.80-86	Desktop-only assumptions shipped to field devices. Validation: Run throttled Puppeteer or equivalent tests for verify-decode-render latency.
ML-C-016	PWA offline bundles need evidence-aware cache groups	SRC-055 pp.268-276	Offline cache serves stale or unpaired assets. Validation: Test first-run online, partial download, offline boot, stale release, and policy mismatch.
ML-C-017	Native/mobile parity remains unverified until drawer and focus behavior match	SRC-055 pp.268-276; SRC-054 p.51	Mobile app as a shortcut around governance. Validation: Acceptance must include Evidence Drawer resolution, abstain paths, and policy mismatch.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.D. D. MapLibre RS / Experimental Renderers

MapLibre RS and experimental renderers remain retained baseline topics. Current sources add Rust PMTiles tooling, but not direct MapLibre RS renderer doctrine.

Status this run: RETAINED. Evidence basis: SRC-MASTER-PREV only in this run. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.E. E. Style Specification, Sources, Layers, Expressions

Styles, sources, layers, filters, and expressions are rendering contracts. They must be bound to manifests and stable source-layer metadata.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-E-064	Tile properties should stay flat, typed, and filterable	SRC-054 pp.36-37	Deep JSON blobs in vector tiles. Validation: Validate tile metadata, field types, source_id, evidence_ref, and temporal fields.
ML-E-065	Style expressions should consume simple worker-set flags	SRC-054 p.37	Complex expressions in the render loop. Validation: Performance tests must distinguish worker cost from style expression cost.

new idea_id	title	evidence	risk / validation consequence
ML-E-066	Style versioning must align with layer release versions	SRC-054 pp.49-50; SRC-055 p.274	Uncontrolled style drift. Validation: Check that style digests match ReleaseManifest and cache invalidation records.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.F. F. Sprites, Glyphs, Fonts, Design Tokens

Sprites, glyphs, fonts, and design tokens are release dependencies because they affect offline rendering, accessibility, and trust-visible states.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-F-010	Glyphs and sprites are part of the release bundle	SRC-055 pp.273-274	Glyph or sprite failure hides trust-visible state. Validation: Offline boot tests must include missing glyph and missing sprite cases.
ML-F-011	Fonts and icons need policy-visible fallback	SRC-055 p.274	Trust state indicated only by optional icon assets. Validation: Visual regression should test fallback glyph/sprite behavior.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.G. G. GeoJSON and Runtime Data

GeoJSON and runtime data are useful for small overlays, fixtures, synthetic tests, and runtime selection. They must not become a large production default without gates.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-G-067	GeoJSON is useful for synthetic fixtures and small overlays	SRC-055 pp.32,88-89	Dense GeoJSON as production default. Validation: Add size, CRS, and topology checks before using GeoJSON in map runtime.
ML-G-068	Runtime data must carry lineage fields before catalog promotion	SRC-055 p.240	Runtime-only data detached from provenance. Validation: Reject payloads missing source_id, temporal fields, or evidence references.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.H. H. Vector Tiles: MVT and MLT

MVT and MLT carry vector features for fast rendering. KFM must validate schema, source-layer names, feature IDs, and evidence references.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-H-052	Vector tile extraction - SRC-054 p.36	SRC-054 p.36	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-H-053	Vector tile extraction - SRC-055 p.191	SRC-055 p.191	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-H-054	Vector tile extraction - SRC-055 p.231	SRC-055 p.231	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.I. I. PMTiles, MBTiles, Static Tiles, Serverless Delivery

PMTiles, MBTiles, static tiles, and serverless delivery are powerful for immutable public-safe artifacts, offline PWAs, and rural/field use.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-I-107	PMTiles/static tile extraction - SRC-054 p.31	SRC-054 p.31	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-108	PMTiles/static tile extraction - SRC-054 p.32	SRC-054 p.32	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-109	PMTiles/static tile extraction - SRC-054 p.35	SRC-054 p.35	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-110	PMTiles/static tile extraction - SRC-054 p.37	SRC-054 p.37	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-111	PMTiles/static tile extraction - SRC-054 p.39	SRC-054 p.39	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-I-112	PMTiles/static tile extraction - SRC-054 p.40	SRC-054 p.40	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-113	PMTiles/static tile extraction - SRC-054 p.41	SRC-054 p.41	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-114	PMTiles/static tile extraction - SRC-054 p.42	SRC-054 p.42	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-115	PMTiles/static tile extraction - SRC-054 p.43	SRC-054 p.43	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-I-116	PMTiles/static tile extraction - SRC-054 p.51	SRC-054 p.51	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
...	96 additional current-run ideas listed in the Master Idea Index.	See Section 5.	Same traceability rules apply.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.J. J. Martin and Server-Mediated Tile Serving

Server-mediated tile serving is needed when dynamic access control, conversion, steward mediation, or PostGIS-backed serving is required.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-J-013	Server-side conversion remains the production path for Cesium consumers	SRC-055 pp.40-43	Prototype adapters promoted to production without review. Validation: Validate converted imagery/3D Tiles against the source PMTiles digest and release manifest.
ML-J-014	Tile server choices need access-model justification	SRC-055 p.87	Tile server introduced as convenience bypass. Validation: Test range behavior, access controls, caching, and audit logs.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.K. K. Raster, COG, DEM, Terrain, Hillshade

COGs, rasters, DEMs, terrain, and hillshade support imagery and surface context, but need overviews, range requests, metadata, and caveats.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-K-079	Raster/COG/terrain extraction - SRC-054 p.19	SRC-054 p.19	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-080	Raster/COG/terrain extraction - SRC-054 p.59	SRC-054 p.59	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-081	Raster/COG/terrain extraction - SRC-054 p.76	SRC-054 p.76	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-082	Raster/COG/terrain extraction - SRC-054 p.84	SRC-054 p.84	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-083	Raster/COG/terrain extraction - SRC-054 p.110	SRC-054 p.110	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-084	Raster/COG/terrain extraction - SRC-054 p.150	SRC-054 p.150	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-085	Raster/COG/terrain extraction - SRC-054 p.155	SRC-054 p.155	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-086	Raster/COG/terrain extraction - SRC-054 p.203	SRC-054 p.203	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-087	Raster/COG/terrain extraction - SRC-054 p.263	SRC-054 p.263	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-K-088	Raster/COG/terrain extraction - SRC-054 p.275	SRC-054 p.275	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
...	6 additional current-run ideas listed in the Master Idea Index.	See Section 5.	Same traceability rules apply.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.L. L. WMS / WMTS / External Map Services

WMS, WMTS, and external map services are useful context layers only when attribution, freshness, rights, and source role are clear.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-L-026	WMTS/TMS is a stable fallback for 2.5D or unverifiable high-res layers	SRC-055 pp.40-43	Fallback layer displayed as equivalent to verified layer. Validation: Test fallback selection when PMTiles verification fails.
ML-L-027	External map services remain contextual unless evidence-linked	SRC-054 pp.49-50; SRC-055 pp.40-43	Context service used as claim proof. Validation: Validate source URI, attribution, temporal fields, and rights before rendering.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.M. M. LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest

Manifests bind releases to artifacts, sources, hashes, receipts, catalog records, promotion decisions, and rollback targets.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-M-113	Manifest/release object extraction - SRC-054 p.54	SRC-054 p.54	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-114	Manifest/release object extraction - SRC-054 p.71	SRC-054 p.71	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-115	Manifest/release object extraction - SRC-054 p.74	SRC-054 p.74	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-116	Manifest/release object extraction - SRC-054 p.96	SRC-054 p.96	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-117	Manifest/release object extraction - SRC-054 p.105	SRC-054 p.105	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-118	Manifest/release object extraction - SRC-054 p.117	SRC-054 p.117	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-119	Manifest/release object extraction - SRC-054 p.123	SRC-054 p.123	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-120	Manifest/release object extraction - SRC-054 p.169	SRC-054 p.169	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-M-121	Manifest/release object extraction - SRC-054 p.218	SRC-054 p.218	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-M-122	Manifest/release object extraction - SRC-054 p.269	SRC-054 p.269	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
...	24 additional current-run ideas listed in the Master Idea Index.	See Section 5.	Same traceability rules apply.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.N. N. Evidence Drawer Payloads and Click Resolution

Evidence Drawer payloads are the mandatory inspection surface for clicked features and consequential map claims.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-N-086	Evidence Drawer extraction - SRC-054 p.20	SRC-054 p.20	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-087	Evidence Drawer extraction - SRC-054 p.24	SRC-054 p.24	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-088	Evidence Drawer extraction - SRC-054 p.30	SRC-054 p.30	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-089	Evidence Drawer extraction - SRC-054 p.49	SRC-054 p.49	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-090	Evidence Drawer extraction - SRC-054 p.94	SRC-054 p.94	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-091	Evidence Drawer extraction - SRC-054 p.119	SRC-054 p.119	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-092	Evidence Drawer extraction - SRC-054 p.131	SRC-054 p.131	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-093	Evidence Drawer extraction - SRC-054 p.205	SRC-054 p.205	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-N-094	Evidence Drawer extraction - SRC-054 p.206	SRC-054 p.206	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-N-095	Evidence Drawer extraction - SRC-054 p.212	SRC-054 p.212	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
...	19 additional current-run ideas listed in the Master Idea Index.	See Section 5.	Same traceability rules apply.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.O. O. Focus Mode and Governed AI Map Context

Focus Mode is governed AI over map context. It can answer only when evidence resolves and citations validate; otherwise it abstains, denies, or errors.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-O-070	Focus Mode extraction - SRC-054 p.26	SRC-054 p.26	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-071	Focus Mode extraction - SRC-054 p.44	SRC-054 p.44	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-072	Focus Mode extraction - SRC-054 p.89	SRC-054 p.89	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-073	Focus Mode extraction - SRC-054 p.111	SRC-054 p.111	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-074	Focus Mode extraction - SRC-054 p.120	SRC-054 p.120	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-075	Focus Mode extraction - SRC-054 p.168	SRC-054 p.168	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-076	Focus Mode extraction - SRC-054 p.183	SRC-054 p.183	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-077	Focus Mode extraction - SRC-054 p.196	SRC-054 p.196	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-O-078	Focus Mode extraction - SRC-054 p.197	SRC-054 p.197	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-O-079	Focus Mode extraction - SRC-054 p.200	SRC-054 p.200	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
...	7 additional current-run ideas listed in the Master Idea Index.	See Section 5.	Same traceability rules apply.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.P. P. Time-Aware Map Interaction and Timeline

Time-aware map interaction covers time sliders, deltas, time-index manifests, valid windows, and temporal filtering.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-P-079	Time/timeline extraction - SRC-054 p.68	SRC-054 p.68	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-080	Time/timeline extraction - SRC-054 p.98	SRC-054 p.98	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-081	Time/timeline extraction - SRC-054 p.104	SRC-054 p.104	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-082	Time/timeline extraction - SRC-054 p.201	SRC-054 p.201	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-083	Time/timeline extraction - SRC-054 p.264	SRC-054 p.264	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-084	Time/timeline extraction - SRC-054 p.268	SRC-054 p.268	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-085	Time/timeline extraction - SRC-055 p.72	SRC-055 p.72	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-086	Time/timeline extraction - SRC-055 p.77	SRC-055 p.77	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-P-087	Time/timeline extraction - SRC-055 p.84	SRC-055 p.84	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-P-088	Time/timeline extraction - SRC-055 p.98	SRC-055 p.98	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
...	6 additional current-run ideas listed in the Master Idea Index.	See Section 5.	Same traceability rules apply.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.Q. Q. Sensitive Geometry, Geoprivacy, Rights, and Policy

Sensitive geometry, geoprivacy, rights, consent, obligations, and public-safe transforms are fail-closed concerns.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-Q-084	Sensitive geometry/policy extraction - SRC-054 p.143	SRC-054 p.143	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-Q-085	Sensitive geometry/policy extraction - SRC-054 p.305	SRC-054 p.305	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-Q-086	Sensitive geometry/policy extraction - SRC-055 p.110	SRC-055 p.110	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-Q-087	Sensitive geometry/policy extraction - SRC-055 p.280	SRC-055 p.280	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-Q-088	Sensitive geometry/policy extraction - SRC-055 p.287	SRC-055 p.287	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.R. R. Plugin, Wrapper, and Dependency Governance

Plugins, wrappers, protocol handlers, adapters, and dependencies must pass license, security, maturity, and policy review.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-R-040	PMTiles protocol handlers require dependency admission	SRC-055 pp.36-38	Protocol plugin adopted without review. Validation: Security, license, version, and fail-closed tests before merge.
ML-R-041	Cesium MVT adapters are prototype-only until hardened	SRC-055 pp.40-43	Adapter maturity overstated. Validation: Test CORS, Range, memory, render drift, and evidence identity.
ML-R-042	Browser verification libraries need explicit trust roots	SRC-055 pp.80-86	Cryptographic UI theater without enforceable roots. Validation: Negative tests for invalid signature, wrong key, stale bundle, and policy mismatch.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.S. S. Accessibility, UX, and Trust-Visible States

Accessibility, UX, and trust-visible states ensure users see denied, stale, missing, corrected, unverifiable, and abstained states.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-S-063	No-data and unverifiable states are UI requirements	SRC-054 pp.39,49-52; SRC-055 pp.274-276	Silent blank maps. Validation: Accessibility and visual-regression tests must cover no-data, stale, deny, abstain, and error states.
ML-S-064	Reviewer ergonomics protects governance quality	SRC-054 p.52	Review fatigue leading to rubber-stamp approval. Validation: User tests should measure review task completion and traceability.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.T. T. Performance, Caching, CDN, Range Requests, and Resource Timing

Performance and caching include Range requests, CORS, CDN headers, workers, memory, tile load, resource timing, and interaction latency.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-T-083	Performance/cache extraction - SRC-054 p.50	SRC-054 p.50	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-T-084	Performance/cache extraction - SRC-054 p.83	SRC-054 p.83	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-T-085	Performance/cache extraction - SRC-054 p.87	SRC-054 p.87	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-T-086	Performance/cache extraction - SRC-054 p.222	SRC-054 p.222	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-T-087	Performance/cache extraction - SRC-054 p.315	SRC-054 p.315	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-T-088	Performance/cache extraction - SRC-055 p.82	SRC-055 p.82	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-T-089	Performance/cache extraction - SRC-055 p.190	SRC-055 p.190	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-T-090	Performance/cache extraction - SRC-055 p.233	SRC-055 p.233	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.U. U. Testing, CI, Validation, Rollback, and Proof Objects

Testing, CI, validation, rollback, and proof objects make the map stack inspectable and reversible.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-U-172	Validation/proof extraction - SRC-054 p.33	SRC-054 p.33	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-U-173	Validation/proof extraction - SRC-054 p.80	SRC-054 p.80	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-U-174	Validation/proof extraction - SRC-054 p.149	SRC-054 p.149	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-U-175	Validation/proof extraction - SRC-054 p.243	SRC-054 p.243	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-U-176	Validation/proof extraction - SRC-055 p.236	SRC-055 p.236	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-U-177	Validation/proof extraction - SRC-055 p.254	SRC-055 p.254	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.V. V. Exports, Screenshots, Reports, and Citation Preservation

Exports, screenshots, reports, and citations must preserve release identity, style, source, and evidence metadata.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-V-049	Uncited exports are not acceptable release artifacts	SRC-054 pp.49-50; SRC-055 p.111	Screenshot treated as proof. Validation: Export tests must fail on missing citations or unresolved EvidenceRefs.
ML-V-050	Catalog closure is mandatory before reports or screenshots	SRC-054 p.50; SRC-055 p.240	Report built from uncataloged artifacts. Validation: Test generated report metadata against catalog and release manifests.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIReceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.W. W. 3D / Cesium / Deck.gl / Overlay Interoperability

3D, Cesium, deck.gl, globe, and terrain are conditional overlays. They are not the default when 2D carries the evidence more clearly.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-W-060	3D/Cesium/deck extraction - SRC-054 p.38	SRC-054 p.38	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-061	3D/Cesium/deck extraction - SRC-054 p.63	SRC-054 p.63	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-062	3D/Cesium/deck extraction - SRC-054 p.142	SRC-054 p.142	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.

new idea_id	title	evidence	risk / validation consequence
ML-W-063	3D/Cesium/deck extraction - SRC-054 p.151	SRC-054 p.151	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-064	3D/Cesium/deck extraction - SRC-054 p.216	SRC-054 p.216	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-065	3D/Cesium/deck extraction - SRC-054 p.217	SRC-054 p.217	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-066	3D/Cesium/deck extraction - SRC-054 p.219	SRC-054 p.219	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-067	3D/Cesium/deck extraction - SRC-054 p.220	SRC-054 p.220	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-068	3D/Cesium/deck extraction - SRC-054 p.221	SRC-054 p.221	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
ML-W-069	3D/Cesium/deck extraction - SRC-054 p.223	SRC-054 p.223	Do not treat the page mention as proof of implementation; it is source evidence for design and backlog. Validation: Add or extend fixtures and fail-closed checks that exercise the cited artifact, source, time, evidence, or UI behavior.
...	32 additional current-run ideas listed in the Master Idea Index.	See Section 5.	Same traceability rules apply.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.X. X. Anti-Patterns and Failure Modes

Anti-patterns and failure modes record what would break KFM trust, even when the UI appears polished.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-X-101	PMTiles is not ideal for highly transactional GIS	SRC-055 pp.272-276	PMTiles as universal default. Validation: Run update-size and rollback-complexity checks.
ML-X-102	Style filters do not protect sensitive geometry	SRC-054 p.43; SRC-055 p.280	Hiding sensitive locations with layer styling. Validation: Test that exact sensitive geometry is absent from artifacts, not only invisible.
ML-X-103	Verification failures must not degrade into high-resolution display	SRC-055 pp.40-43,80-86	Fallback logic that shows unverified data. Validation: Inject bad signature and digest fixtures.
ML-X-104	3D is not the default when 2D explains the evidence	SRC-054 p.51; SRC-055 pp.40-43	3D/globe as visual polish. Validation: Test 3D opt-in, evidence binding, and performance budgets.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.Y. Y. Implementation Backlog and PR Plan

Implementation backlog and PR plan converts discoveries into small reversible build steps.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-Y-111	Next PR should add verified mobile PMTiles fixtures	SRC-055 pp.33-39,80-86	Starting with live hosting before fixture proof. Validation: Run mobile-emulated verify-decode-render tests.
ML-Y-112	Next PR should add tile manifest and layer registry binding schemas	SRC-054 pp.41-43; SRC-055 pp.40-43	Parallel schema homes without ADR. Validation: Schema tests: required fields, digests, source-layer match, rollback target.
ML-Y-113	Next PR should add Evidence Drawer click-resolution fixture	SRC-054 pp.38,42-43; SRC-055 pp.44-46	Popup substitutes for Evidence Drawer. Validation: Test feature click to EvidenceBundle, ABSTAIN on missing evidence, and DENY on policy failure.
ML-Y-114	Next PR should add PMTiles service-worker cache tests	SRC-055 pp.268-276	Offline cache ungoverned by release state. Validation: Test offline boot, missing glyphs/sprites, stale release, and deny load.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

7.Z. Z. Open Questions and Verification Backlog

Open questions and verification backlog preserve uncertainty rather than turning proposals into false facts.

Status this run: EXPANDED. Evidence basis: SRC-054/SRC-055 current-run evidence. CONFIRMED means evidence in attached documents or generated scan; PROPOSED means implementation placement or behavior not verified in a mounted repository.

new idea_id	title	evidence	risk / validation consequence
ML-Z-143	Which MapLibre GL JS version should KFM pin after current verification?	SRC-003 p.3; SRC-055 pp.36-38	NEEDS VERIFICATION Validation: Verify official release, compatibility with PMTiles protocol, and browser support.
ML-Z-144	What browser-side signature verification stack is acceptable?	SRC-055 pp.80-86	NEEDS VERIFICATION Validation: Test valid, invalid, stale, wrong-key, and missing transparency proof cases.
ML-Z-145	What are final mobile tile budgets by device tier?	SRC-055 pp.34-39,80-86	NEEDS VERIFICATION Validation: Run calibrated traces and store budget receipts.
ML-Z-146	Which PMTiles hosting environments satisfy Range and CORS requirements?	SRC-055 pp.40-43,271-274	NEEDS VERIFICATION Validation: Automated header tests before release.
ML-Z-147	When should KFM choose PMTiles overlays instead of republishing a base archive?	SRC-055 pp.29-38,272-276	NEEDS VERIFICATION Validation: Run delta-manifest and rollback simulations.
ML-Z-148	What is the authoritative schema home for map-layer binding objects in the mounted repo?	SRC-004 pp.1-3; SRC-054 p.43	NEEDS VERIFICATION Validation: Open drift/ADR if repo conflicts with Directory Rules.
ML-Z-149	What sensitive-geometry thresholds apply per domain and zoom level?	SRC-054 p.43; SRC-055 pp.69-78,280	NEEDS VERIFICATION Validation: Negative fixtures for exact geometry and missing receipts.

new idea_id	title	evidence	risk / validation consequence
ML-Z-150	What release rollback behavior should the map shell perform when a PMTiles delta is rejected?	SRC-055 pp.29-38,273-276	NEEDS VERIFICATION Validation: Test rejected delta, stale cache, and rollback manifest selection.

Required objects/contracts: SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, PolicyDecision, PromotionDecision, CitationValidationReport, AIRceipt, RunReceipt, rollback target, cache invalidation record, plus category-specific fixtures. All homes are PROPOSED until repo evidence is mounted.

Public UI implication: standard clients must use governed APIs, released artifacts, catalog records, tile services, and EvidenceBundle resolution. They must not read RAW/WORK/QUARANTINE/candidate data or direct model output.

Policy implication: fail closed for unknown rights, sensitive geometry, stale evidence, missing catalog closure, unresolved EvidenceRef, invalid digest, or missing rollback target.

Validation tests: schema validation, source-ledger completeness, artifact digest checks, no-public-RAW route, no-unreleased-tile load, Evidence Drawer resolution, Focus Mode citation/abstain behavior, visual regression, accessibility, tile load budget, and rollback restoration.

8. Components-Functions-Features Matrix

component	function / feature	inputs / outputs	governed dependencies	runtime behavior / tests / status
MapLibre GL JS	2D web renderer and interaction runtime Feature: Map camera, layer toggles, click selection, timeline filters	Inputs: LayerManifest, StyleManifest, TileArtifactManifest, released PMTiles/COG/GeoJSON where allowed Outputs: Rendered map state and click candidates	API: governed layer/style/tile API Evidence: EvidenceRef/EvidenceBundle on click Policy: PolicyDecision and release state Tile/style: Style JSON, sources, layers, PMTiles/MVT/COG	Runtime/UI: Public-safe 2D shell with trust states Tests: style validity, click resolution, no RAW load, visual regression Status: CONFIRMED doctrine / PROPOSED implementation
MapLibre Native	Native/mobile renderer candidate Feature: Platform parity for field or offline users	Inputs: Released mobile-safe artifacts and cache manifest Outputs: Mobile/offline rendered state	API: governed API or cached EvidenceBundle Evidence: Cached evidence or abstain Policy: Public-safe/offline policy Tile/style: PMTiles/tiles, style/glyph/sprite bundle	Runtime/UI: No claims when evidence/API unavailable Tests: parity, offline denial, cache tests Status: NEEDS VERIFICATION
MapLibre RS	Experimental renderer/tooling candidate Feature: Potential Rust-native renderer ecosystem	Inputs: Unknown Outputs: Unknown	API: Unknown Evidence: Unknown Policy: Unknown Tile/style: Unknown	Runtime/UI: No production role yet Tests: version/maturity check Status: RETAINED / NEEDS VERIFICATION
Style JSON	Renderer style contract Feature: Defines visual layers and expressions	Inputs: StyleManifest, sprites, glyphs, source metadata Outputs: Versioned style artifact	API: style manifest endpoint Evidence: Citation fields not in style alone Policy: Policy labels from manifest Tile/style: source-layer, filters, expressions	Runtime/UI: Trust-visible styles and fallbacks Tests: style spec validation, visual regression Status: PROPOSED
source definition	Binds source to tile/data endpoint Feature: Vector/raster/geojson source config	Inputs: TileArtifactManifest, source descriptor Outputs: Map source object	API: layer source API Evidence: source_id/evidence_ref linkage Policy: release/policy gate Tile/style: PMTiles/MVT/COG/GeoJSON	Runtime/UI: Data access through released URL only Tests: source URL/digest checks Status: PROPOSED
layer definition	Binds visual layer to source-layer Feature: Fill/line/symbol/raster/hillshade layers	Inputs: LayerManifest, style layer Outputs: Rendered visual layer	API: layer catalog API Evidence: Evidence Drawer fields Policy: policy labels/sensitivity Tile/style: style source-layer	Runtime/UI: Layer toggles with trust state Tests: source-layer and promoted tests Status: PROPOSED
expressions	Runtime styling logic Feature: Filters, color ramps, visibility	Inputs: StyleManifest, typed tile properties Outputs: Paint/layout behavior	API: style API Evidence: No evidence claims alone Policy: no sensitive style-only hiding Tile/style: style JSON	Runtime/UI: Simple worker-set flags preferred Tests: expression performance tests Status: PROPOSED
filters	Map and time filtering Feature: Feature selection by properties/time	Inputs: Tile properties, time index Outputs: Filtered visible set	API: governed context API Evidence: MapContextEnvelope Policy: policy-safe filtering only Tile/style: MVT/PMTiles properties	Runtime/UI: Visible evidence subset Tests: filter/time tests Status: PROPOSED
sprites	Icon sheet for symbols/trust states Feature: Icons, markers, badges	Inputs: Sprite manifest/release bundle Outputs: Sprite assets	API: style asset API Evidence: n/a Policy: must not hide policy Tile/style: style dependency	Runtime/UI: Fallback icons Tests: offline missing-sprite test Status: PROPOSED
glyphs	Text rendering assets Feature: Labels and status text	Inputs: Glyph manifest/release bundle Outputs: Glyph assets	API: style asset API Evidence: n/a Policy: must not hide policy Tile/style: style dependency	Runtime/UI: Fallback labels Tests: offline missing-glyph test Status: PROPOSED
MVT	Vector tile format Feature: Compact vector delivery	Inputs: TileArtifactManifest, source-layer metadata Outputs: Vector tile stream	API: tile URL or PMTiles Evidence: feature id/evidence_ref Policy: public-safe only Tile/style: MVT tiles	Runtime/UI: Fast decode and render Tests: schema, digest, source-layer tests Status: PROPOSED
MLT	Emerging tile format Feature: Potential future vector tile optimization	Inputs: Unknown toolchain Outputs: Unknown	API: Unknown Evidence: Unknown Policy: policy before adoption Tile/style: MLT artifacts	Runtime/UI: Pilot only Tests: toolchain and renderer tests Status: NEEDS VERIFICATION

component	function / feature	inputs / outputs	governed dependencies	runtime behavior / tests / status
PMTiles	Single-file tile archive Feature: Static/serverless/offline tile delivery	Inputs: PMTiles archive, delta manifest, ReleaseManifest Outputs: Released archive	API: tile service/static URL Evidence: feature evidence fields Policy: public-safe archive policy Tile/style: MVT/raster tiles in PMTiles	Runtime/UI: MapLibre protocol and PWA cache Tests: pmtiles verify, Range/CORS, digest Status: CONFIRMED source idea / PROPOSED implementation
MBTiles	Local/legacy tile archive Feature: Packaging or conversion source	Inputs: MBTiles artifact Outputs: Converted tiles	API: server/convert API Evidence: must retain evidence refs Policy: publication policy Tile/style: MVT/raster tiles	Runtime/UI: Usually internal or conversion step Tests: conversion digest tests Status: PROPOSED
COG	Cloud Optimized GeoTIFF Feature: Raster/imagery with overviews and range reads	Inputs: COG asset, STAC item Outputs: Raster source	API: COG protocol/tile API Evidence: source/citation metadata Policy: rights/sensitivity Tile/style: COG/raster tiles	Runtime/UI: Raster context or analysis layer Tests: range, overview, checksum tests Status: CONFIRMED source idea / PROPOSED implementation
Martin	PostGIS tile server option Feature: Dynamic/server-mediated MVT serving	Inputs: PostGIS tables, source registry Outputs: MVT tiles	API: tile server API Evidence: feature refs from DB Policy: access-control policy Tile/style: MVT	Runtime/UI: Use when static tiles insufficient Tests: access/log/version tests Status: RETAINED / NEEDS VERIFICATION
Maputnik	Style authoring tool Feature: Style JSON editing	Inputs: Style JSON, tokens Outputs: Style artifact	API: style registry Evidence: n/a Policy: style review policy Tile/style: style spec	Runtime/UI: Author styles before validation Tests: style diff and review tests Status: RETAINED
layer catalog	Governed layer inventory Feature: Discovery, toggles, trust badges	Inputs: LayerManifest, release/correction state Outputs: Layer list/payload	API: layer catalog API Evidence: EvidenceDrawerPayload links Policy: PolicyDecision labels Tile/style: tile/style deps	Runtime/UI: Shows released/stale/denied states Tests: catalog closure tests Status: PROPOSED
Evidence Drawer	Trust inspection surface Feature: Evidence, source, rights, receipts, caveats	Inputs: EvidenceDrawerPayload, EvidenceBundle Outputs: Drawer content	API: governed evidence API Evidence: EvidenceBundle required Policy: PolicyDecision shown Tile/style: clicked feature evidence_ref	Runtime/UI: Required before claim display Tests: click-to-bundle tests Status: CONFIRMED doctrine / PROPOSED implementation
Focus Mode	Governed AI map context Feature: Answer/abstain/deny/error with citations	Inputs: MapContextEnvelope, FocusModeRequest Outputs: FocusModeResponse, AIReceipt	API: governed AI API Evidence: resolved evidence only Policy: policy pre/post checks Tile/style: visible filtered map state	Runtime/UI: Cite-or-abstain synthesis Tests: citation validation tests Status: PROPOSED
timeline / time slider	Time-aware interaction Feature: Filter by valid/source/release time	Inputs: time index, time fields Outputs: filtered layer state	API: timeline API Evidence: evidence temporal scope Policy: stale/embargo policy Tile/style: time-sliced tiles	Runtime/UI: Trust-visible time context Tests: temporal filter tests Status: PROPOSED
MapContextEnvelope	Bounded map context object Feature: Pass visible layers, bounds, time, filters	Inputs: Map state, layer manifests Outputs: AI/API context payload	API: context API Evidence: EvidenceRefs only Policy: policy-safe context Tile/style: map state	Runtime/UI: No raw features to model Tests: scope/serialization tests Status: PROPOSED
LayerManifest	Layer contract Feature: Layer identity, source, fields, trust labels	Inputs: SourceDescriptor, tile/style refs Outputs: Layer registry row	API: catalog/layer API Evidence: evidence field mapping Policy: sensitivity/release Tile/style: tile/style deps	Runtime/UI: Layer load and drawer bind Tests: schema/source-layer tests Status: PROPOSED
StyleManifest	Style contract Feature: Style identity, style JSON, assets	Inputs: Style JSON, sprites, glyphs Outputs: Style registry row	API: style API Evidence: n/a Policy: style review Tile/style: style deps	Runtime/UI: Controlled style versions Tests: style digest tests Status: PROPOSED
TileArtifactManifest	Tile artifact contract Feature: Tile URL, digest, type, zooms, metadata	Inputs: PMTiles/MVT/COG/MBTiles Outputs: Tile registry row	API: tile API Evidence: feature evidence fields Policy: public-safe policy Tile/style: tile deps	Runtime/UI: Verified tile load Tests: pmtiles/cog/mvt checks Status: PROPOSED
MapReleaseManifest	Map publication contract Feature: Released layer/style/tile set	Inputs: manifests, receipts, policy decisions Outputs: Published map release	API: release API Evidence: evidence closure Policy: PromotionDecision Tile/style: all map deps	Runtime/UI: Rollback-ready release Tests: release/rollback tests Status: PROPOSED

component	function / feature	inputs / outputs	governed dependencies	runtime behavior / tests / status
PolicyDecision	Policy outcome Feature: allow/deny/hold/abstain/error	Inputs: policy inputs, evidence, sensitivity Outputs: decision envelope	API: policy API Evidence: evidence refs Policy: central dependency Tile/style: all deps	Runtime/UI: Trust-state labels Tests: policy fixture tests Status: PROPOSED
CitationValidationReport	Citation check Feature: Validate citations and evidence refs	Inputs: EvidenceBundle, generated answer Outputs: pass/fail report	API: citation API Evidence: required Policy: policy postcheck Tile/style: n/a	Runtime/UI: Focus Mode gate Tests: citation pass/fail tests Status: PROPOSED
AIReceipt	AI execution receipt Feature: Record model/provider/context/citations/outcome	Inputs: FocusModeRequest/Response Outputs: receipt	API: governed AI API Evidence: evidence bundle ids Policy: policy decisions Tile/style: n/a	Runtime/UI: Audit trail Tests: receipt schema tests Status: PROPOSED
rollback target	Reversal anchor Feature: Restore prior release/artifact/cache	Inputs: prior manifest/version Outputs: rollback record	API: release API Evidence: evidence lineage Policy: correction policy Tile/style: tile/style deps	Runtime/UI: Visible correction path Tests: rollback restoration tests Status: PROPOSED
cache invalidation record	Cache governance Feature: Invalidate stale artifacts safely	Inputs: release manifest/cache keys Outputs: cache receipt	API: release/cache API Evidence: n/a Policy: policy mismatch deny Tile/style: PMTiles/style/glyph/sprite	Runtime/UI: Stale badge or deny load Tests: cache invalidation tests Status: PROPOSED

9. Tile Strategy Chapter

Comparison	When it wins	When it fails	KFM trust consequence	Release / rollback consequence	Validation needed
GeoJSON vs vector tiles	GeoJSON wins for small fixtures, debug overlays, bounded runtime edits, and transparent examples. Vector tiles win for larger public layers, repeated use, and client performance.	GeoJSON fails as a dense production default; vector tiles fail when schema is wrong or evidence refs are stripped.	Neither is proof; both must resolve to evidence.	GeoJSON fixtures can change quickly; vector tiles need manifests and rollback.	CRS, size, topology, field, evidence_ref, and performance tests.
MVT vs MLT	MVT wins today for mature tooling and MapLibre support. MLT may win when KFM proves renderer/toolchain benefits.	MLT fails as a default before KFM validation and support evidence.	MVT/MLT are carriers only.	MLT adoption needs pilot release manifests and rollback to MVT.	Toolchain, rendering, schema parity, browser, mobile, and regression tests.
PMTiles vs MBTiles	PMTiles wins for static hosting, object storage, Range requests, and PWAs. MBTiles wins for local conversion, desktop, or legacy workflows.	PMTiles fails for highly transactional updates; MBTiles fails as public serverless delivery without mediation.	Both are derived artifacts.	PMTiles release pairs with manifests; MBTiles often becomes an internal build input.	pmtiles verify, metadata, digest, cache, conversion, and provenance checks.
Static tiles vs dynamic tile server	Static tiles win for immutable public-safe releases and CDN scale. Dynamic servers win for access control, steward mediation, fresh PostGIS data, or conversion.	Static tiles fail for fine-grained access; dynamic servers fail if they bypass promotion.	Public clients still need governed release state.	Static rollback swaps release manifests; dynamic rollback requires service config/version rollback.	Access logs, auth, release gates, no RAW path, and tile digest tests.
PMTiles serverless vs Martin server-mediated	PMTiles wins for offline/field/public slices. Martin or server-mediated serving wins for dynamic PostGIS-backed layers.	PMTiles fails for per-user access; Martin fails if exposed without governance.	Choose by access model, not convenience.	PMTiles rollback by artifact version; Martin rollback by schema/table/service version.	Range/CORS for PMTiles; auth, SQL, source-role, and log checks for Martin.
Vector vs raster	Vector wins for filterable features, symbols, labels, and Evidence Drawer click identity. Raster wins for imagery, modeled surfaces, and continuous fields.	Vector fails for huge complex attributes; raster fails when users need feature-level evidence.	Vector features need evidence_ref; rasters need source/cell/scene metadata and caveats.	Vector releases need schema/source-layer manifests; raster releases need COG/STAC/overviews.	MVT schema, COG range/overview, STAC, citation, and visual tests.
COG vs raster tiles	COG wins for large imagery/raster assets with range reads and overviews. Raster tiles win for broad compatibility or simple WMTS.	COG fails if hosting lacks Range; raster tiles fail with tile explosion or stale tiles.	Both are downstream carriers.	COG rollback by asset digest; raster tile rollback by tile set/version.	Range headers, overview, checksum, STAC, cache invalidation.
Terrain-enabled vs non-terrain	Terrain wins when elevation/vertical context carries evidence burden. Non-terrain wins for clarity and speed.	Terrain fails when it is decoration or obscures evidence.	3D/terrain must preserve evidence identity.	Terrain releases need DEM/tileset manifests and fallback.	3D performance, camera sync, attribution, and evidence tests.
2D default vs 3D conditional	2D default wins for clarity, accessibility, and trust. 3D conditional wins for vertical structures, volumes, terrain, or line-of-sight evidence.	3D fails when it replaces evidence with spectacle.	MapLibre 2D is default; Cesium/deck overlays conditional.	3D rollback must not break 2D release.	Justification record, performance budget, Drawer parity, visual regression.

Comparison	When it wins	When it fails	KFM trust consequence	Release / rollback consequence	Validation needed
Public-safe tiles vs restricted/internal tiles	Public-safe tiles win for normal clients. Restricted/internal tiles remain behind steward access and never substitute for public redaction.	Public-safe fails if exact sensitive geometry remains; restricted fails if reachable by normal clients.	Sensitivity transformed before publication.	Release manifests must distinguish public and restricted artifacts.	Sensitive geometry deny fixtures, redaction receipts, access-control tests.

10. Governance and Trust-Membrane Chapter

MapLibre fits into KFM as a renderer and interaction runtime downstream of trust. The governed flow is: released layer -> user click -> governed API -> EvidenceBundle -> Evidence Drawer -> Focus Mode answer / abstain / deny / error. The flow is CONFIRMED doctrine from SRC-MASTER-PREV and reinforced by current-run sources SRC-054 pp.38-43 and SRC-055 pp.44-46, pp.268-276.

Rule	Required behavior	Reason / evidence
No public RAW path	Public clients and ordinary UI surfaces never read RAW, WORK, QUARANTINE, candidate, unpublished, canonical/internal stores, or direct model output.	KFM invariant in prompt, SRC-004, SRC-005; current sources repeat clients receive released artifacts only.
No direct model client	Focus Mode uses governed API and bounded context, not direct browser traffic to model runtimes.	SRC-008, SRC-009, SRC-054 p.49.
No unreleased tile load	MapLibre loads only released, verified, public-safe tile/style/sprite/glyph artifacts.	SRC-054 pp.42-43; SRC-055 pp.36-43, pp.271-276.
No sensitive geometry hidden only by style	Sensitive geometry must be generalized/redacted before publication, with receipts.	SRC-054 p.43; SRC-055 pp.69-78, p.280.
No popup as Evidence Drawer substitute	Popups may summarize; Evidence Drawer resolves evidence and trust state.	SRC-MASTER-PREV; SRC-054 pp.38,42; SRC-055 p.4.
No uncited export	Reports, screenshots, and exports preserve citations, release manifests, and evidence metadata.	SRC-054 p.50; SRC-055 p.111, p.240.
Verification before render	Digest/signature/spec_hash checks must pass or fail closed to fallback/deny.	SRC-055 pp.29-38, pp.80-87.

11. Implementation-Ready Object Map

All object families below are PROPOSED unless a mounted repository later proves them. Directory Rules basis: SRC-004 states that folder placement encodes responsibility and lifecycle, root folders are authority boundaries, and the default schema-home convention is schemas/contracts/v1/<...>. Therefore path examples below are PROPOSED, responsibility-rooted, and subject to ADR/drift verification.

Object family	Field intent	KFM role	PROPOSED home (Directory Rules basis)
SourceDescriptor	source_id, name, source_uri, authority_role, rights, sensitivity, cadence, owner, limitations	Defines admissible source identity and role.	schemas/contracts/v1/sources/source_descriptor.schema.json
LayerManifest	layer_id, title, geometry_type, source_id, source_layer, evidence_ref_field, temporal_fields, policy_label, release_state	Binds a UI layer to governed source/evidence semantics.	schemas/contracts/v1/map/layer_manifest.schema.json
StyleManifest	style_id, version, style_json_url, sprites, glyphs, style_digest, layer_ids, release_id	Controls style identity and dependencies.	schemas/contracts/v1/map/style_manifest.schema.json
TileArtifactManifest	artifact_id, type, url, digest, zooms, bounds, format, source_layers, metadata, range_required, cors_required	Controls PMTiles/MVT/COG/MBTiles release artifacts.	schemas/contracts/v1/map/tile_artifact_manifest.schema.json
MapReleaseManifest	release_id, layer_manifests, style_manifest, tile_artifacts, release_time, supersedes, rollback_target, cache_keys	Defines a complete published map release.	schemas/contracts/v1/map/map_release_manifest.schema.json
EvidenceBundle	bundle_id, source_refs, claims, citations, spec_hash, rights_status, sensitivity, limitations, receipts	Truth-bearing evidence object that outranks generated language.	schemas/contracts/v1/evidence/evidence_bundle.schema.json
EvidenceDrawerPayload	feature_id, layer_id, evidence_bundle_refs, source summary, citations, policy state, release state, limitations	Payload shown after click/selection.	schemas/contracts/v1/ui/evidence_drawer_payload.schema.json
MapContextEnvelope	visible_layers, bounds, zoom, pitch, bearing, filters, time_window, selected_features, evidence_refs	Bounded map context sent to governed API/Focus Mode.	schemas/contracts/v1/ui/map_context_envelope.schema.json
FocusModeRequest	question, map_context_envelope, evidence_refs, policy_context, user_role	Structured request for governed synthesis.	schemas/contracts/v1/ai/focus_mode_request.schema.json
FocusModeResponse	outcome, answer, citations, abstain_reason, deny_reason, evidence_used, policy_decisions, ai_receipt_id	Finite result envelope: ANSWER/ABSTAIN/DENY/ERROR.	schemas/contracts/v1/ai/focus_mode_response.schema.json
AIReceipt	receipt_id, model_provider, model_id, context_hash, evidence_ids, citation_report_id, policy_ids, runtime, outcome	Audit trail for model execution without storing private reasoning.	schemas/contracts/v1/ai/ai_receipt.schema.json
CitationValidationReport	report_id, answer_id, citation_refs, resolved, missing, unsupported, verdict	Validates cite-or-abstain before display/export.	schemas/contracts/v1/evidence/citation_validation_report.schema.json
PolicyDecision	decision_id, input_ref, policy_id, outcome, obligations, reasons, timestamps, reviewer	Policy verdict and obligations.	schemas/contracts/v1/policy/policy_decision.schema.json
PromotionDecision	promotion_id, candidate_ref, gates, outcome, reviewer, rollback_target, reasons	Governed state transition into release.	schemas/contracts/v1/release/promotion_decision.schema.json
RunReceipt	run_id, inputs, outputs, spec_hash, tool_versions, actor, timestamps, signatures	Process memory proving what ran.	schemas/contracts/v1/proofs/run_receipt.schema.json
rollback target	release_id, prior_release_id, artifact_digests, cache_keys, correction_notice	Defines reversible release target.	schemas/contracts/v1/release/rollback_target.schema.json
stale-source fixture	source_id, stale_timestamp, expected_badge, expected_abstain_or_deny	Test fixture for stale source state.	tests/fixtures/map/stale_source_fixture.json
sensitive-geometry deny fixture	geometry, sensitivity_label, attempted_public_artifact, expected_policy_outcome	Test fixture proving exact sensitive geometry cannot publish.	tests/fixtures/map/sensitive_geometry_deny_fixture.json

12. Validation and Test Plan

Test area	Required check	Status
schema validation	Validate all SourceDescriptor, LayerManifest, StyleManifest, TileArtifactManifest, MapReleaseManifest, EvidenceBundle, EvidenceDrawerPayload, MapContextEnvelope, FocusModeRequest/Response, AIReceipt, CitationValidationReport, PolicyDecision, PromotionDecision, RunReceipt schemas.	PROPOSED
source ledger completeness	Every public layer source_id resolves to a ledgered SourceDescriptor with rights, sensitivity, authority role, cadence, and limitations.	PROPOSED
style source/layer validity	Style JSON sources and source-layer names match TileArtifactManifest vector_layers and LayerManifest source_layer.	PROPOSED
PMTiles / MVT / MLT manifest checks	Archive opens, metadata present, tile index valid, digest/spec_hash match, vector layer schema valid; MLT remains pilot until verified.	PROPOSED
COG / range request requirements	COG has overviews, correct range-readable hosting, checksum, STAC asset, and caveats.	PROPOSED
no public raw path	Browser tests confirm no RAW/WORK/QUARANTINE/candidate/canonical direct fetch.	PROPOSED
no direct model client	Frontend cannot call model runtime directly; only governed API endpoints are allowed.	PROPOSED
no unreleased tile load	Map shell refuses tile/style/sprite/glyph artifacts not listed in MapReleaseManifest.	PROPOSED
sensitive geometry deny	Exact sensitive geometry fixture fails before tile build or public release.	PROPOSED
source rights unknown blocks release	Unknown rights or missing attribution produces DENY/HOLD and no published artifact.	PROPOSED
stale source badge / abstain	Stale fixture renders stale badge and Focus Mode abstains or marks staleness.	PROPOSED
clicked feature resolves EvidenceBundle	Map click with feature_id/evidence_ref returns EvidenceDrawerPayload and resolved EvidenceBundle.	PROPOSED
citation validation pass/fail	Focus answers with missing or unsupported citations fail before display/export.	PROPOSED
Focus Mode answer with citations	ANSWER outcome requires citations and policy pass.	PROPOSED
Focus Mode abstain on missing evidence	Missing EvidenceBundle or unresolved EvidenceRef returns ABSTAIN, not fluent guess.	PROPOSED
rollback restores prior manifest	Rollback test restores previous MapReleaseManifest and cache keys.	PROPOSED
cache invalidation recorded	Release change emits cache invalidation record for PMTiles, style, glyphs, sprites, and service worker.	PROPOSED
visual regression for styles	Golden screenshots across trust states and layer toggles.	PROPOSED
keyboard navigation / accessibility	Keyboard access for layer catalog, Evidence Drawer, Focus Mode, timeline, and trust states.	PROPOSED
tile load budget	Verify tile decode/render budgets, especially mobile/PWA constraints.	PROPOSED
interaction-to-drawer latency budget	Click-to-drawer and timeline filter latency stay within budget or expose loading/abstain state.	PROPOSED

13. Anti-Patterns Register

Anti-pattern	Why it is dangerous / required mitigation
Treating MapLibre as truth authority	MapLibre renders; it does not decide truth, policy, release, or citation.
Treating tiles as proof	PMTiles/MVT/COG/MLT/raster tiles are downstream carriers and must resolve to EvidenceBundle.
Treating layer toggle as publication	Publication is a governed PromotionDecision and ReleaseManifest, not a UI state.
Hiding exact sensitive geometry with style filters	Sensitive geometry must be transformed before publication and proven absent from public artifacts.
Dense GeoJSON as production default for large datasets	Use vector tiles or server-mediated delivery for scale; keep GeoJSON bounded.
Uncontrolled style drift	Style JSON, sprites, glyphs, and icons must be versioned and tied to release manifests.
Plugin adoption without review	Protocol handlers, adapters, wrappers, and verification libraries need allowlist review.
Direct browser access to model runtime	Focus Mode must use governed API with evidence and policy checks.
AI answers based only on rendered map features	Rendered features are candidates; AI must resolve EvidenceBundles or abstain.
Uncited popups or exports	Popups/reports/screenshots need citations or abstain/deny.
MLT as default before KFM validation	MLT remains NEEDS VERIFICATION until toolchain and renderer support are proven.
3D/globe as default when 2D is clearer	3D is conditional on evidence burden, performance, and trust parity.
Mapbox-era backports without legal/security review	Legacy code or plugins require license/security review before adoption.
PMTiles as universal default	PMTiles is poor for highly transactional, real-time, fine-grained access-controlled data.
Verification theater	A trust badge or signature icon is not enough; failures must deny/fallback before render.
No-data silence	Missing tiles, stale release, or policy mismatch must be explicit, not blank.
Source-layer mismatch	A style source-layer mismatch can misrender or hide features; validate before release.
Catalog optionality	Missing DCAT/STAC/PROV closure must fail release.

14. Open Questions and Verification Backlog

Open item	Status / next verification
Package versions: MapLibre GL JS, pmtils JS, PMTiles CLI, pmtils-rs, tippecanoe, planetiler, GDAL, CesiumJS, sigstore browser libraries, test harness packages.	NEEDS VERIFICATION before implementation or public release.
Schema home: verify mounted repo ADRs before creating schemas/contracts/v1/map or alternate homes.	NEEDS VERIFICATION before implementation or public release.
Actual repo topology: apps/web, apps/governed-api, ui, web, jsonschema, policies, styles, viewer_templates, artifacts remain NEEDS VERIFICATION.	NEEDS VERIFICATION before implementation or public release.
MapLibre GL JS target version and compatibility with PMTiles protocol, worker filters, style spec, and browser security constraints.	NEEDS VERIFICATION before implementation or public release.
MLT support status, renderer/toolchain maturity, and whether it belongs in a KFM pilot only.	NEEDS VERIFICATION before implementation or public release.
Mobile/native parity: whether native or PWA clients can provide Evidence Drawer, Focus Mode, denial, abstain, and offline cache behavior equal to web.	NEEDS VERIFICATION before implementation or public release.
PMTiles hosting Range/CORS behavior for S3, CloudFront, R2, static nginx, GitHub Pages, local filesystem, and PWA bundle use.	NEEDS VERIFICATION before implementation or public release.
Martin deployment/access model and whether server-mediated tiles are justified for dynamic PostGIS layers.	NEEDS VERIFICATION before implementation or public release.
Plugin allowlist: PMTiles protocol, Cesium MVT adapters, deck.gl overlays, MapLibre wrappers, signature verification libraries.	NEEDS VERIFICATION before implementation or public release.
Style/sprite/glyph hosting and cache invalidation strategy across releases and offline service workers.	NEEDS VERIFICATION before implementation or public release.
Attribution and rights fields for external WMS/WMTS/context layers and source-specific caveats.	NEEDS VERIFICATION before implementation or public release.
Sensitive geometry transform rules by domain, zoom level, precision, aggregation, embargo, and steward review.	NEEDS VERIFICATION before implementation or public release.
Release manifest location, cache invalidation record home, and rollback target object naming after Directory Rules and repo ADR verification.	NEEDS VERIFICATION before implementation or public release.
CI workflow: exact test runner, OPA/Conftest/Cosign/Sigstore availability, and no-network fixture strategy.	NEEDS VERIFICATION before implementation or public release.
Accessibility acceptance: keyboard, screen reader, high-contrast, trust-state labels, and drawer/focus navigation.	NEEDS VERIFICATION before implementation or public release.
Performance budgets: tile load, verify-decode-render, worker processing, click-to-drawer latency, timeline filter latency, and memory caps by device tier.	NEEDS VERIFICATION before implementation or public release.
Browser-side signature verification: trusted roots, bundle format, worker boundary, detached signatures vs signed time-index, transparency proof handling.	NEEDS VERIFICATION before implementation or public release.
PMTiles base/delta strategy: thresholds for overlay, regional split, full republish, rollback complexity, and review churn.	NEEDS VERIFICATION before implementation or public release.
Cesium/3D: when production should use server-side conversion to WMTS/TMS/3D Tiles instead of client adapters.	NEEDS VERIFICATION before implementation or public release.
Export governance: screenshot/report metadata and citation preservation acceptance criteria.	NEEDS VERIFICATION before implementation or public release.

15. How to Continue This Master PDF Next Run

Attach this PDF again with any new documents.

The next run should read this PDF first as SRC-MASTER-PREV.

Preserve all idea IDs unless merged or superseded.

Update the delta table.

Add new sources to the source ledger.

Add new discoveries to the idea index.

Recompute counts.

Keep uncertainty labels visible.

16. Final PDF Self-Check

Check	Result
No obvious cut-off text	Tables are wrapped and split by category; wide matrices are carded into narrow columns. Render QA was planned after generation.
No missing source ledger	Section 4 contains SRC-MASTER-PREV, SRC-RUN-PROMPT, 39 retained PDFs, and SRC-054/SRC-055.
No missing delta section	Section 3 records prior version, current version, retained/expanded counts, conflicts, and source status.
No missing open questions	Section 14 lists open verification items including versions, schema home, hosting, plugins, sensitivity, CI, accessibility, and performance.
No unsupported implementation claims	All paths, schemas, validators, APIs, and runtime behavior are PROPOSED or NEEDS VERIFICATION unless supported by attached doctrine/source evidence.
No forgotten prior master-PDF ideas	1802 prior ideas were parsed from SRC-MASTER-PREV and carried forward as RETAINED or EXPANDED.
No external research silently introduced	No EXT-* sources were used in this run; version-sensitive operational facts are marked NEEDS VERIFICATION.

Final PDF Self-Check - v1.4

Check	Result
No obvious cut-off text in generated v1.4 packet	PASS after render inspection of generated pages. Wide tables were wrapped/cards, not squeezed.
No missing source ledger	PASS. Current-run source ledger is in Section 4; prior v1.3 ledger is preserved in Appendix A.
No missing delta section	PASS. Section 3 records the v1.3 -> v1.4 delta.
No missing open questions	PASS. Section 14 lists cumulative and SRC-056-specific verification backlog.
No unsupported implementation claims	PASS with honesty note: no mounted repo was available; paths and object families are PROPOSED unless verified elsewhere.
No forgotten prior master-PDF ideas	PASS. The v1.3 master baseline is appended unchanged as SRC-MASTER-PREV.
Prior master does not outrank original evidence	PASS. Section 3 states source authority boundary.
PDF-only output	PASS. Exactly one final user-facing PDF is written to /mnt/data with the requested filename.
Important honesty note	The current New Ideas 4-25 and 4-30 PDFs were already included in v1.3 by title and scan profile; the only newly attached source materially changing the atlas in this run is SRC-056 Pass 15 Part 2. Current repo/runtime implementation remains UNKNOWN.

End of v1.4 cumulative update.

Final PDF Self-Check - v1.5

Check	Result
No obvious cut-off text	PASS after rendering selected generated pages including cover, source ledger, idea cards, component matrix, appendix boundary, prior baseline start and end.
No missing source ledger	PASS. Section 4 includes current-run source ledger; SRC-MASTER-PREV preserves prior ledger lineage.
No missing delta section	PASS. Section 3 records prior version, new sources, retained/expanded/superseded counts and open carry-forward questions.
No missing open questions	PASS. Section 14 lists required open verification items plus new runtime, hosting, consent, probe and release-manifest checks.
No unsupported implementation claims	PASS with honesty note: no mounted repo, tests, workflows, dashboards, logs or runtime proof were available; paths and implementation objects are PROPOSED unless future repo evidence verifies them.
No forgotten prior master-PDF ideas	PASS. The complete v1.4 baseline is appended unchanged as SRC-MASTER-PREV, preserving 2162 prior idea records.
Prior master does not outrank original evidence	PASS. The continuity section says source evidence can refine, conflict with or supersede prior entries.
PDF-only output	PASS. Exactly one user-facing PDF with the requested filename is in /mnt/data.
Important honesty note	The current run adds 95 delta records from New Ideas 5.pdf and New Ideas 5-8-26.pdf. Current repo/runtime implementation remains UNKNOWN.

End of v1.5 cumulative update.